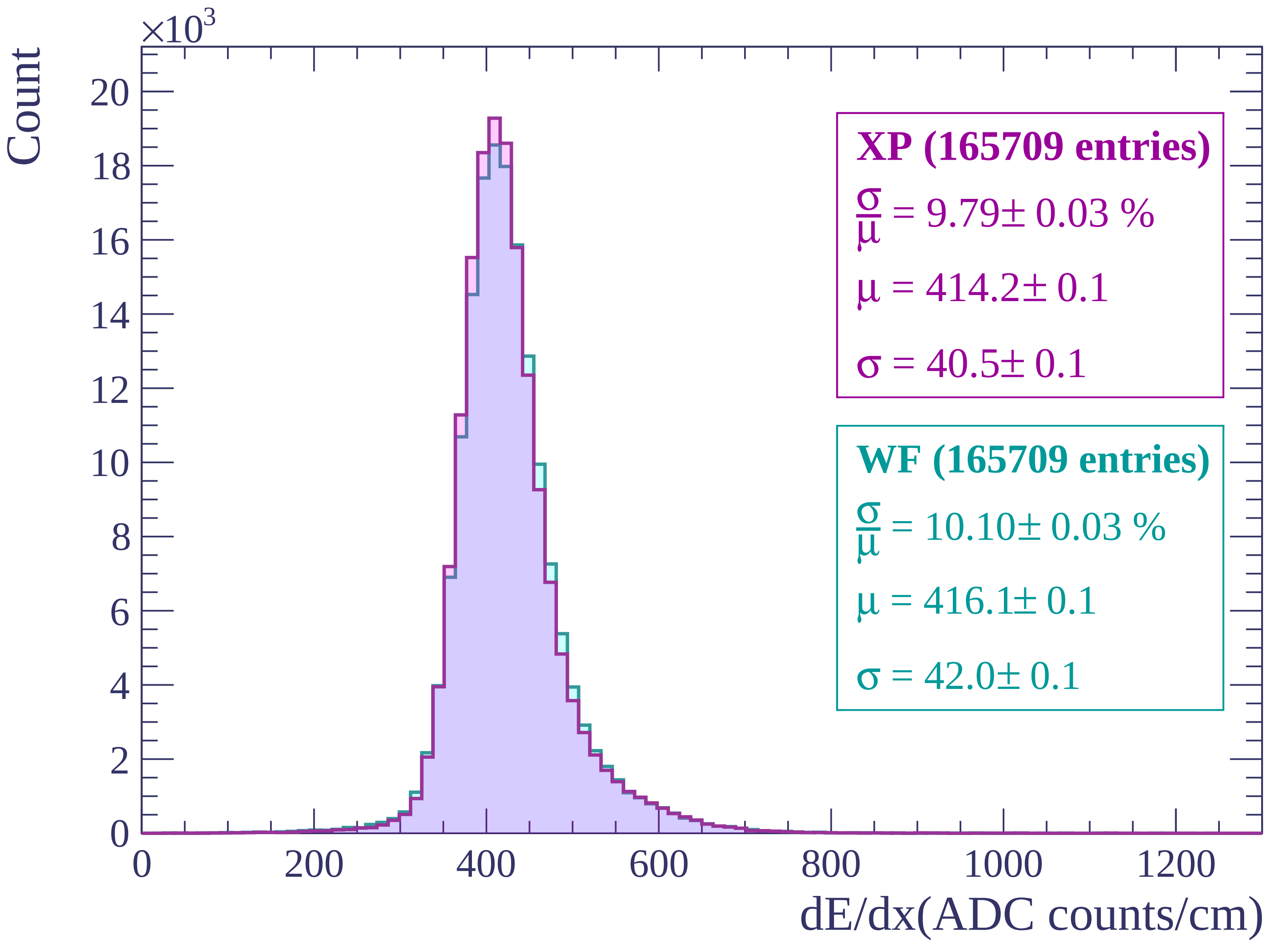
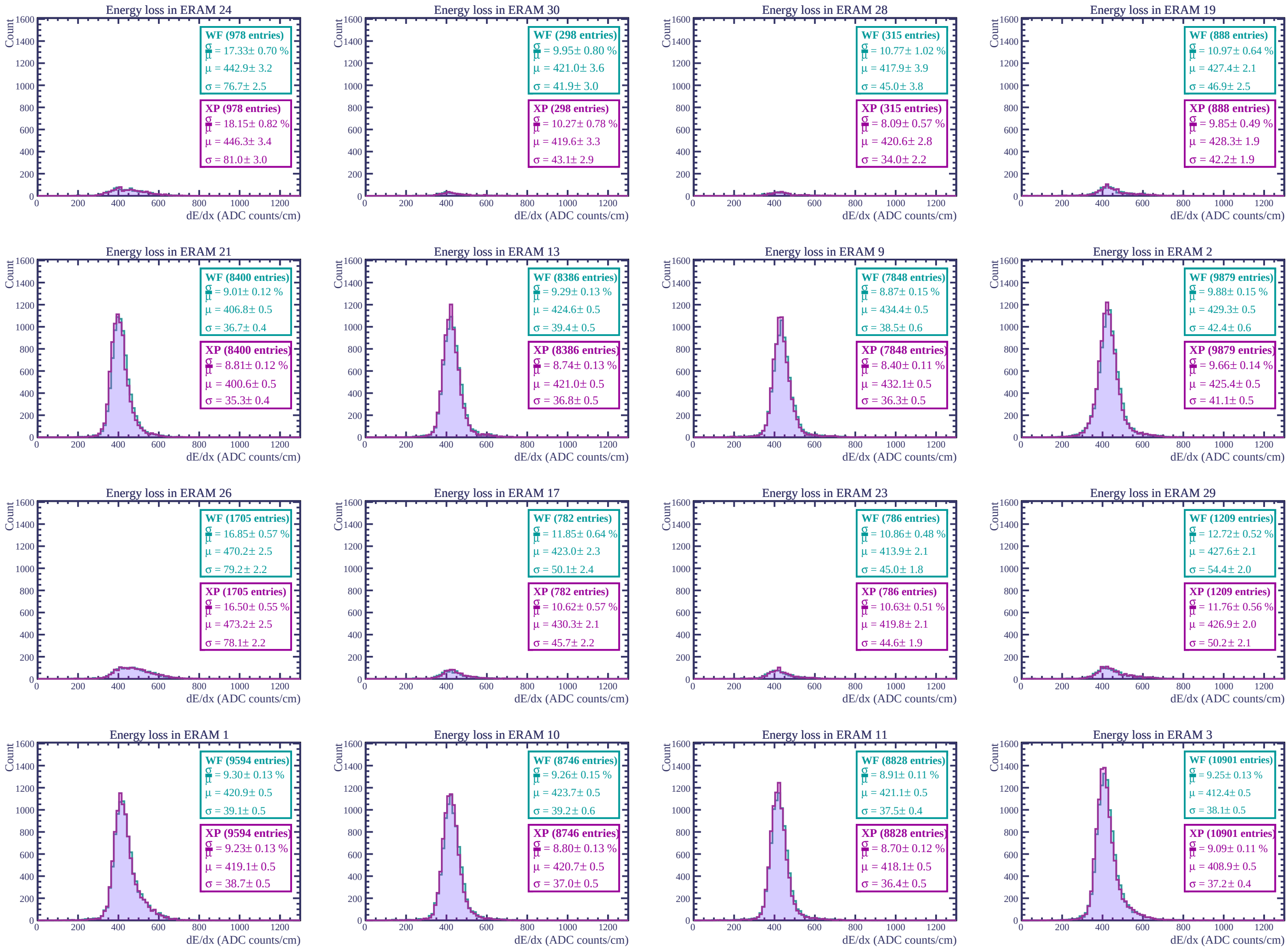
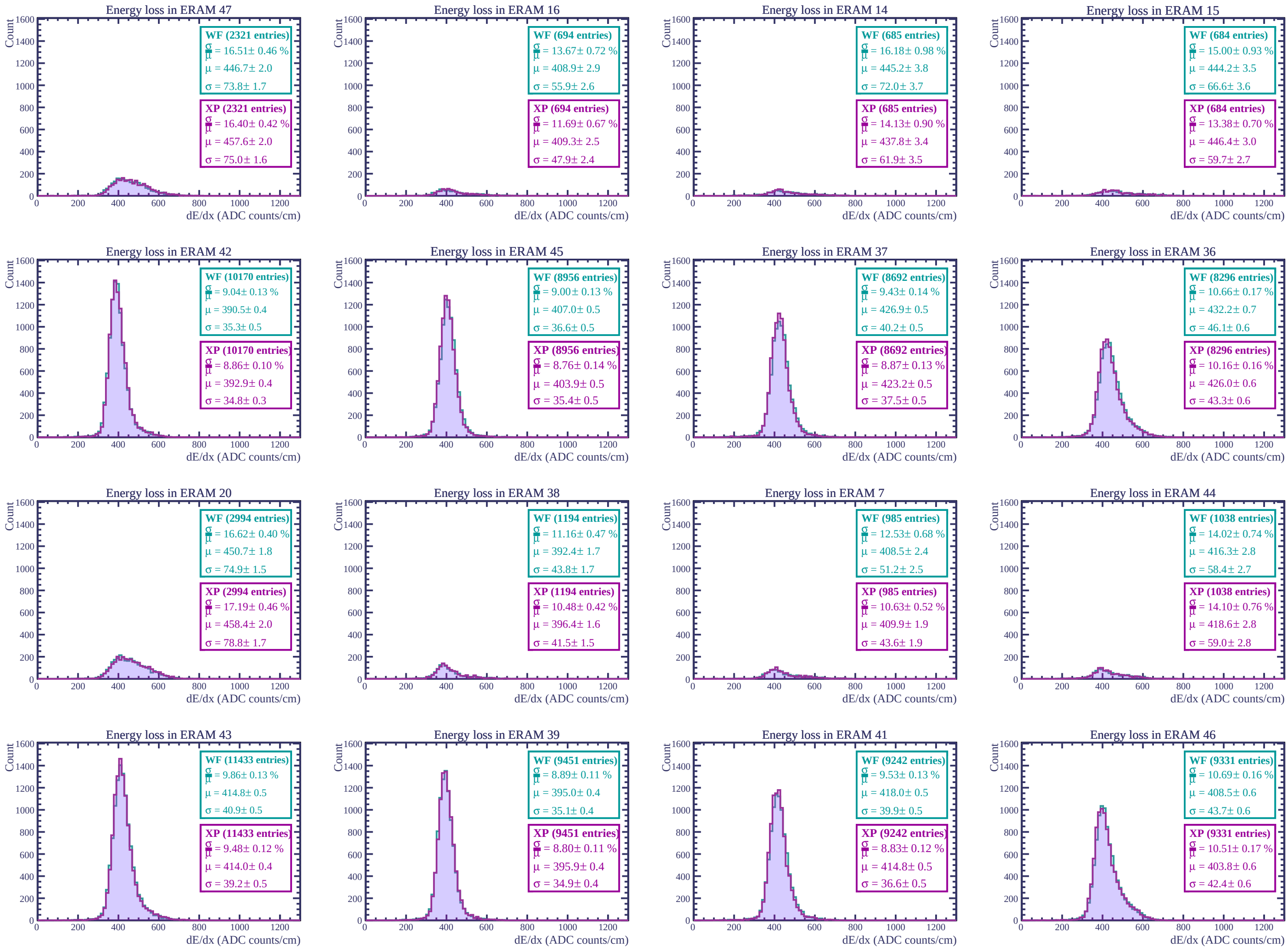
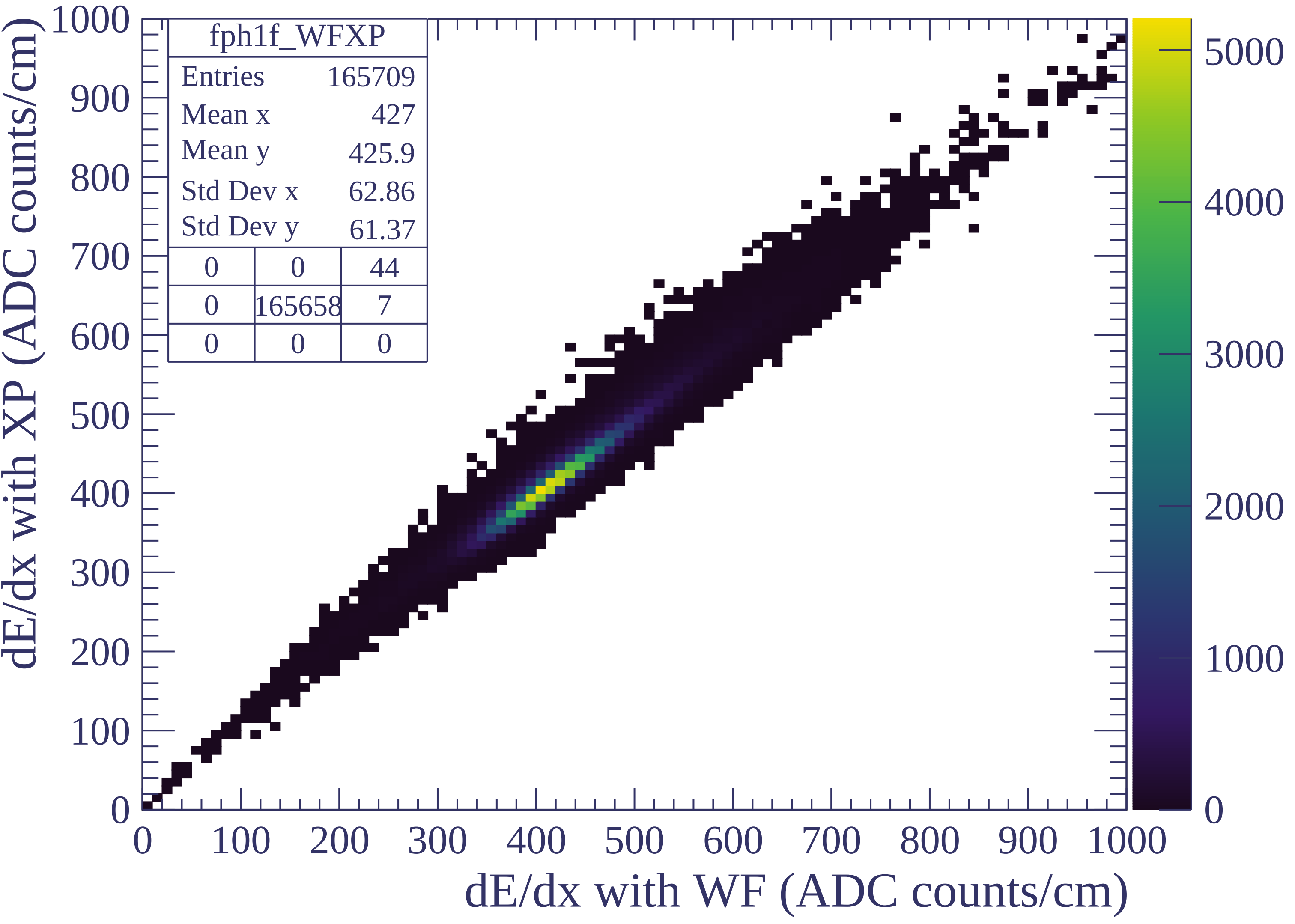
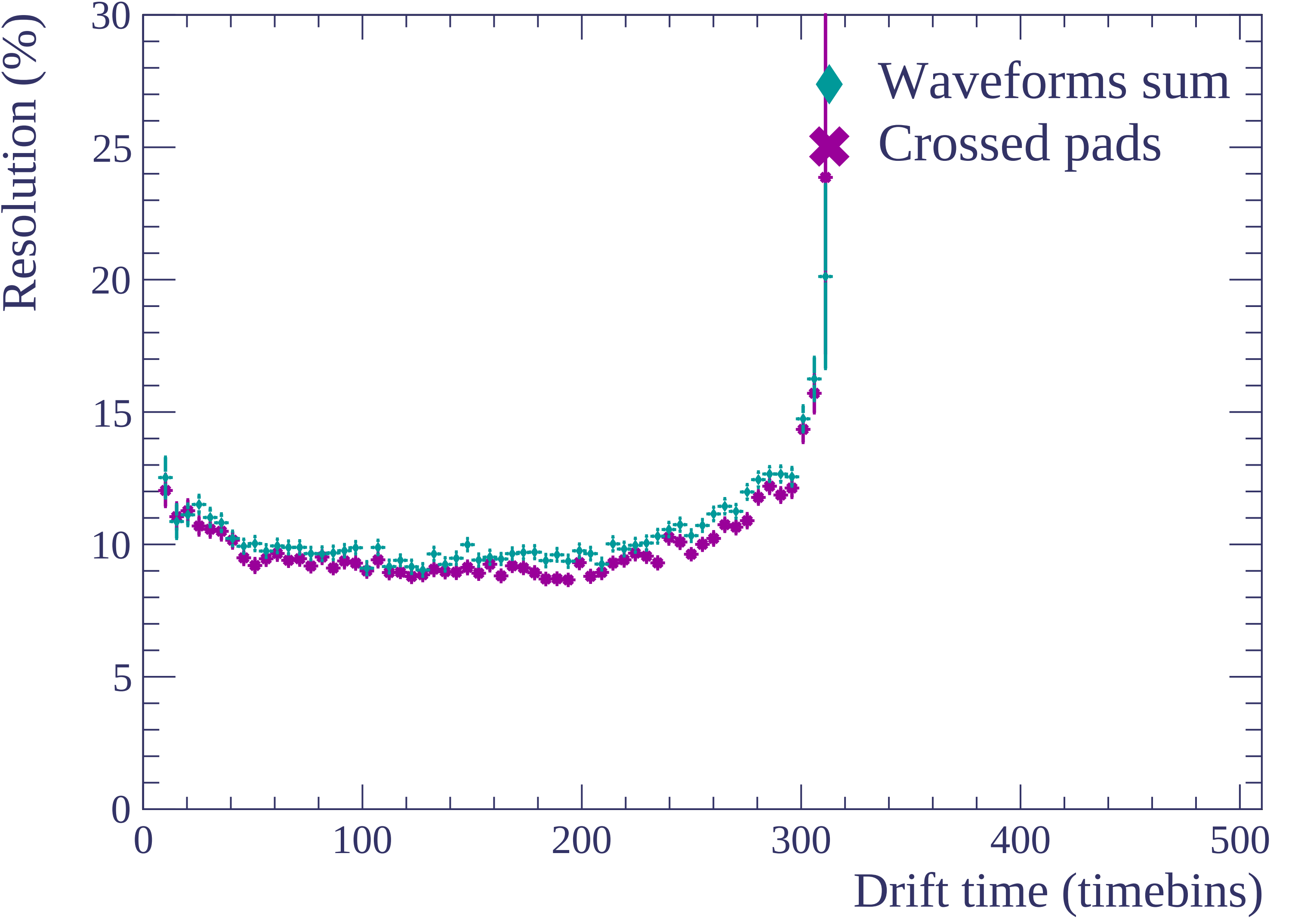

Count
×10^3
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (165709 entries)
σ/μ = 9.79± 0.03 %
μ = 414.2± 0.1
σ = 40.5± 0.1
WF (165709 entries)
σ/μ = 10.10± 0.03 %
μ = 416.1± 0.1
σ = 42.0± 0.1

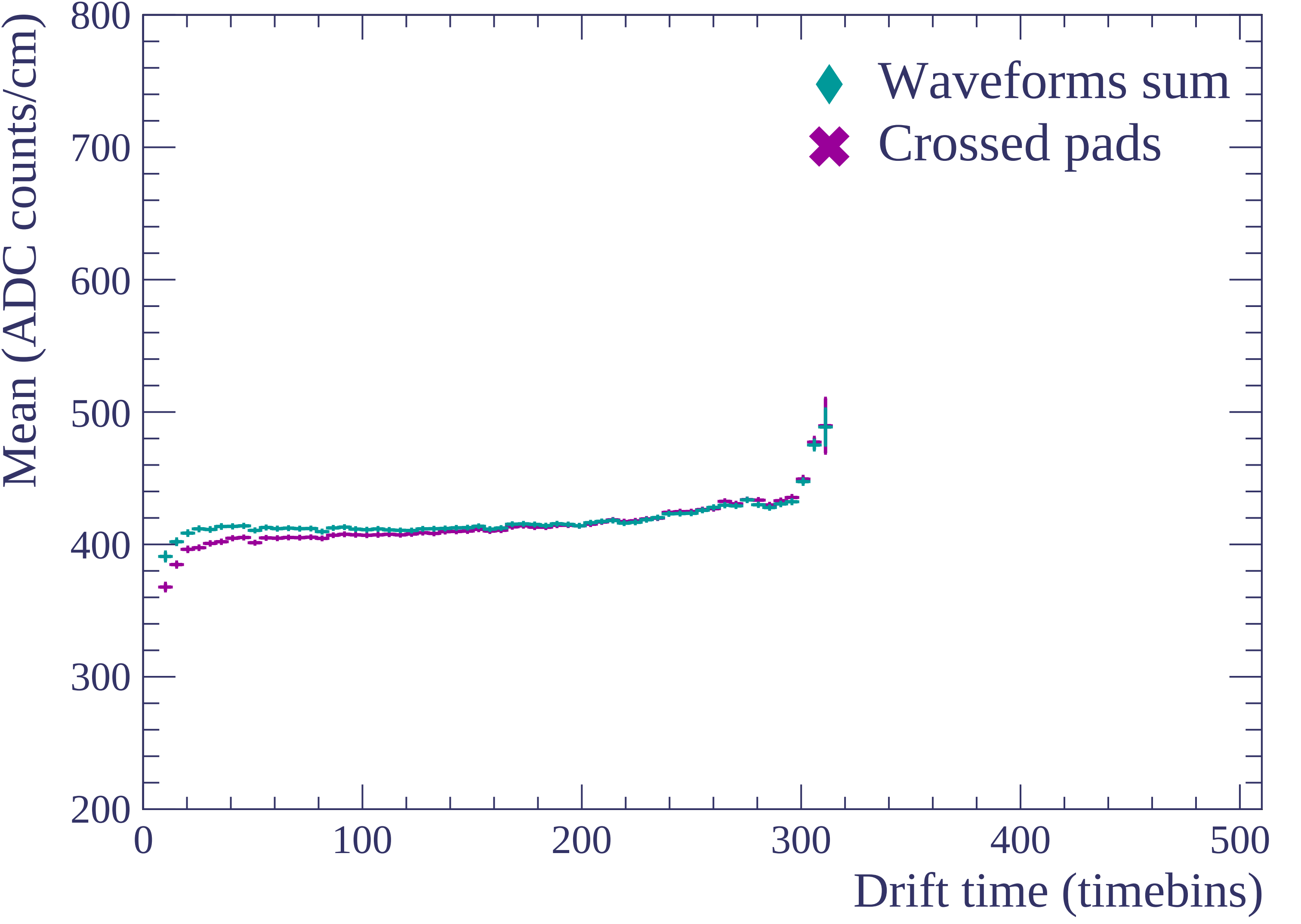

Energy loss in ERAM 24
WF (978 entries)
σ/μ = 17.33± 0.70 %
μ = 442.9± 3.2
σ = 76.7± 2.5
XP (978 entries)
σ/μ = 18.15± 0.82 %
μ = 446.3± 3.4
σ = 81.0± 3.0
Energy loss in ERAM 30
WF (298 entries)
σ/μ = 9.95± 0.80 %
μ = 421.0± 3.6
σ = 41.9± 3.0
XP (298 entries)
σ/μ = 10.27± 0.78 %
μ = 419.6± 3.3
σ = 43.1± 2.9
Energy loss in ERAM 28
WF (315 entries)
σ/μ = 10.77± 1.02 %
μ = 417.9± 3.9
σ = 45.0± 3.8
XP (315 entries)
σ/μ = 8.09± 0.57 %
μ = 420.6± 2.8
σ = 34.0± 2.2
Energy loss in ERAM 19
WF (888 entries)
σ/μ = 10.97± 0.64 %
μ = 427.4± 2.1
σ = 46.9± 2.5
XP (888 entries)
σ/μ = 9.85± 0.49 %
μ = 428.3± 1.9
σ = 42.2± 1.9
Energy loss in ERAM 21
WF (8400 entries)
σ/μ = 9.01± 0.12 %
μ = 406.8± 0.5
σ = 36.7± 0.4
XP (8400 entries)
σ/μ = 8.81± 0.12 %
μ = 400.6± 0.5
σ = 35.3± 0.4
Energy loss in ERAM 13
WF (8386 entries)
σ/μ = 9.29± 0.13 %
μ = 424.6± 0.5
σ = 39.4± 0.5
XP (8386 entries)
σ/μ = 8.74± 0.13 %
μ = 421.0± 0.5
σ = 36.8± 0.5
Energy loss in ERAM 9
WF (7848 entries)
σ/μ = 8.87± 0.15 %
μ = 434.4± 0.5
σ = 38.5± 0.6
XP (7848 entries)
σ/μ = 8.40± 0.11 %
μ = 432.1± 0.5
σ = 36.3± 0.5
Energy loss in ERAM 2
WF (9879 entries)
σ/μ = 9.88± 0.15 %
μ = 429.3± 0.5
σ = 42.4± 0.6
XP (9879 entries)
σ/μ = 9.66± 0.14 %
μ = 425.4± 0.5
σ = 41.1± 0.5
Energy loss in ERAM 26
WF (1705 entries)
σ/μ = 16.85± 0.57 %
μ = 470.2± 2.5
σ = 79.2± 2.2
XP (1705 entries)
σ/μ = 16.50± 0.55 %
μ = 473.2± 2.5
σ = 78.1± 2.2
Energy loss in ERAM 17
WF (782 entries)
σ/μ = 11.85± 0.64 %
μ = 423.0± 2.3
σ = 50.1± 2.4
XP (782 entries)
σ/μ = 10.62± 0.57 %
μ = 430.3± 2.1
σ = 45.7± 2.2
Energy loss in ERAM 23
WF (786 entries)
σ/μ = 10.86± 0.48 %
μ = 413.9± 2.1
σ = 45.0± 1.8
XP (786 entries)
σ/μ = 10.63± 0.51 %
μ = 419.8± 2.1
σ = 44.6± 1.9
Energy loss in ERAM 29
WF (1209 entries)
σ/μ = 12.72± 0.52 %
μ = 427.6± 2.1
σ = 54.4± 2.0
XP (1209 entries)
σ/μ = 11.76± 0.56 %
μ = 426.9± 2.0
σ = 50.2± 2.1
Energy loss in ERAM 1
WF (9594 entries)
σ/μ = 9.30± 0.13 %
μ = 420.9± 0.5
σ = 39.1± 0.5
XP (9594 entries)
σ/μ = 9.23± 0.13 %
μ = 419.1± 0.5
σ = 38.7± 0.5
Energy loss in ERAM 10
WF (8746 entries)
σ/μ = 9.26± 0.15 %
μ = 423.7± 0.5
σ = 39.2± 0.6
XP (8746 entries)
σ/μ = 8.80± 0.13 %
μ = 420.7± 0.5
σ = 37.0± 0.5
Energy loss in ERAM 11
WF (8828 entries)
σ/μ = 8.91± 0.11 %
μ = 421.1± 0.5
σ = 37.5± 0.4
XP (8828 entries)
σ/μ = 8.70± 0.12 %
μ = 418.1± 0.5
σ = 36.4± 0.5
Energy loss in ERAM 3
WF (10901 entries)
σ/μ = 9.25± 0.13 %
μ = 412.4± 0.5
σ = 38.1± 0.5
XP (10901 entries)
σ/μ = 9.09± 0.11 %
μ = 408.9± 0.5
σ = 37.2± 0.4
Count
dE/dx (ADC counts/cm)



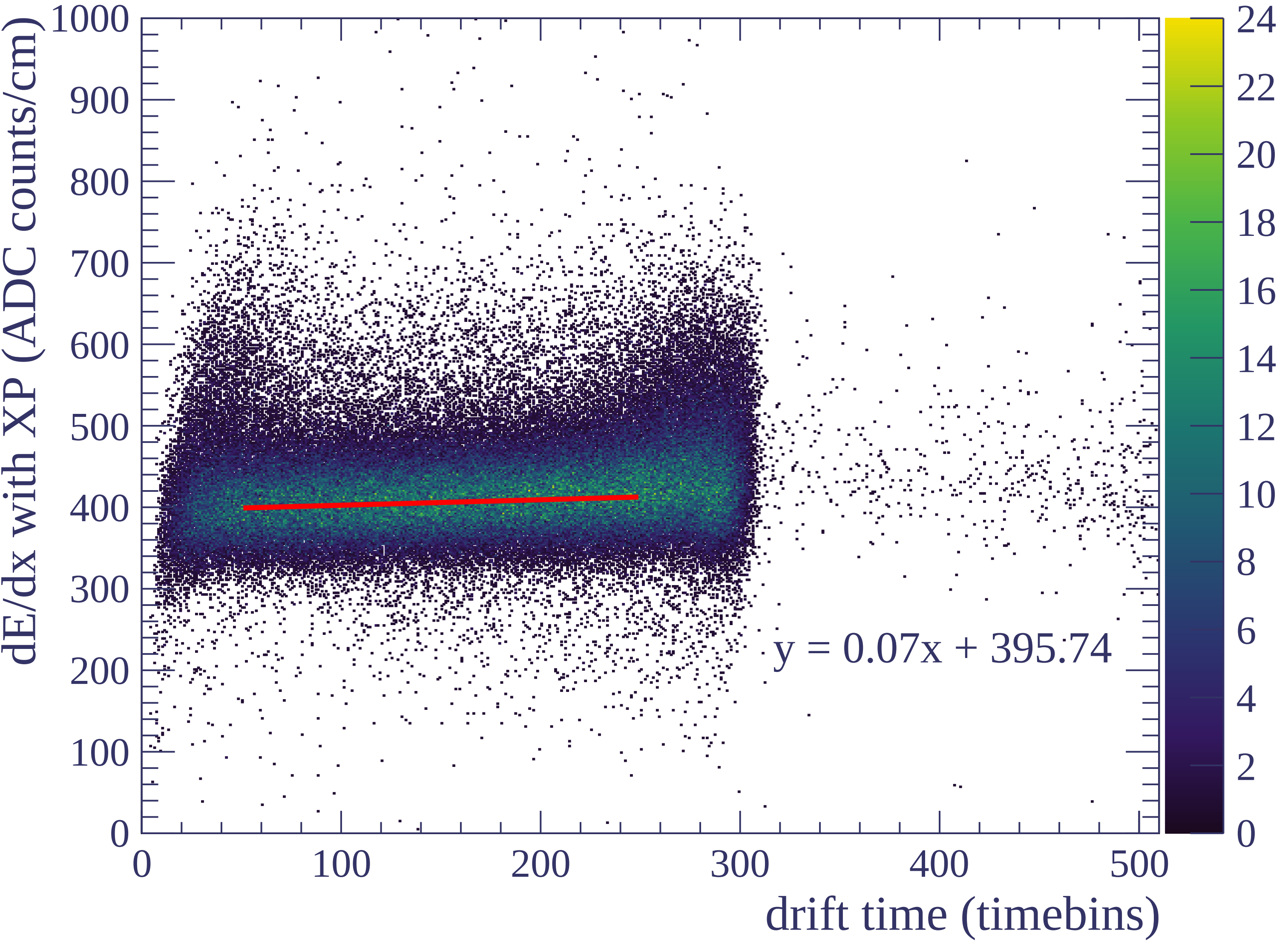
Energy loss in ERAM 47
WF (2321 entries)
σ/μ = 16.51± 0.46 %
μ = 446.7± 2.0
σ = 73.8± 1.7
XP (2321 entries)
σ/μ = 16.40± 0.42 %
μ = 457.6± 2.0
σ = 75.0± 1.6
Energy loss in ERAM 16
WF (694 entries)
σ/μ = 13.67± 0.72 %
μ = 408.9± 2.9
σ = 55.9± 2.6
XP (694 entries)
σ/μ = 11.69± 0.67 %
μ = 409.3± 2.5
σ = 47.9± 2.4
Energy loss in ERAM 14
WF (685 entries)
σ/μ = 16.18± 0.98 %
μ = 445.2± 3.8
σ = 72.0± 3.7
XP (685 entries)
σ/μ = 14.13± 0.90 %
μ = 437.8± 3.4
σ = 61.9± 3.5
Energy loss in ERAM 15
WF (684 entries)
σ/μ = 15.00± 0.93 %
μ = 444.2± 3.5
σ = 66.6± 3.6
XP (684 entries)
σ/μ = 13.38± 0.70 %
μ = 446.4± 3.0
σ = 59.7± 2.7
Energy loss in ERAM 42
WF (10170 entries)
σ/μ = 9.04± 0.13 %
μ = 390.5± 0.4
σ = 35.3± 0.5
XP (10170 entries)
σ/μ = 8.86± 0.10 %
μ = 392.9± 0.4
σ = 34.8± 0.3
Energy loss in ERAM 45
WF (8956 entries)
σ/μ = 9.00± 0.13 %
μ = 407.0± 0.5
σ = 36.6± 0.5
XP (8956 entries)
σ/μ = 8.76± 0.14 %
μ = 403.9± 0.5
σ = 35.4± 0.5
Energy loss in ERAM 37
WF (8692 entries)
σ/μ = 9.43± 0.14 %
μ = 426.9± 0.5
σ = 40.2± 0.5
XP (8692 entries)
σ/μ = 8.87± 0.13 %
μ = 423.2± 0.5
σ = 37.5± 0.5
Energy loss in ERAM 36
WF (8296 entries)
σ/μ = 10.66± 0.17 %
μ = 432.2± 0.7
σ = 46.1± 0.6
XP (8296 entries)
σ/μ = 10.16± 0.16 %
μ = 426.0± 0.6
σ = 43.3± 0.6
Energy loss in ERAM 20
WF (2994 entries)
σ/μ = 16.62± 0.40 %
μ = 450.7± 1.8
σ = 74.9± 1.5
XP (2994 entries)
σ/μ = 17.19± 0.46 %
μ = 458.4± 2.0
σ = 78.8± 1.7
Energy loss in ERAM 38
WF (1194 entries)
σ/μ = 11.16± 0.47 %
μ = 392.4± 1.7
σ = 43.8± 1.7
XP (1194 entries)
σ/μ = 10.48± 0.42 %
μ = 396.4± 1.6
σ = 41.5± 1.5
Energy loss in ERAM 7
WF (985 entries)
σ/μ = 12.53± 0.68 %
μ = 408.5± 2.4
σ = 51.2± 2.5
XP (985 entries)
σ/μ = 10.63± 0.52 %
μ = 409.9± 1.9
σ = 43.6± 1.9
Energy loss in ERAM 44
WF (1038 entries)
σ/μ = 14.02± 0.74 %
μ = 416.3± 2.8
σ = 58.4± 2.7
XP (1038 entries)
σ/μ = 14.10± 0.76 %
μ = 418.6± 2.8
σ = 59.0± 2.8
Energy loss in ERAM 43
WF (11433 entries)
σ/μ = 9.86± 0.13 %
μ = 414.8± 0.5
σ = 40.9± 0.5
XP (11433 entries)
σ/μ = 9.48± 0.12 %
μ = 414.0± 0.4
σ = 39.2± 0.5
Energy loss in ERAM 39
WF (9451 entries)
σ/μ = 8.89± 0.11 %
μ = 395.0± 0.4
σ = 35.1± 0.4
XP (9451 entries)
σ/μ = 8.80± 0.11 %
μ = 395.9± 0.4
σ = 34.9± 0.4
Energy loss in ERAM 41
WF (9242 entries)
σ/μ = 9.53± 0.13 %
μ = 418.0± 0.5
σ = 39.9± 0.5
XP (9242 entries)
σ/μ = 8.83± 0.12 %
μ = 414.8± 0.5
σ = 36.6± 0.5
Energy loss in ERAM 46
WF (9331 entries)
σ/μ = 10.69± 0.16 %
μ = 408.5± 0.6
σ = 43.7± 0.6
XP (9331 entries)
σ/μ = 10.51± 0.17 %
μ = 403.8± 0.6
σ = 42.4± 0.6
Count
dE/dx (ADC counts/cm)


fph1f_WFXP
Entries 165709
Mean x 427
Mean y 425.9
Std Dev x 62.86
Std Dev y 61.37
0 0 44
0 165658 7
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
0
1000
2000
3000
4000
5000

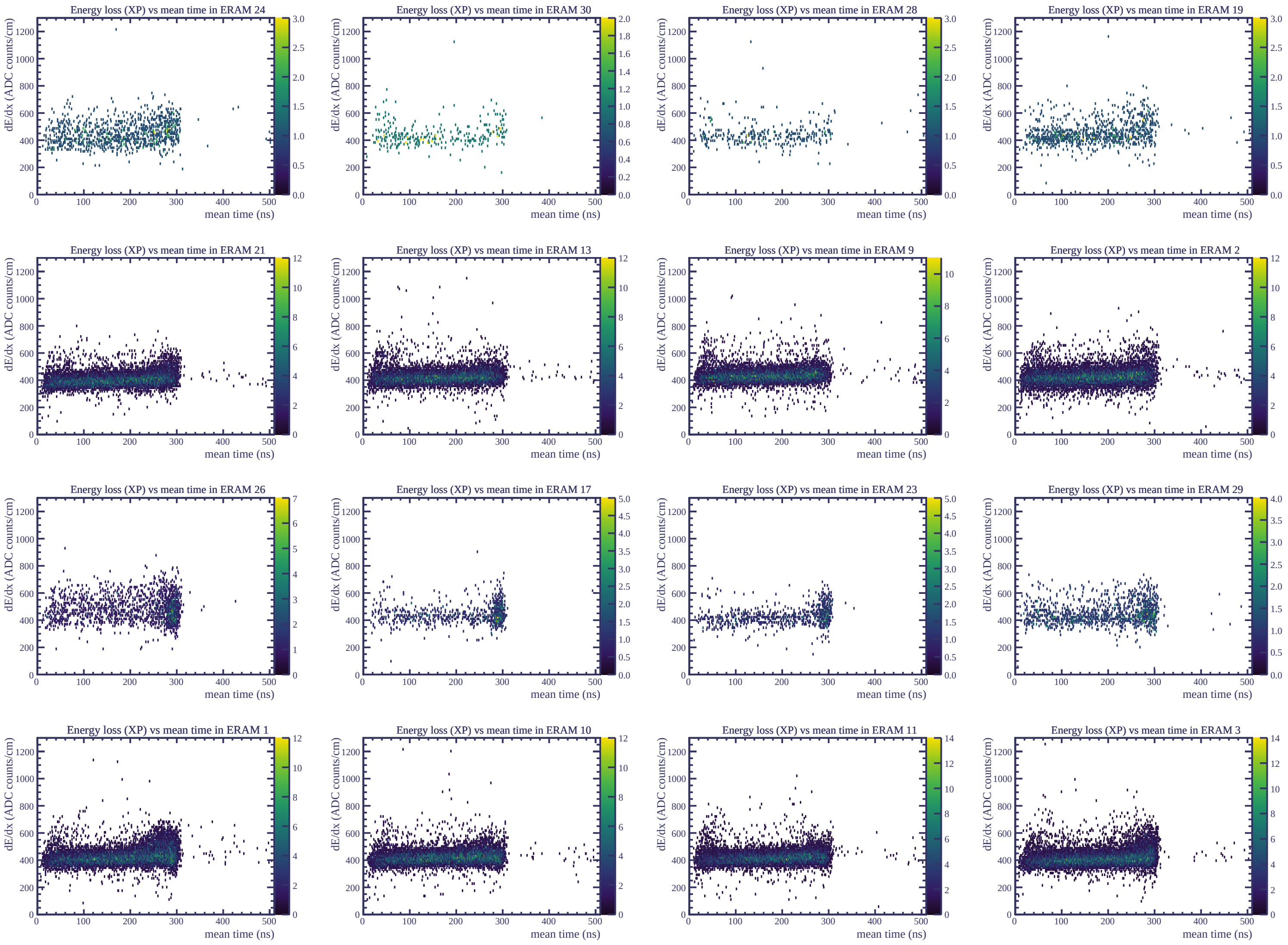

Resolution (%)
Drift time (timebins)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
0
100
200
300
400
500

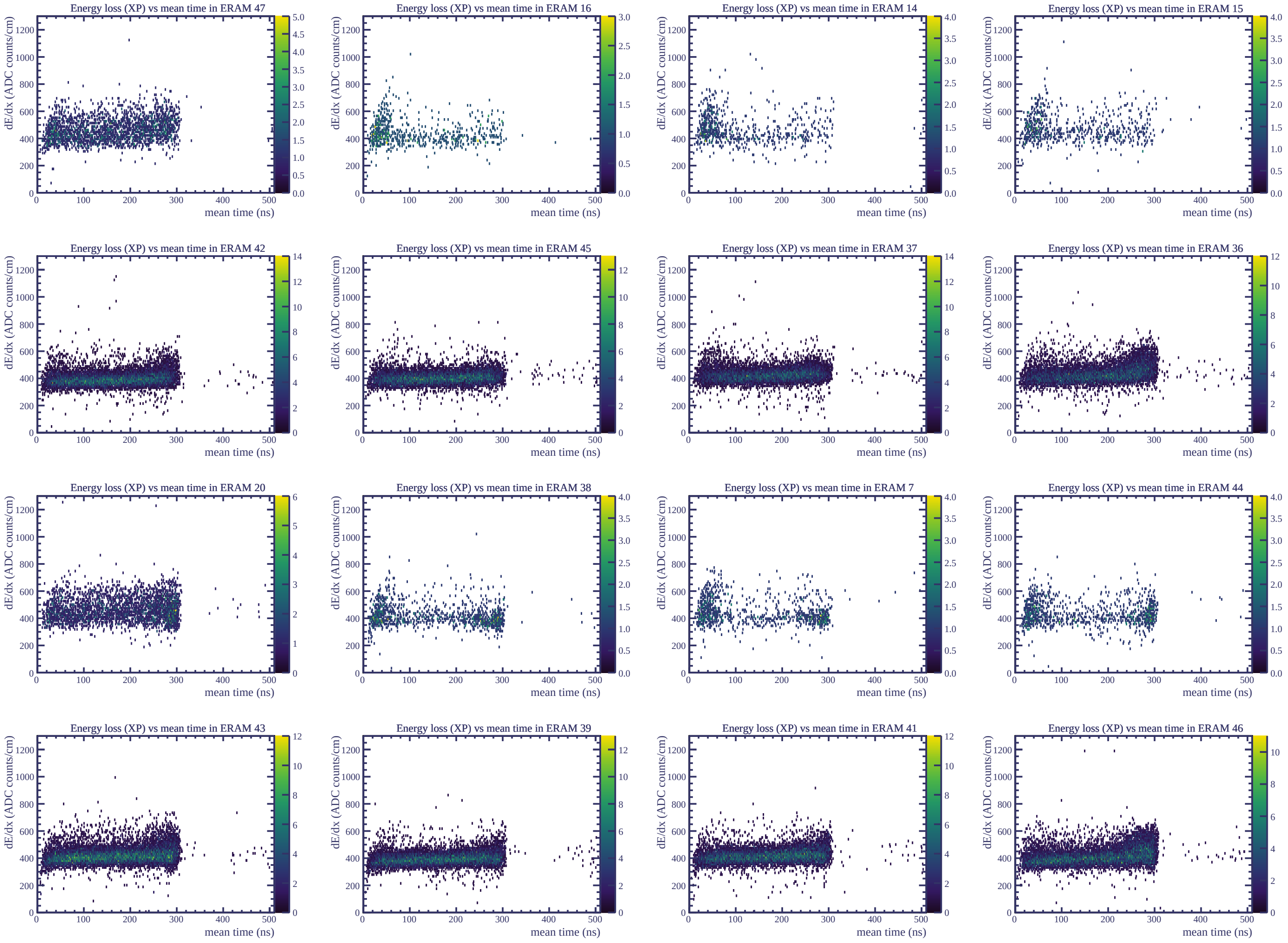


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

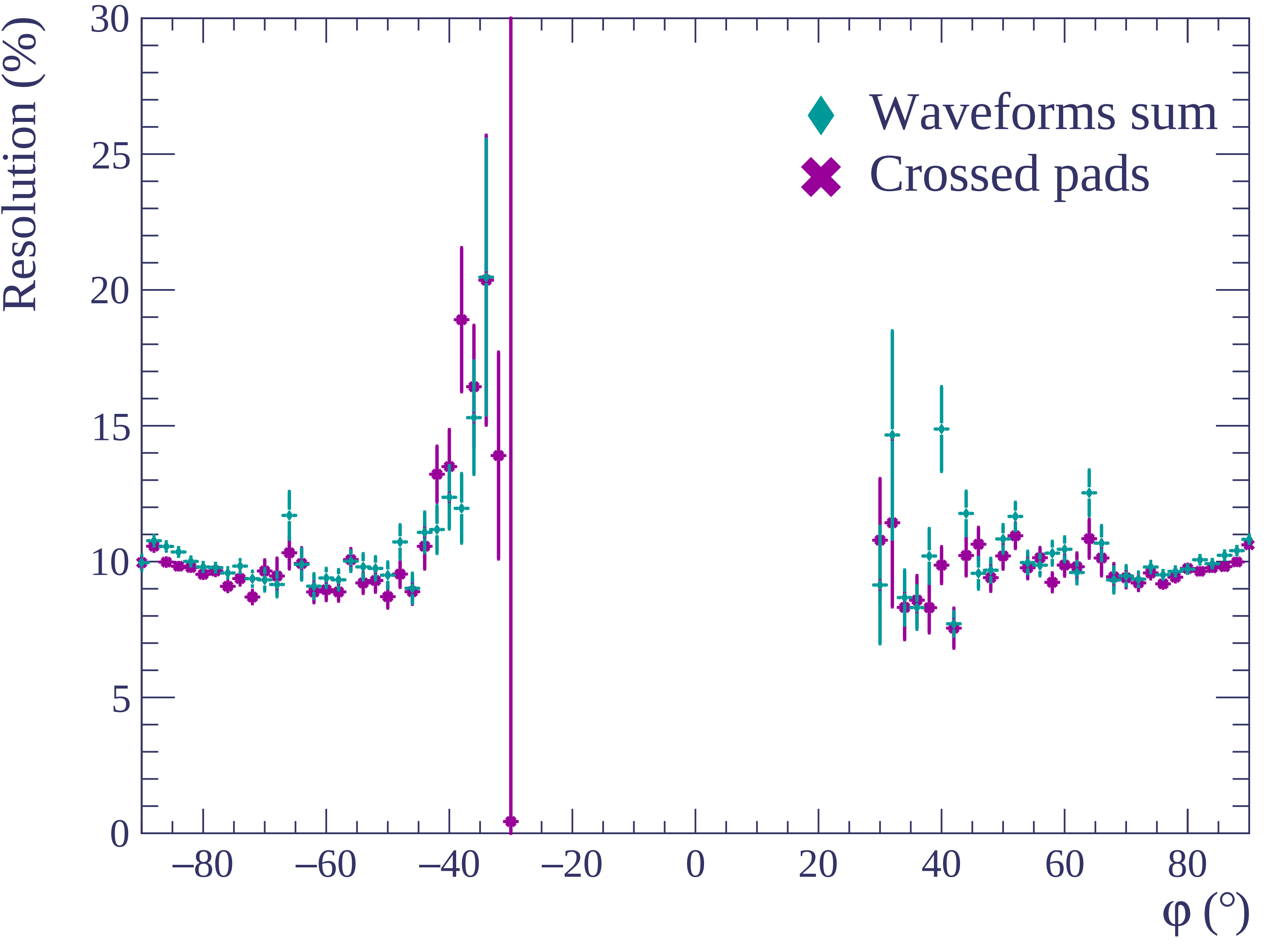


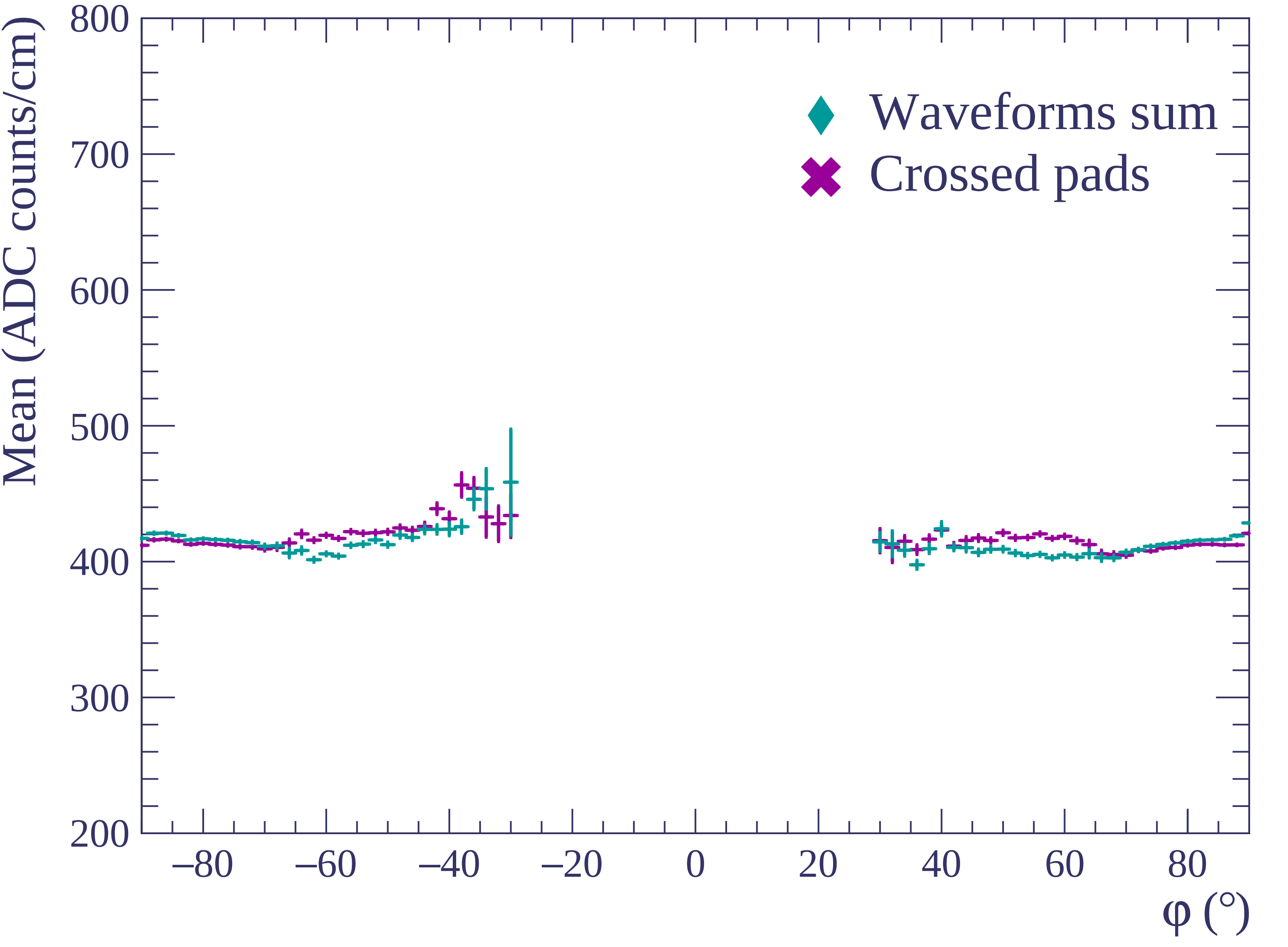
dE/dx with XP (ADC counts/cm)
drift time (timebins)
y = 0.07x + 395.74
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16
18
20
22
24

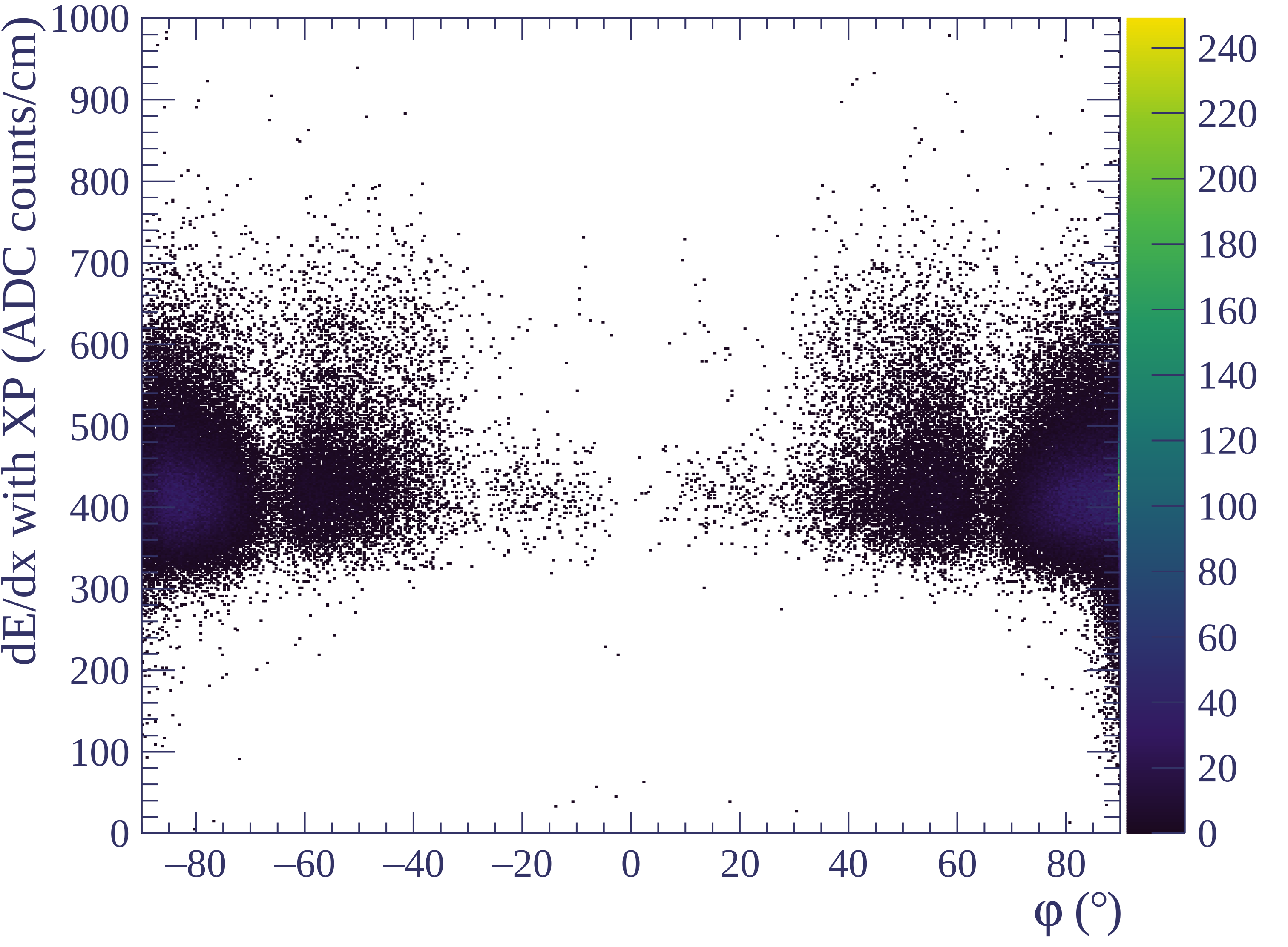

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)



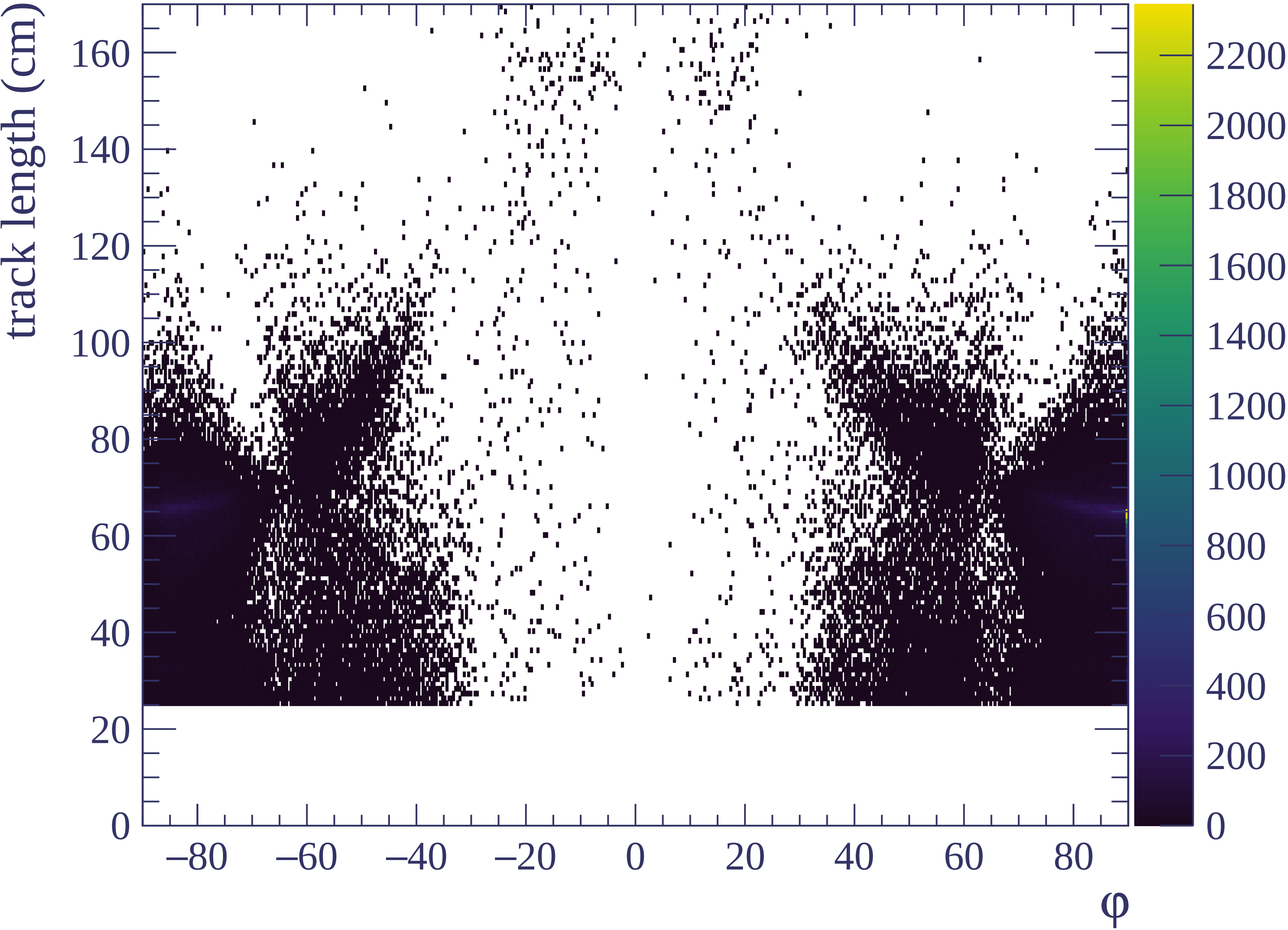
Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

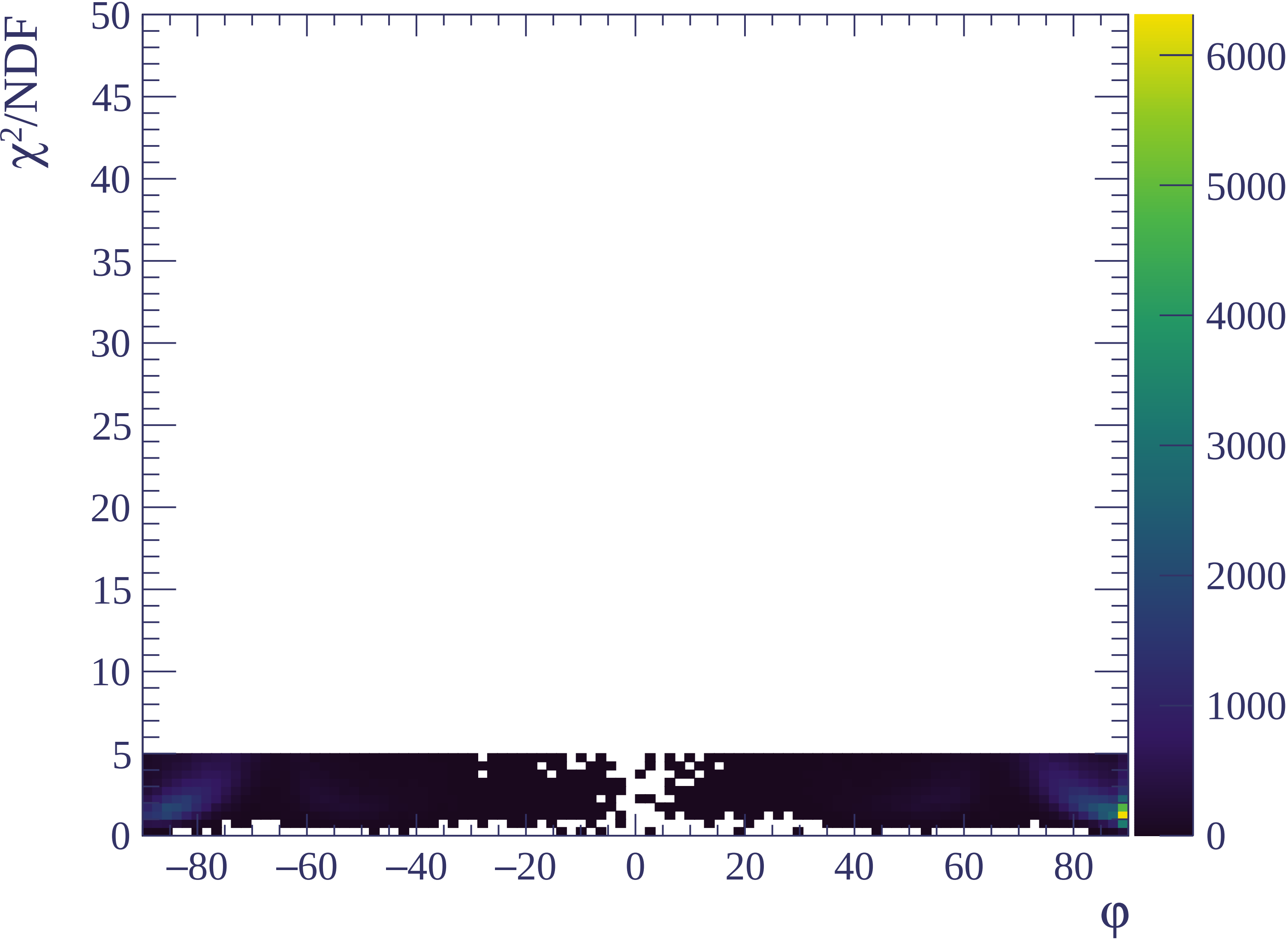

Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

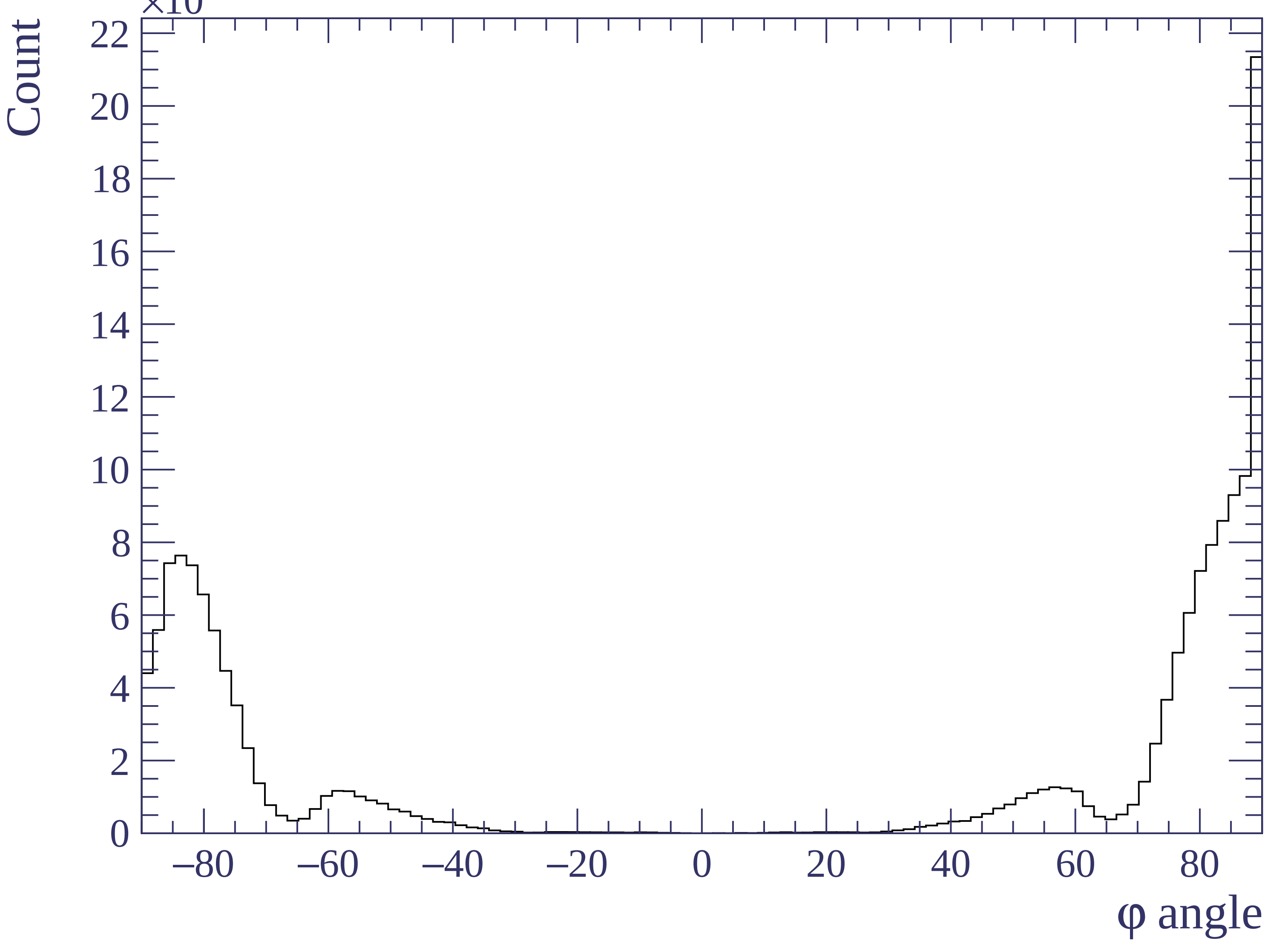


dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
20
40
60
80
100
120
140
160
180
200
220
240

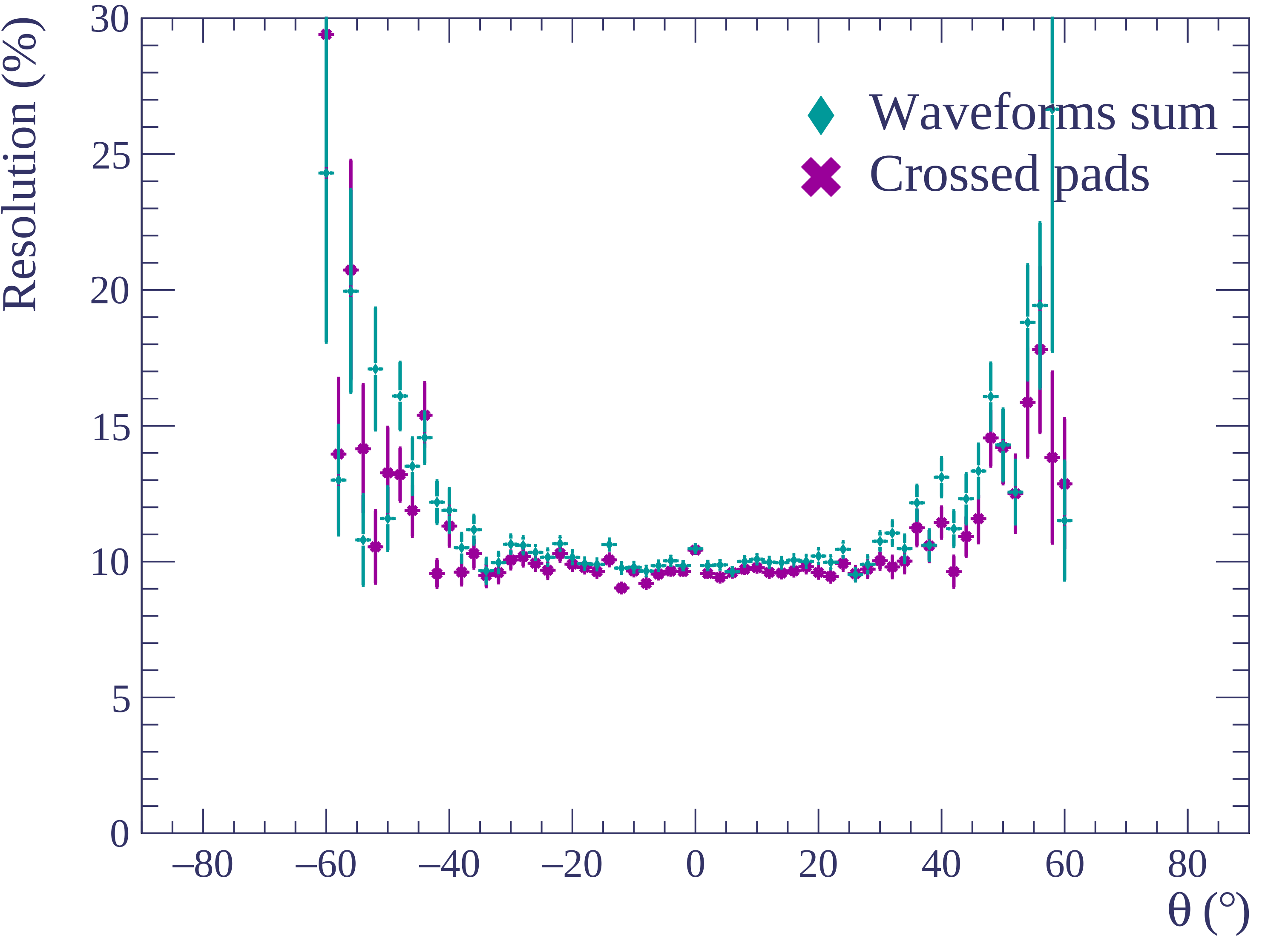

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

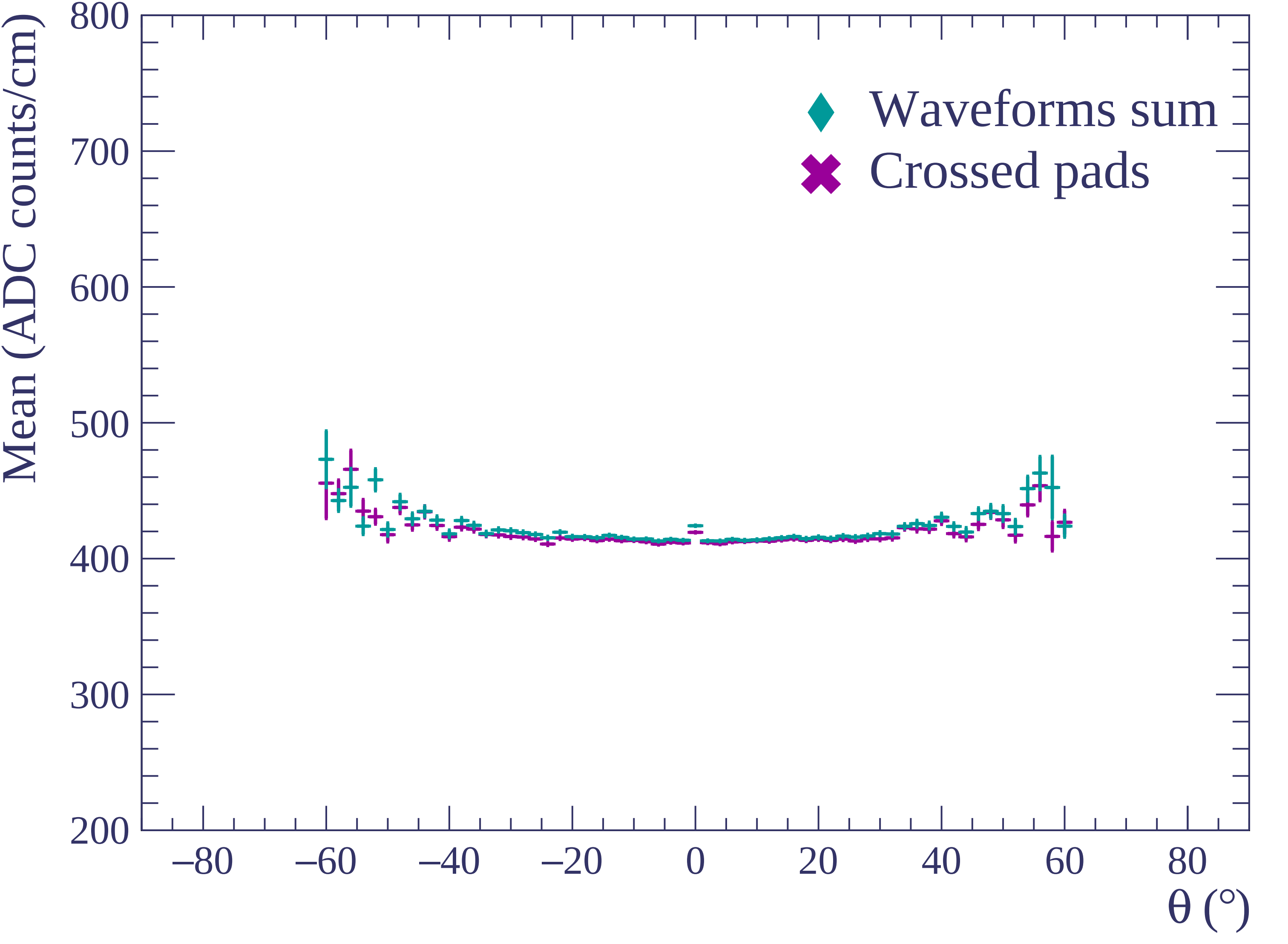


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
6000
5000
4000
3000
2000
1000
0

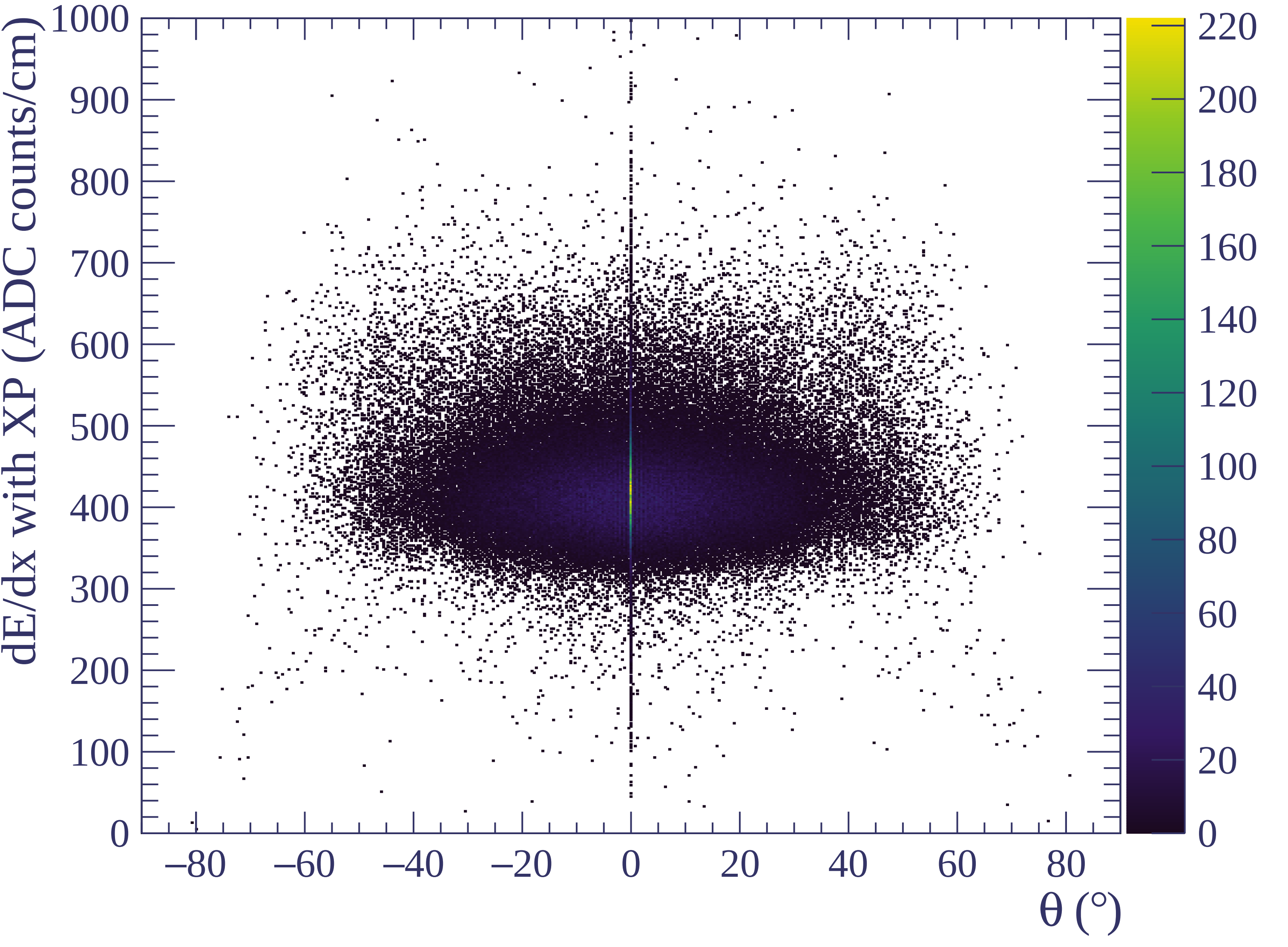


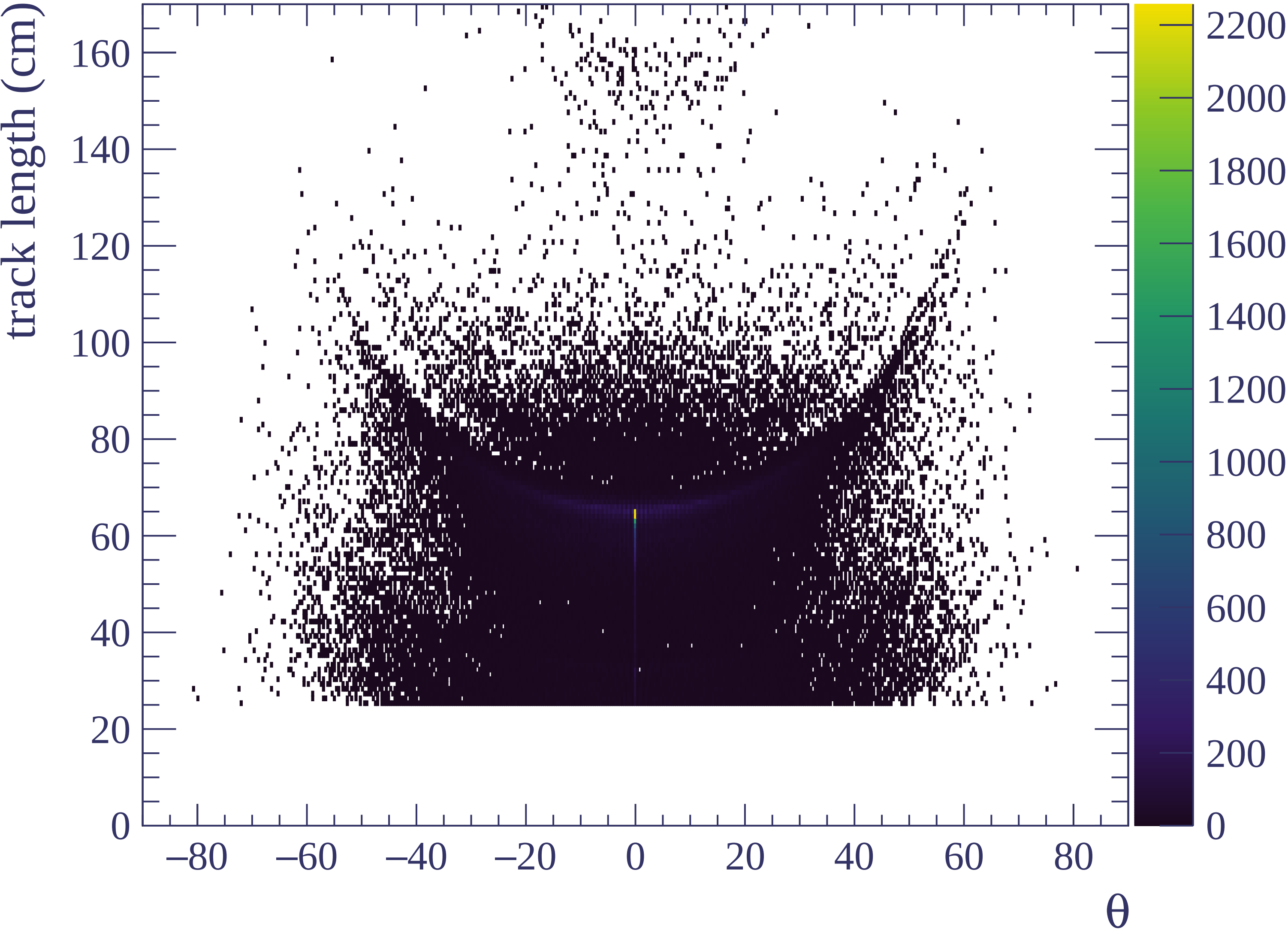
Count
×10
22
20
18
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

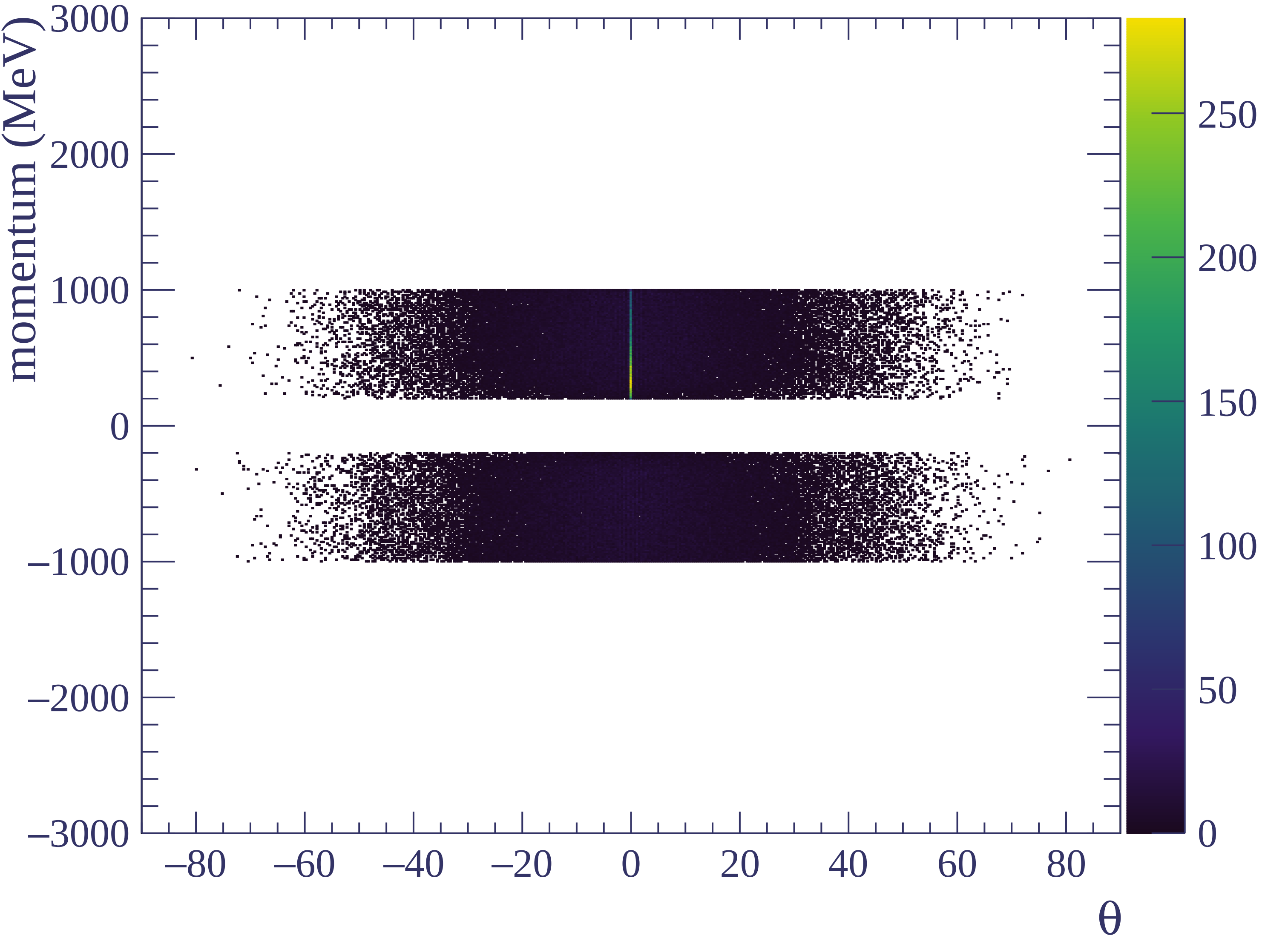

Resolution (%)
30
25
20
15
10
5
0
Waveforms sum
Crossed pads
−80
−60
−40
−20
0
20
40
60
80
θ (°)

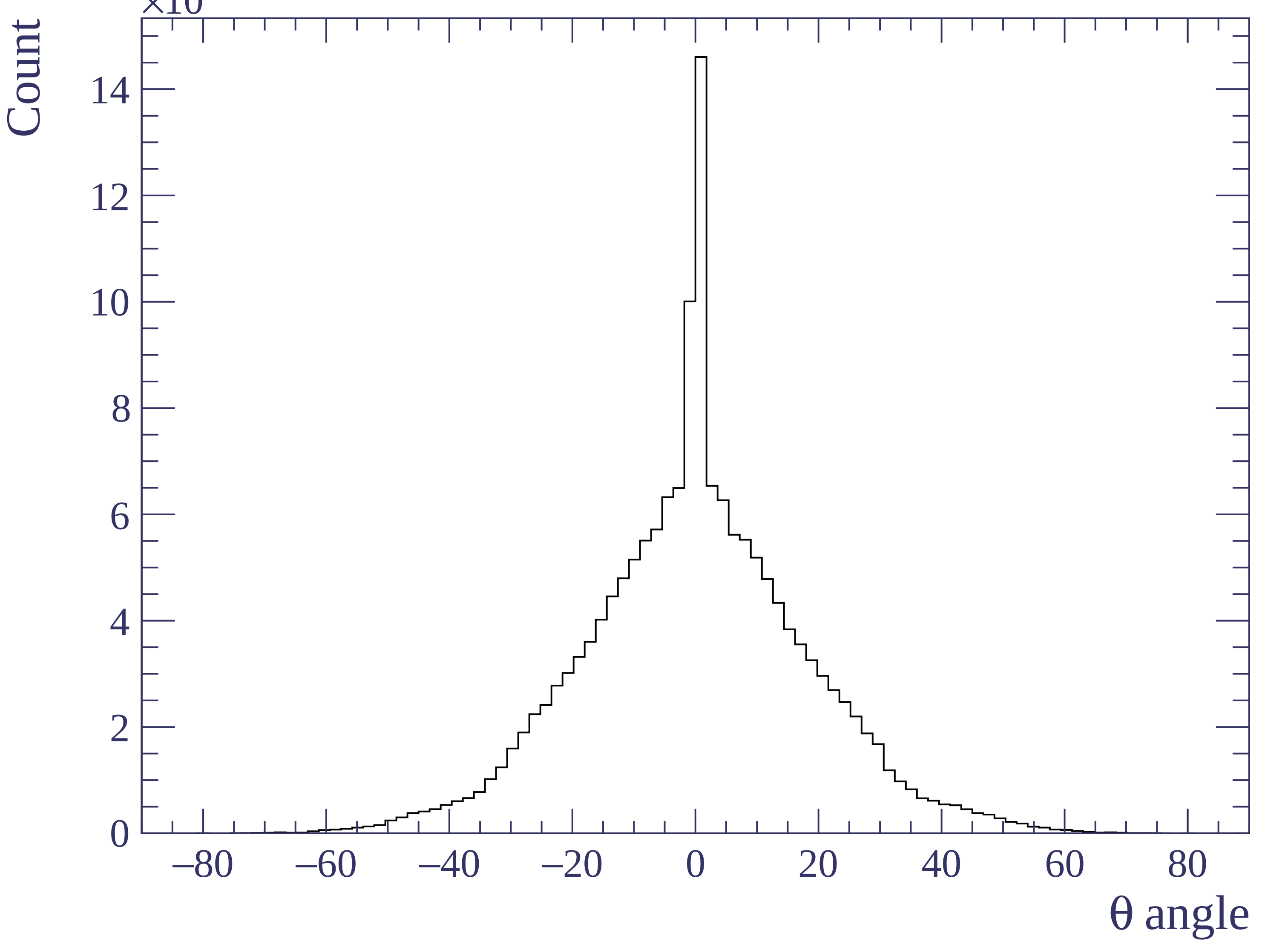


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads

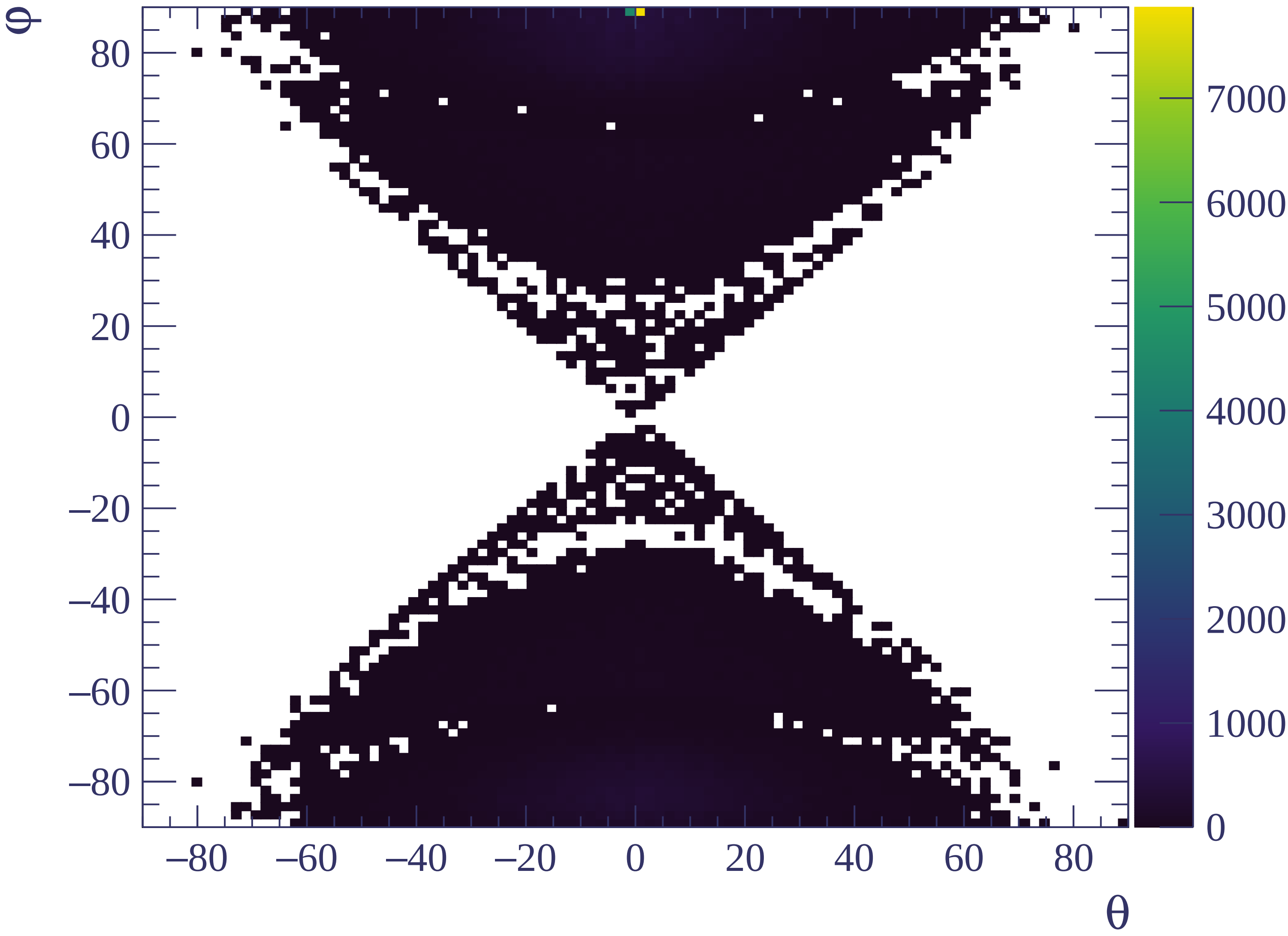

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
220
200
180
160
140
120
100
80
60
40
20
0


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

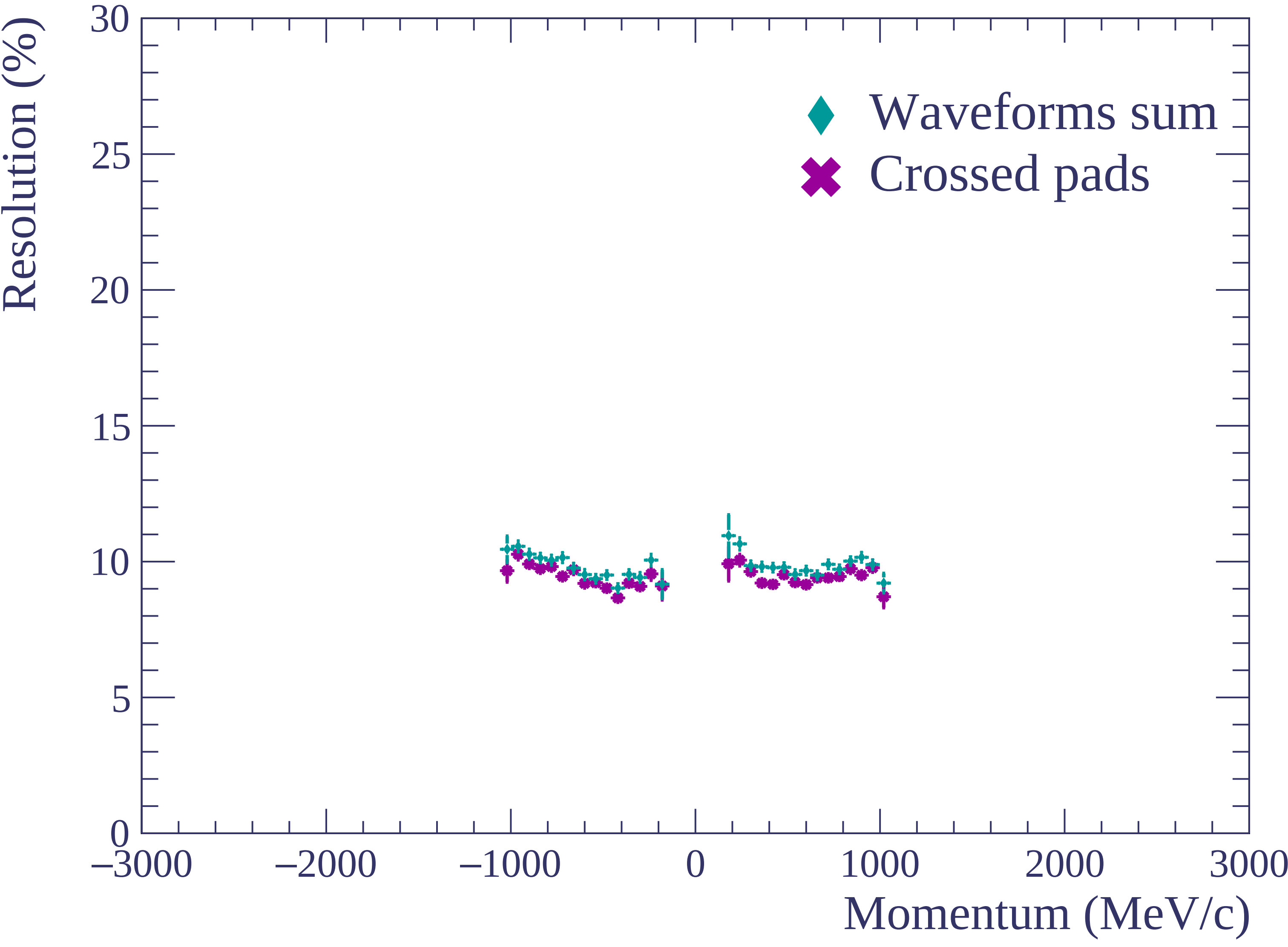

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
250
200
150
100
50
0

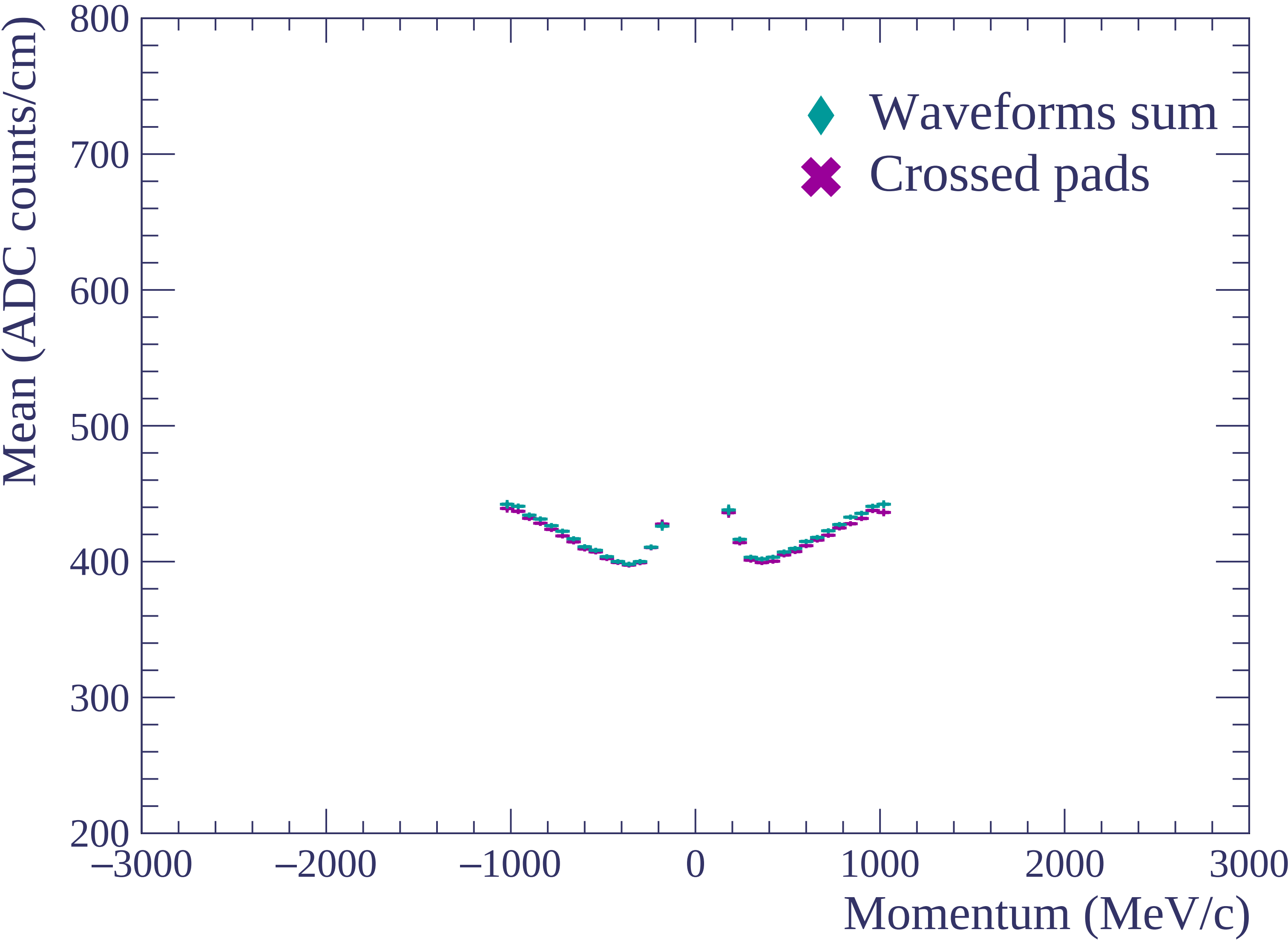


×10
Count
0
2
4
6
8
10
12
14
−80
−60
−40
−20
0
20
40
60
80
θ angle

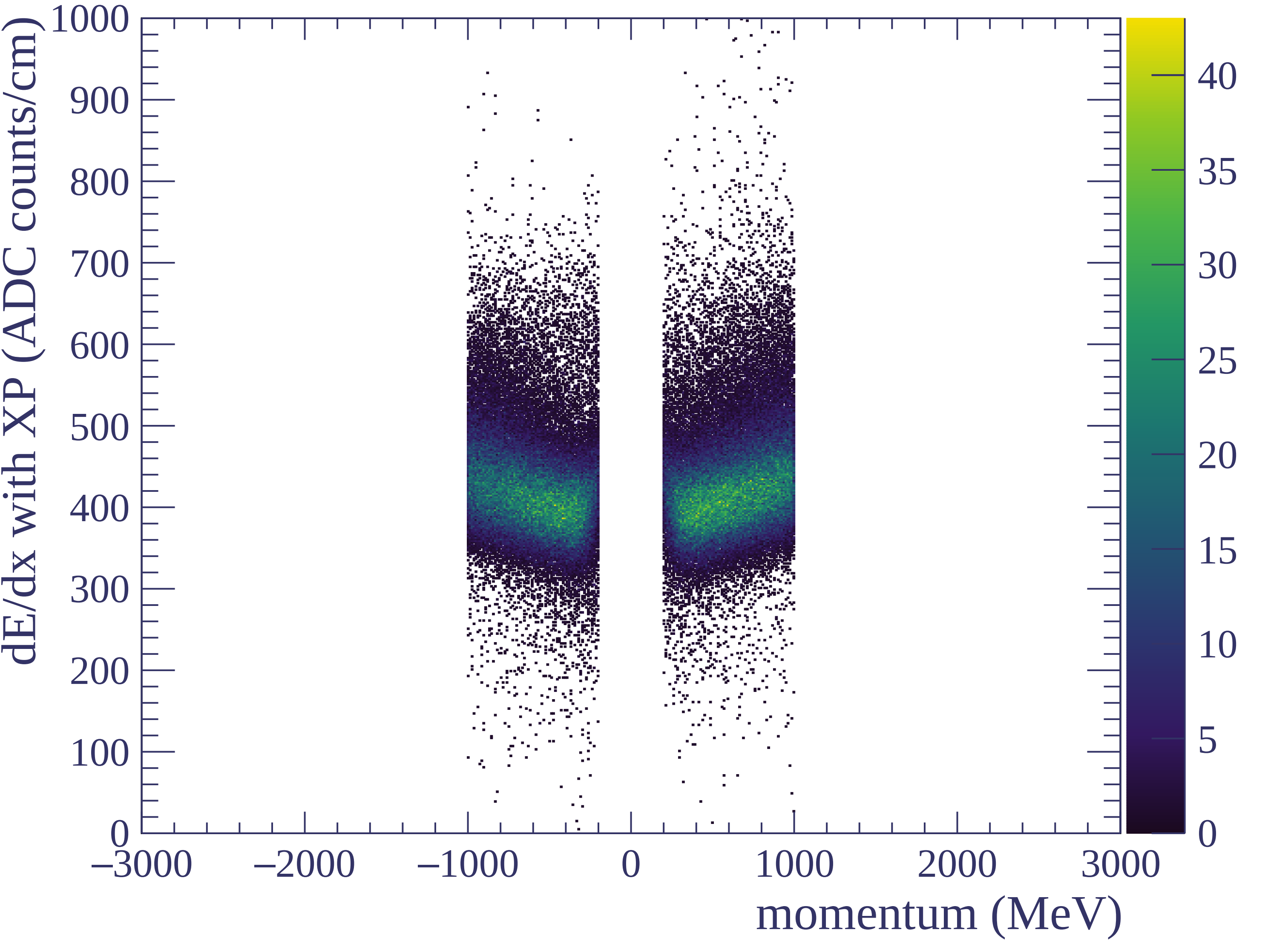


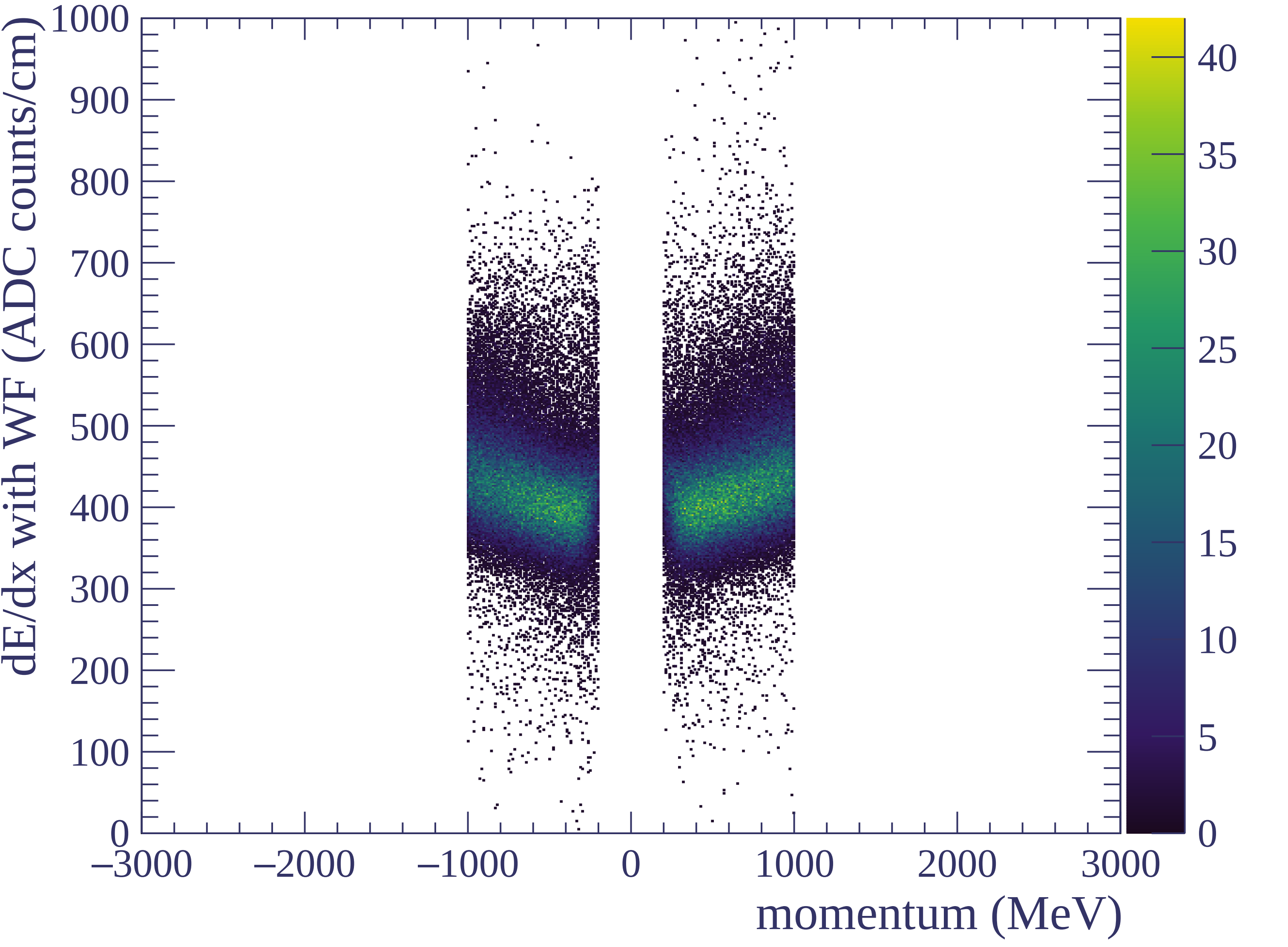
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
7000
6000
5000
4000
3000
2000
1000
0

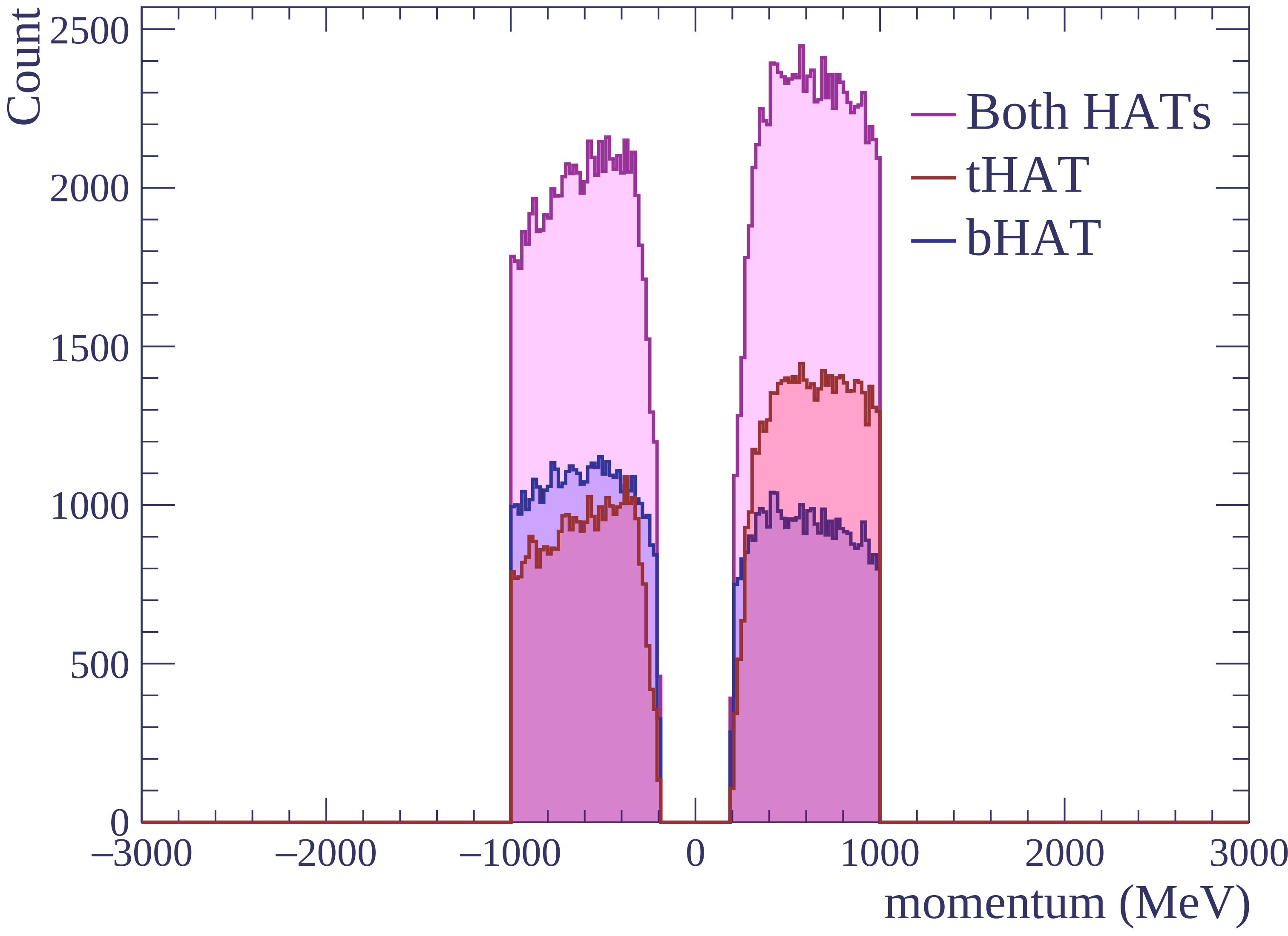

Resolution (%)
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



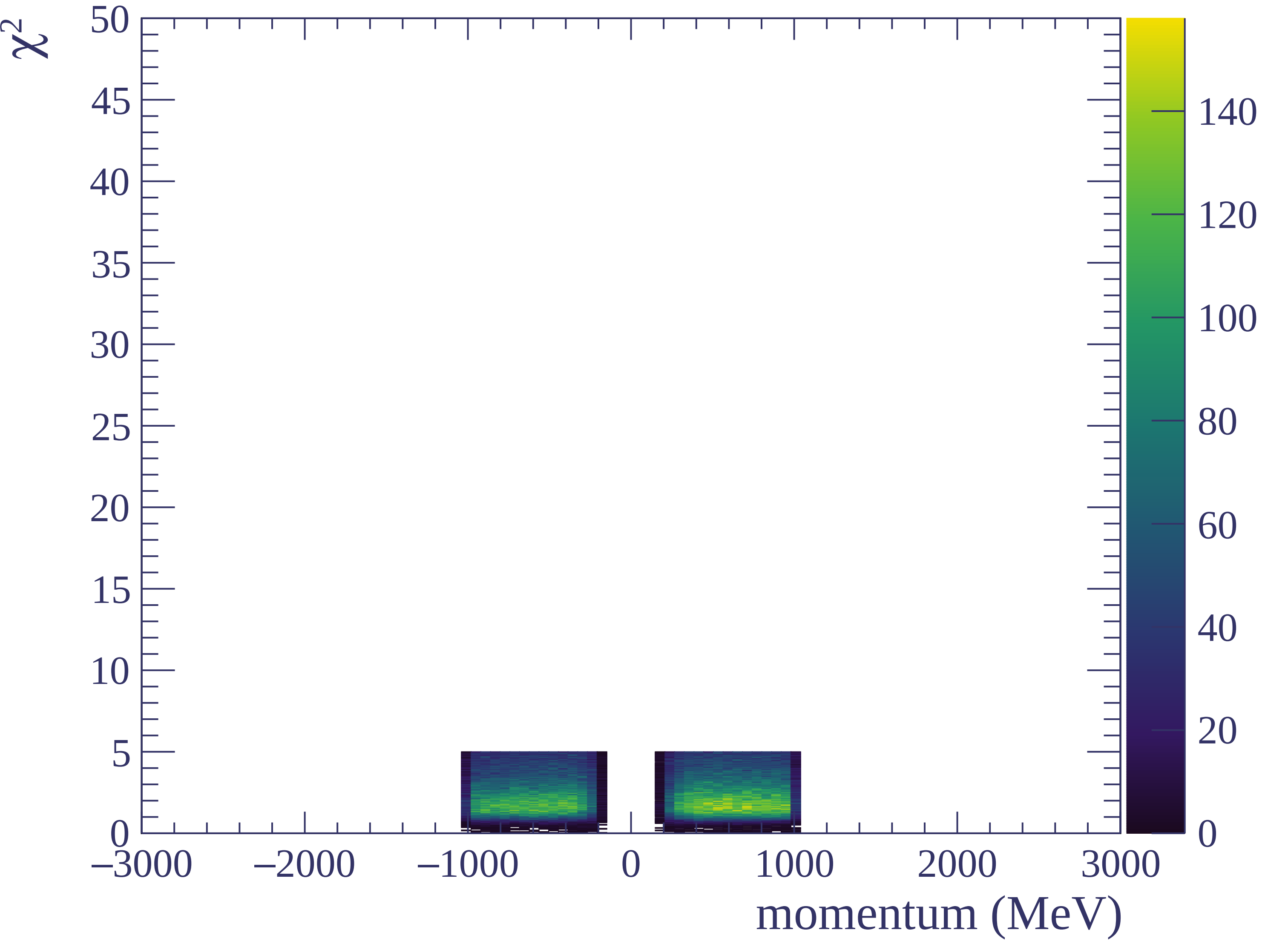
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads


dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
5
10
15
20
25
30
35
40

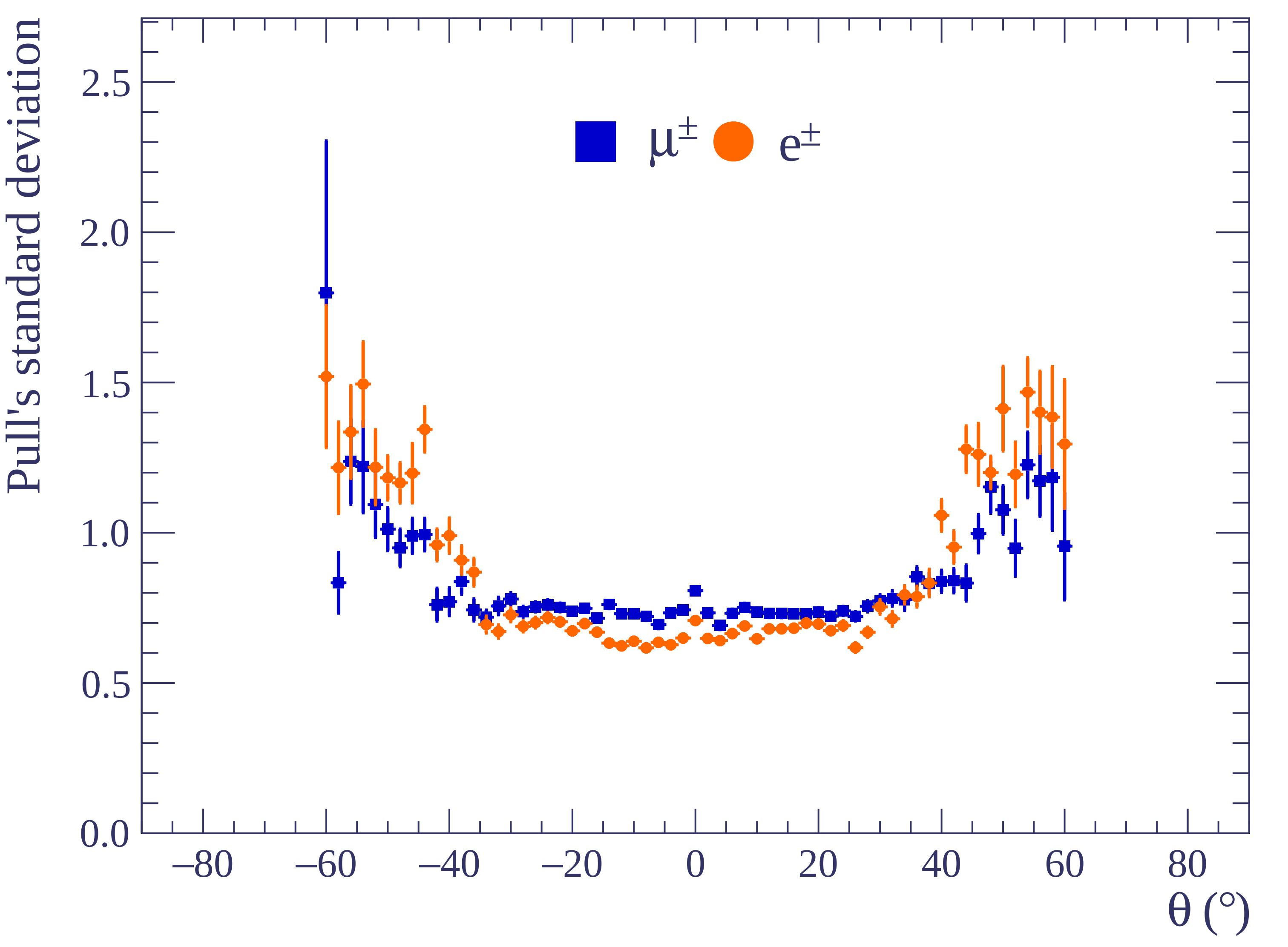

dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
0
5
10
15
20
25
30
35
40



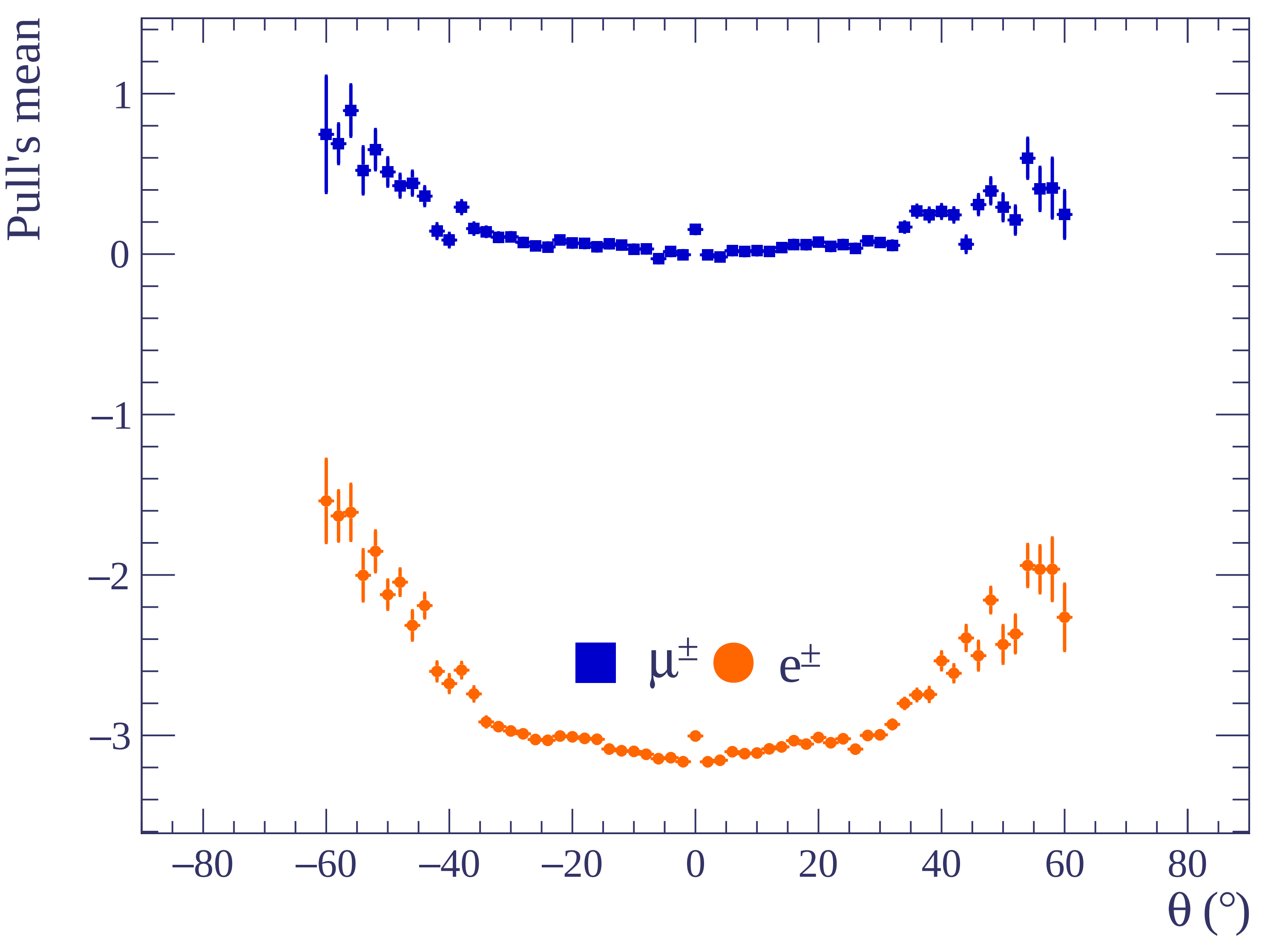
Count
2500
2000
1500
1000
500
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



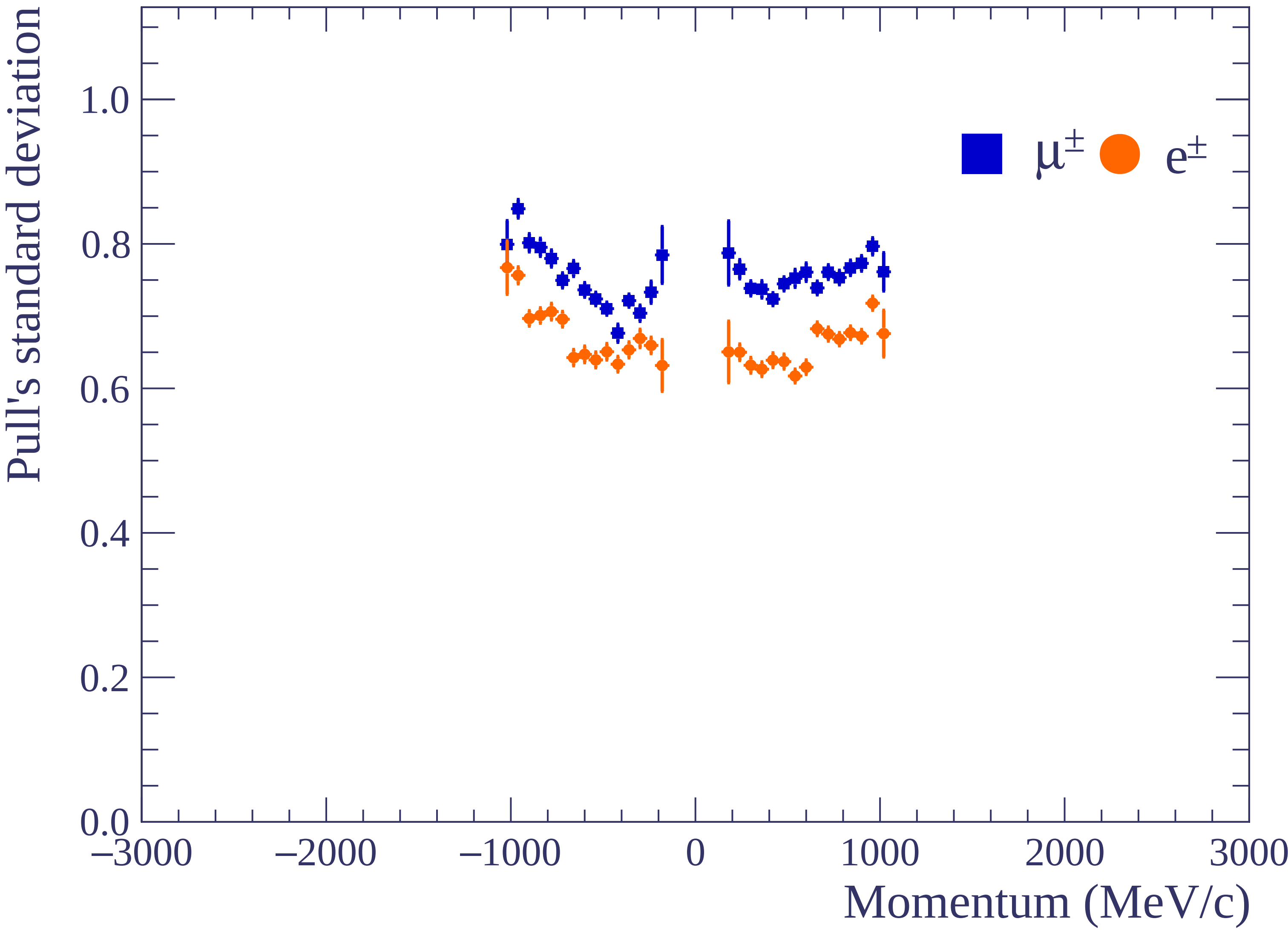
χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
140
120
100
80
60
40
20
0



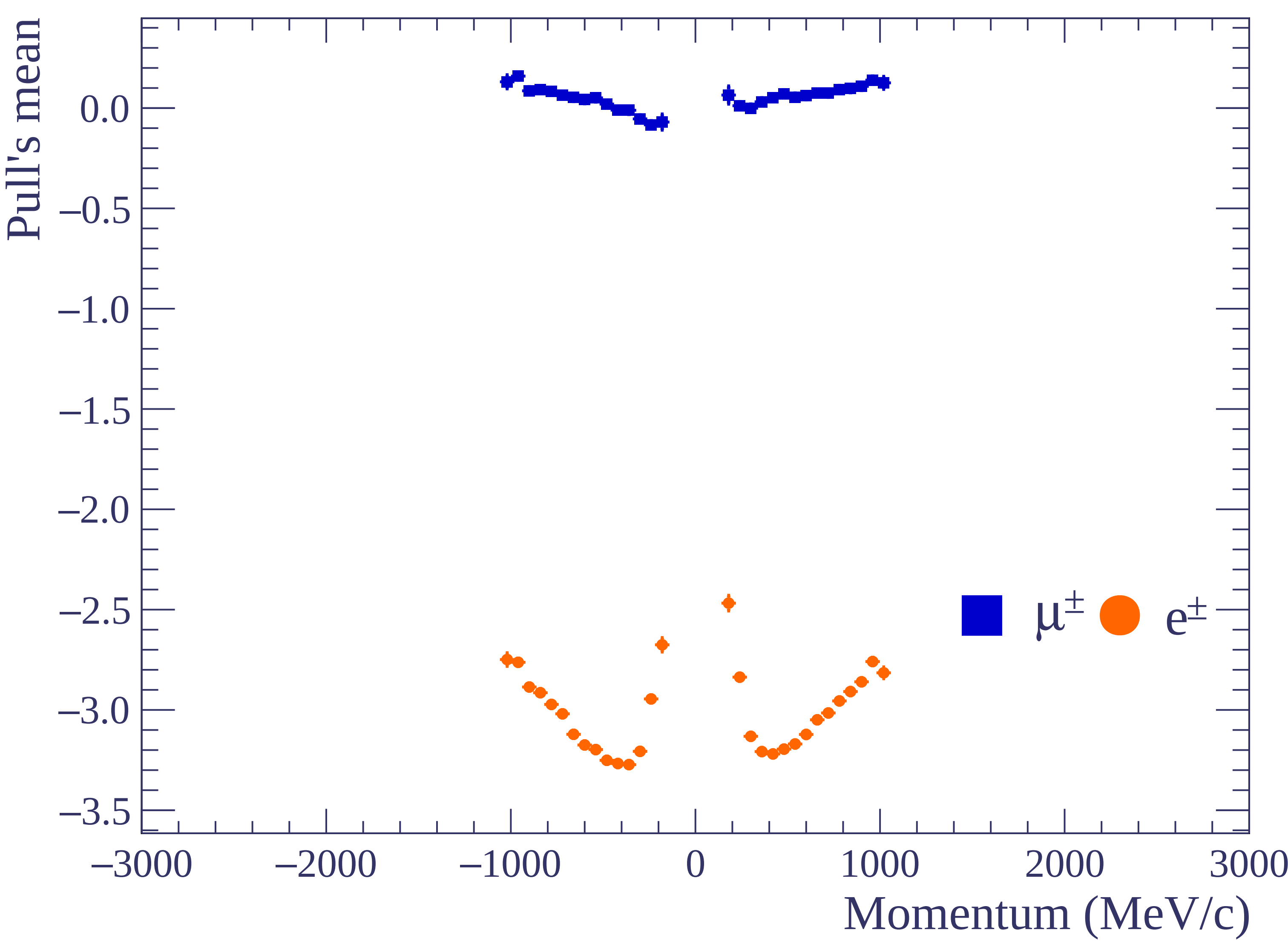
Pull's standard deviation
2.5
2.0
1.5
1.0
0.5
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



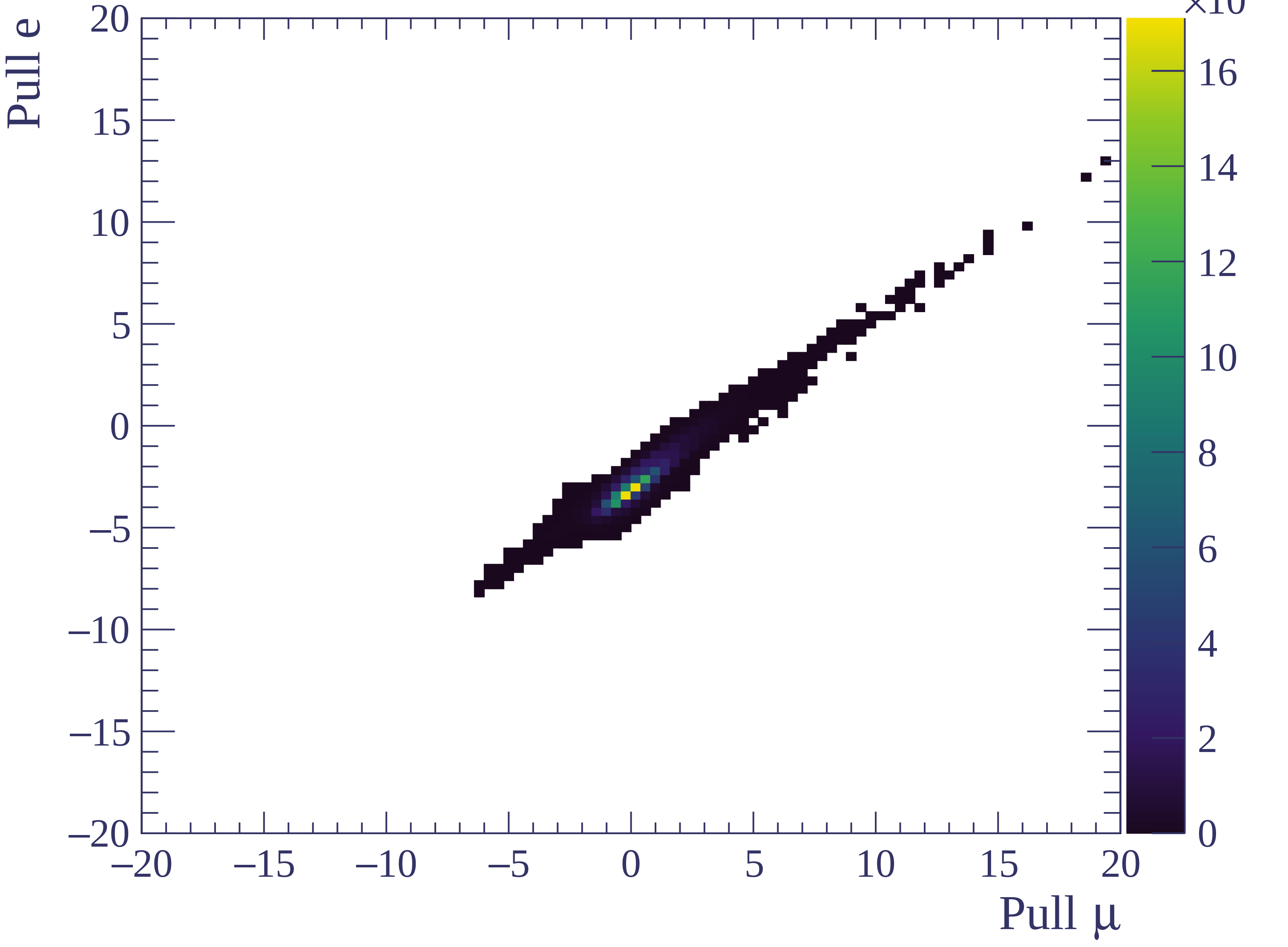
Pull's mean
1
0
−1
−2
−3
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)



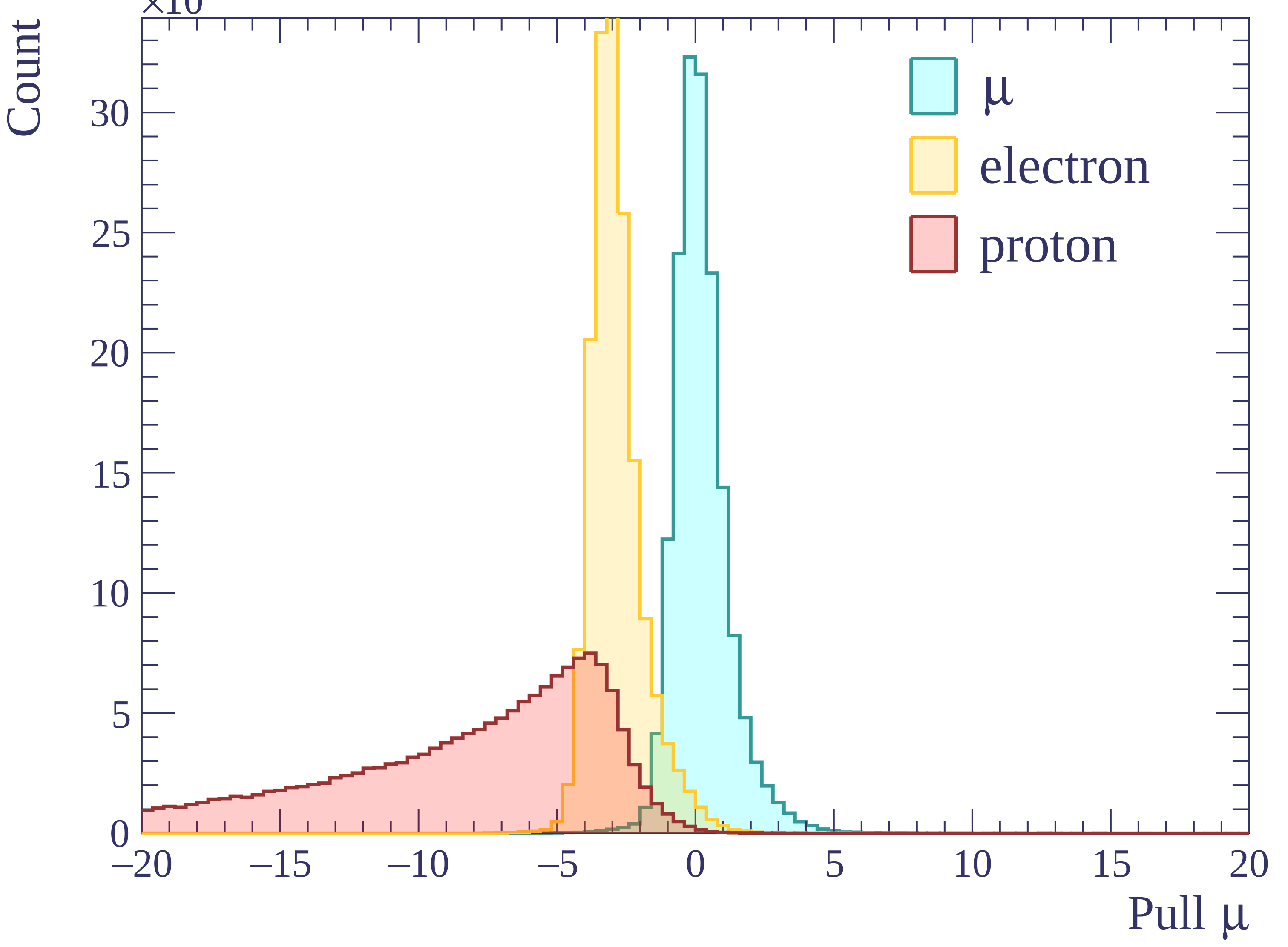
Pull's standard deviation
1.0
0.8
0.6
0.4
0.2
0.0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
μ±
e±



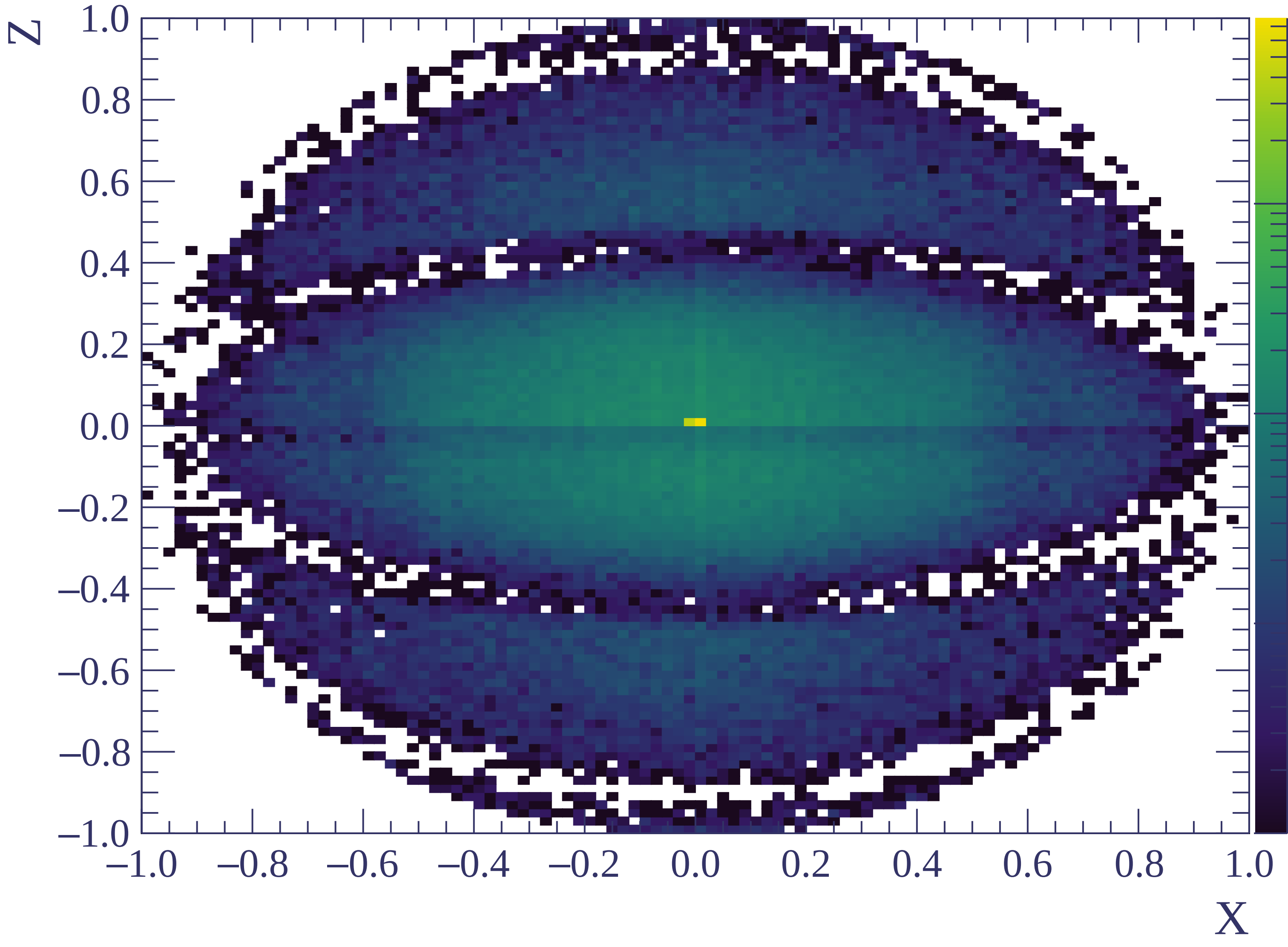
Pull's mean
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



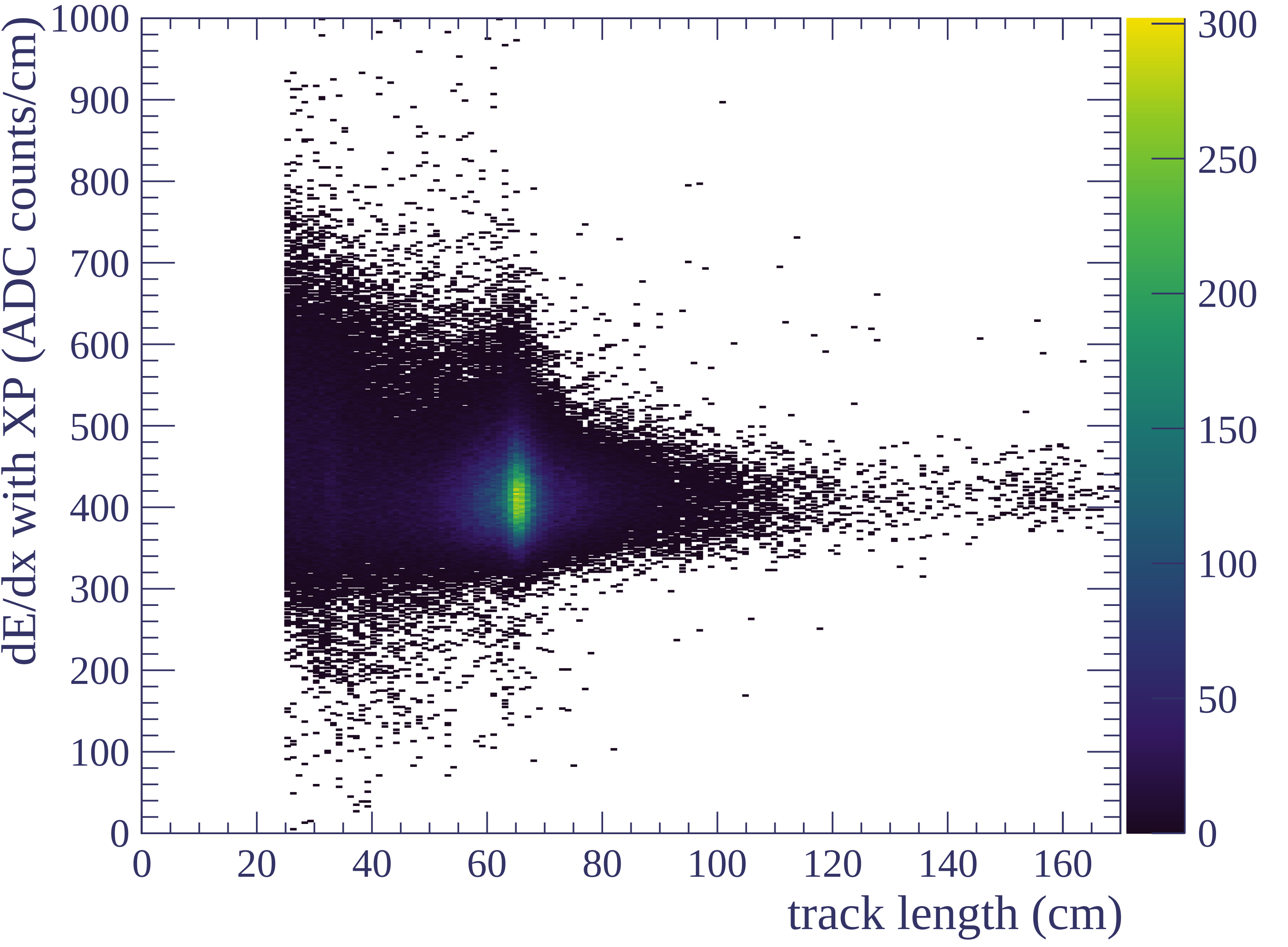
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
×10
16
14
12
10
8
6
4
2
0

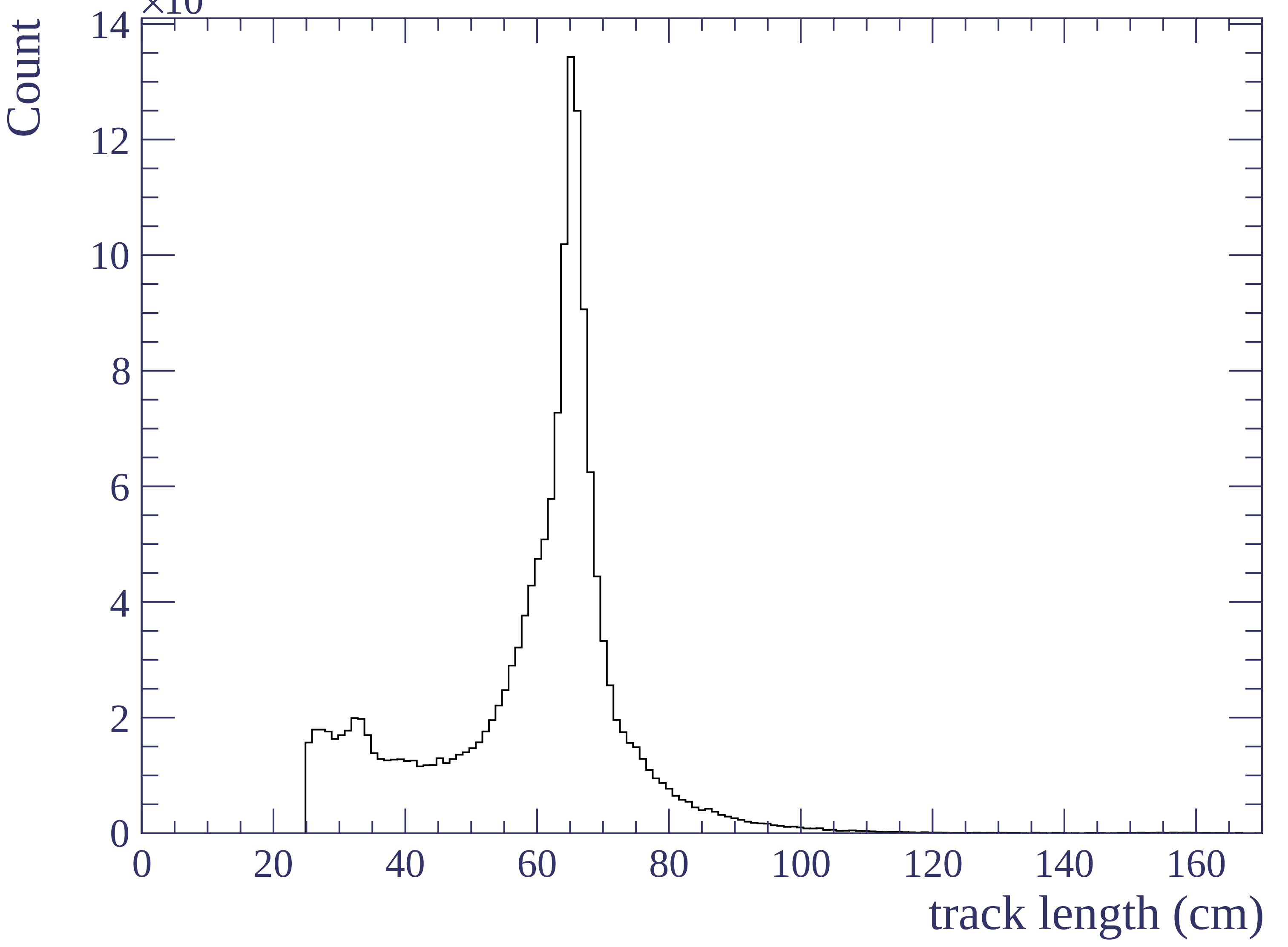


×10
Count
30
25
20
15
10
5
0
−20
−15
−10
−5
0
5
10
15
20
Pull μ
μ
electron
proton

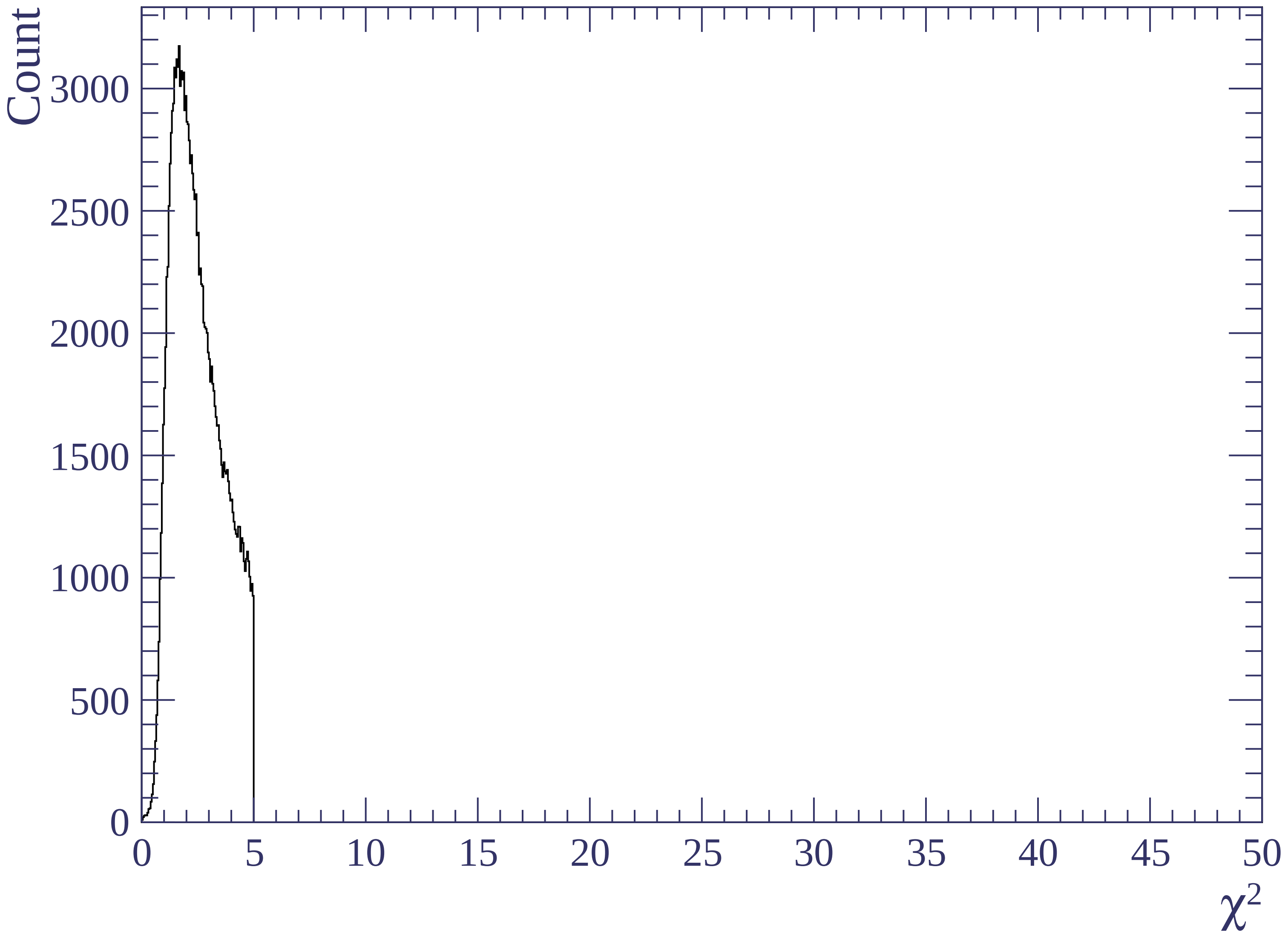


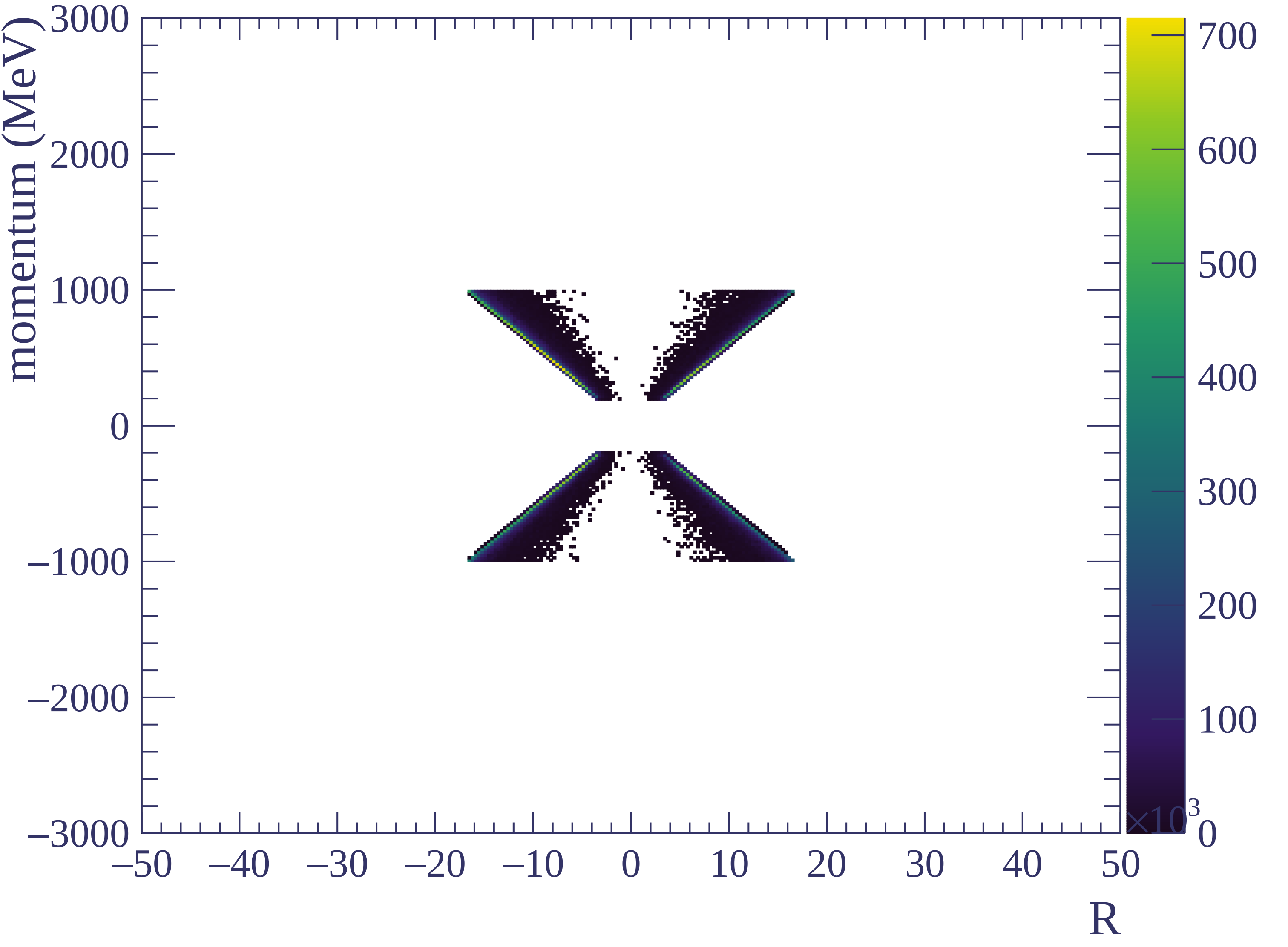
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X


dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300

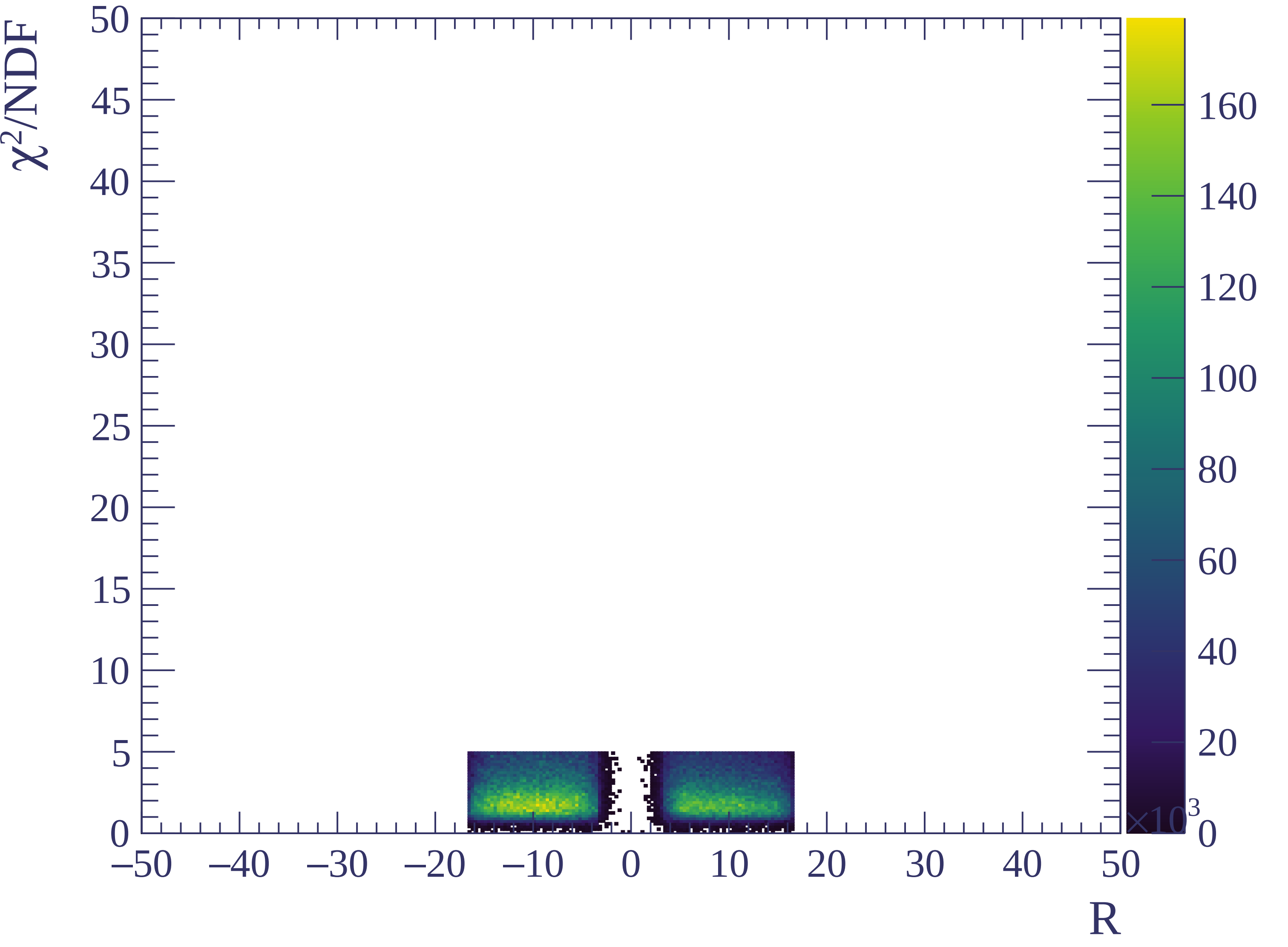


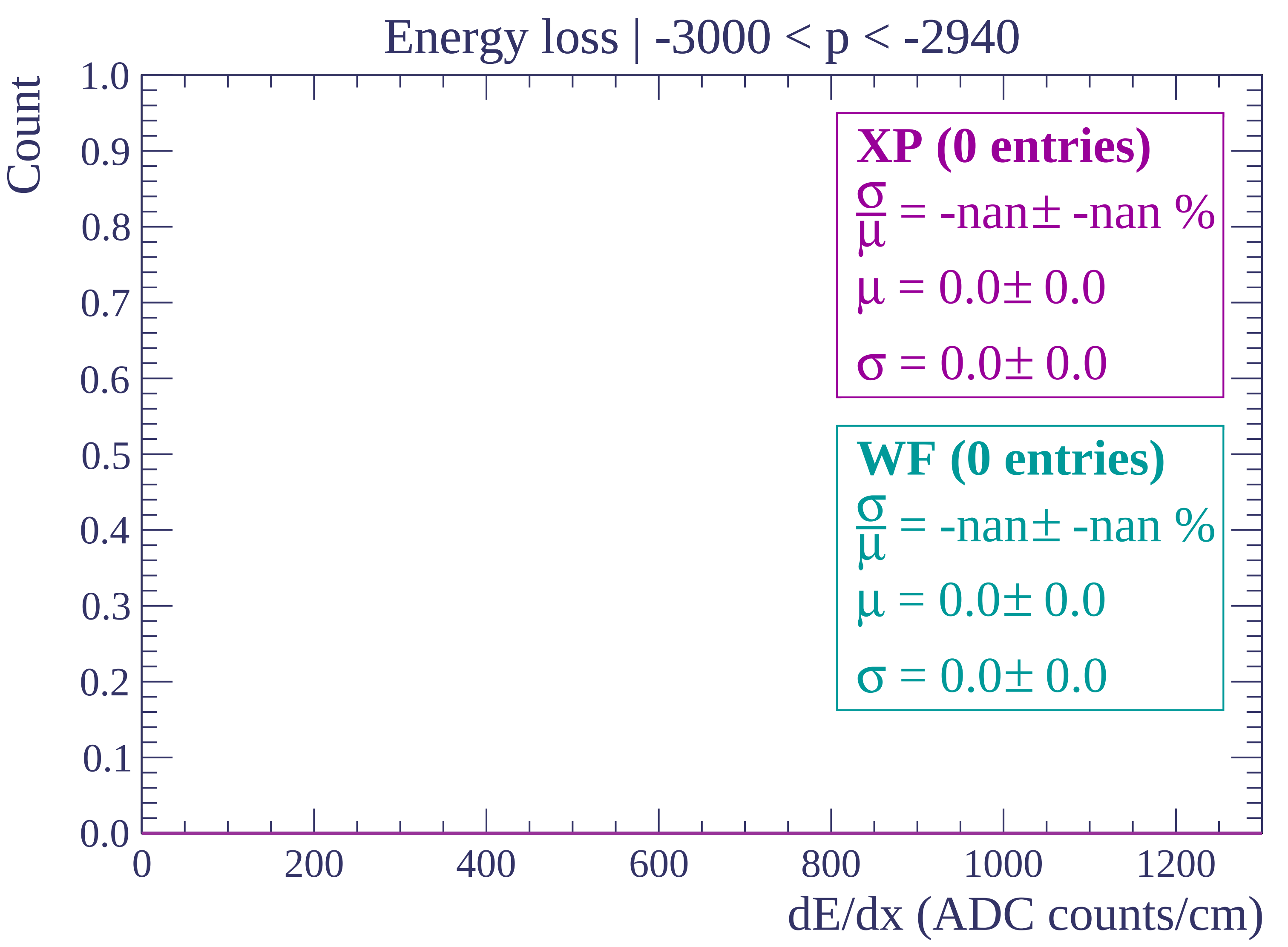
Count
×10
14
12
10
8
6
4
2
0
0
20
40
60
80
100
120
140
160
track length (cm)


Count
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

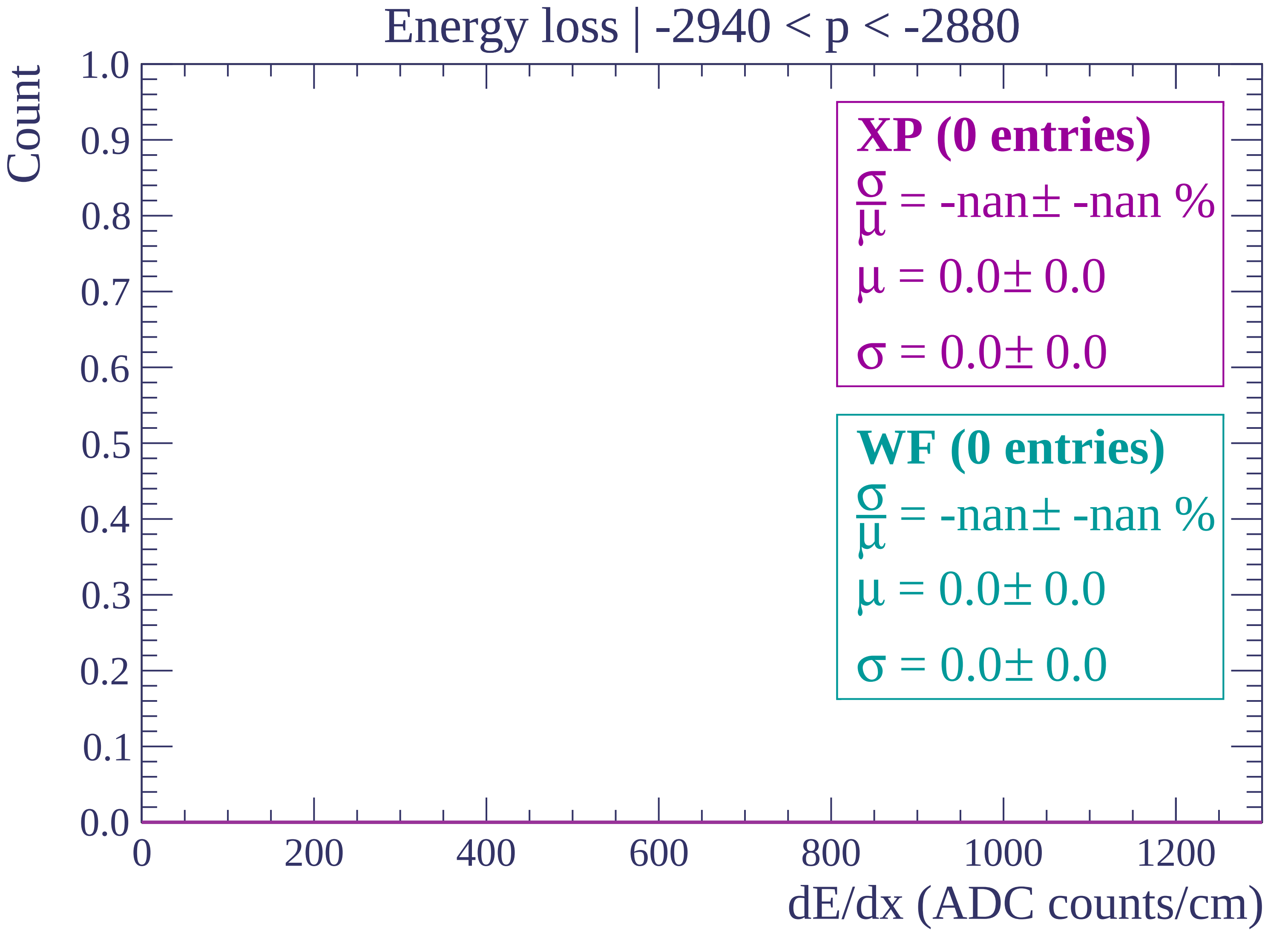


momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
700
600
500
400
300
200
100
0
×10^3

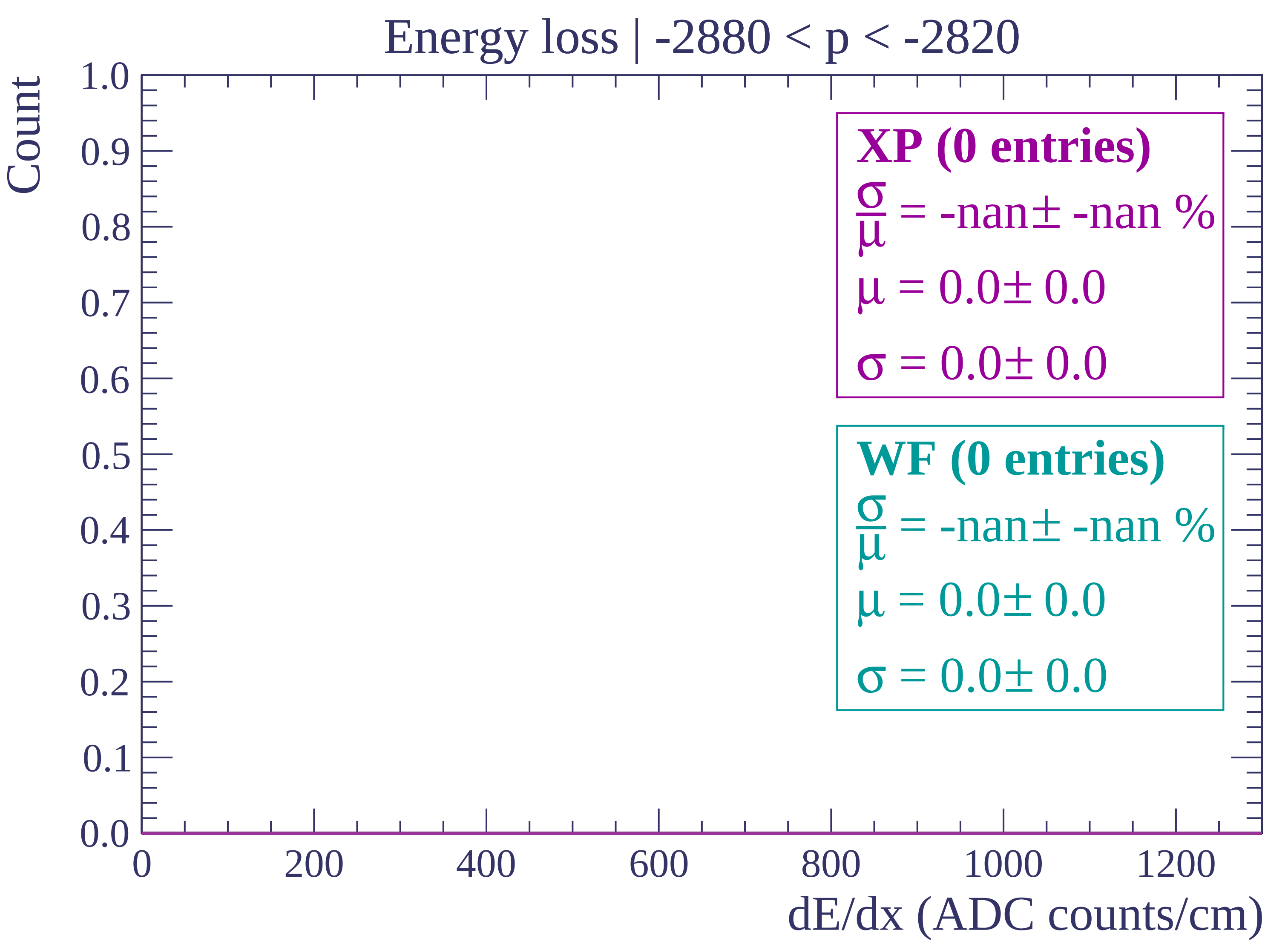

χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
160
140
120
100
80
60
40
20
0
$\times 10^3$

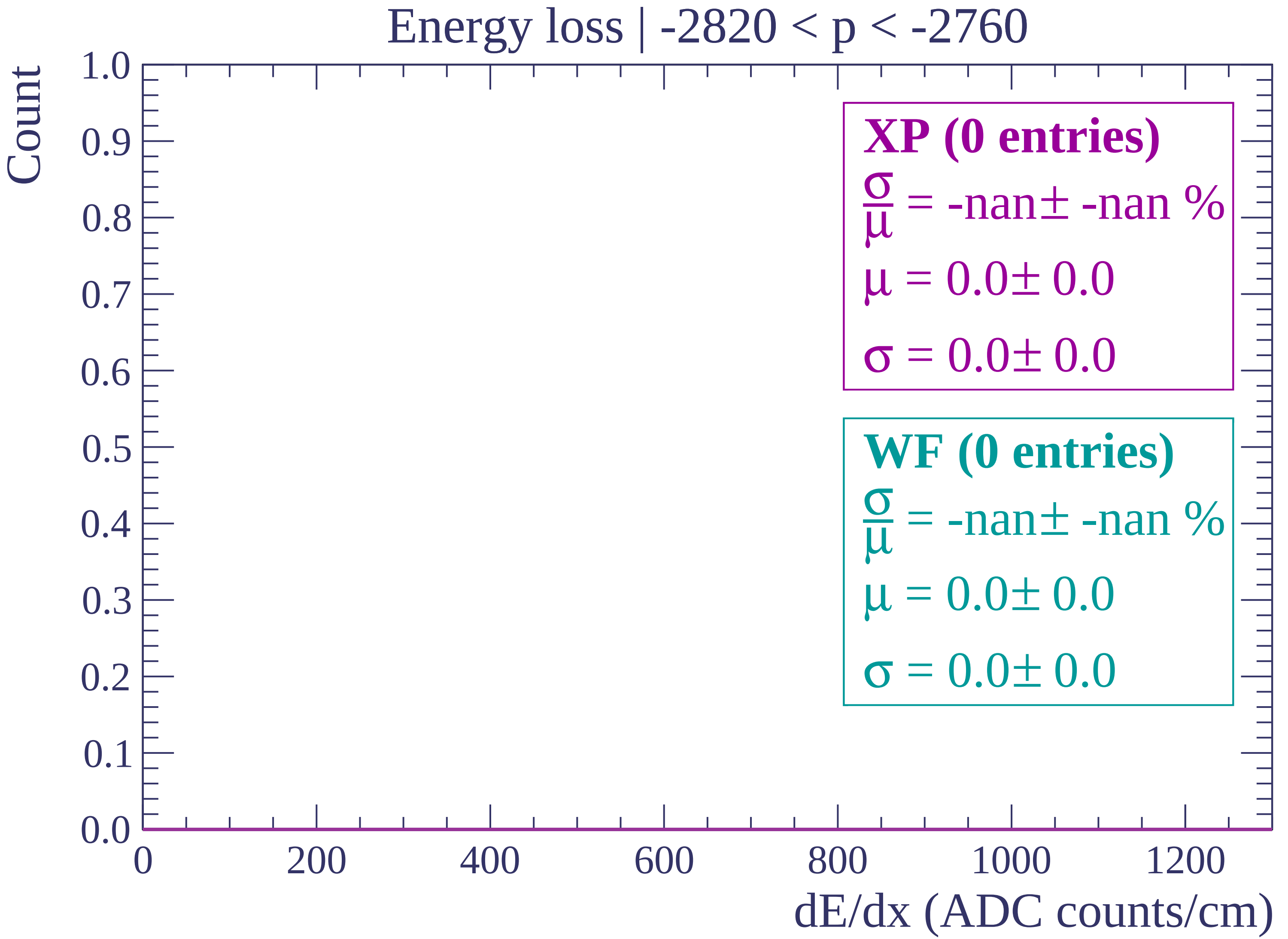


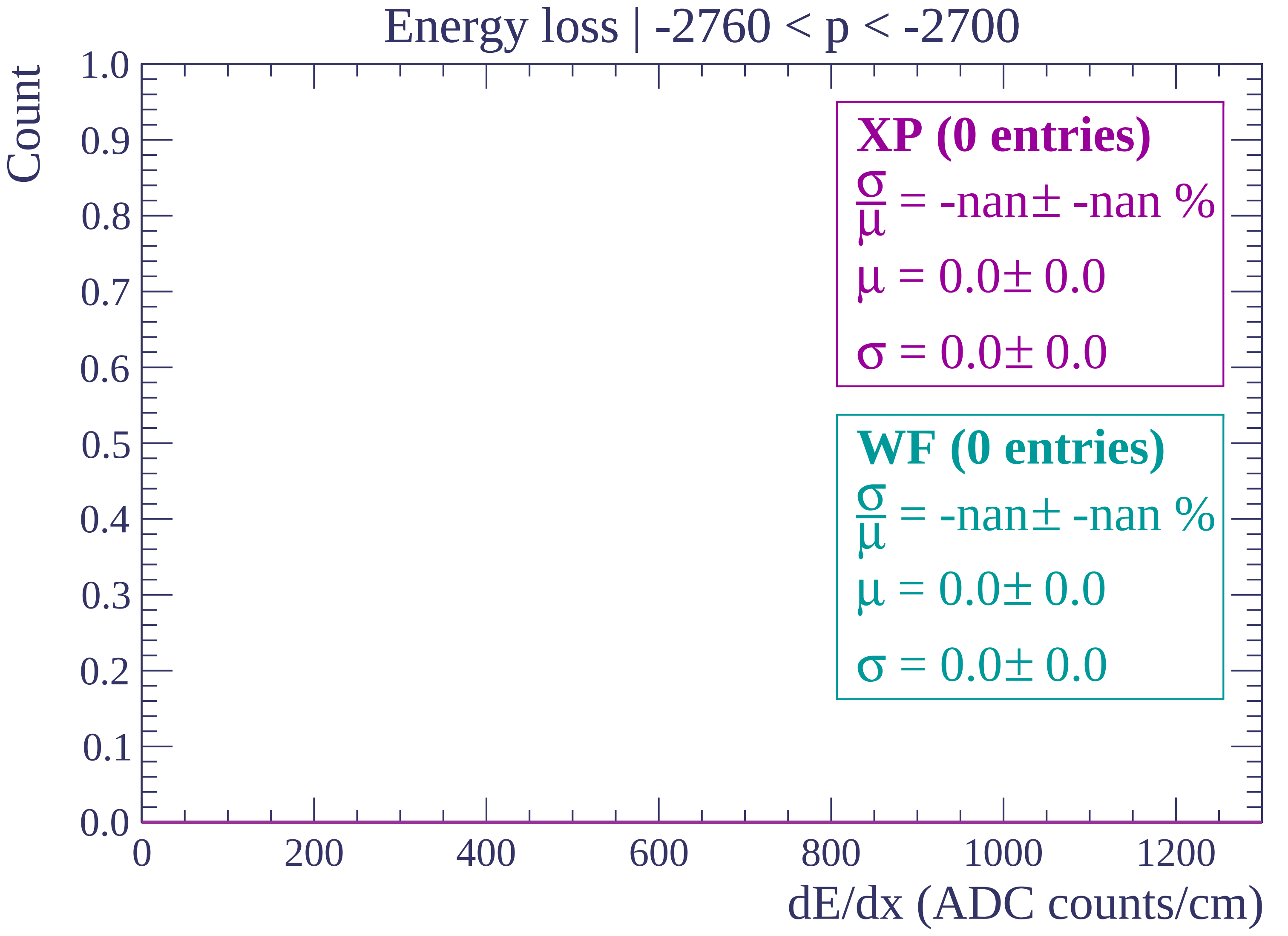
Energy loss | -3000 < p < -2940
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



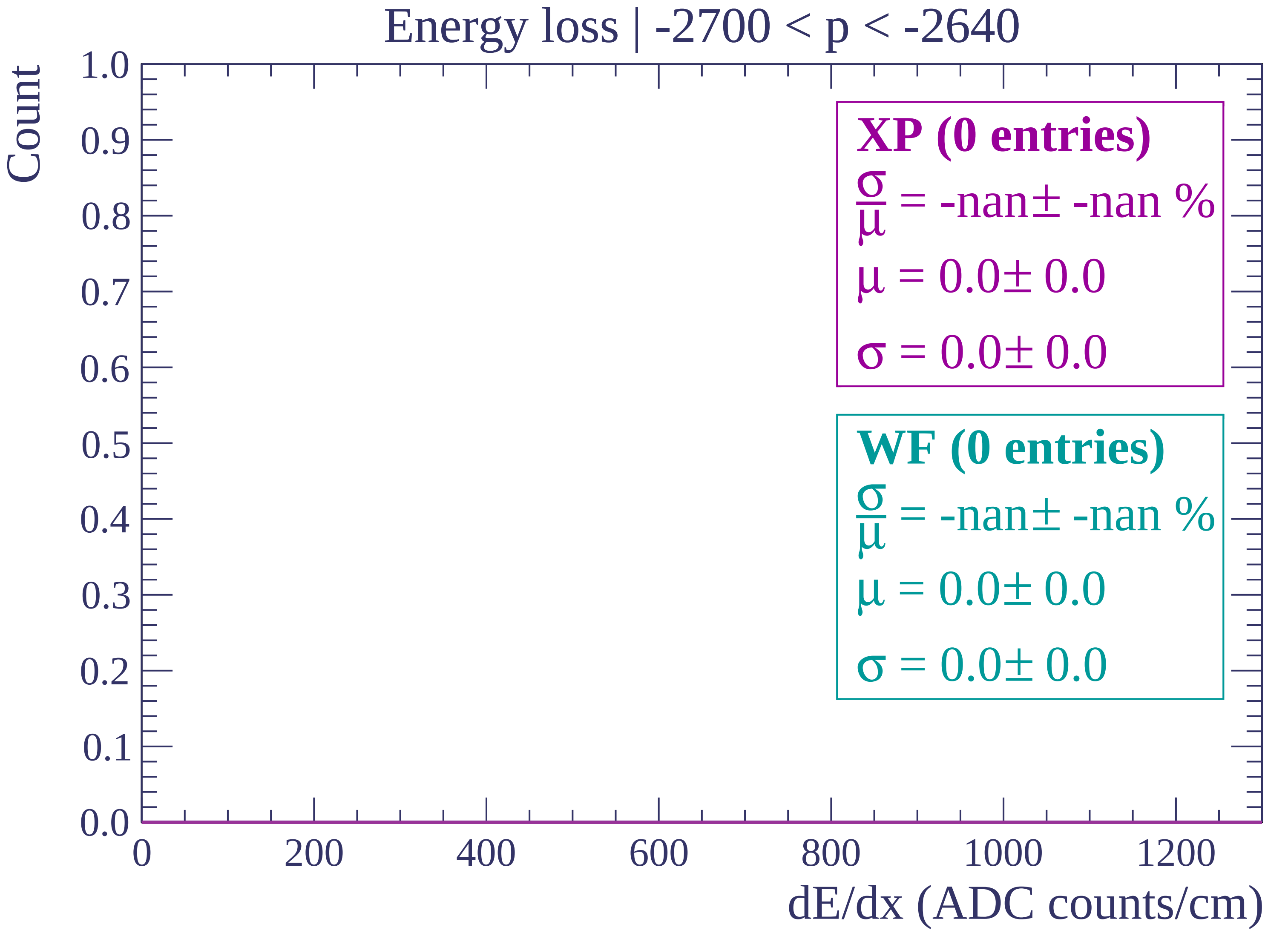
Energy loss | -2940 < p < -2880
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

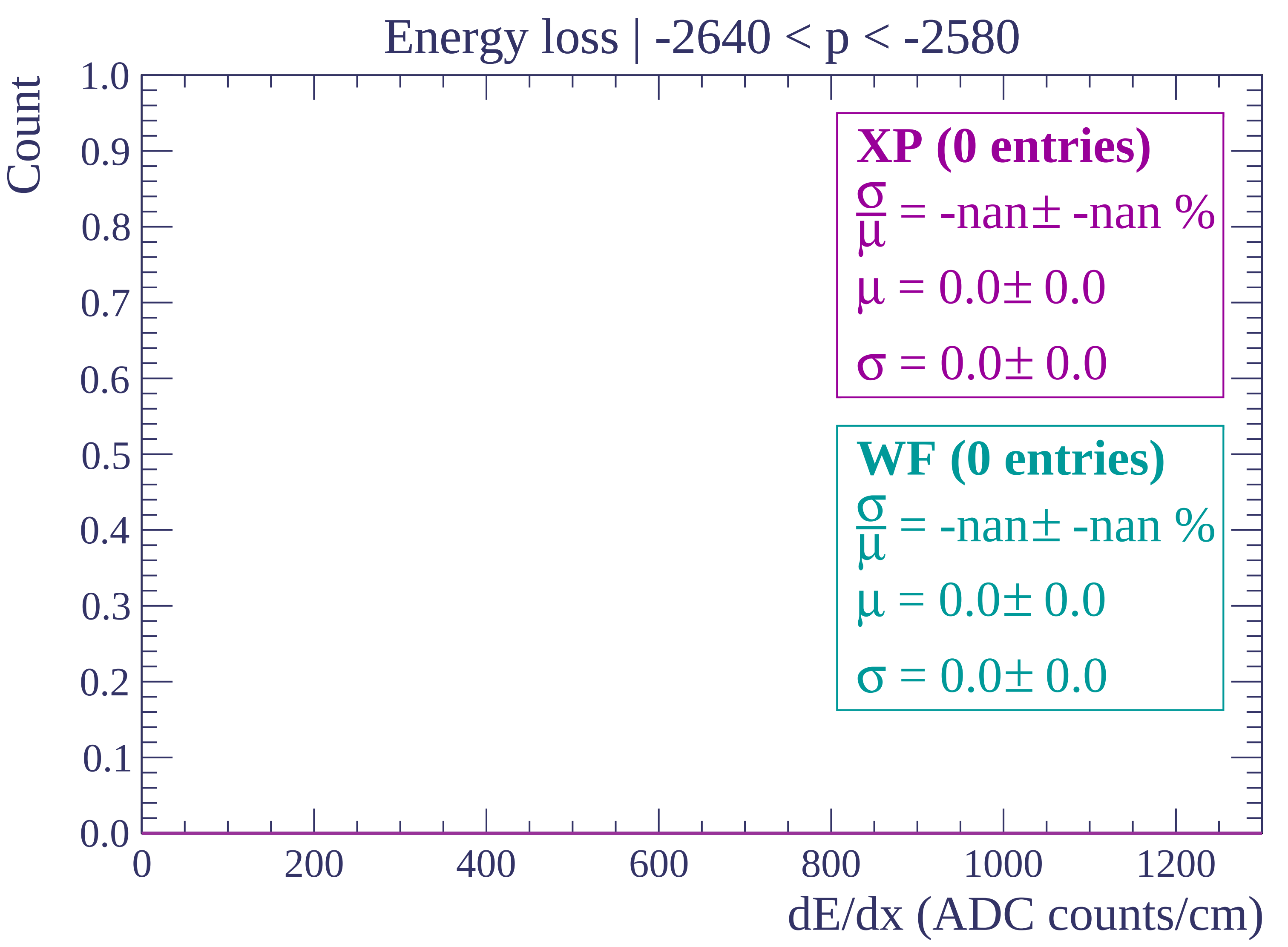

Energy loss | -2820 < p < -2760
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



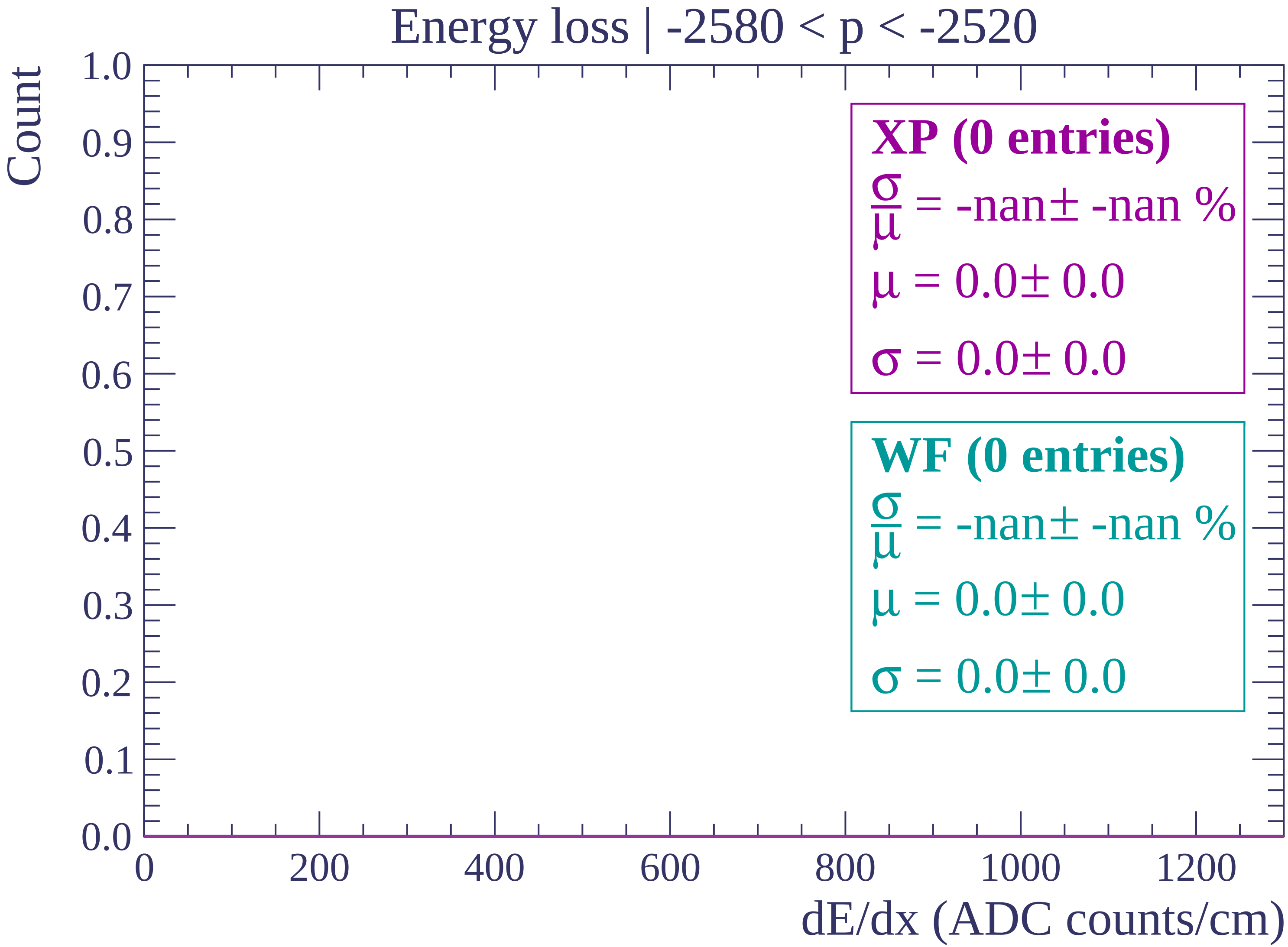
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



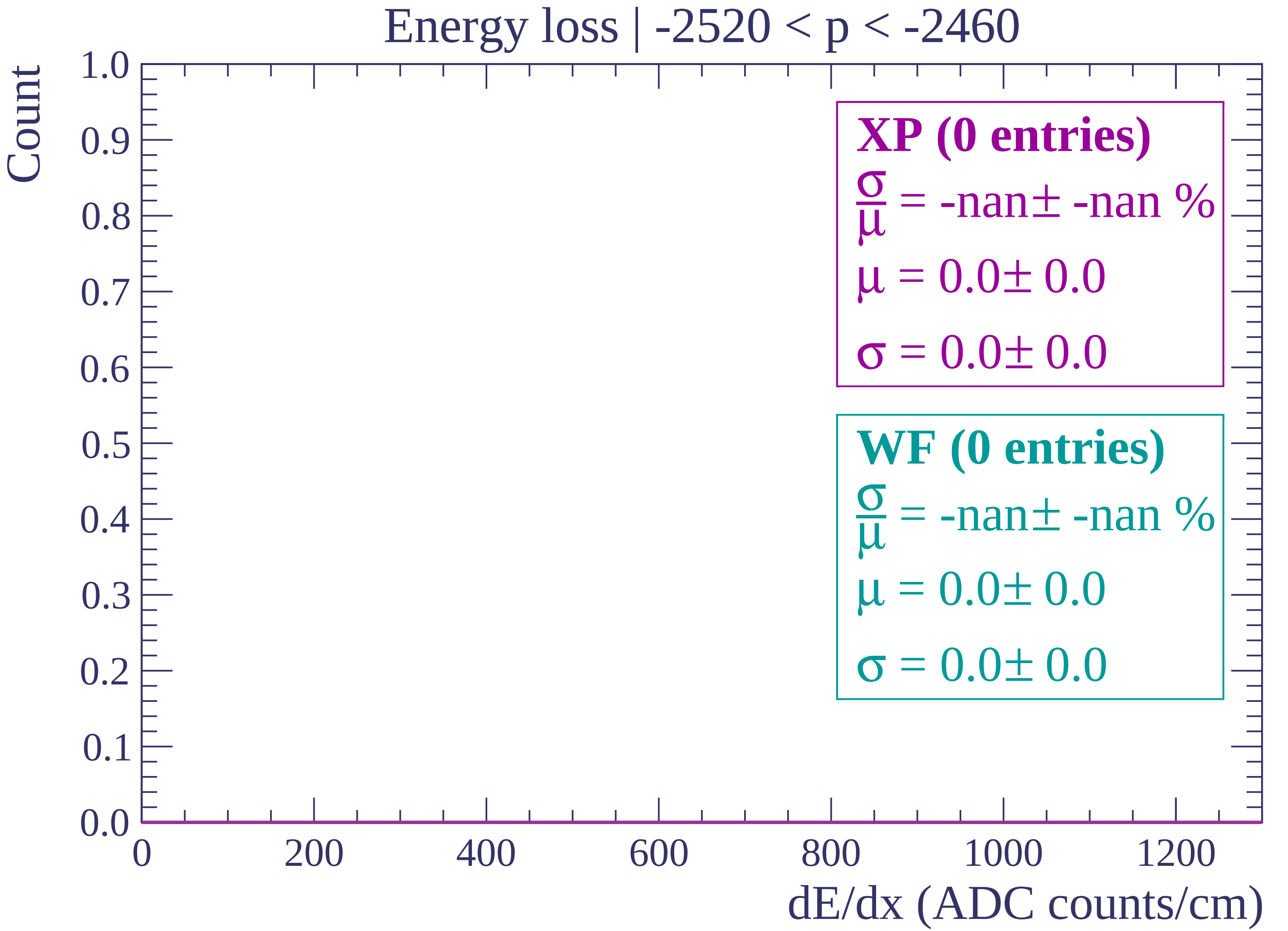
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

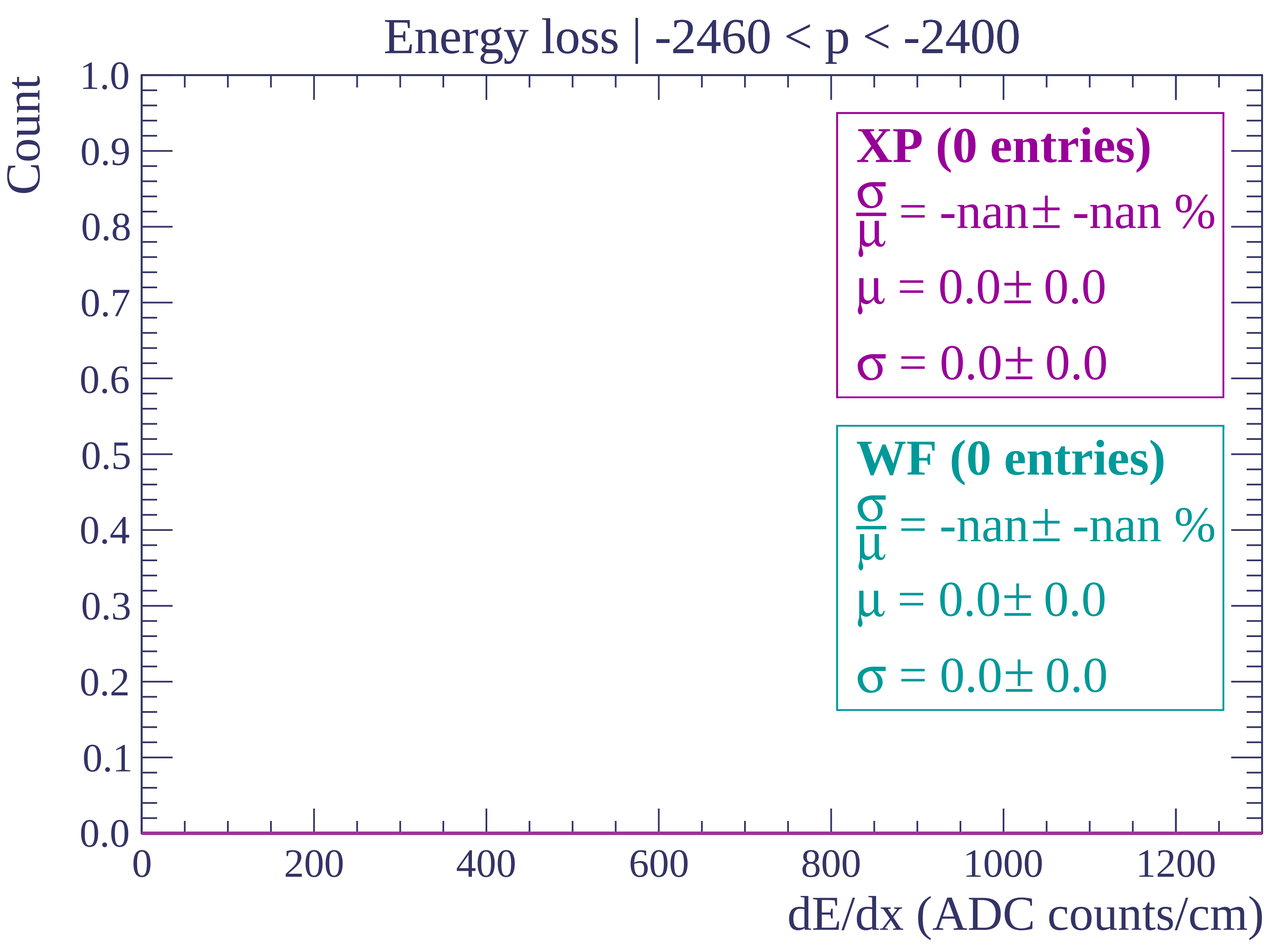


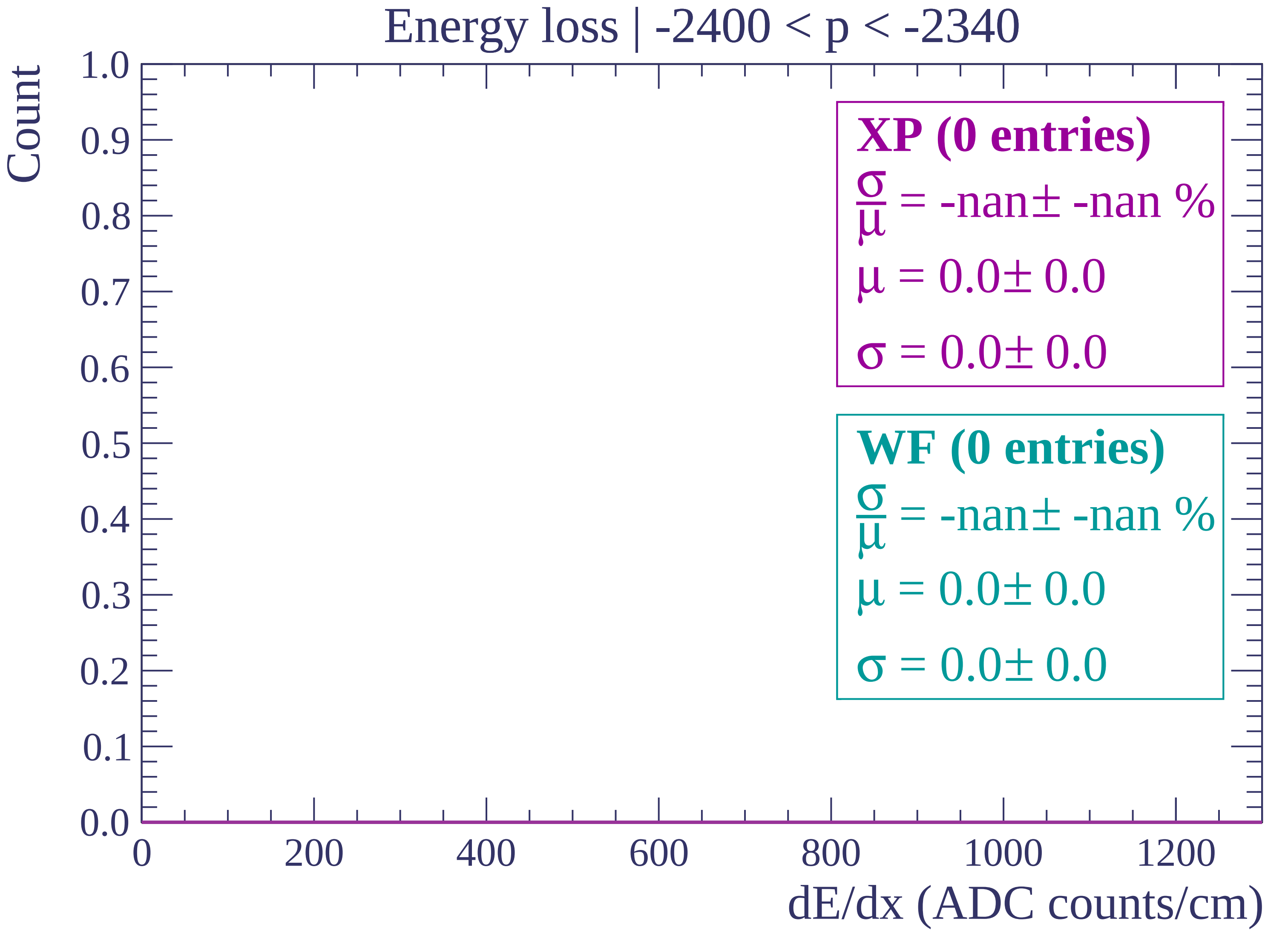
Energy loss | -2640 < p < -2580
Count
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
dE/dx (ADC counts/cm)

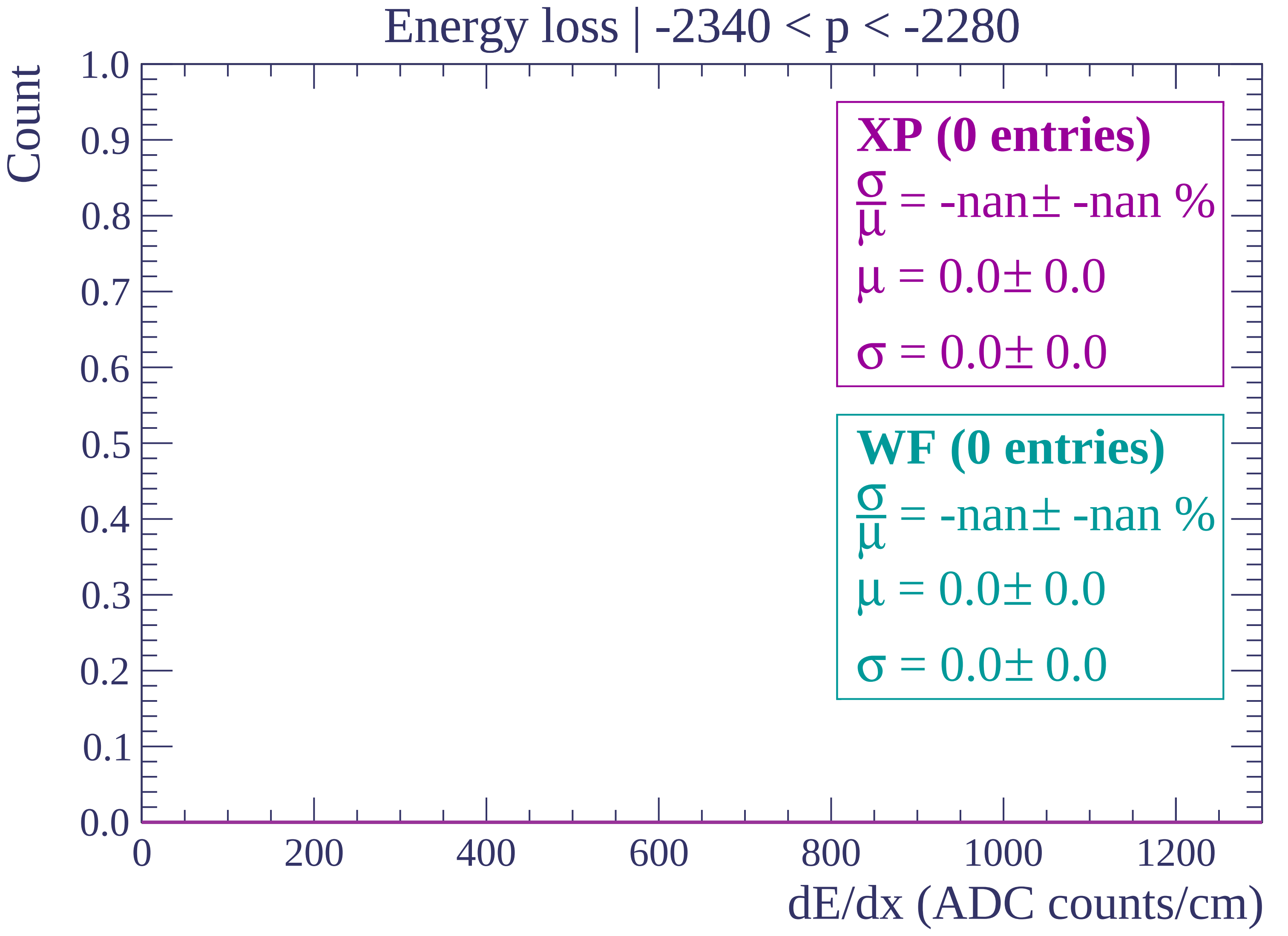


Energy loss | -2580 < p < -2520
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

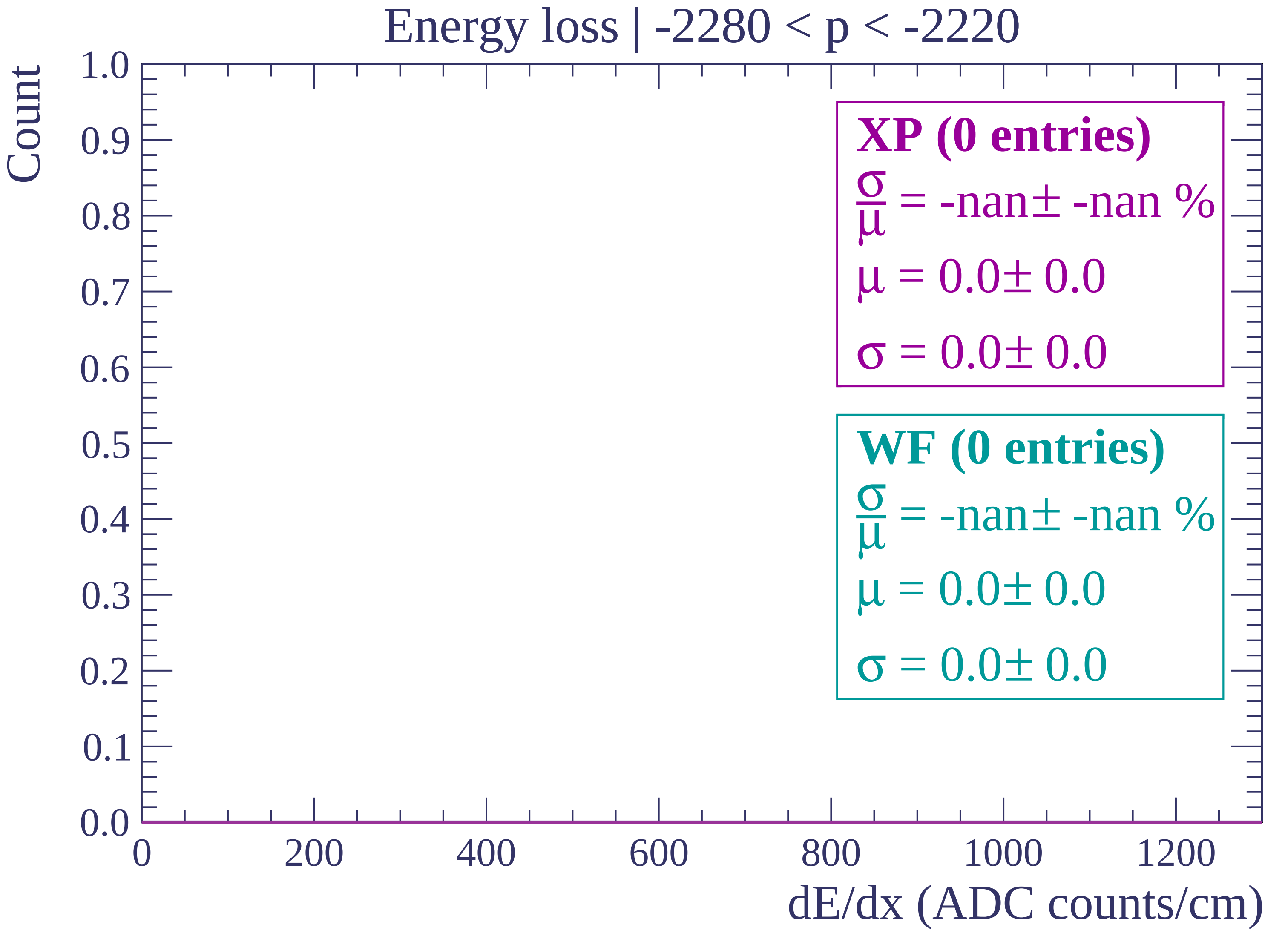

Energy loss | -2520 < p < -2460
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

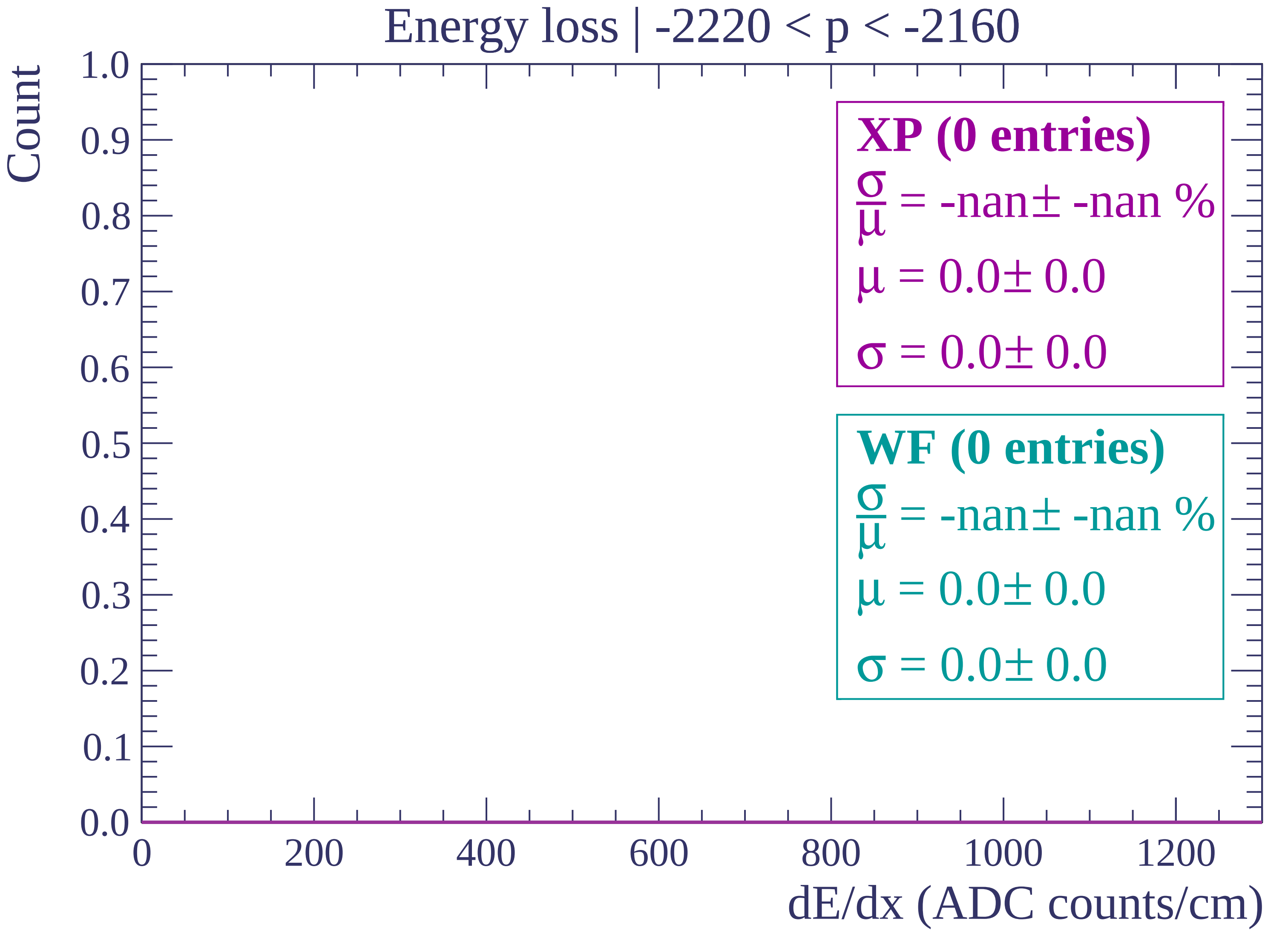

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

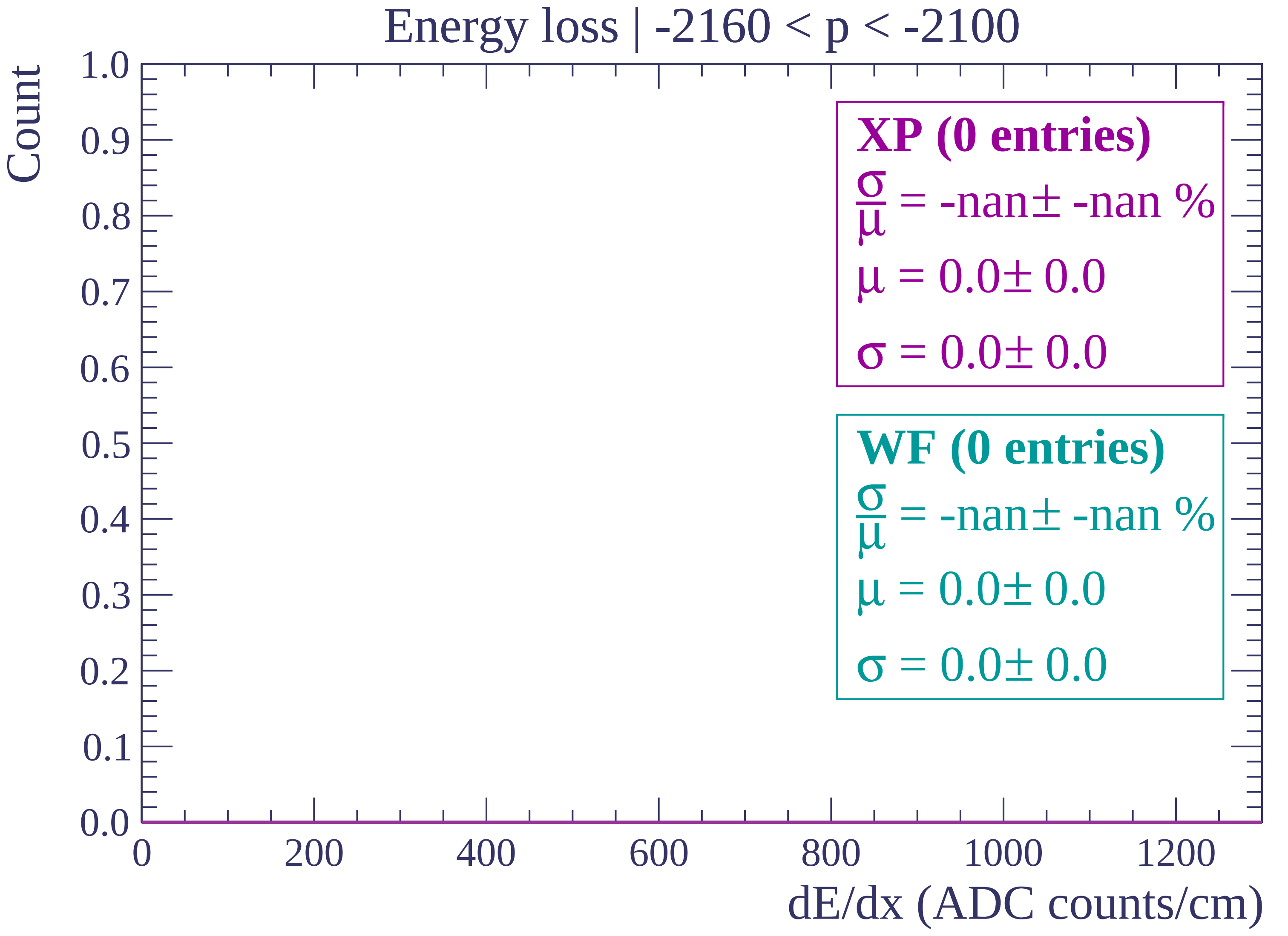

Energy loss | -2400 < p < -2340
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

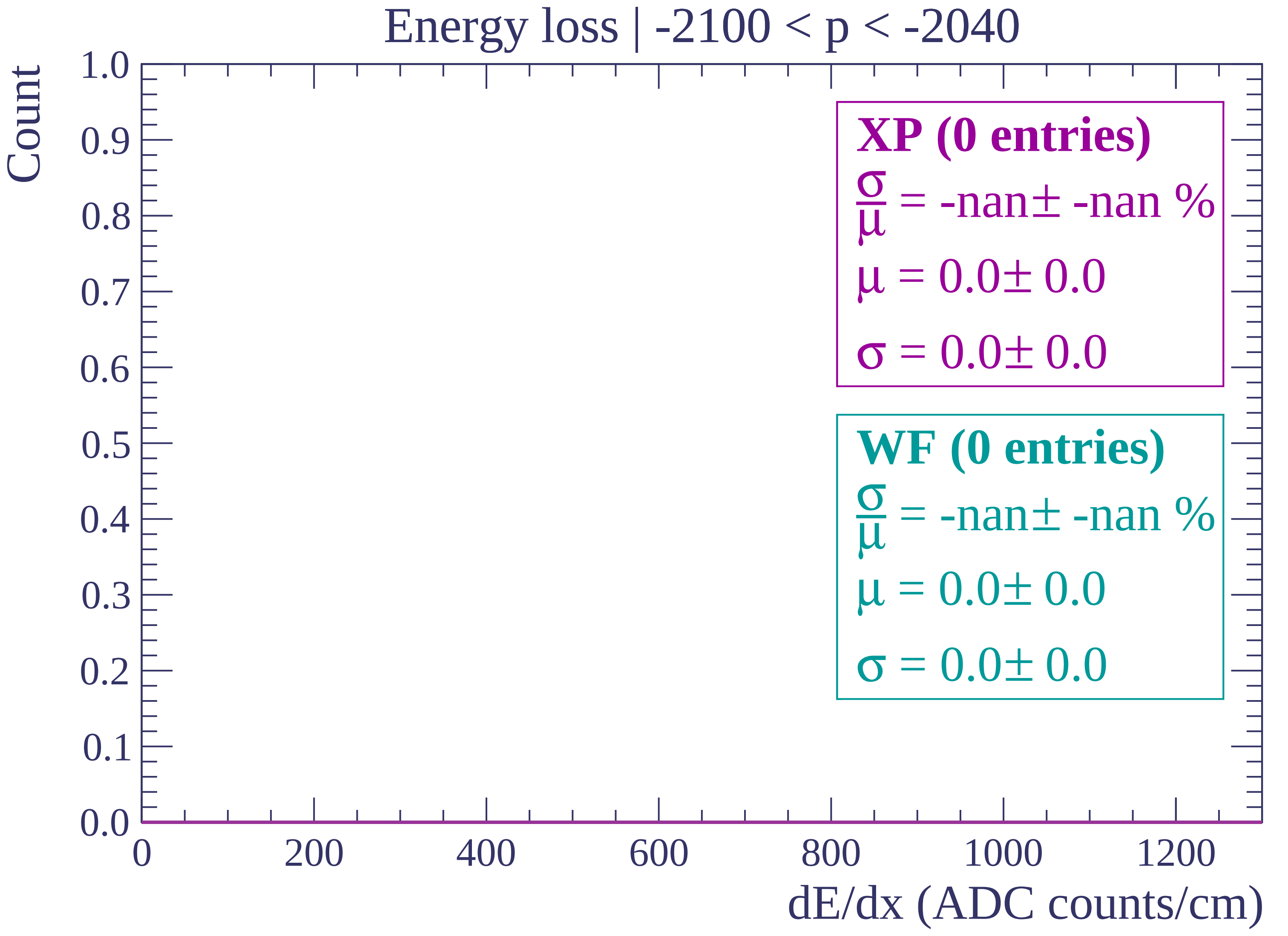


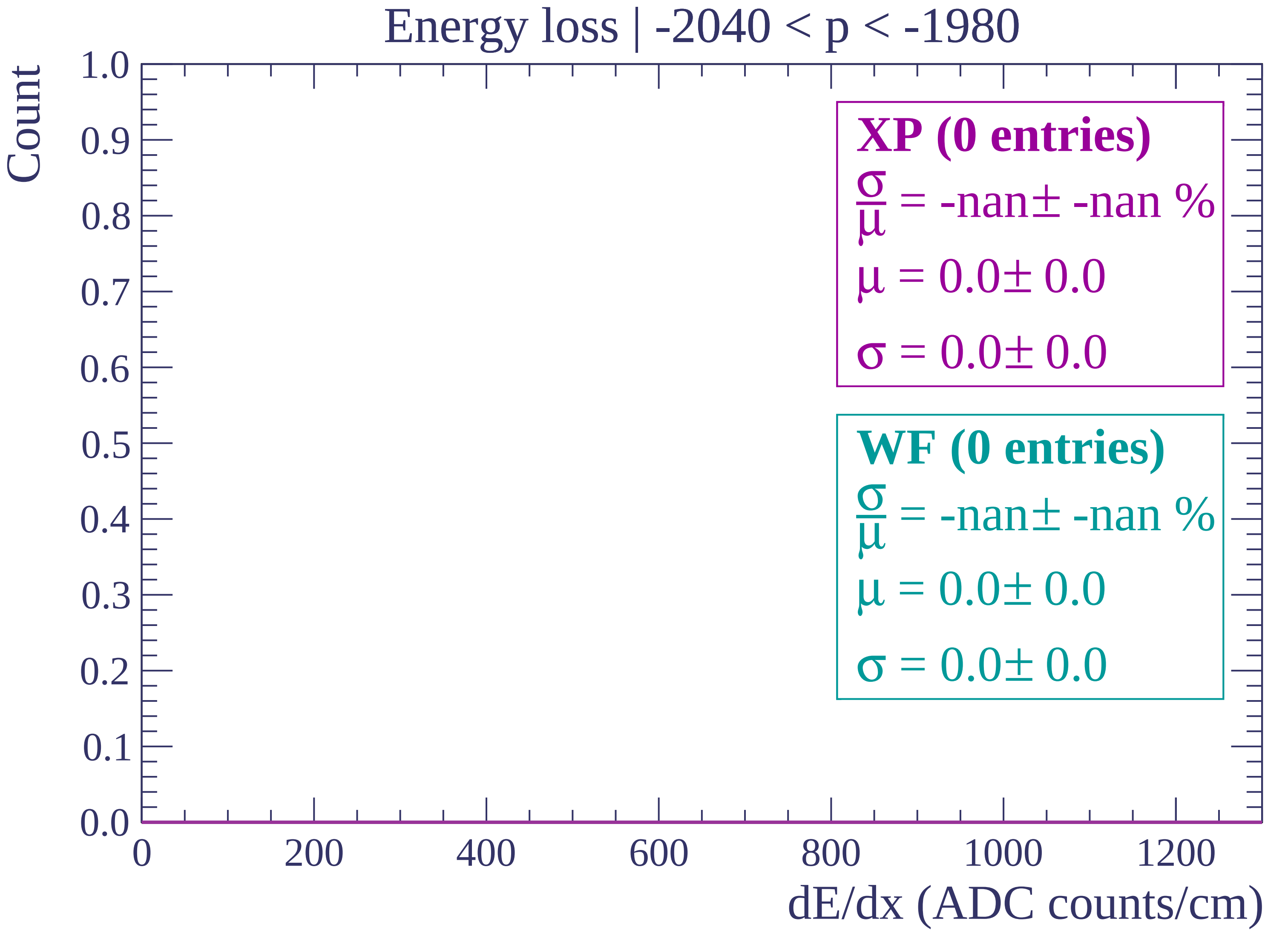
Energy loss | -2340 < p < -2280
Count
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -2280 < p < -2220
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

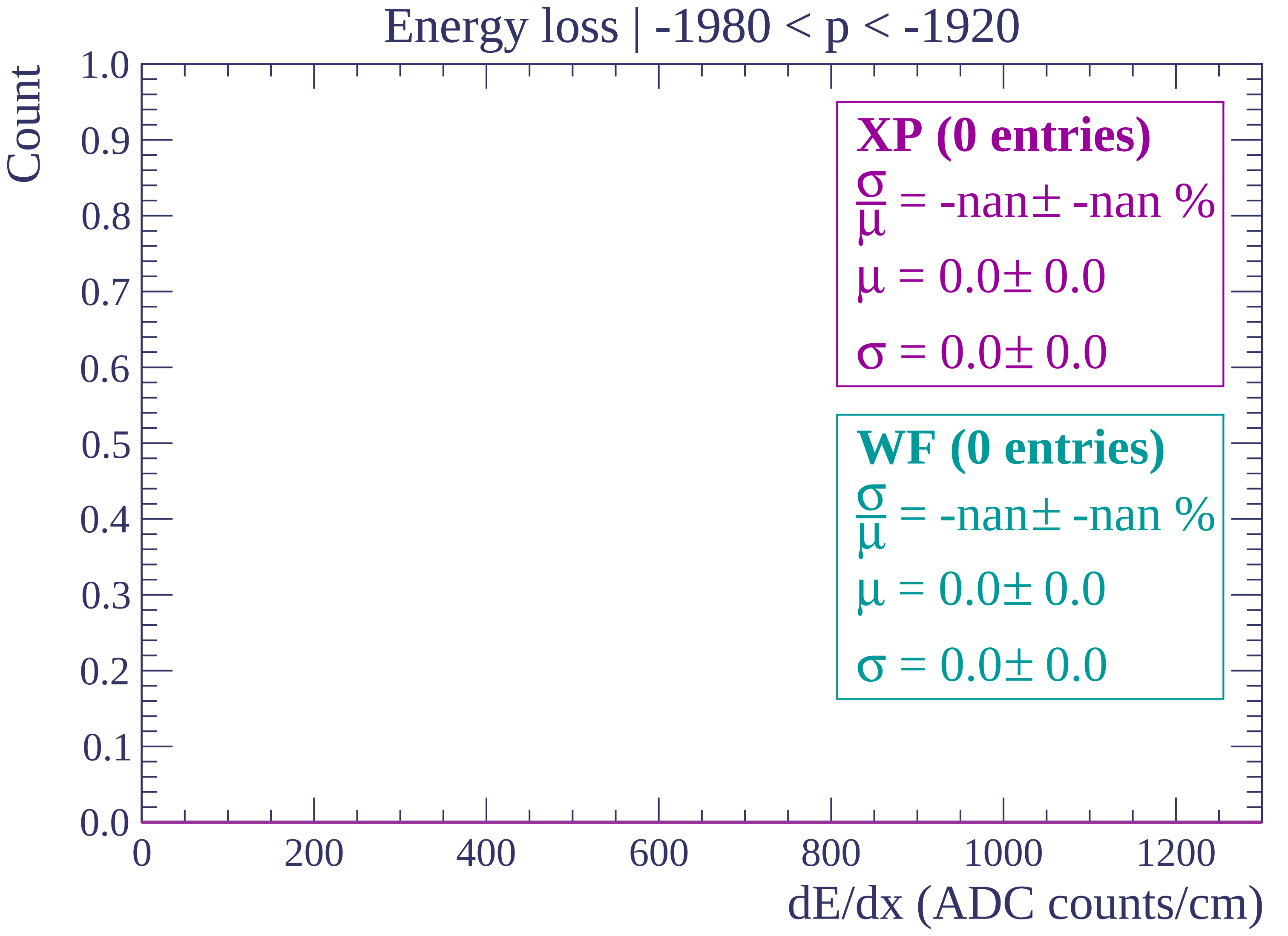

Energy loss | -2220 < p < -2160
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

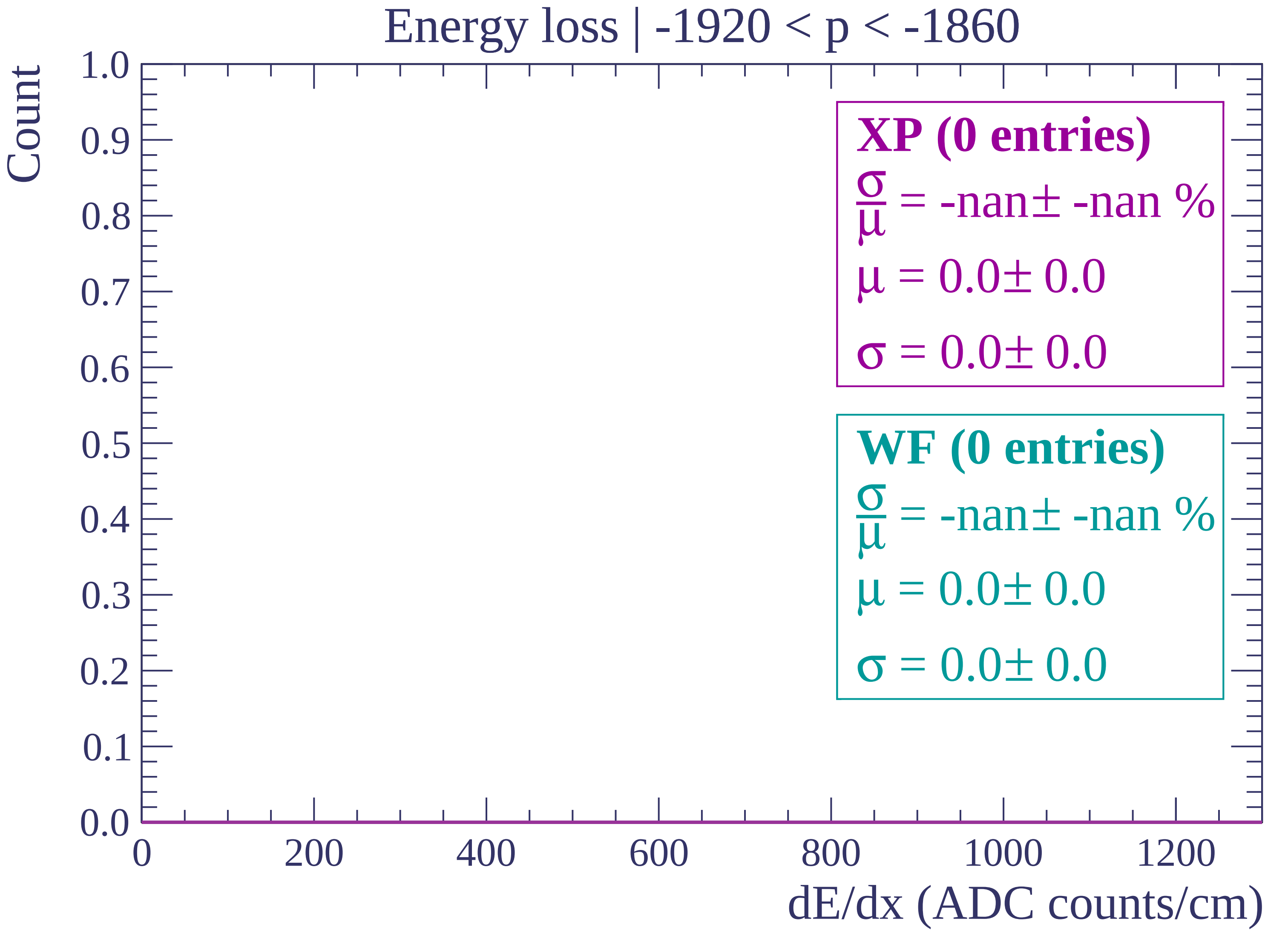

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

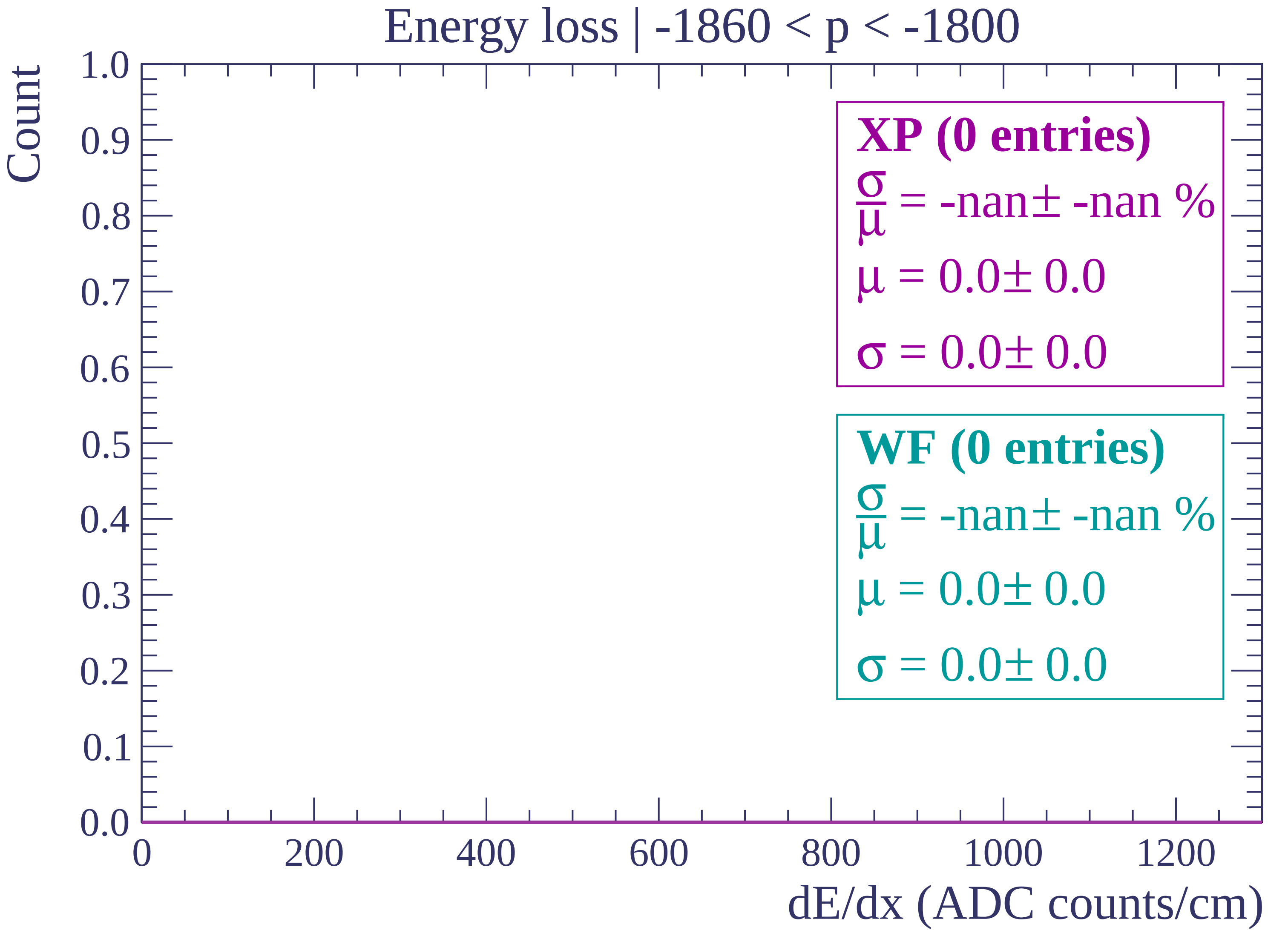

Energy loss | -2100 < p < -2040
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



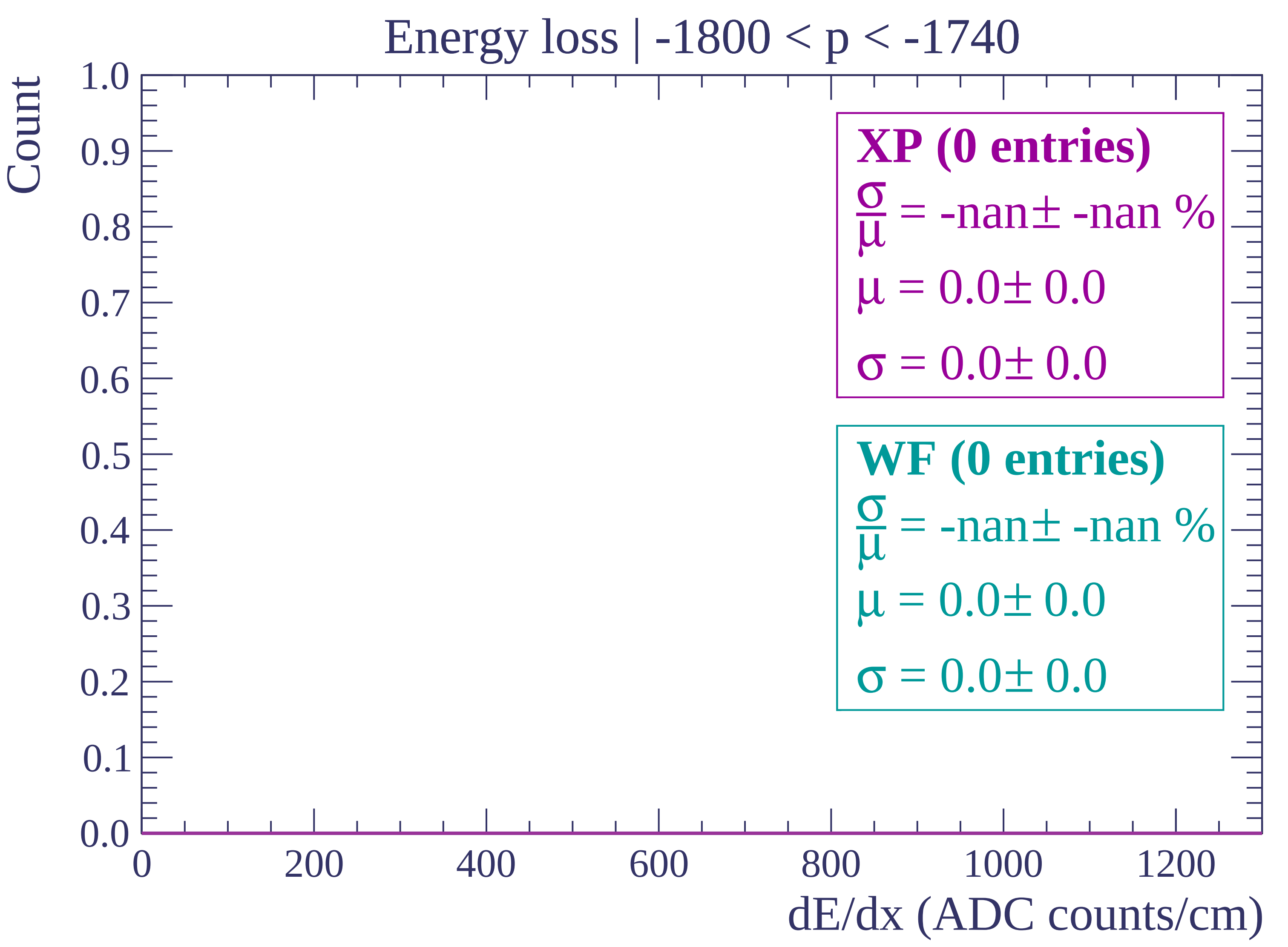
Energy loss | -2040 < p < -1980
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -1980 < p < -1920
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

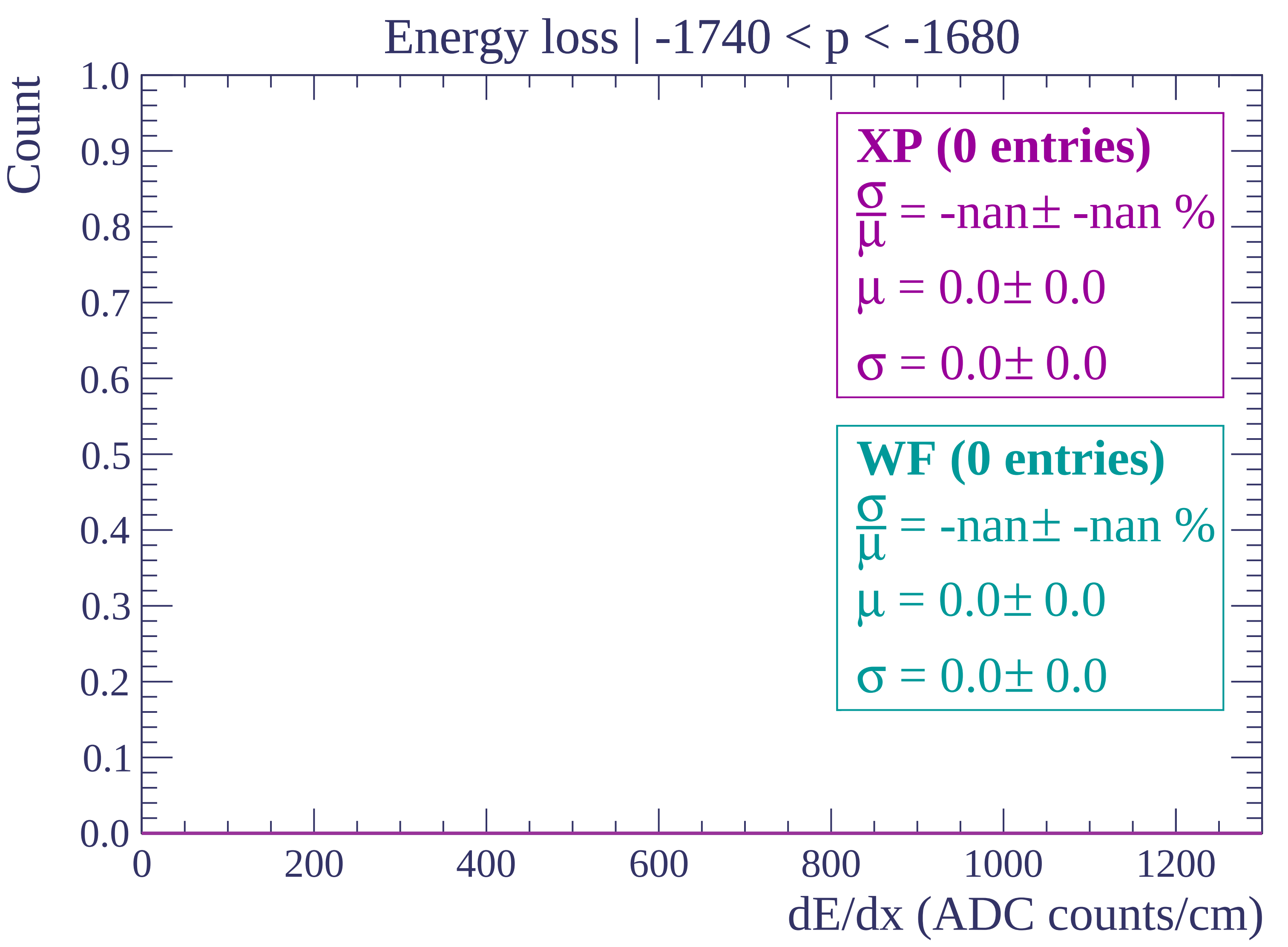

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

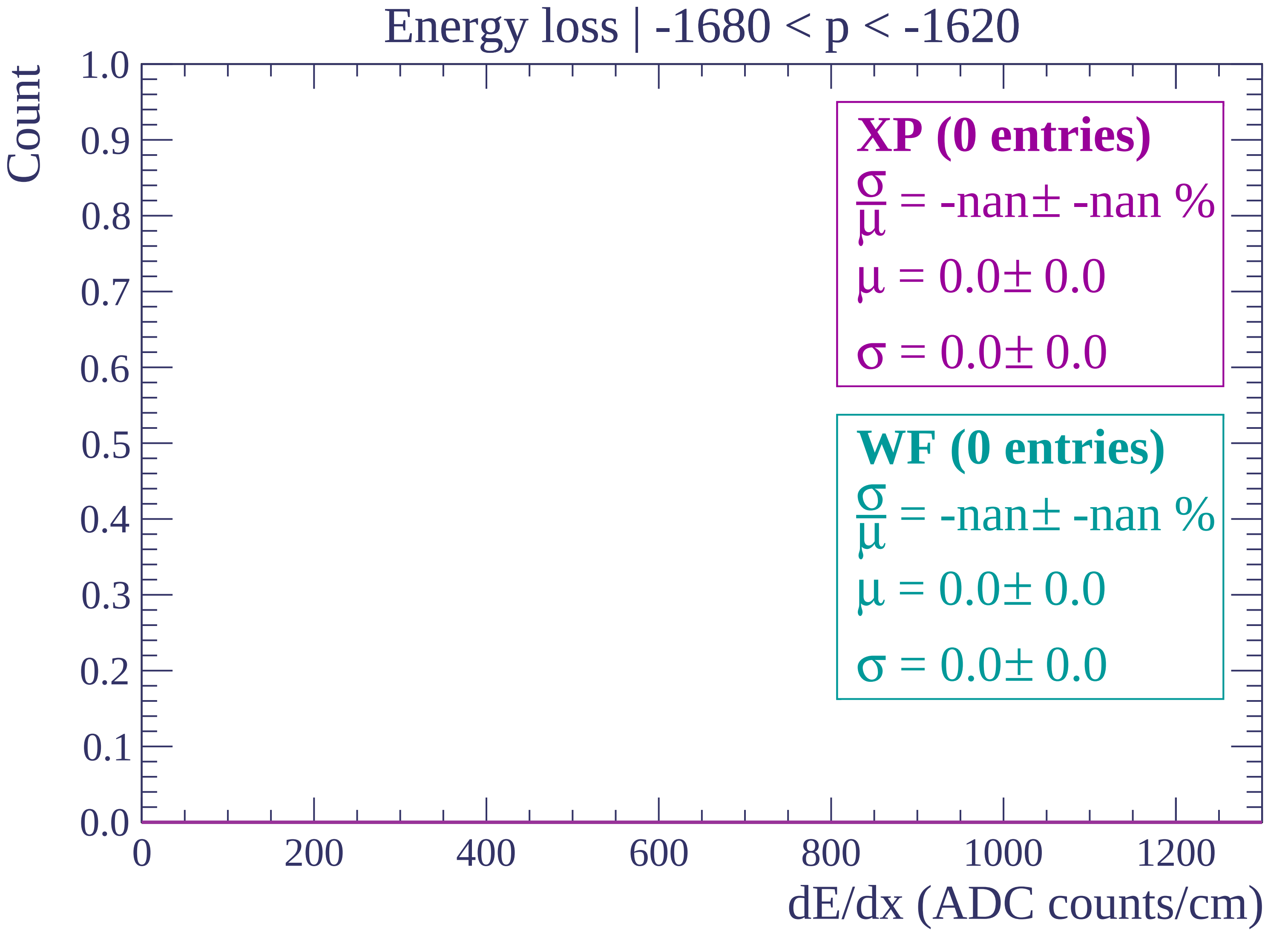


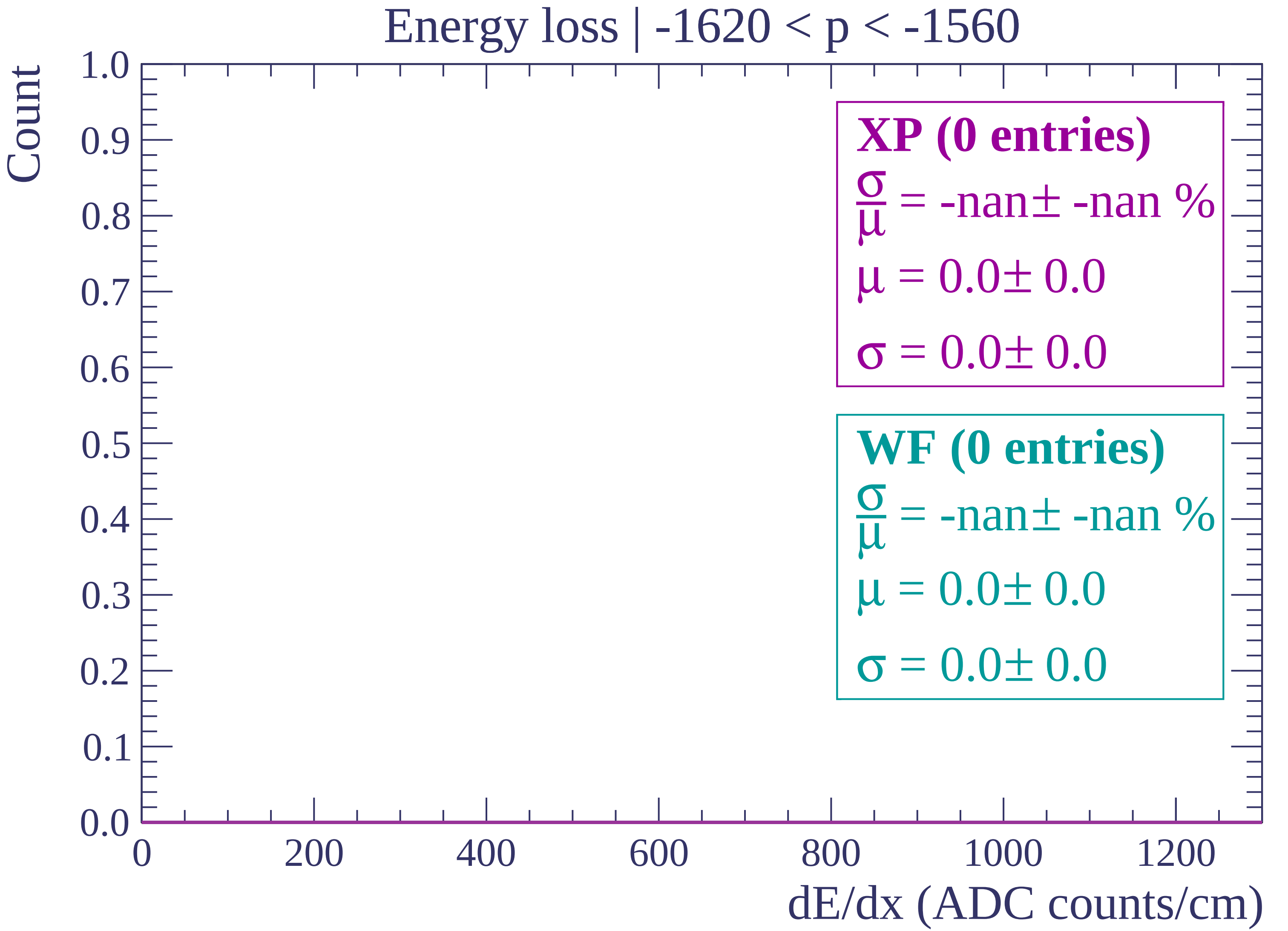
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

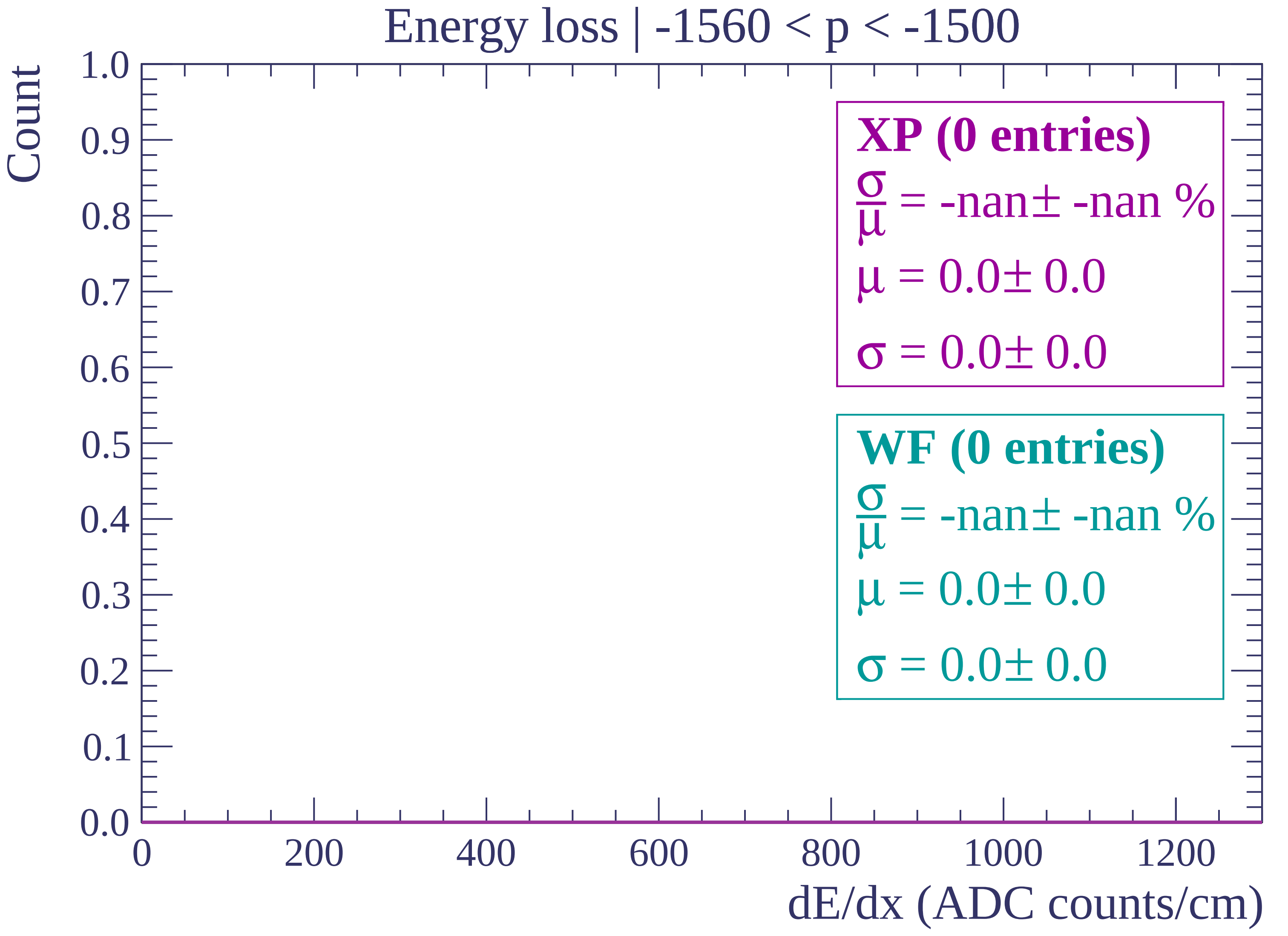


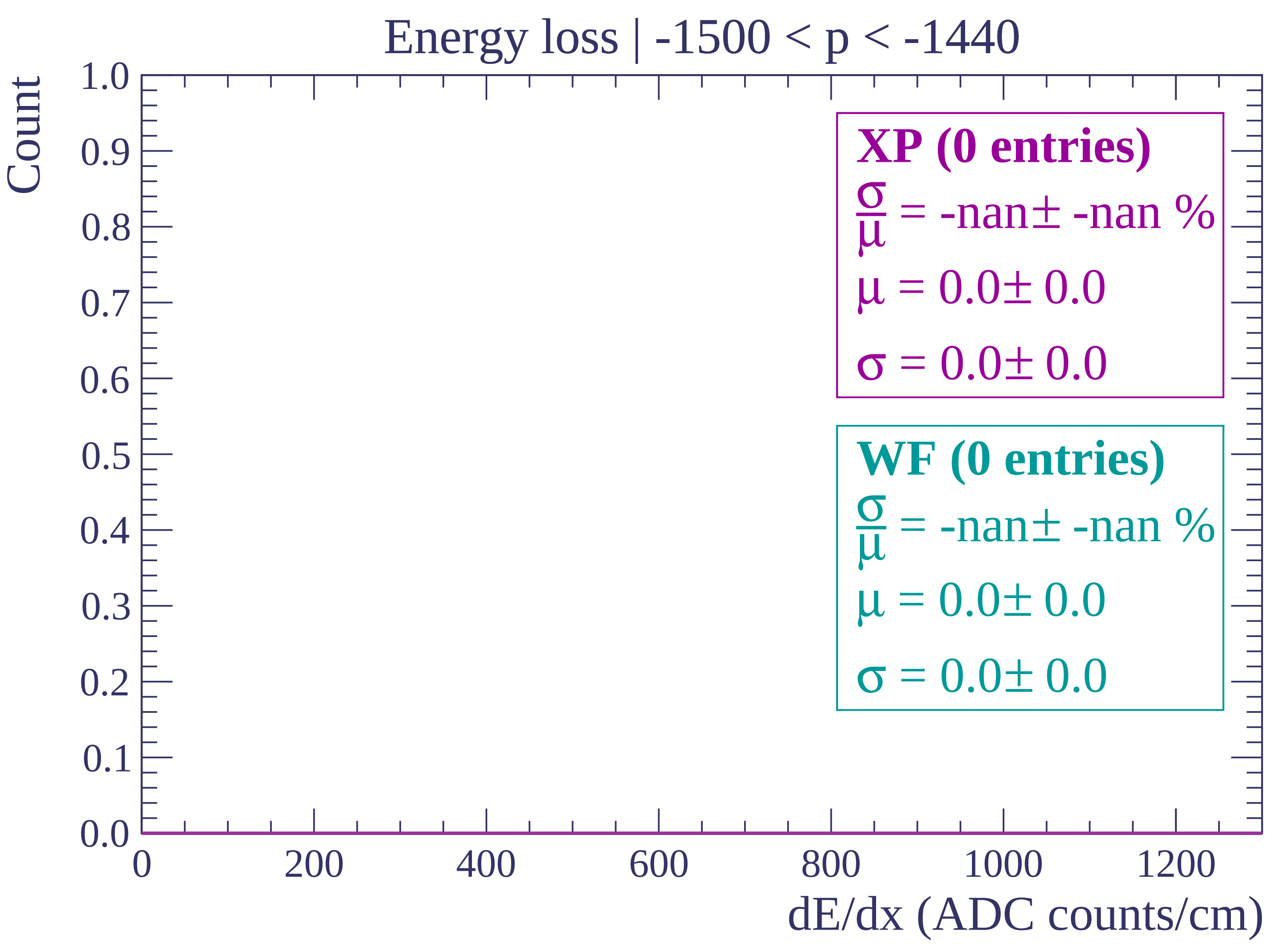
Energy loss | -1800 < p < -1740
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -1740 < p < -1680
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

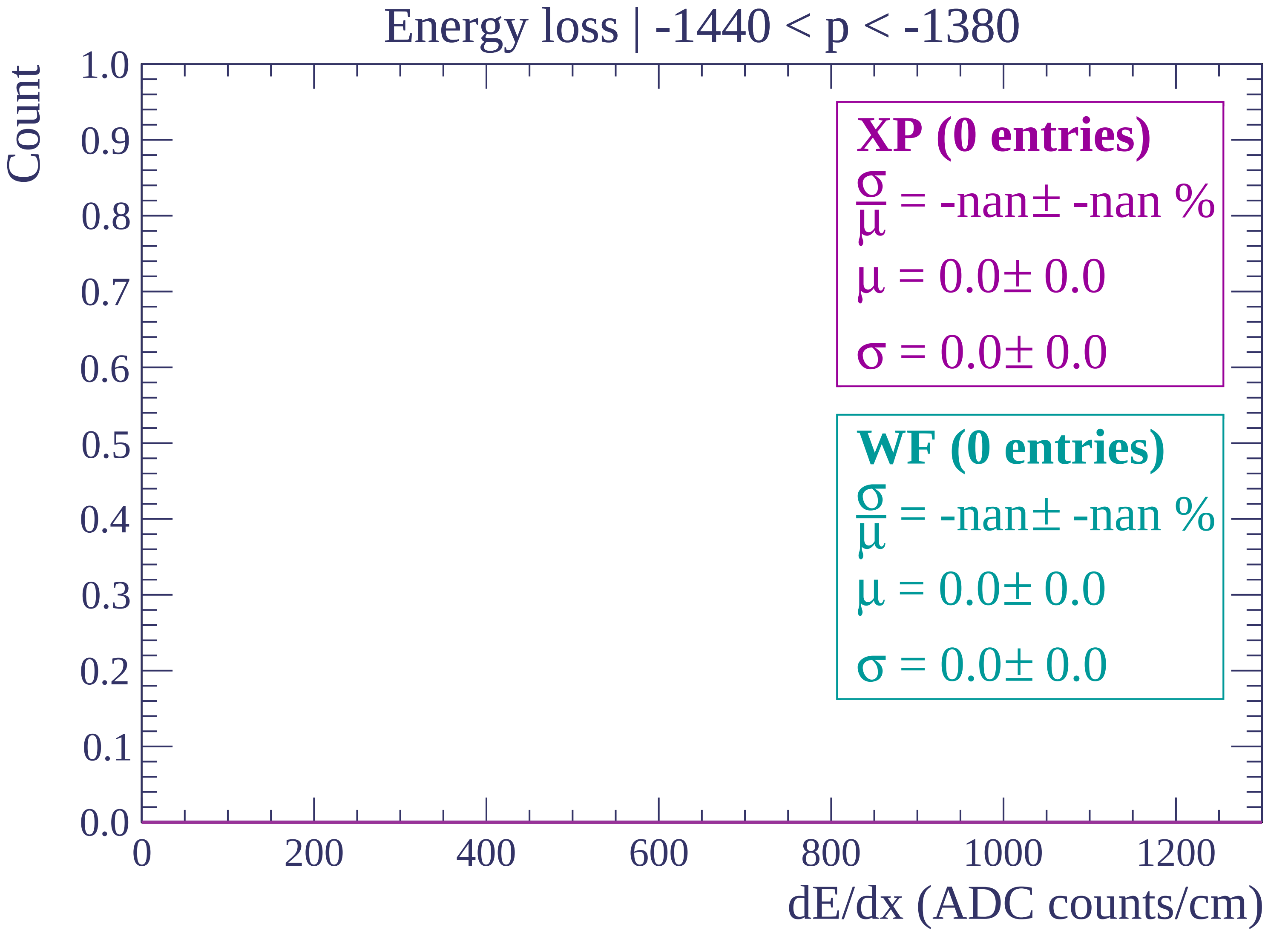

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

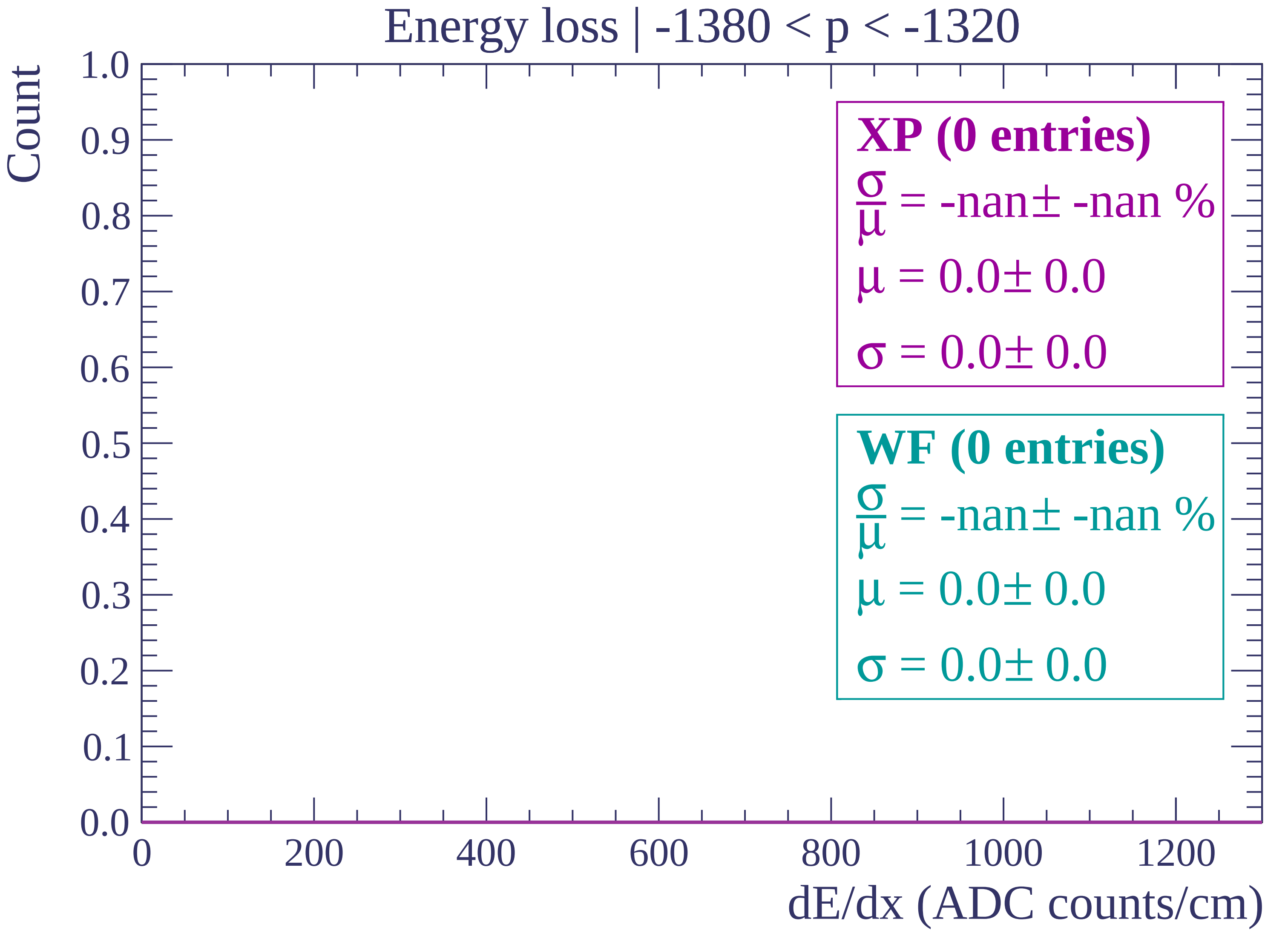

Energy loss | -1620 < p < -1560
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -1560 < p < -1500
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

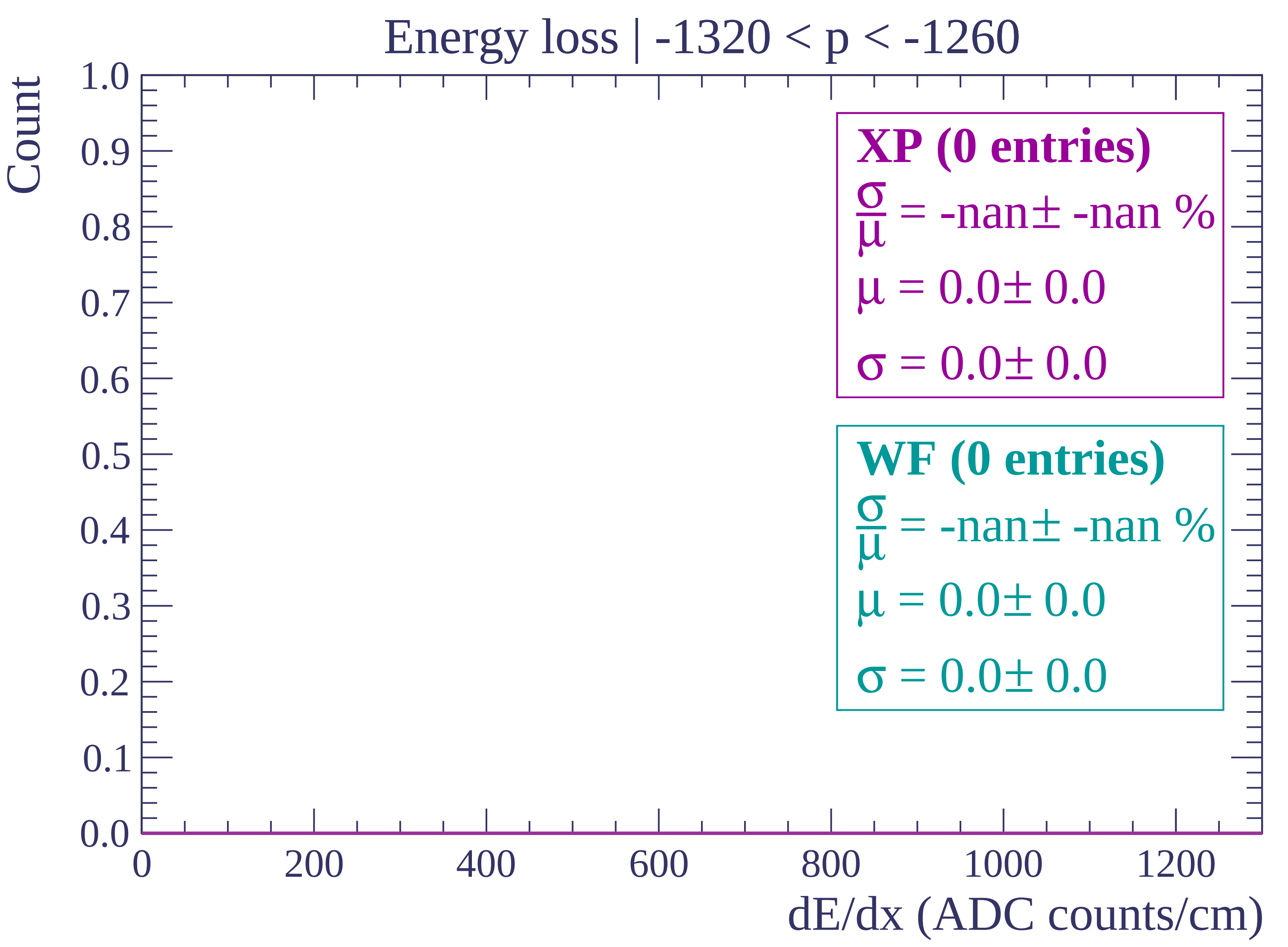


Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

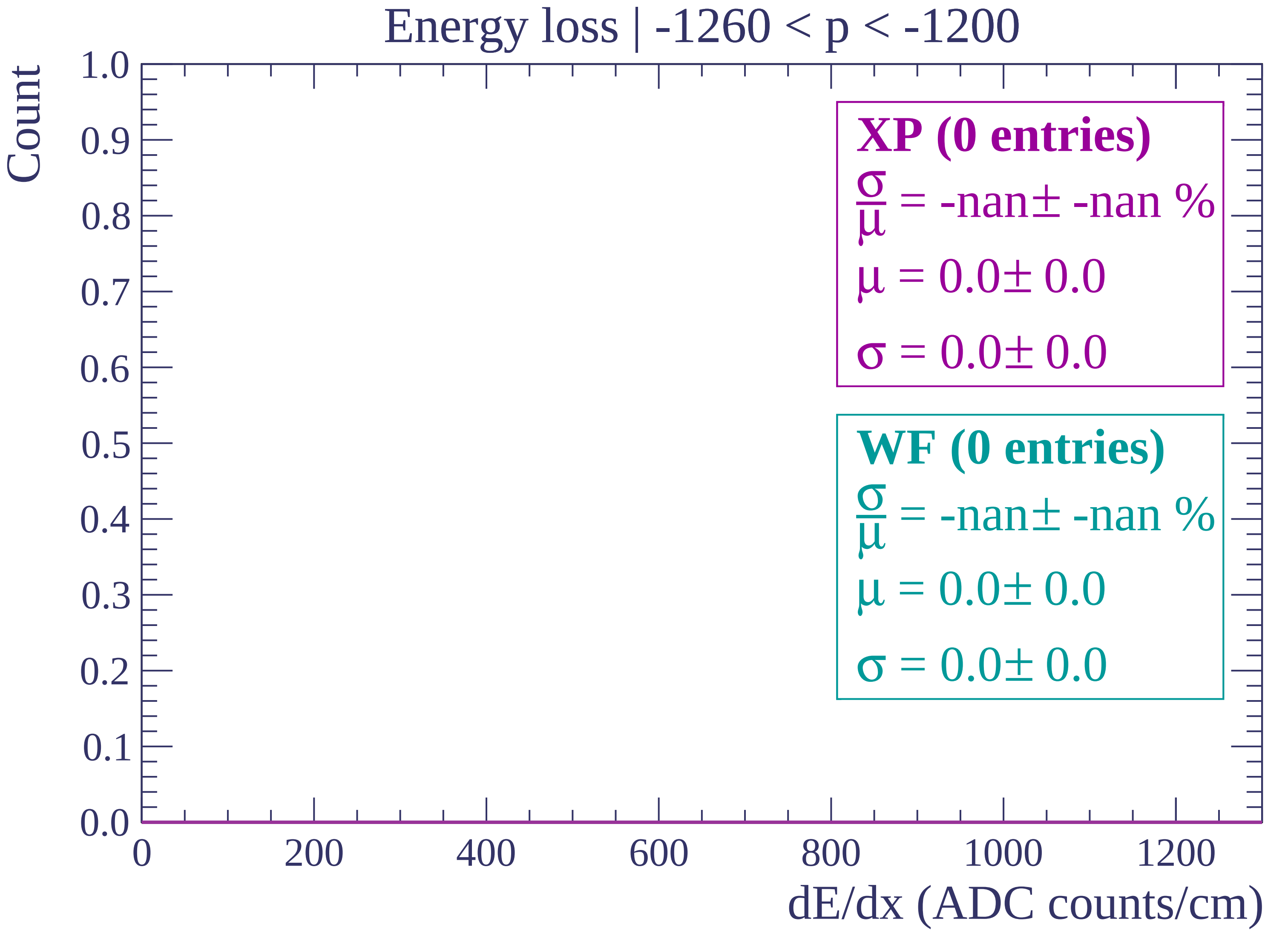

Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0



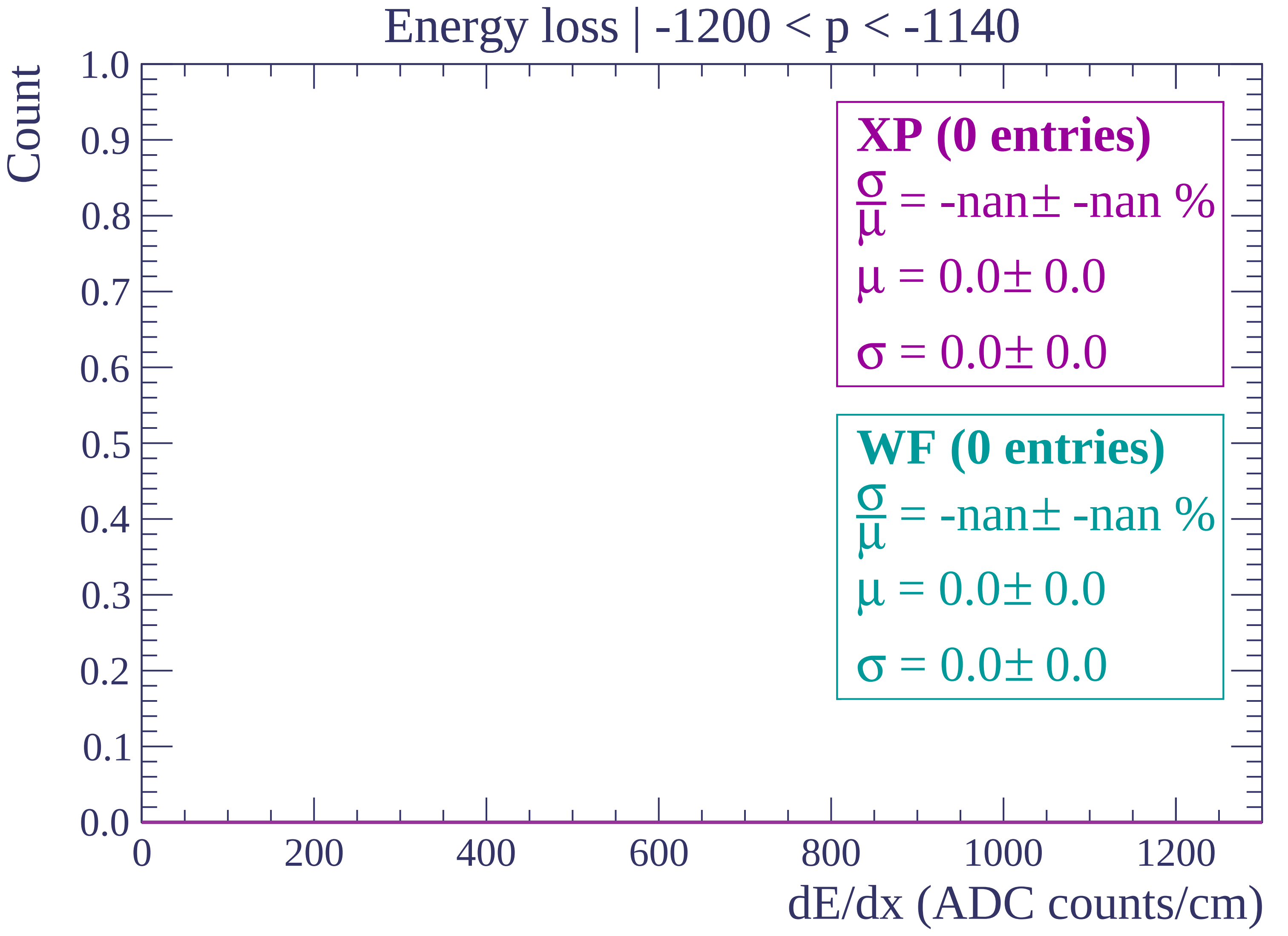
Energy loss | -1380 < p < -1320
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



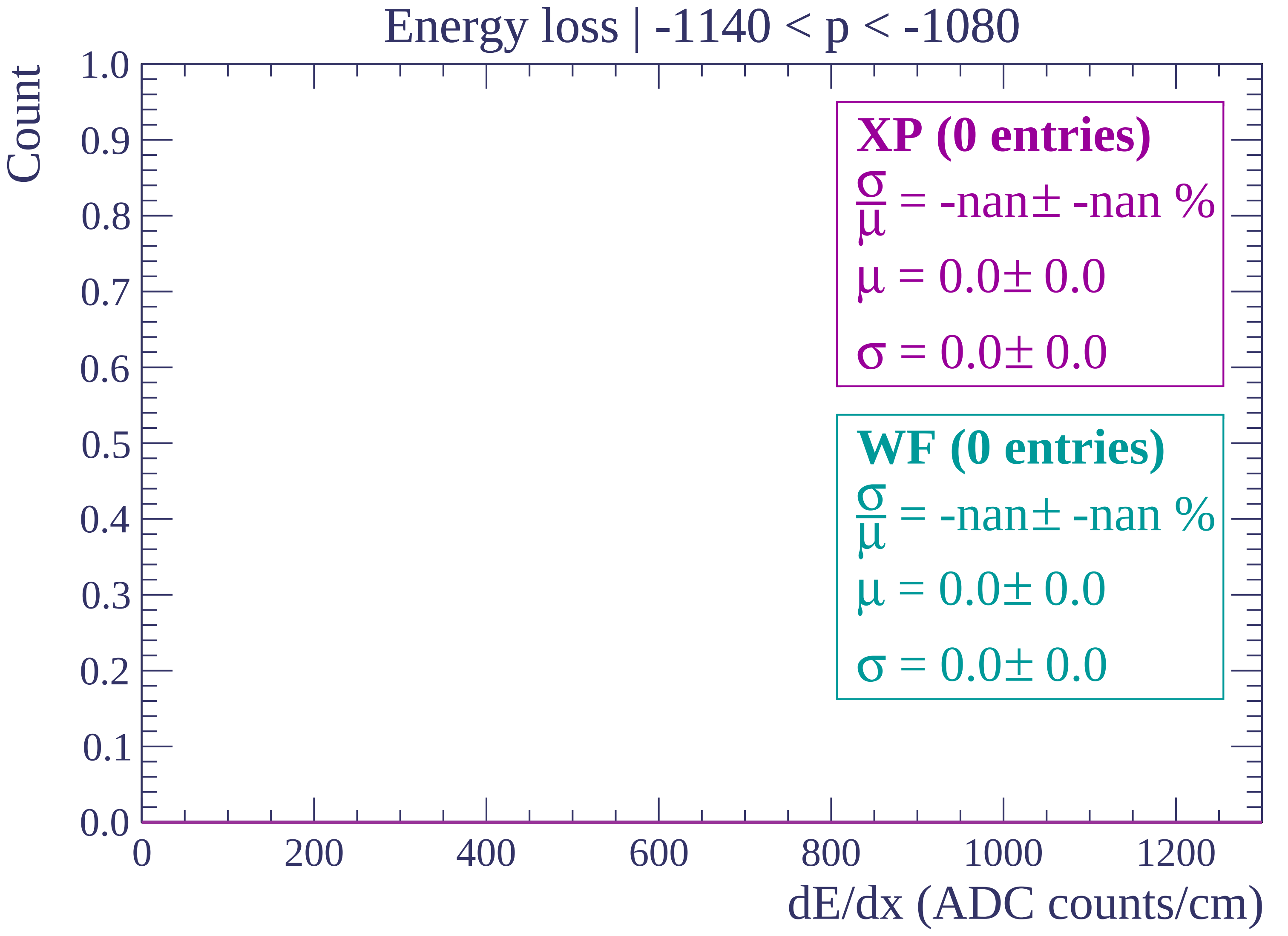
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

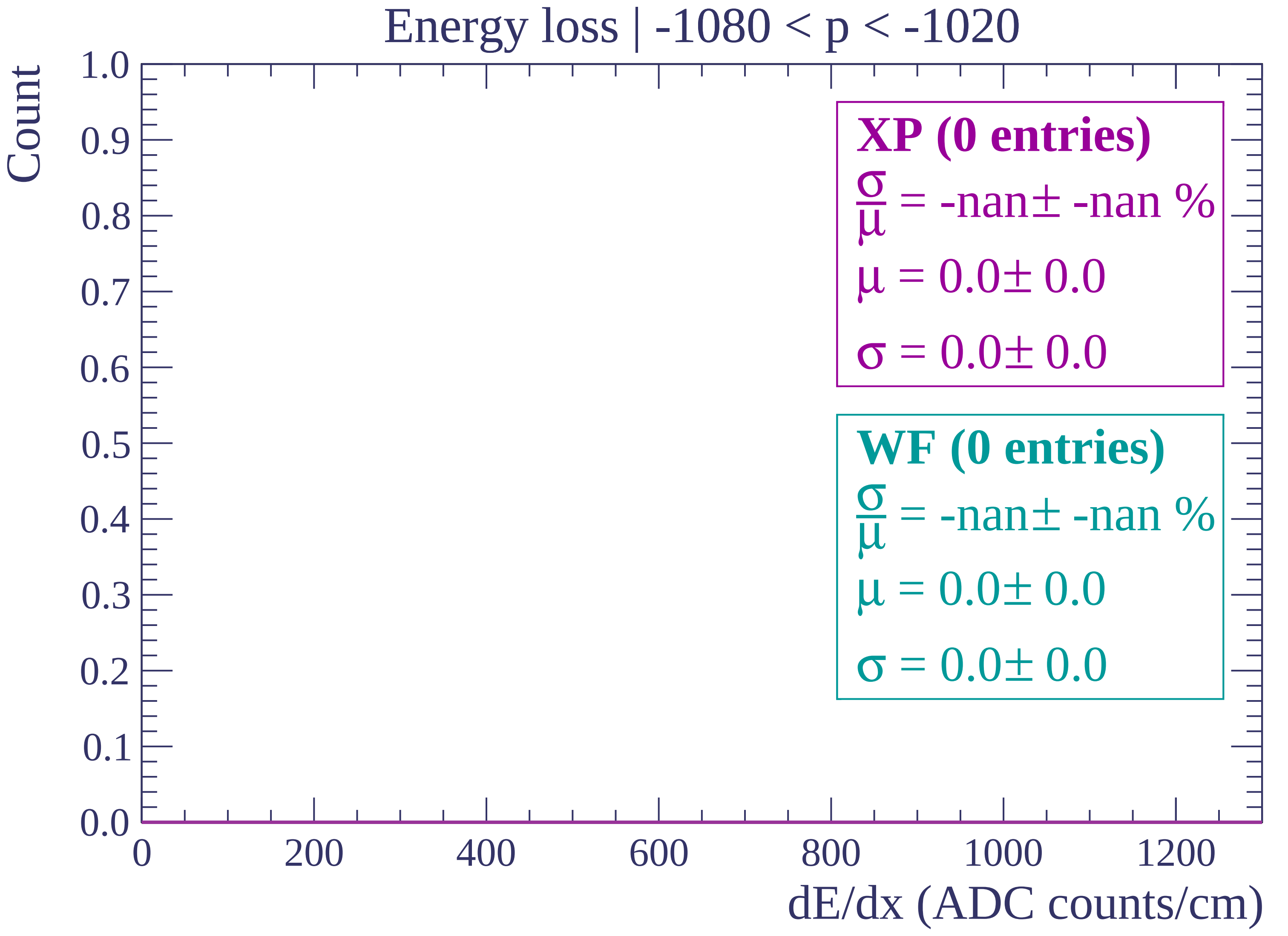


Energy loss | -1260 < p < -1200
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

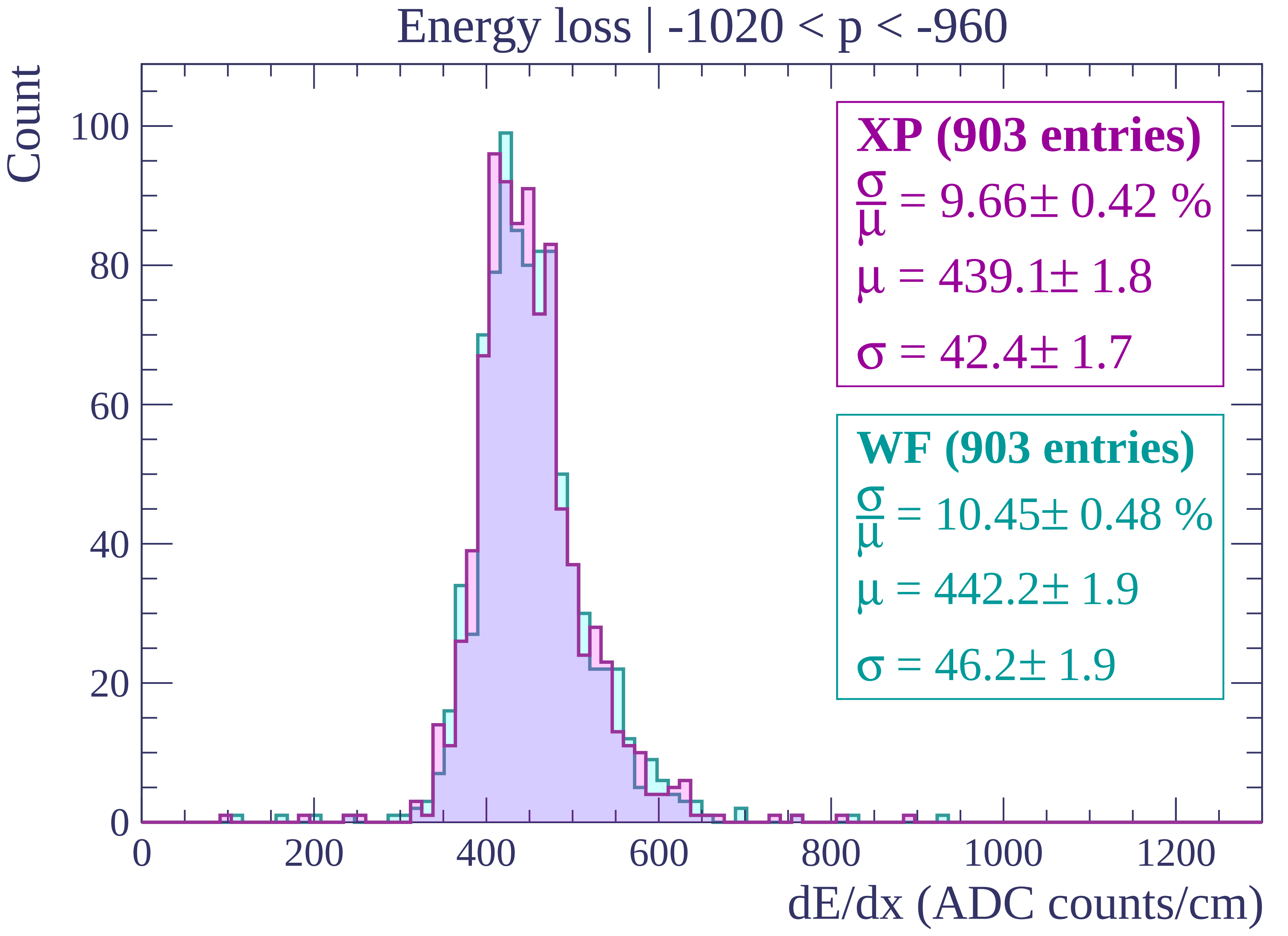


Energy loss | -1200 < p < -1140
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

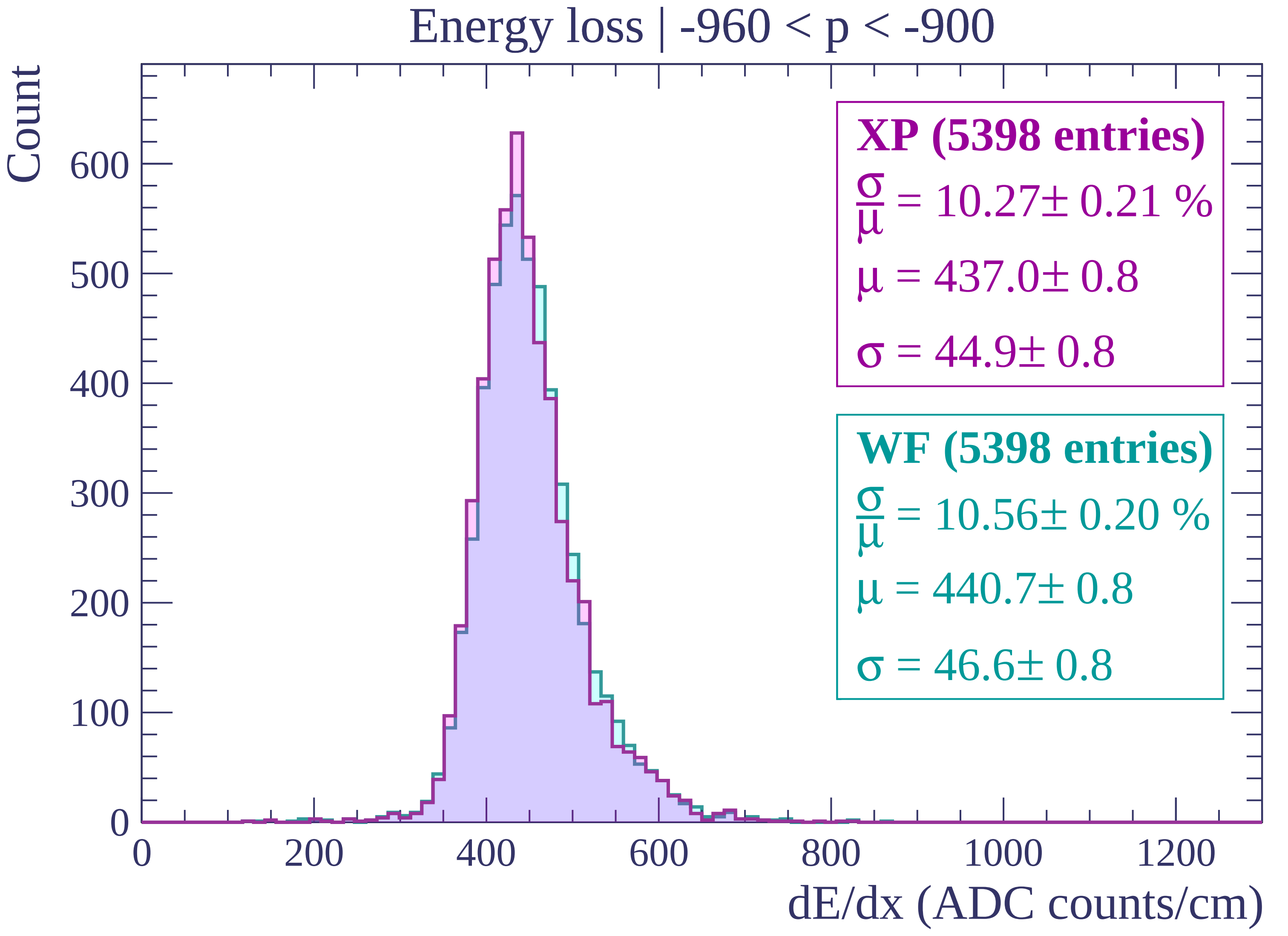

Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

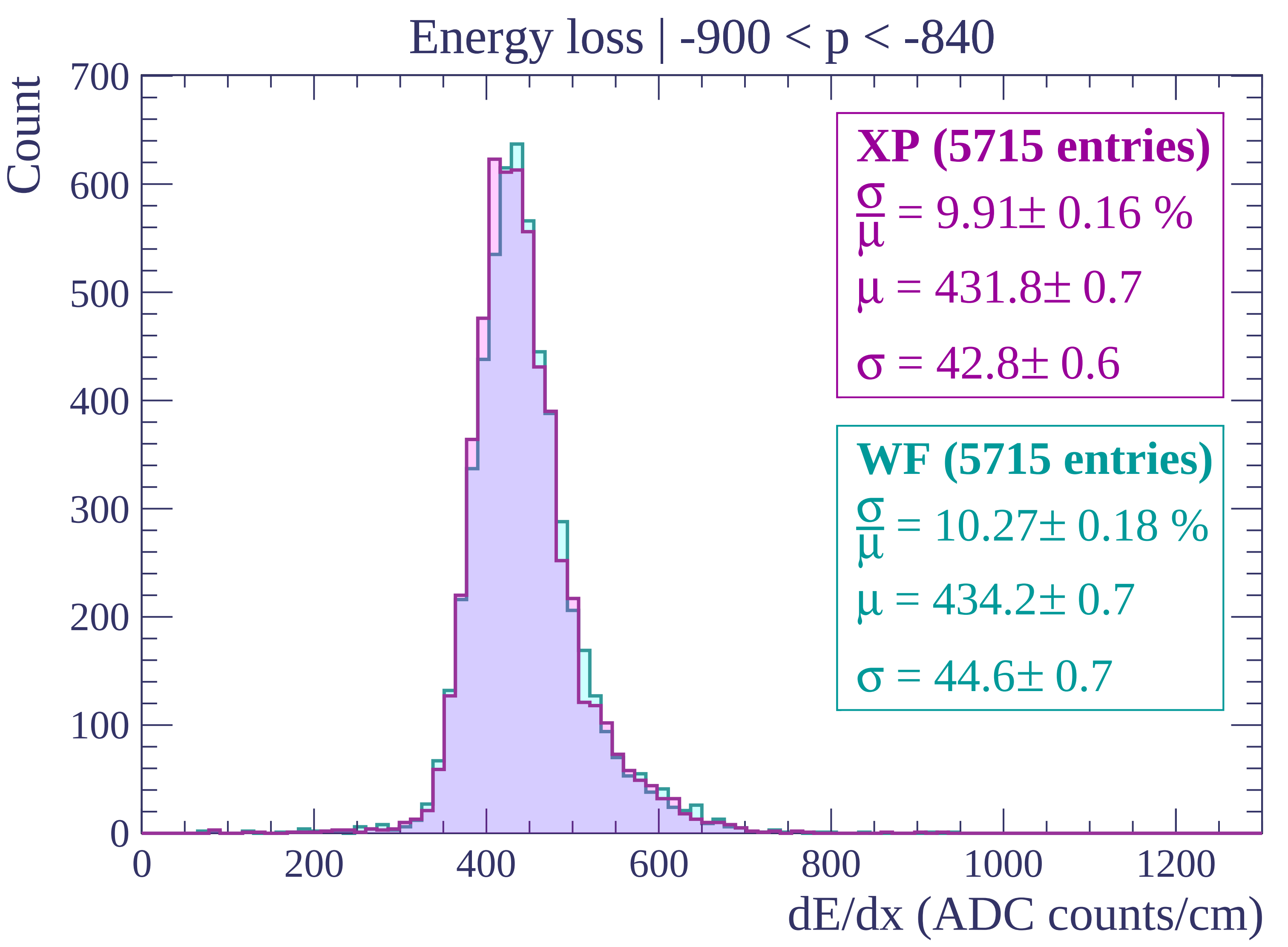


Energy loss | -1080 < p < -1020
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

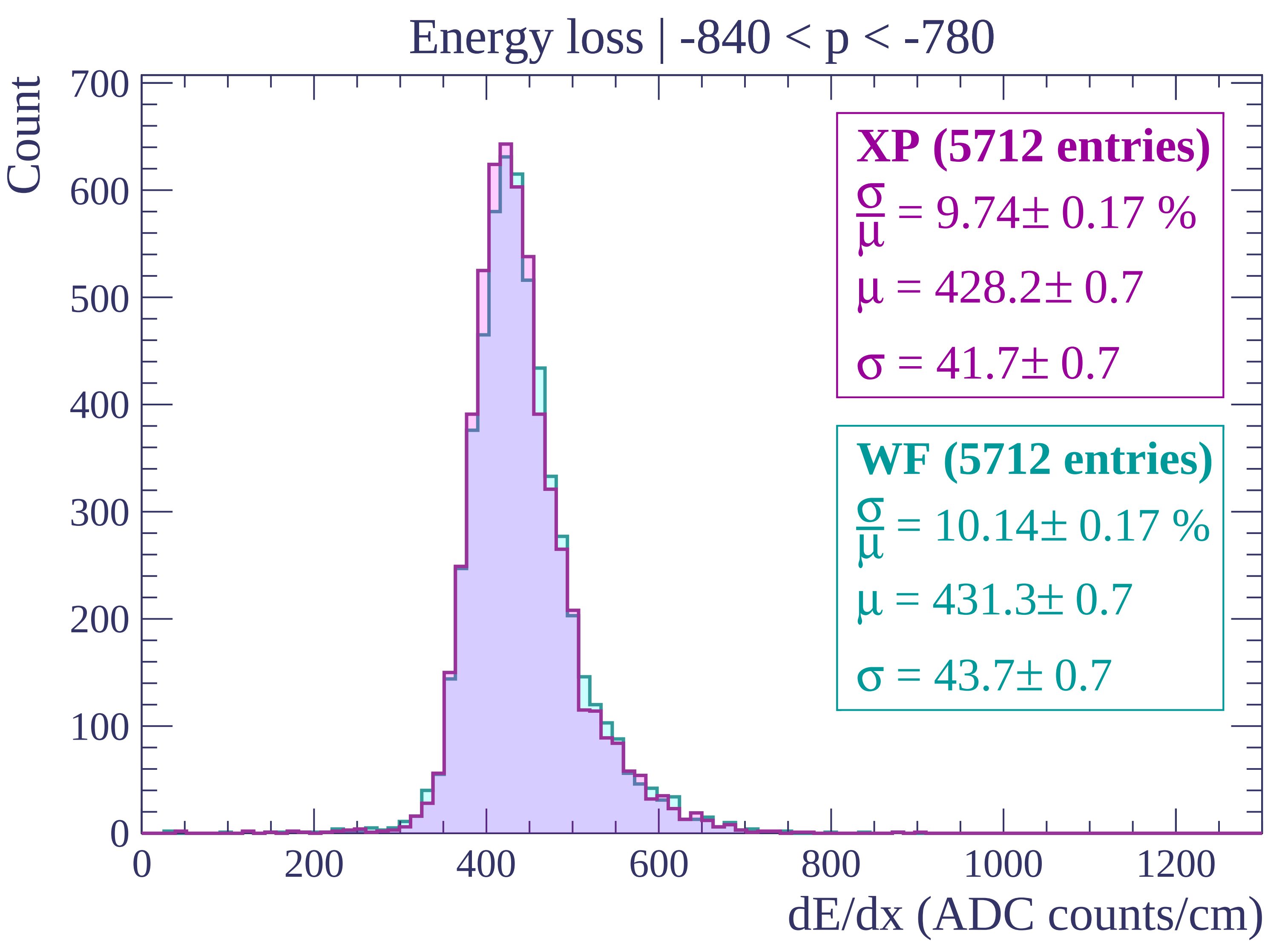


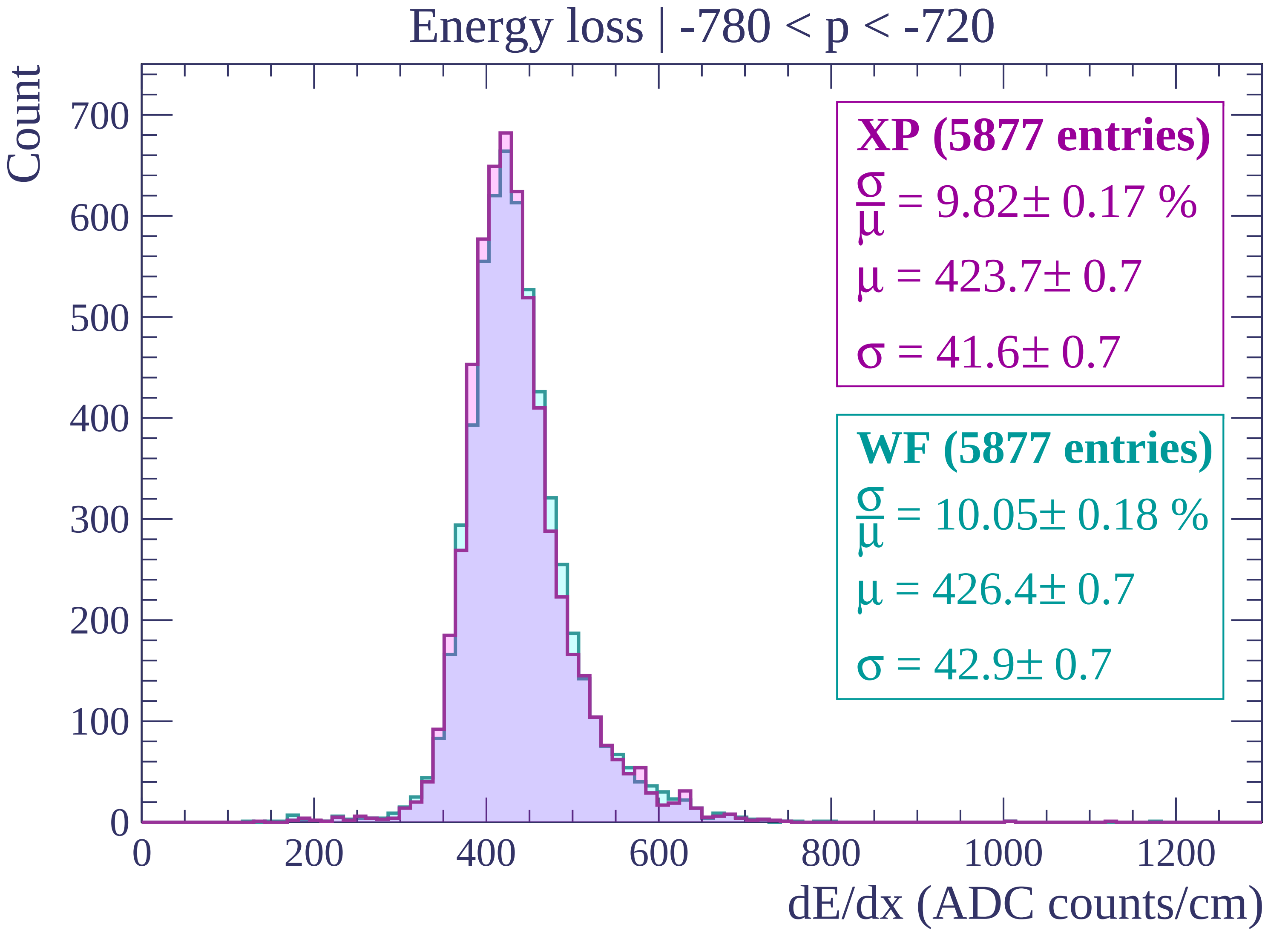
Energy loss | -1020 < p < -960
Count
XP (903 entries)
σ/μ = 9.66± 0.42 %
μ = 439.1± 1.8
σ = 42.4± 1.7
WF (903 entries)
σ/μ = 10.45± 0.48 %
μ = 442.2± 1.9
σ = 46.2± 1.9
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
XP (5398 entries)
$\frac{\sigma}{\mu}$ = 10.27± 0.21 %
μ = 437.0± 0.8
σ = 44.9± 0.8
WF (5398 entries)
$\frac{\sigma}{\mu}$ = 10.56± 0.20 %
μ = 440.7± 0.8
σ = 46.6± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

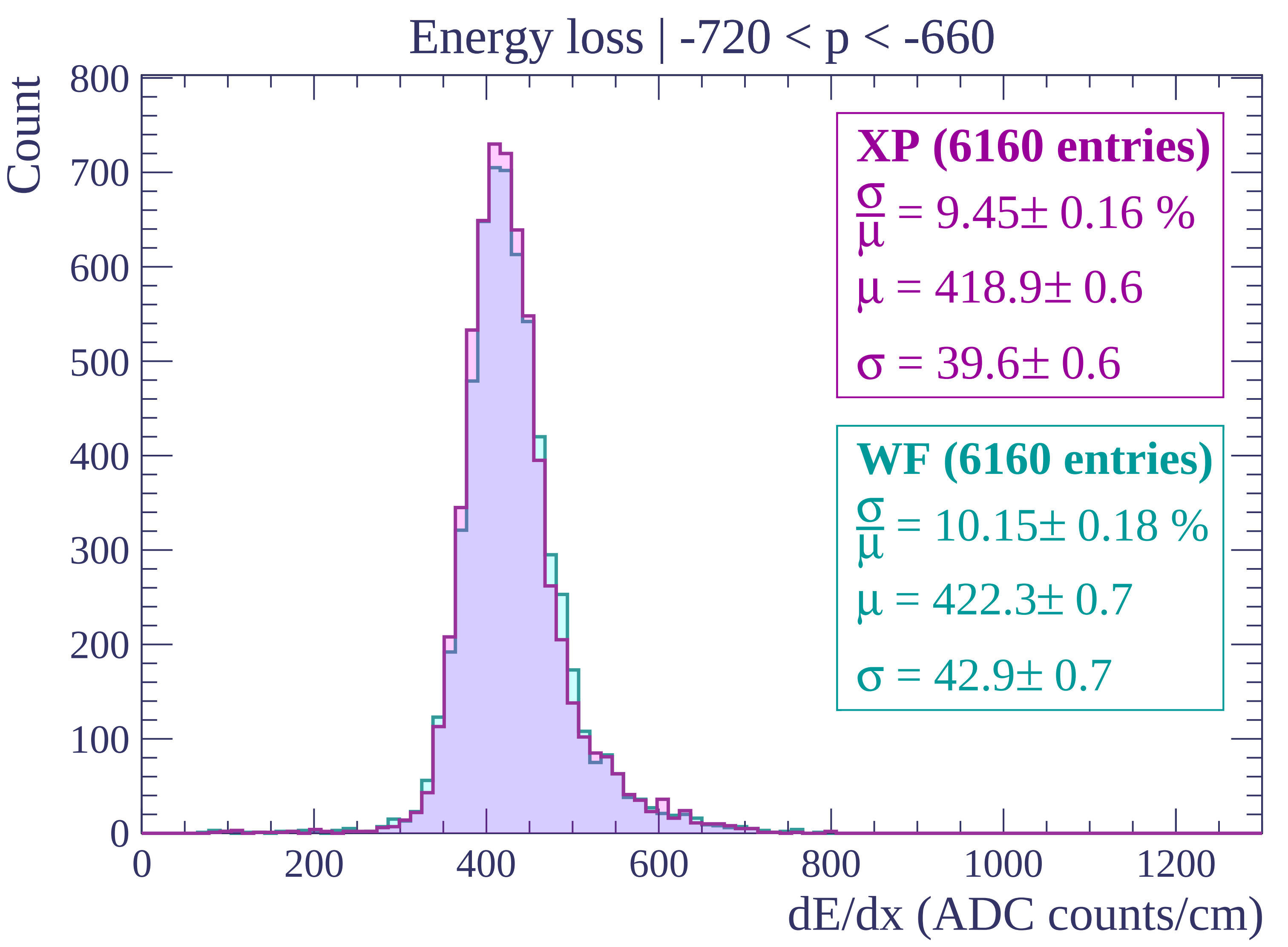


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (5715 entries)
σ/μ = 9.91± 0.16 %
μ = 431.8± 0.7
σ = 42.8± 0.6
WF (5715 entries)
σ/μ = 10.27± 0.18 %
μ = 434.2± 0.7
σ = 44.6± 0.7

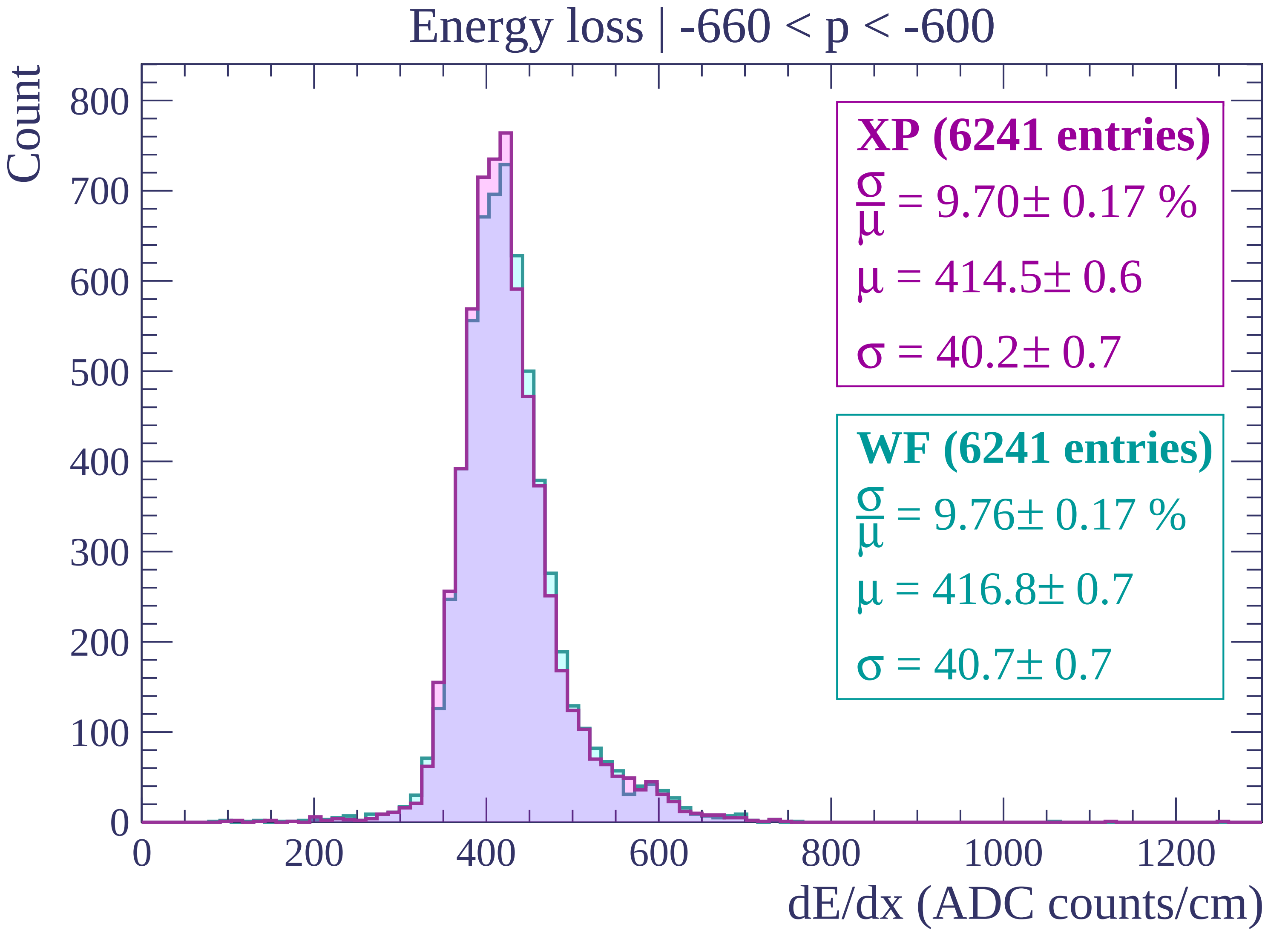

Energy loss | -840 < p < -780
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5712 entries)
σ/μ = 9.74± 0.17 %
μ = 428.2± 0.7
σ = 41.7± 0.7
WF (5712 entries)
σ/μ = 10.14± 0.17 %
μ = 431.3± 0.7
σ = 43.7± 0.7

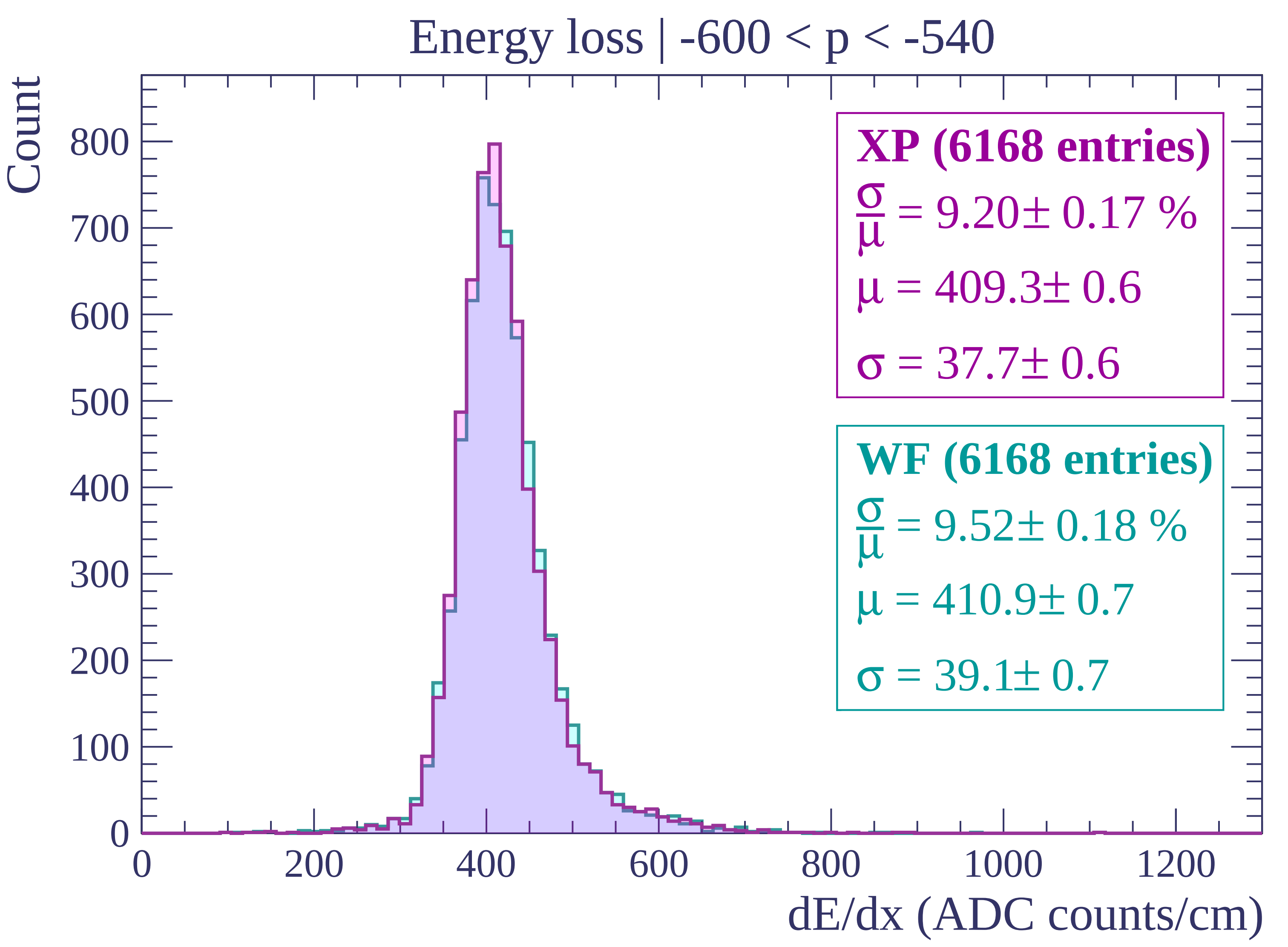


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5877 entries)
$\frac{\sigma}{\mu}$ = 9.82± 0.17 %
μ = 423.7± 0.7
σ = 41.6± 0.7
WF (5877 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.18 %
μ = 426.4± 0.7
σ = 42.9± 0.7

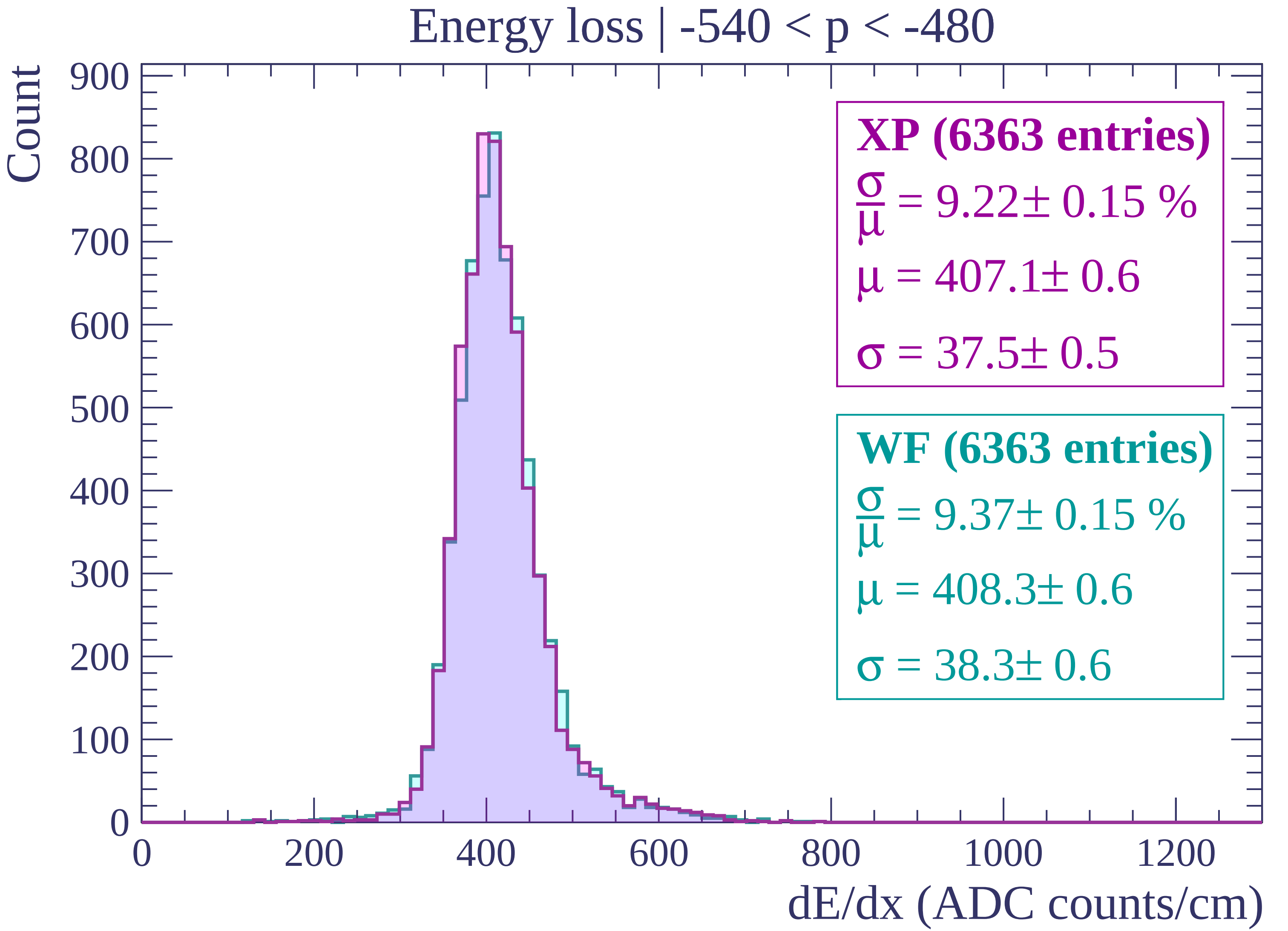


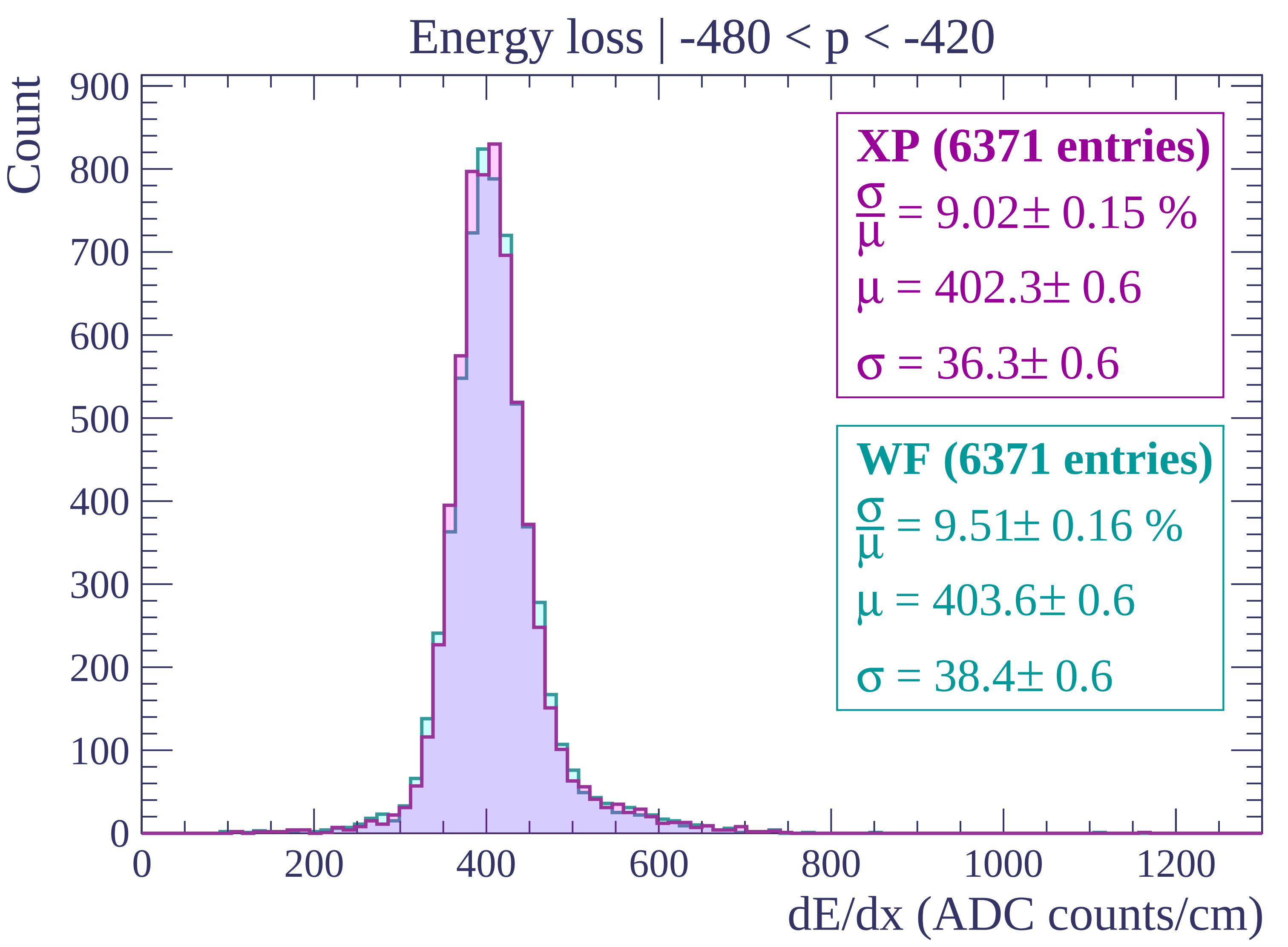
Energy loss | -720 < p < -660
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6160 entries)
$\frac{\sigma}{\mu}$ = 9.45± 0.16 %
μ = 418.9± 0.6
σ = 39.6± 0.6
WF (6160 entries)
$\frac{\sigma}{\mu}$ = 10.15± 0.18 %
μ = 422.3± 0.7
σ = 42.9± 0.7


Energy loss | -660 < p < -600
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6241 entries)
σ/μ = 9.70± 0.17 %
μ = 414.5± 0.6
σ = 40.2± 0.7
WF (6241 entries)
σ/μ = 9.76± 0.17 %
μ = 416.8± 0.7
σ = 40.7± 0.7

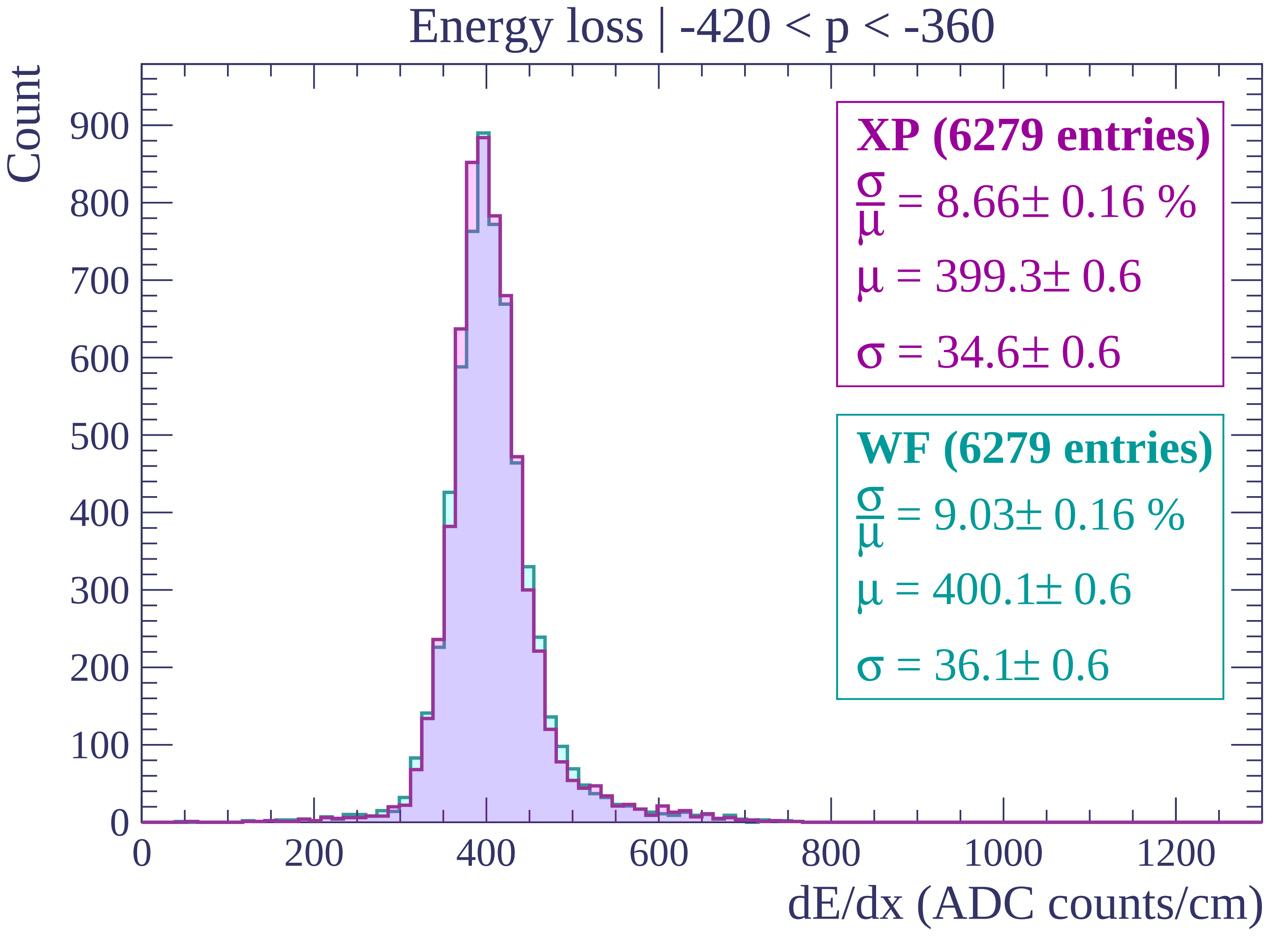


Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6168 entries)
σ/μ = 9.20± 0.17 %
μ = 409.3± 0.6
σ = 37.7± 0.6
WF (6168 entries)
σ/μ = 9.52± 0.18 %
μ = 410.9± 0.7
σ = 39.1± 0.7

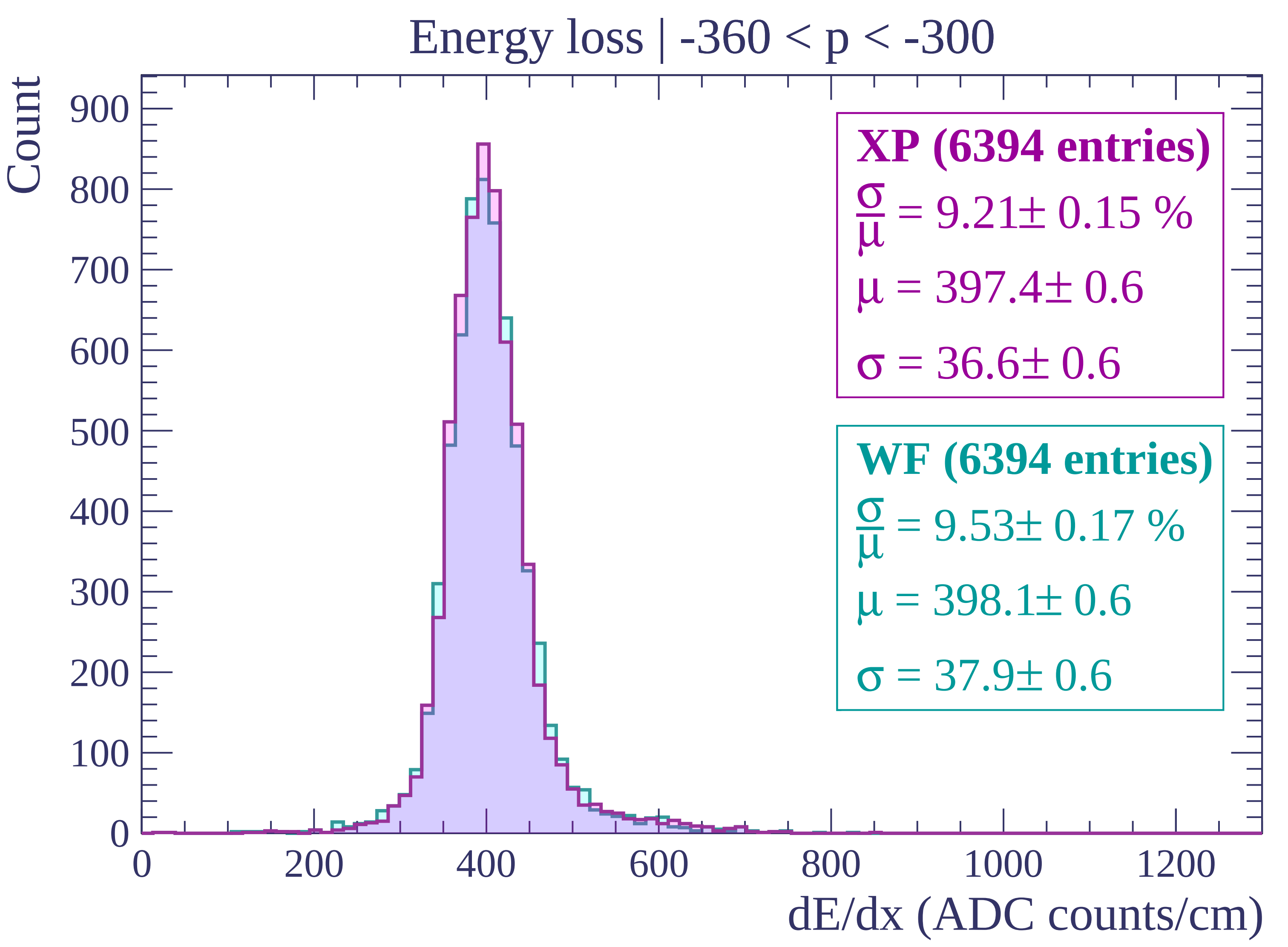

Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (6363 entries)
σ/μ = 9.22± 0.15 %
μ = 407.1± 0.6
σ = 37.5± 0.5
WF (6363 entries)
σ/μ = 9.37± 0.15 %
μ = 408.3± 0.6
σ = 38.3± 0.6

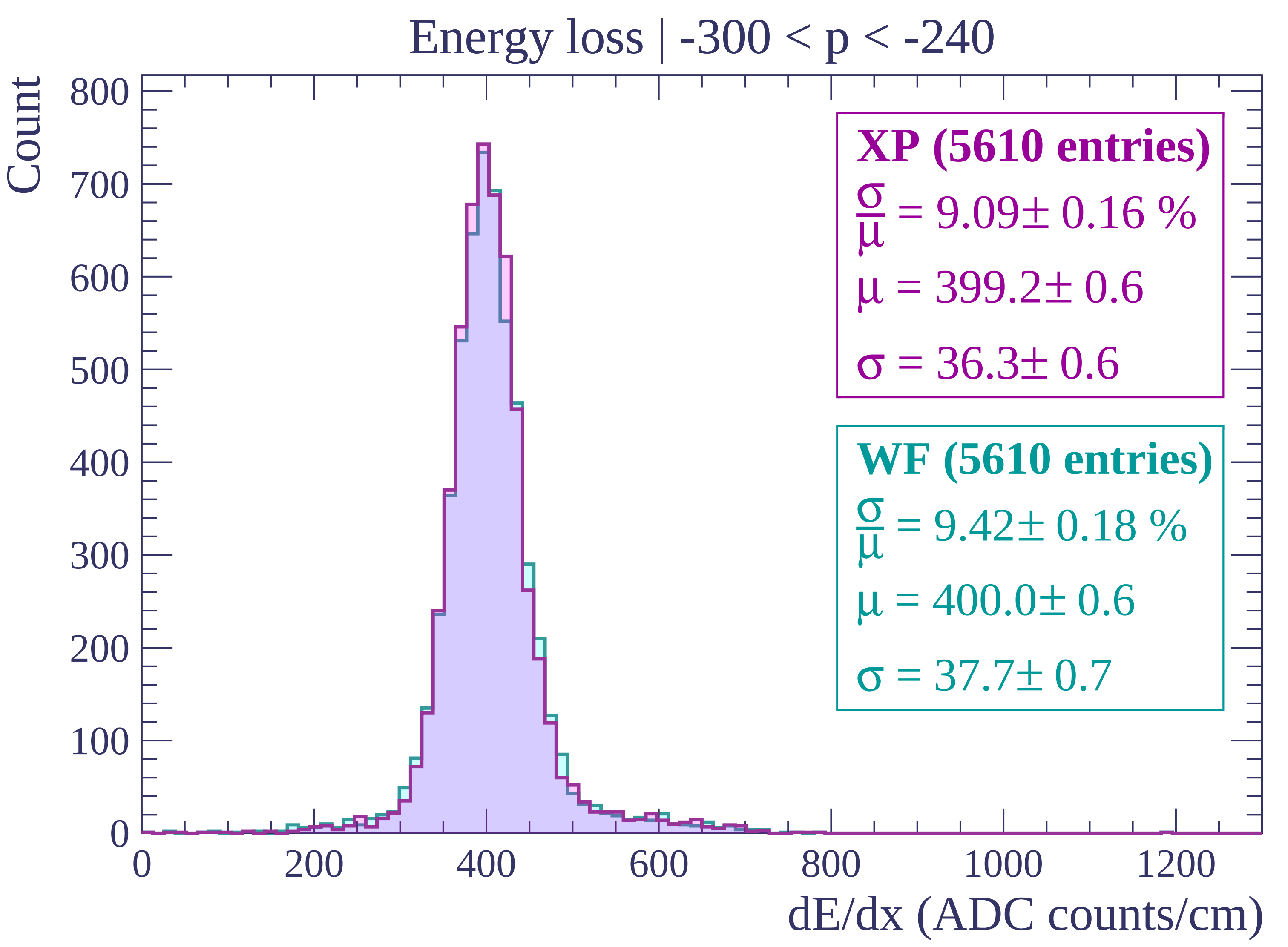


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6371 entries)
σ/μ = 9.02± 0.15 %
μ = 402.3± 0.6
σ = 36.3± 0.6
WF (6371 entries)
σ/μ = 9.51± 0.16 %
μ = 403.6± 0.6
σ = 38.4± 0.6

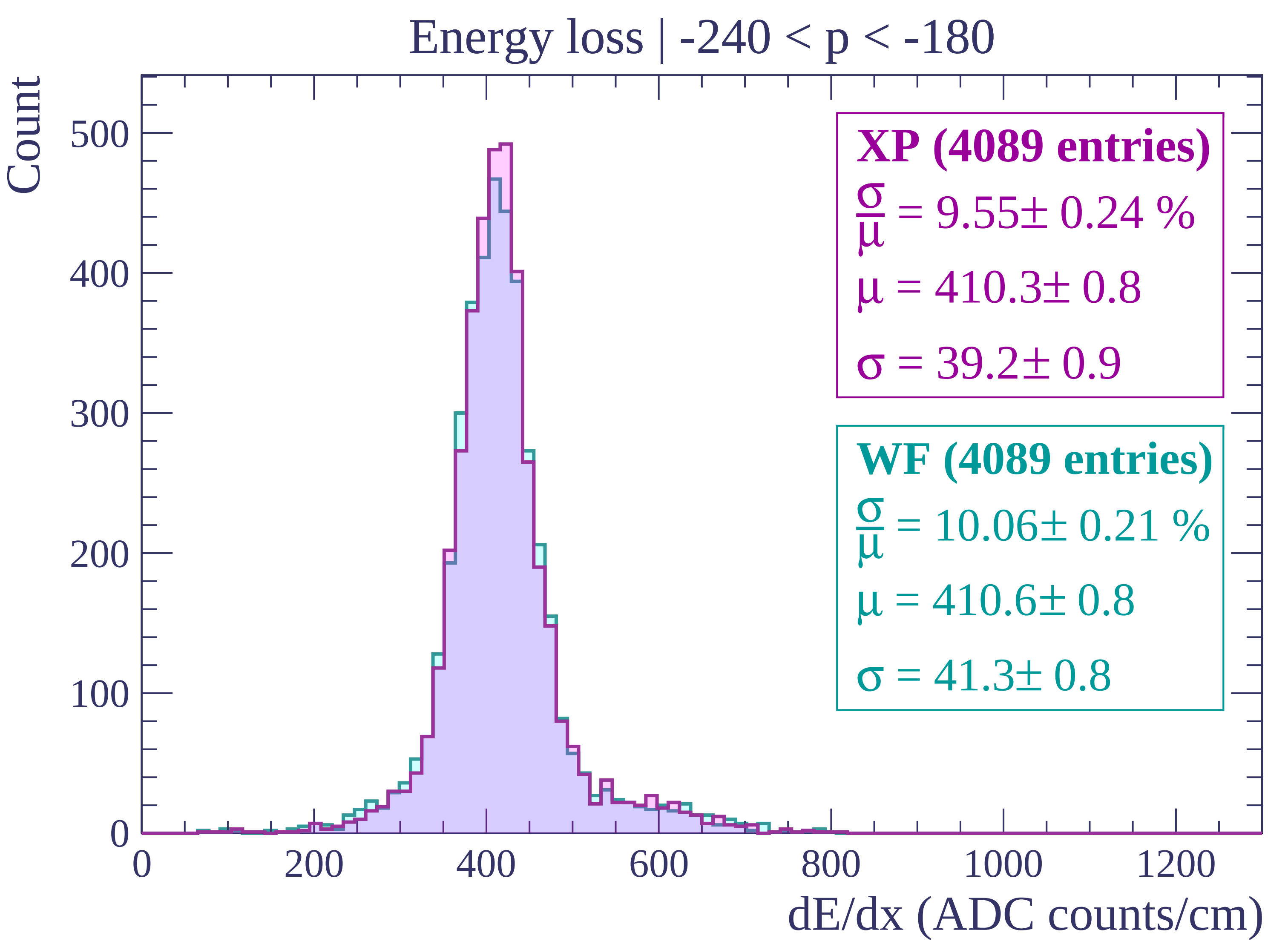


Energy loss | -420 < p < -360
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6279 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.16 %
μ = 399.3± 0.6
σ = 34.6± 0.6
WF (6279 entries)
$\frac{\sigma}{\mu}$ = 9.03± 0.16 %
μ = 400.1± 0.6
σ = 36.1± 0.6

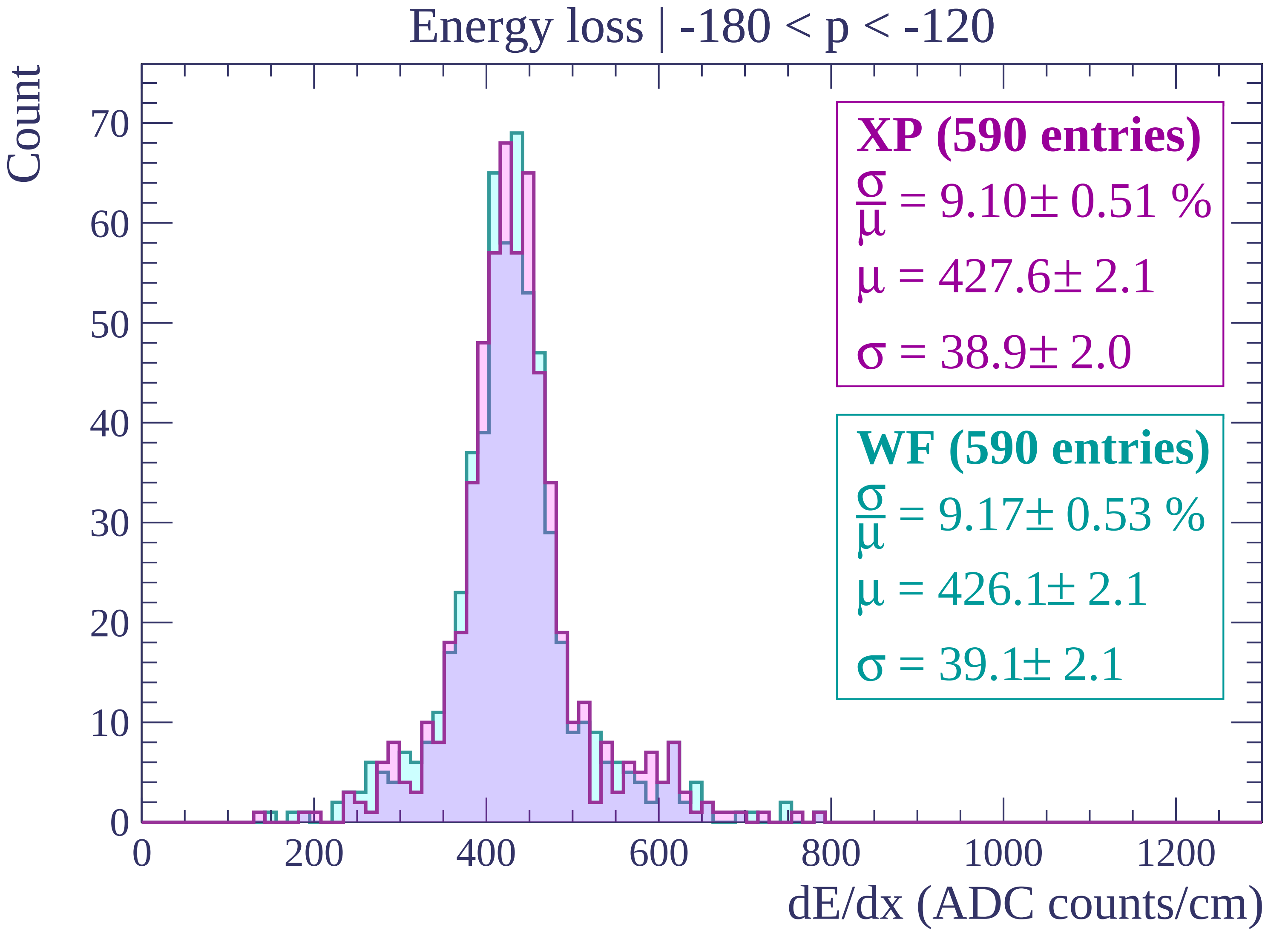

Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6394 entries)
σ/μ = 9.21± 0.15 %
μ = 397.4± 0.6
σ = 36.6± 0.6
WF (6394 entries)
σ/μ = 9.53± 0.17 %
μ = 398.1± 0.6
σ = 37.9± 0.6



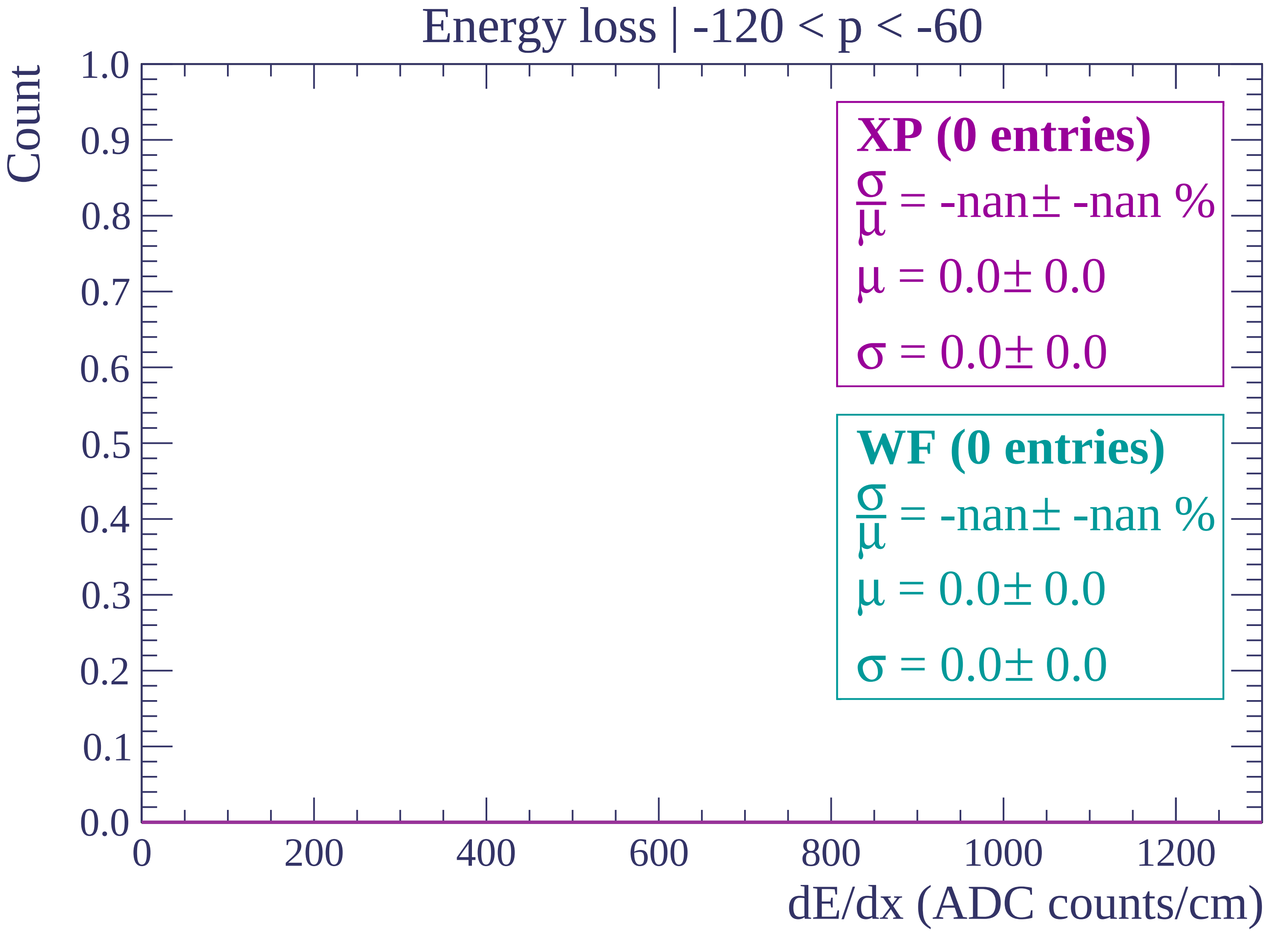
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (5610 entries)
σ/μ = 9.09± 0.16 %
μ = 399.2± 0.6
σ = 36.3± 0.6
WF (5610 entries)
σ/μ = 9.42± 0.18 %
μ = 400.0± 0.6
σ = 37.7± 0.7



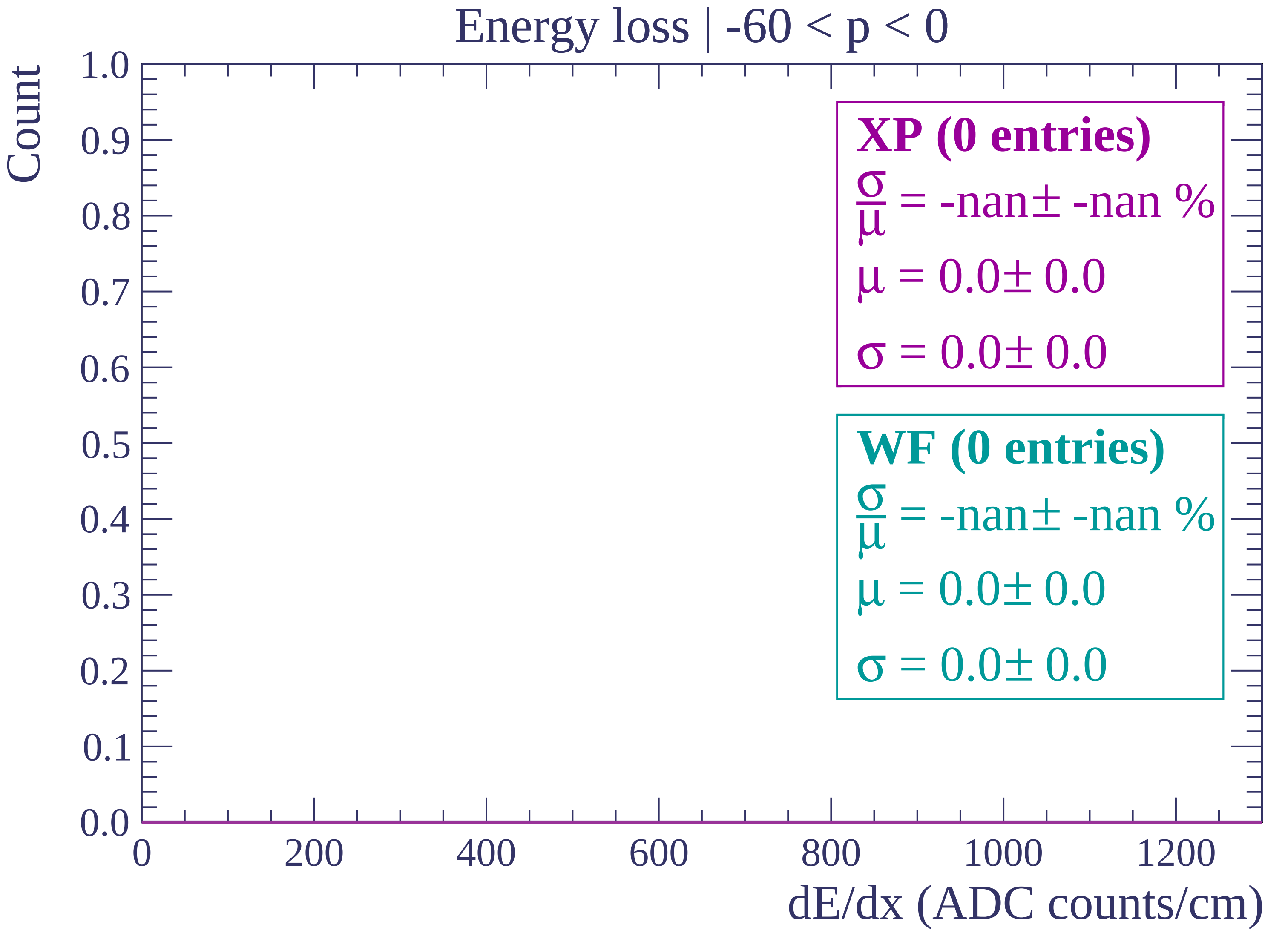
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
XP (4089 entries)
$\frac{\sigma}{\mu}$ = 9.55± 0.24 %
μ = 410.3± 0.8
σ = 39.2± 0.9
WF (4089 entries)
$\frac{\sigma}{\mu}$ = 10.06± 0.21 %
μ = 410.6± 0.8
σ = 41.3± 0.8
0
100
200
300
400
500
0
200
400
600
800
1000
1200

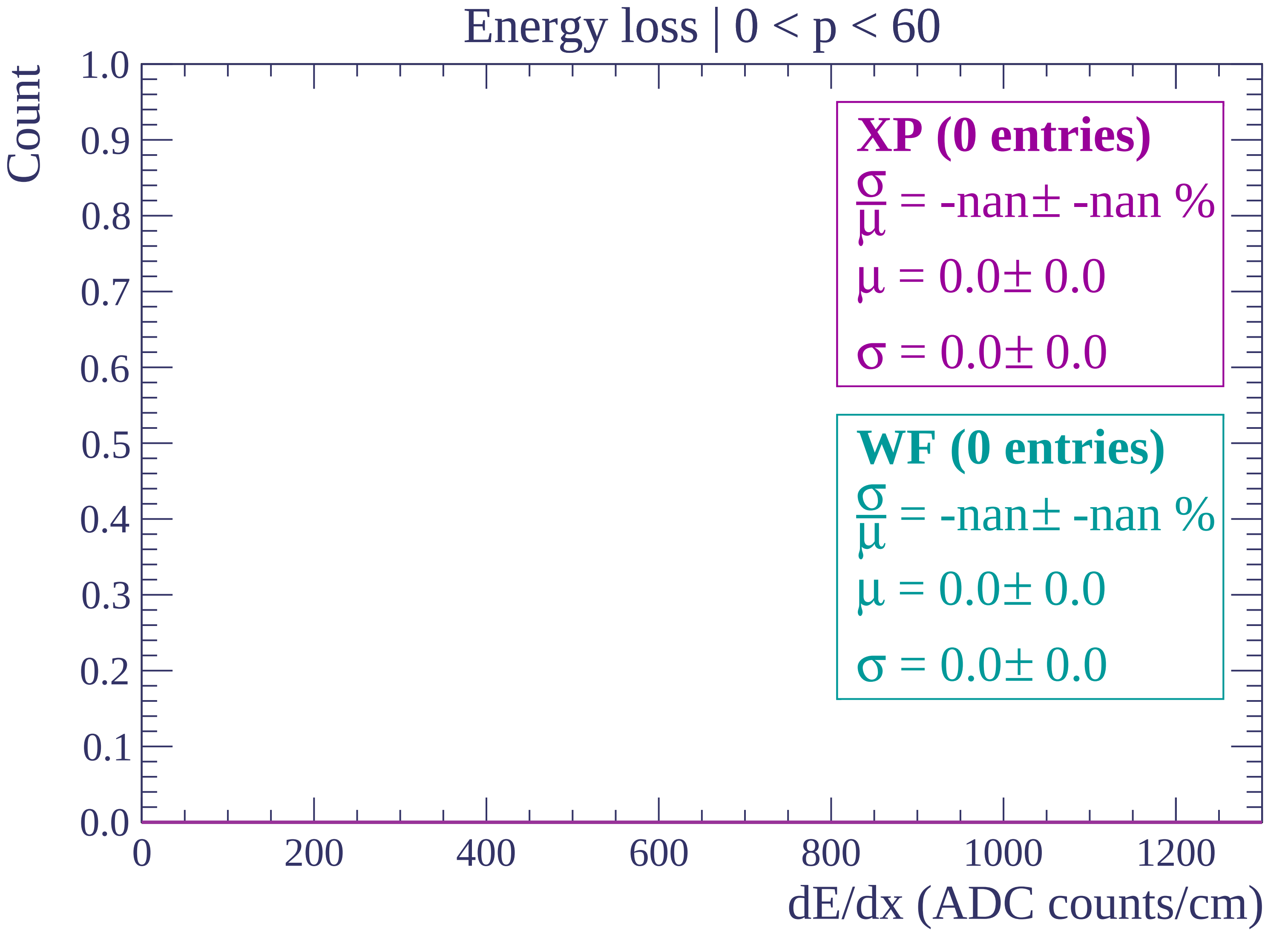


Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (590 entries)
σ/μ = 9.10± 0.51 %
μ = 427.6± 2.1
σ = 38.9± 2.0
WF (590 entries)
σ/μ = 9.17± 0.53 %
μ = 426.1± 2.1
σ = 39.1± 2.1

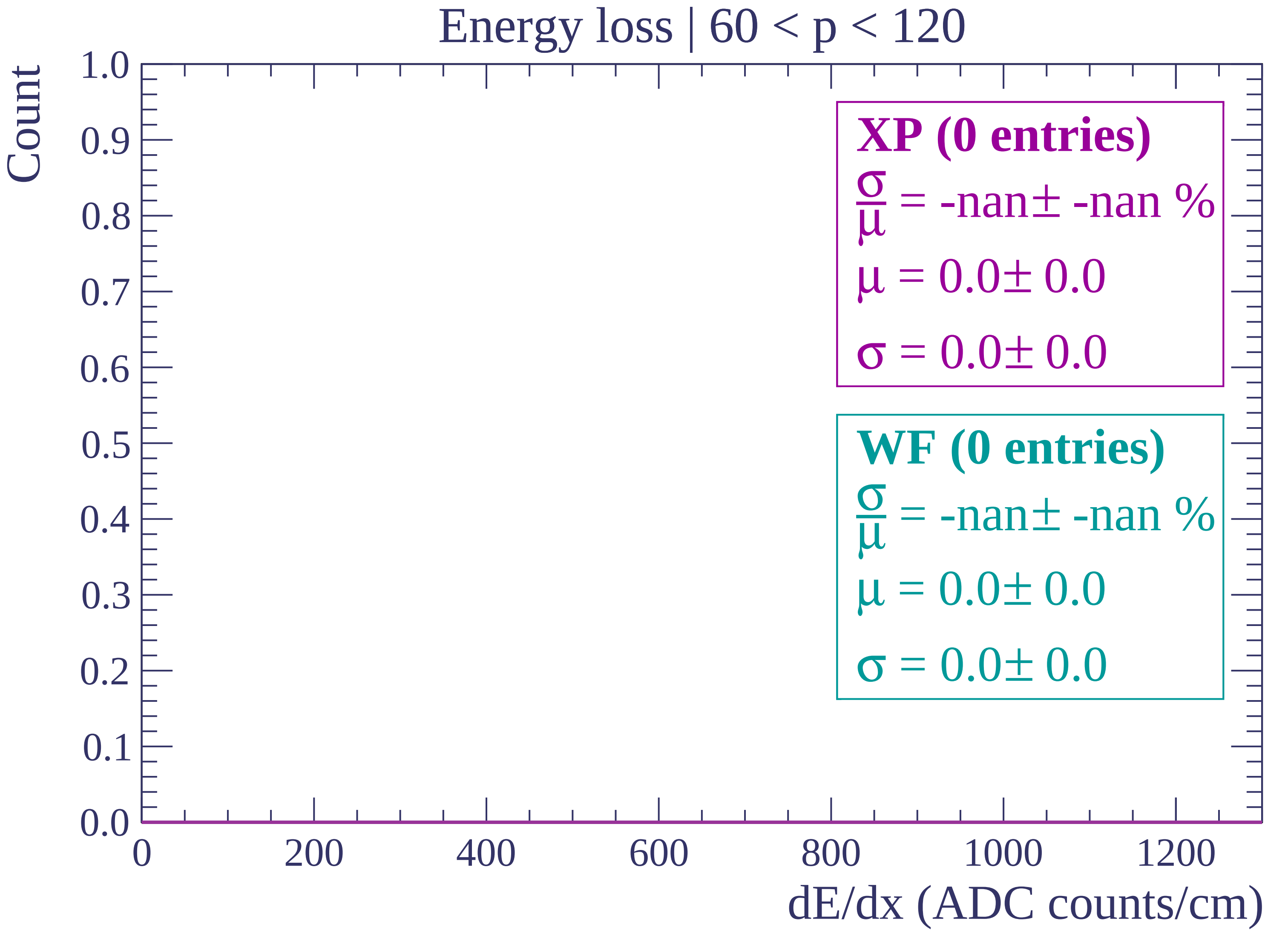


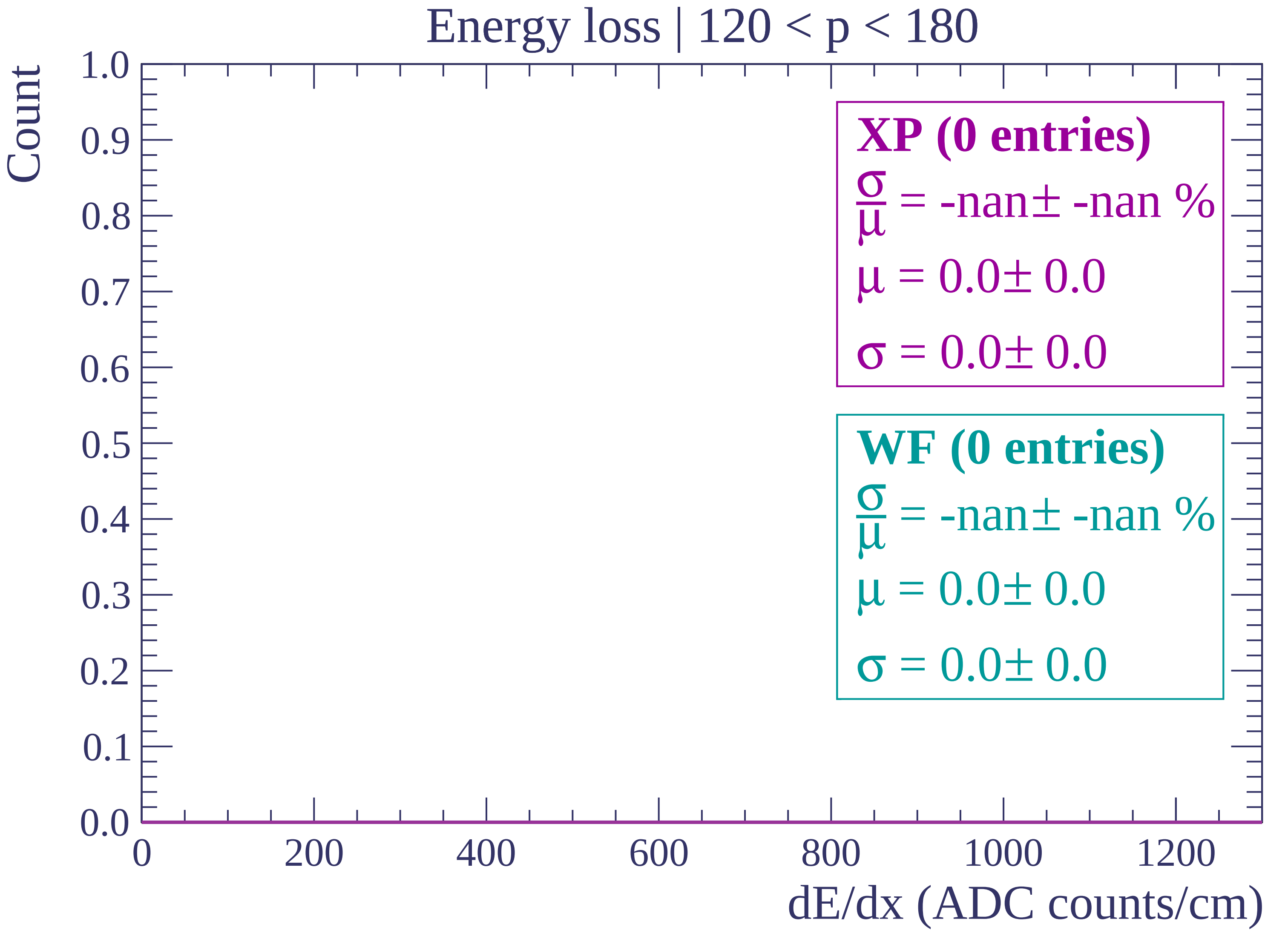
Energy loss | -120 < p < -60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

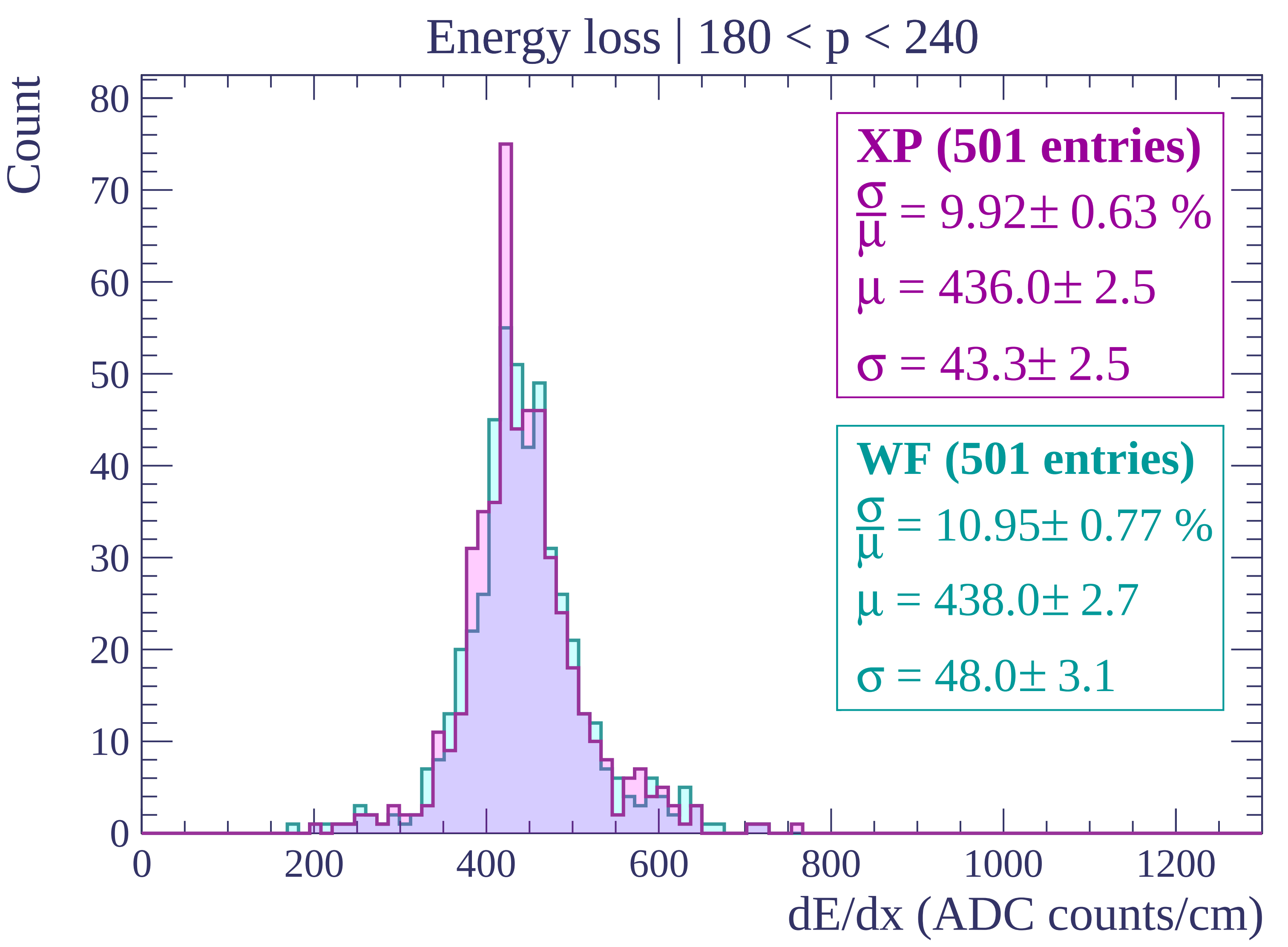


Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

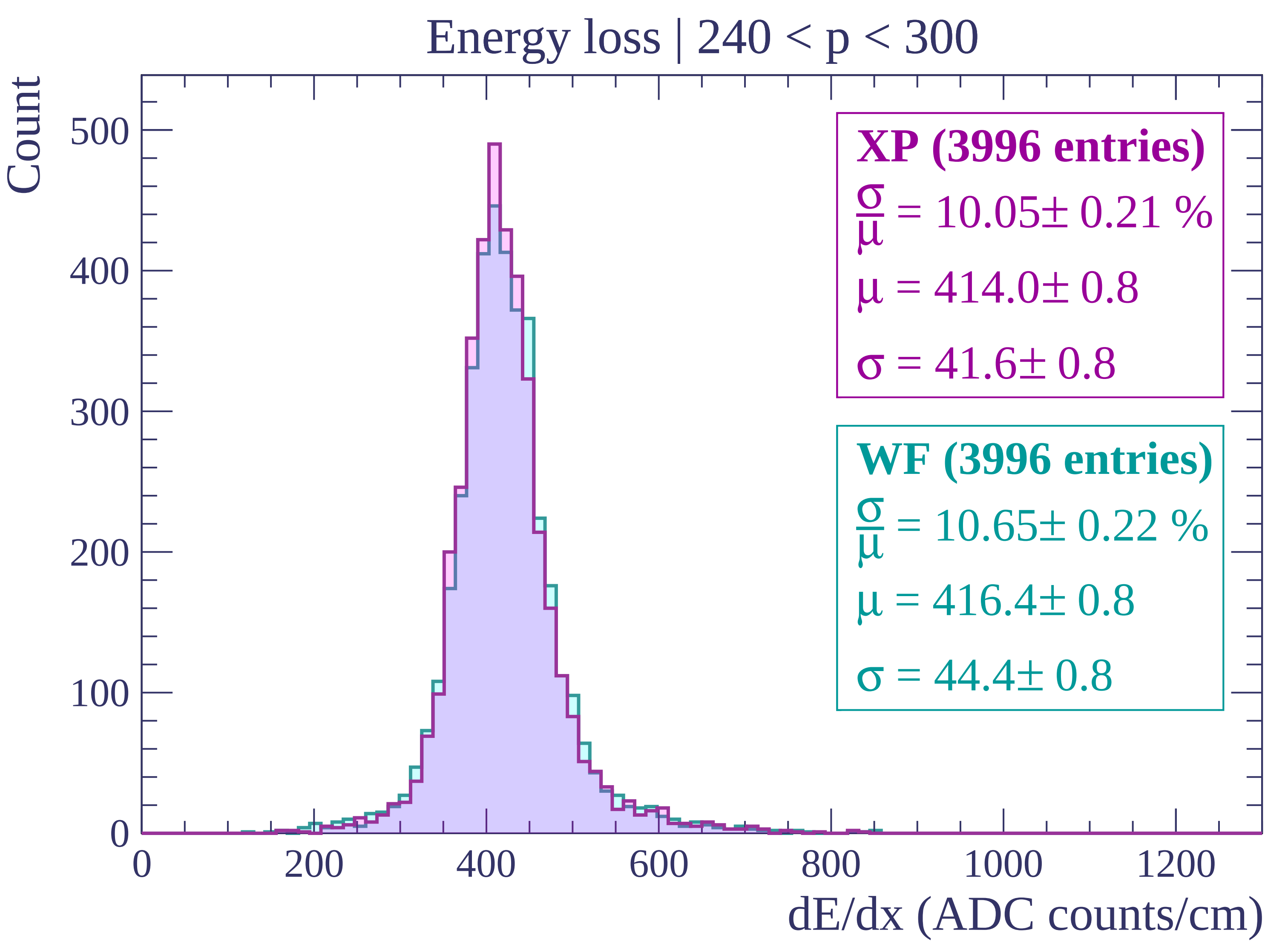

Energy loss | 60 < p < 120
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

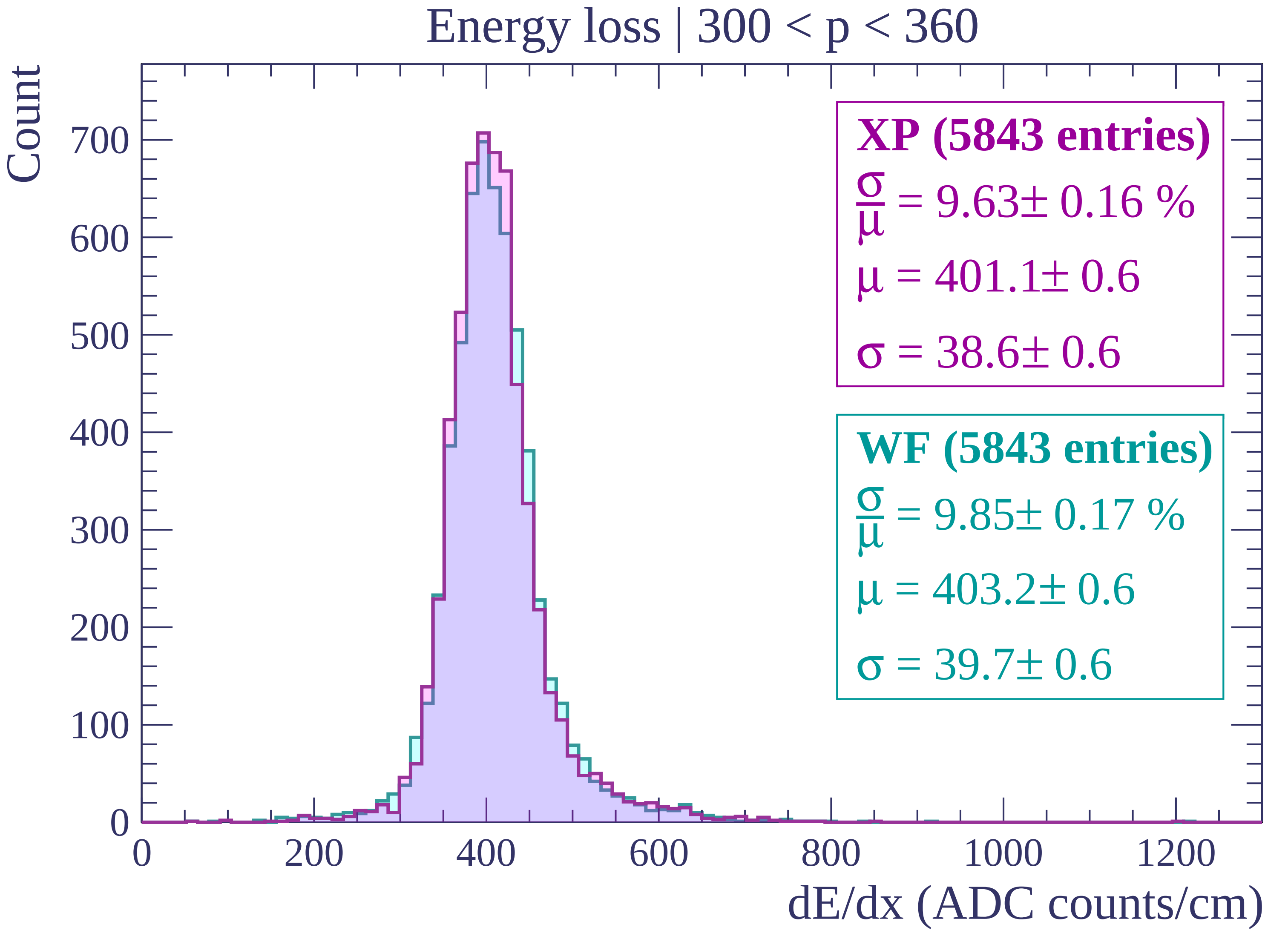


Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

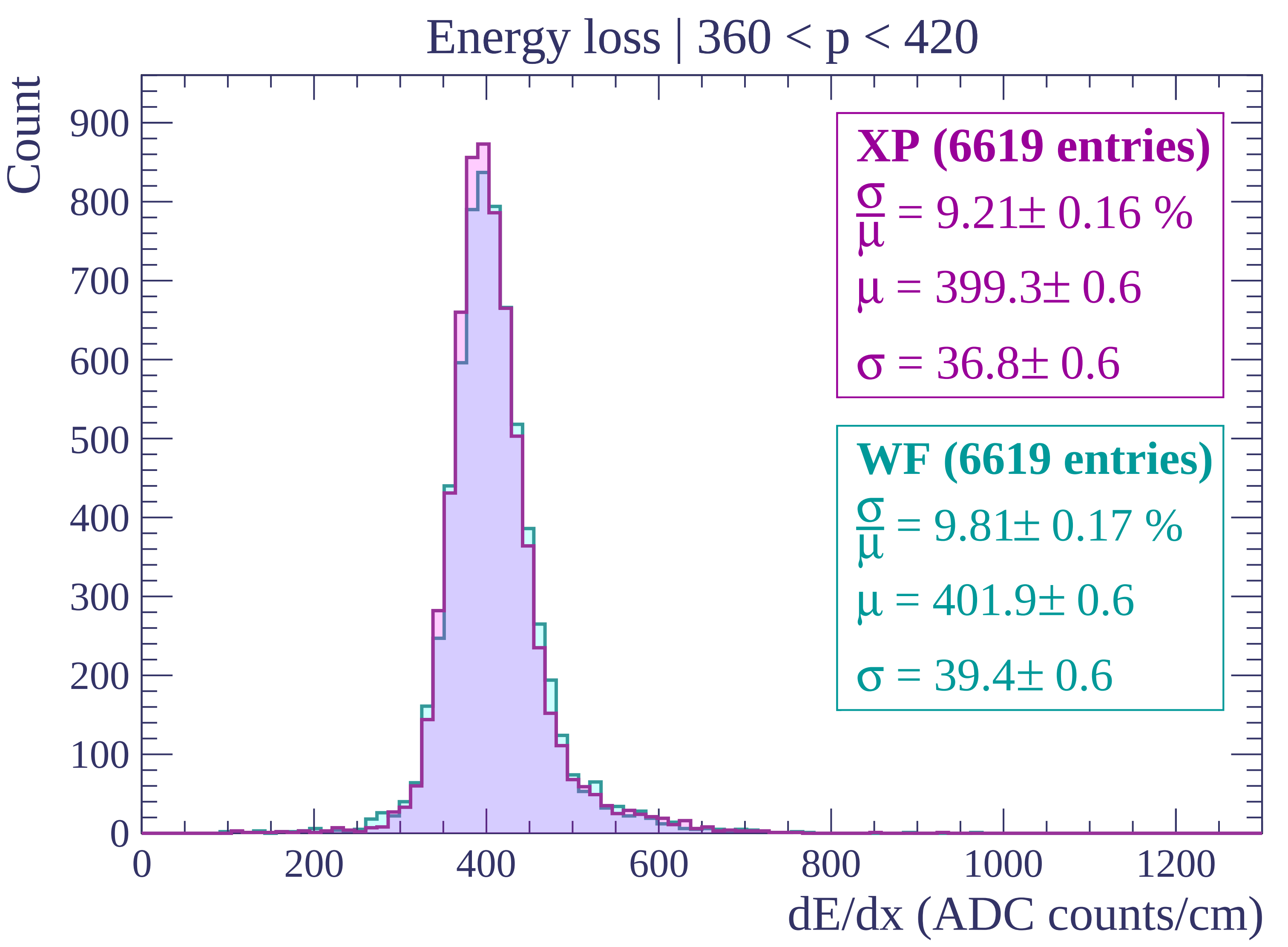


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (501 entries)
σ/μ = 9.92± 0.63 %
μ = 436.0± 2.5
σ = 43.3± 2.5
WF (501 entries)
σ/μ = 10.95± 0.77 %
μ = 438.0± 2.7
σ = 48.0± 3.1

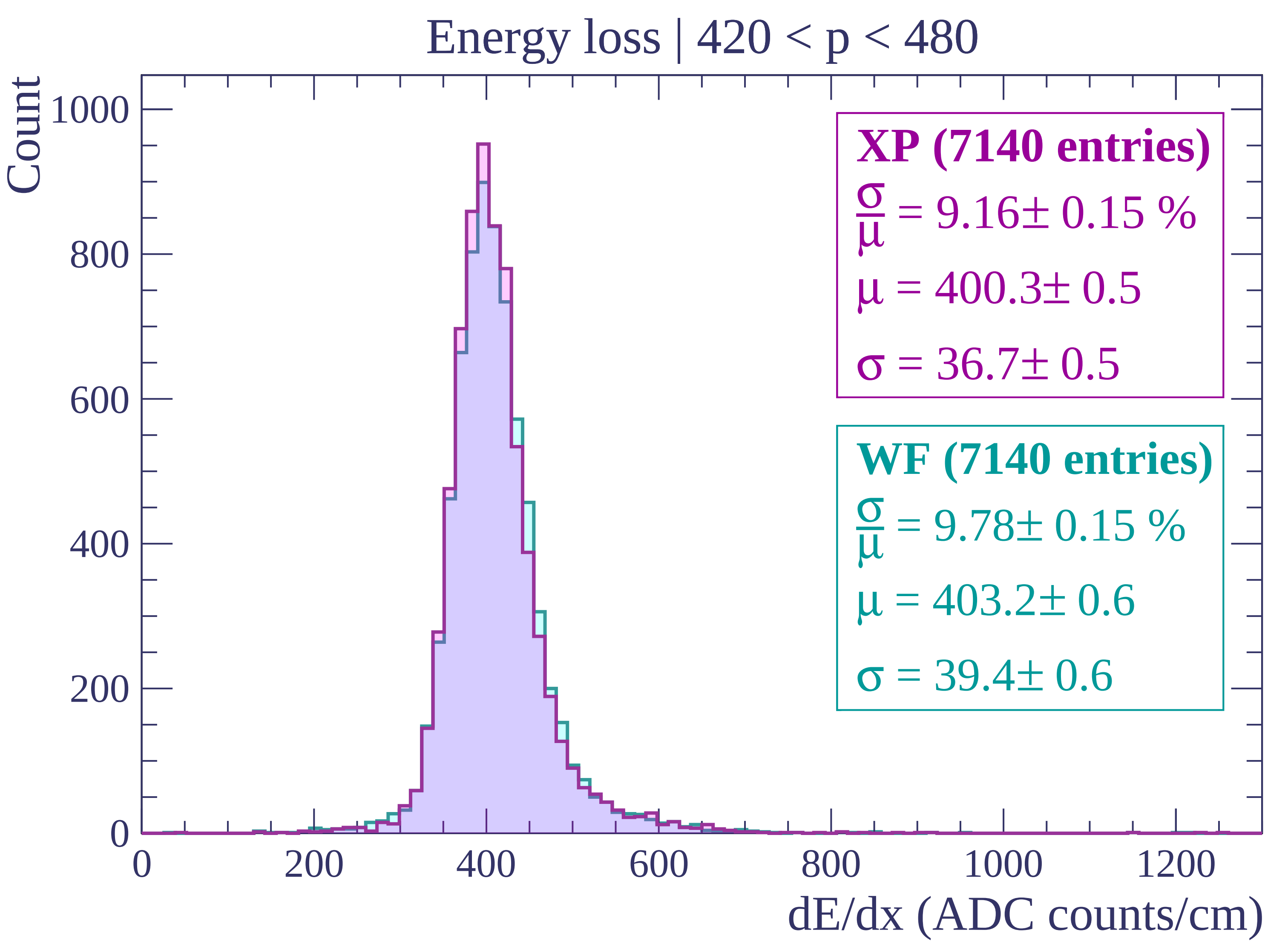

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3996 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.21 %
μ = 414.0± 0.8
σ = 41.6± 0.8
WF (3996 entries)
$\frac{\sigma}{\mu}$ = 10.65± 0.22 %
μ = 416.4± 0.8
σ = 44.4± 0.8

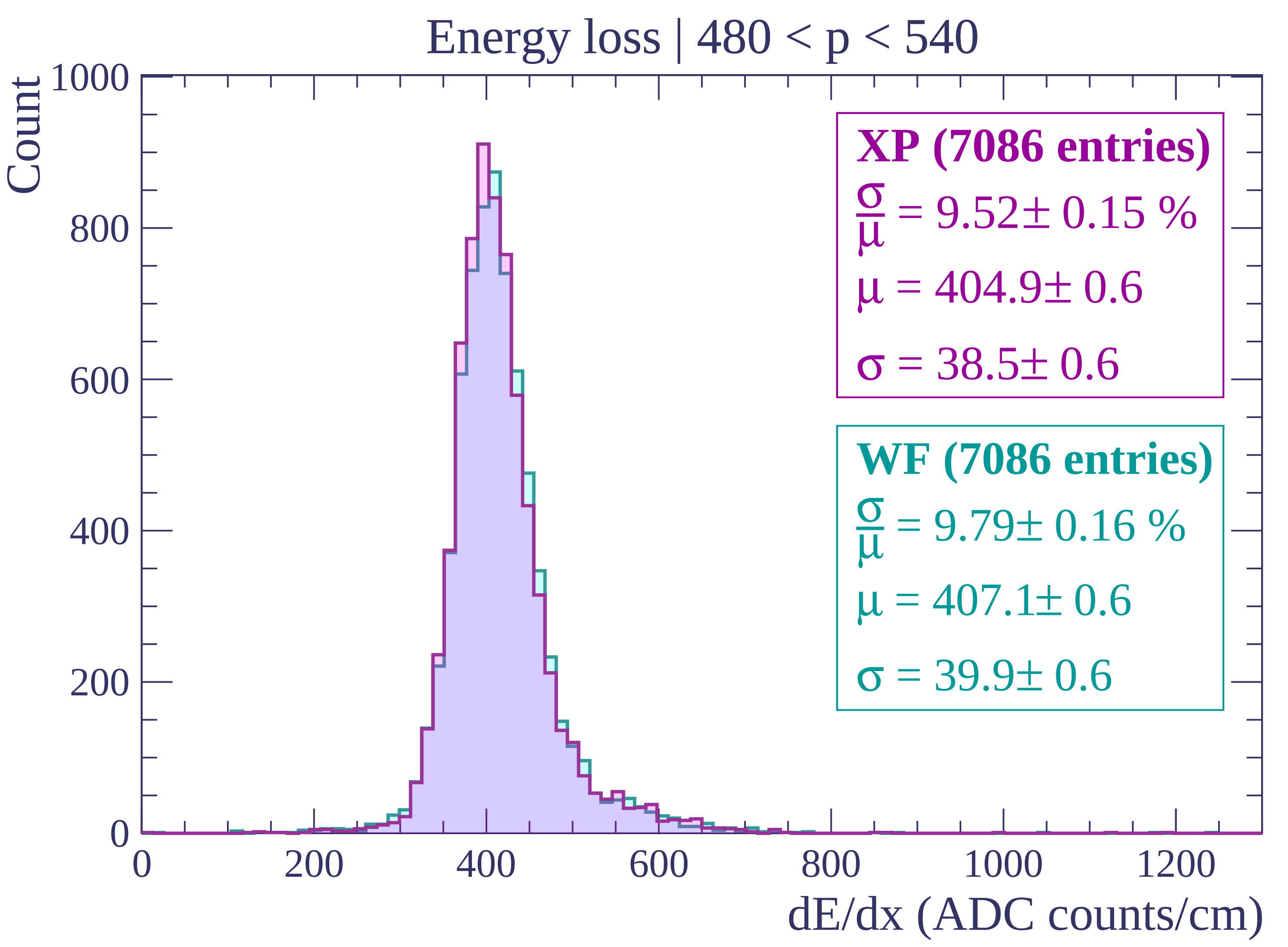


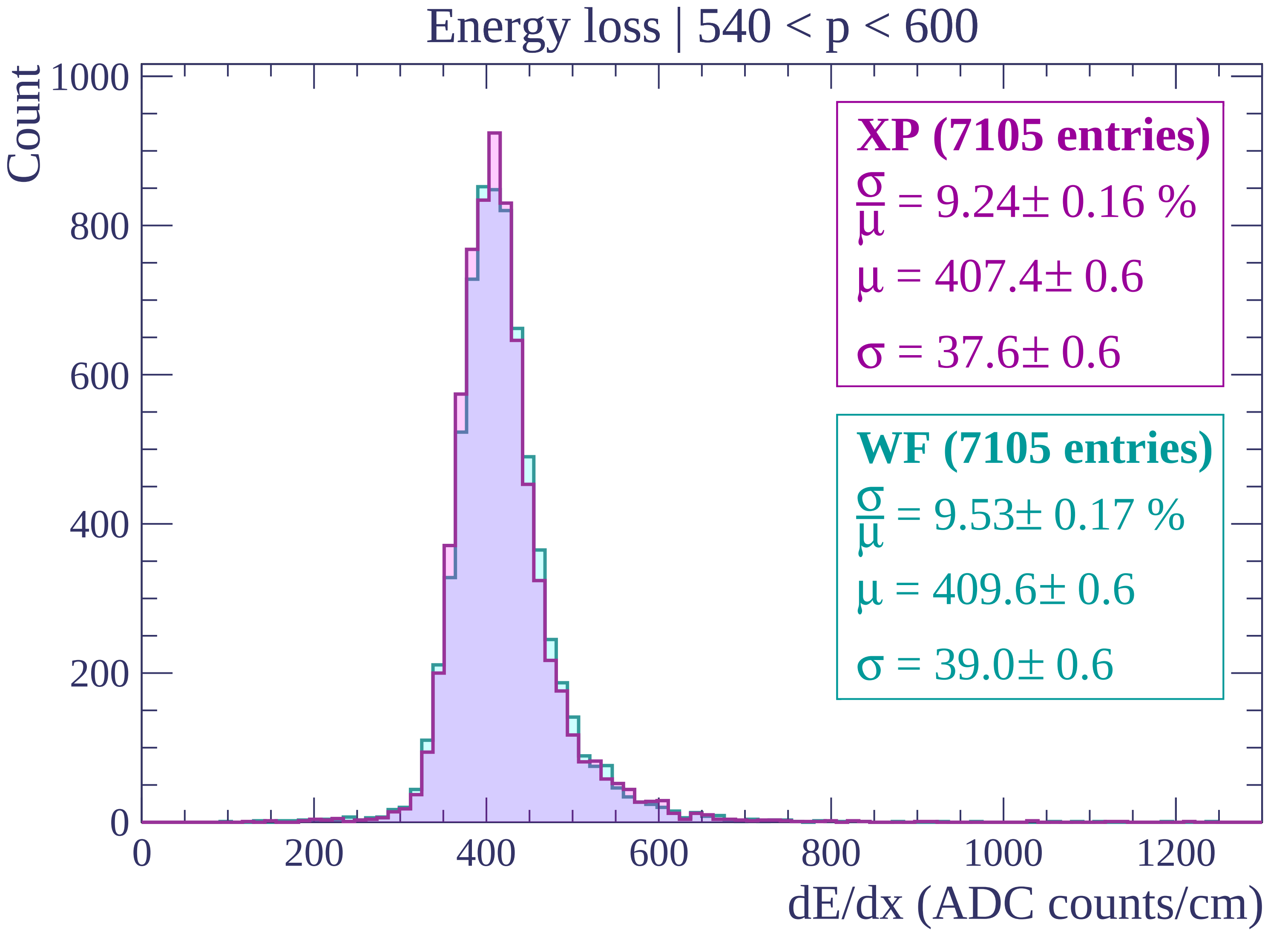
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5843 entries)
σ/μ = 9.63± 0.16 %
μ = 401.1± 0.6
σ = 38.6± 0.6
WF (5843 entries)
σ/μ = 9.85± 0.17 %
μ = 403.2± 0.6
σ = 39.7± 0.6



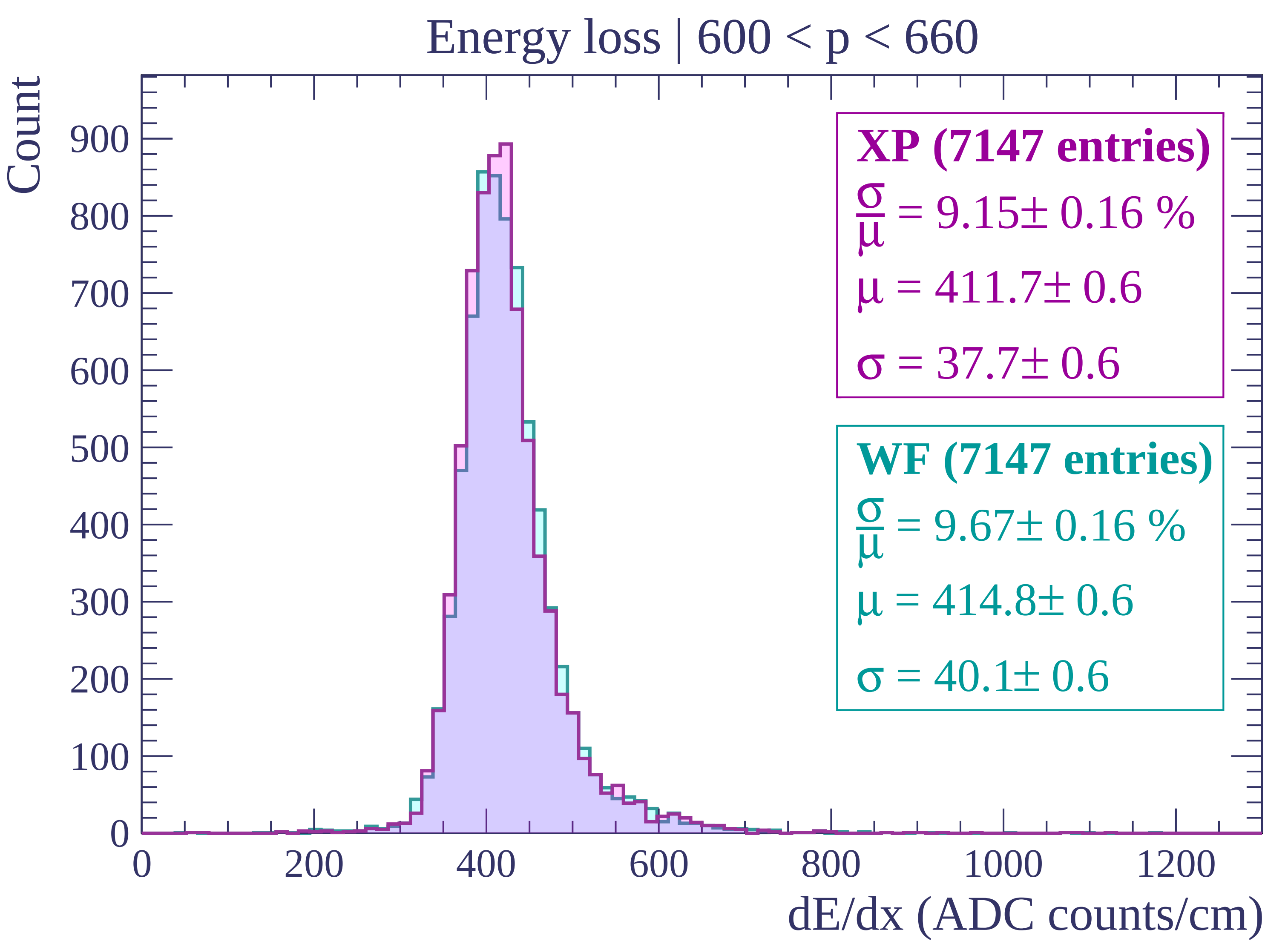
Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6619 entries)
σ/μ = 9.21± 0.16 %
μ = 399.3± 0.6
σ = 36.8± 0.6
WF (6619 entries)
σ/μ = 9.81± 0.17 %
μ = 401.9± 0.6
σ = 39.4± 0.6

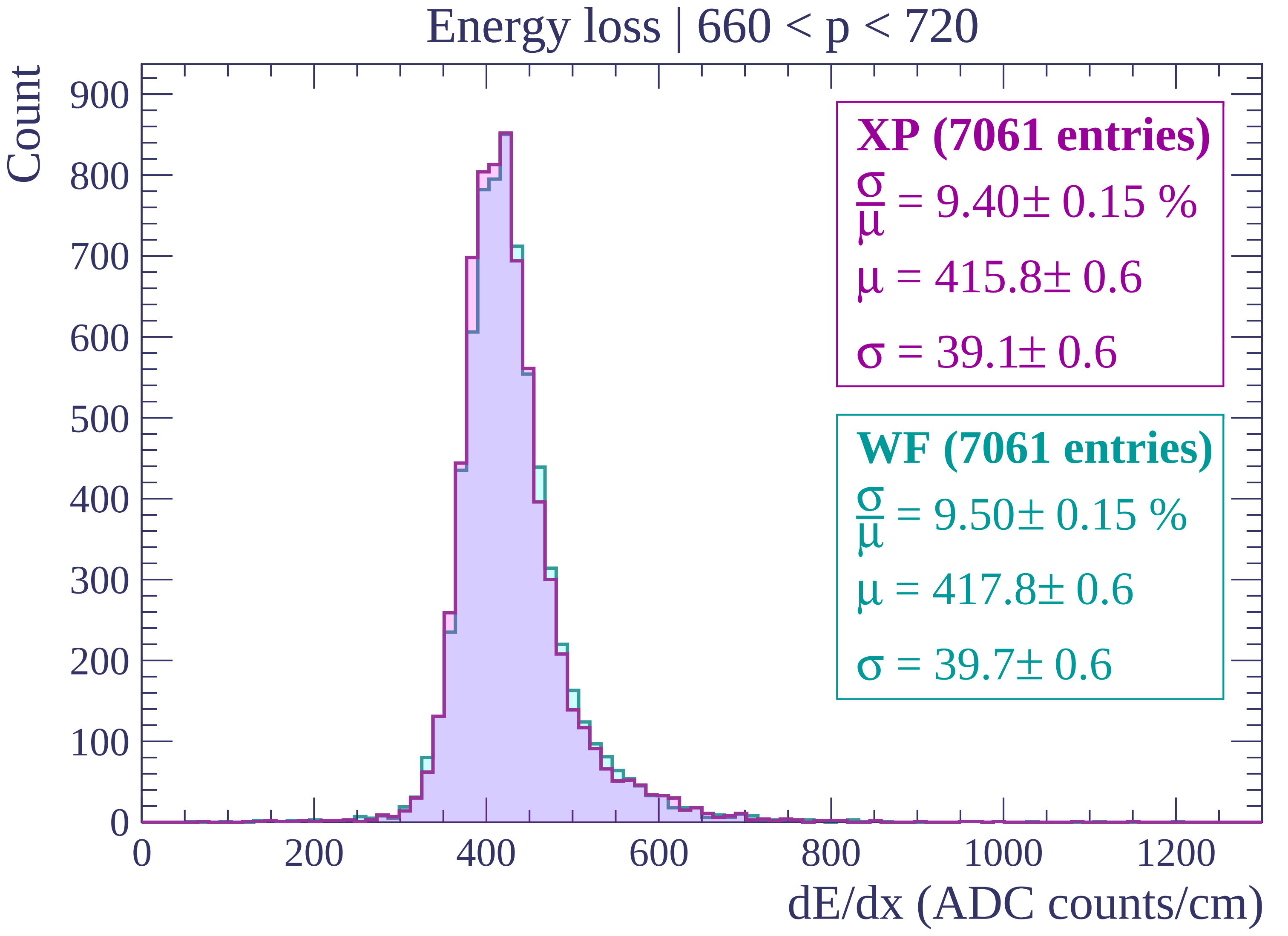

Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
XP (7140 entries)
$\frac{\sigma}{\mu}$ = 9.16± 0.15 %
μ = 400.3± 0.5
σ = 36.7± 0.5
WF (7140 entries)
$\frac{\sigma}{\mu}$ = 9.78± 0.15 %
μ = 403.2± 0.6
σ = 39.4± 0.6
0
200
400
600
800
1000
1200


Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7086 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.15 %
μ = 404.9± 0.6
σ = 38.5± 0.6
WF (7086 entries)
$\frac{\sigma}{\mu}$ = 9.79± 0.16 %
μ = 407.1± 0.6
σ = 39.9± 0.6

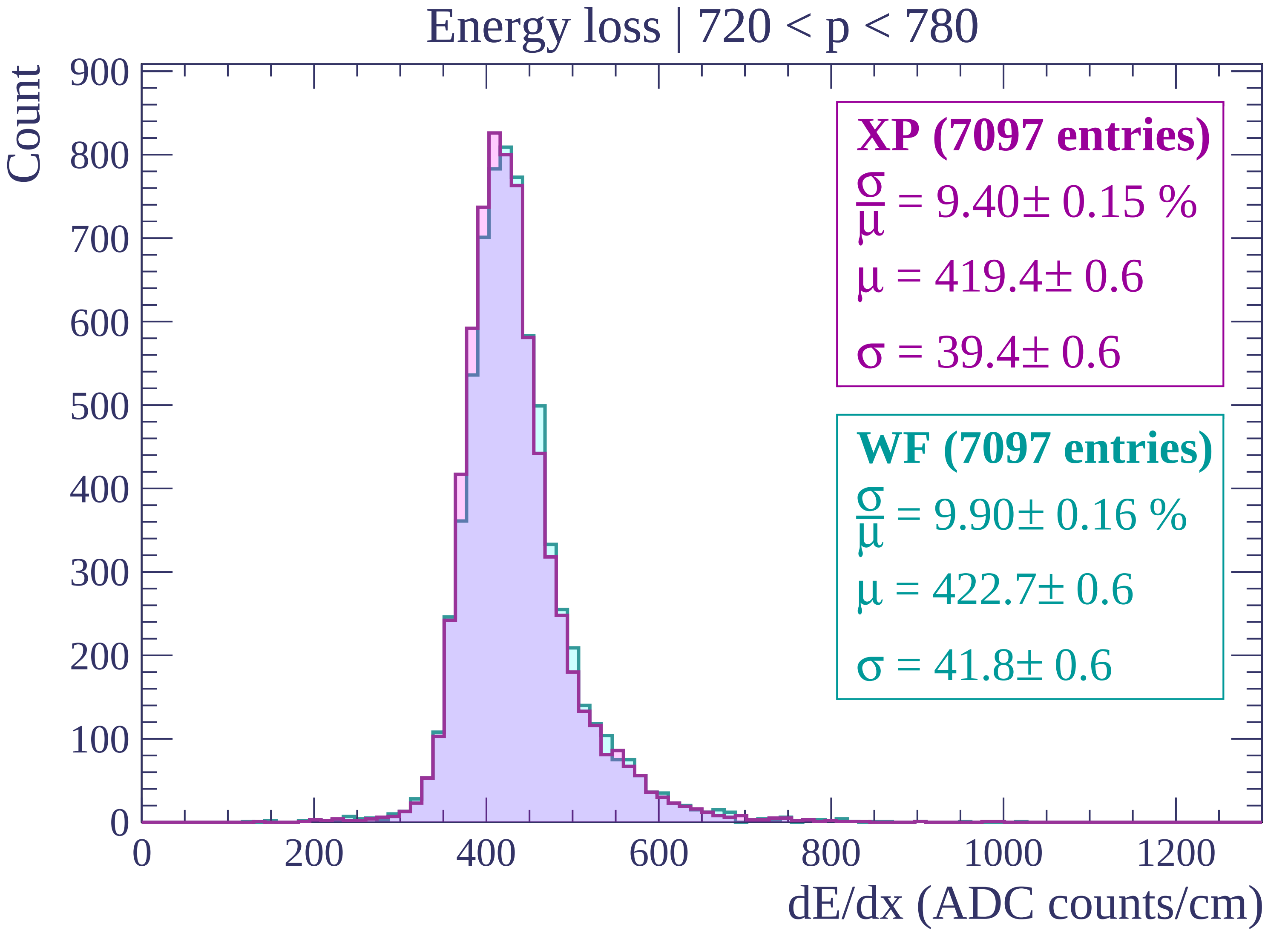


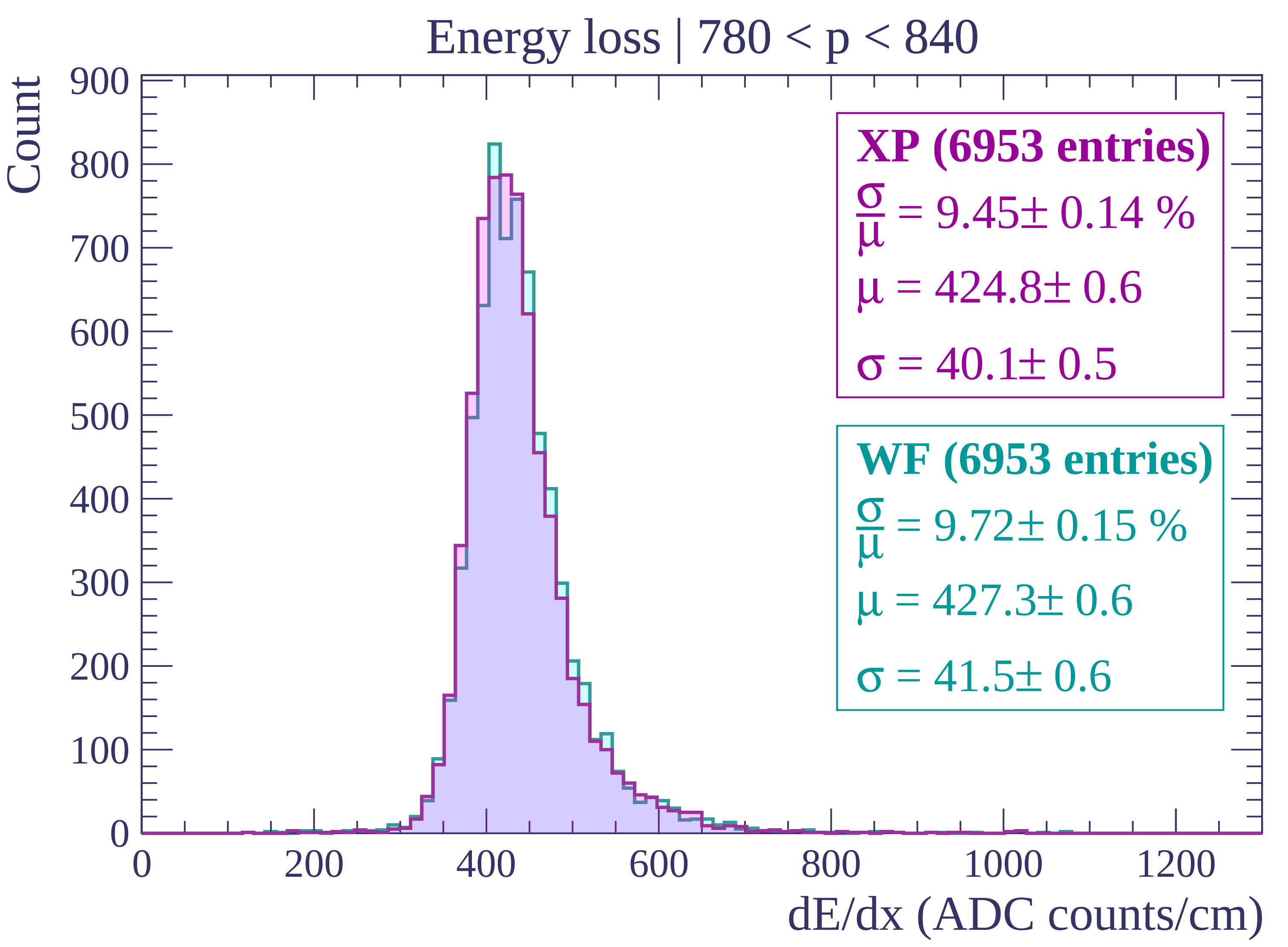
Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
XP (7105 entries)
$\frac{\sigma}{\mu}$ = 9.24± 0.16 %
μ = 407.4± 0.6
σ = 37.6± 0.6
WF (7105 entries)
$\frac{\sigma}{\mu}$ = 9.53± 0.17 %
μ = 409.6± 0.6
σ = 39.0± 0.6
0
200
400
600
800
1000
1200


Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (7147 entries)
σ/μ = 9.15± 0.16 %
μ = 411.7± 0.6
σ = 37.7± 0.6
WF (7147 entries)
σ/μ = 9.67± 0.16 %
μ = 414.8± 0.6
σ = 40.1± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

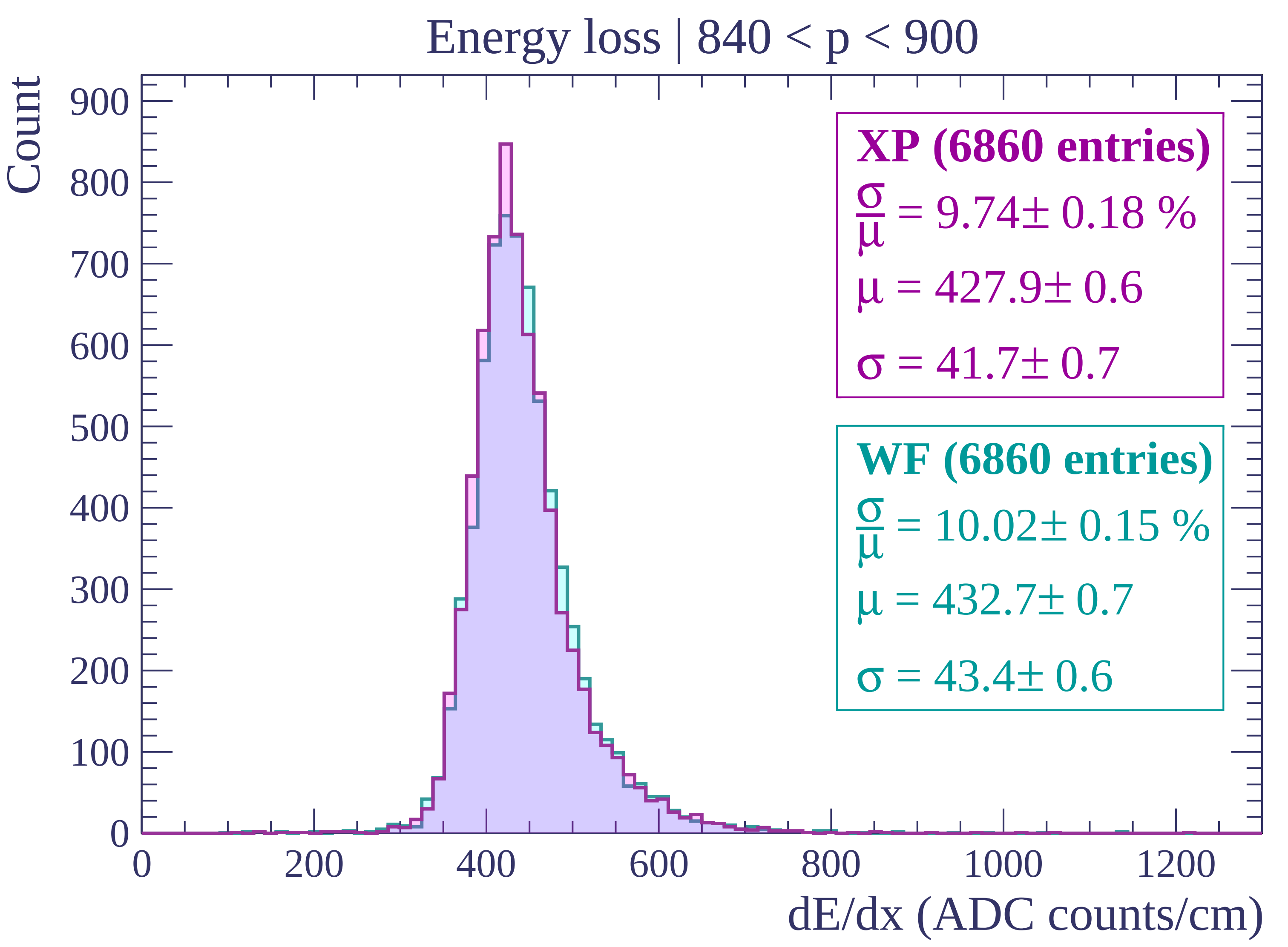


Energy loss | 660 < p < 720
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7061 entries)
σ/μ = 9.40± 0.15 %
μ = 415.8± 0.6
σ = 39.1± 0.6
WF (7061 entries)
σ/μ = 9.50± 0.15 %
μ = 417.8± 0.6
σ = 39.7± 0.6

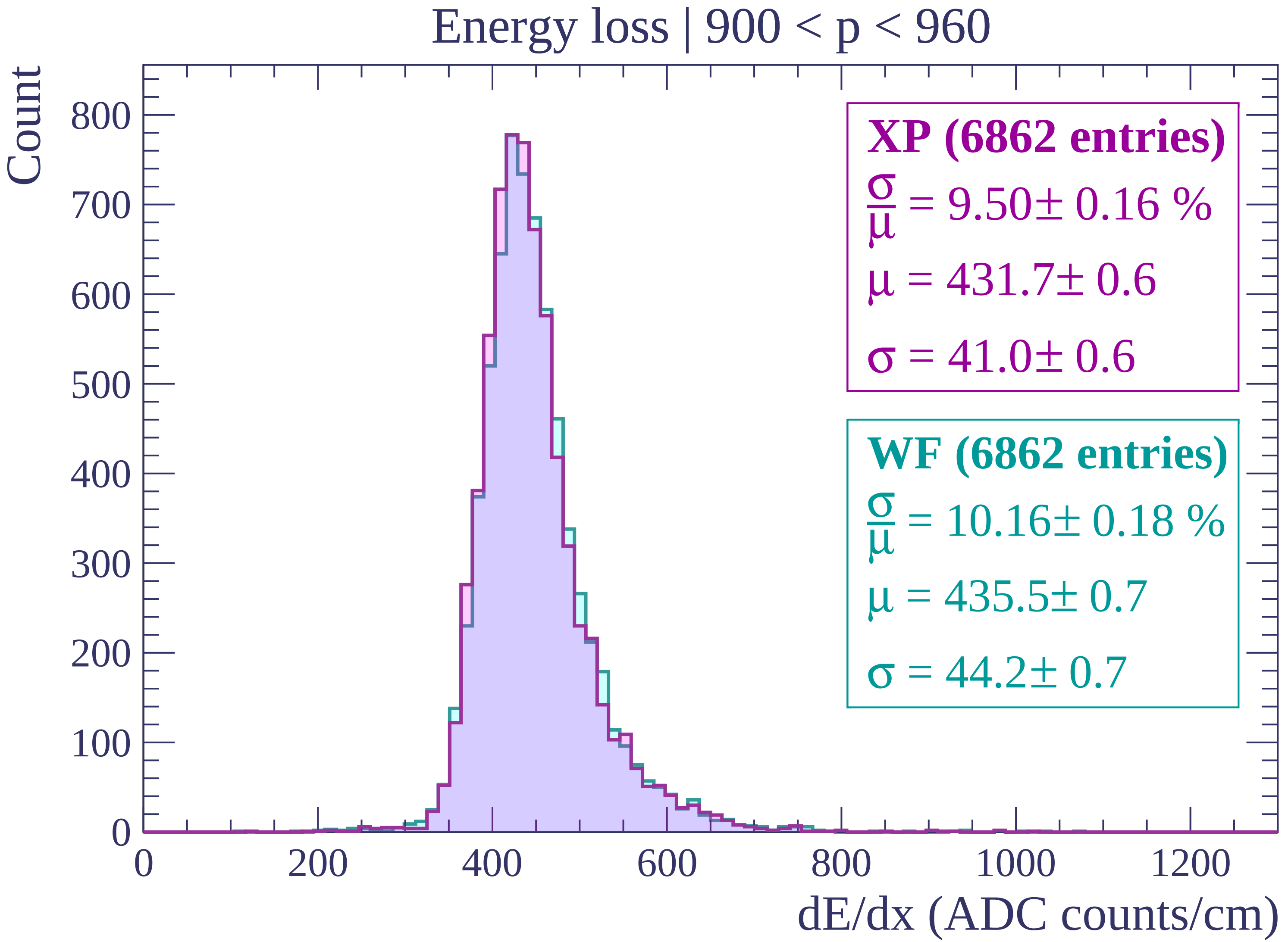

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7097 entries)
$\frac{\sigma}{\mu}$ = 9.40± 0.15 %
μ = 419.4± 0.6
σ = 39.4± 0.6
WF (7097 entries)
$\frac{\sigma}{\mu}$ = 9.90± 0.16 %
μ = 422.7± 0.6
σ = 41.8± 0.6

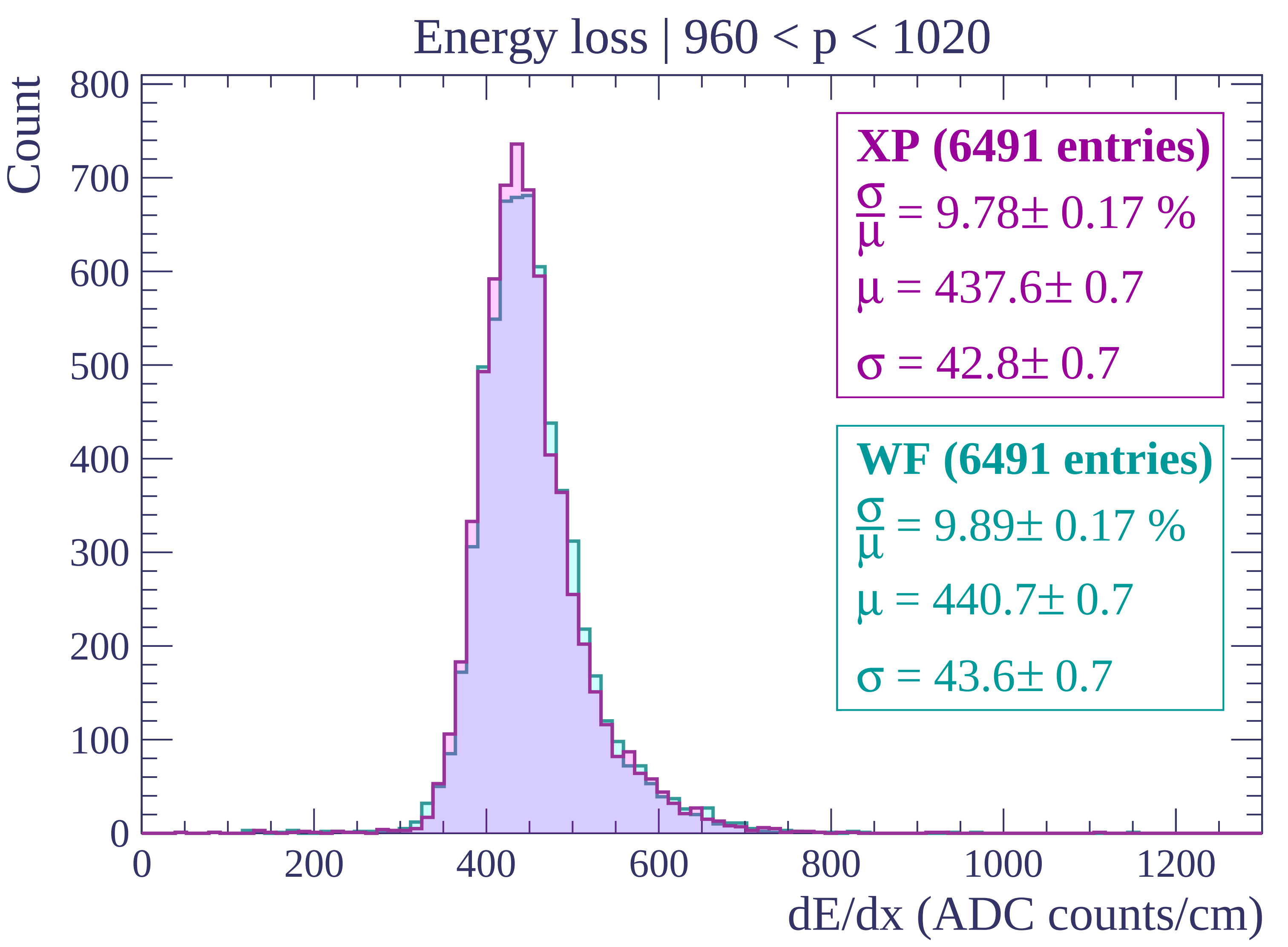


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6953 entries)
σ/μ = 9.45± 0.14 %
μ = 424.8± 0.6
σ = 40.1± 0.5
WF (6953 entries)
σ/μ = 9.72± 0.15 %
μ = 427.3± 0.6
σ = 41.5± 0.6

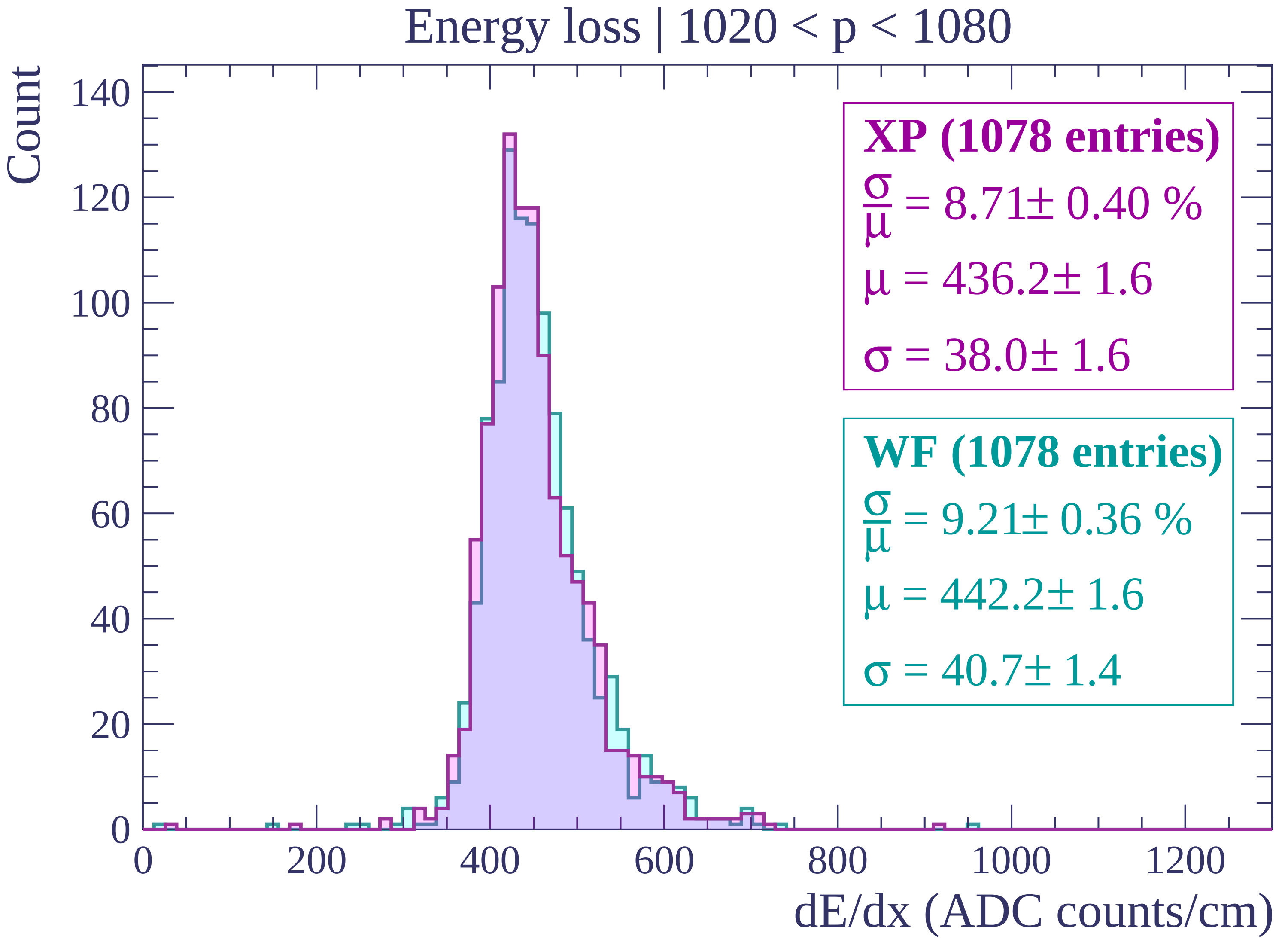


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6860 entries)
$\frac{\sigma}{\mu}$ = 9.74± 0.18 %
μ = 427.9± 0.6
σ = 41.7± 0.7
WF (6860 entries)
$\frac{\sigma}{\mu}$ = 10.02± 0.15 %
μ = 432.7± 0.7
σ = 43.4± 0.6

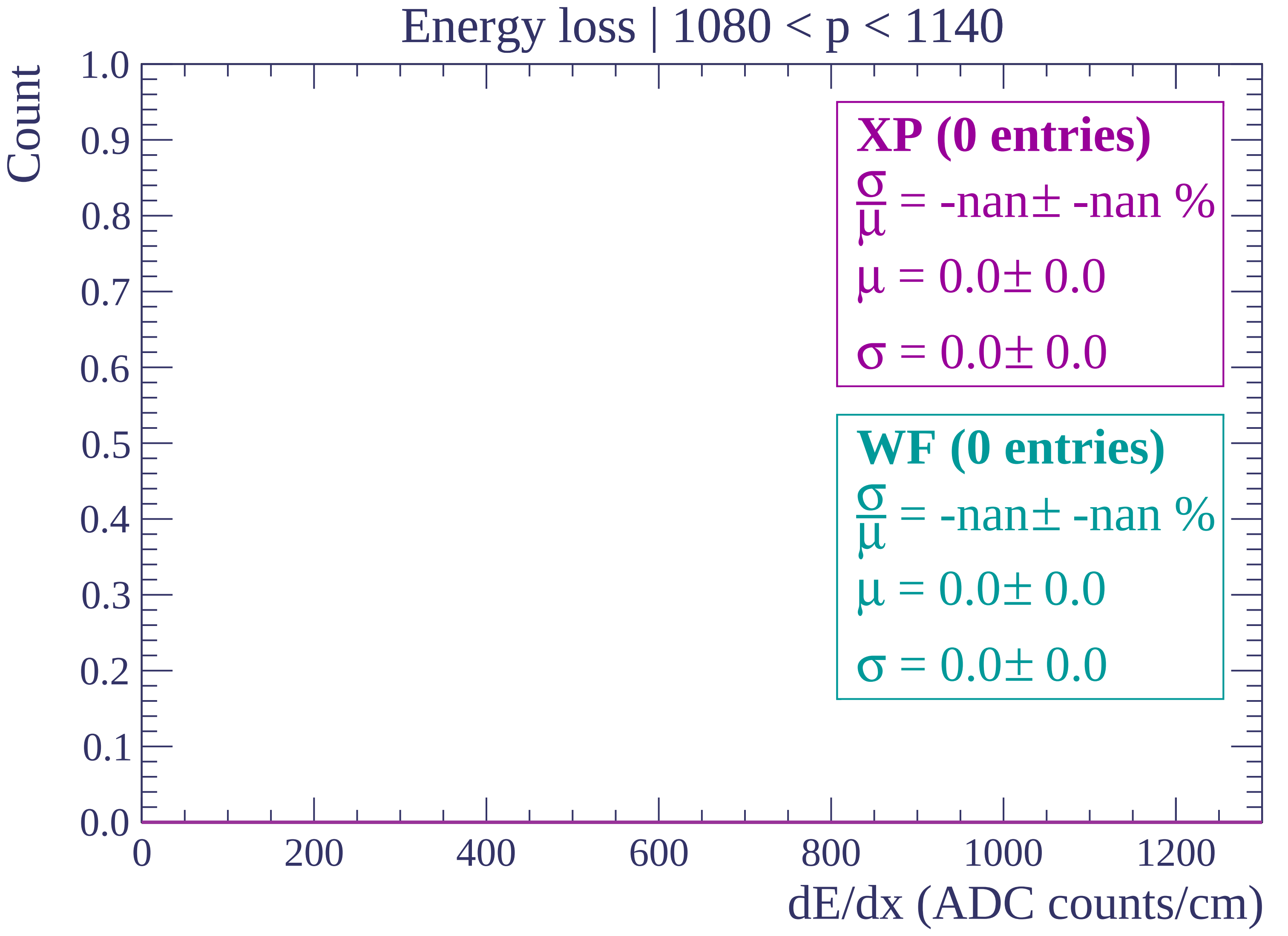

Energy loss | 900 < p < 960
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6862 entries)
σ/μ = 9.50± 0.16 %
μ = 431.7± 0.6
σ = 41.0± 0.6
WF (6862 entries)
σ/μ = 10.16± 0.18 %
μ = 435.5± 0.7
σ = 44.2± 0.7

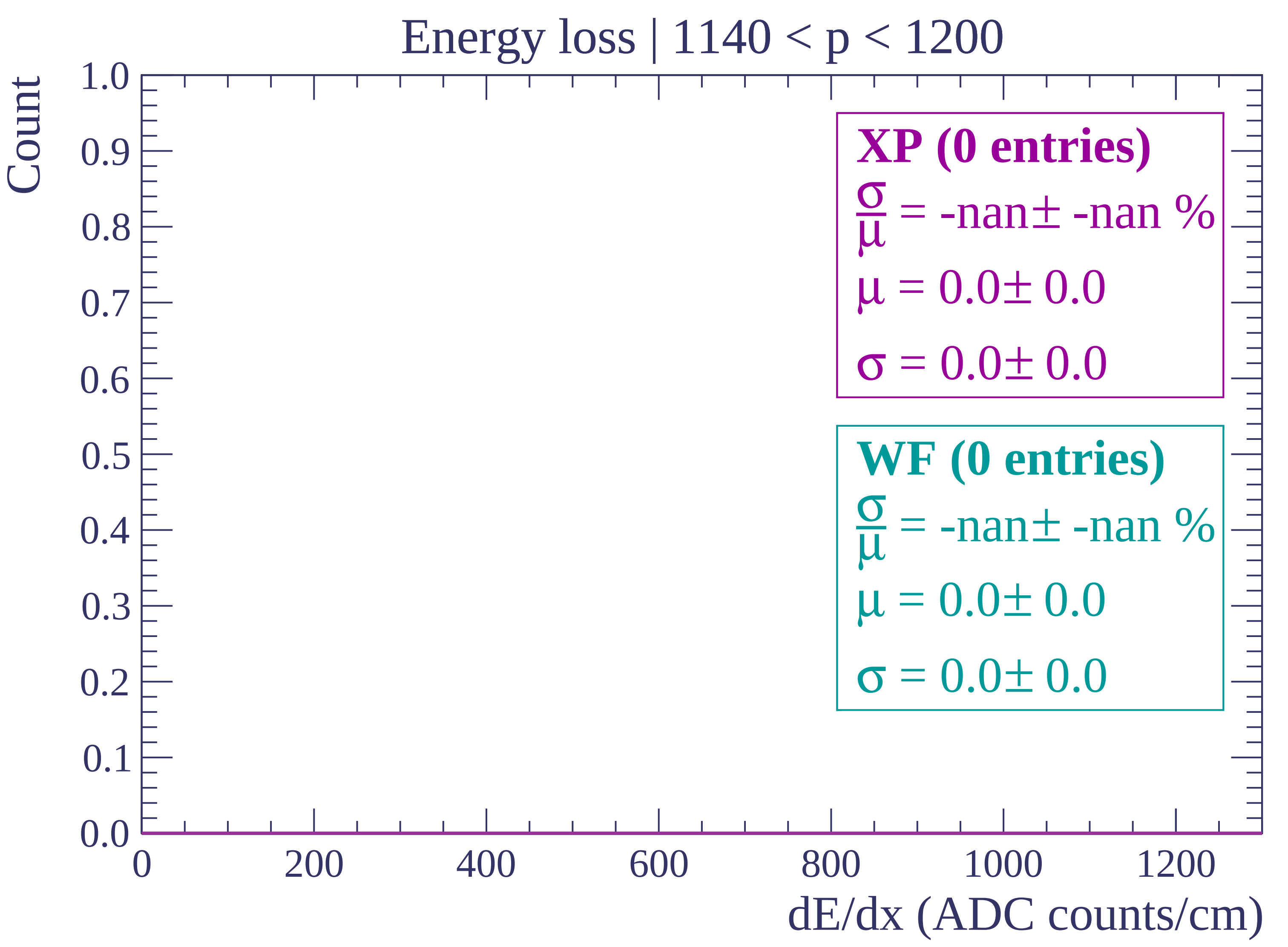


Energy loss | 960 < p < 1020
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6491 entries)
σ/μ = 9.78± 0.17 %
μ = 437.6± 0.7
σ = 42.8± 0.7
WF (6491 entries)
σ/μ = 9.89± 0.17 %
μ = 440.7± 0.7
σ = 43.6± 0.7

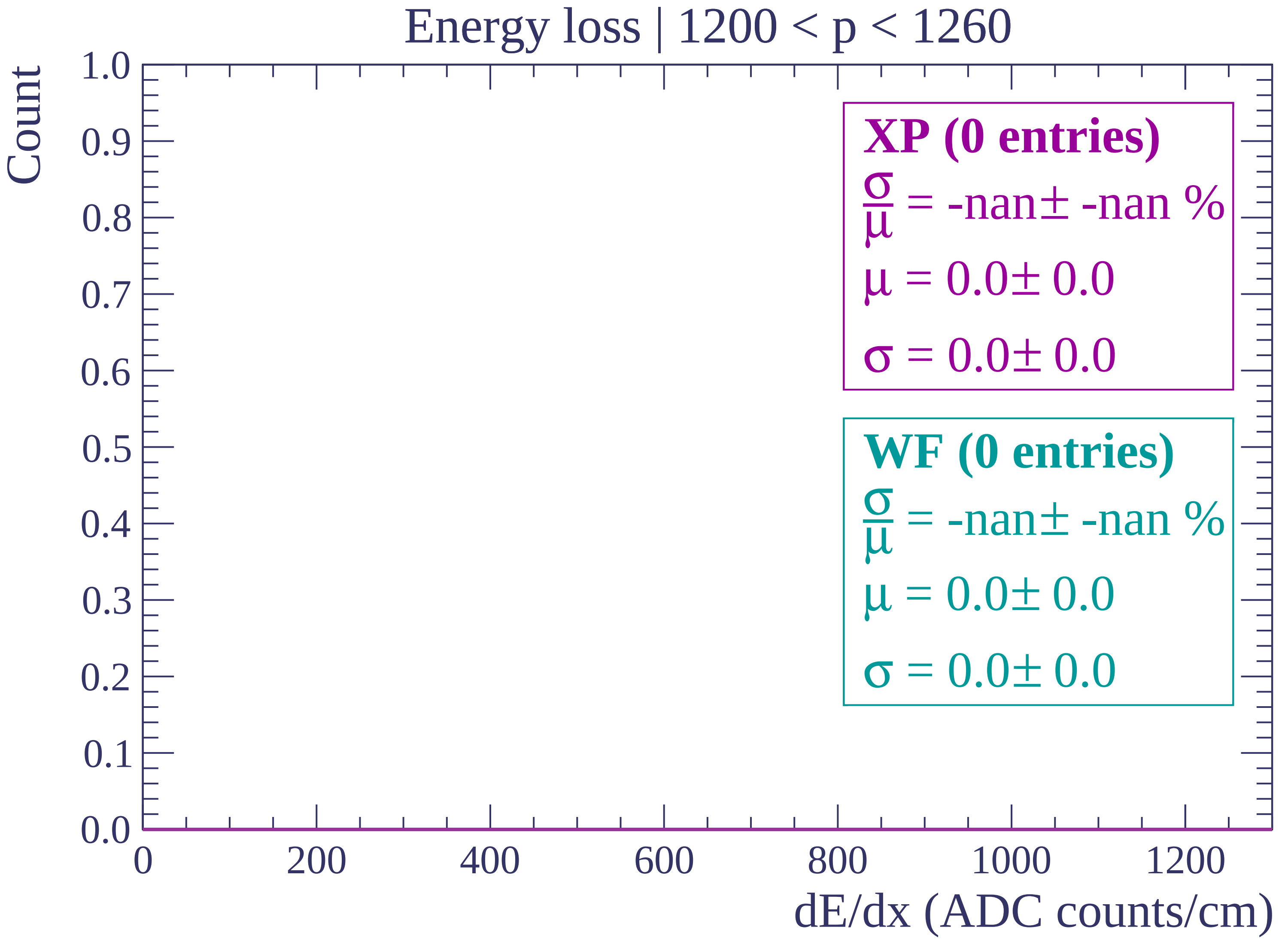


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1078 entries)
$\frac{\sigma}{\mu}$ = 8.71± 0.40 %
μ = 436.2± 1.6
σ = 38.0± 1.6
WF (1078 entries)
$\frac{\sigma}{\mu}$ = 9.21± 0.36 %
μ = 442.2± 1.6
σ = 40.7± 1.4

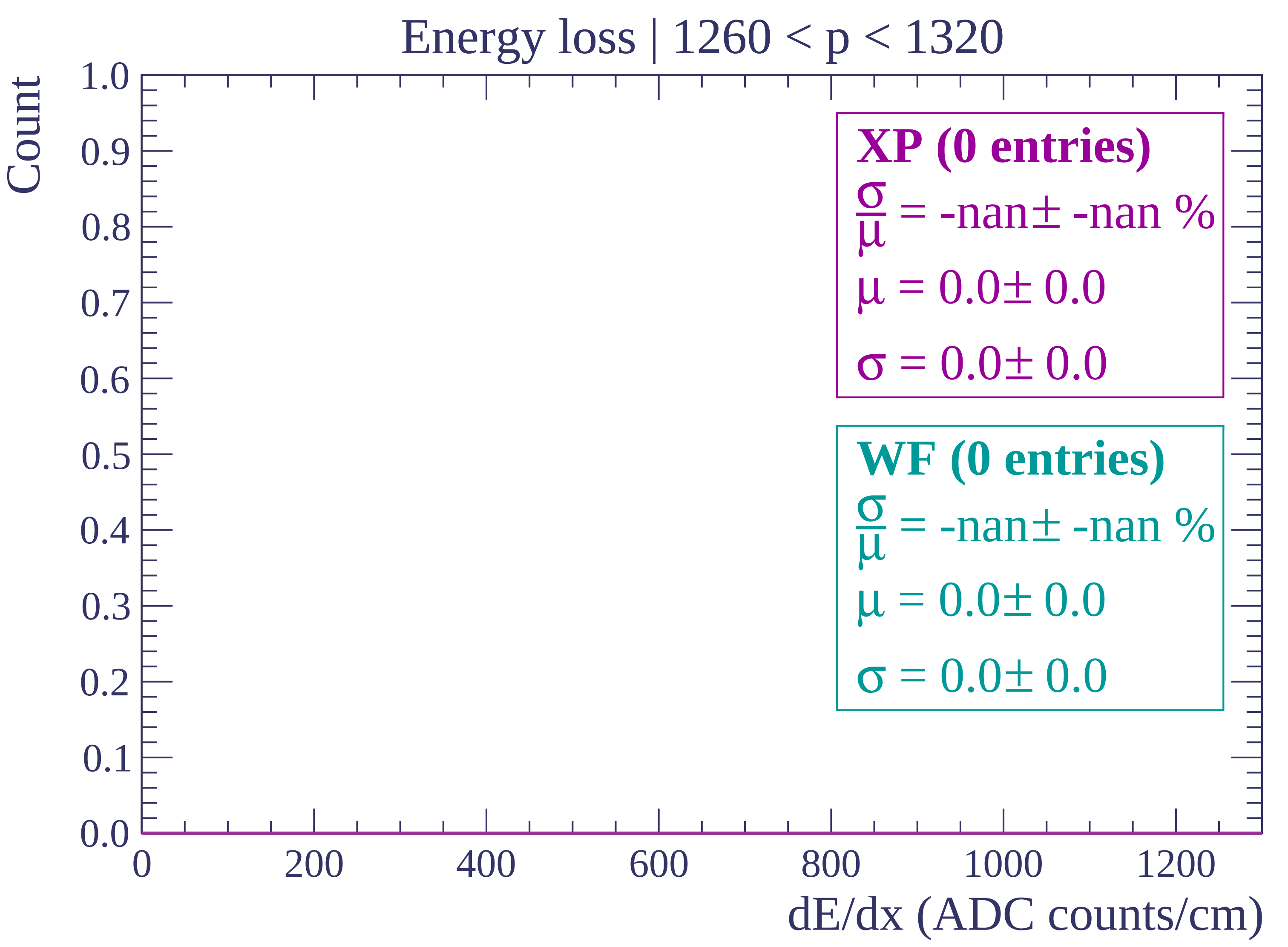

Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



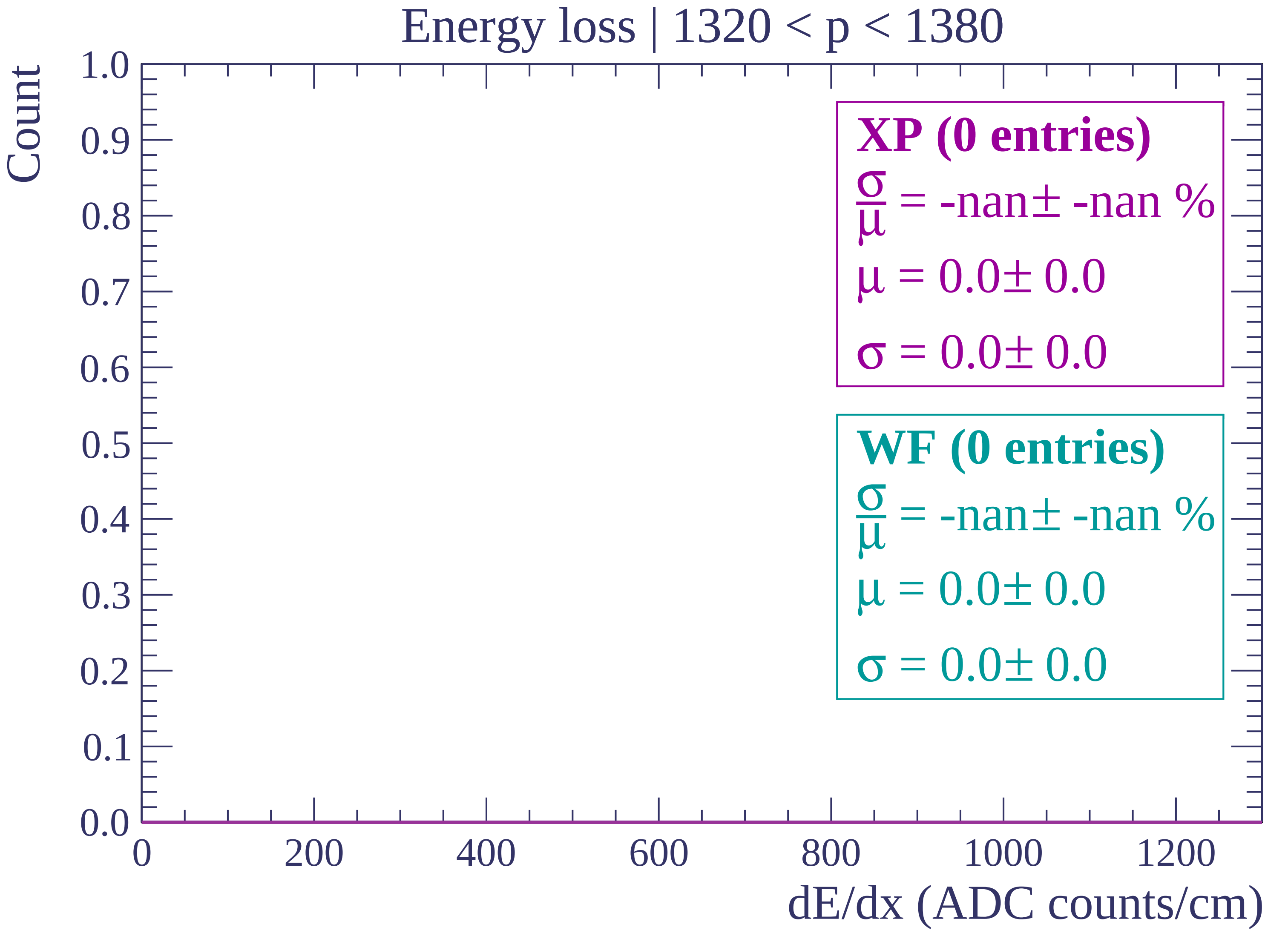
Energy loss | 1140 < p < 1200
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



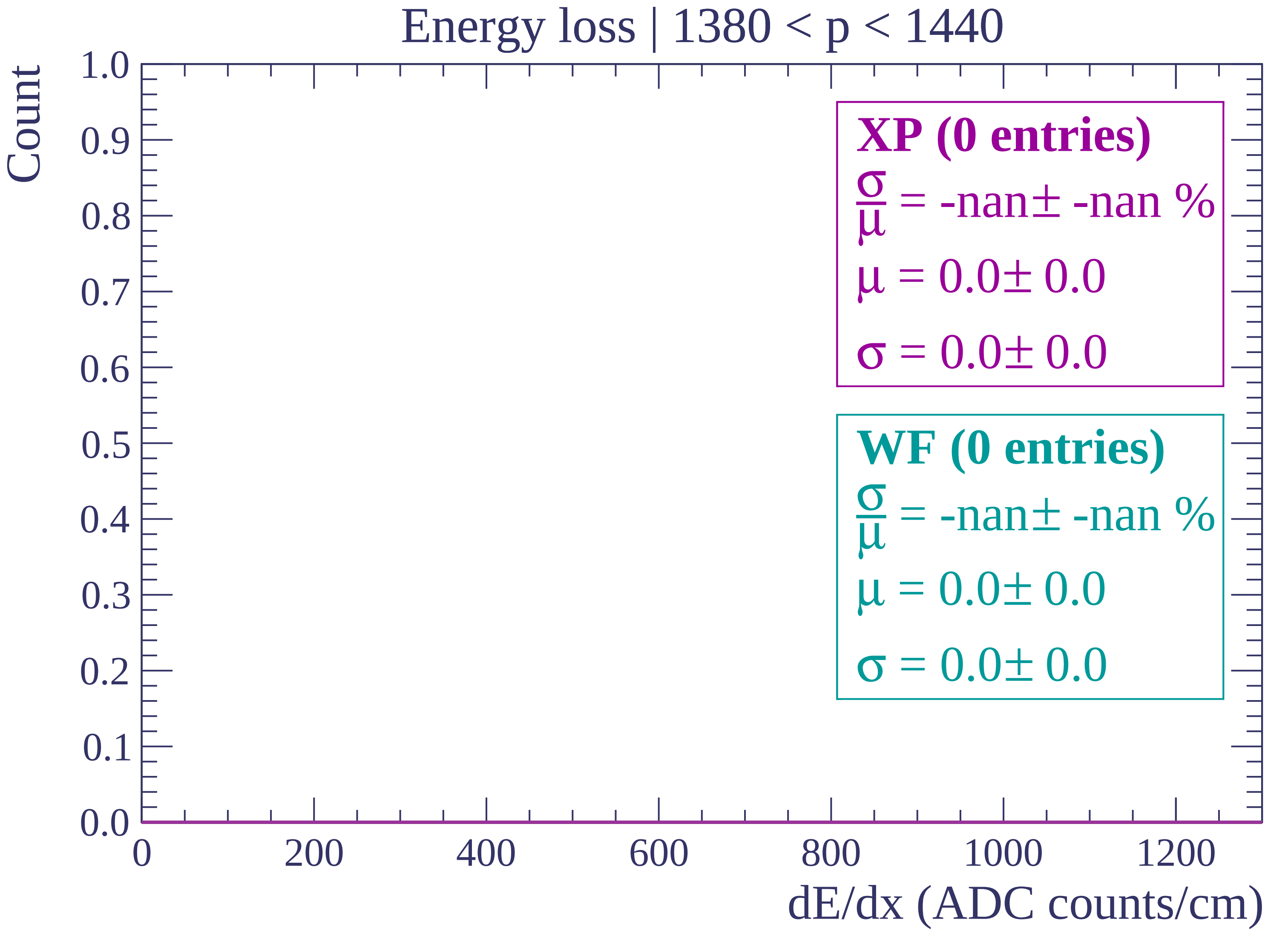
Energy loss | 1200 < p < 1260
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



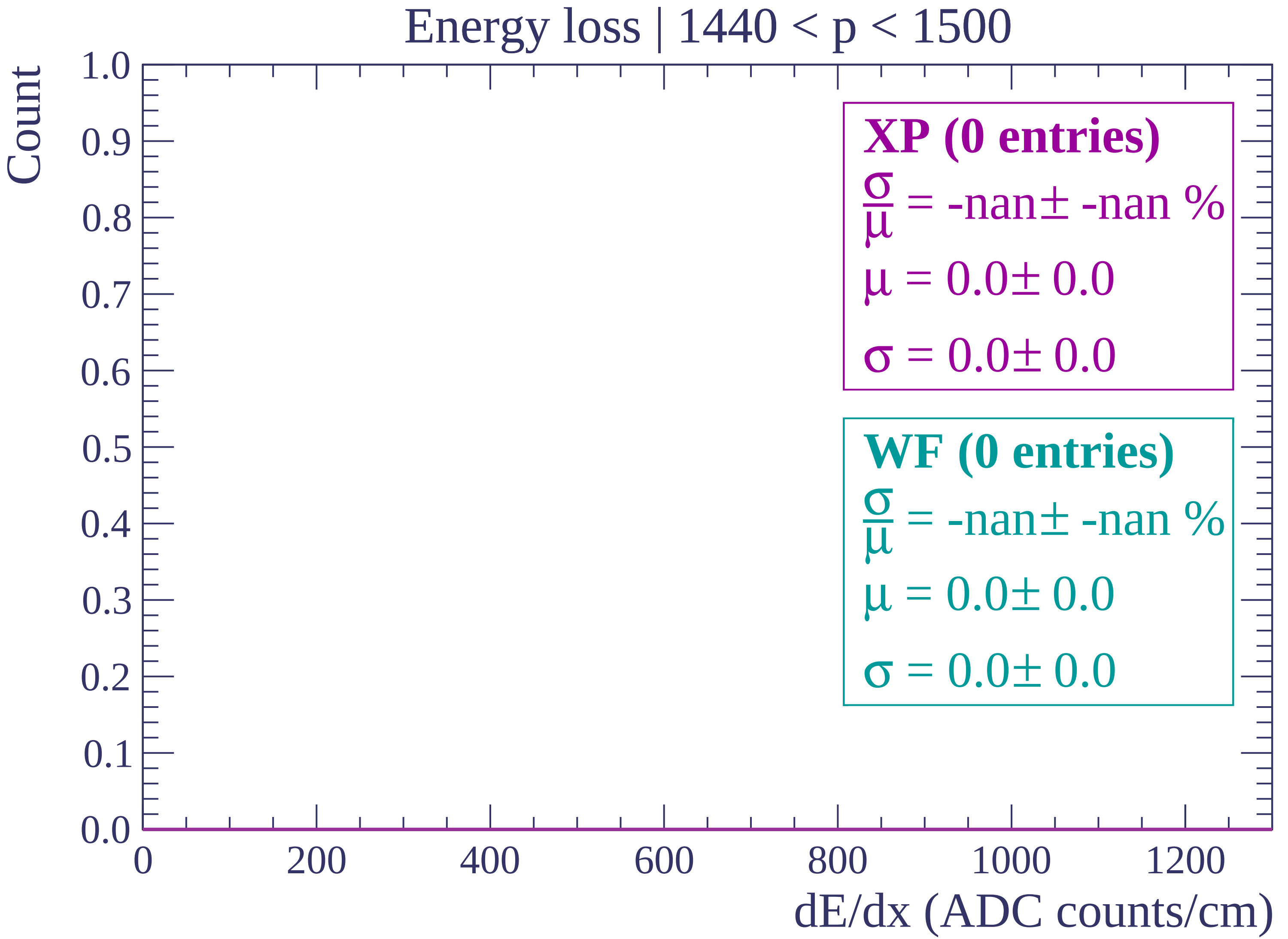
Energy loss | 1260 < p < 1320
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

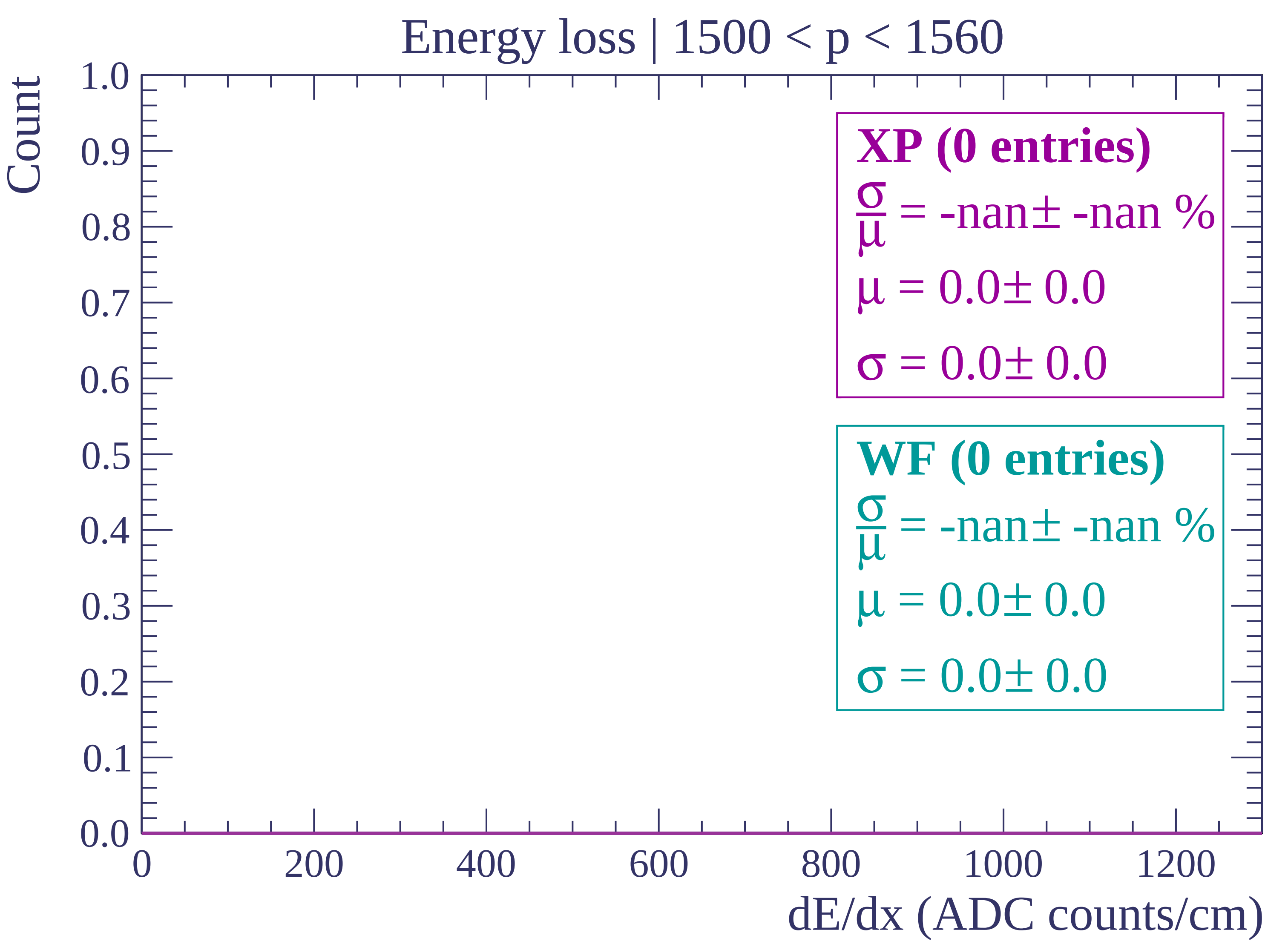


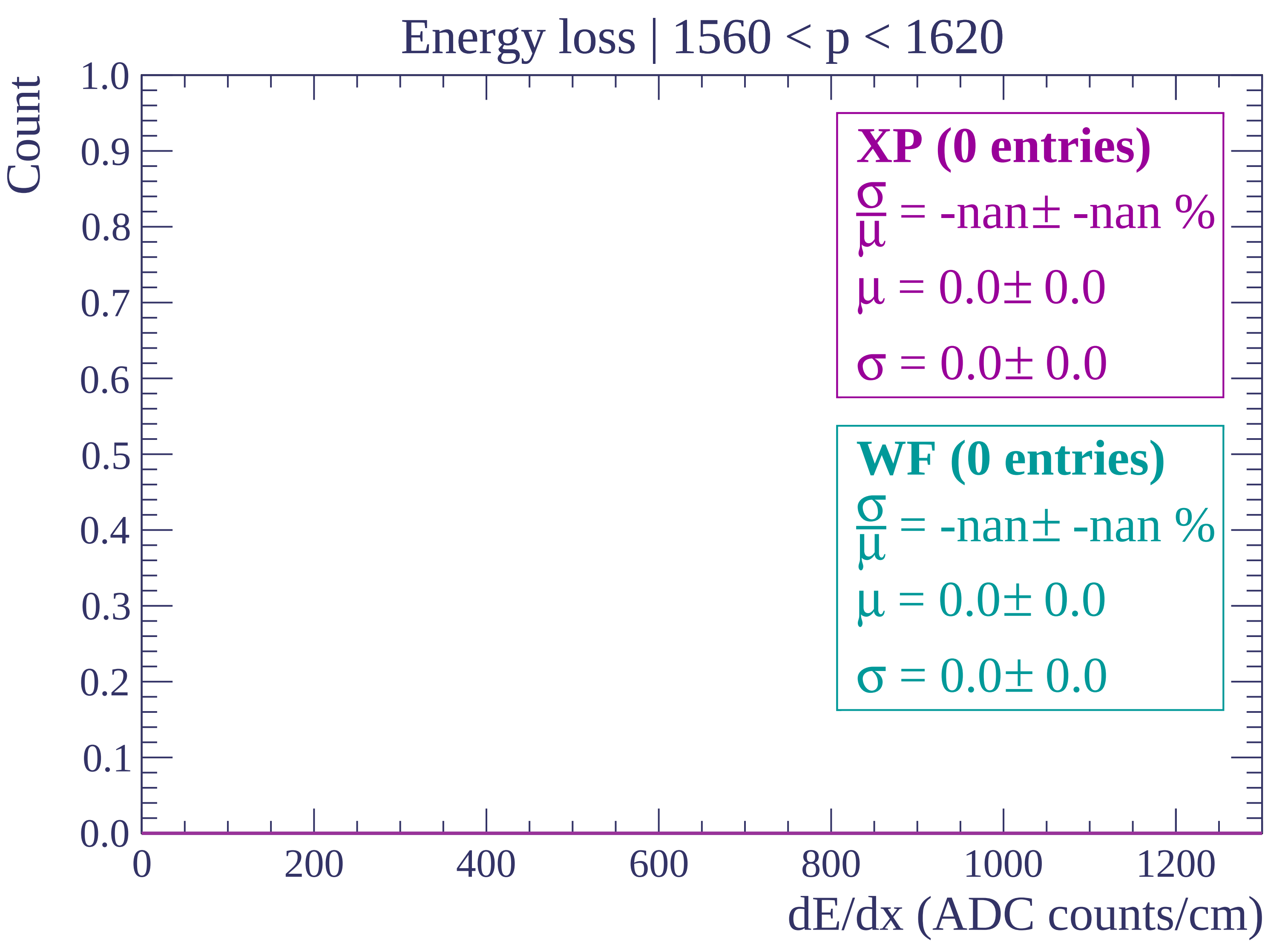
Energy loss | 1320 < p < 1380
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

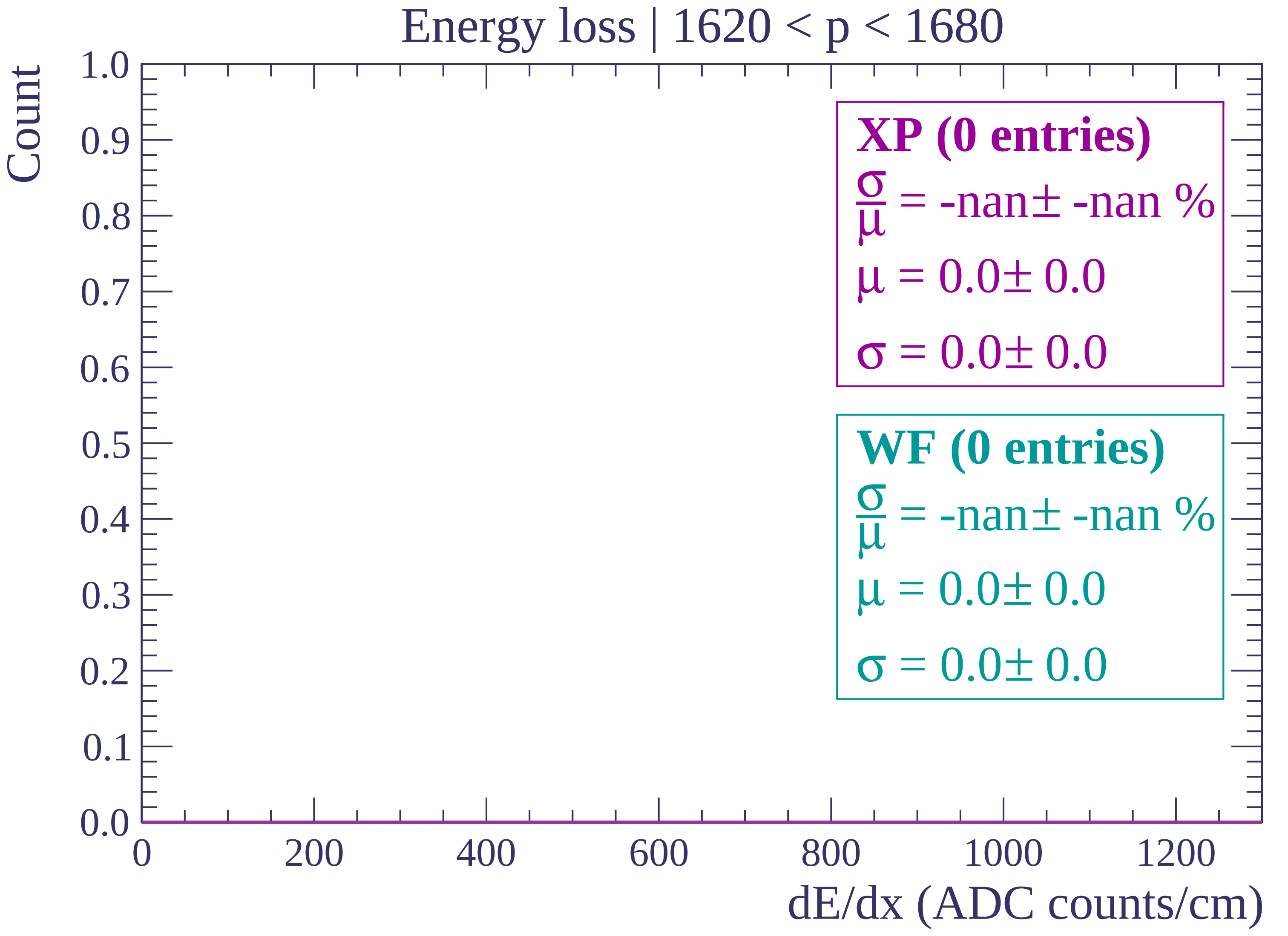


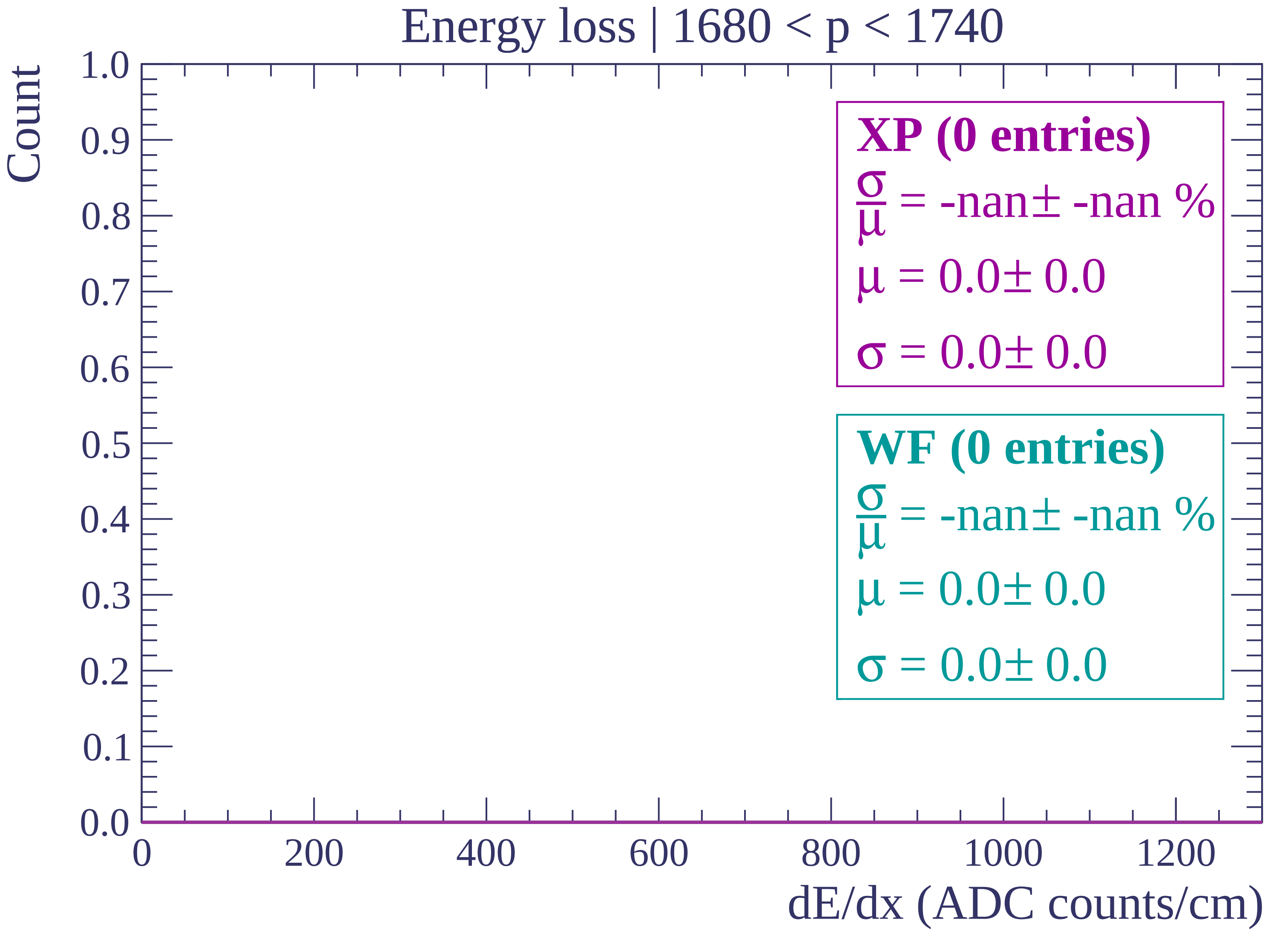
Energy loss | 1380 < p < 1440
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1440 < p < 1500
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

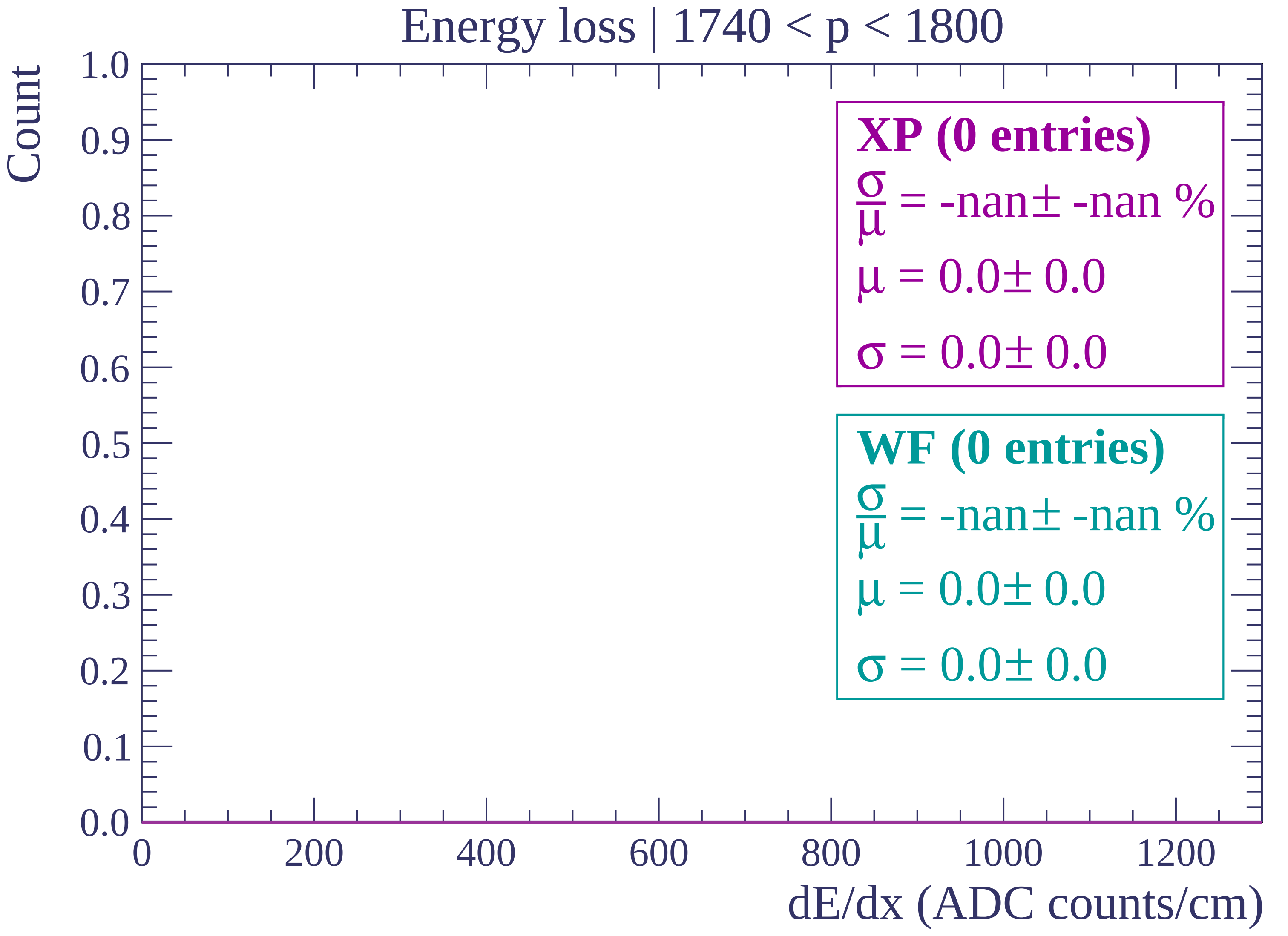

Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

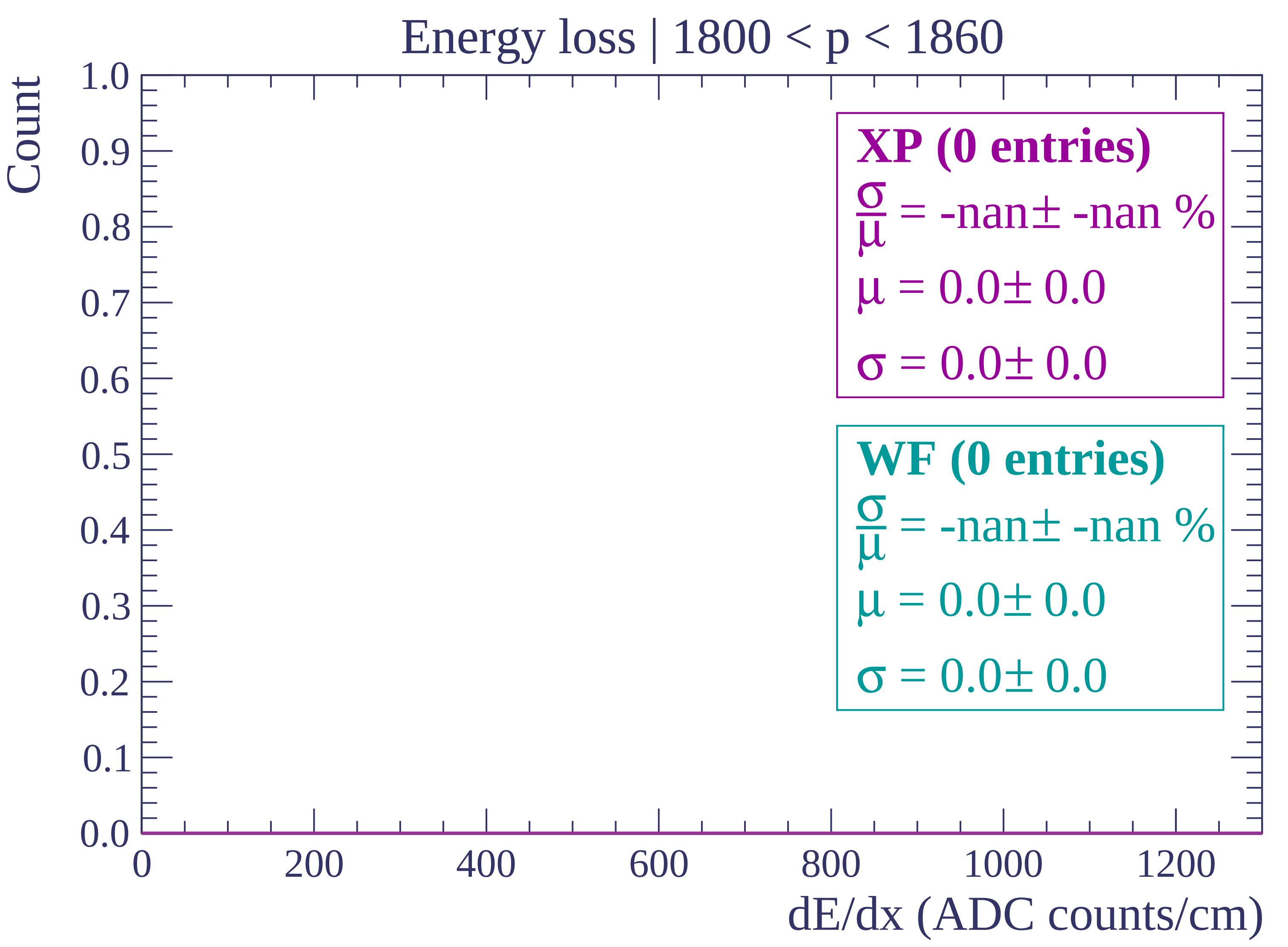

Energy loss | 1620 < p < 1680
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

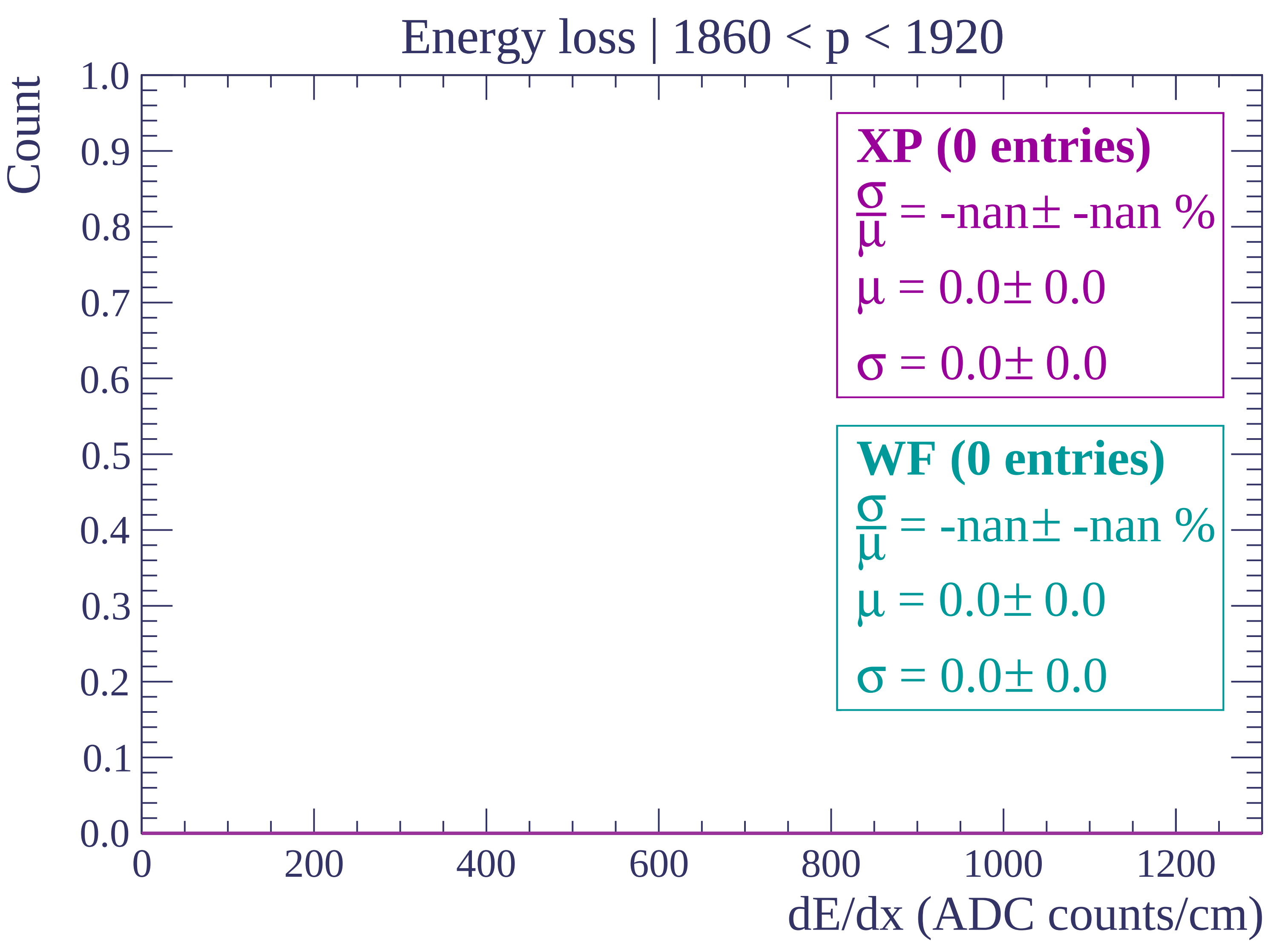


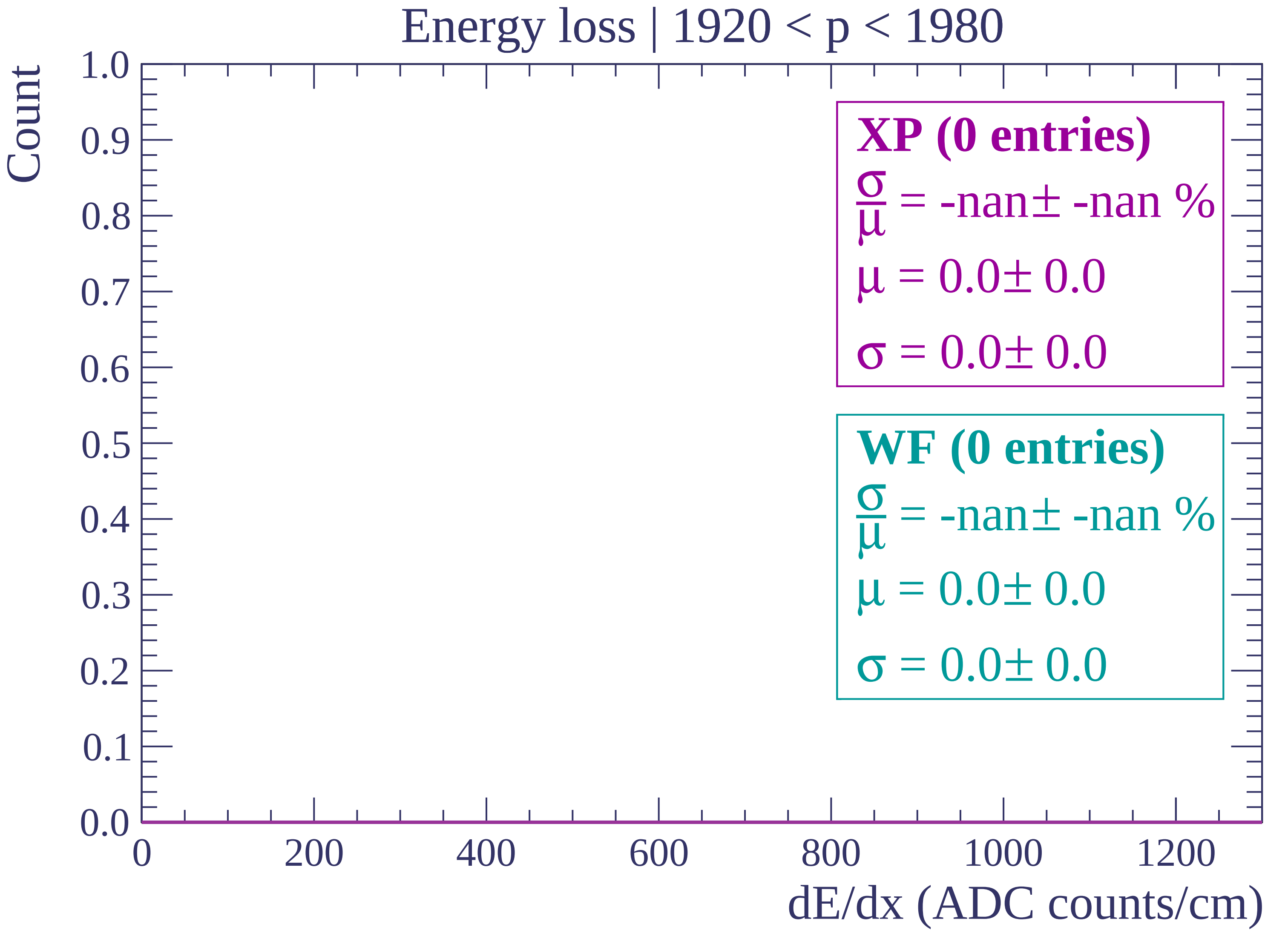
Energy loss | 1680 < p < 1740
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



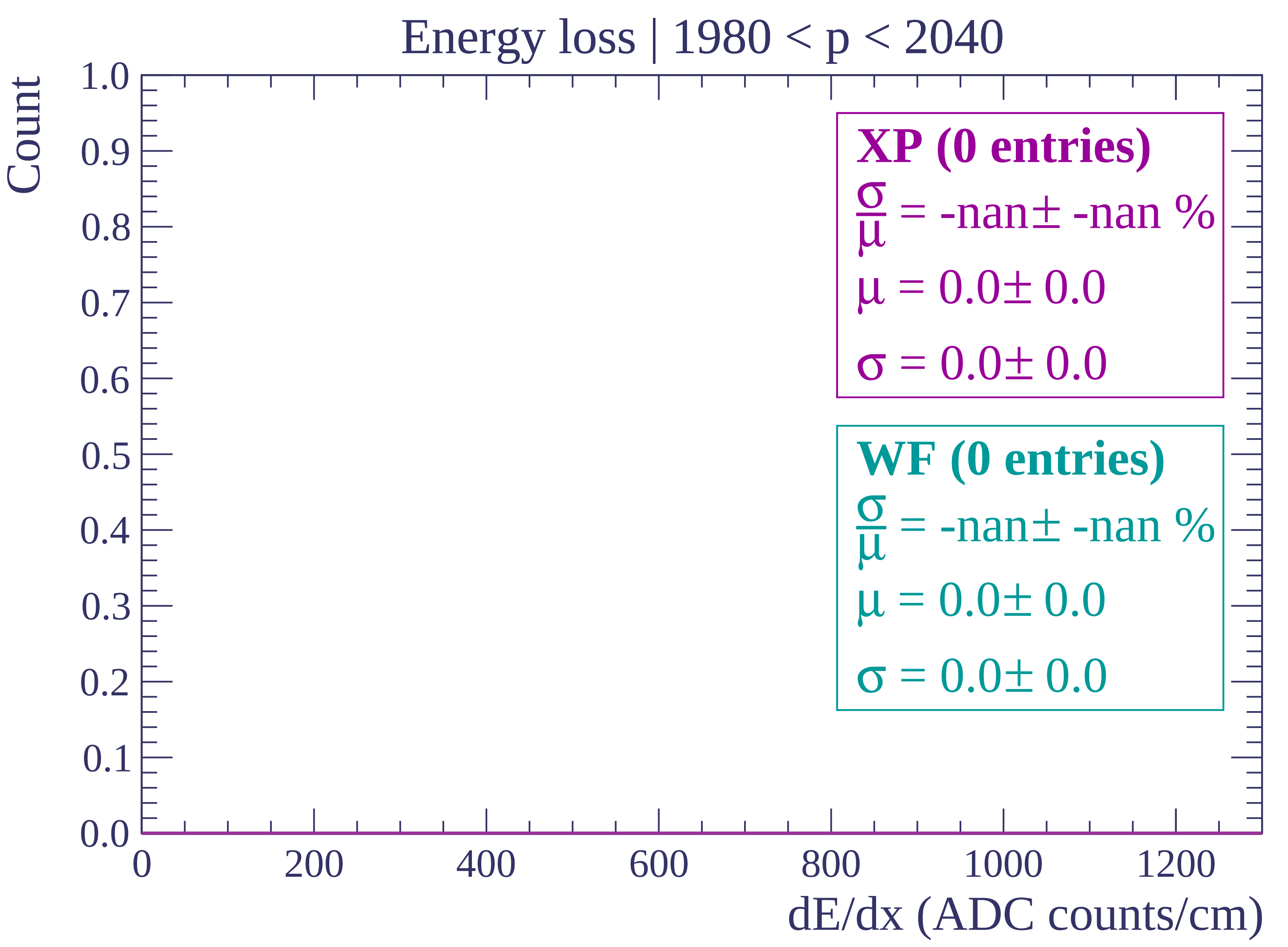
Energy loss | 1740 < p < 1800
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

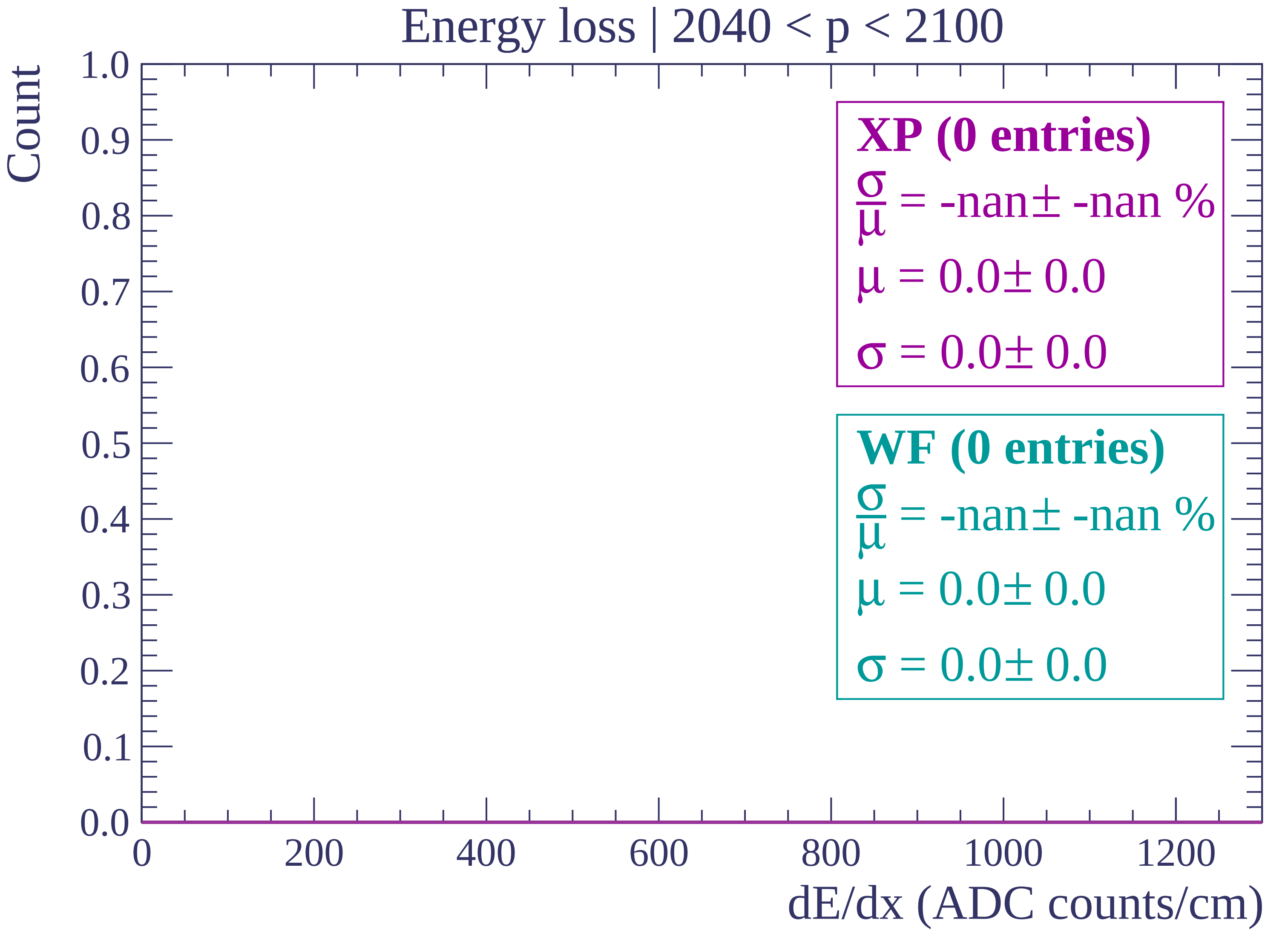

Energy loss | 1860 < p < 1920
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



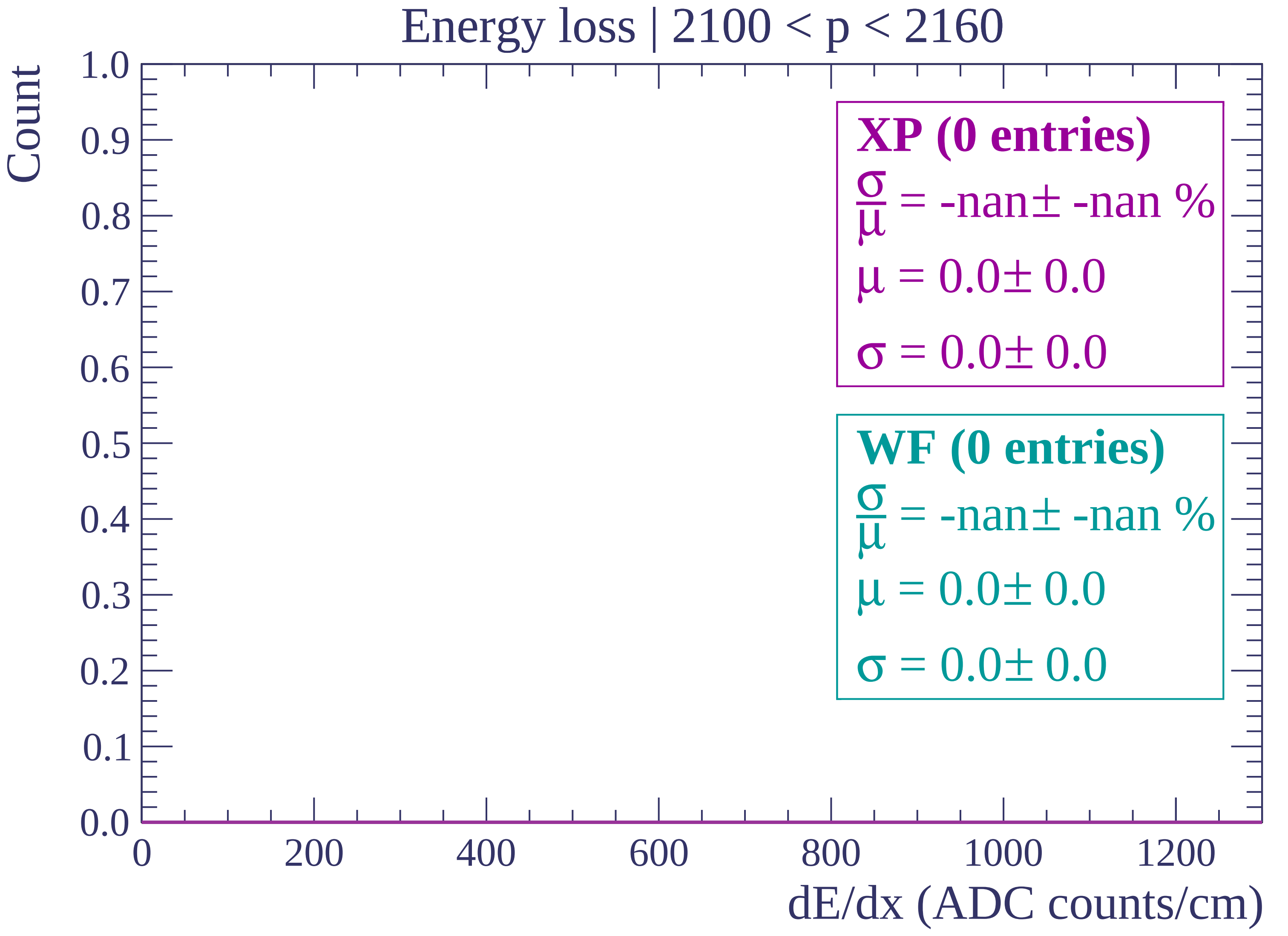
Energy loss | 1920 < p < 1980
Count
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



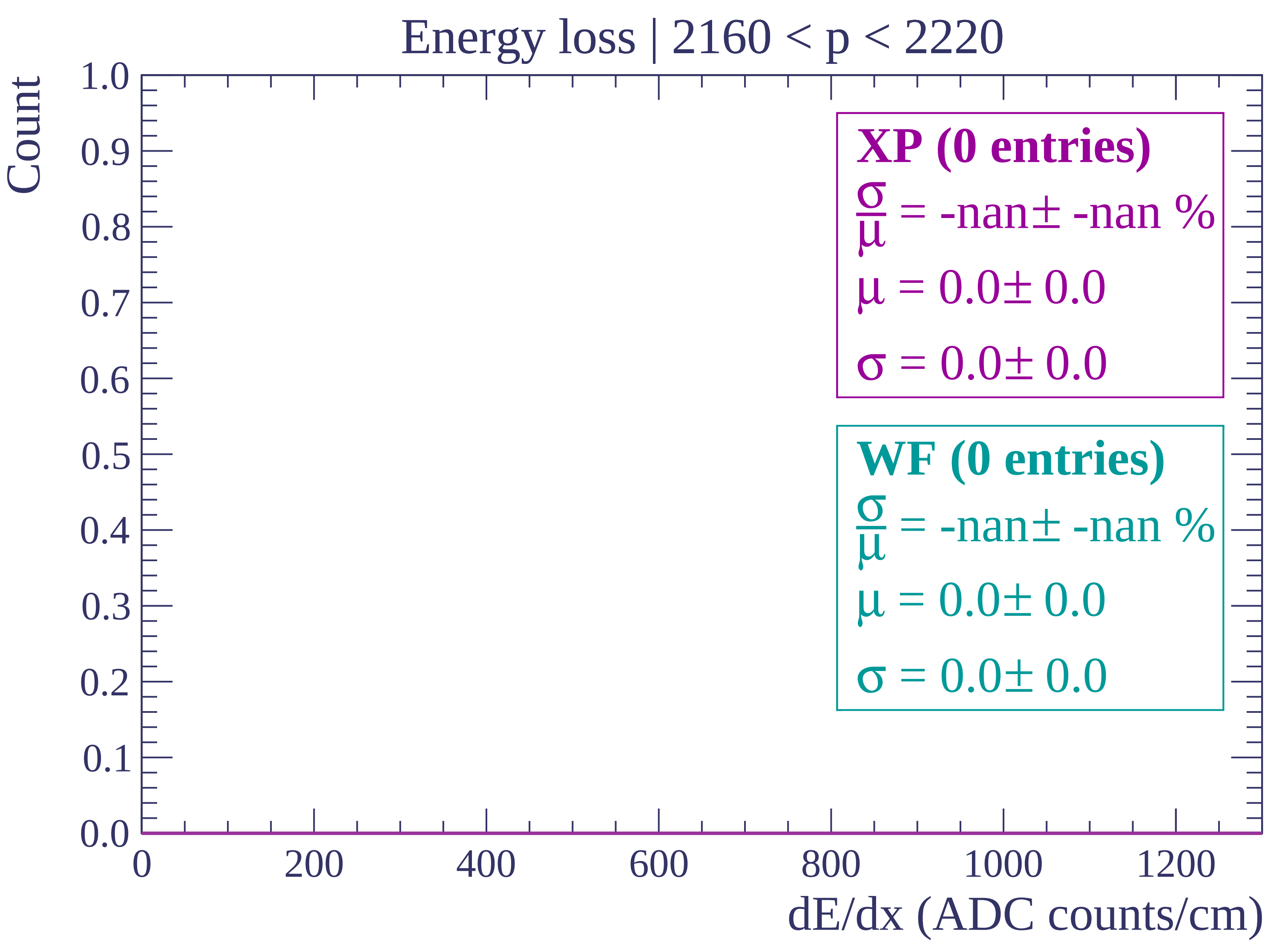
Energy loss | 1980 < p < 2040
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

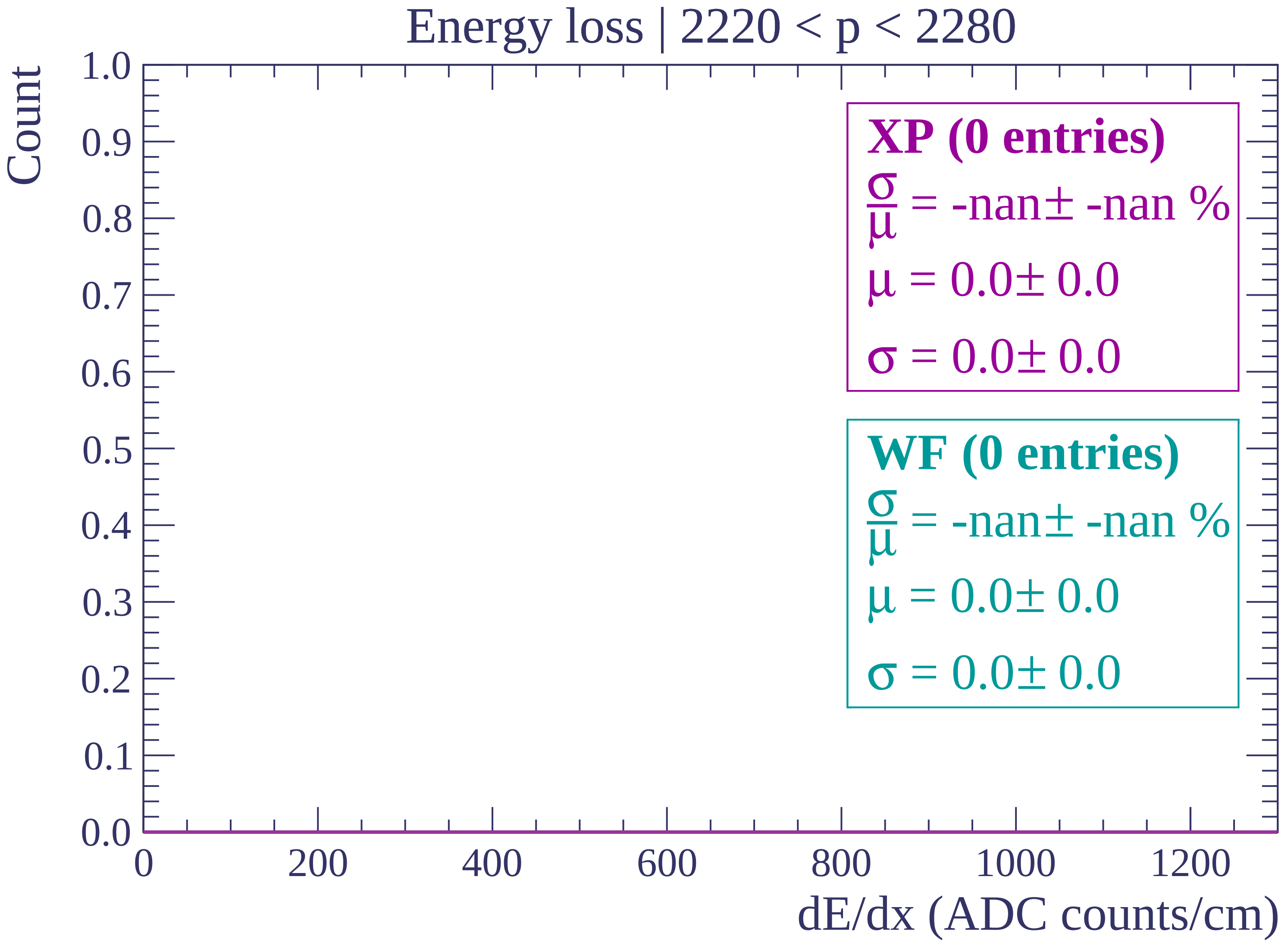


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

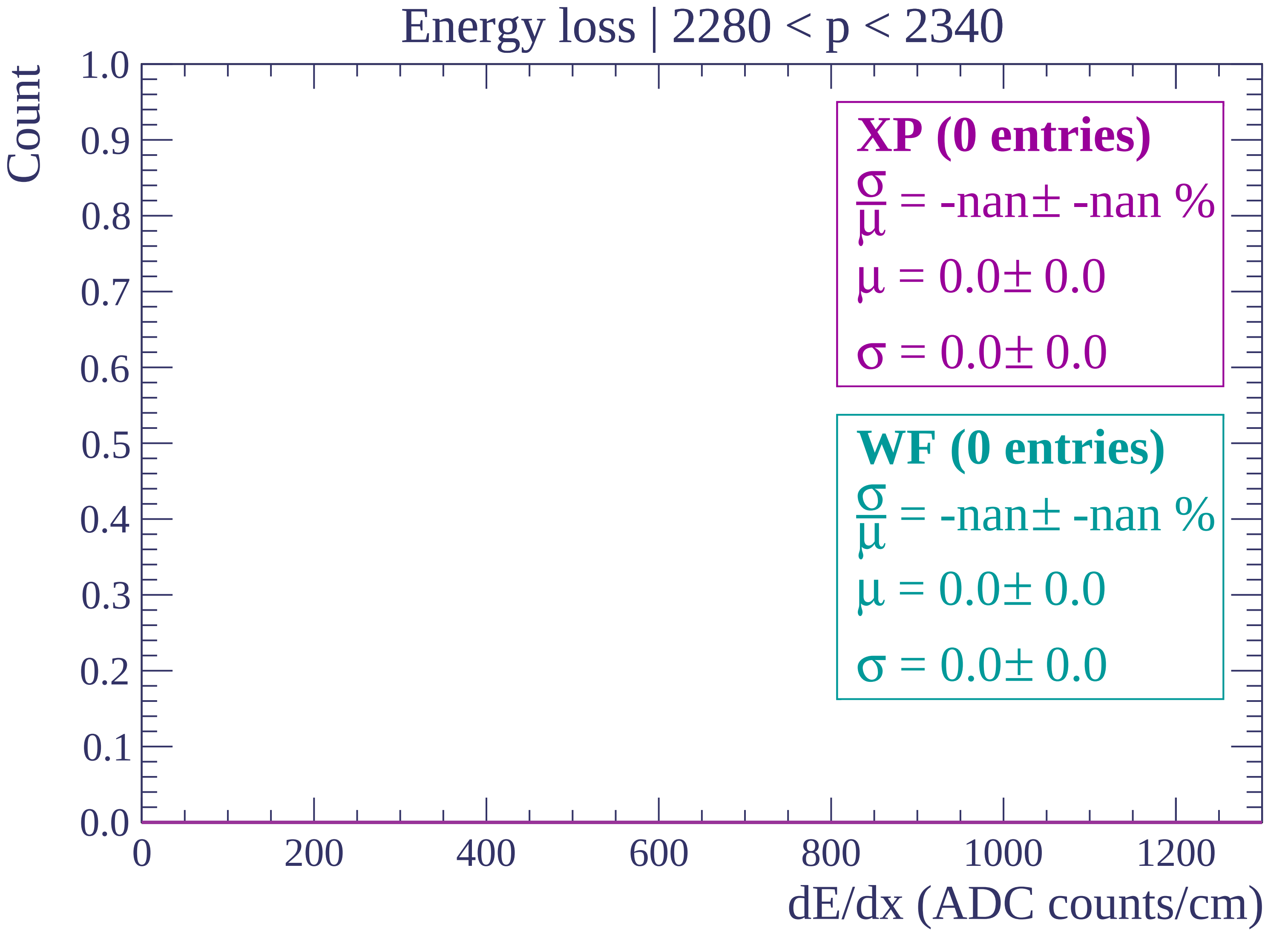


Energy loss | 2100 < p < 2160
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

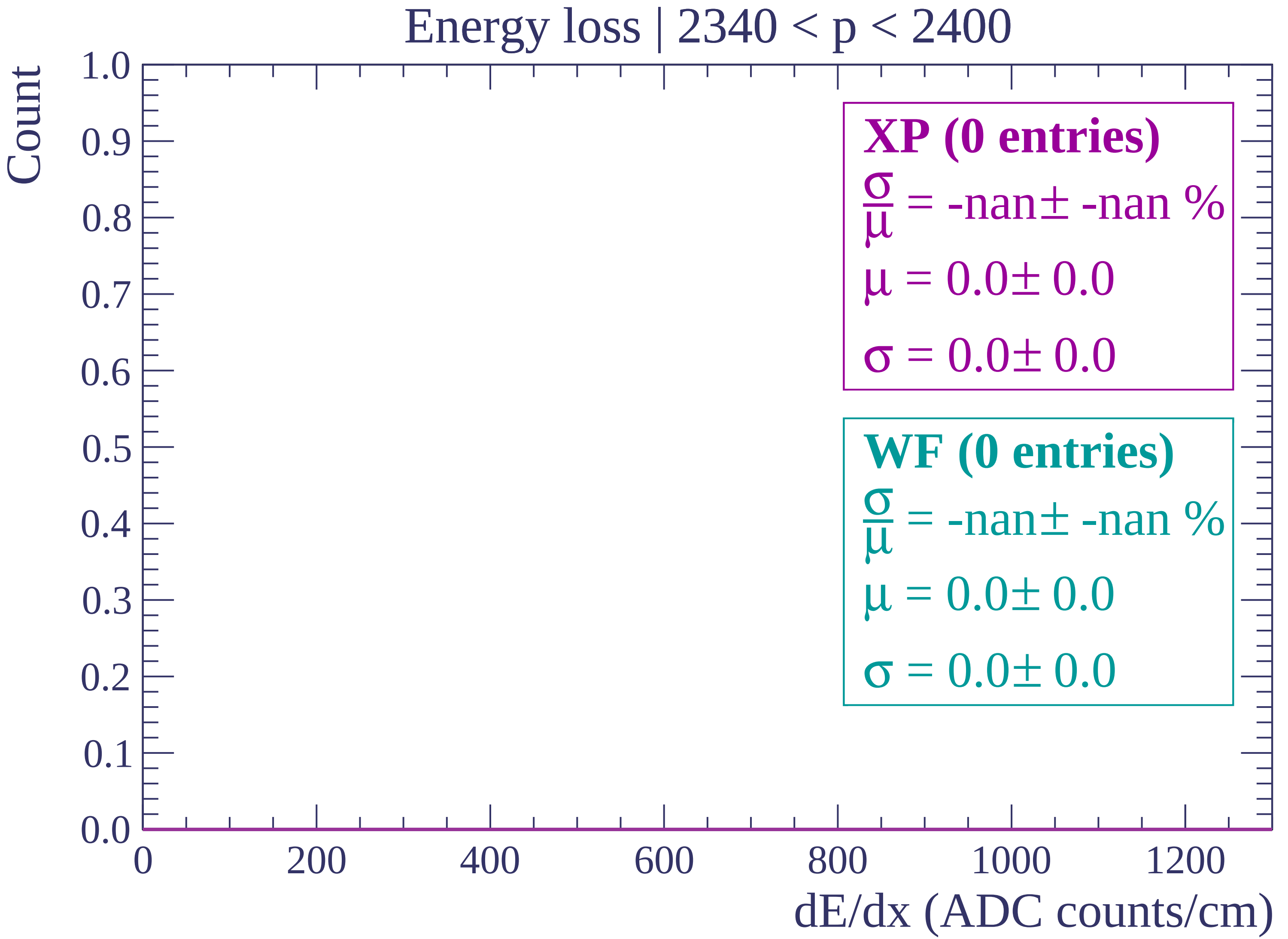

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



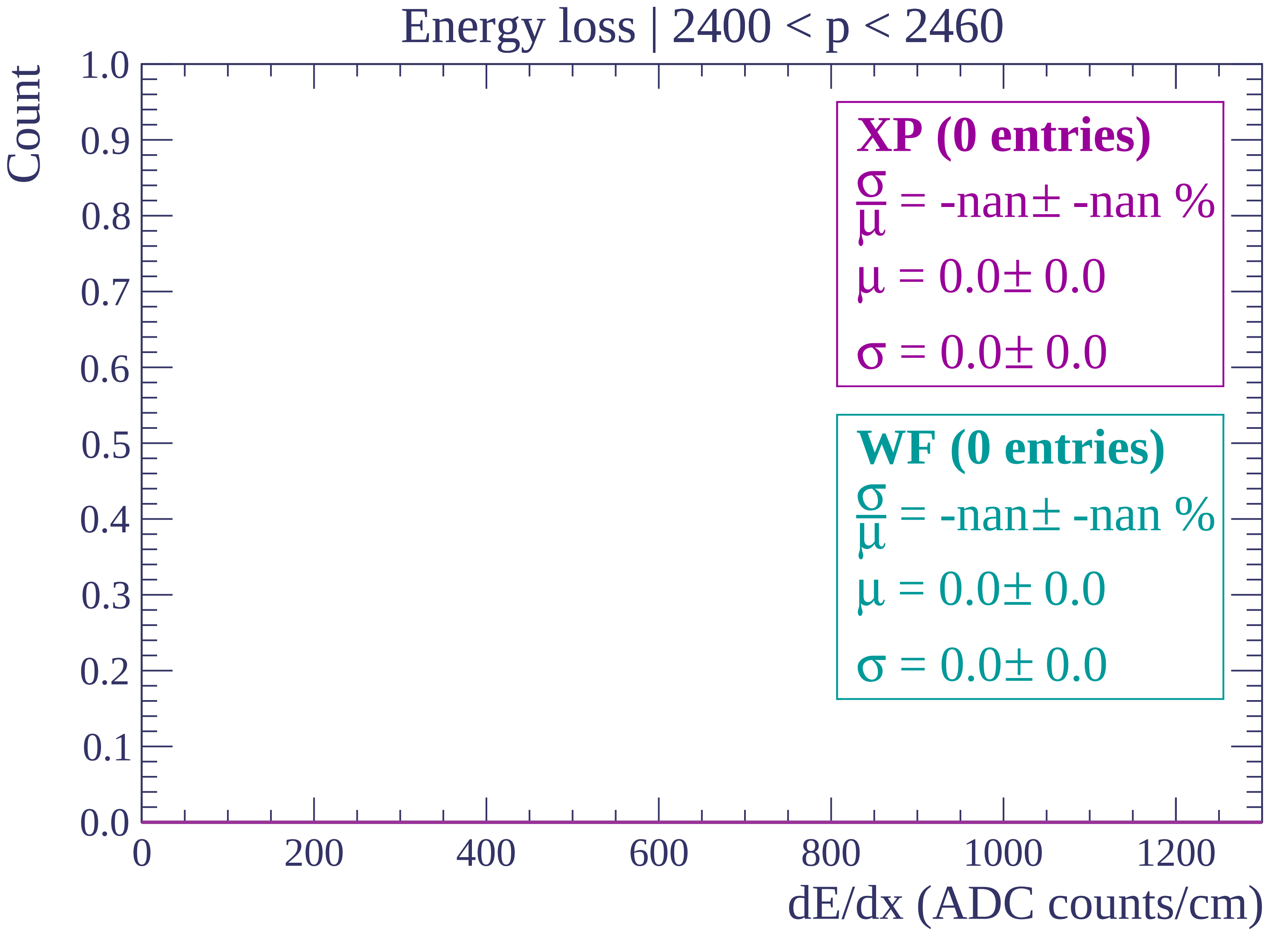
Energy loss | 2220 < p < 2280
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



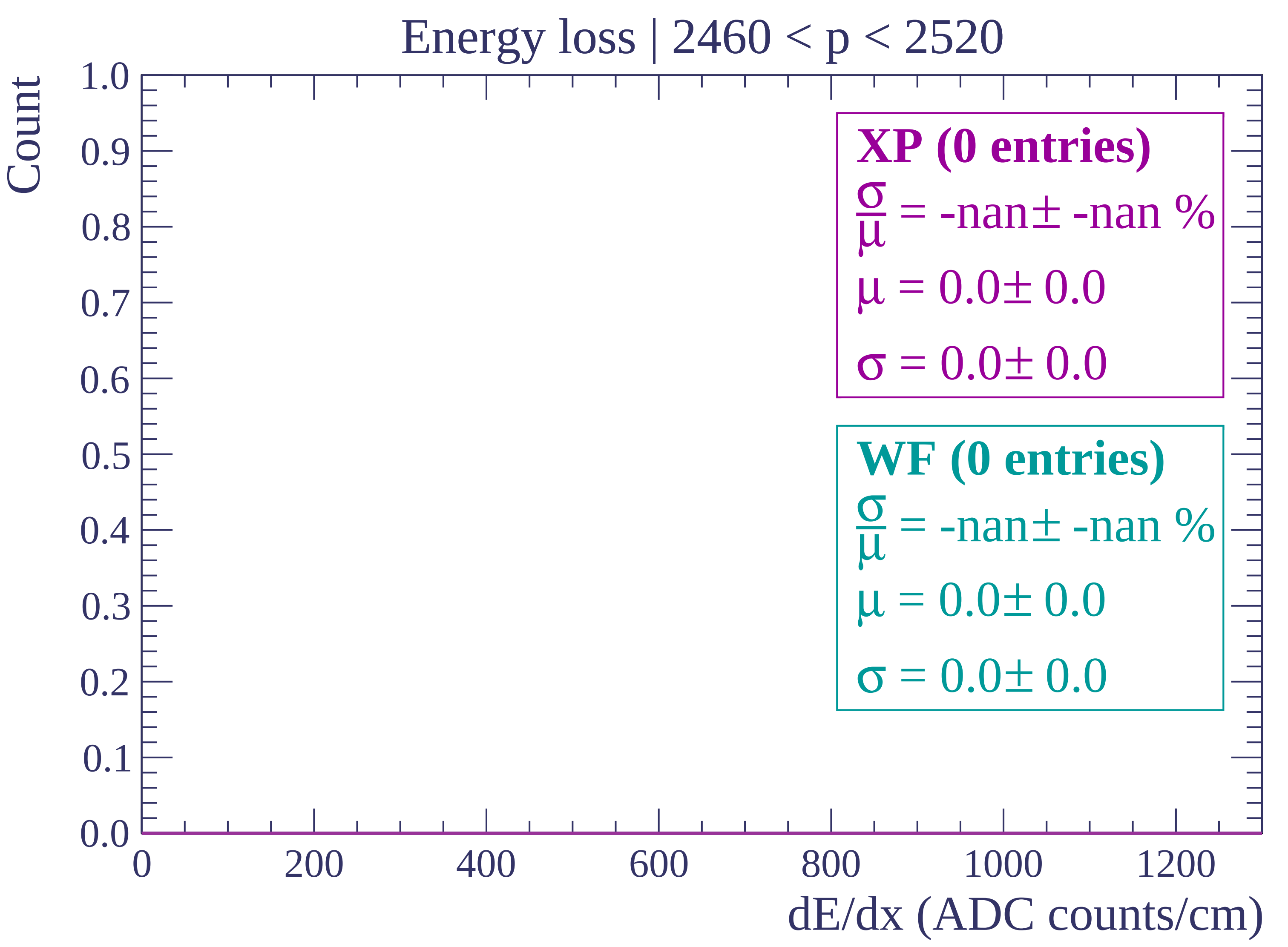
Energy loss | 2280 < p < 2340
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

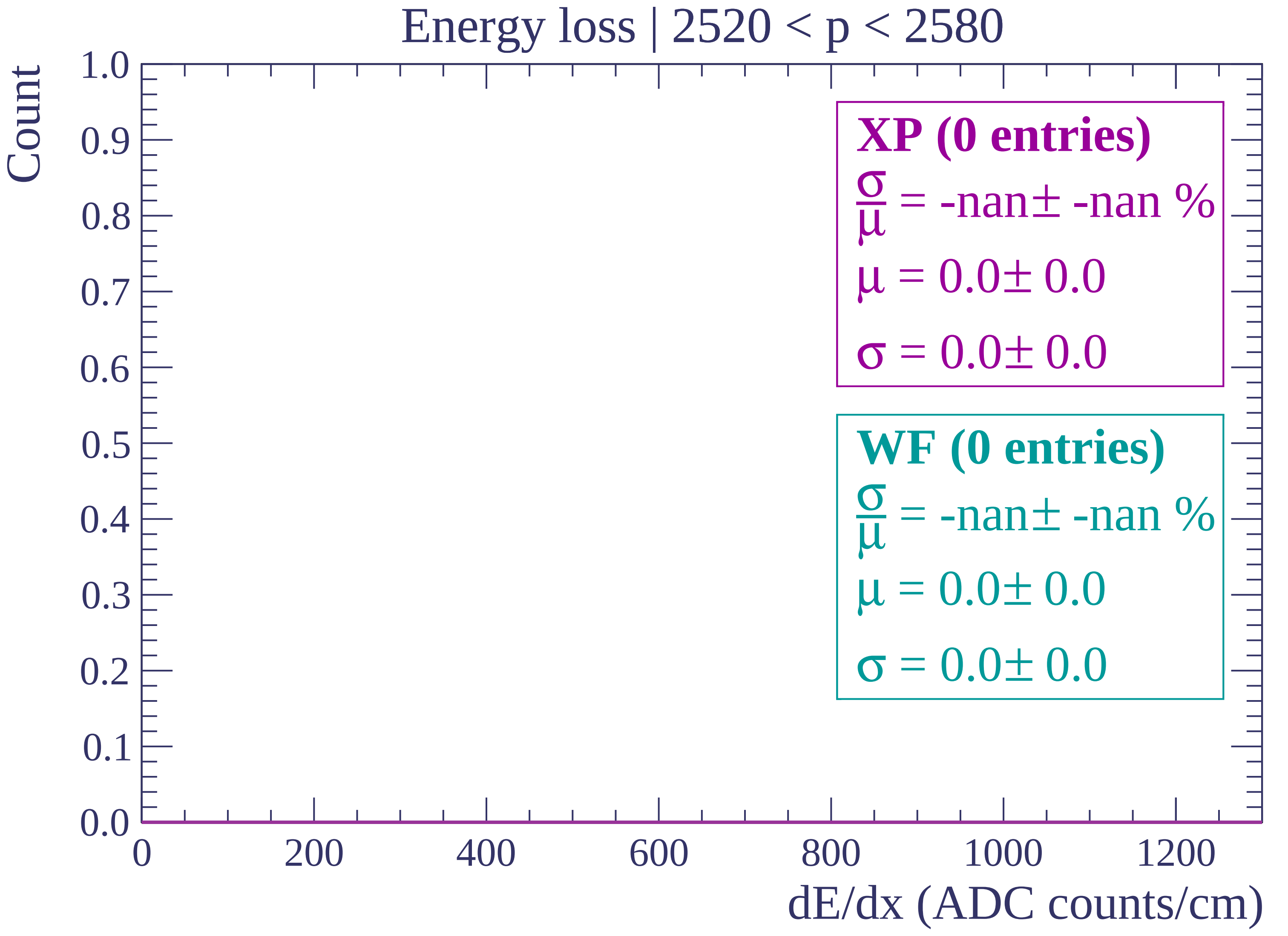


Energy loss | 2340 < p < 2400
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

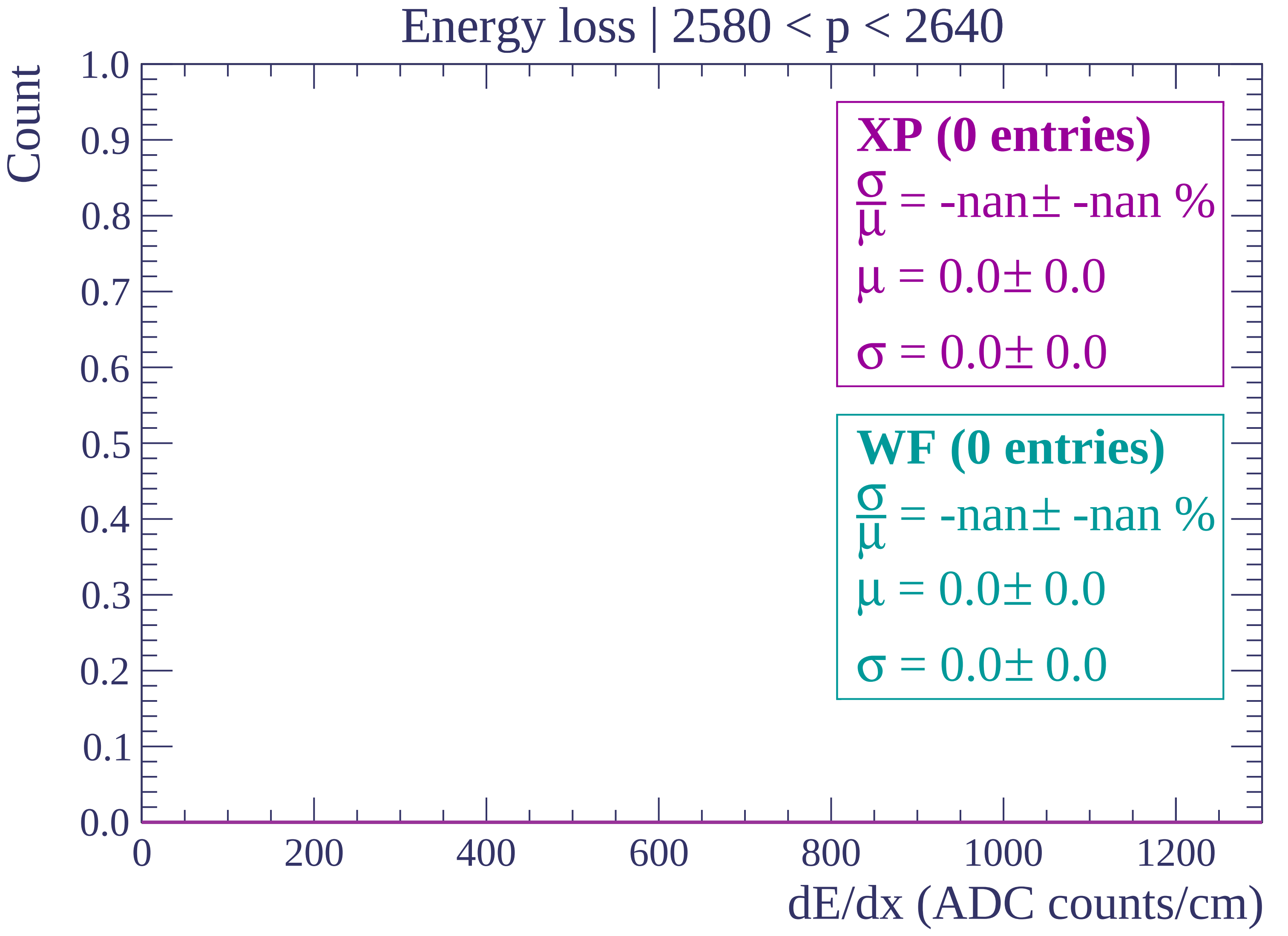


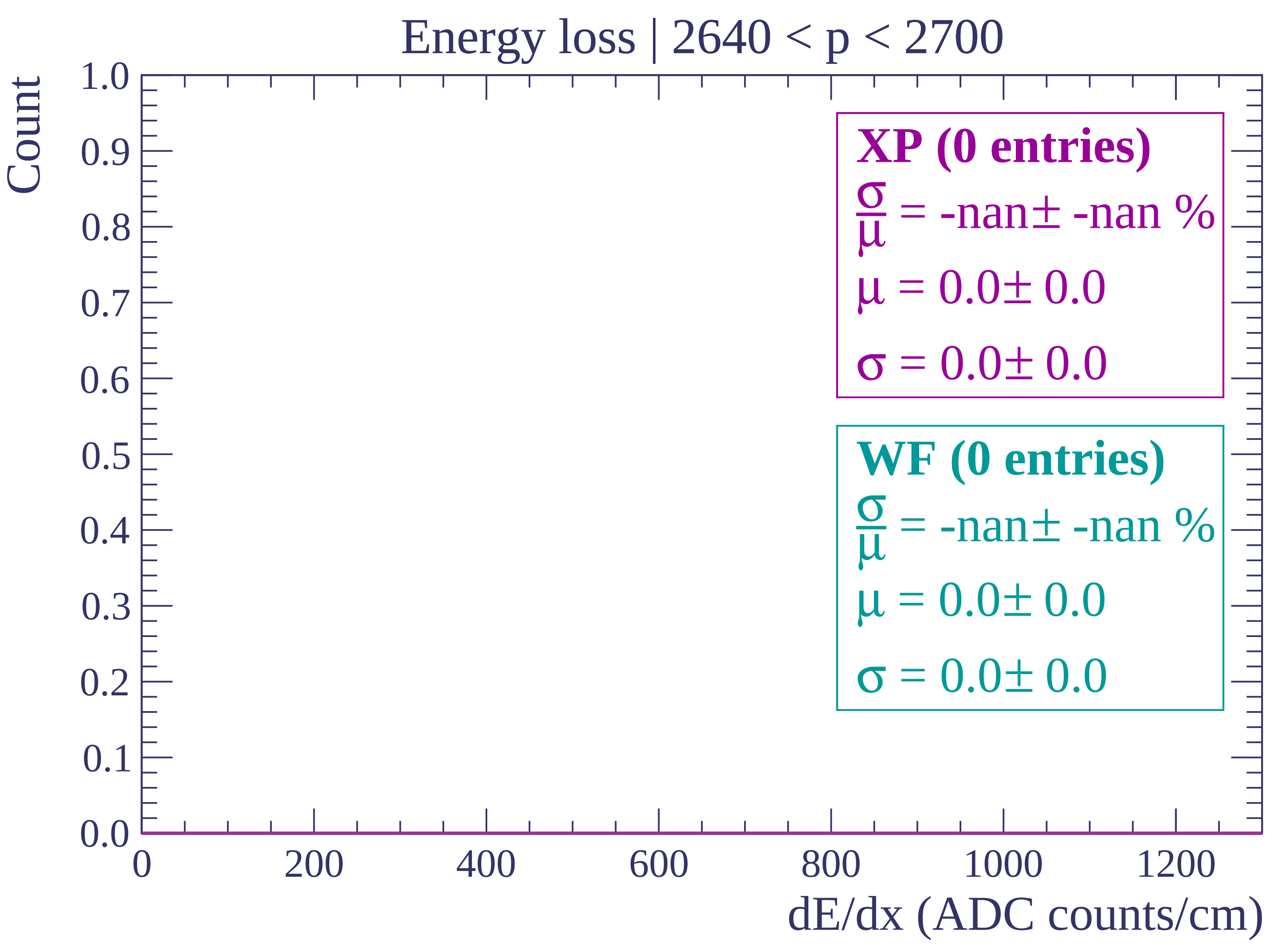
Energy loss | 2400 < p < 2460
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 2460 < p < 2520
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

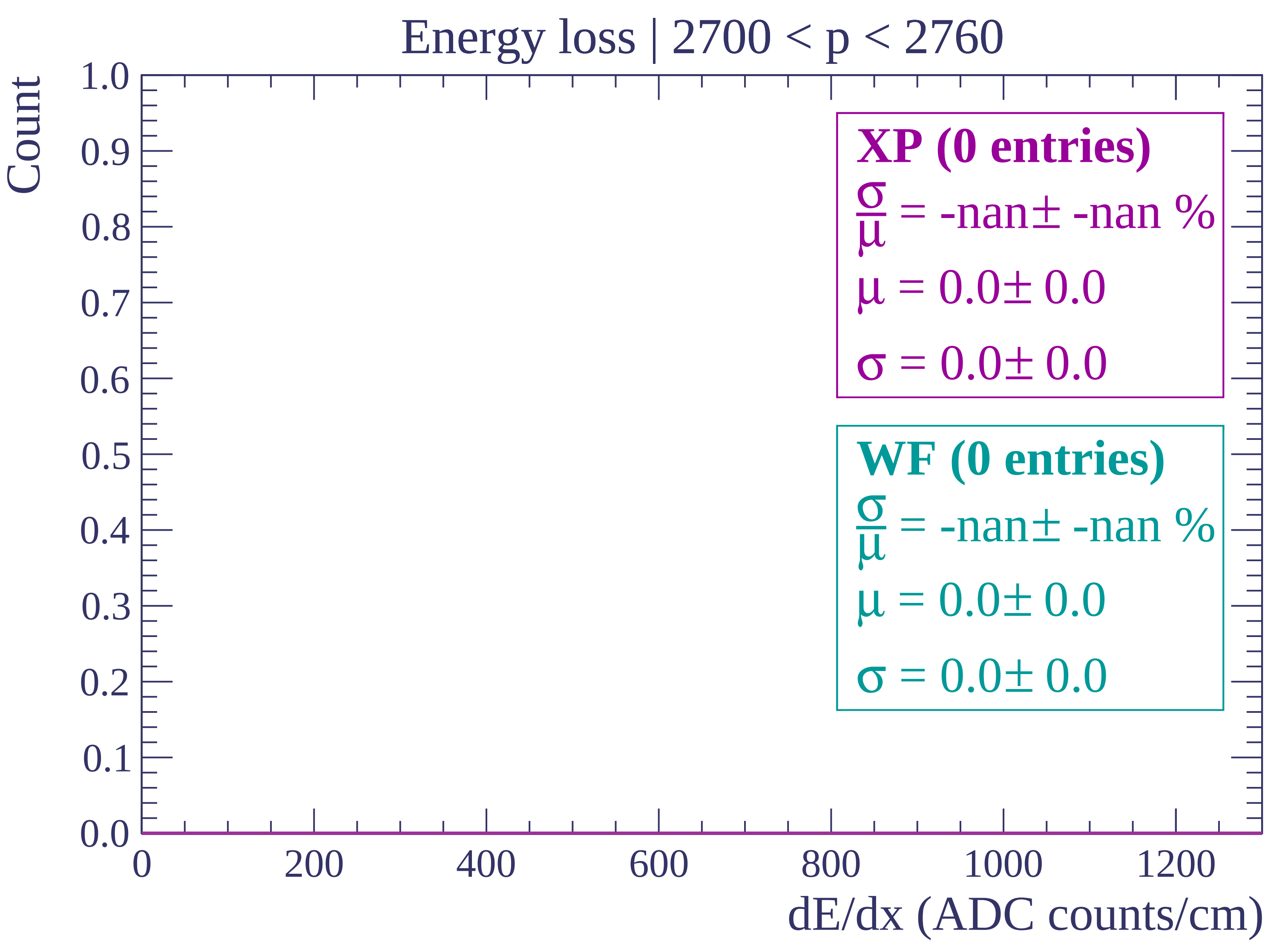


Energy loss | 2520 < p < 2580
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

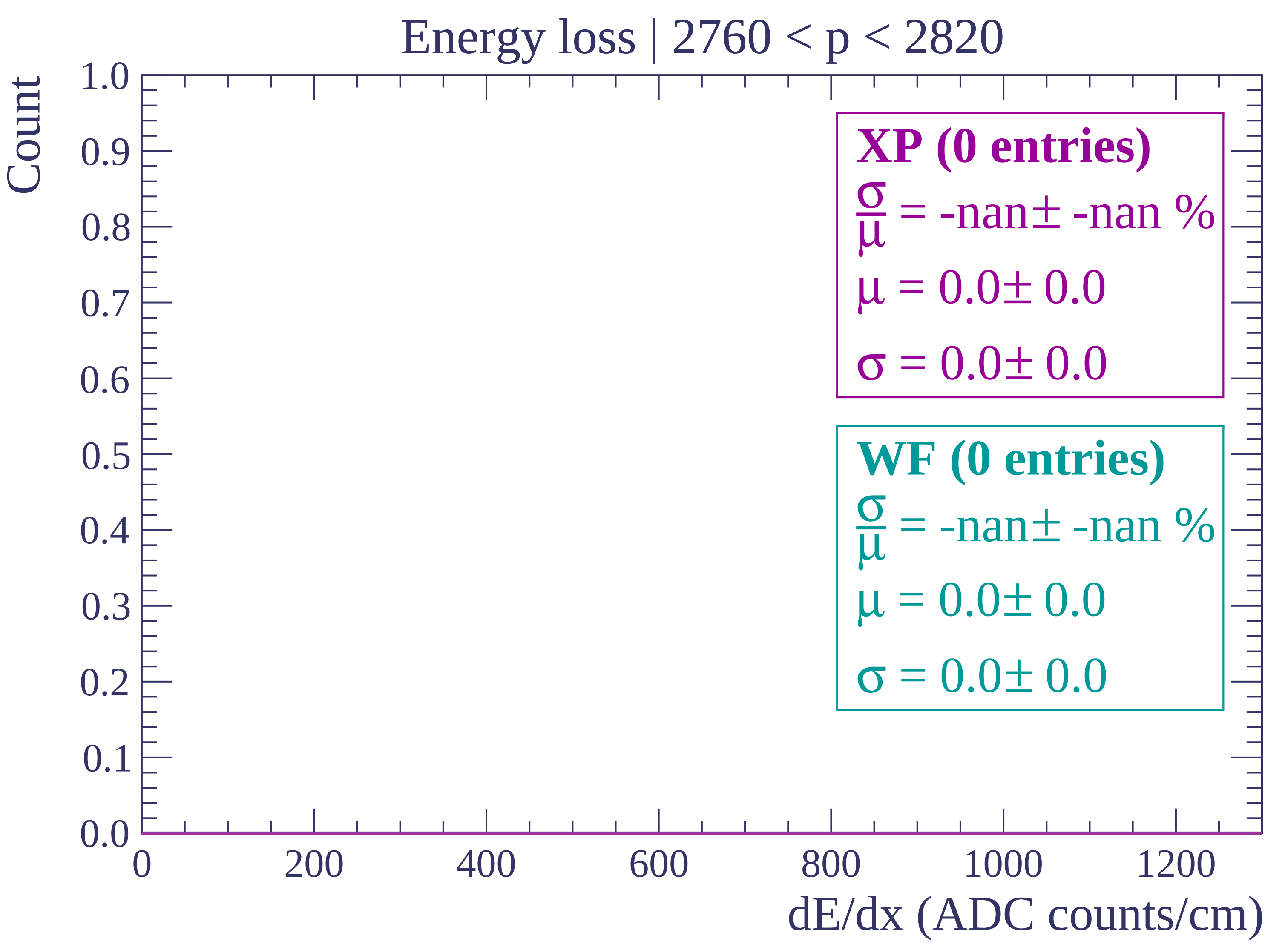

Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200



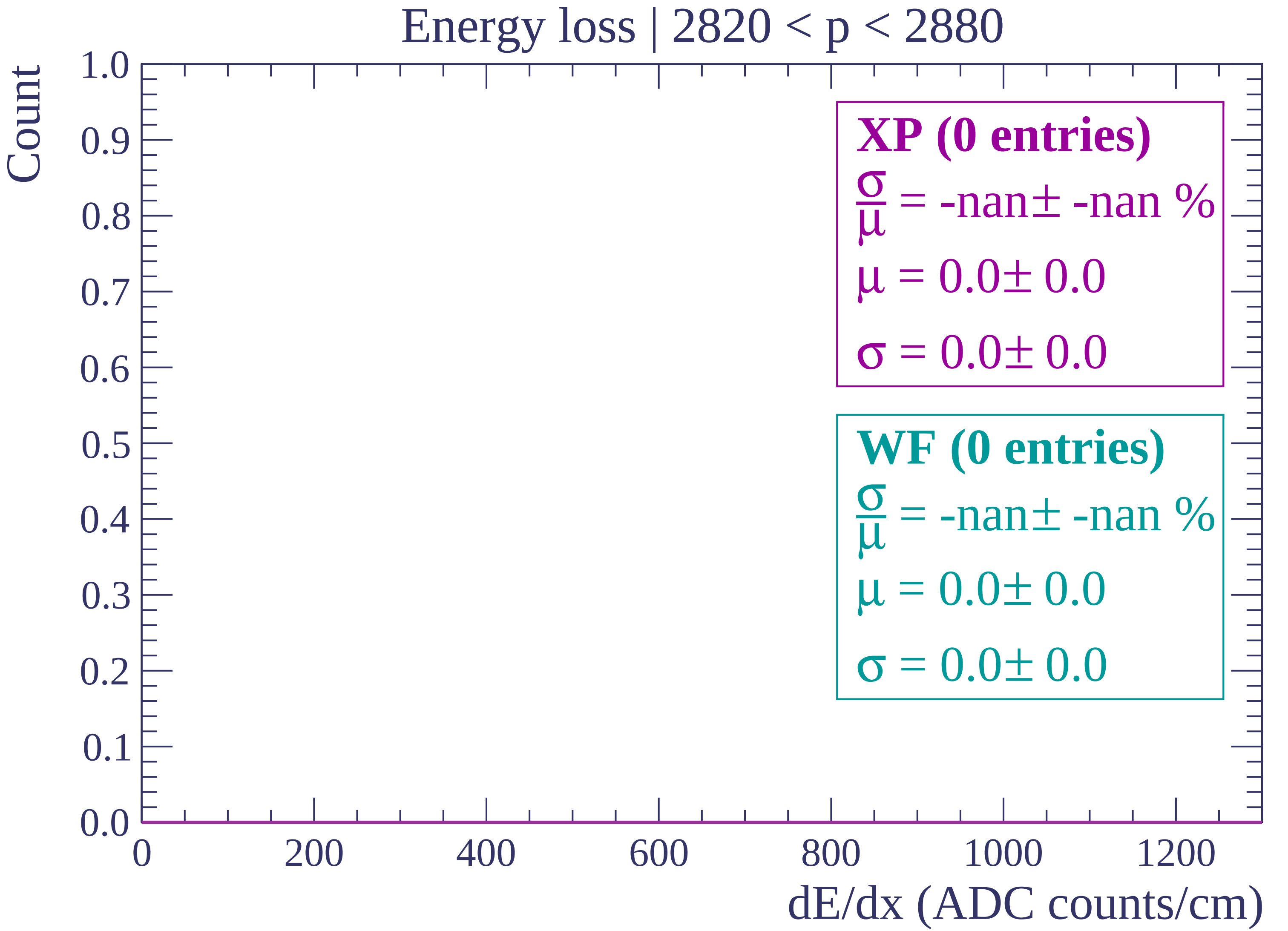
Energy loss | 2640 < p < 2700
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

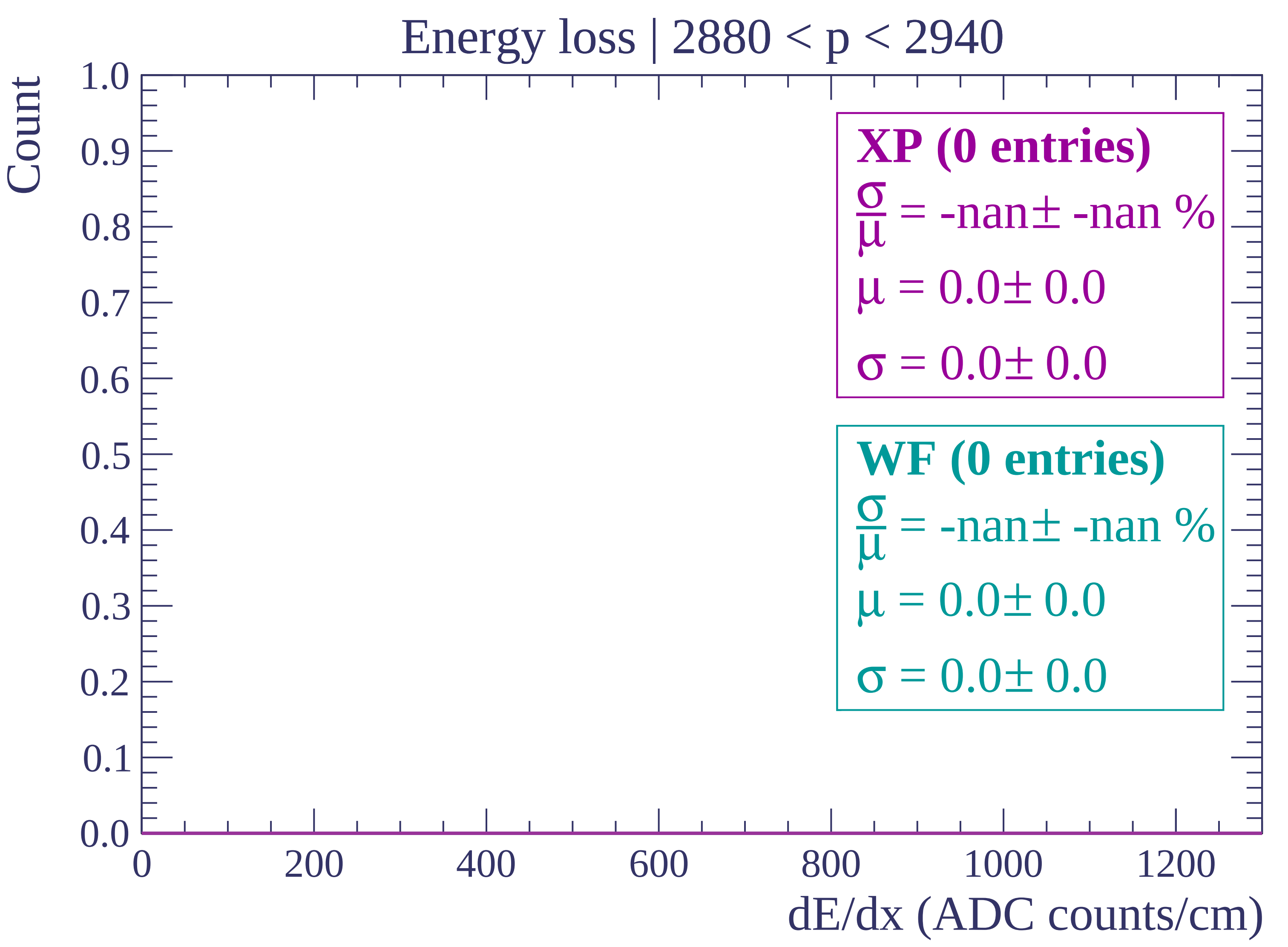


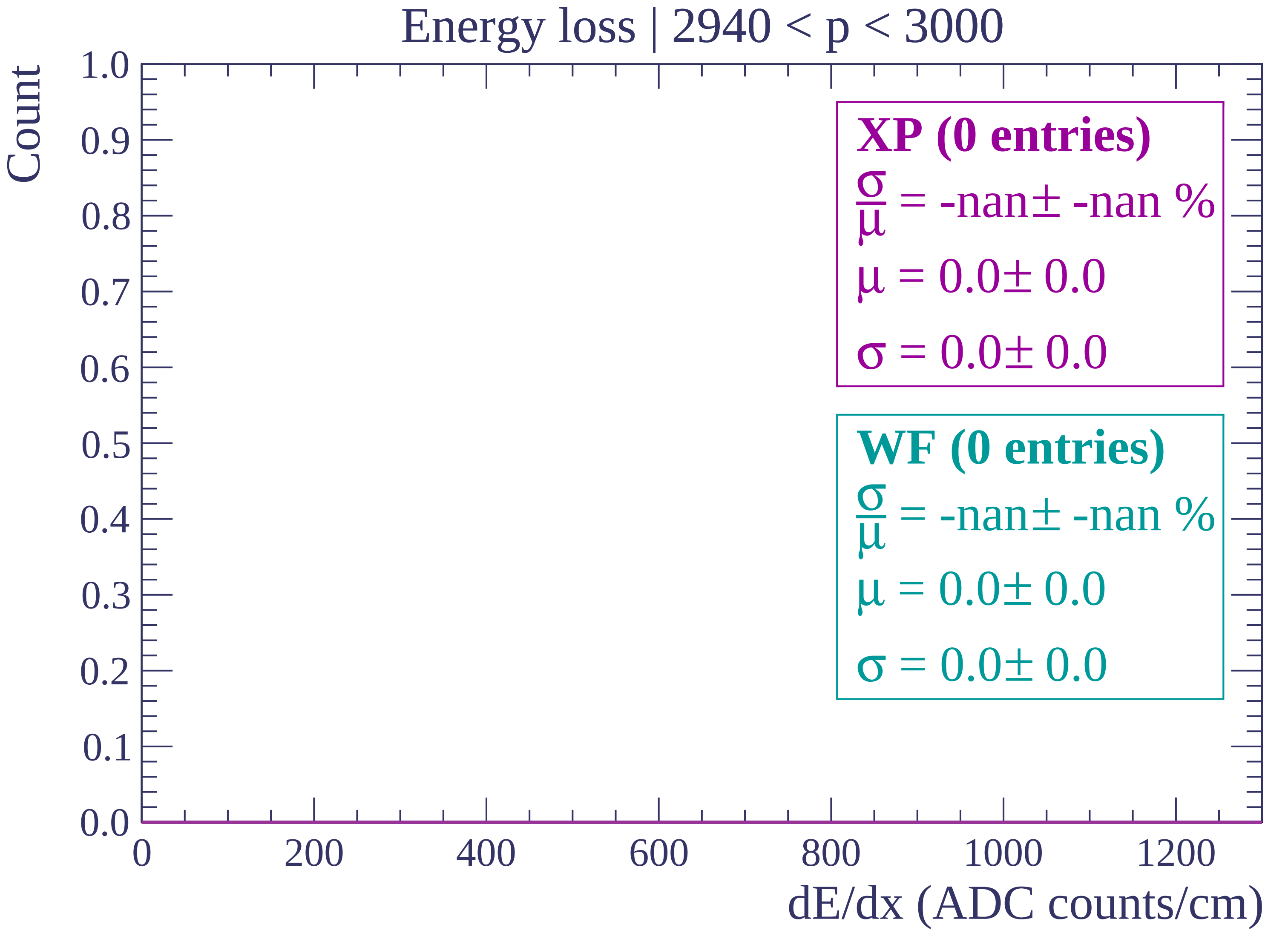
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

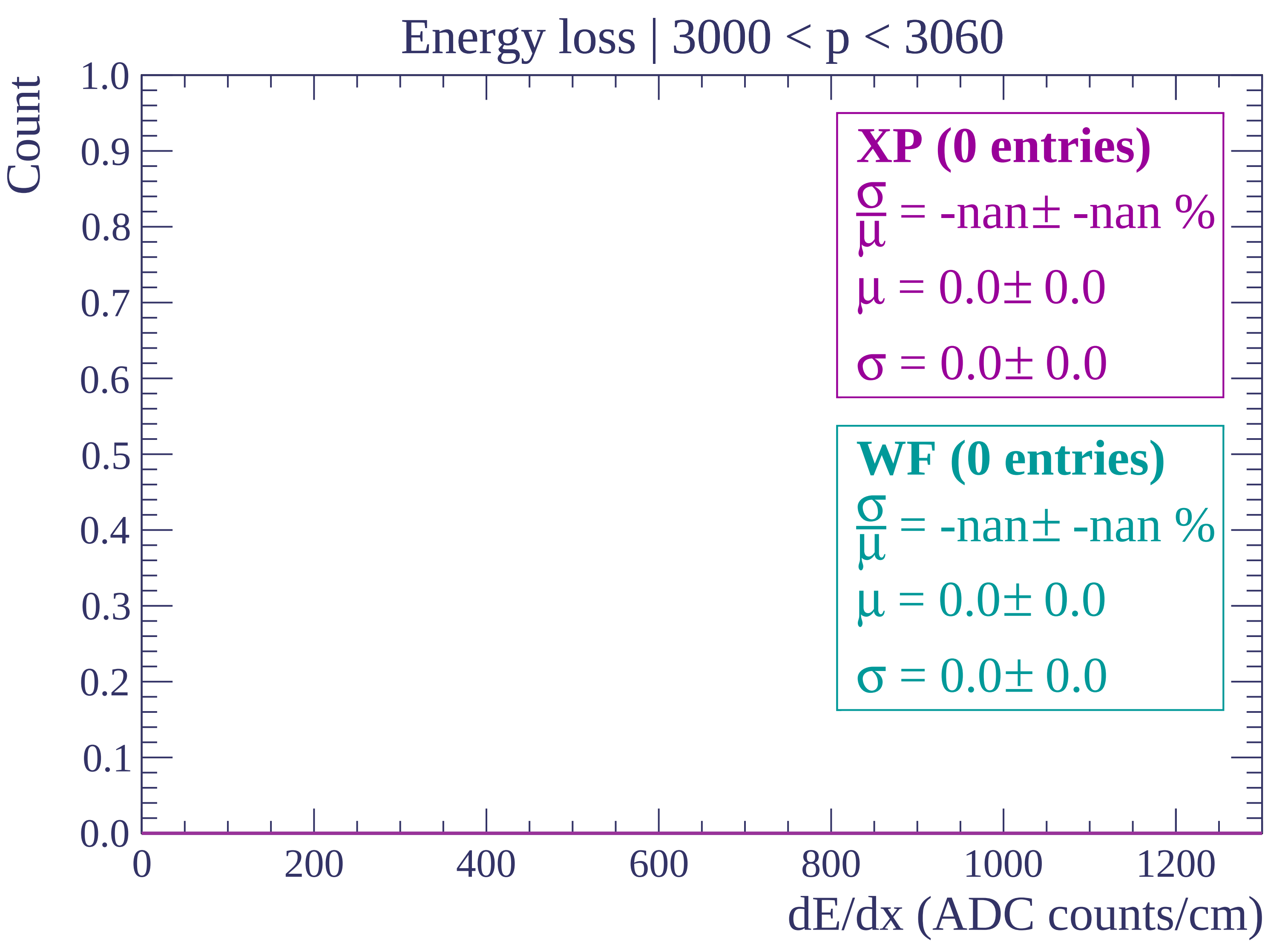


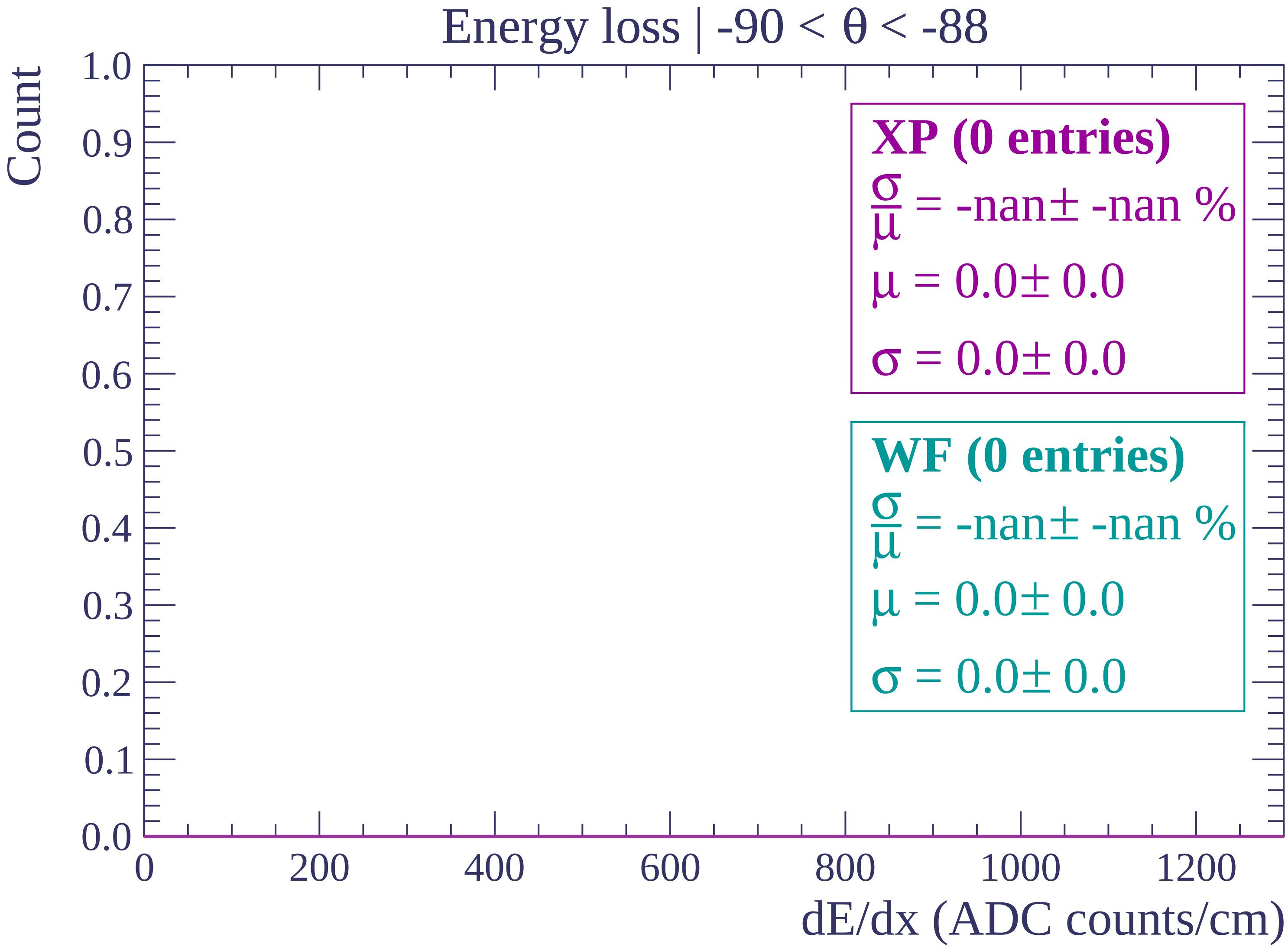
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

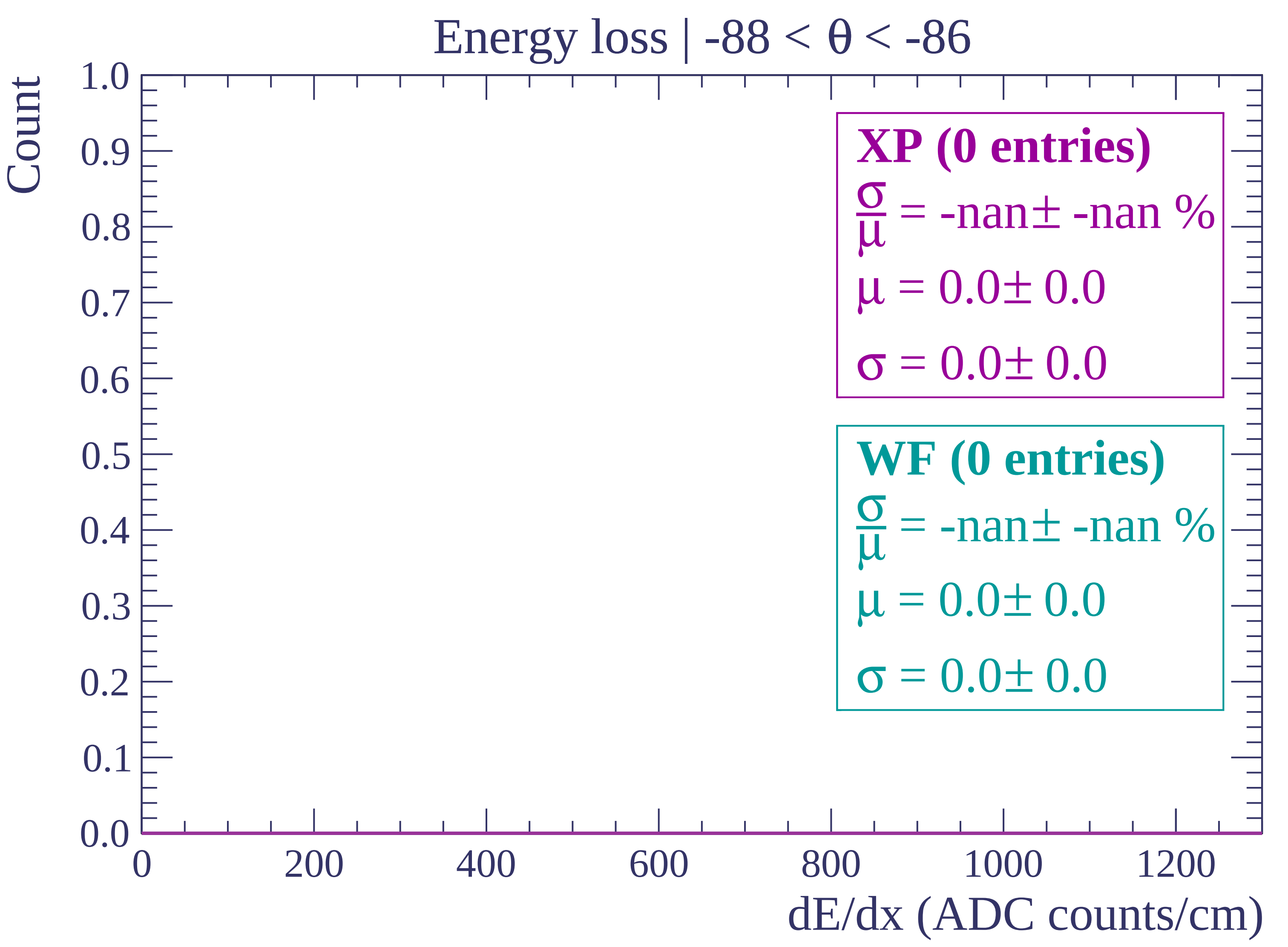

Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 2940 < p < 3000
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

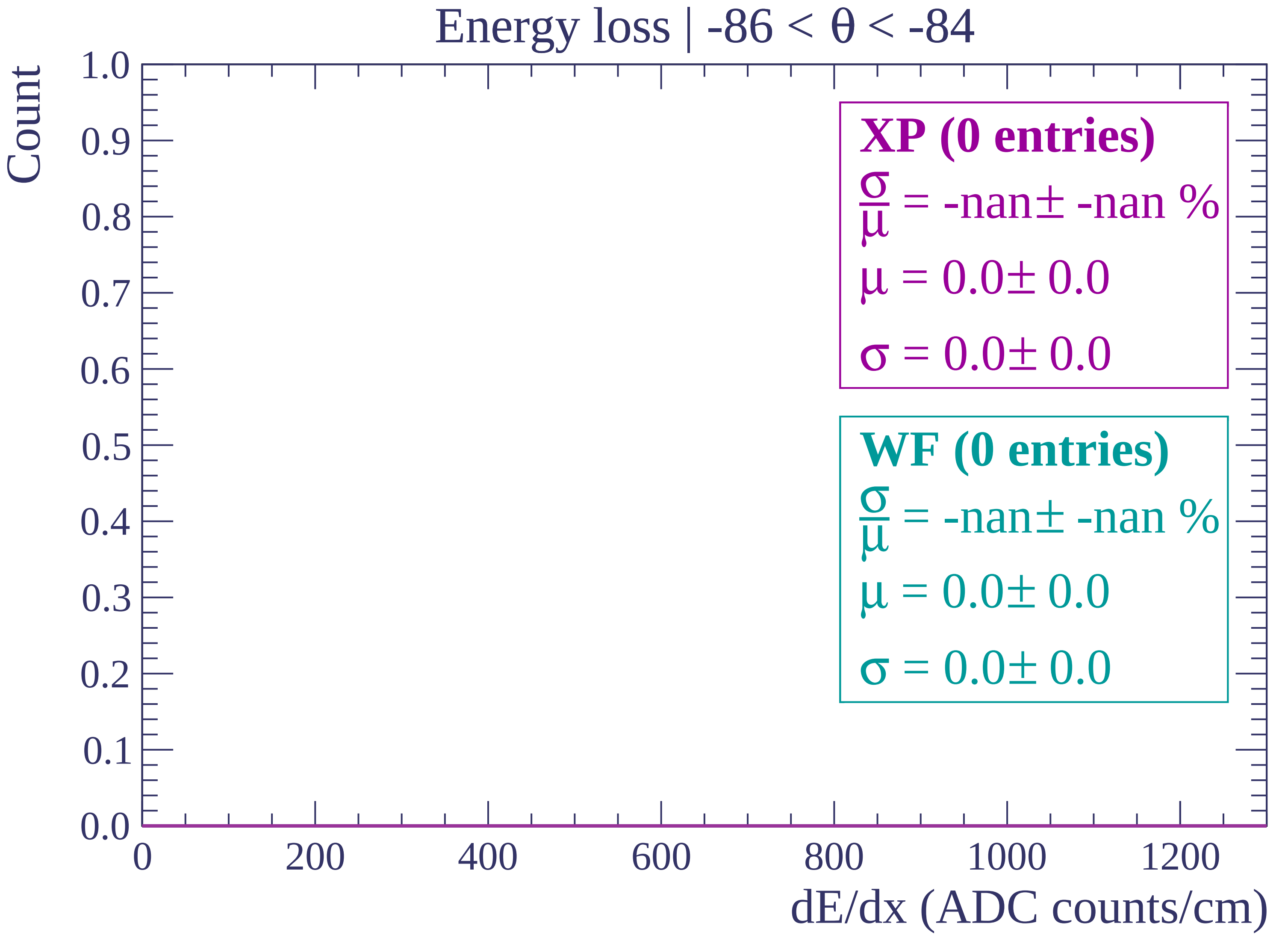

Energy loss | 3000 < p < 3060
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

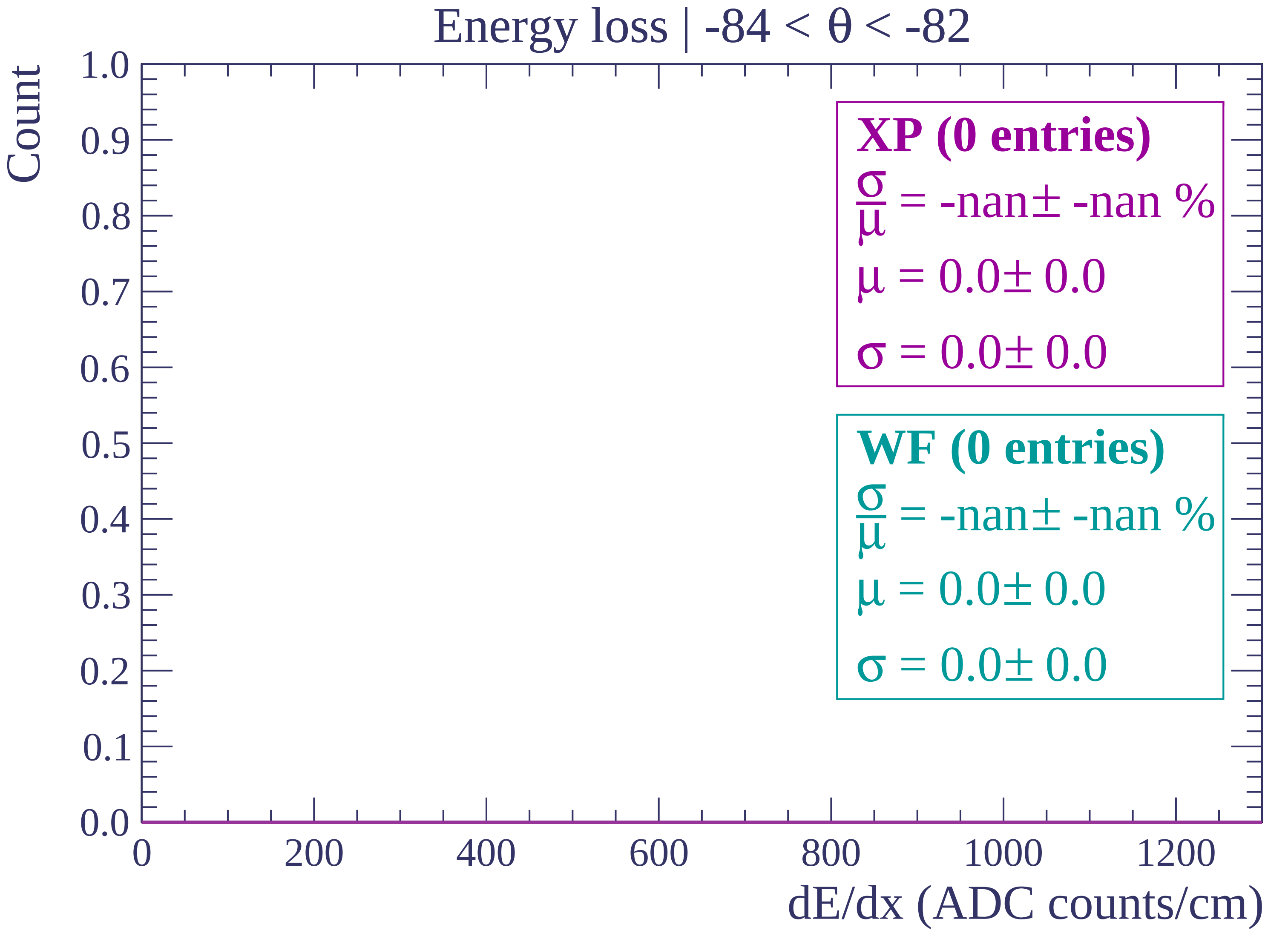


Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

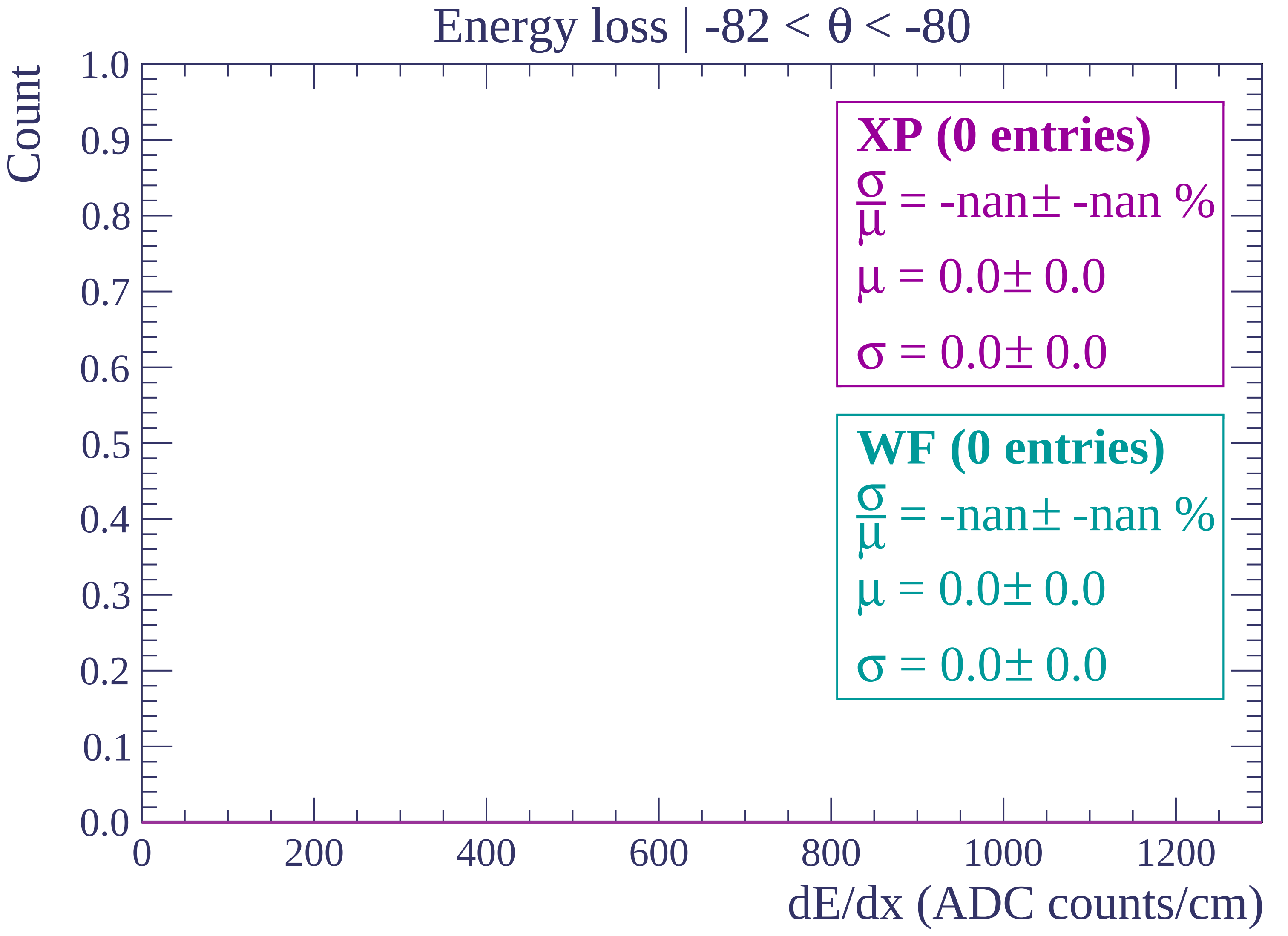


Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

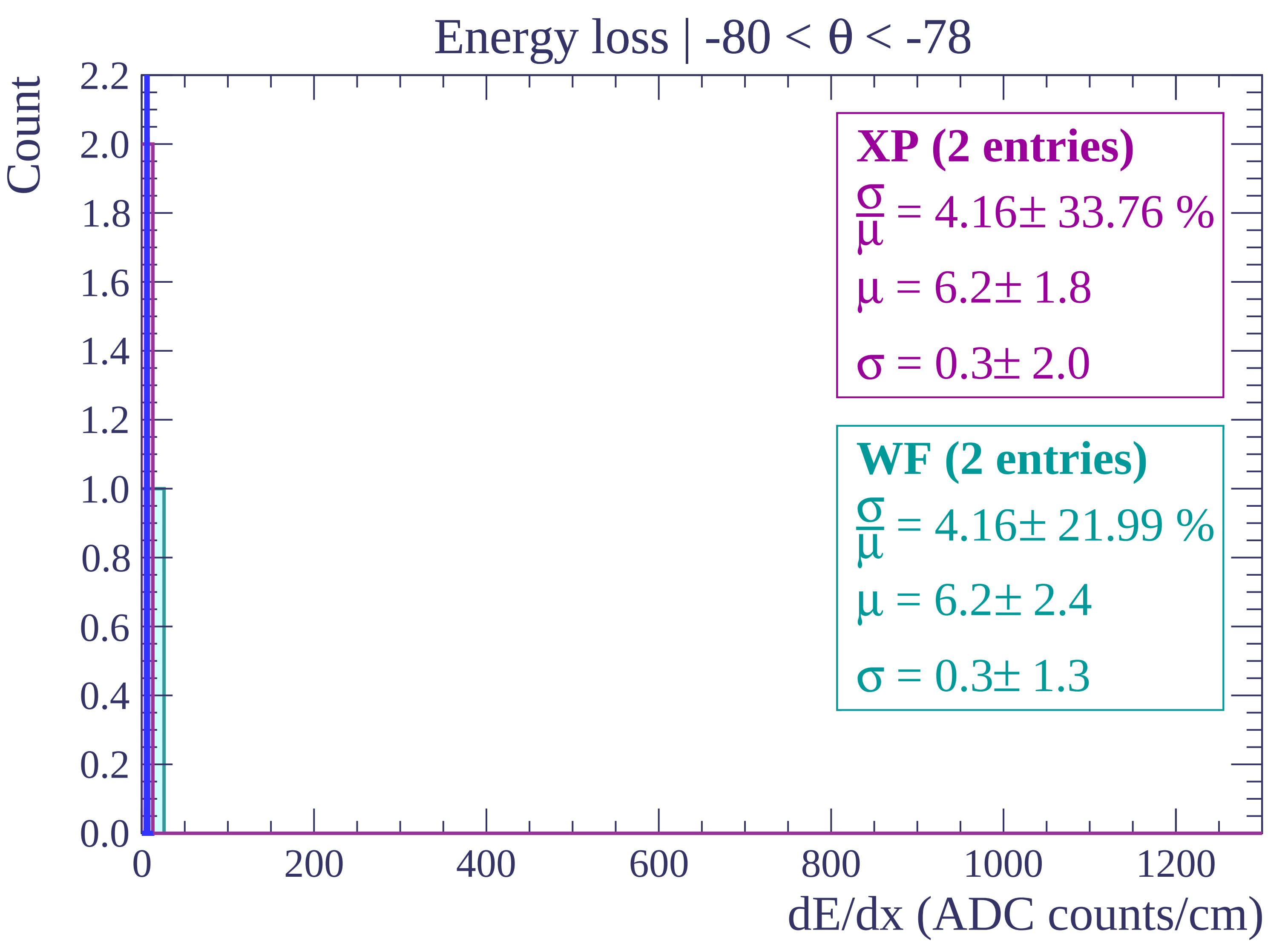

Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



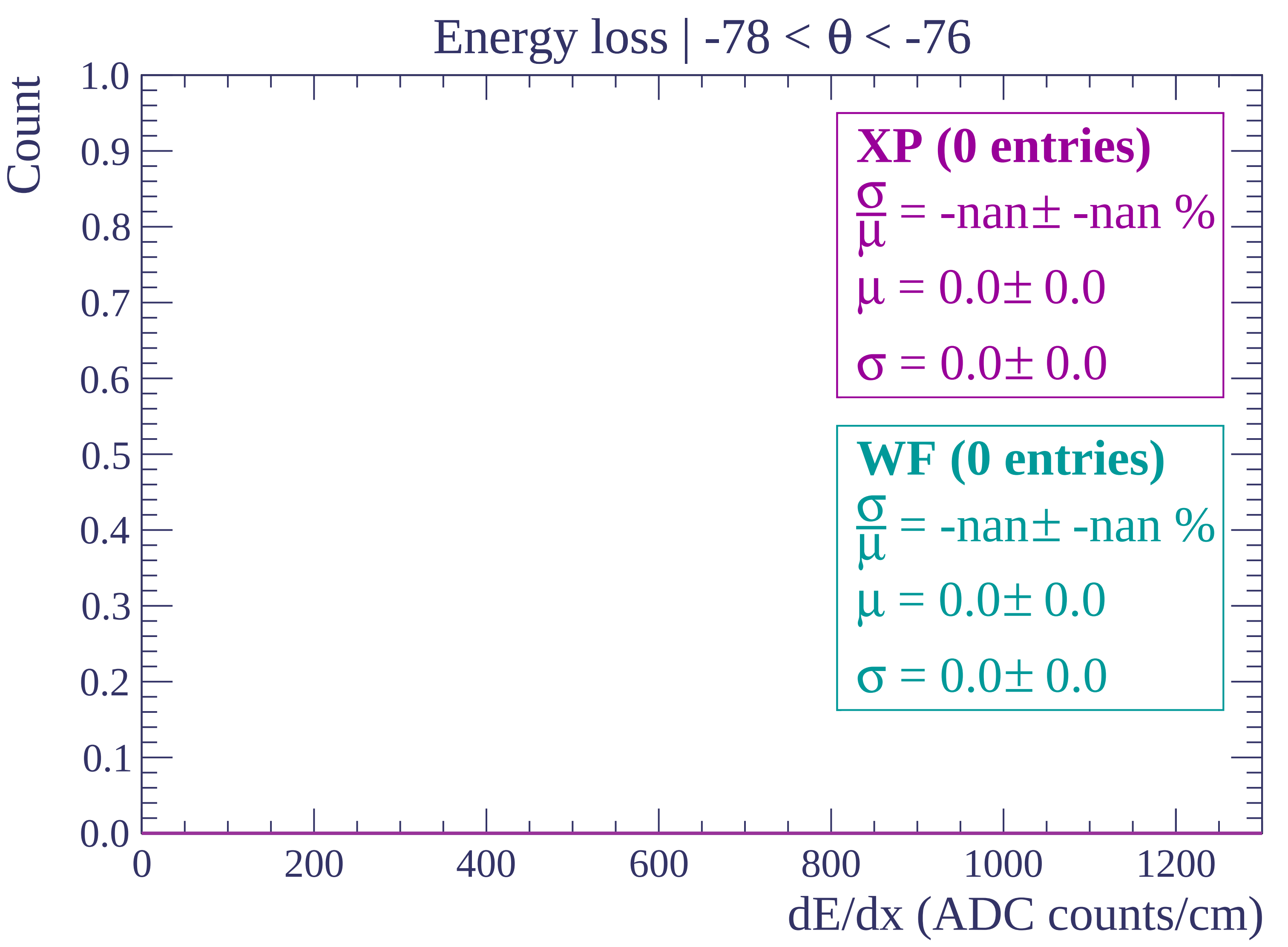
Energy loss | -84 < θ < -82
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



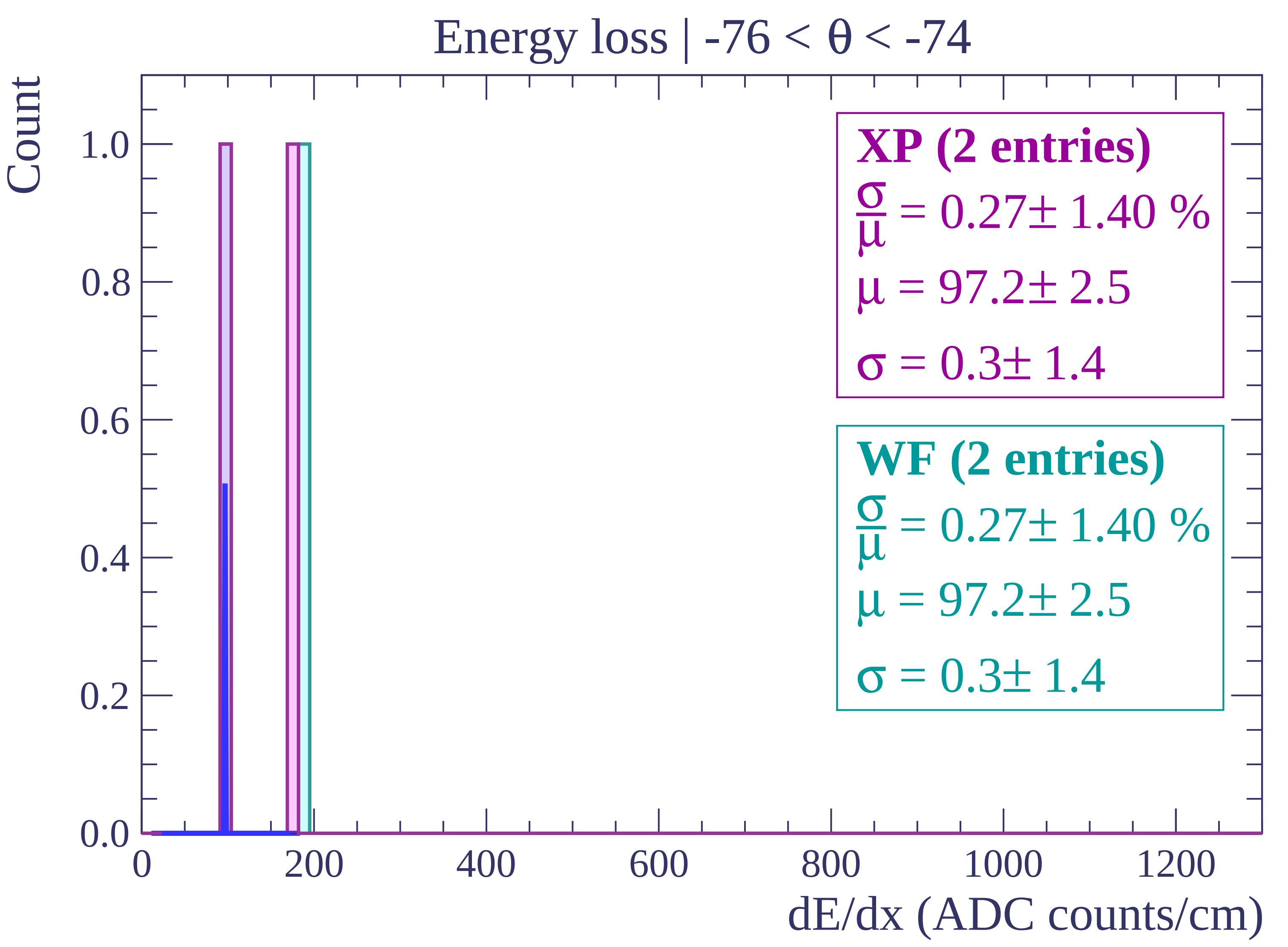
Energy loss | -82 < θ < -80
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



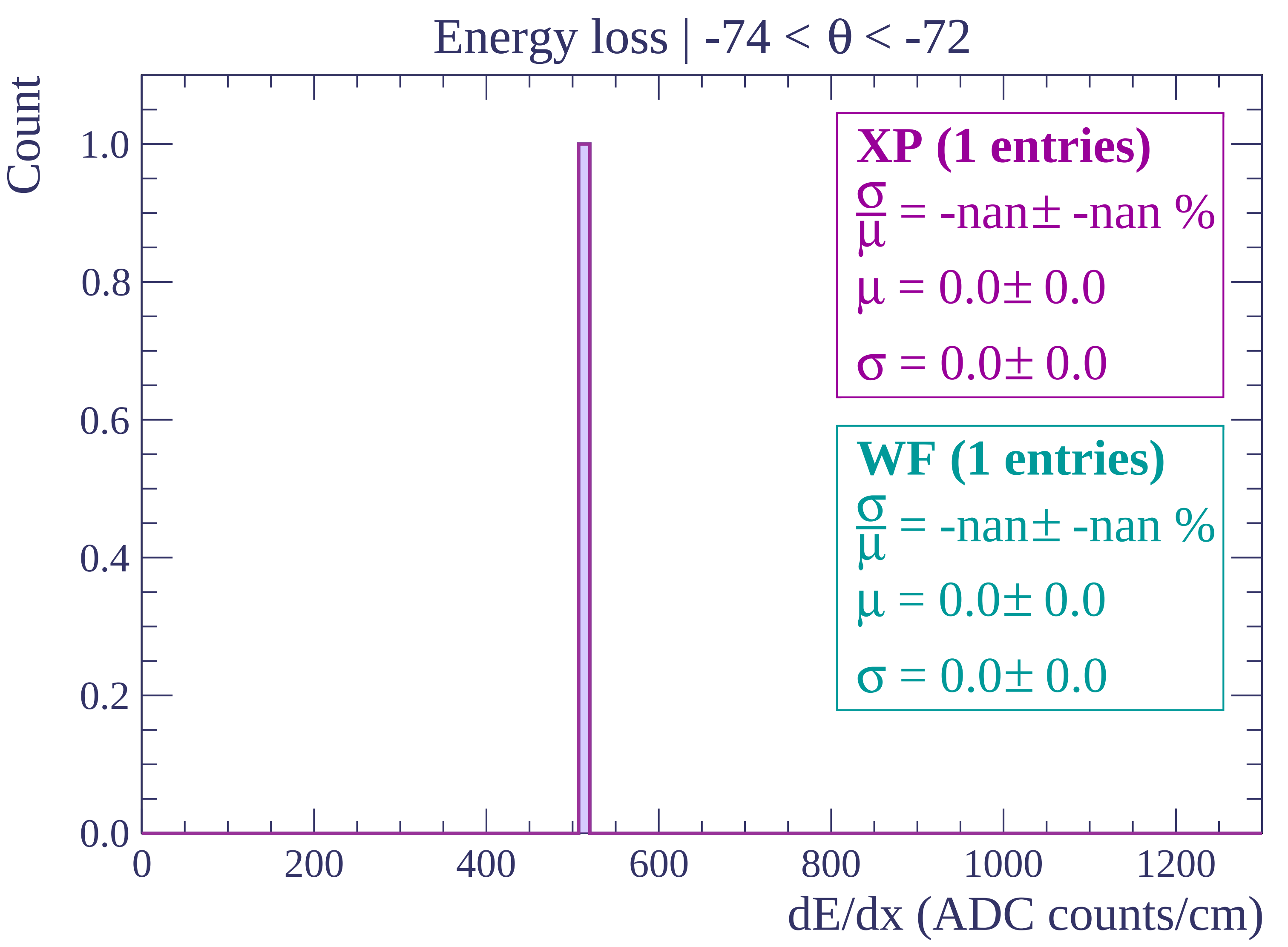
Energy loss | -80 < θ < -78
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2 entries)
σ/μ = 4.16± 33.76 %
μ = 6.2± 1.8
σ = 0.3± 2.0
WF (2 entries)
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3



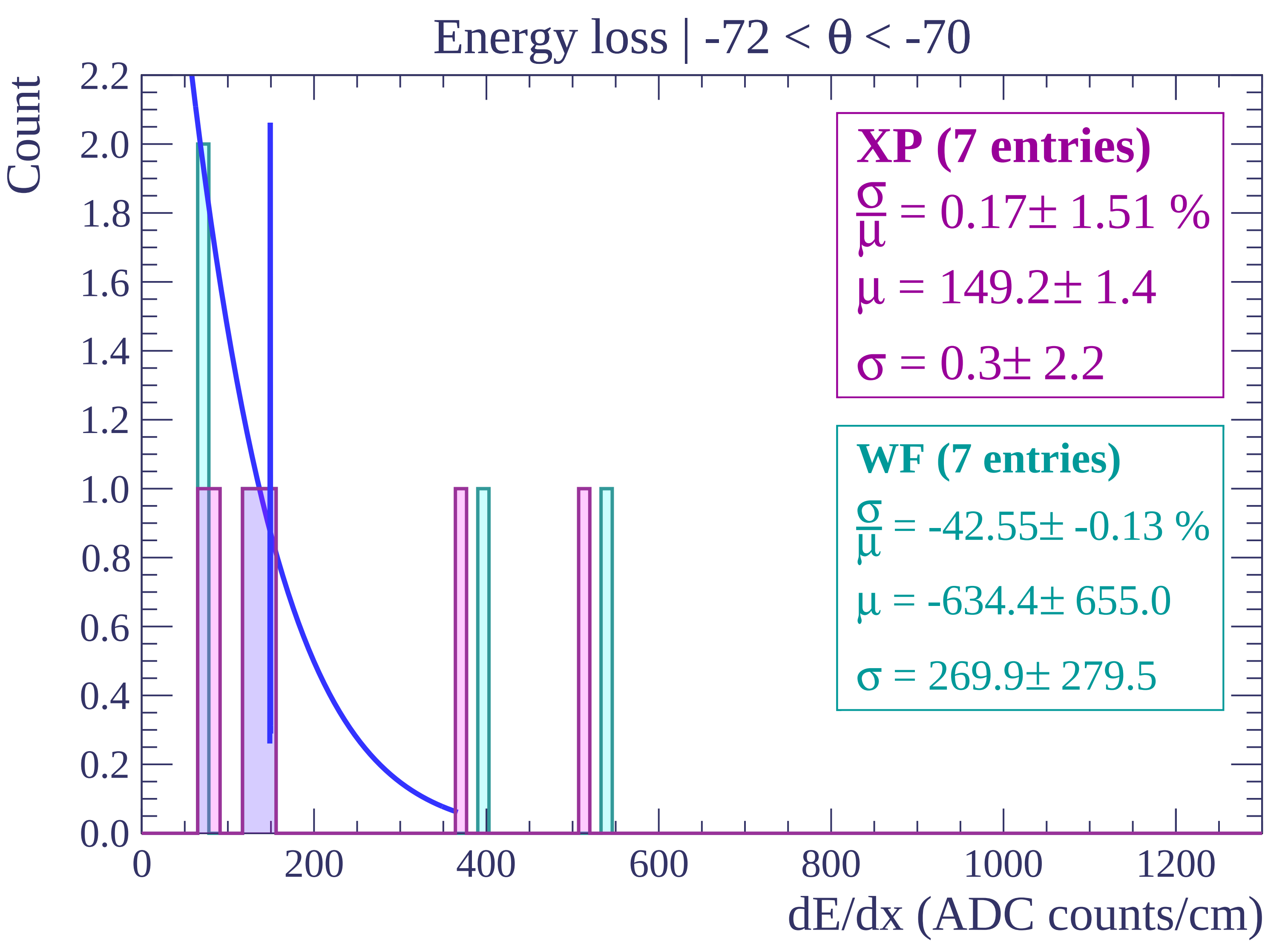
Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

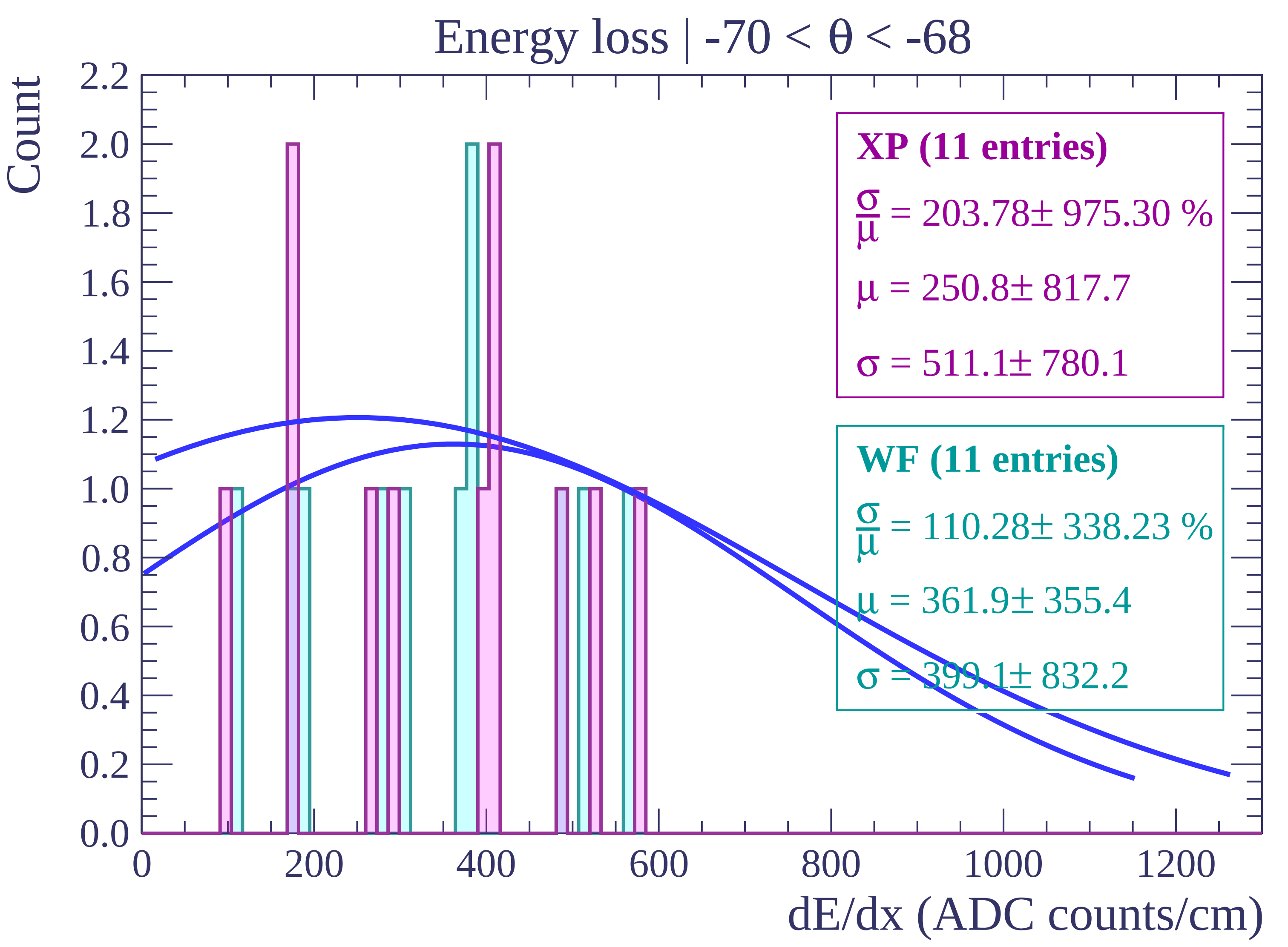


Energy loss | -76 < θ < -74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2 entries)
$\frac{\sigma}{\mu}$ = 0.27± 1.40 %
μ = 97.2± 2.5
σ = 0.3± 1.4
WF (2 entries)
$\frac{\sigma}{\mu}$ = 0.27± 1.40 %
μ = 97.2± 2.5
σ = 0.3± 1.4

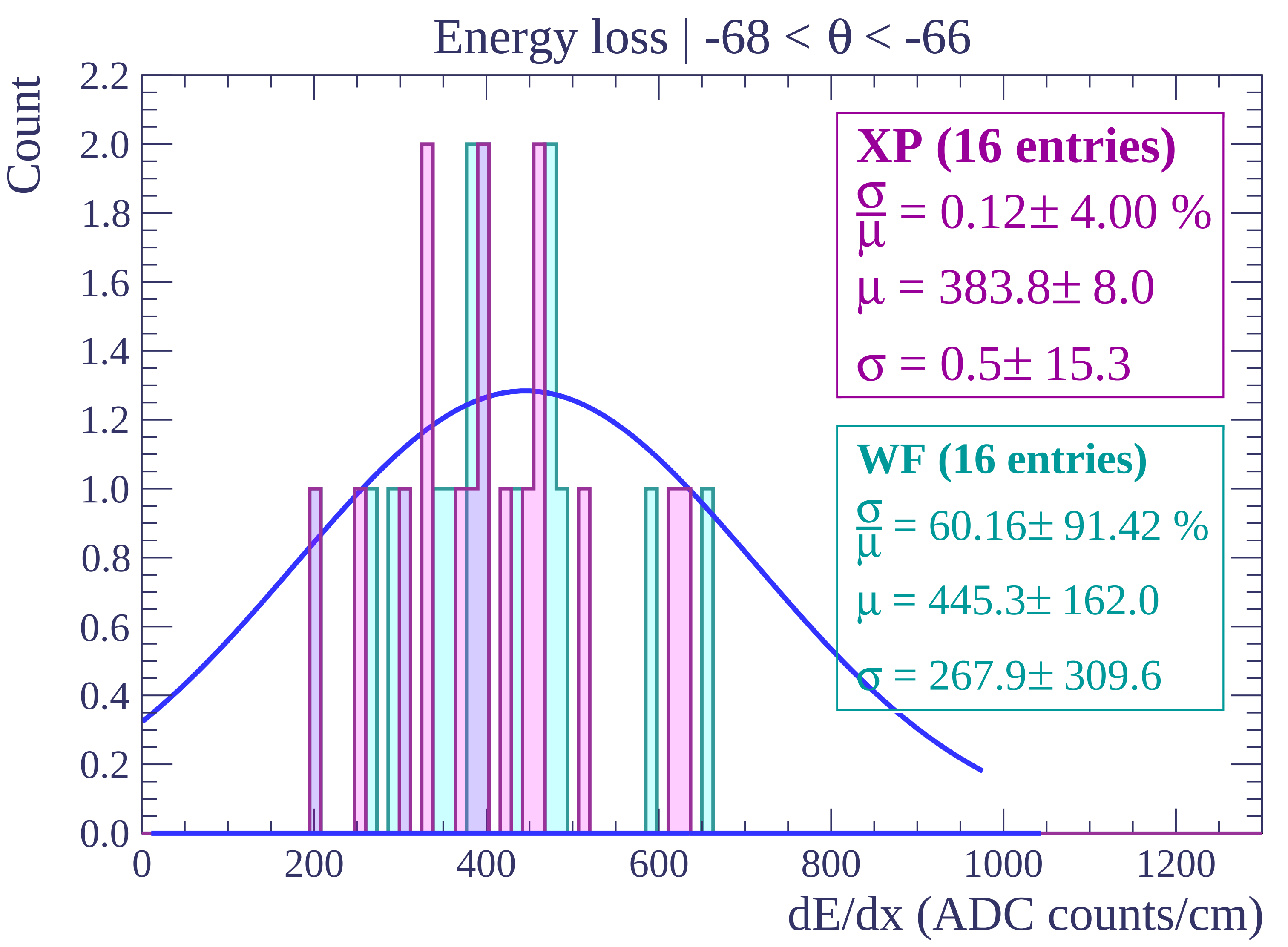


Energy loss | -74 < θ < -72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

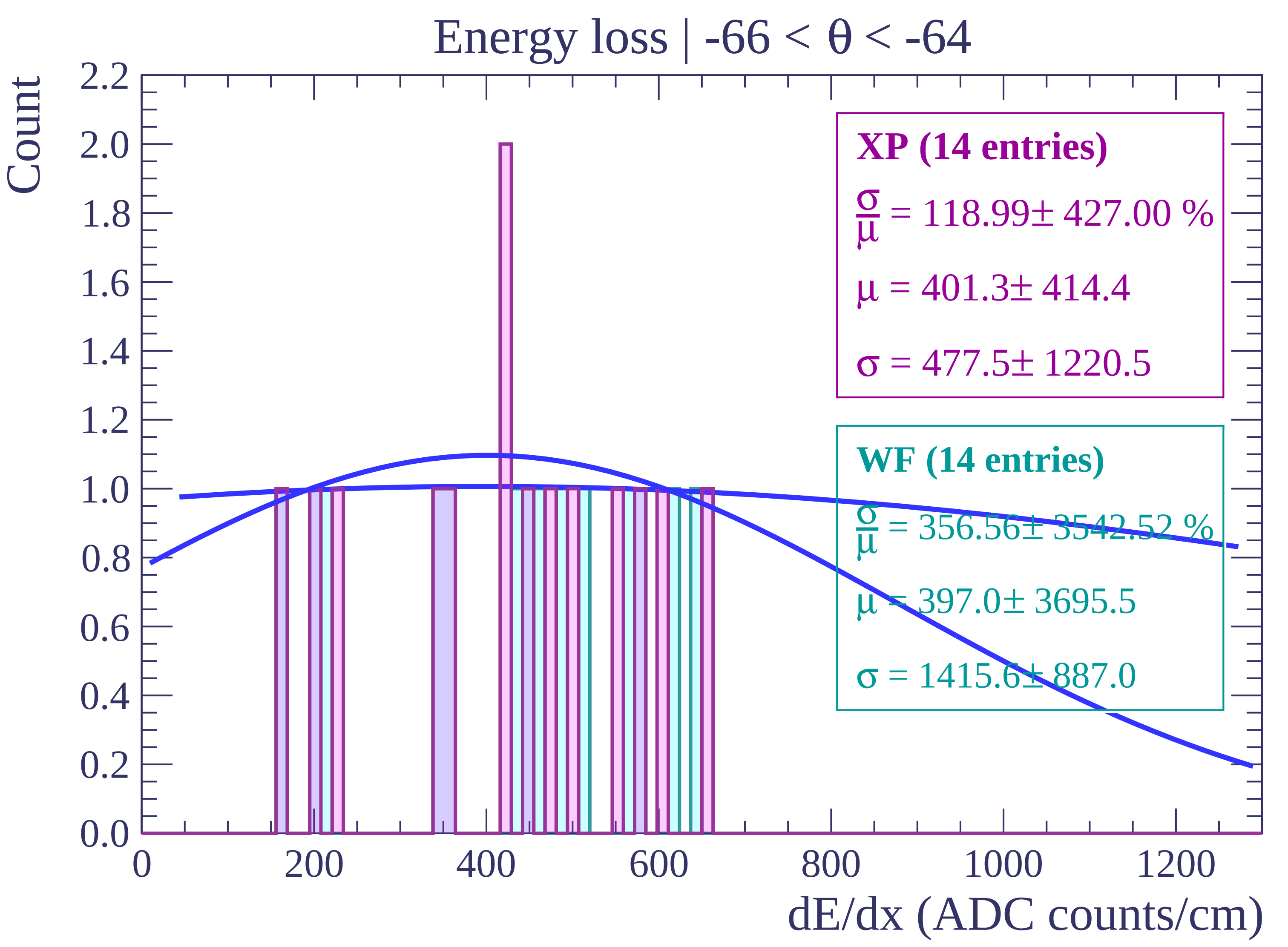

Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
XP (7 entries)
σ/μ = 0.17± 1.51 %
μ = 149.2± 1.4
σ = 0.3± 2.2
WF (7 entries)
σ/μ = -42.55± -0.13 %
μ = -634.4± 655.0
σ = 269.9± 279.5



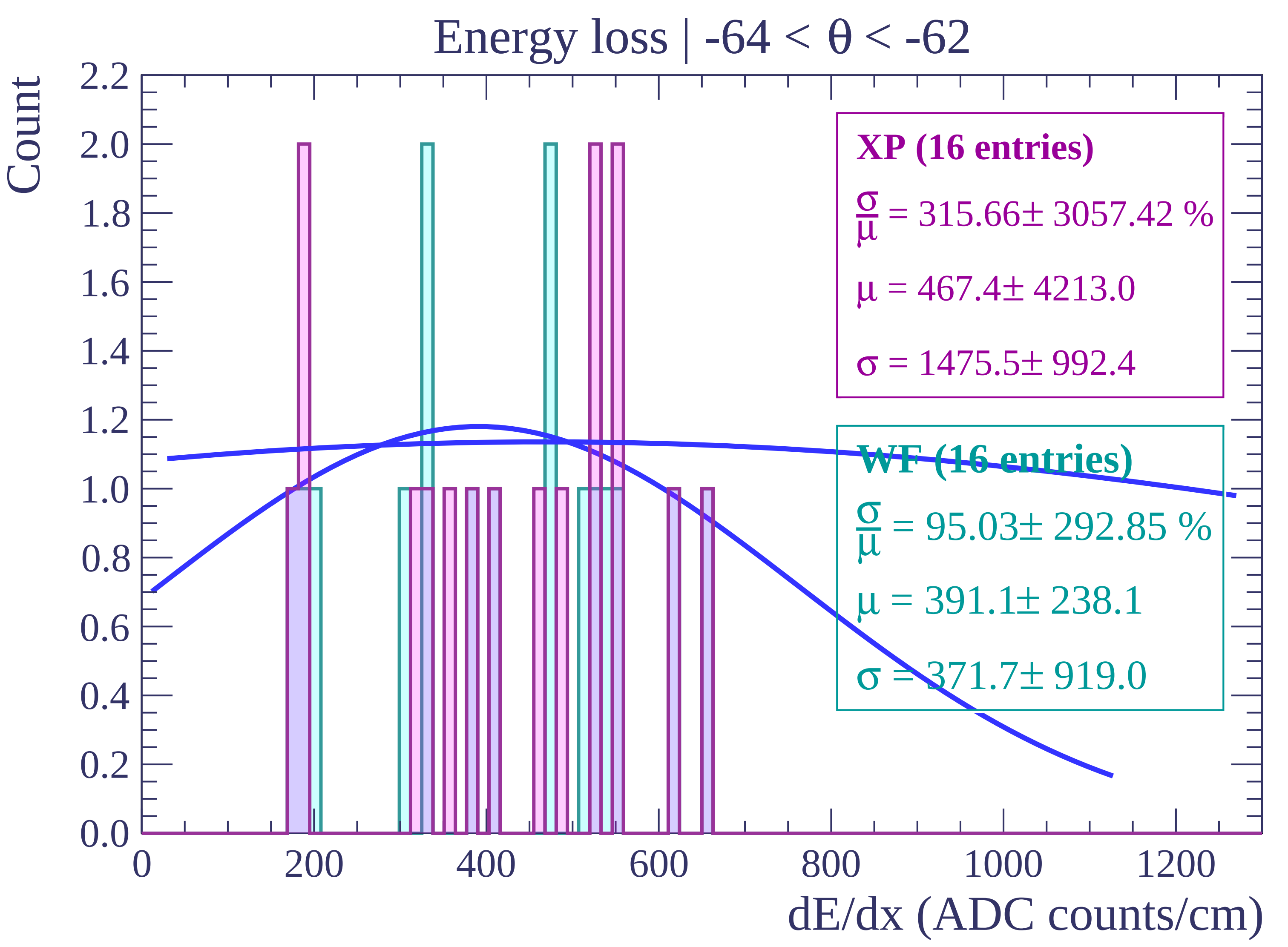
Energy loss | -70 < θ < -68
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (11 entries)
σ/μ = 203.78± 975.30 %
μ = 250.8± 817.7
σ = 511.1± 780.1
WF (11 entries)
σ/μ = 110.28± 338.23 %
μ = 361.9± 355.4
σ = 399.1± 832.2



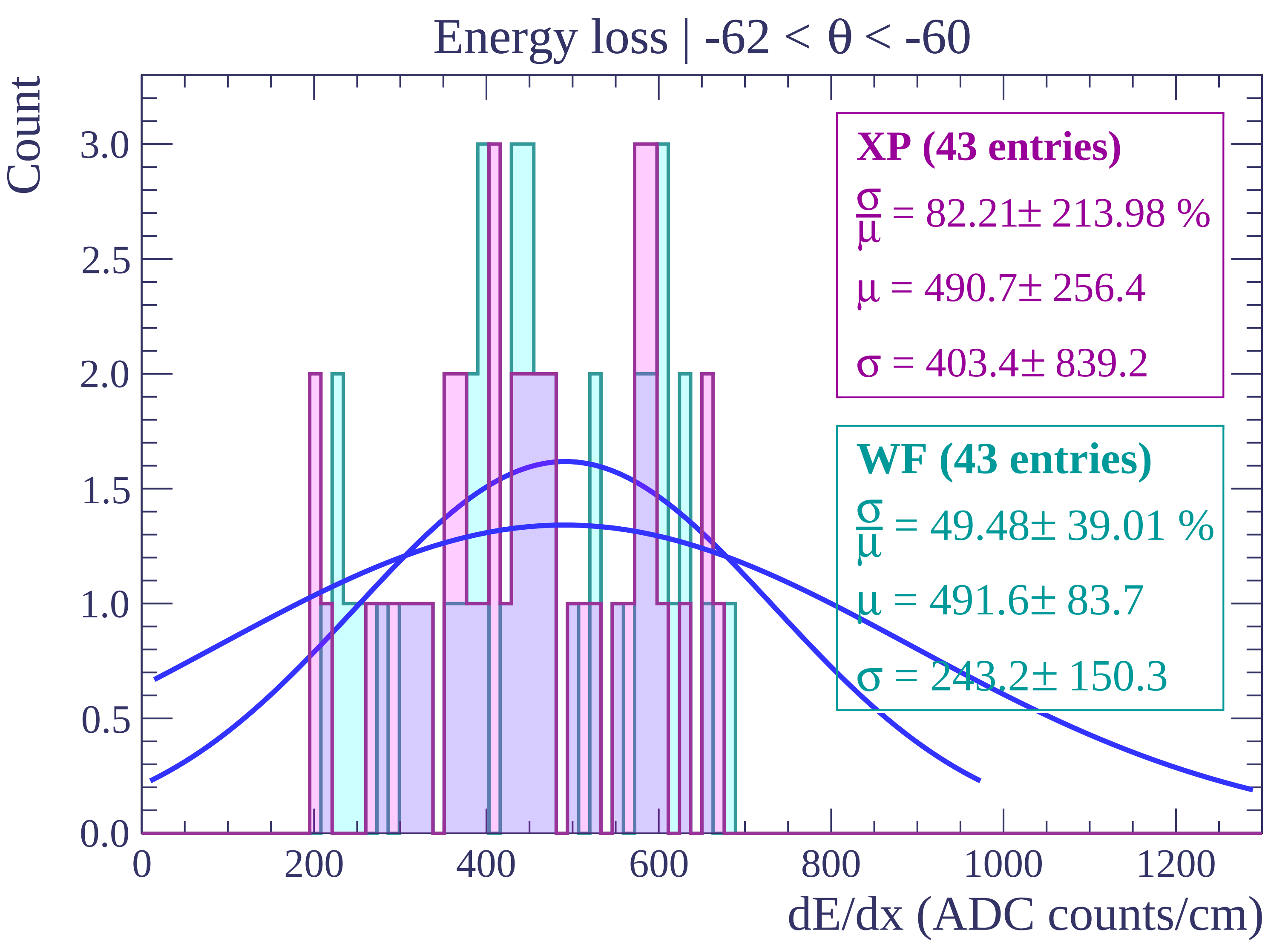
Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
XP (16 entries)
σ/μ = 0.12± 4.00 %
μ = 383.8± 8.0
σ = 0.5± 15.3
WF (16 entries)
σ/μ = 60.16± 91.42 %
μ = 445.3± 162.0
σ = 267.9± 309.6



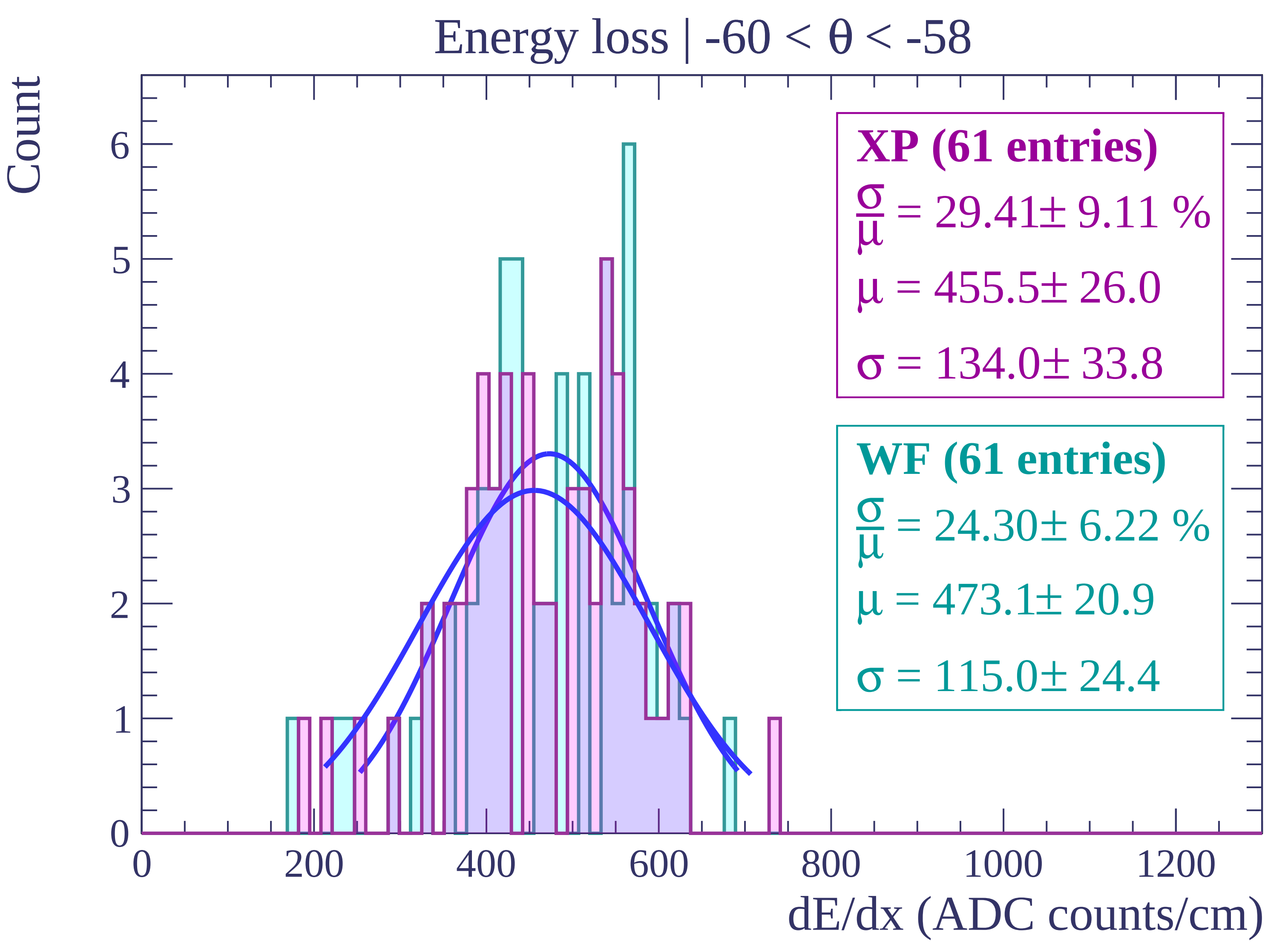
Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
XP (14 entries)
σ/μ = 118.99± 427.00 %
μ = 401.3± 414.4
σ = 477.5± 1220.5
WF (14 entries)
σ/μ = 356.56± 3542.52 %
μ = 397.0± 3695.5
σ = 1415.6± 887.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



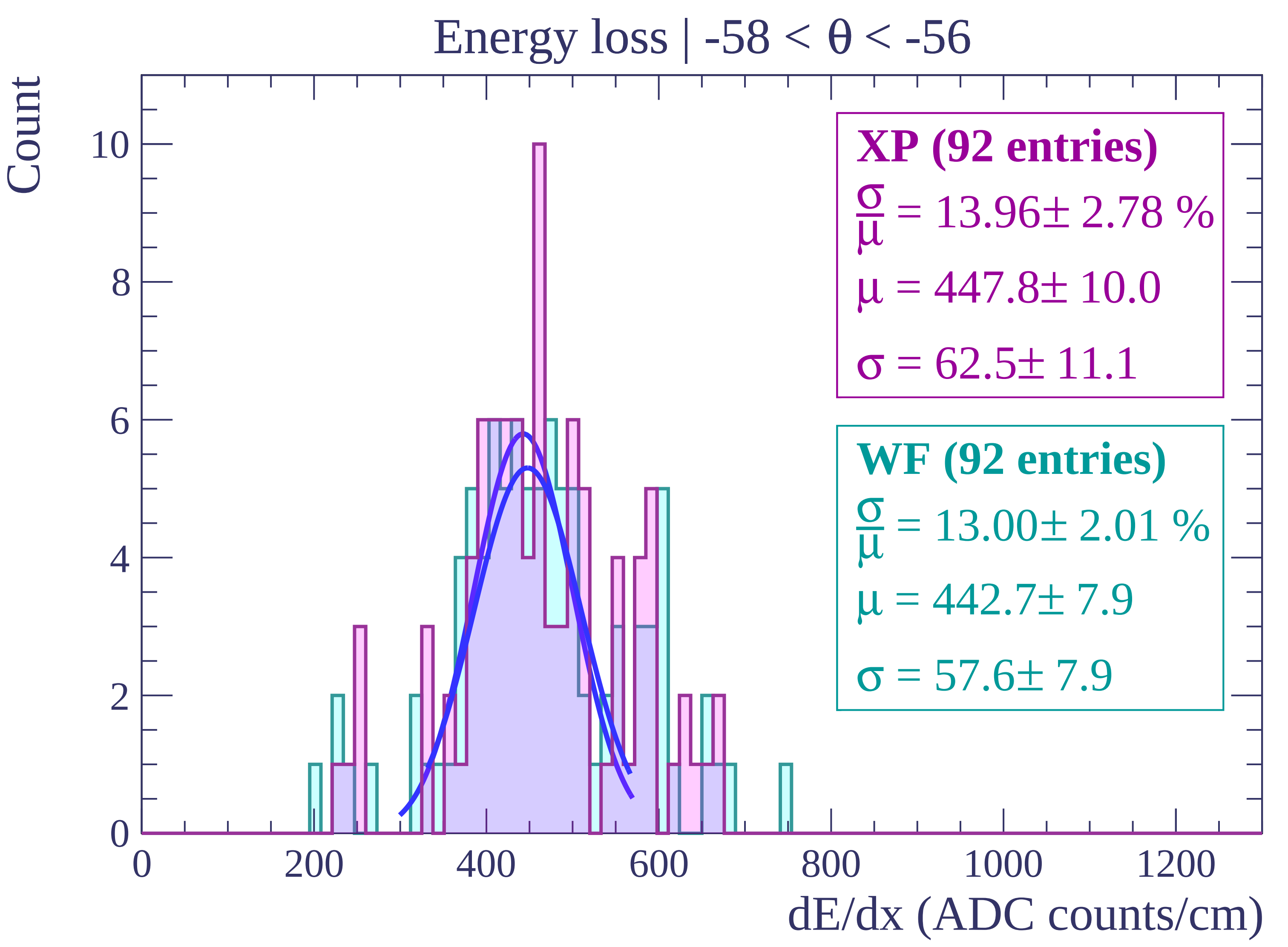
Energy loss | -64 < θ < -62
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (16 entries)
σ/μ = 315.66± 3057.42 %
μ = 467.4± 4213.0
σ = 1475.5± 992.4
WF (16 entries)
σ/μ = 95.03± 292.85 %
μ = 391.1± 238.1
σ = 371.7± 919.0



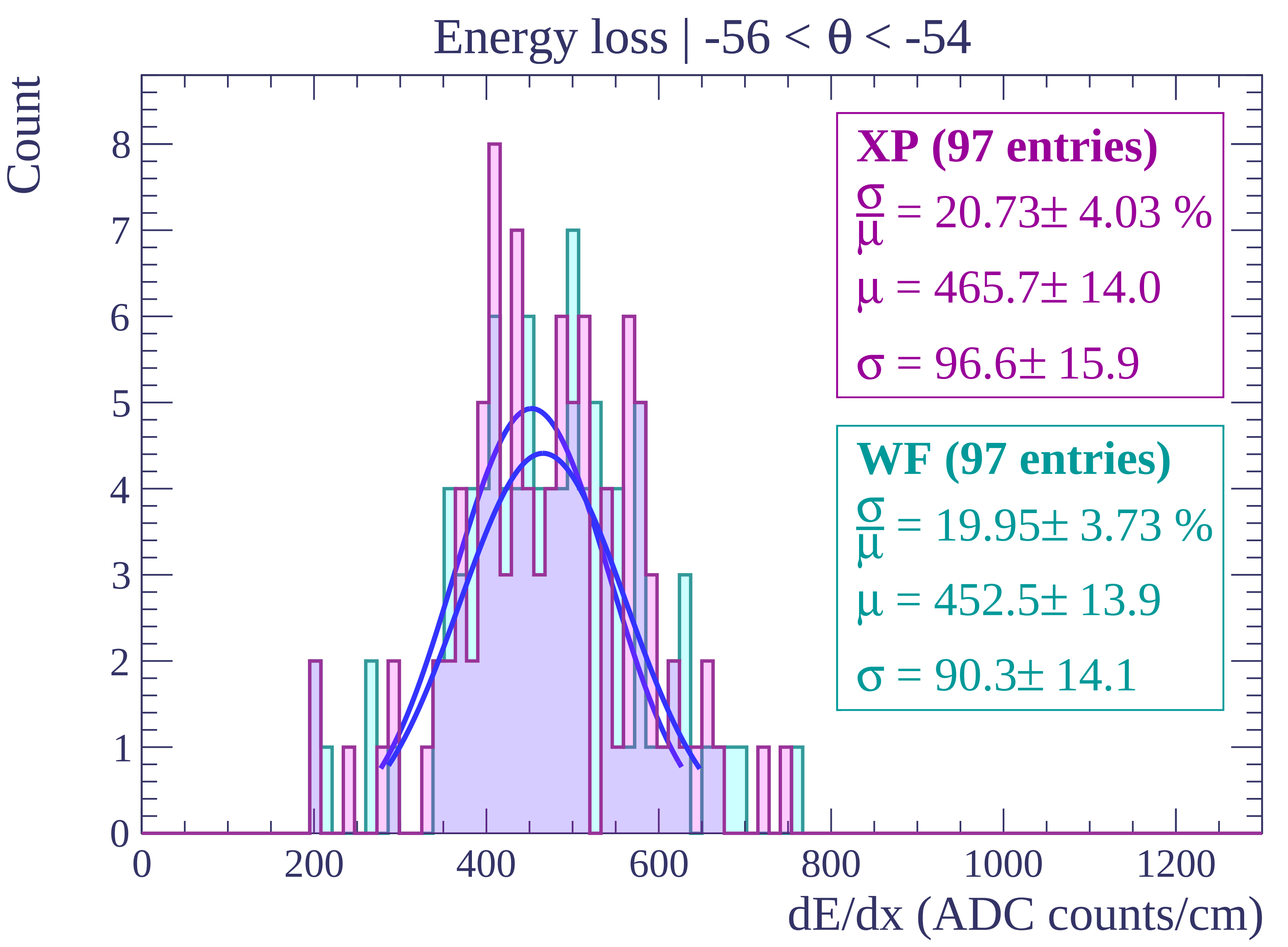
Energy loss | -62 < θ < -60
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (43 entries)
σ/μ = 82.21± 213.98 %
μ = 490.7± 256.4
σ = 403.4± 839.2
WF (43 entries)
σ/μ = 49.48± 39.01 %
μ = 491.6± 83.7
σ = 243.2± 150.3



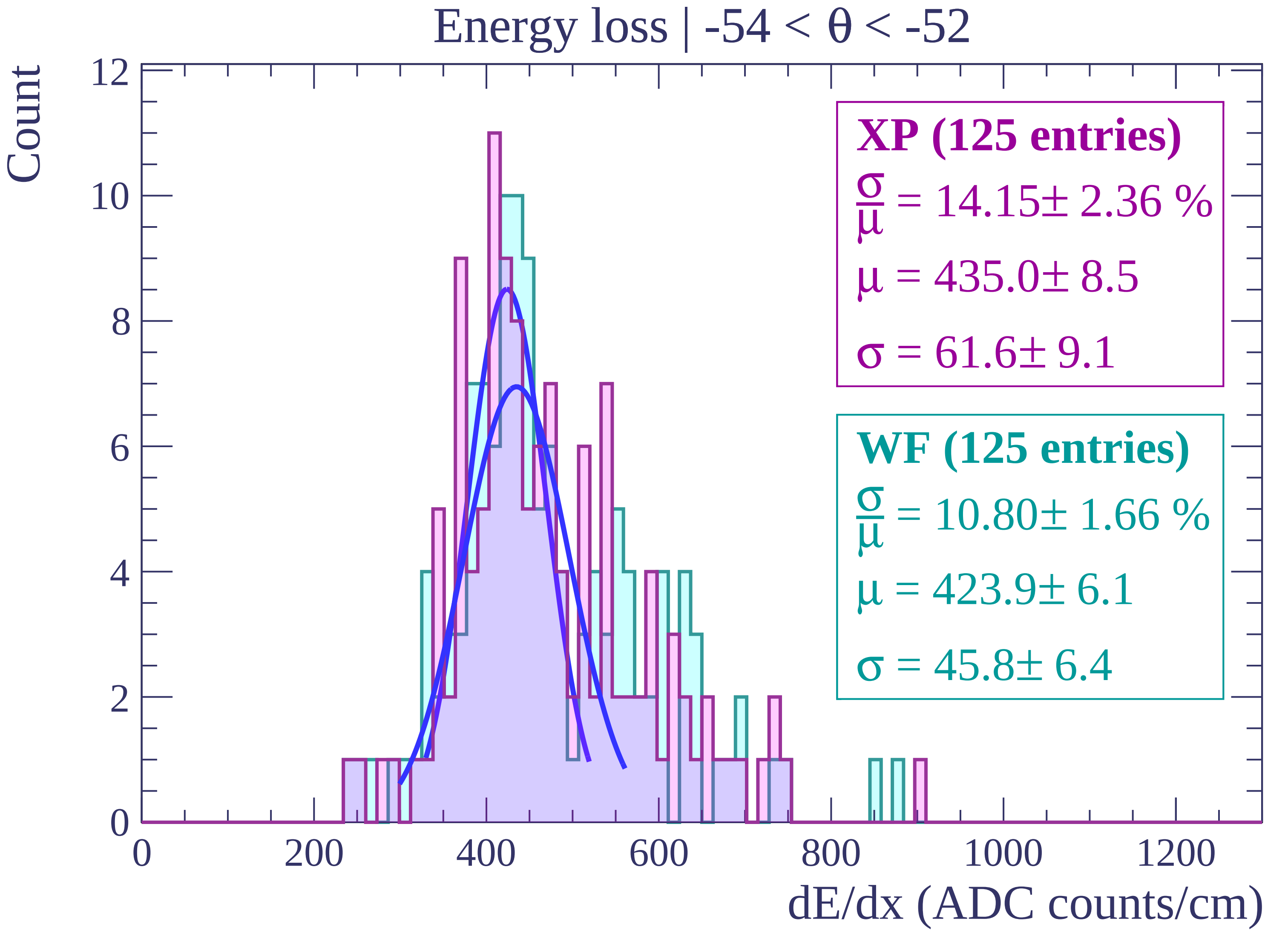
Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP (61 entries)
σ/μ = 29.41± 9.11 %
μ = 455.5± 26.0
σ = 134.0± 33.8
WF (61 entries)
σ/μ = 24.30± 6.22 %
μ = 473.1± 20.9
σ = 115.0± 24.4



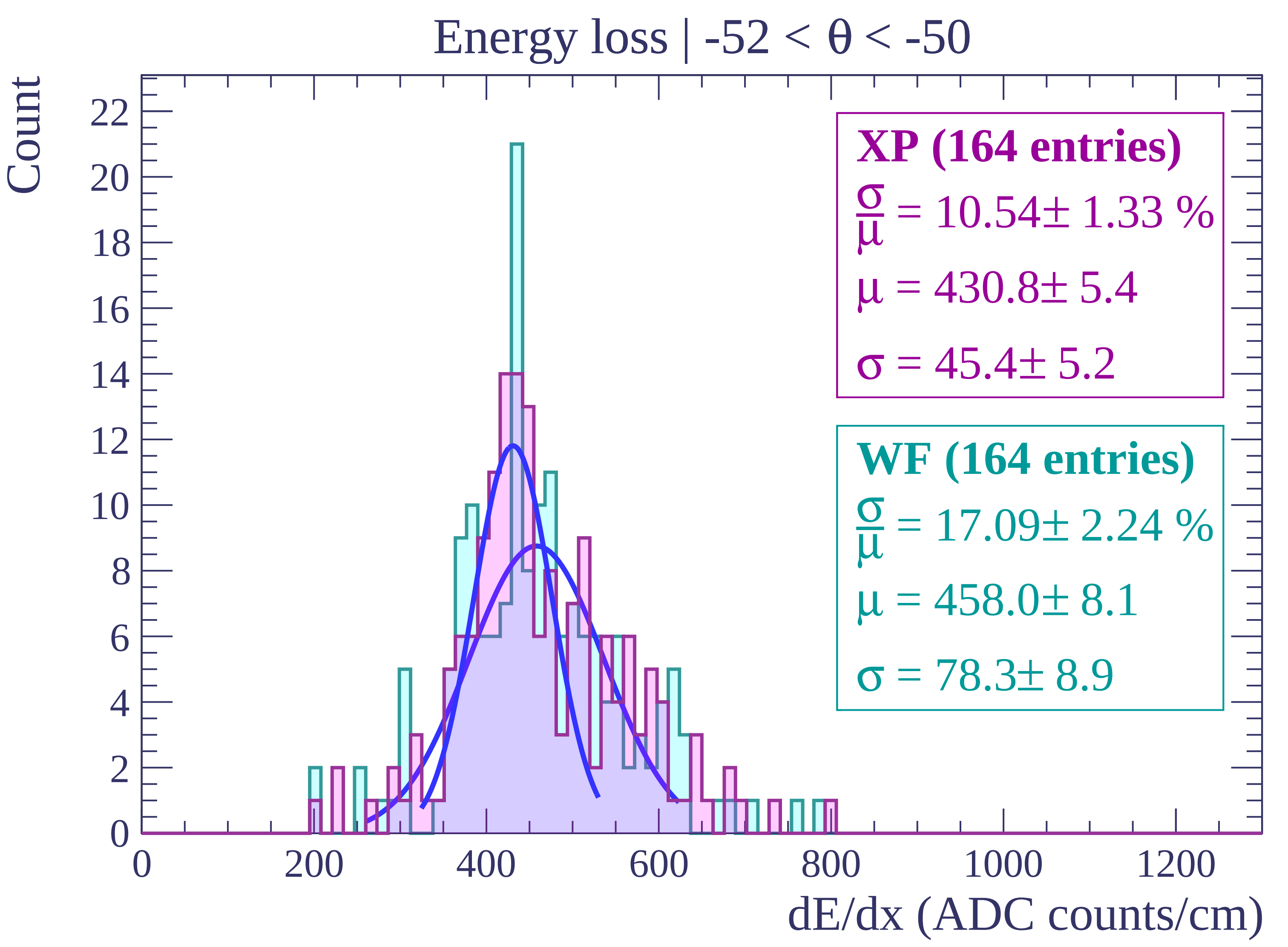
Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP (92 entries)
σ/μ = 13.96± 2.78 %
μ = 447.8± 10.0
σ = 62.5± 11.1
WF (92 entries)
σ/μ = 13.00± 2.01 %
μ = 442.7± 7.9
σ = 57.6± 7.9



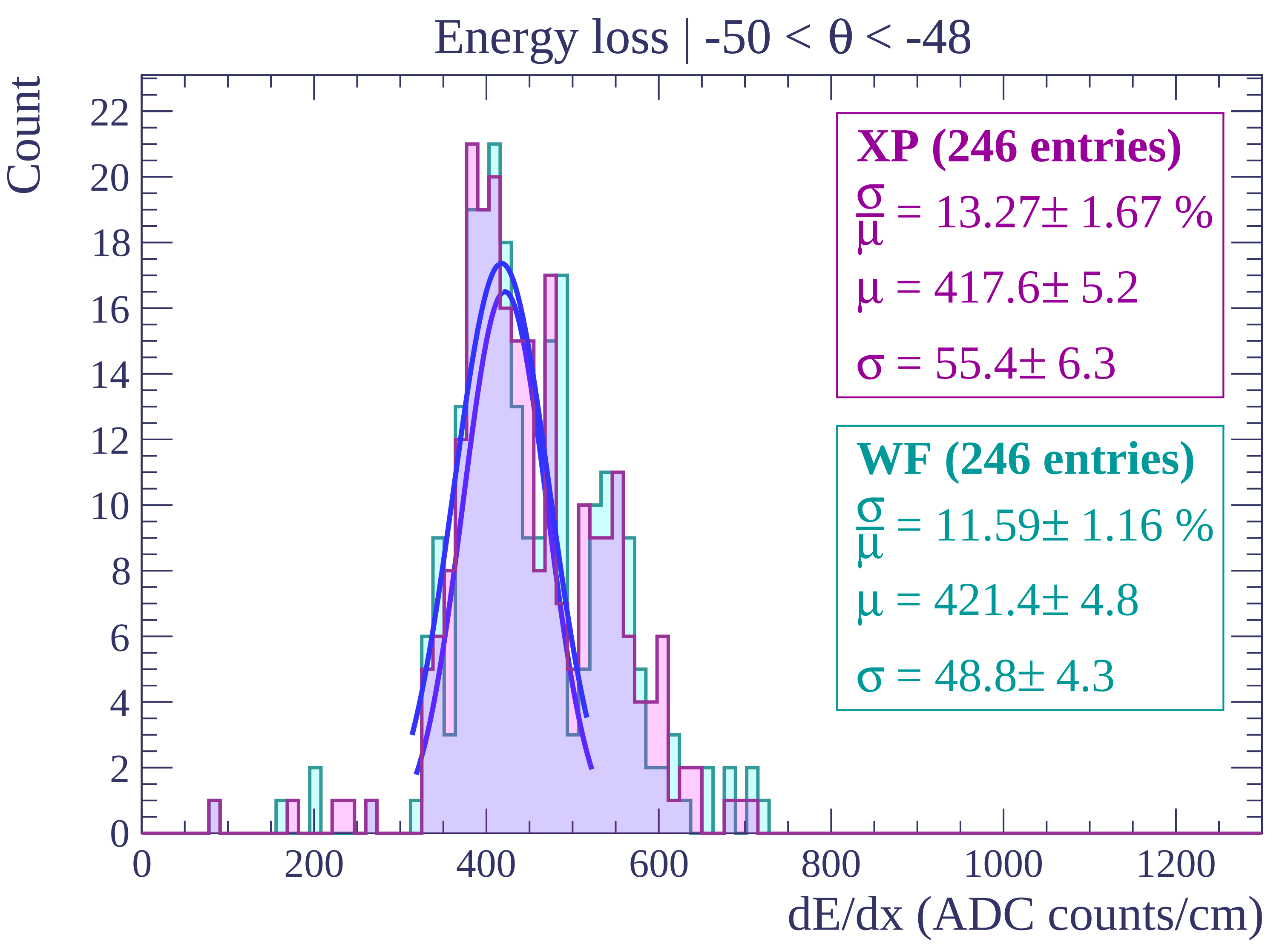
Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP (97 entries)
σ/μ = 20.73± 4.03 %
μ = 465.7± 14.0
σ = 96.6± 15.9
WF (97 entries)
σ/μ = 19.95± 3.73 %
μ = 452.5± 13.9
σ = 90.3± 14.1

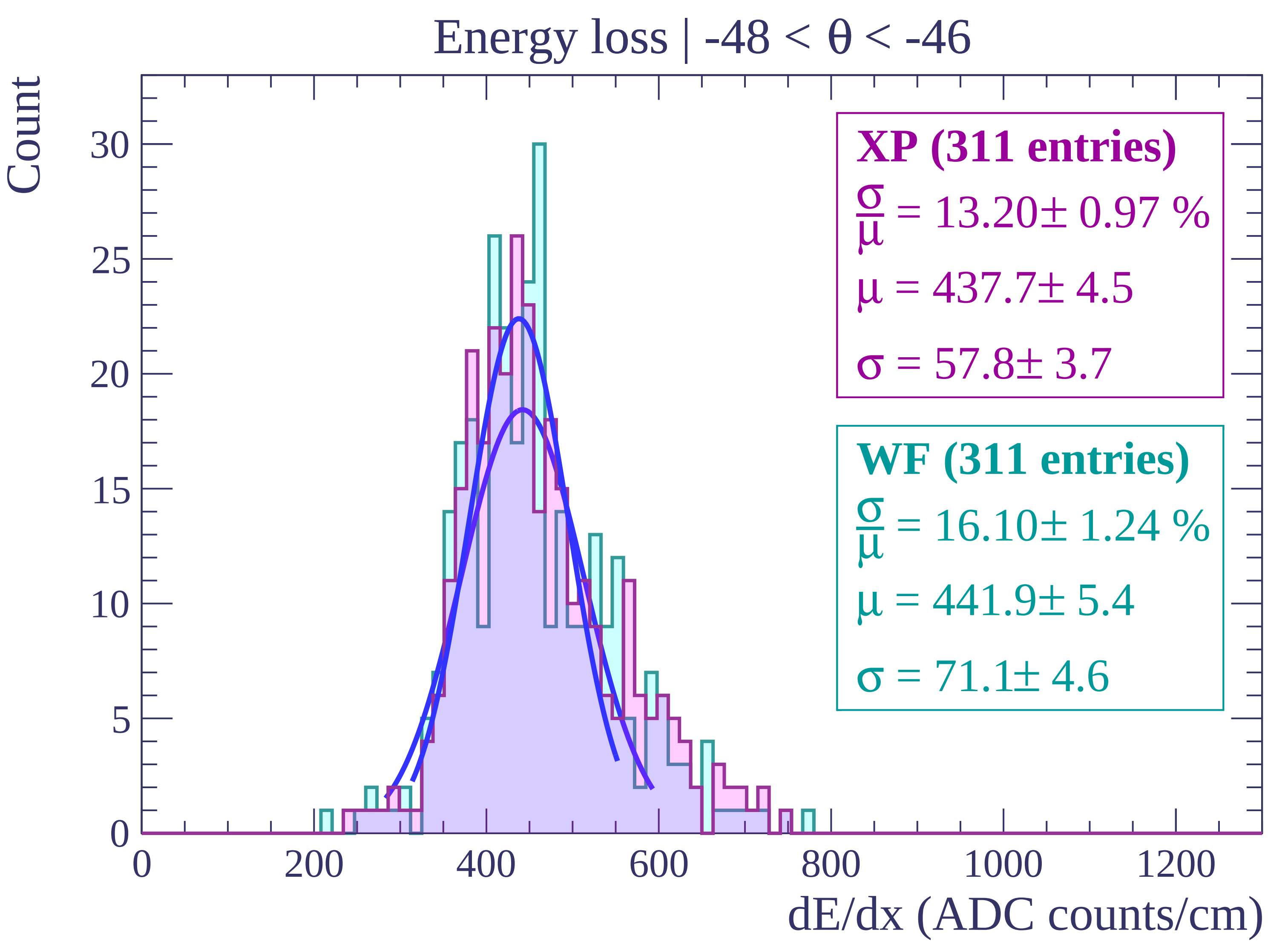


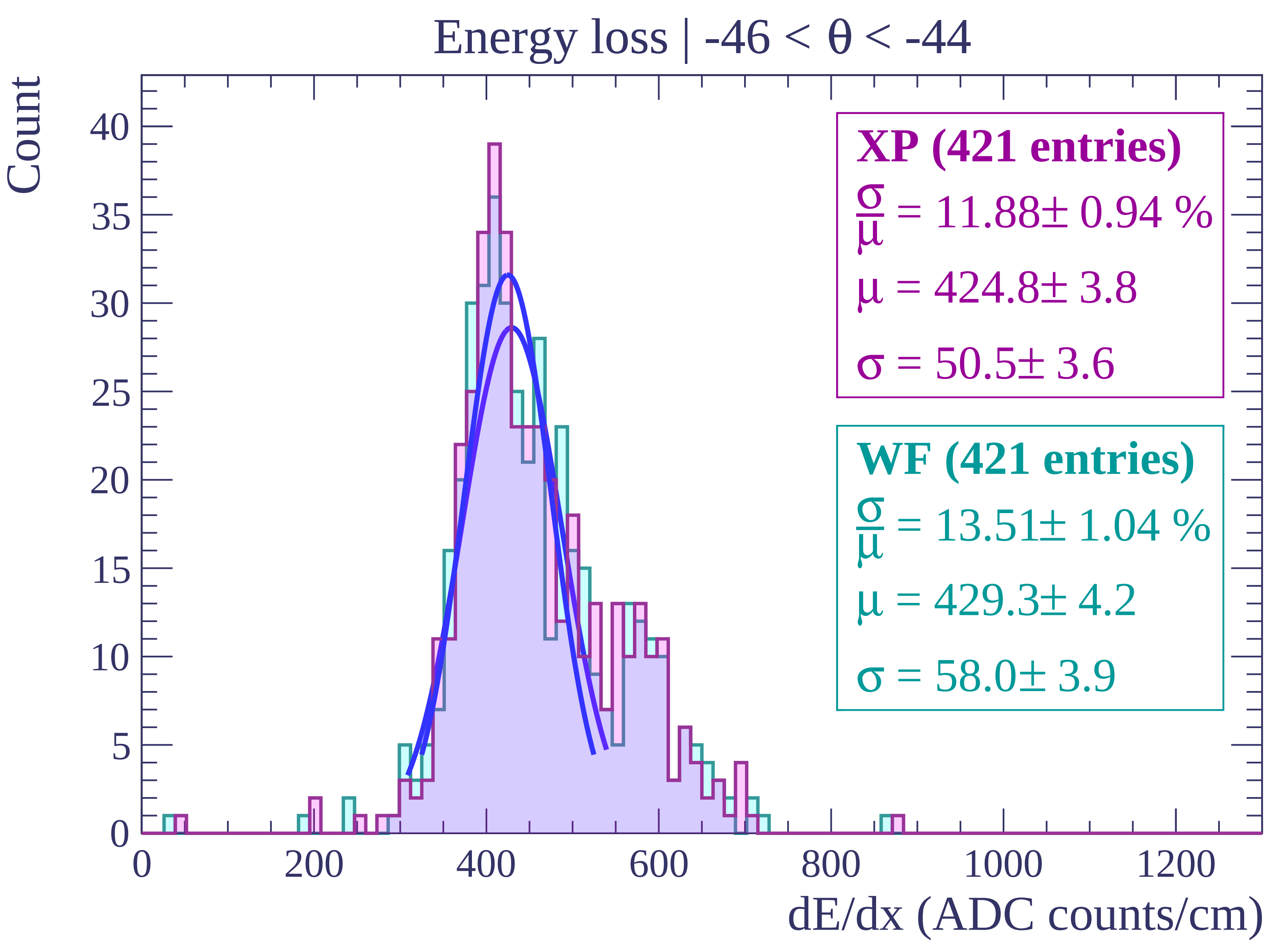
Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP (125 entries)
σ/μ = 14.15± 2.36 %
μ = 435.0± 8.5
σ = 61.6± 9.1
WF (125 entries)
σ/μ = 10.80± 1.66 %
μ = 423.9± 6.1
σ = 45.8± 6.4

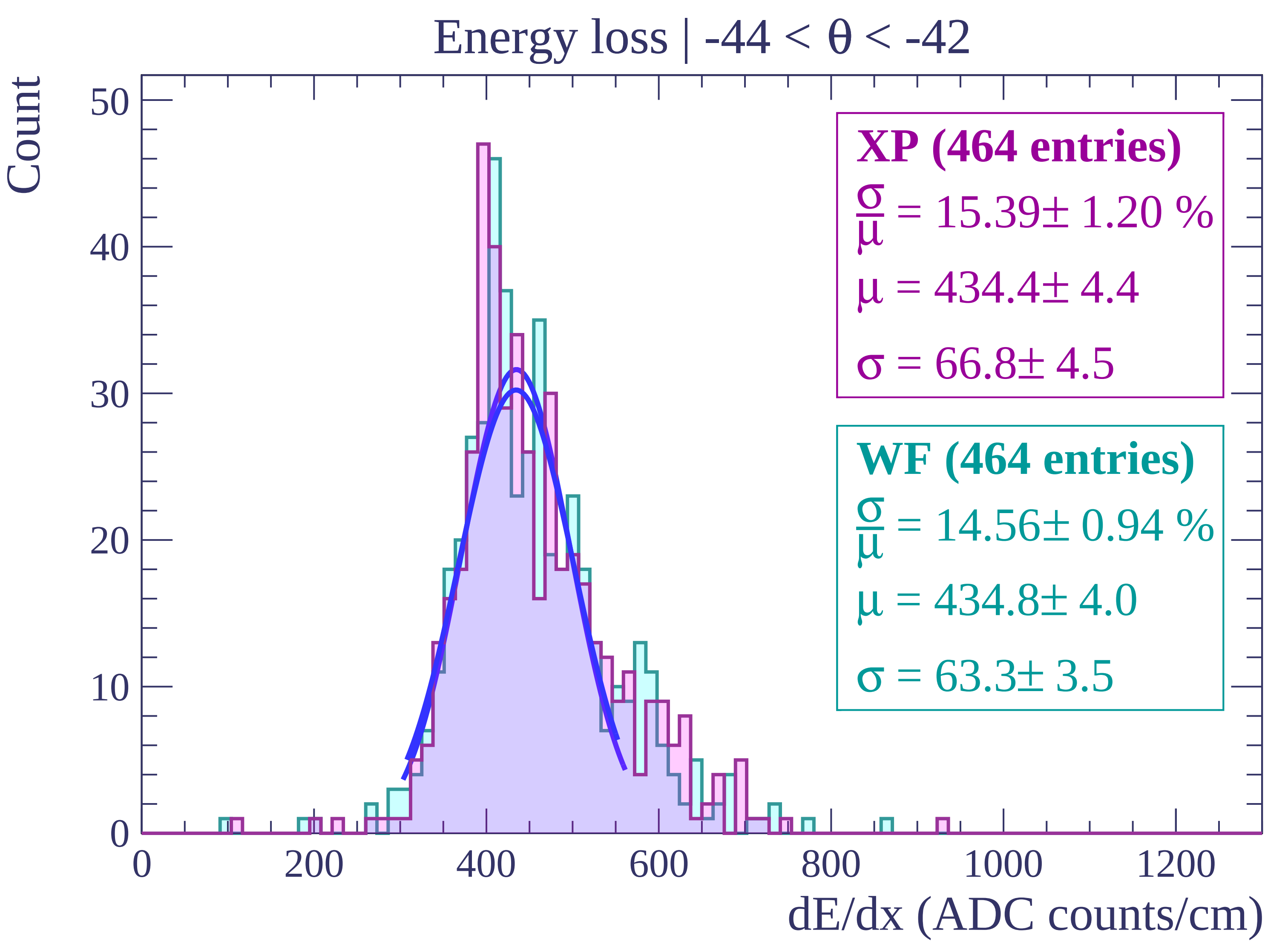


Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22
XP (164 entries)
σ/μ = 10.54± 1.33 %
μ = 430.8± 5.4
σ = 45.4± 5.2
WF (164 entries)
σ/μ = 17.09± 2.24 %
μ = 458.0± 8.1
σ = 78.3± 8.9

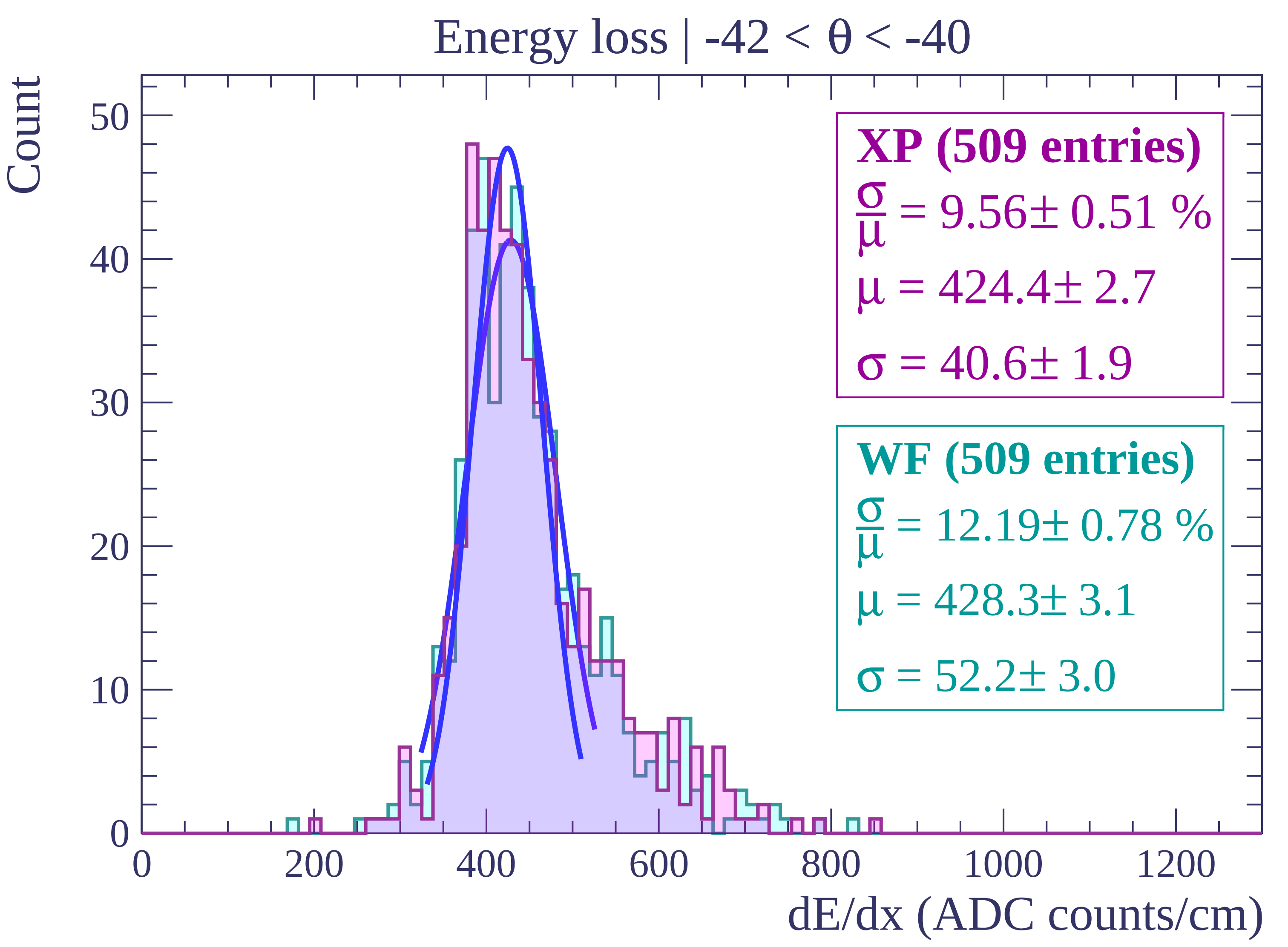

Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
XP (246 entries)
σ/μ = 13.27± 1.67 %
μ = 417.6± 5.2
σ = 55.4± 6.3
WF (246 entries)
σ/μ = 11.59± 1.16 %
μ = 421.4± 4.8
σ = 48.8± 4.3


Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP (311 entries)
σ/μ = 13.20± 0.97 %
μ = 437.7± 4.5
σ = 57.8± 3.7
WF (311 entries)
σ/μ = 16.10± 1.24 %
μ = 441.9± 5.4
σ = 71.1± 4.6

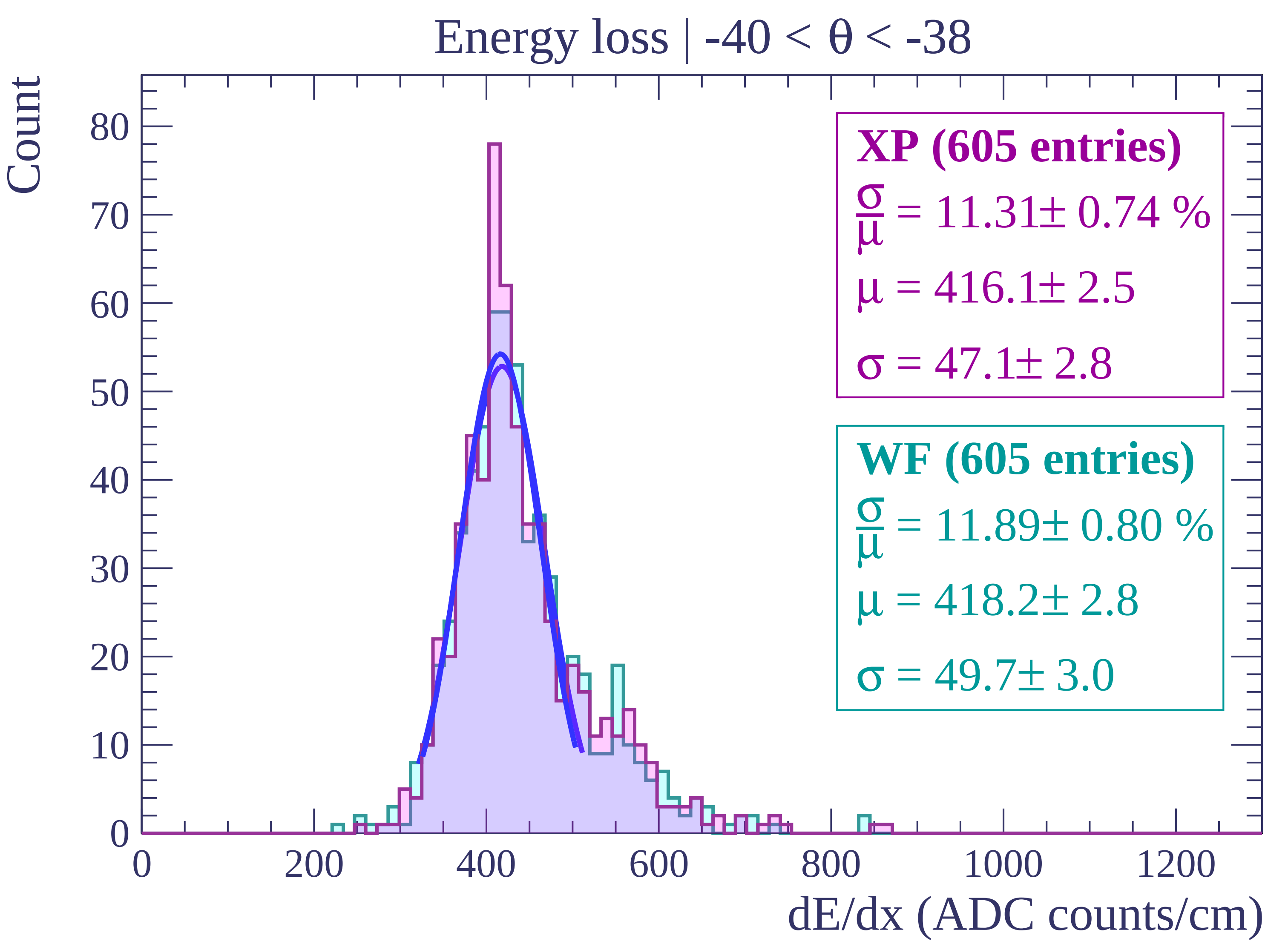

Energy loss | -46 < θ < -44
Count
XP (421 entries)
σ/μ = 11.88± 0.94 %
μ = 424.8± 3.8
σ = 50.5± 3.6
WF (421 entries)
σ/μ = 13.51± 1.04 %
μ = 429.3± 4.2
σ = 58.0± 3.9
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200

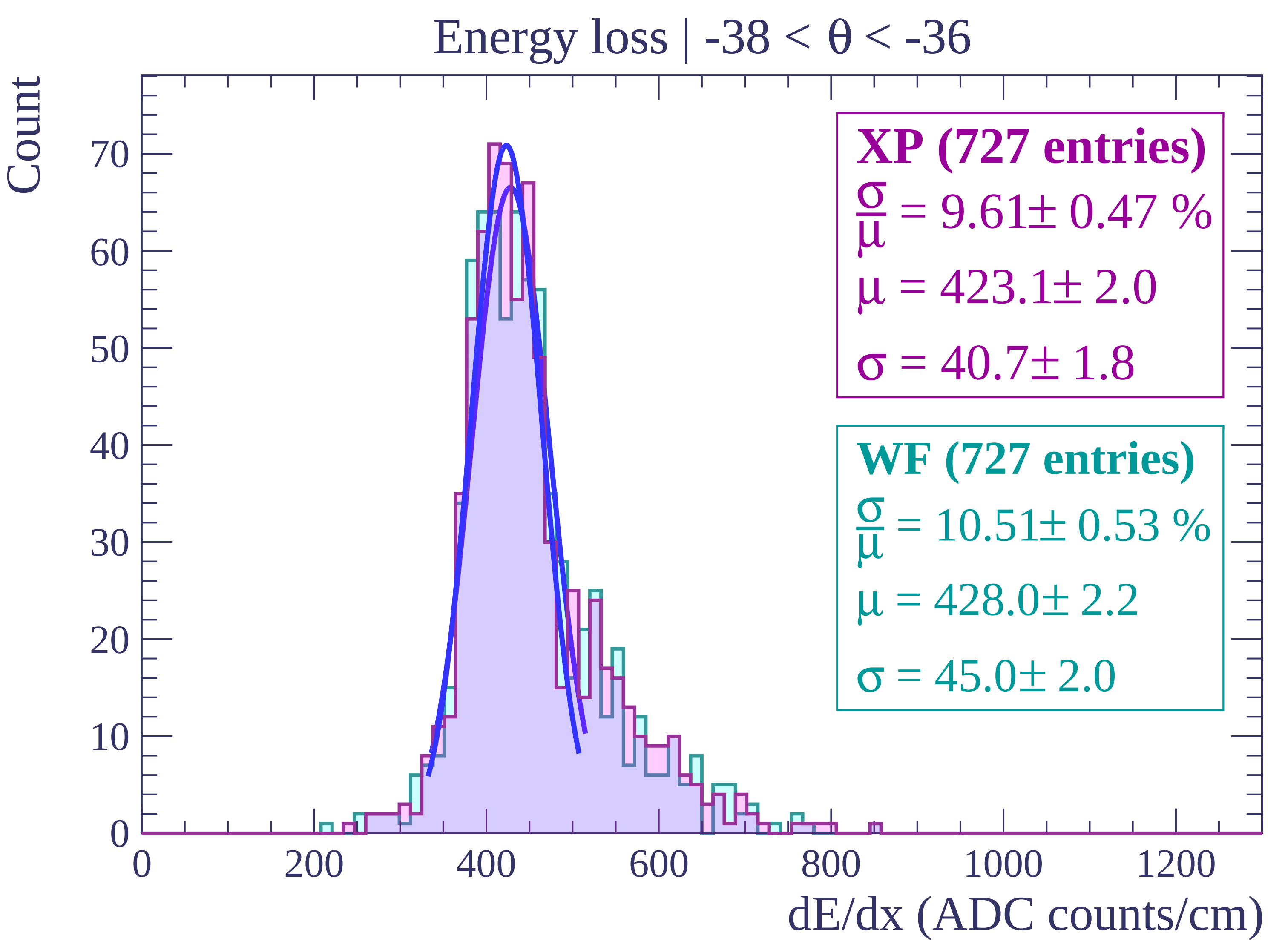


Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (464 entries)
σ/μ = 15.39± 1.20 %
μ = 434.4± 4.4
σ = 66.8± 4.5
WF (464 entries)
σ/μ = 14.56± 0.94 %
μ = 434.8± 4.0
σ = 63.3± 3.5

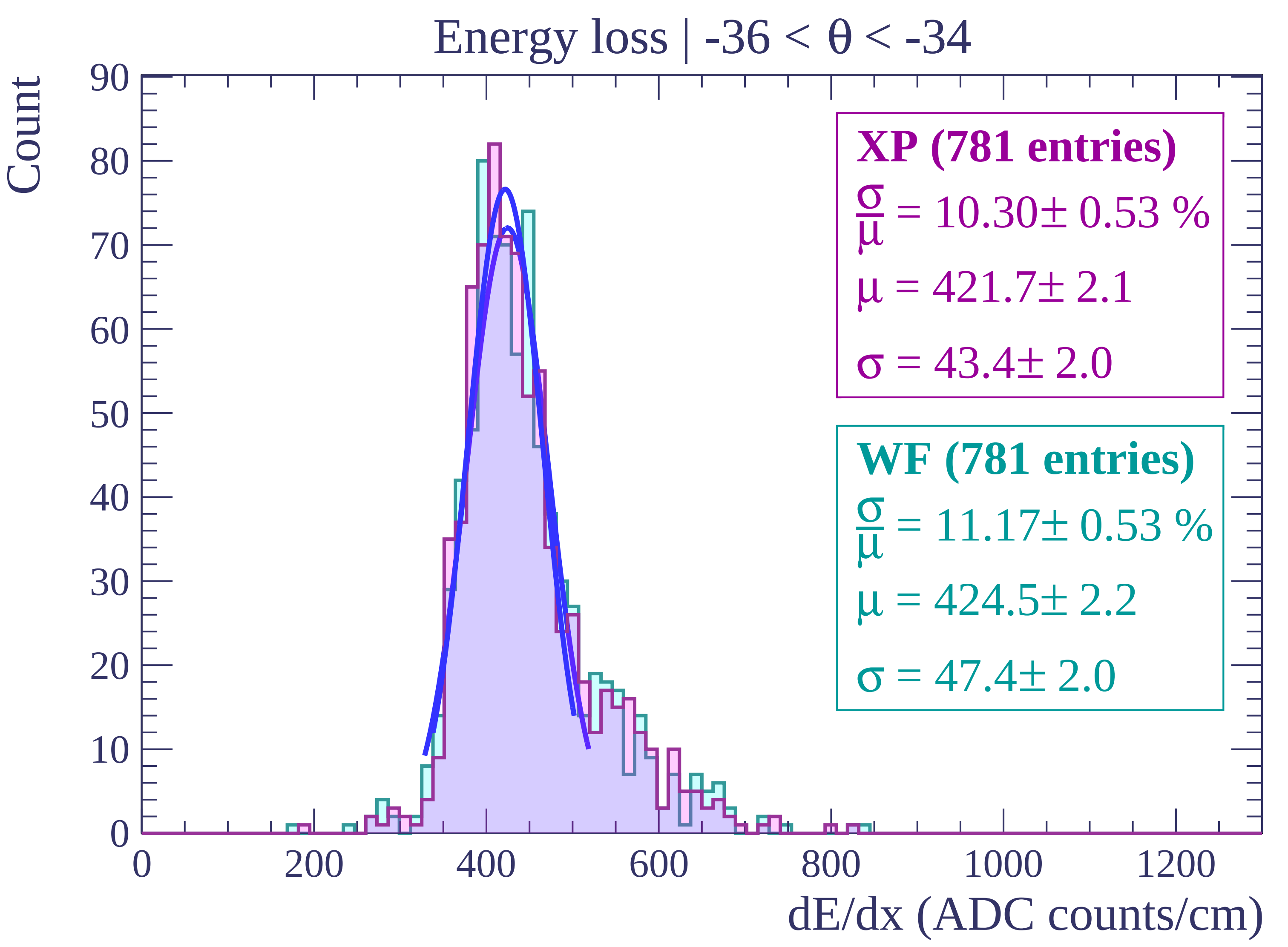


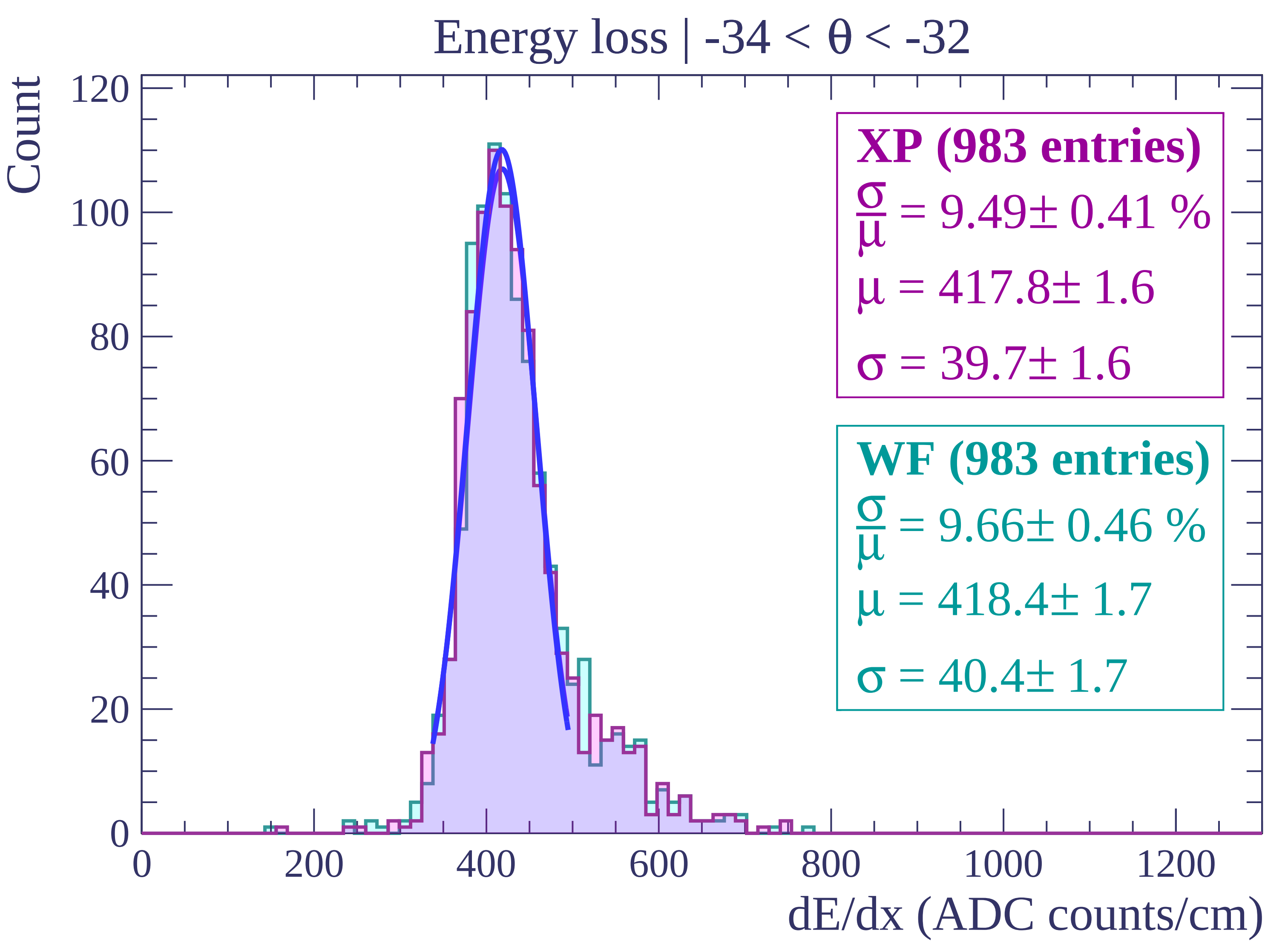
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (509 entries)
σ/μ = 9.56± 0.51 %
μ = 424.4± 2.7
σ = 40.6± 1.9
WF (509 entries)
σ/μ = 12.19± 0.78 %
μ = 428.3± 3.1
σ = 52.2± 3.0

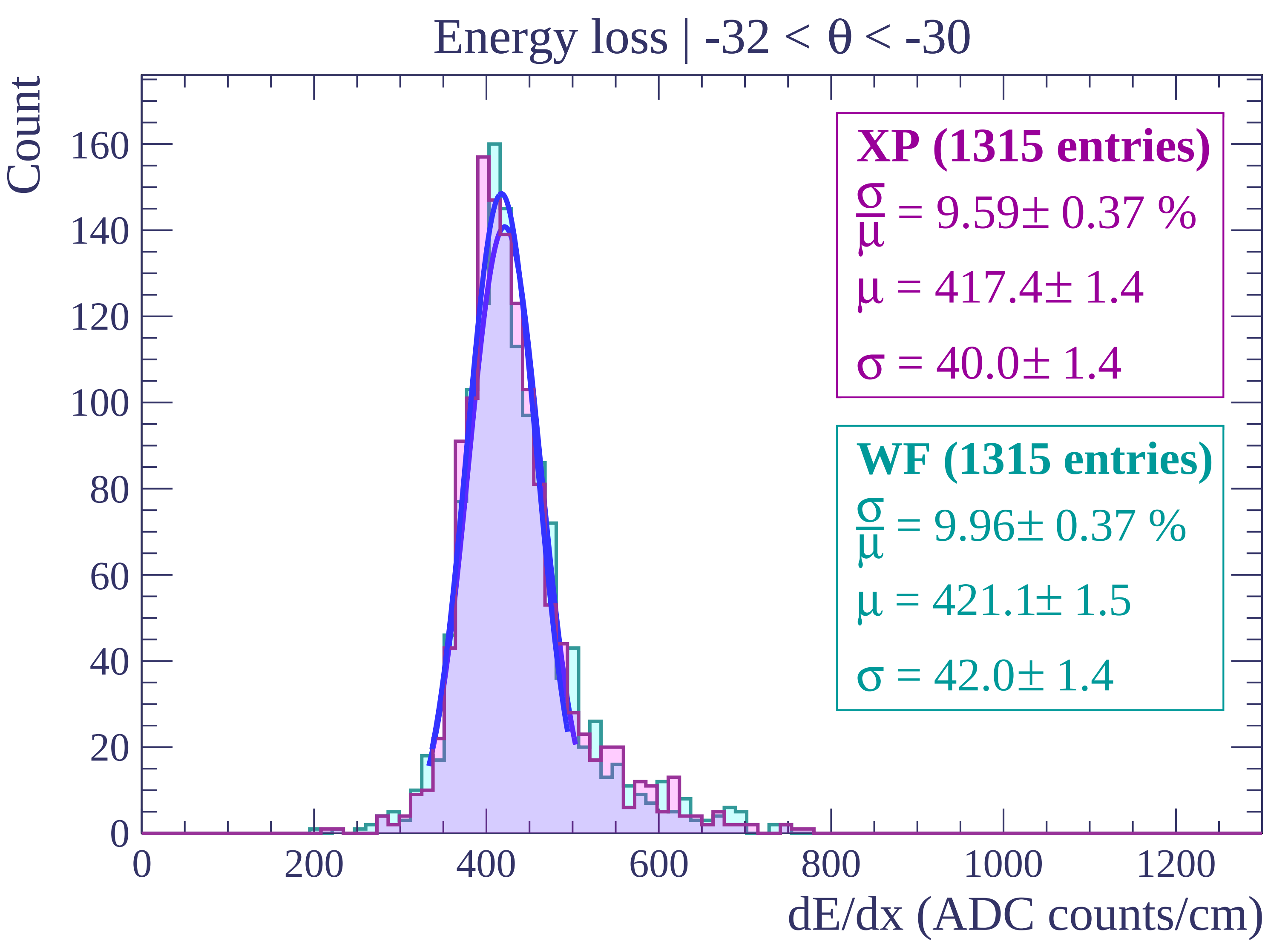

Energy loss | -40 < θ < -38
Count
XP (605 entries)
σ/μ = 11.31± 0.74 %
μ = 416.1± 2.5
σ = 47.1± 2.8
WF (605 entries)
σ/μ = 11.89± 0.80 %
μ = 418.2± 2.8
σ = 49.7± 3.0
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



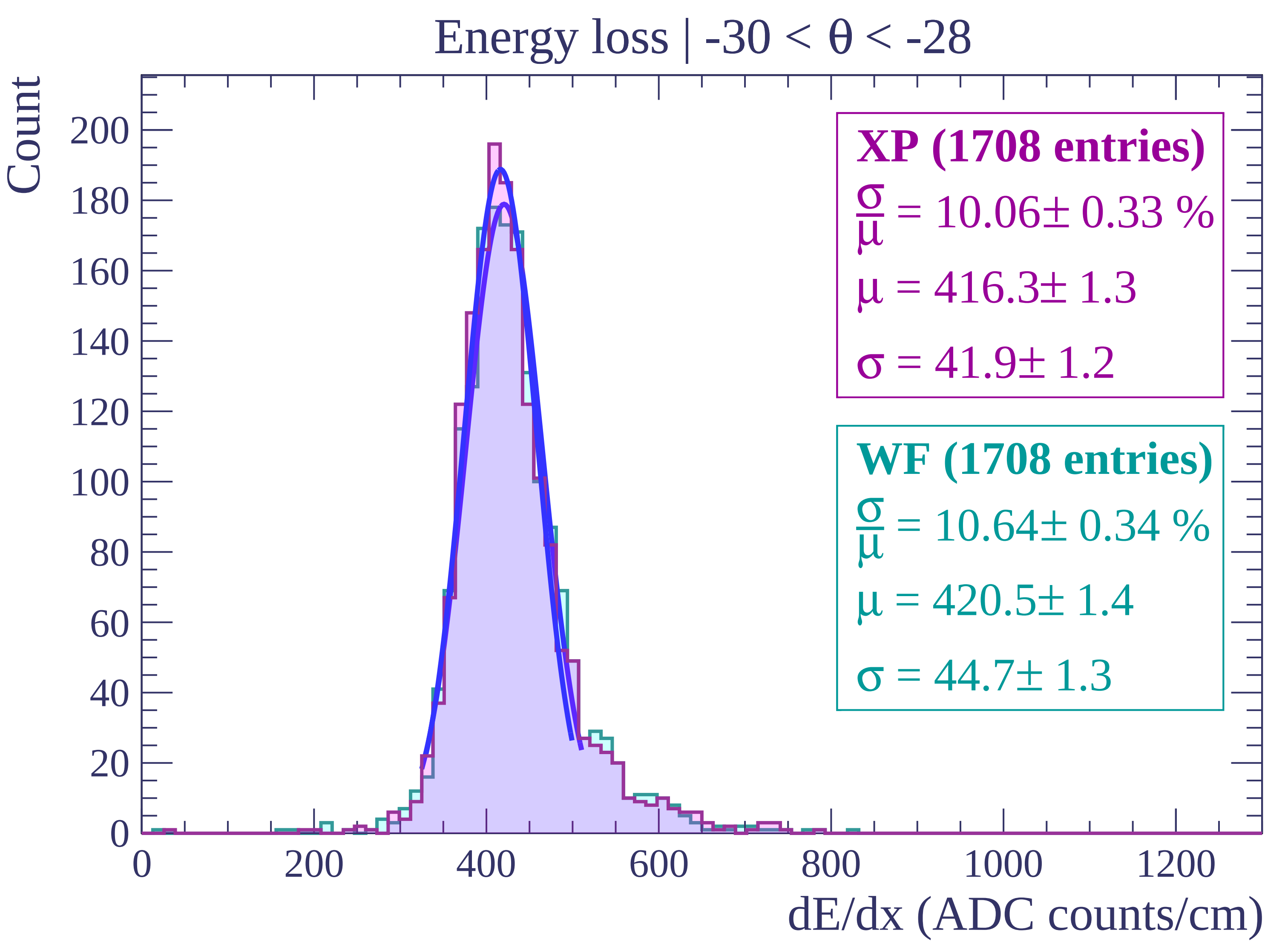
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (727 entries)
σ/μ = 9.61± 0.47 %
μ = 423.1± 2.0
σ = 40.7± 1.8
WF (727 entries)
σ/μ = 10.51± 0.53 %
μ = 428.0± 2.2
σ = 45.0± 2.0


Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
XP (781 entries)
σ/μ = 10.30± 0.53 %
μ = 421.7± 2.1
σ = 43.4± 2.0
WF (781 entries)
σ/μ = 11.17± 0.53 %
μ = 424.5± 2.2
σ = 47.4± 2.0

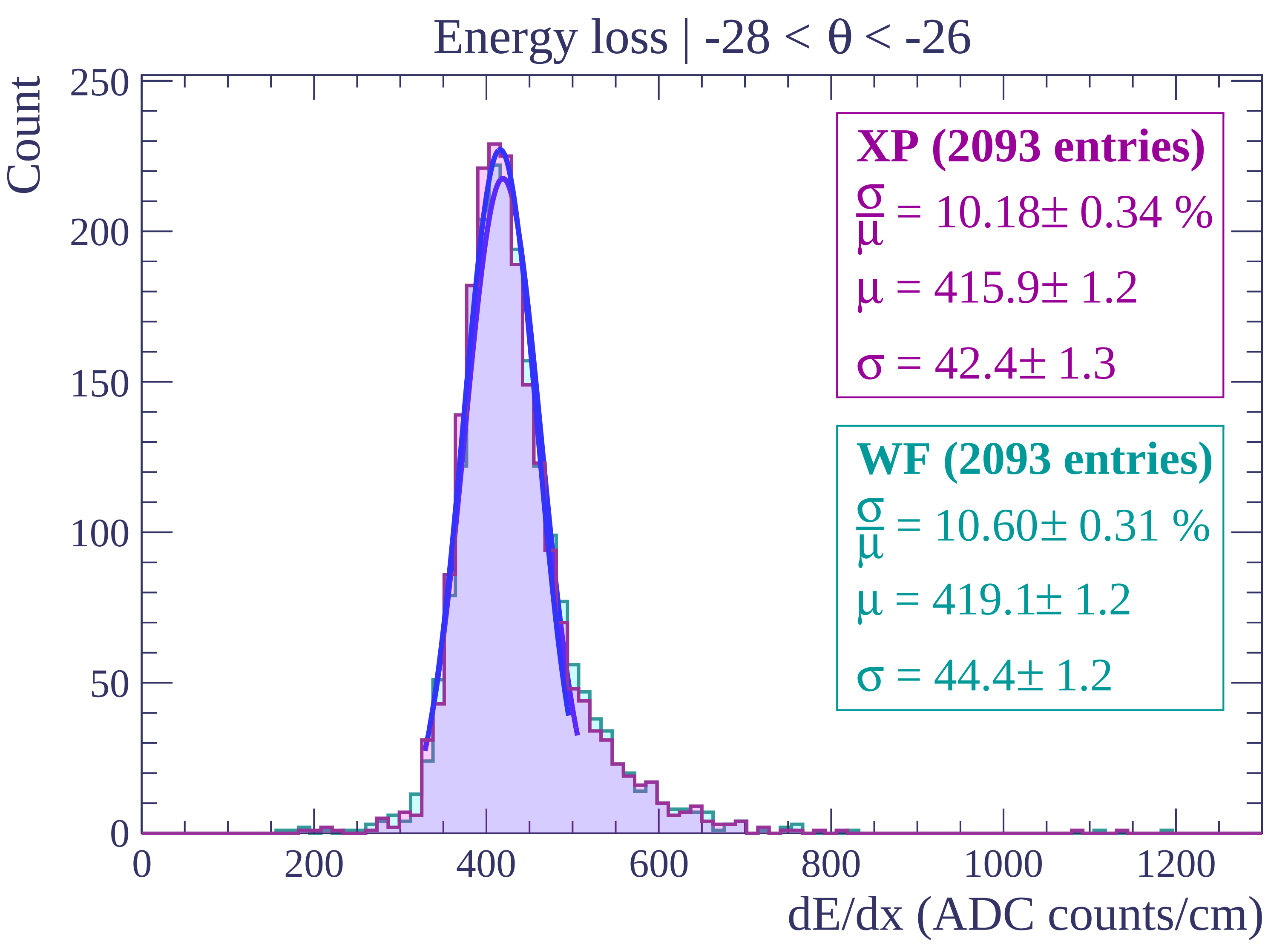

Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (983 entries)
$\frac{\sigma}{\mu}$ = 9.49± 0.41 %
μ = 417.8± 1.6
σ = 39.7± 1.6
WF (983 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.46 %
μ = 418.4± 1.7
σ = 40.4± 1.7



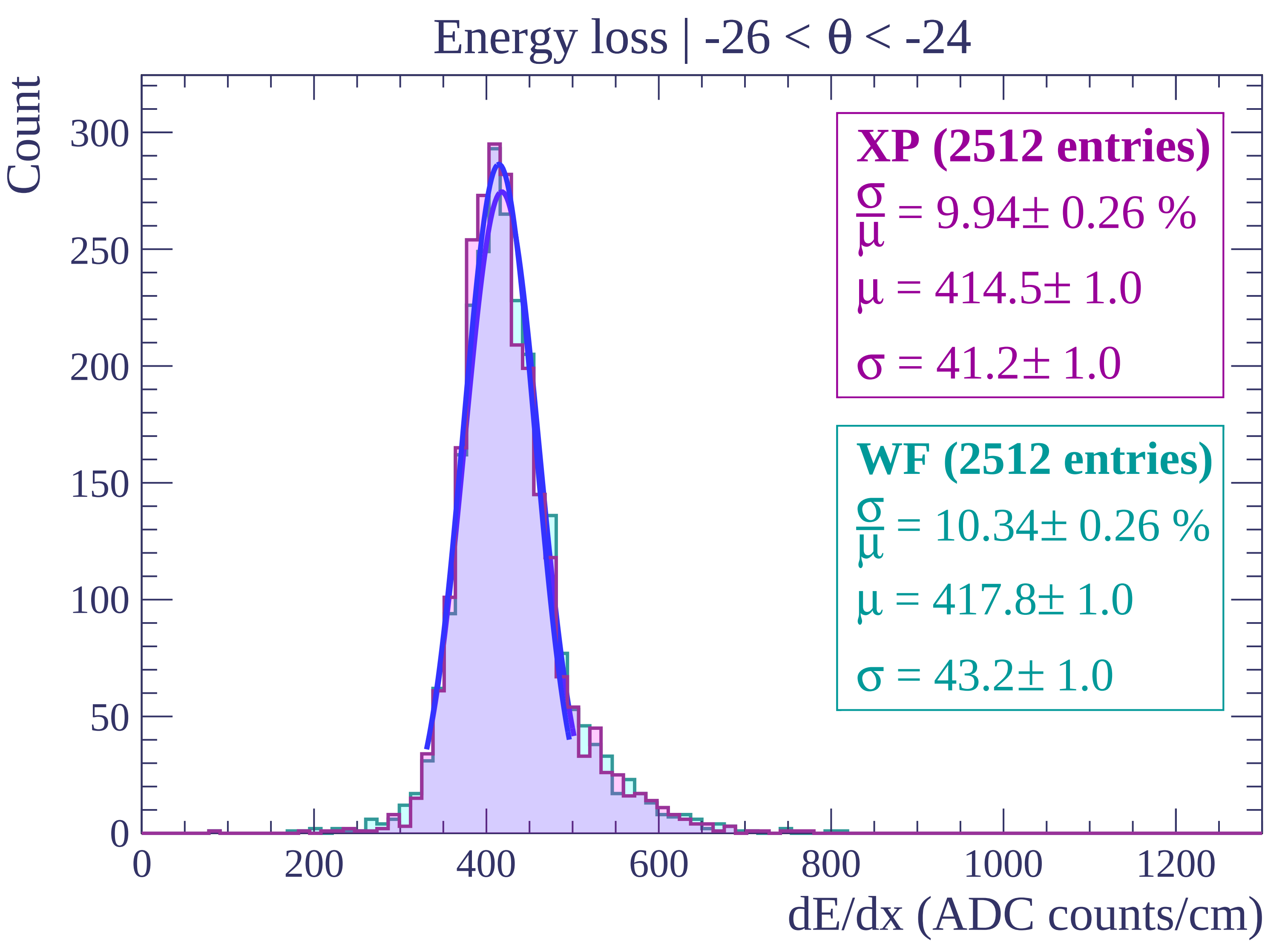
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
XP (1315 entries)
σ/μ = 9.59± 0.37 %
μ = 417.4± 1.4
σ = 40.0± 1.4
WF (1315 entries)
σ/μ = 9.96± 0.37 %
μ = 421.1± 1.5
σ = 42.0± 1.4
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



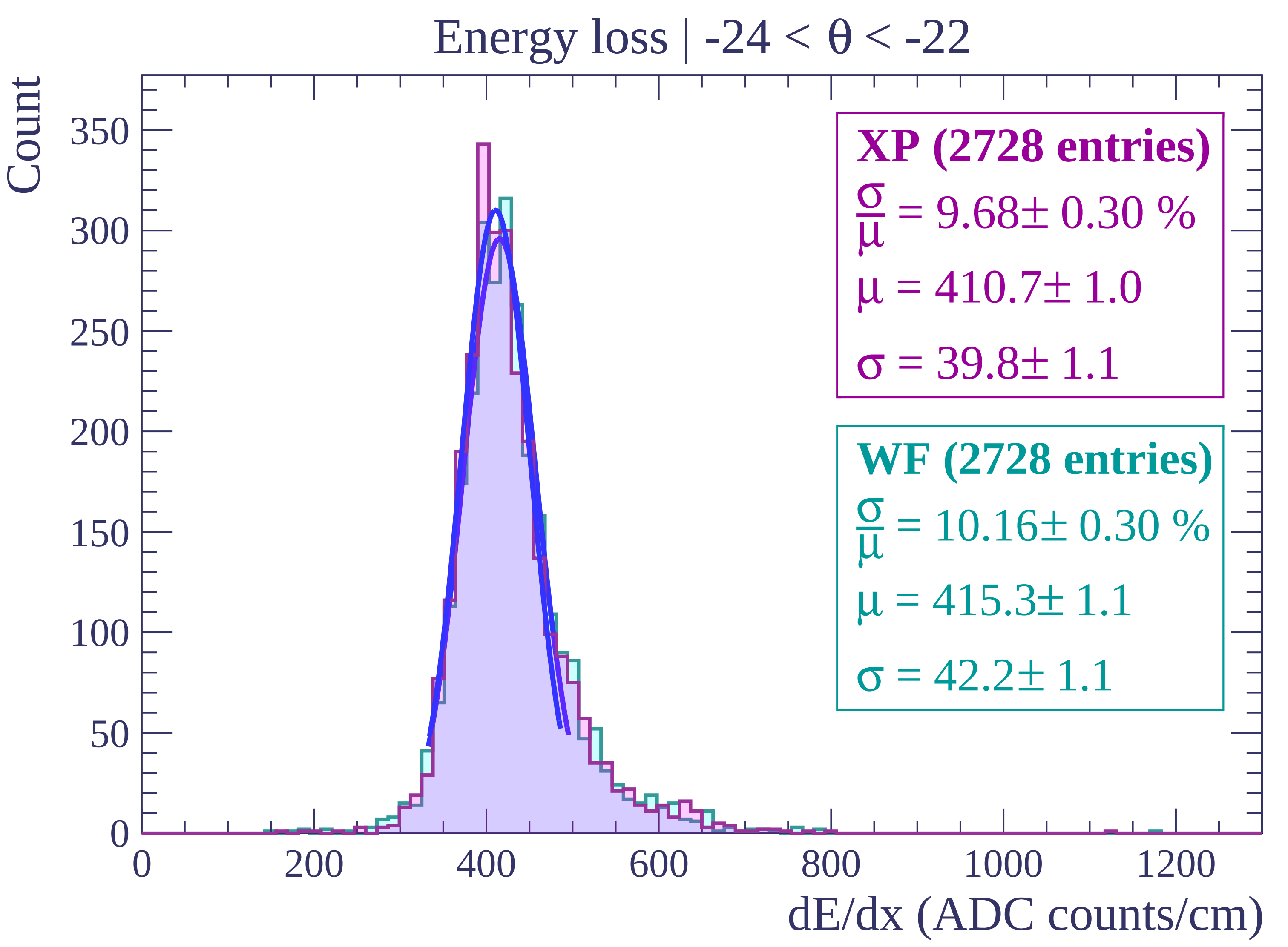
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1708 entries)
σ/μ = 10.06± 0.33 %
μ = 416.3± 1.3
σ = 41.9± 1.2
WF (1708 entries)
σ/μ = 10.64± 0.34 %
μ = 420.5± 1.4
σ = 44.7± 1.3



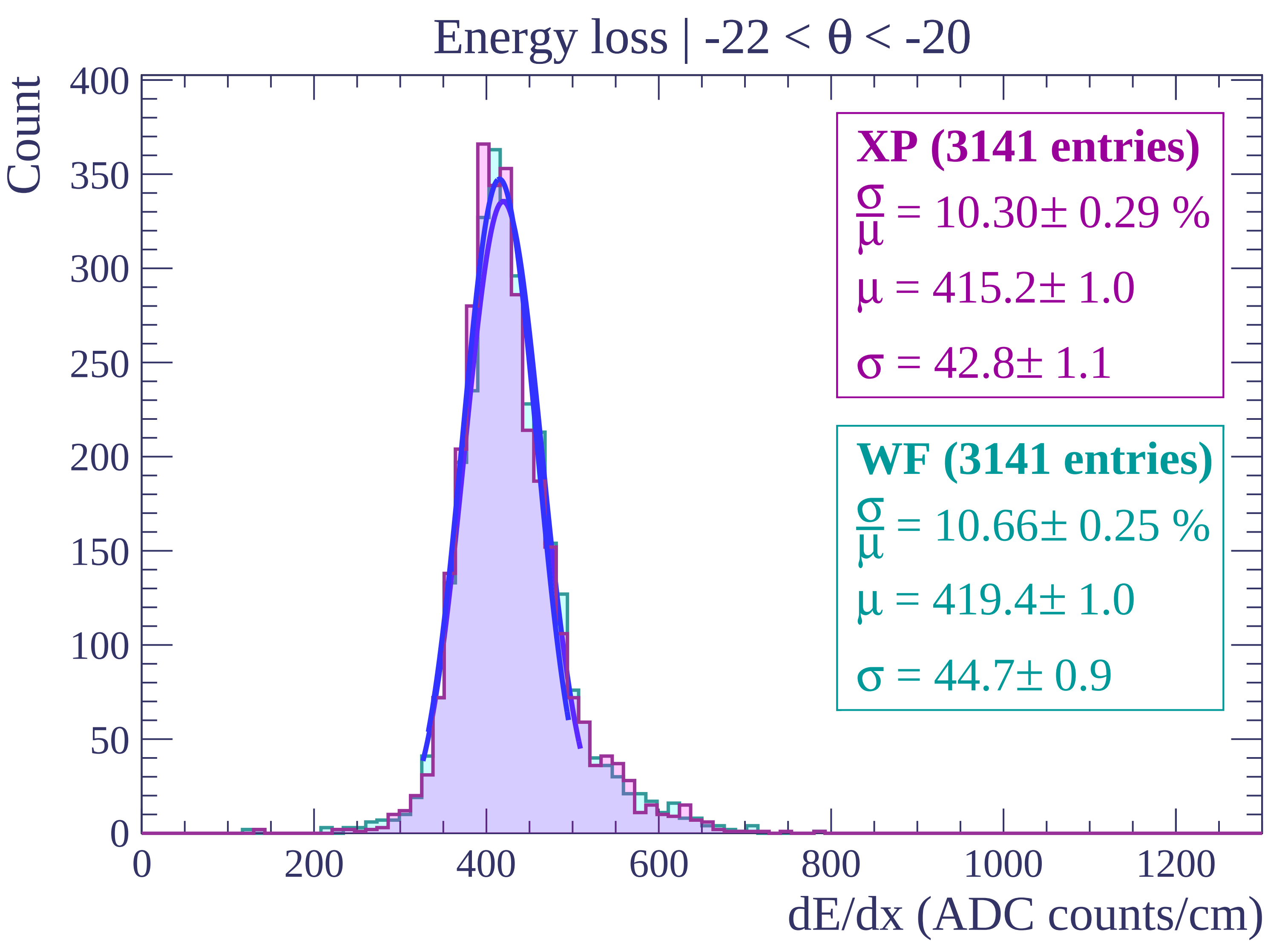
Energy loss | -28 < θ < -26
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2093 entries)
σ/μ = 10.18± 0.34 %
μ = 415.9± 1.2
σ = 42.4± 1.3
WF (2093 entries)
σ/μ = 10.60± 0.31 %
μ = 419.1± 1.2
σ = 44.4± 1.2

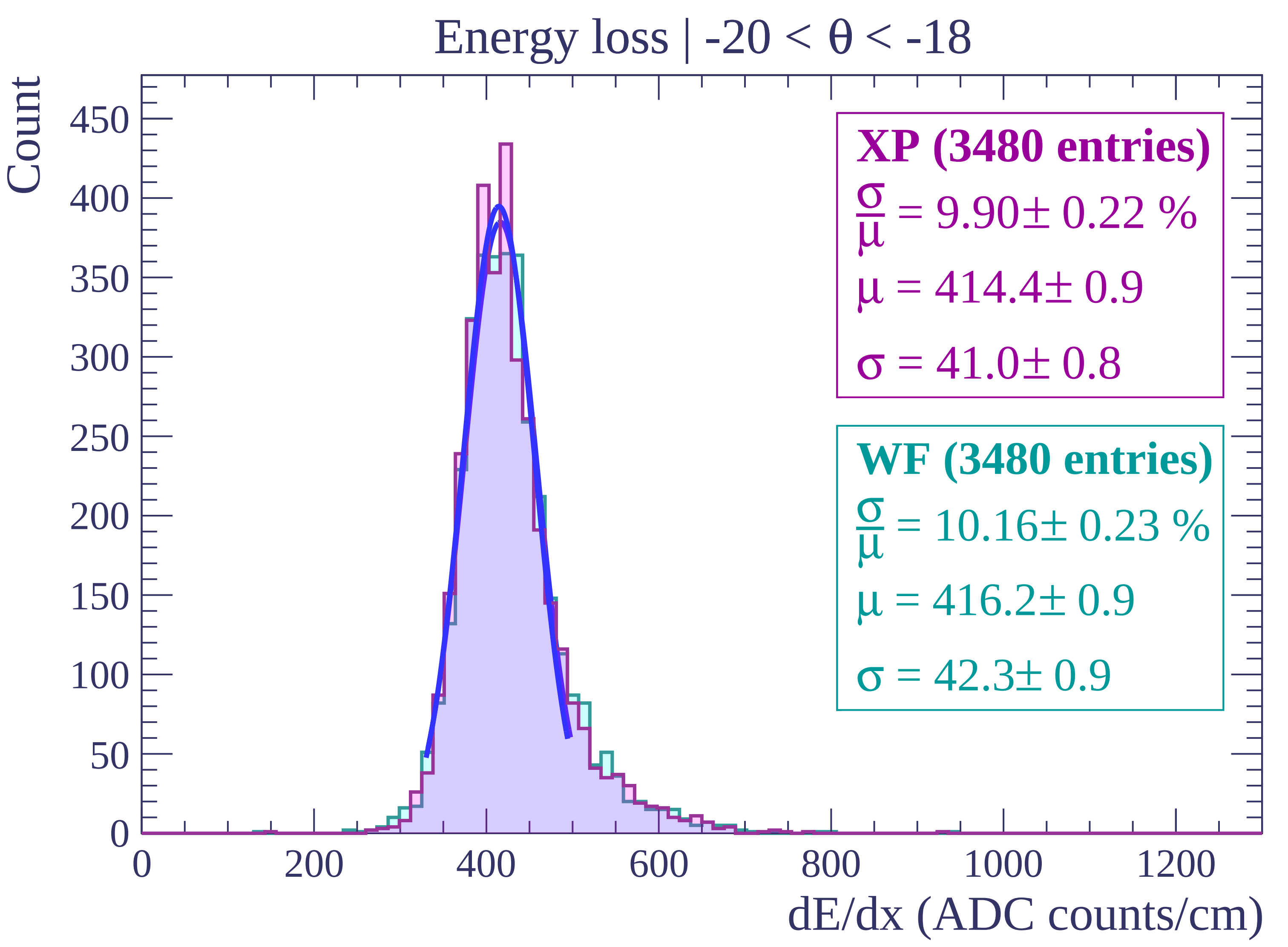


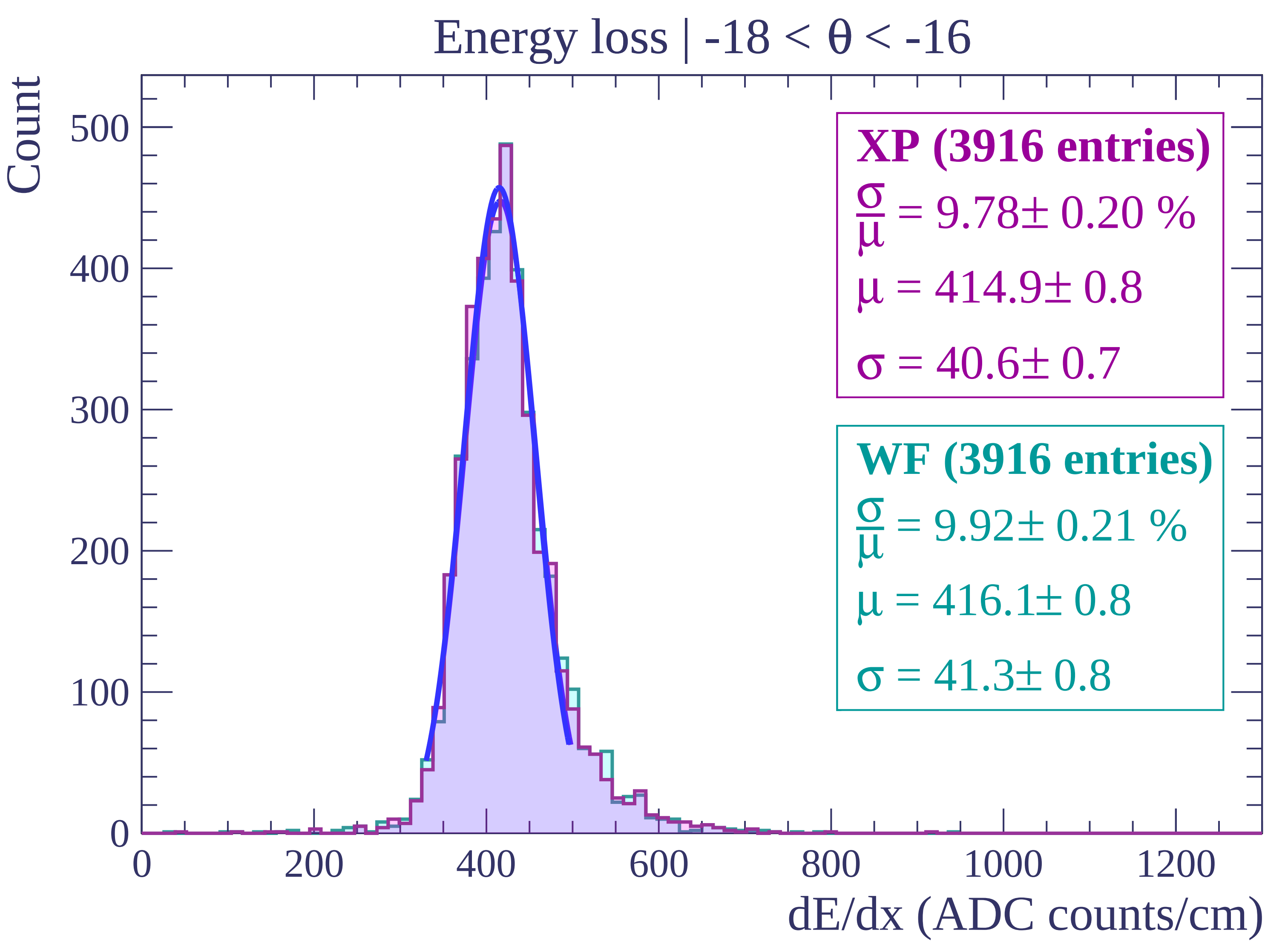
Energy loss | -26 < θ < -24
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2512 entries)
σ/μ = 9.94± 0.26 %
μ = 414.5± 1.0
σ = 41.2± 1.0
WF (2512 entries)
σ/μ = 10.34± 0.26 %
μ = 417.8± 1.0
σ = 43.2± 1.0

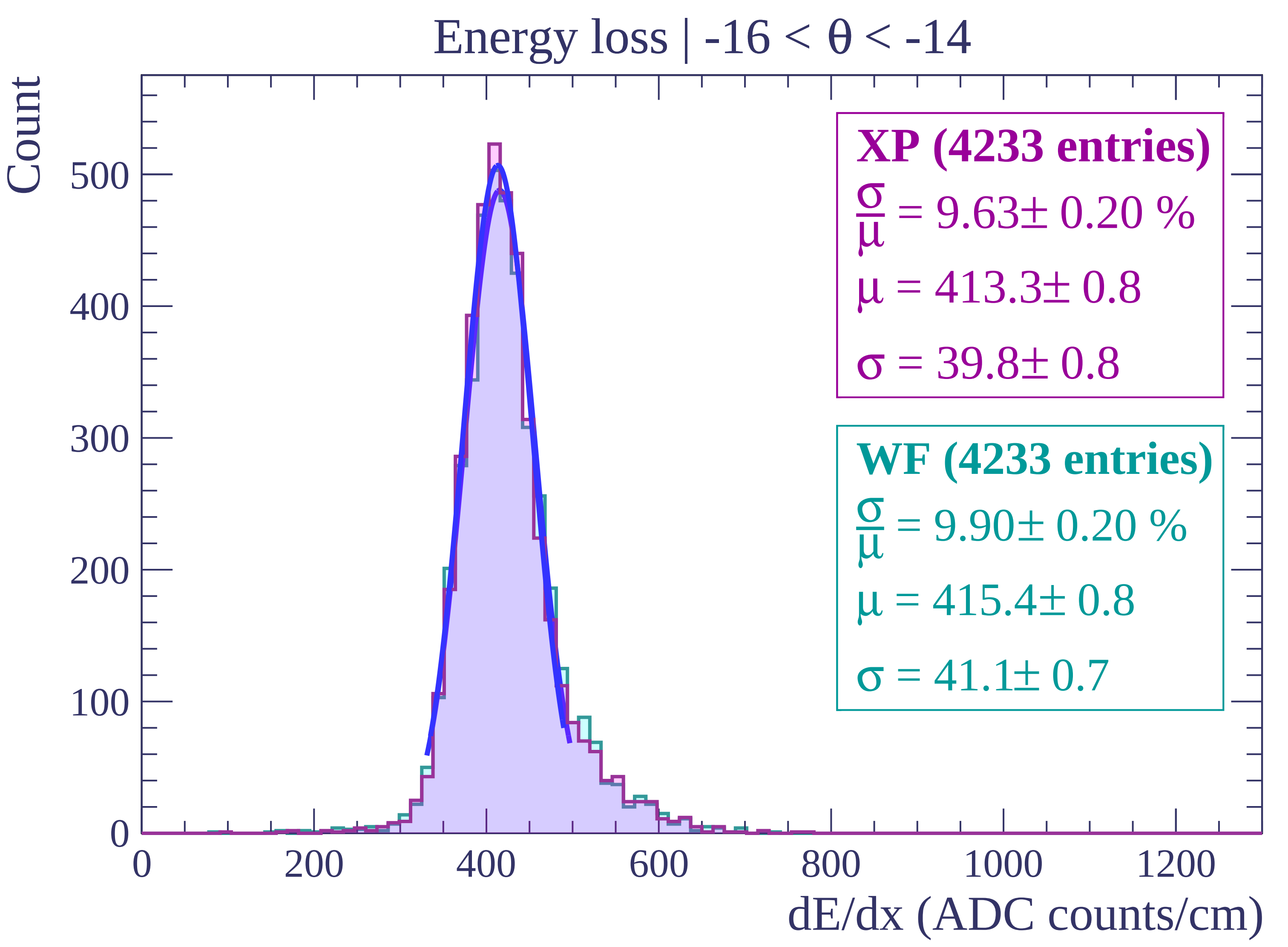


Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
XP (2728 entries)
σ/μ = 9.68± 0.30 %
μ = 410.7± 1.0
σ = 39.8± 1.1
WF (2728 entries)
σ/μ = 10.16± 0.30 %
μ = 415.3± 1.1
σ = 42.2± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

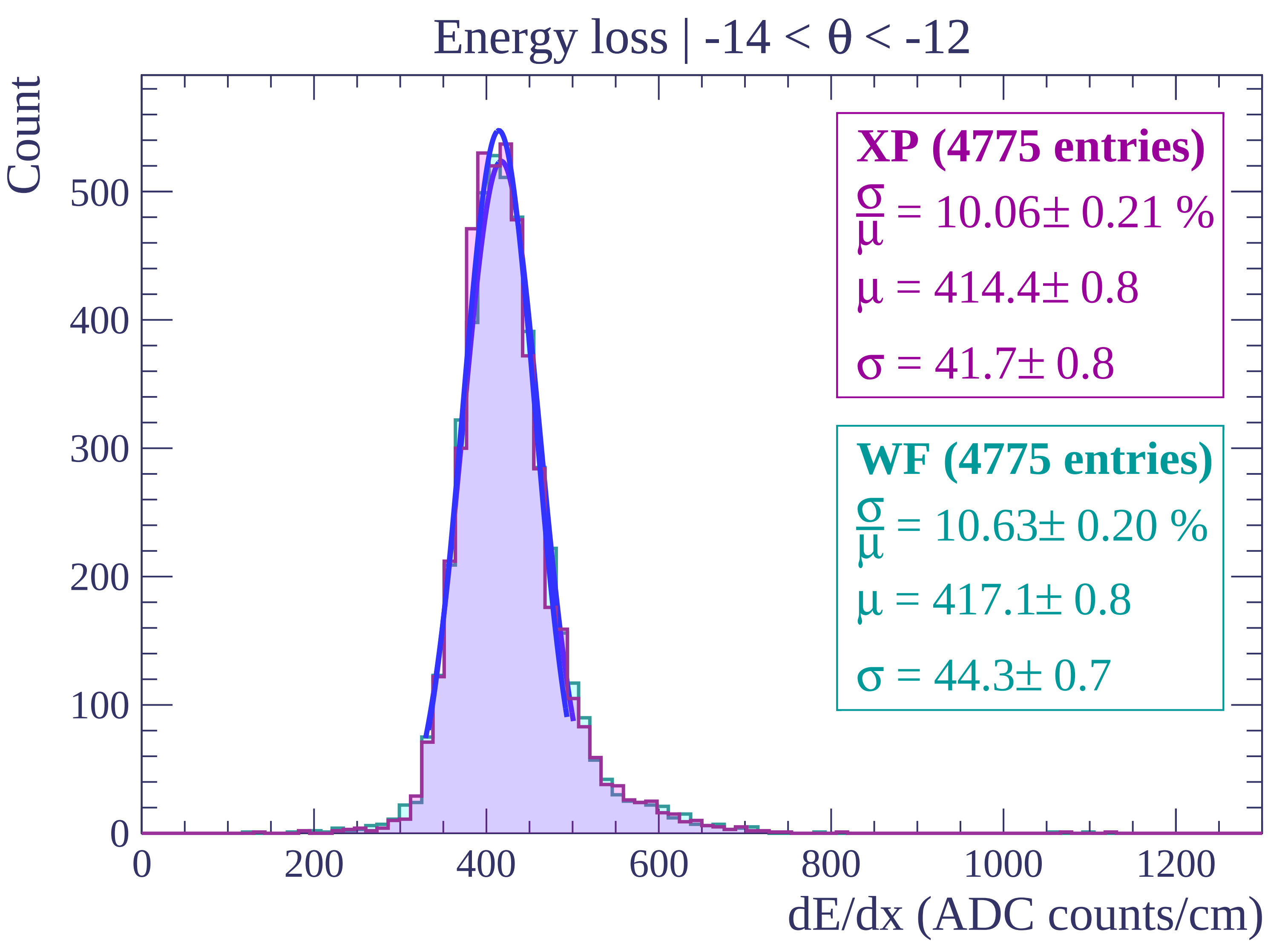

Energy loss | -22 < θ < -20
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3141 entries)
σ/μ = 10.30± 0.29 %
μ = 415.2± 1.0
σ = 42.8± 1.1
WF (3141 entries)
σ/μ = 10.66± 0.25 %
μ = 419.4± 1.0
σ = 44.7± 0.9

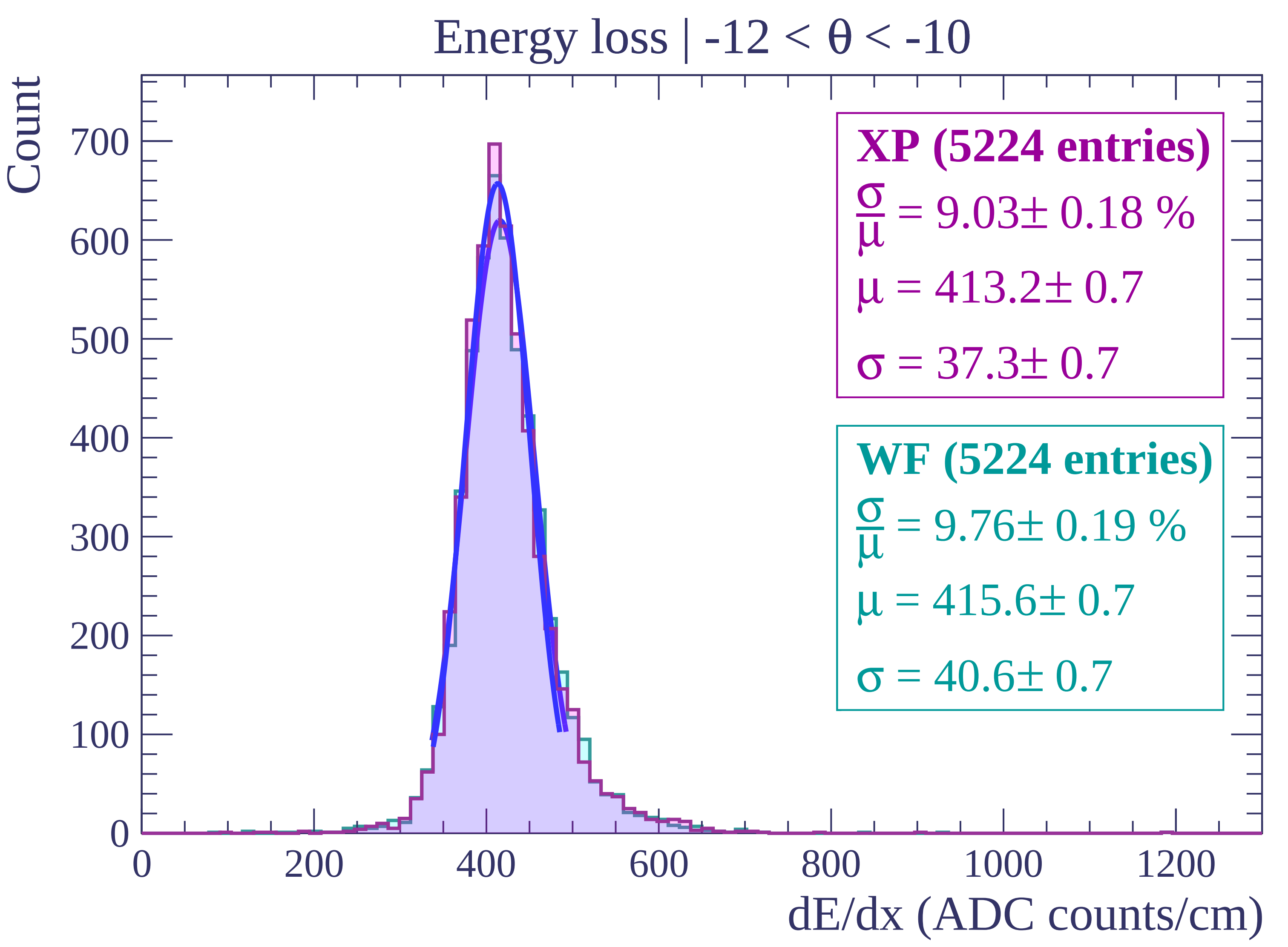

Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP (3480 entries)
σ/μ = 9.90± 0.22 %
μ = 414.4± 0.9
σ = 41.0± 0.8
WF (3480 entries)
σ/μ = 10.16± 0.23 %
μ = 416.2± 0.9
σ = 42.3± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

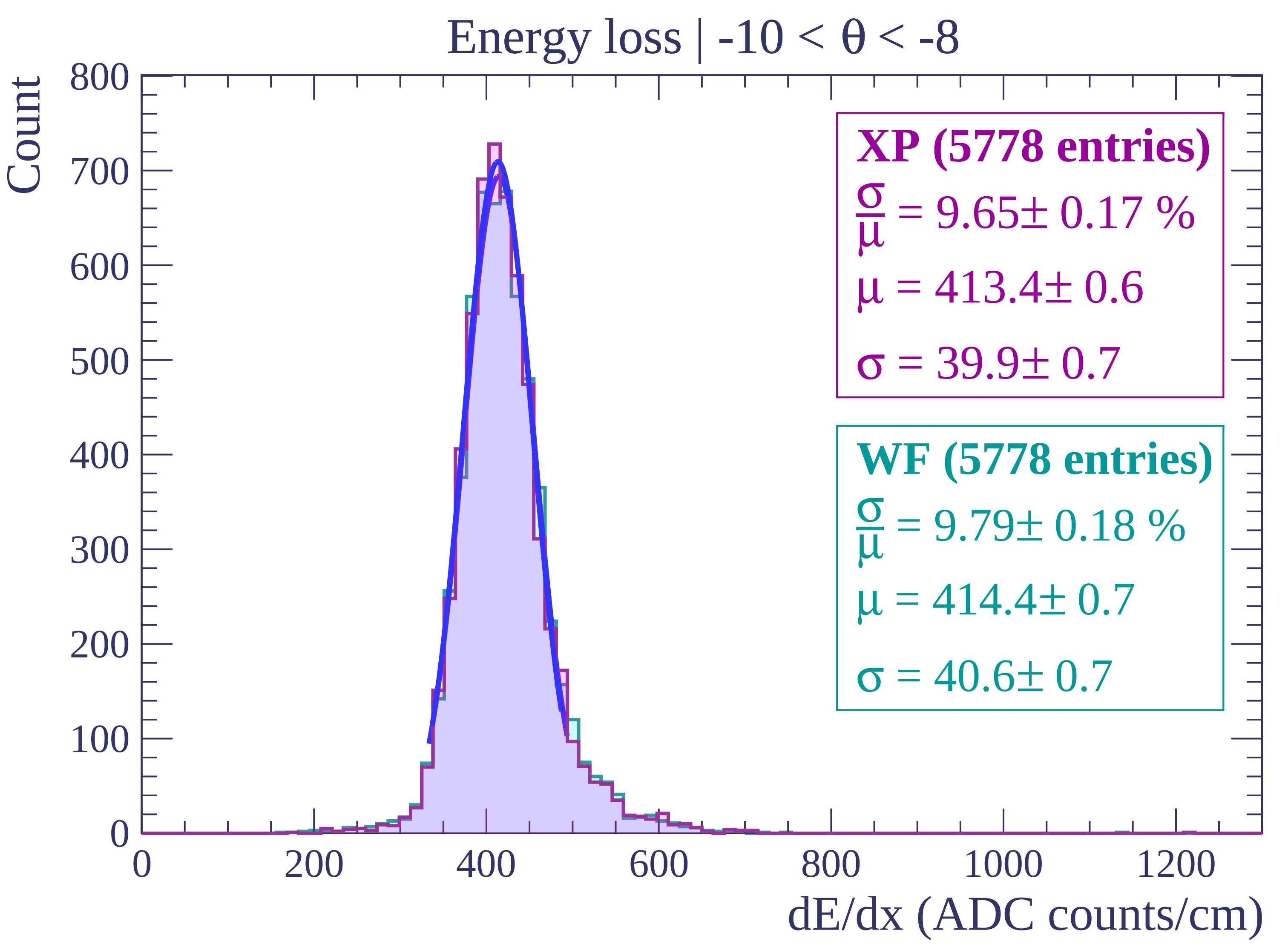

Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3916 entries)
σ/μ = 9.78± 0.20 %
μ = 414.9± 0.8
σ = 40.6± 0.7
WF (3916 entries)
σ/μ = 9.92± 0.21 %
μ = 416.1± 0.8
σ = 41.3± 0.8

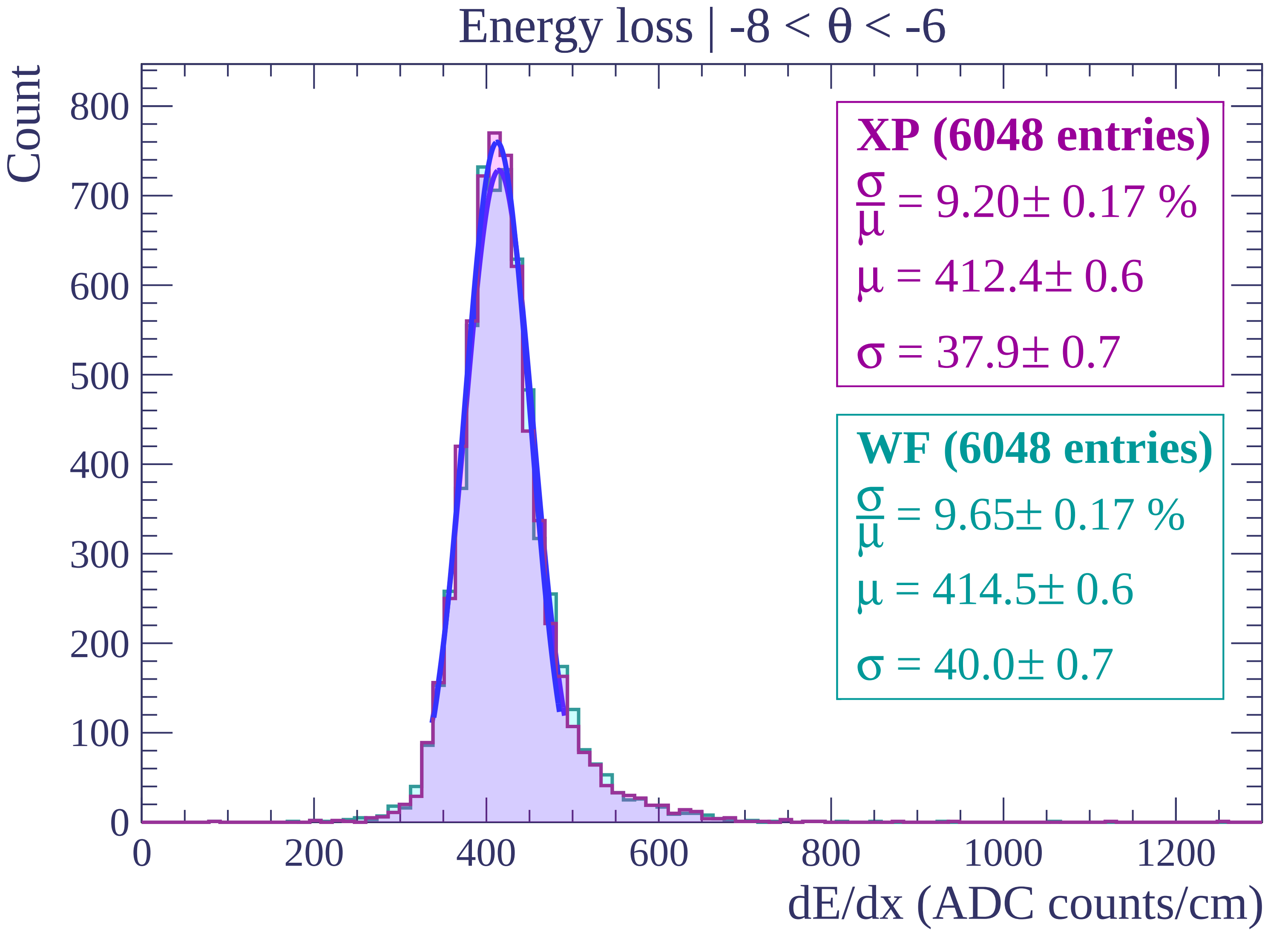


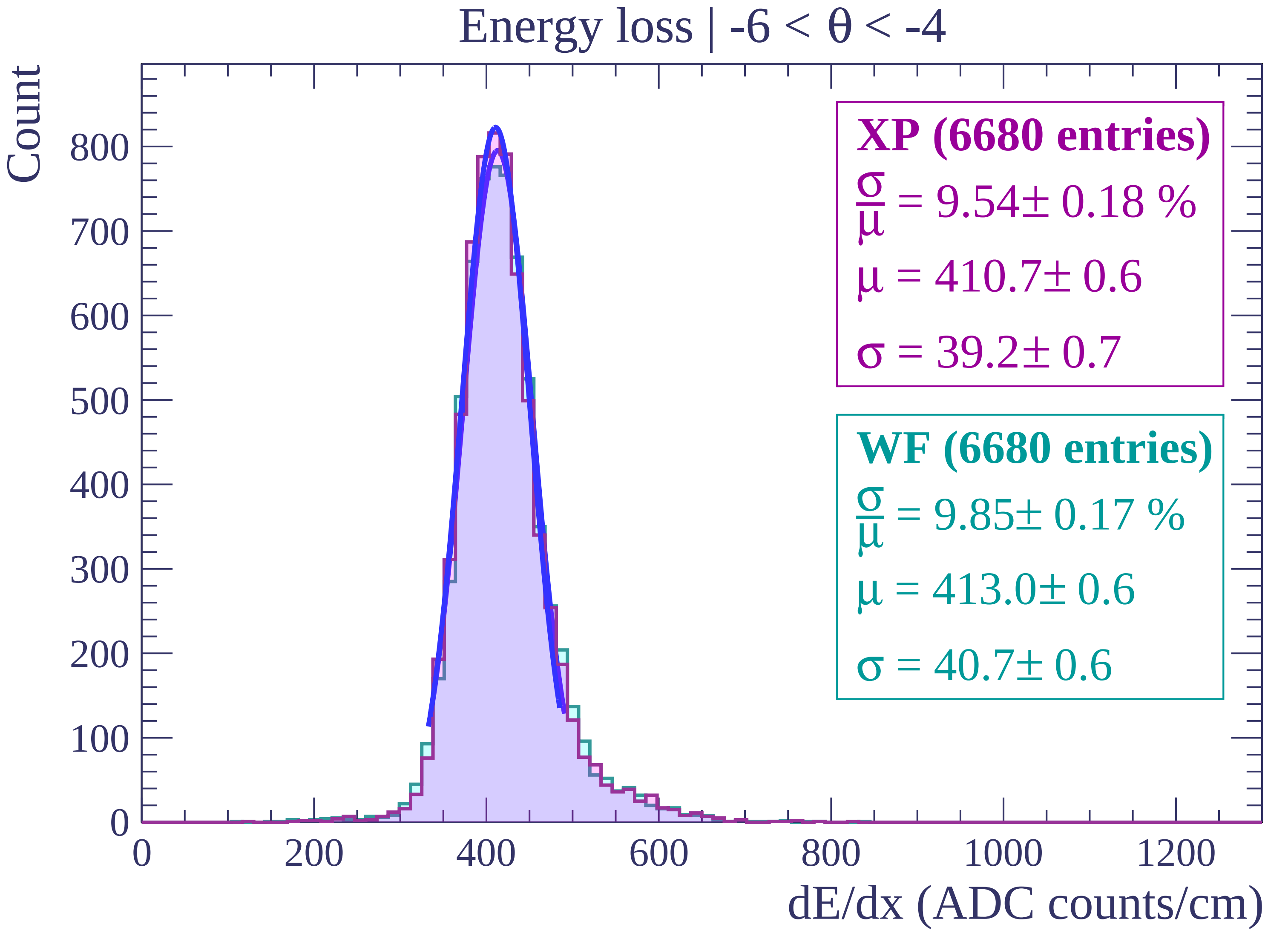
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4233 entries)
σ/μ = 9.63± 0.20 %
μ = 413.3± 0.8
σ = 39.8± 0.8
WF (4233 entries)
σ/μ = 9.90± 0.20 %
μ = 415.4± 0.8
σ = 41.1± 0.7


Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4775 entries)
σ/μ = 10.06± 0.21 %
μ = 414.4± 0.8
σ = 41.7± 0.8
WF (4775 entries)
σ/μ = 10.63± 0.20 %
μ = 417.1± 0.8
σ = 44.3± 0.7

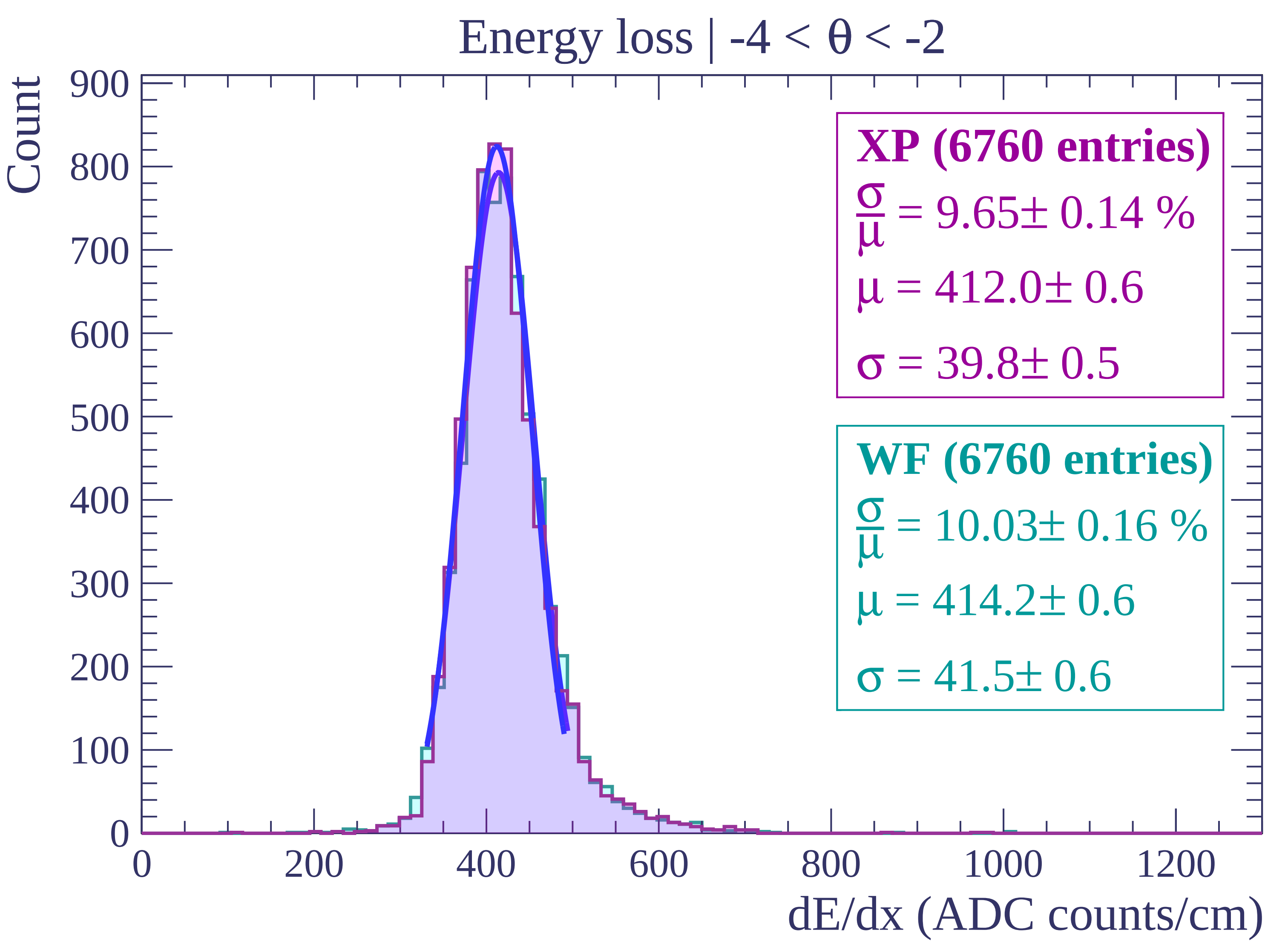

Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5224 entries)
σ/μ = 9.03± 0.18 %
μ = 413.2± 0.7
σ = 37.3± 0.7
WF (5224 entries)
σ/μ = 9.76± 0.19 %
μ = 415.6± 0.7
σ = 40.6± 0.7

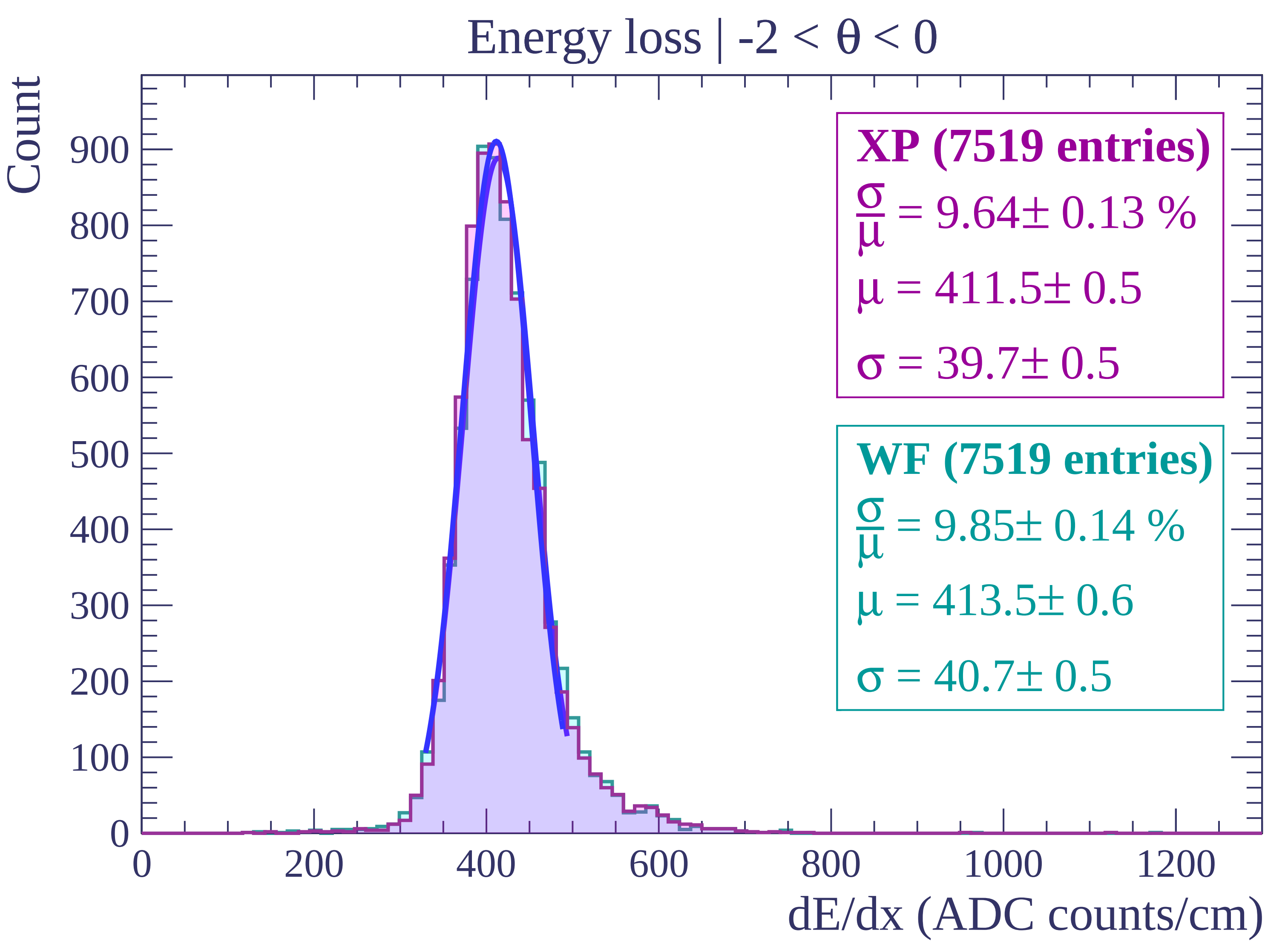

Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5778 entries)
σ/μ = 9.65± 0.17 %
μ = 413.4± 0.6
σ = 39.9± 0.7
WF (5778 entries)
σ/μ = 9.79± 0.18 %
μ = 414.4± 0.7
σ = 40.6± 0.7


Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6048 entries)
σ/μ = 9.20± 0.17 %
μ = 412.4± 0.6
σ = 37.9± 0.7
WF (6048 entries)
σ/μ = 9.65± 0.17 %
μ = 414.5± 0.6
σ = 40.0± 0.7

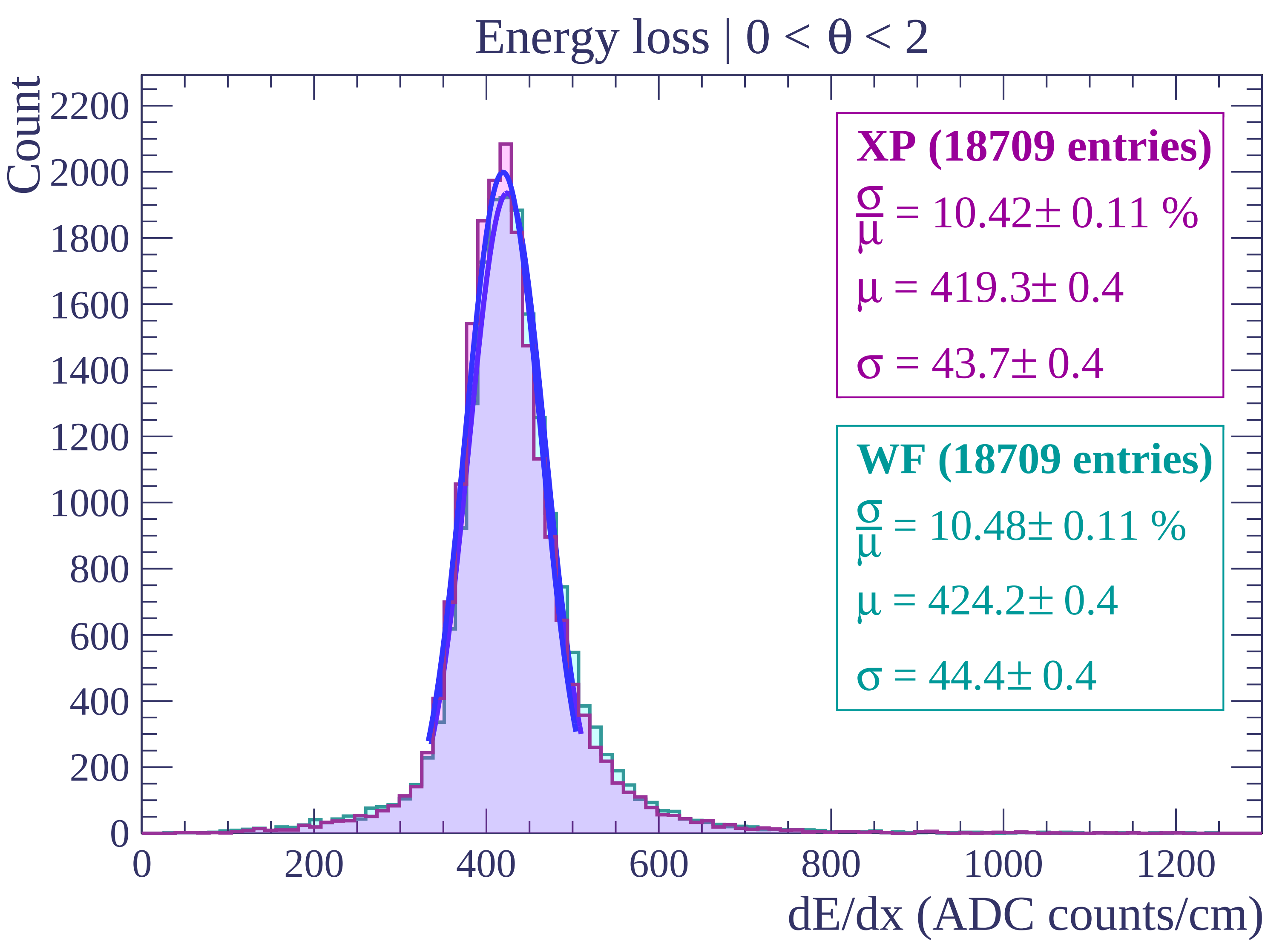


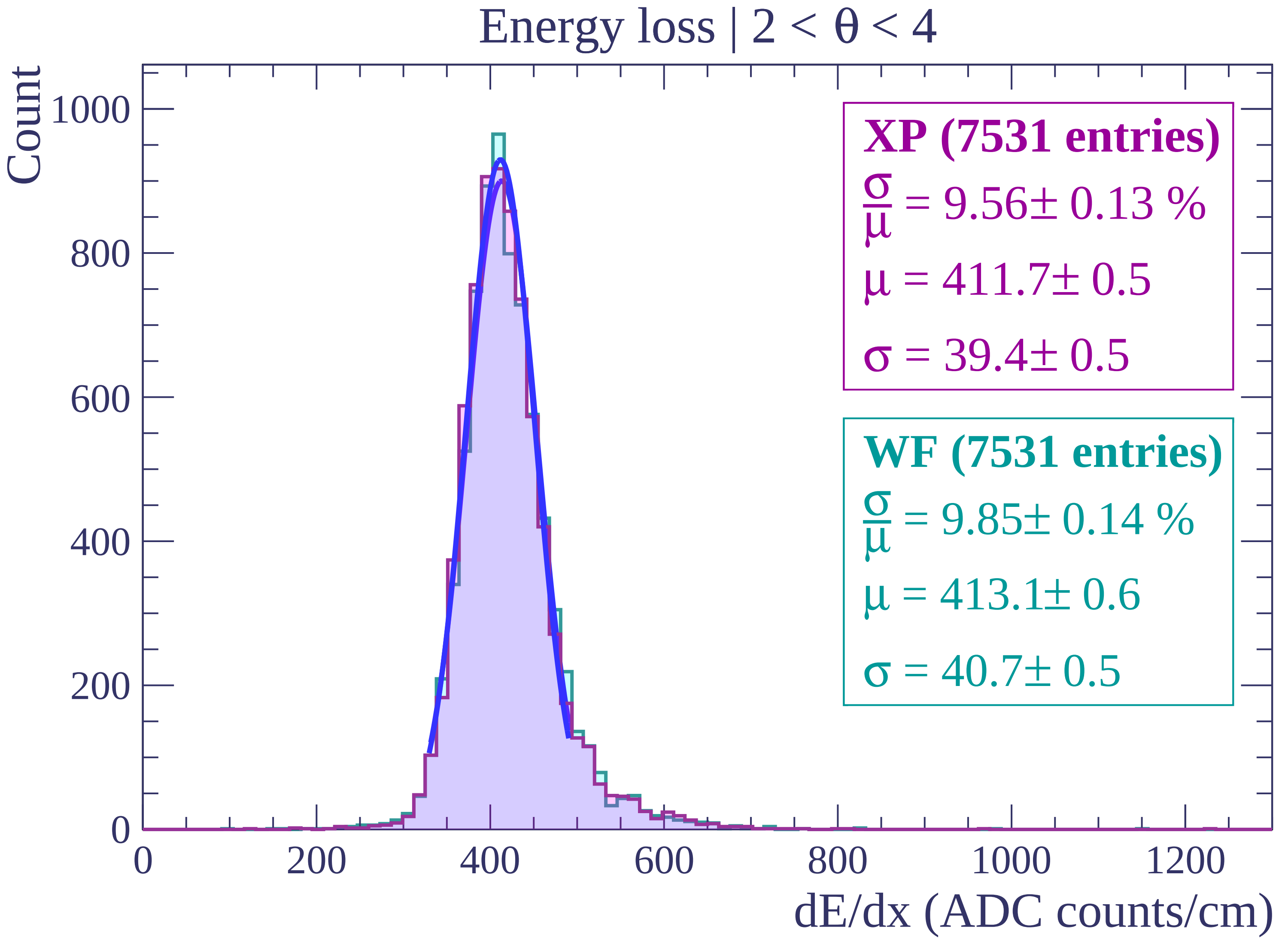
Energy loss | -6 < θ < -4
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6680 entries)
σ/μ = 9.54± 0.18 %
μ = 410.7± 0.6
σ = 39.2± 0.7
WF (6680 entries)
σ/μ = 9.85± 0.17 %
μ = 413.0± 0.6
σ = 40.7± 0.6

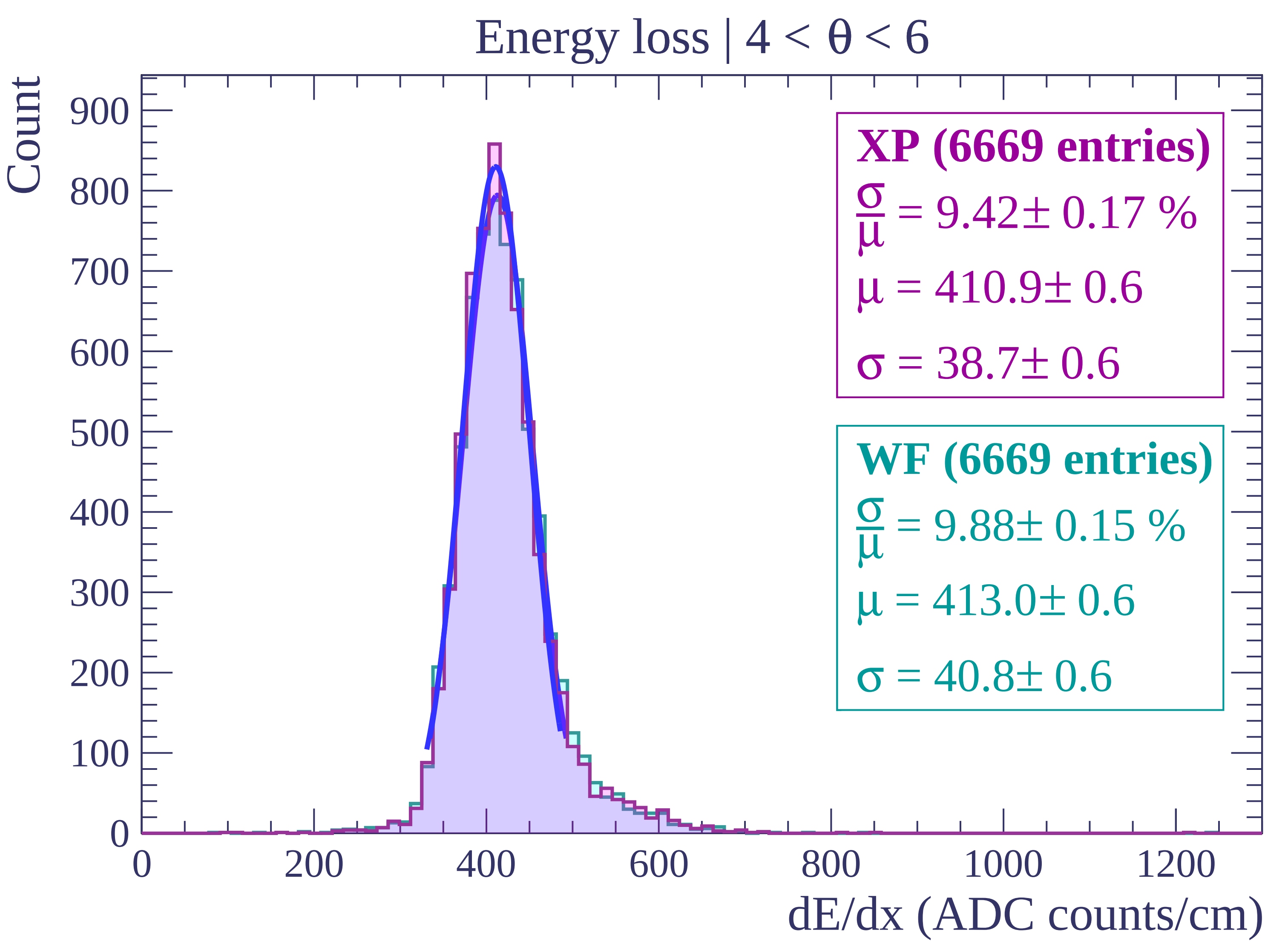

Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
XP (6760 entries)
σ/μ = 9.65± 0.14 %
μ = 412.0± 0.6
σ = 39.8± 0.5
WF (6760 entries)
σ/μ = 10.03± 0.16 %
μ = 414.2± 0.6
σ = 41.5± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

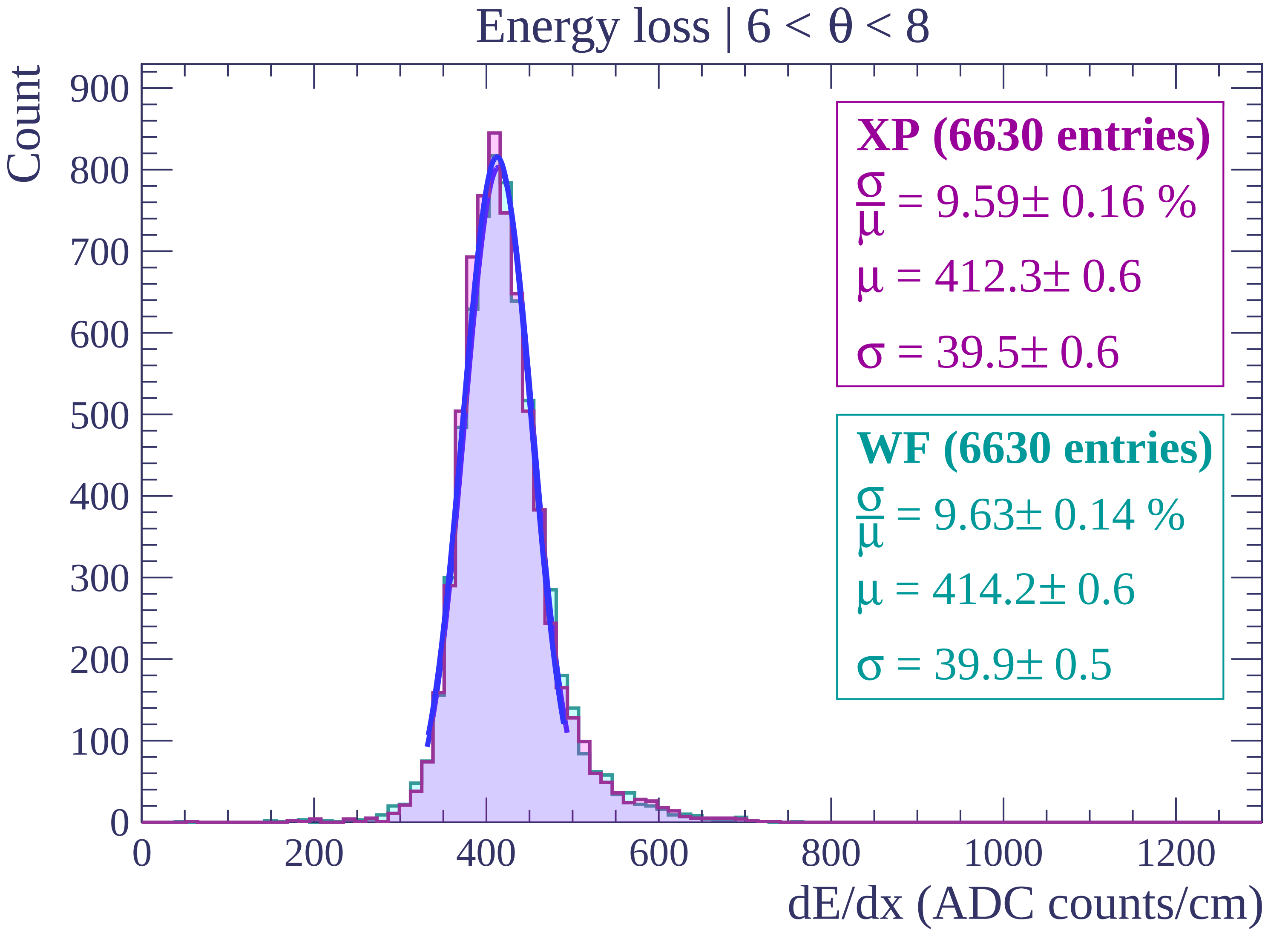


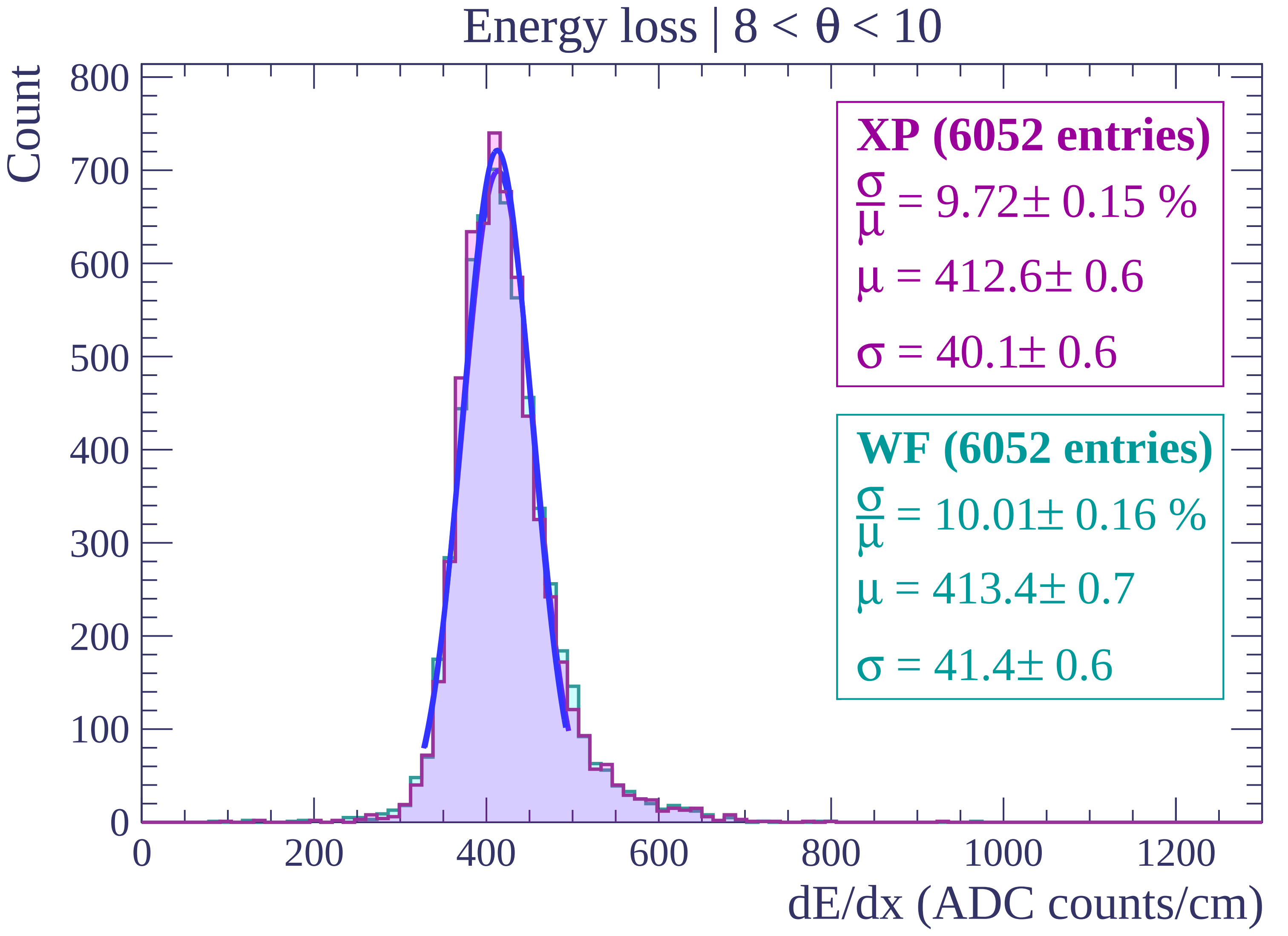
Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7519 entries)
σ/μ = 9.64± 0.13 %
μ = 411.5± 0.5
σ = 39.7± 0.5
WF (7519 entries)
σ/μ = 9.85± 0.14 %
μ = 413.5± 0.6
σ = 40.7± 0.5


Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP (18709 entries)
σ/μ = 10.42± 0.11 %
μ = 419.3± 0.4
σ = 43.7± 0.4
WF (18709 entries)
σ/μ = 10.48± 0.11 %
μ = 424.2± 0.4
σ = 44.4± 0.4

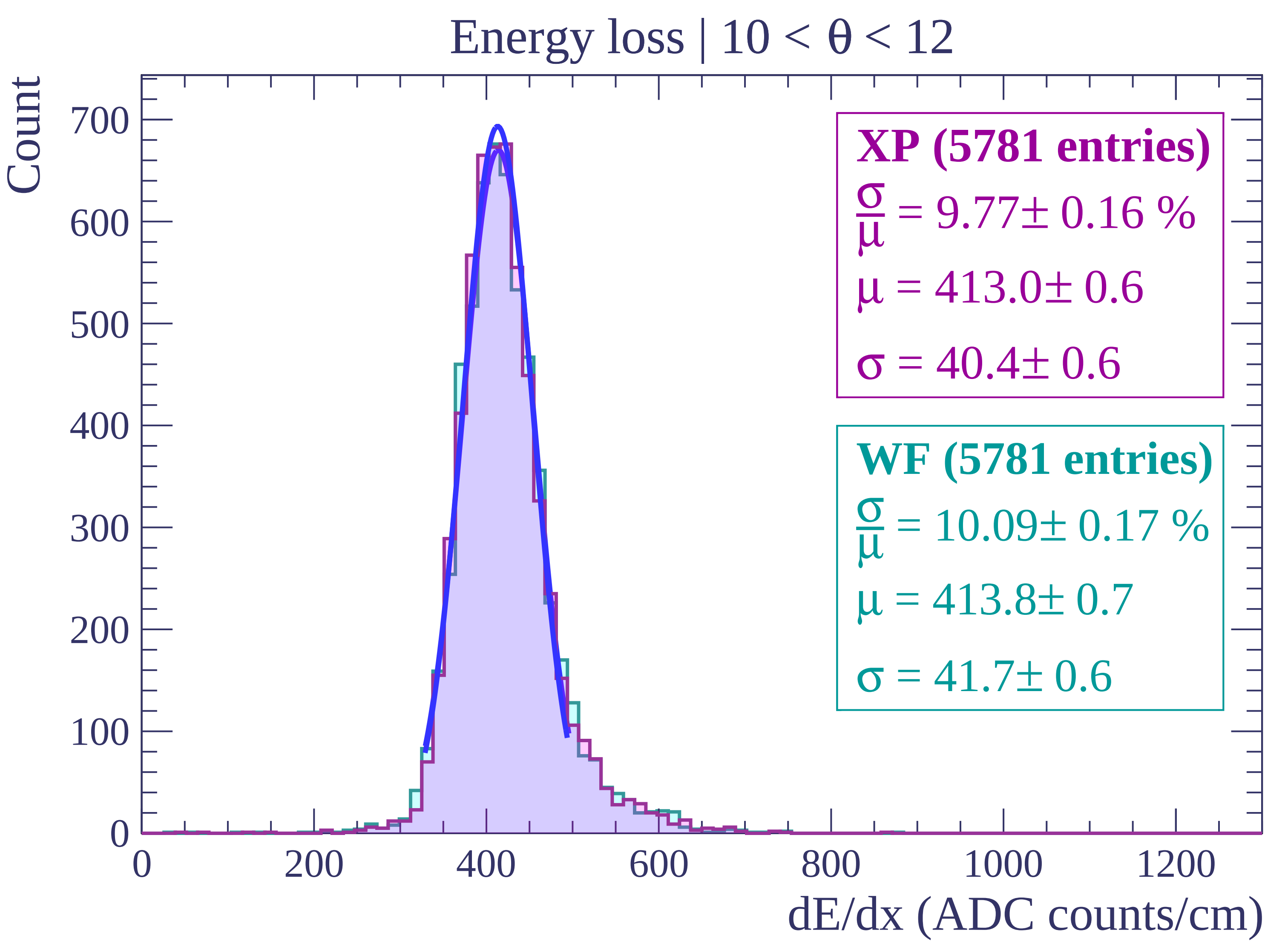

Energy loss | 2 < θ < 4
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7531 entries)
σ/μ = 9.56± 0.13 %
μ = 411.7± 0.5
σ = 39.4± 0.5
WF (7531 entries)
σ/μ = 9.85± 0.14 %
μ = 413.1± 0.6
σ = 40.7± 0.5


Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6669 entries)
σ/μ = 9.42± 0.17 %
μ = 410.9± 0.6
σ = 38.7± 0.6
WF (6669 entries)
σ/μ = 9.88± 0.15 %
μ = 413.0± 0.6
σ = 40.8± 0.6

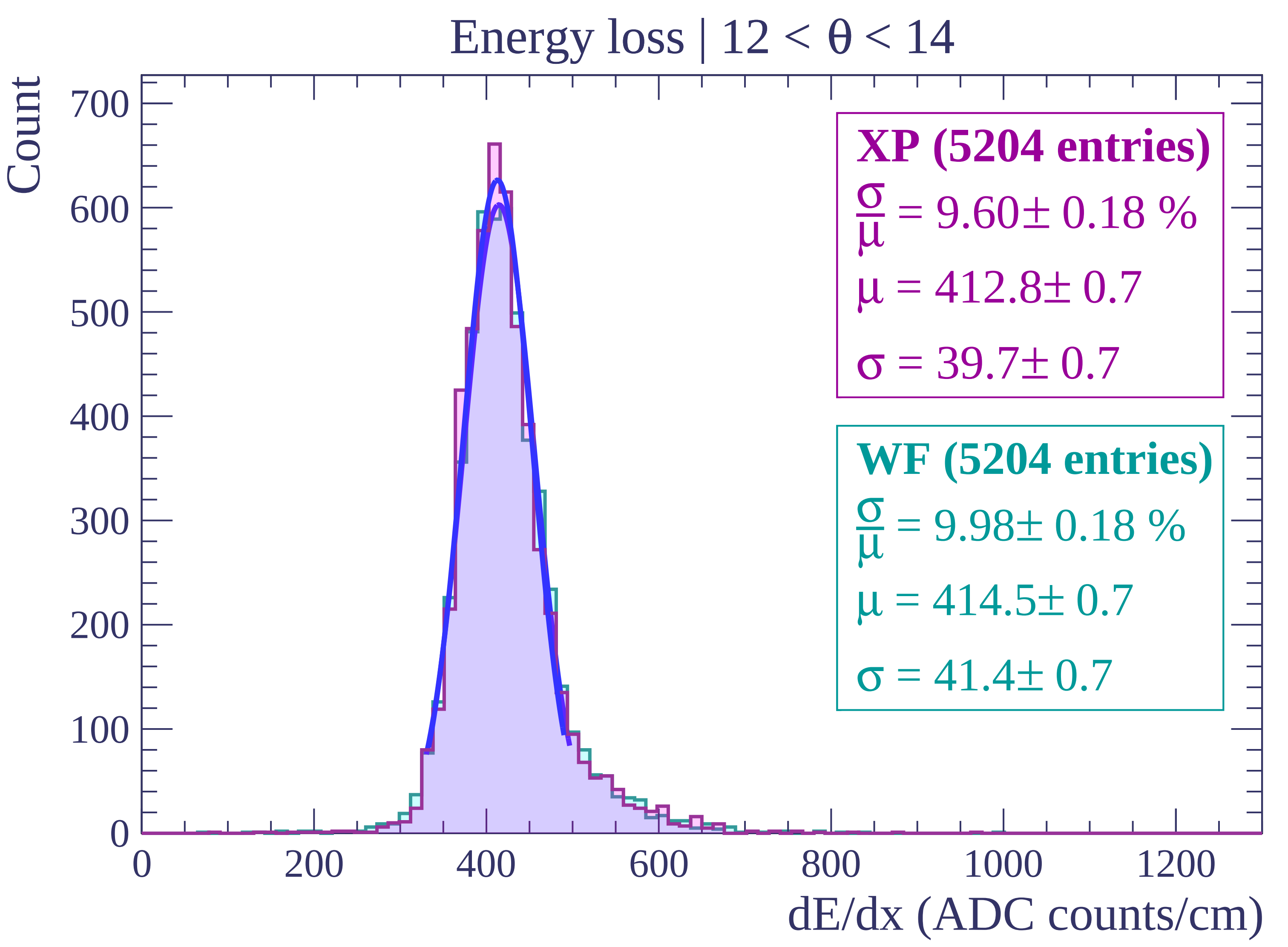

Energy loss | 6 < θ < 8
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6630 entries)
σ/μ = 9.59± 0.16 %
μ = 412.3± 0.6
σ = 39.5± 0.6
WF (6630 entries)
σ/μ = 9.63± 0.14 %
μ = 414.2± 0.6
σ = 39.9± 0.5



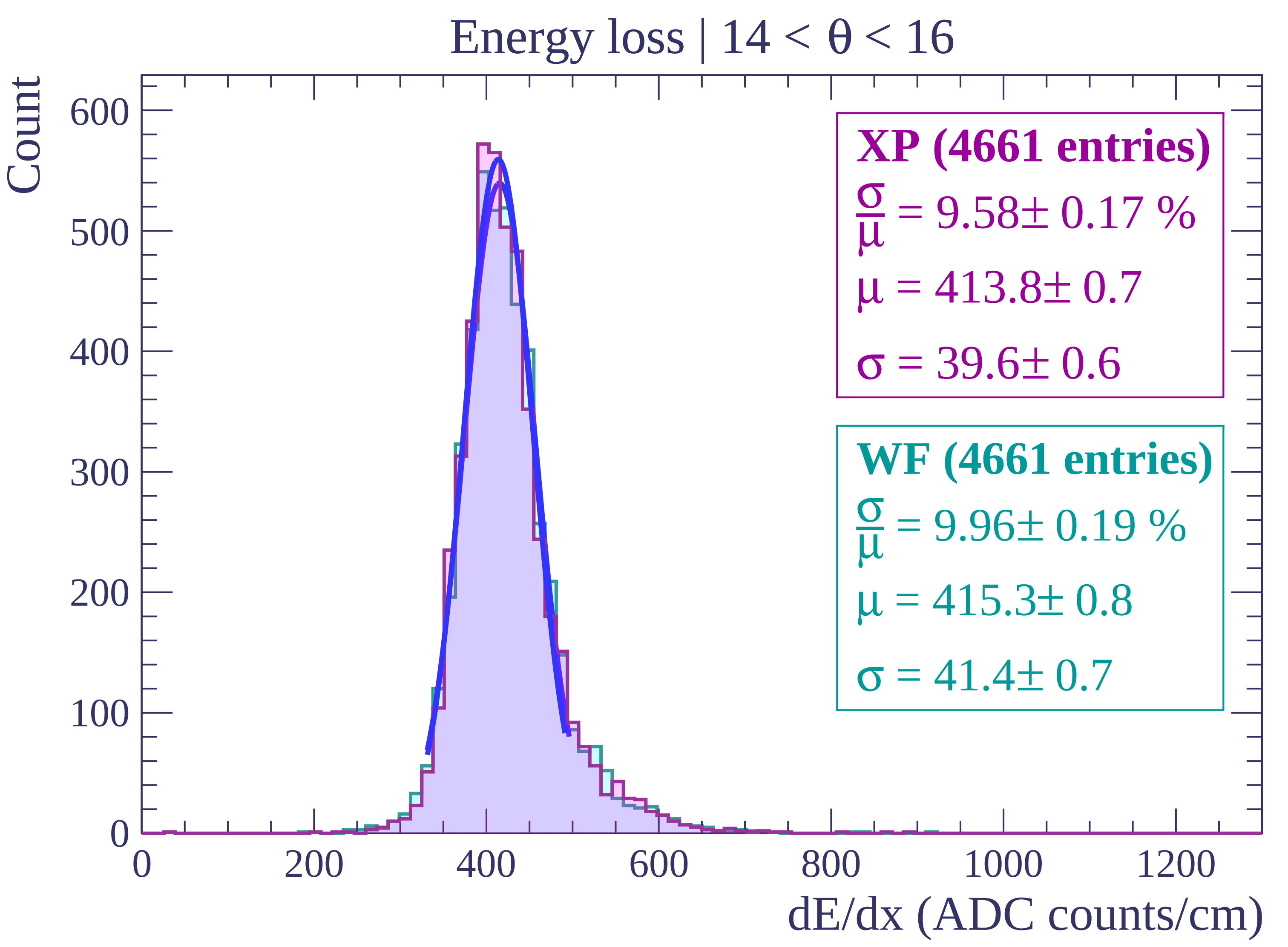
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6052 entries)
σ/μ = 9.72± 0.15 %
μ = 412.6± 0.6
σ = 40.1± 0.6
WF (6052 entries)
σ/μ = 10.01± 0.16 %
μ = 413.4± 0.7
σ = 41.4± 0.6



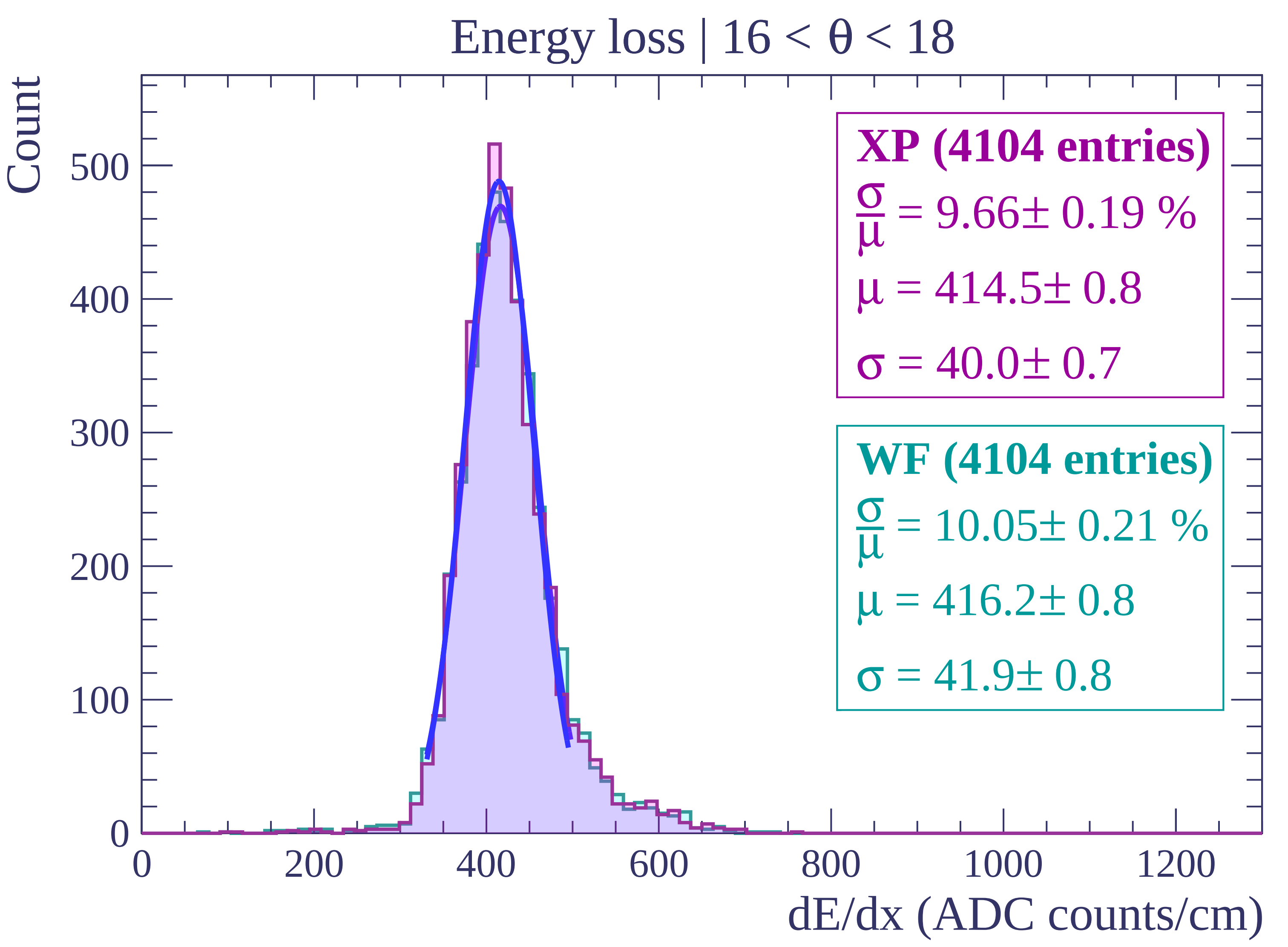
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5781 entries)
σ/μ = 9.77± 0.16 %
μ = 413.0± 0.6
σ = 40.4± 0.6
WF (5781 entries)
σ/μ = 10.09± 0.17 %
μ = 413.8± 0.7
σ = 41.7± 0.6



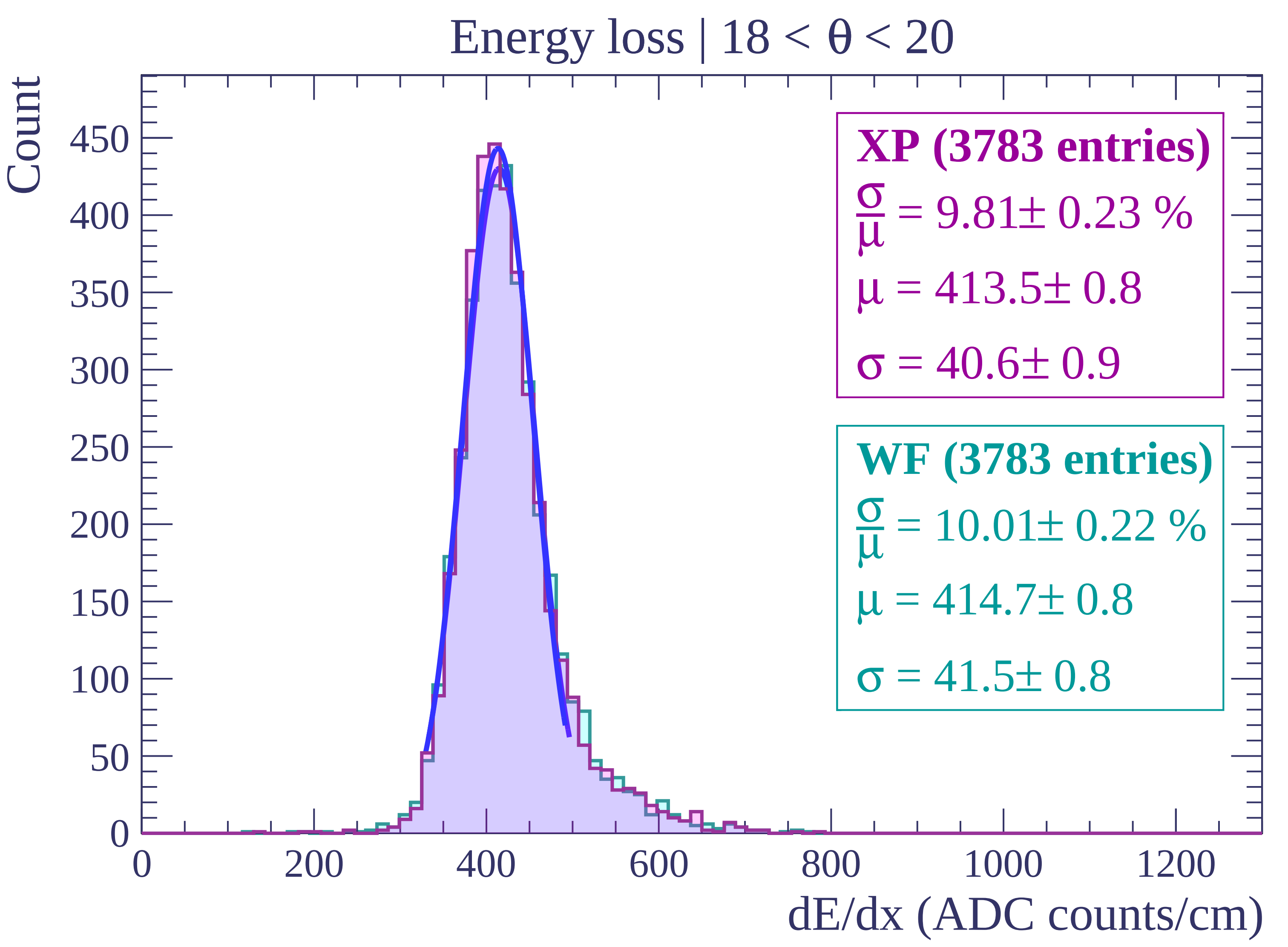
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5204 entries)
σ/μ = 9.60± 0.18 %
μ = 412.8± 0.7
σ = 39.7± 0.7
WF (5204 entries)
σ/μ = 9.98± 0.18 %
μ = 414.5± 0.7
σ = 41.4± 0.7

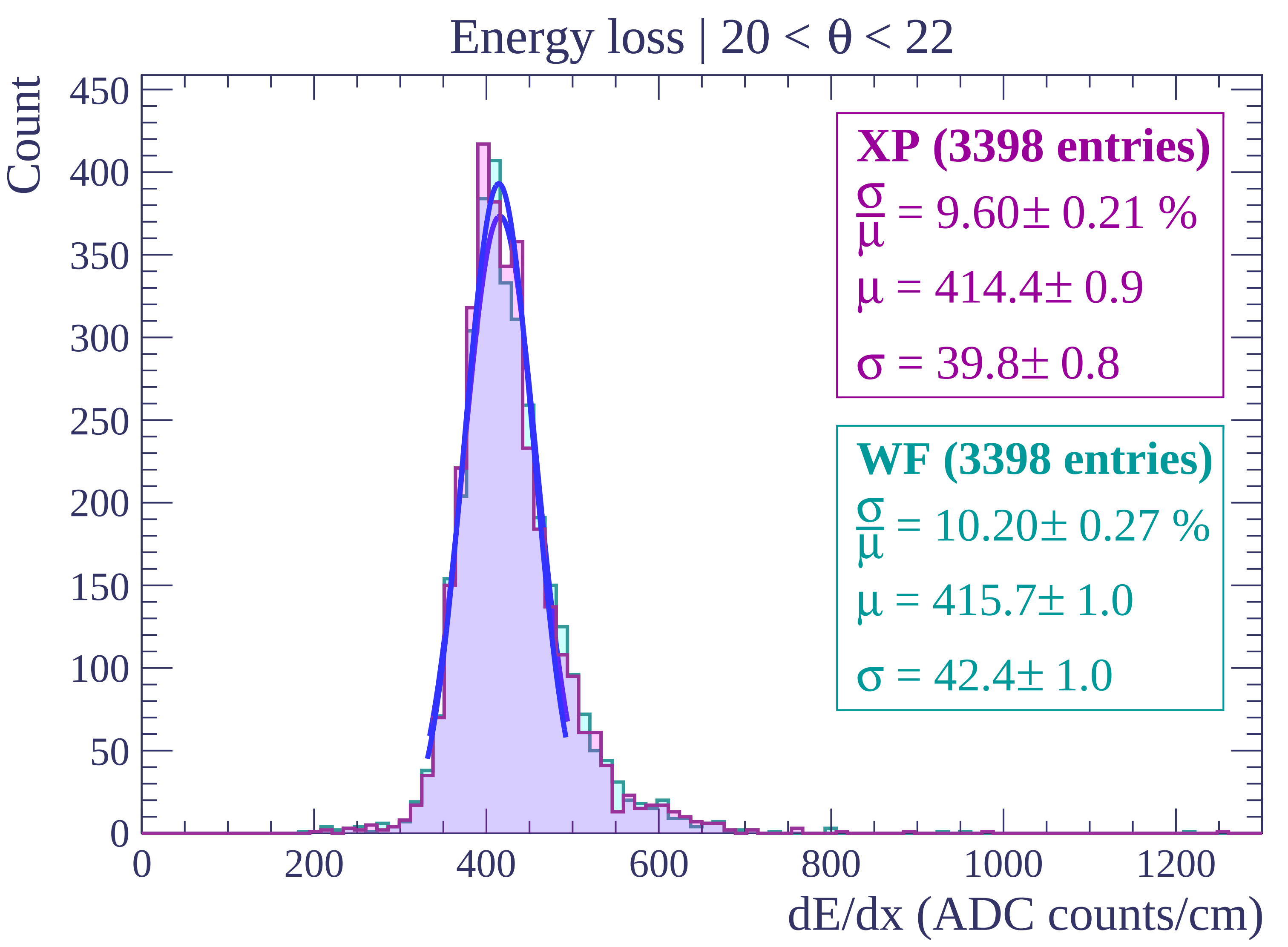


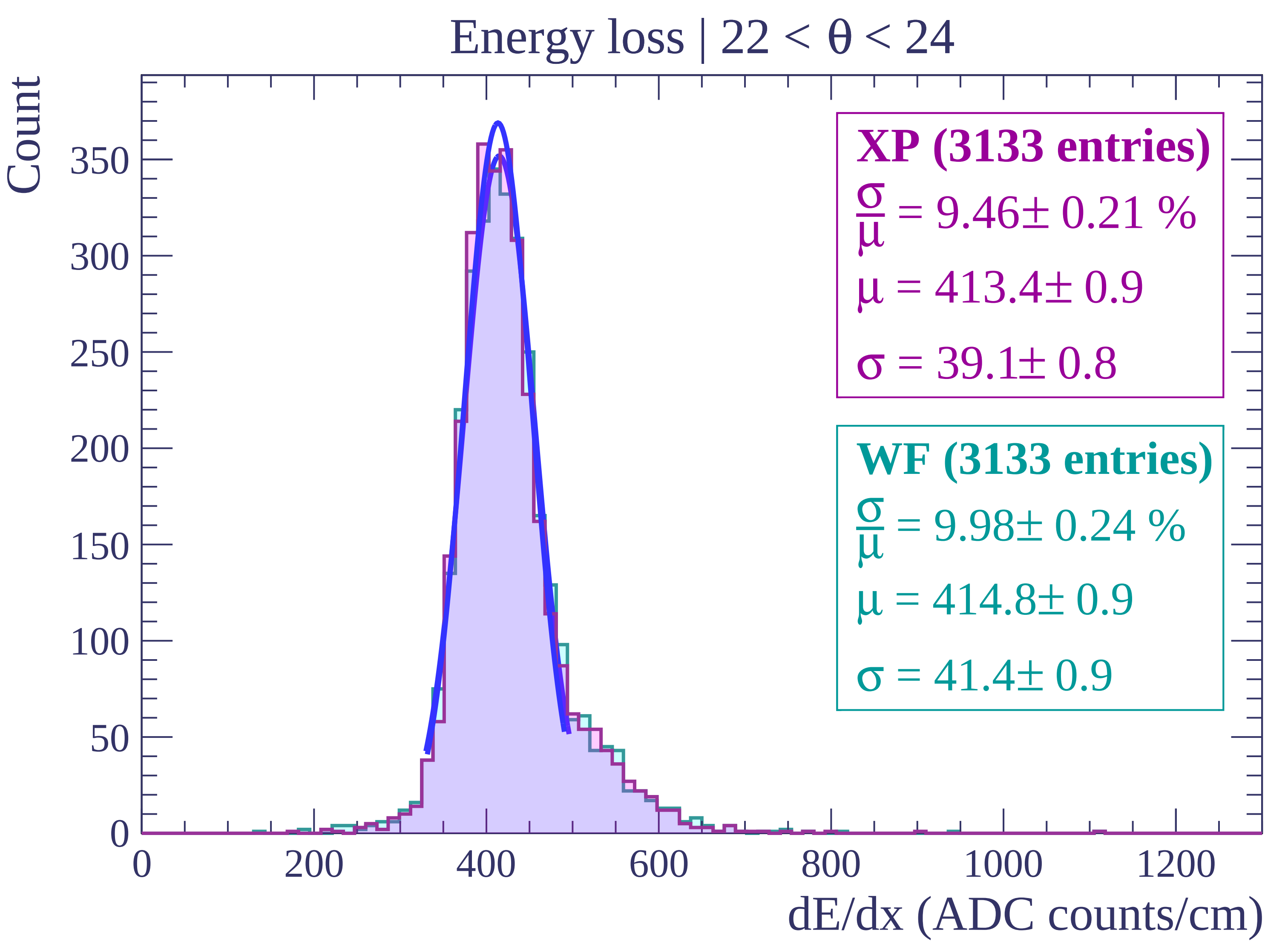
Energy loss | 14 < θ < 16
Count
XP (4661 entries)
σ/μ = 9.58± 0.17 %
μ = 413.8± 0.7
σ = 39.6± 0.6
WF (4661 entries)
σ/μ = 9.96± 0.19 %
μ = 415.3± 0.8
σ = 41.4± 0.7
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



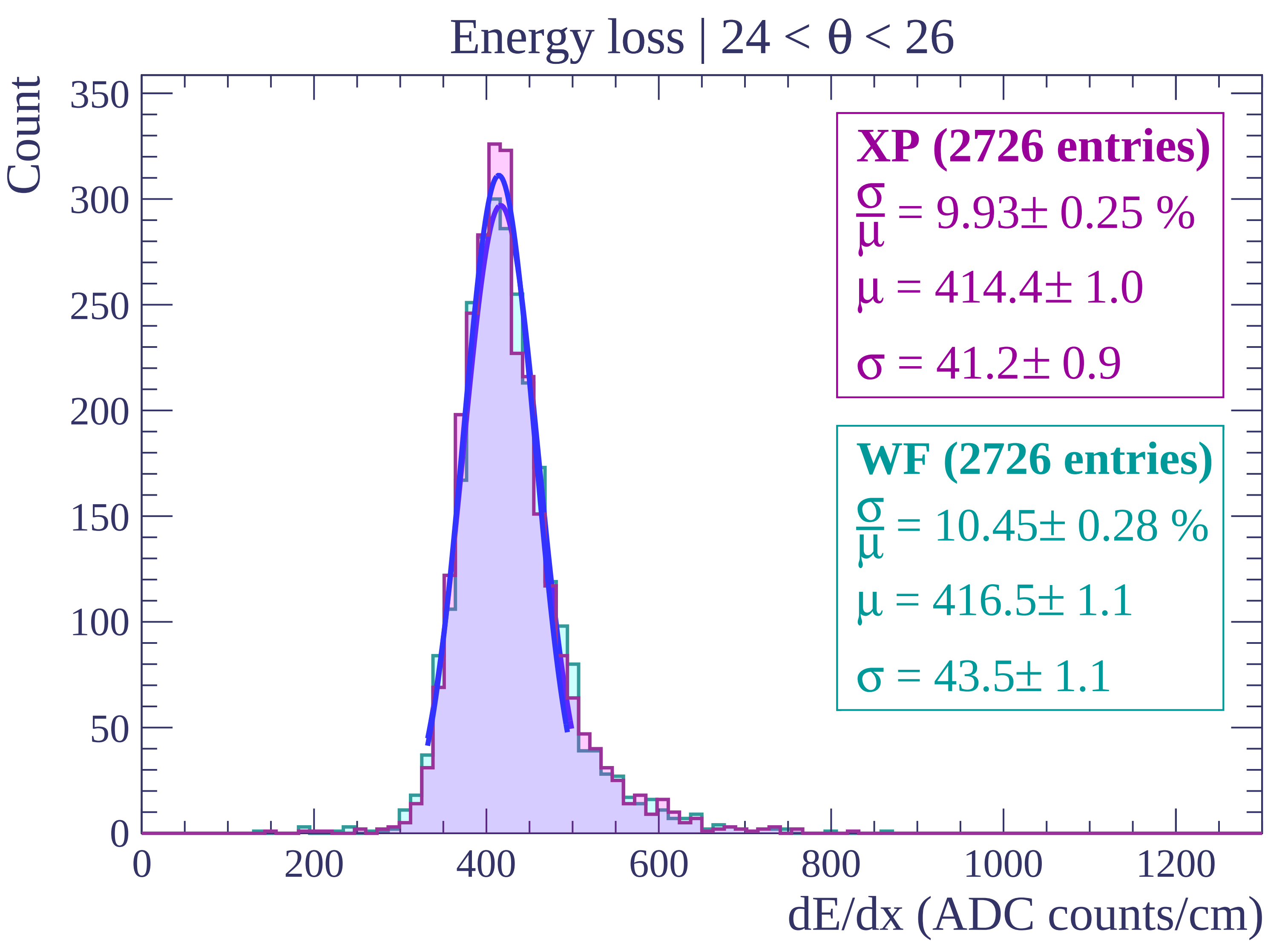
Energy loss | 16 < θ < 18
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4104 entries)
σ/μ = 9.66± 0.19 %
μ = 414.5± 0.8
σ = 40.0± 0.7
WF (4104 entries)
σ/μ = 10.05± 0.21 %
μ = 416.2± 0.8
σ = 41.9± 0.8

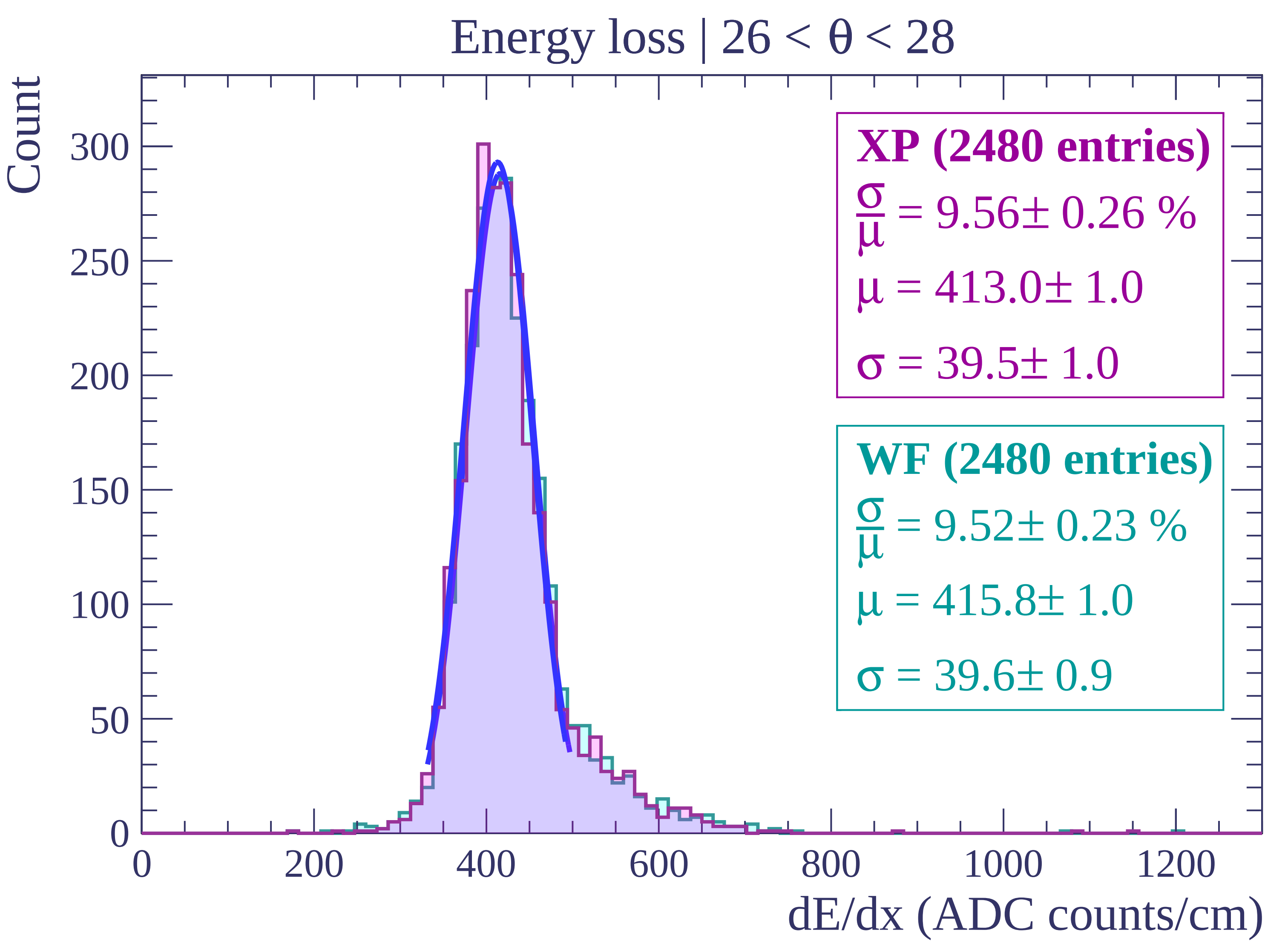

Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP (3783 entries)
σ/μ = 9.81± 0.23 %
μ = 413.5± 0.8
σ = 40.6± 0.9
WF (3783 entries)
σ/μ = 10.01± 0.22 %
μ = 414.7± 0.8
σ = 41.5± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200


Energy loss | 20 < θ < 22
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3398 entries)
σ/μ = 9.60± 0.21 %
μ = 414.4± 0.9
σ = 39.8± 0.8
WF (3398 entries)
σ/μ = 10.20± 0.27 %
μ = 415.7± 1.0
σ = 42.4± 1.0

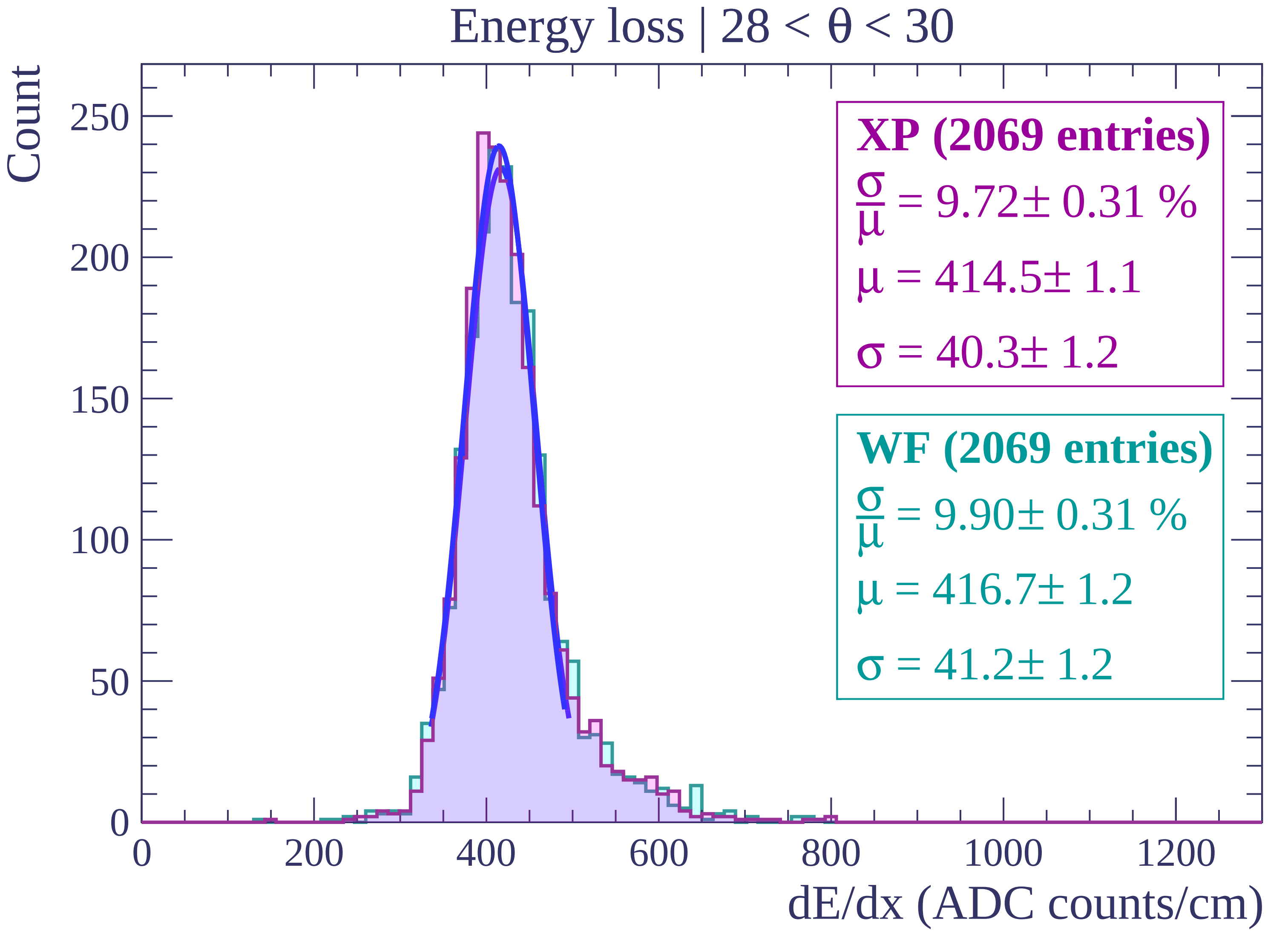


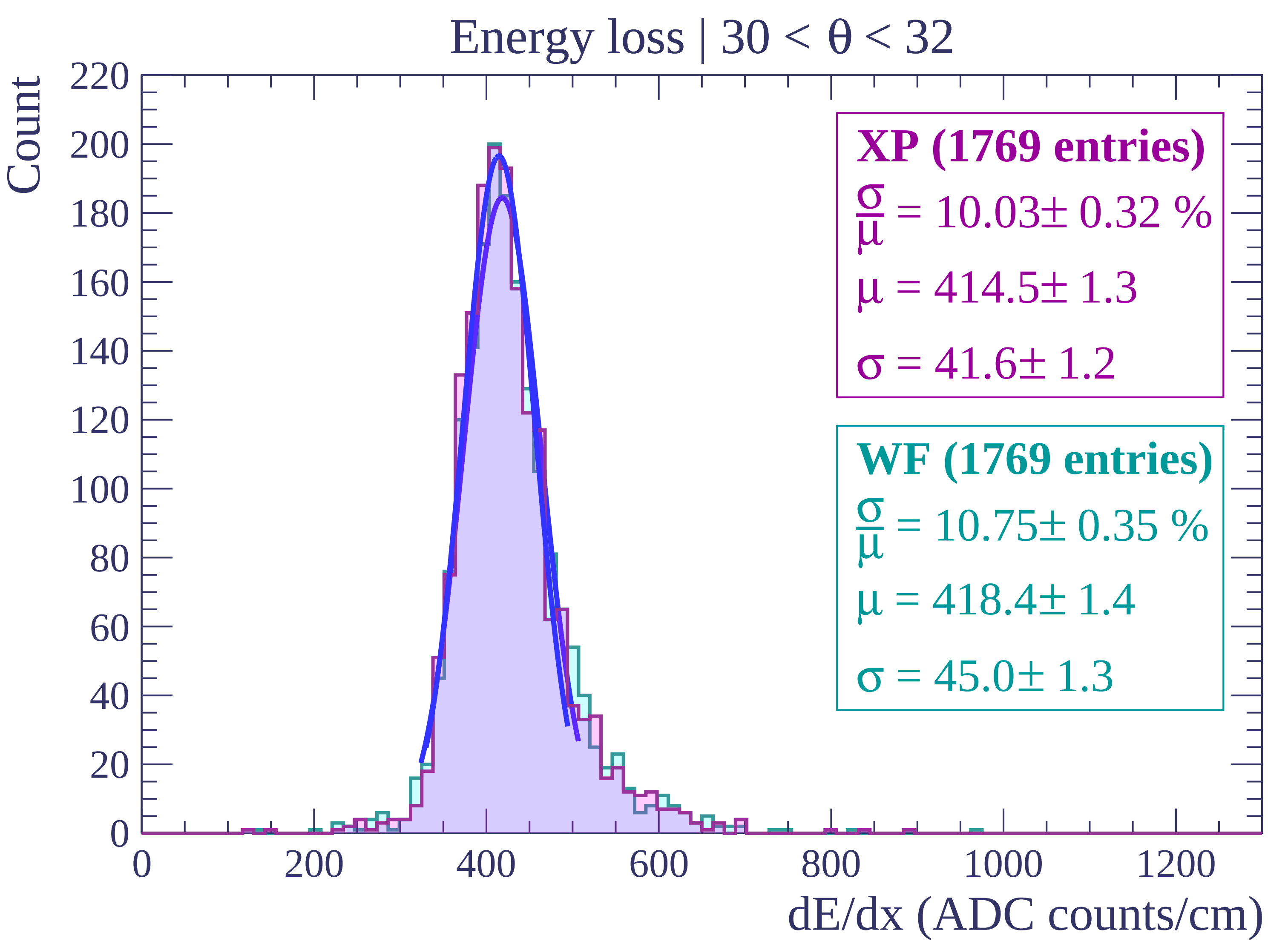
Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3133 entries)
σ/μ = 9.46± 0.21 %
μ = 413.4± 0.9
σ = 39.1± 0.8
WF (3133 entries)
σ/μ = 9.98± 0.24 %
μ = 414.8± 0.9
σ = 41.4± 0.9

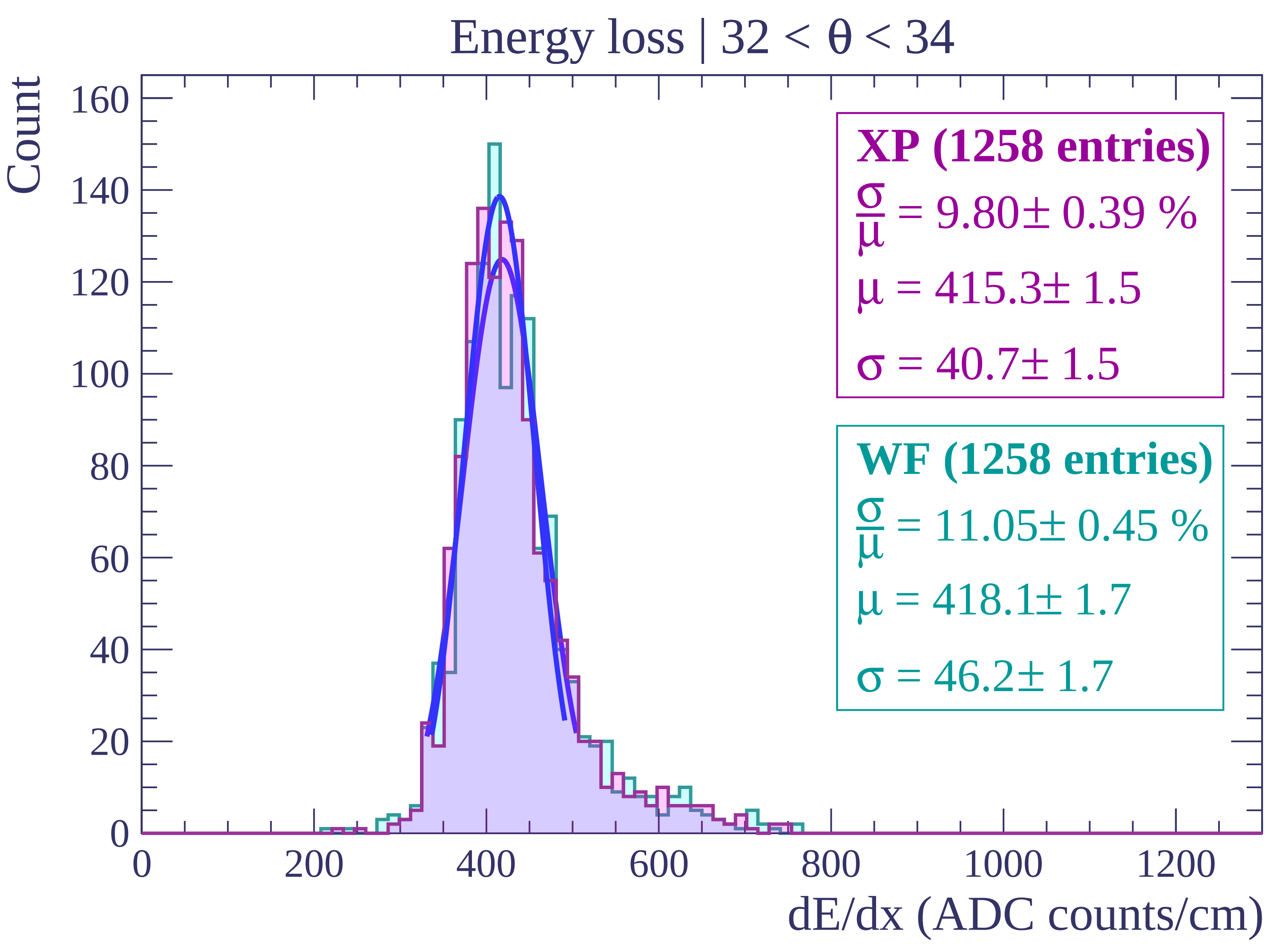

Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP (2726 entries)
σ/μ = 9.93± 0.25 %
μ = 414.4± 1.0
σ = 41.2± 0.9
WF (2726 entries)
σ/μ = 10.45± 0.28 %
μ = 416.5± 1.1
σ = 43.5± 1.1



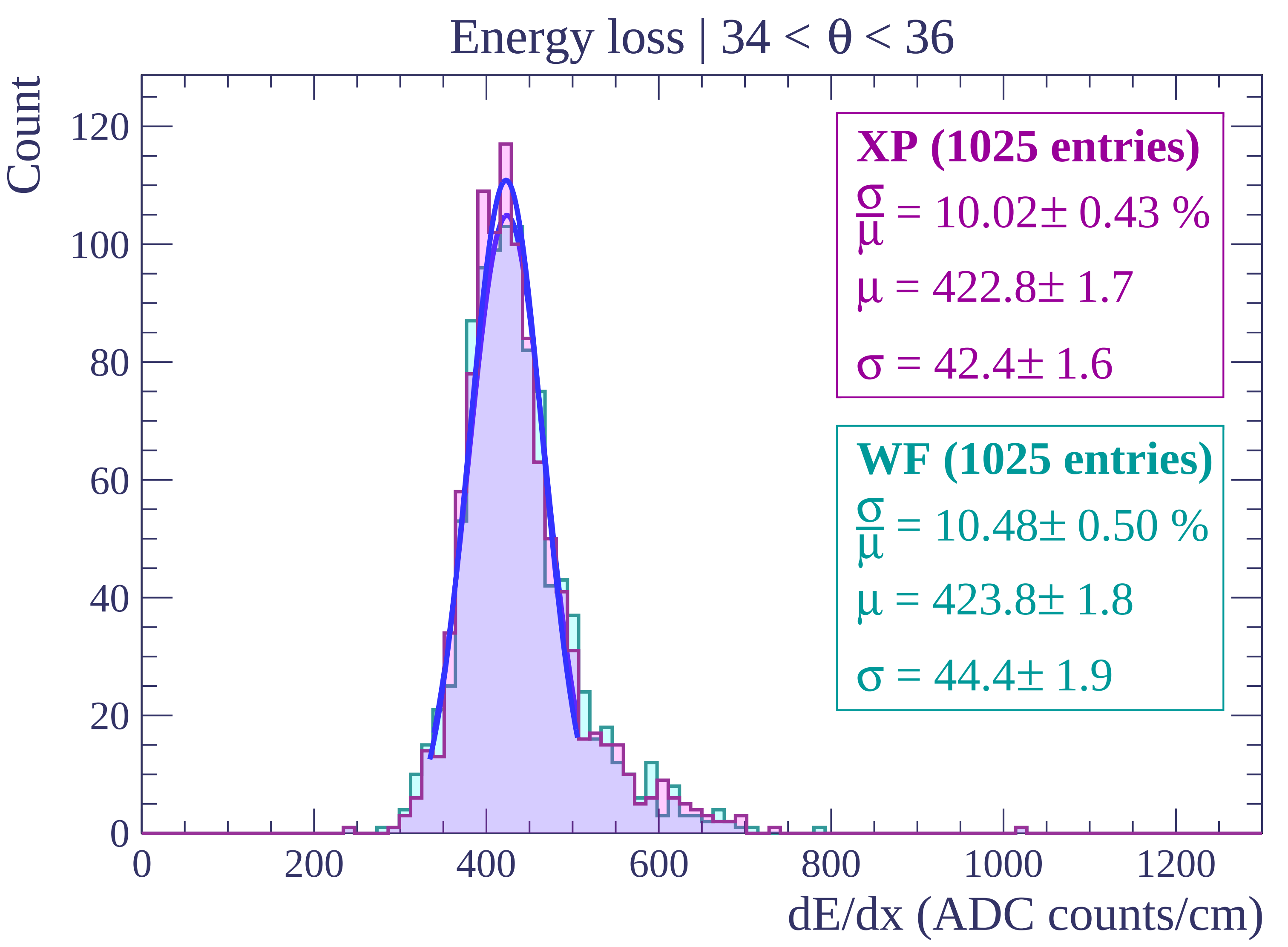
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2480 entries)
$\frac{\sigma}{\mu}$ = 9.56± 0.26 %
μ = 413.0± 1.0
σ = 39.5± 1.0
WF (2480 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.23 %
μ = 415.8± 1.0
σ = 39.6± 0.9


Energy loss | 28 < θ < 30
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2069 entries)
σ/μ = 9.72± 0.31 %
μ = 414.5± 1.1
σ = 40.3± 1.2
WF (2069 entries)
σ/μ = 9.90± 0.31 %
μ = 416.7± 1.2
σ = 41.2± 1.2

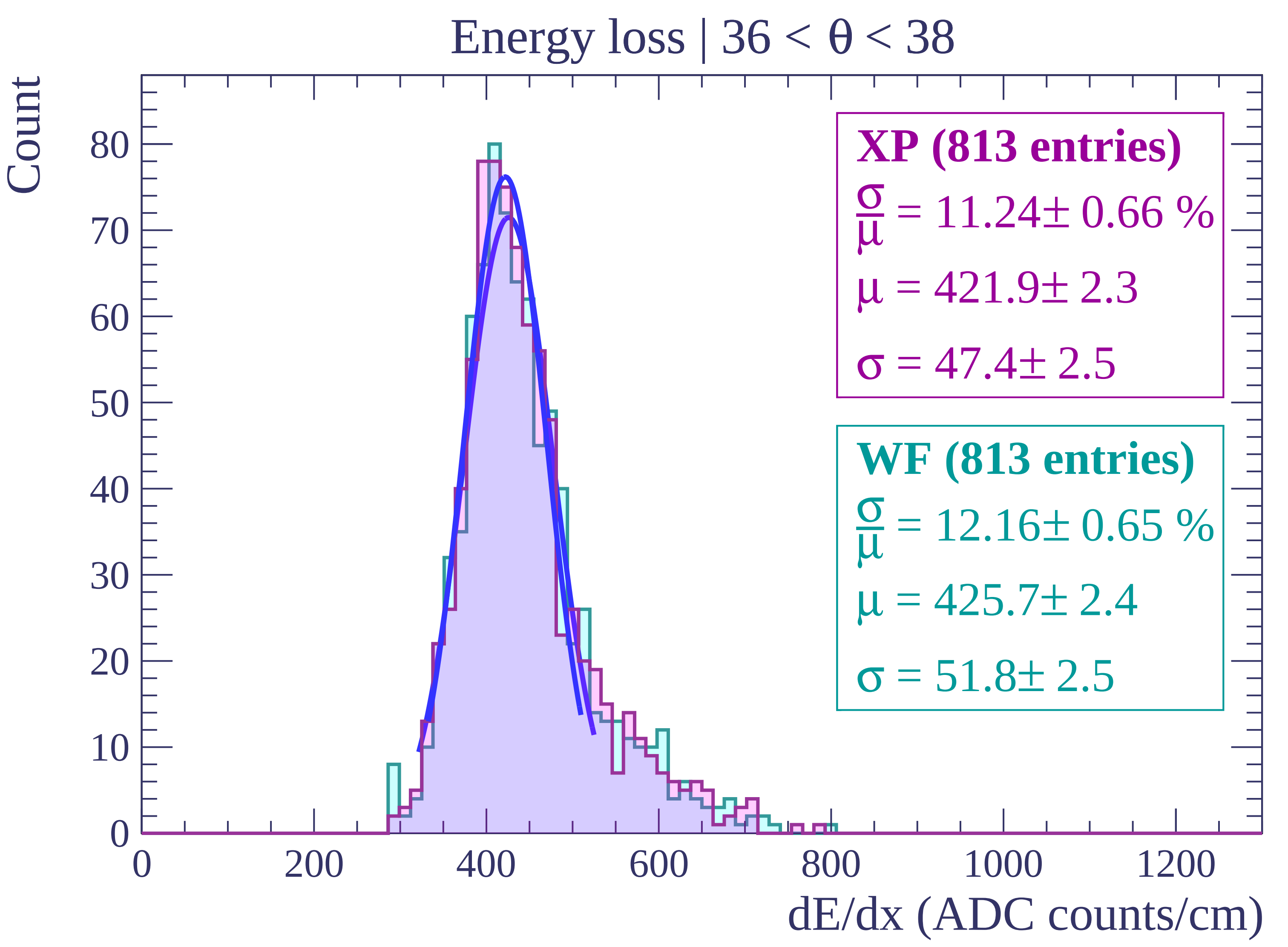

Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
XP (1769 entries)
σ/μ = 10.03± 0.32 %
μ = 414.5± 1.3
σ = 41.6± 1.2
WF (1769 entries)
σ/μ = 10.75± 0.35 %
μ = 418.4± 1.4
σ = 45.0± 1.3



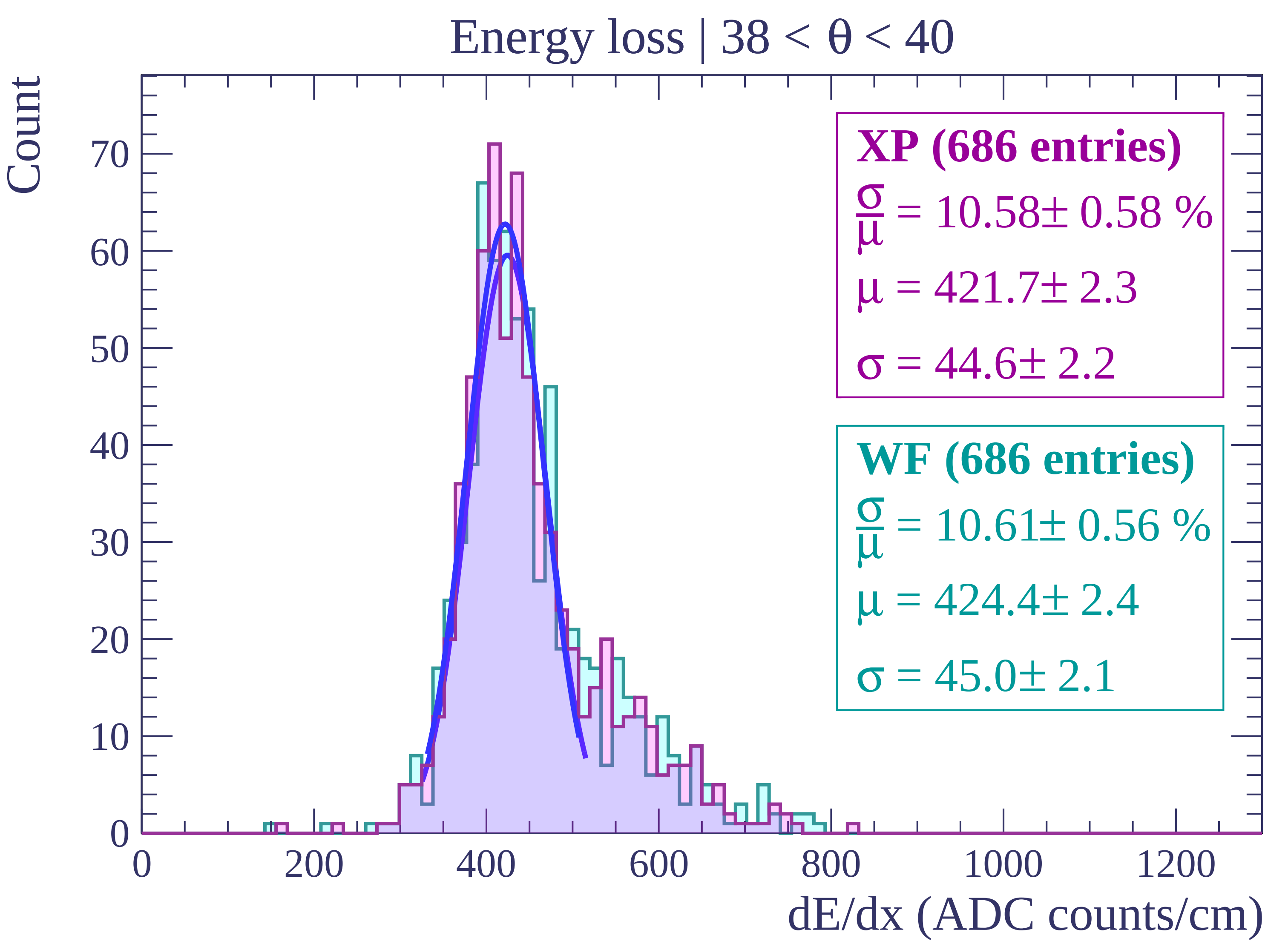
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1258 entries)
$\frac{\sigma}{\mu}$ = 9.80± 0.39 %
μ = 415.3± 1.5
σ = 40.7± 1.5
WF (1258 entries)
$\frac{\sigma}{\mu}$ = 11.05± 0.45 %
μ = 418.1± 1.7
σ = 46.2± 1.7



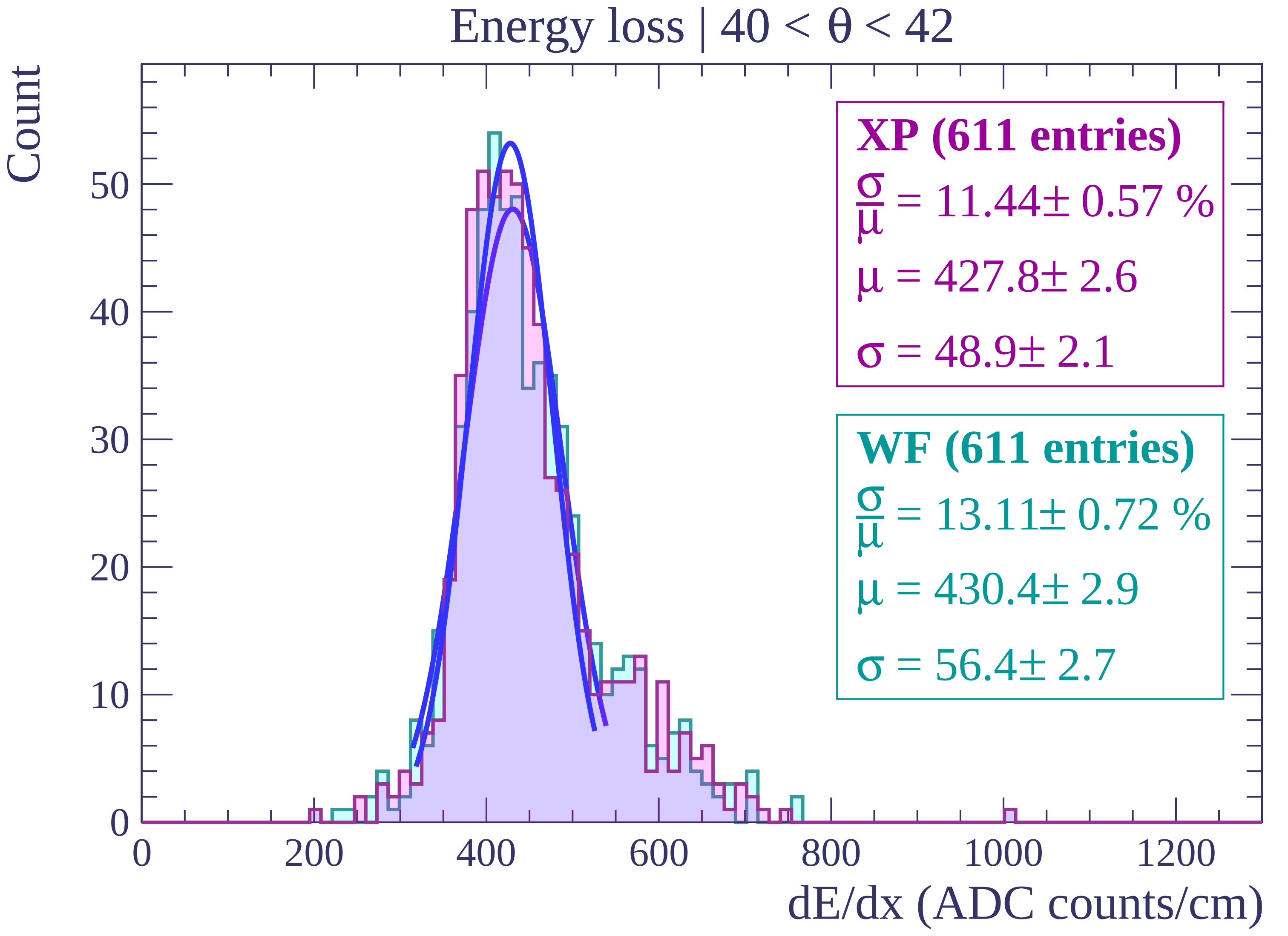
Energy loss | 34 < θ < 36
Count
XP (1025 entries)
σ/μ = 10.02± 0.43 %
μ = 422.8± 1.7
σ = 42.4± 1.6
WF (1025 entries)
σ/μ = 10.48± 0.50 %
μ = 423.8± 1.8
σ = 44.4± 1.9
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

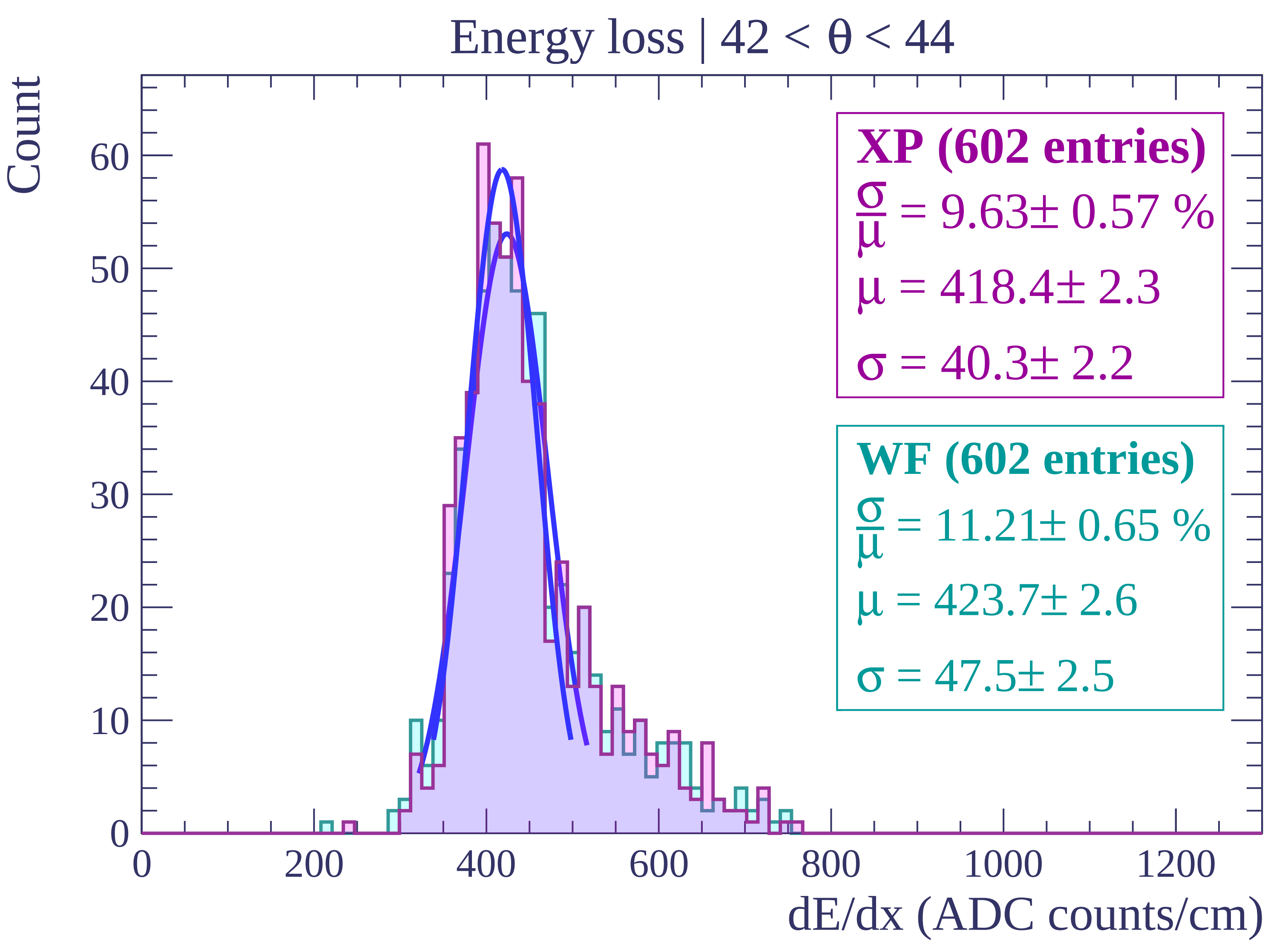


Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
XP (813 entries)
σ/μ = 11.24± 0.66 %
μ = 421.9± 2.3
σ = 47.4± 2.5
WF (813 entries)
σ/μ = 12.16± 0.65 %
μ = 425.7± 2.4
σ = 51.8± 2.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

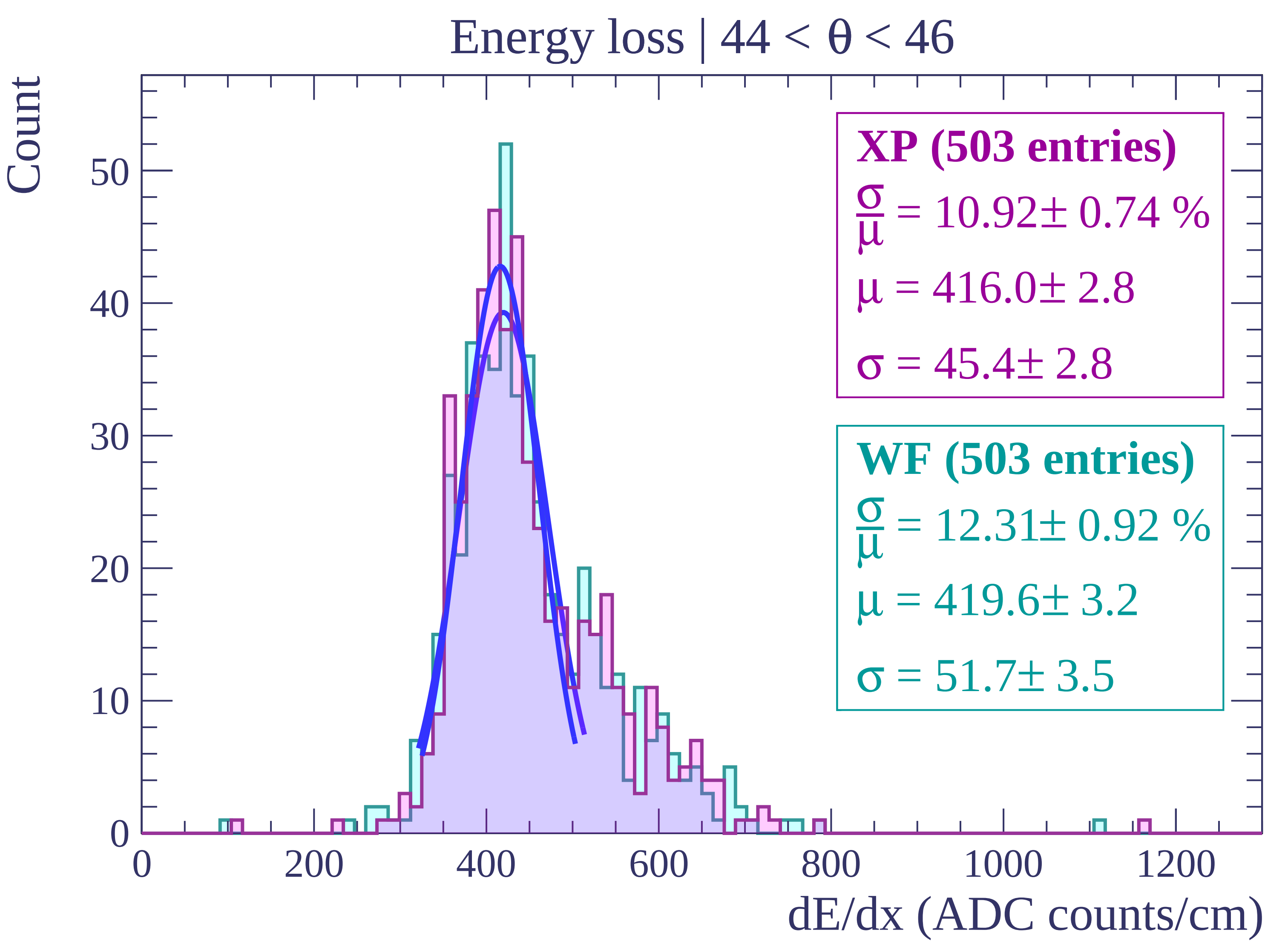


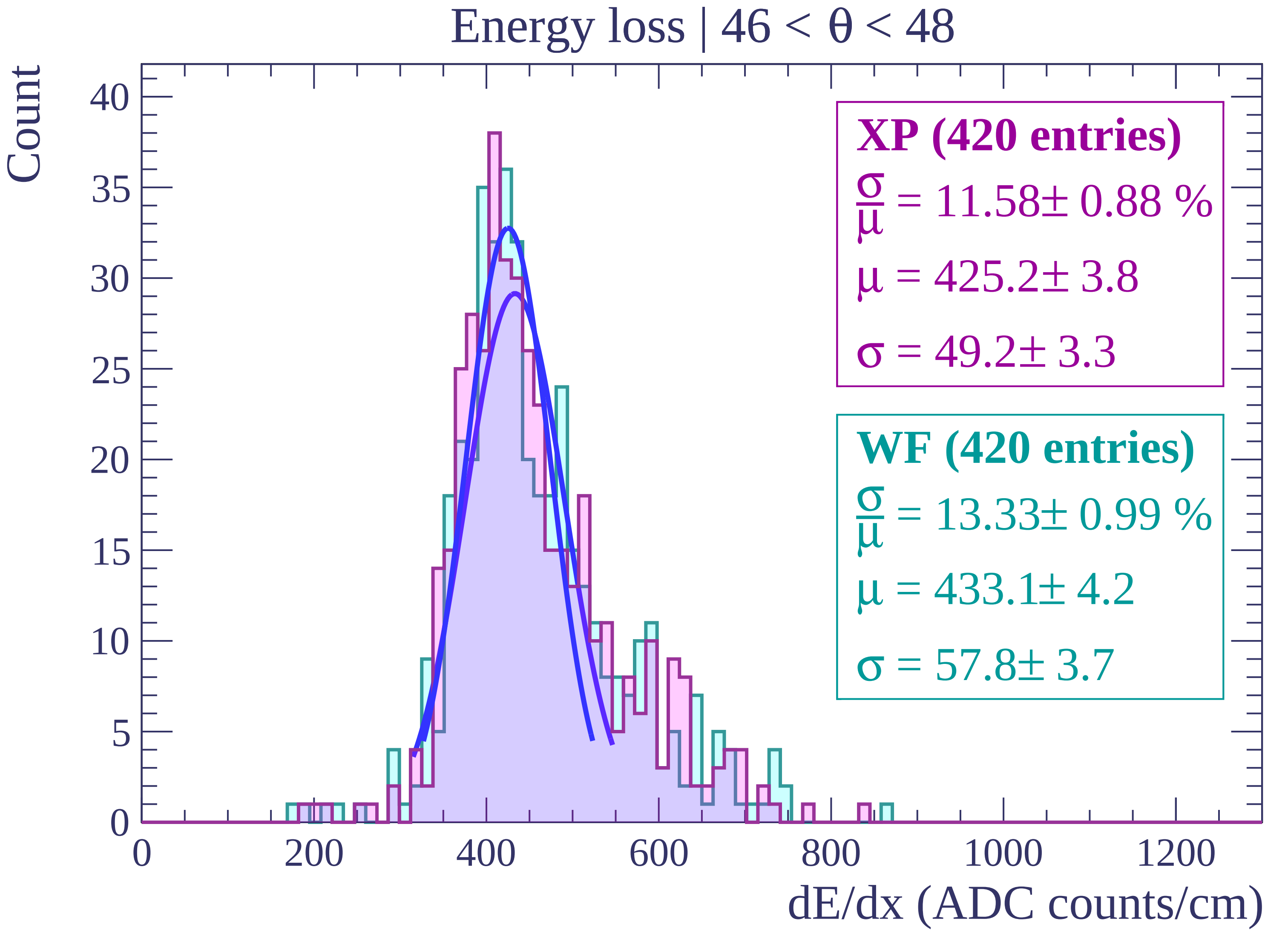
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP (686 entries)
σ/μ = 10.58± 0.58 %
μ = 421.7± 2.3
σ = 44.6± 2.2
WF (686 entries)
σ/μ = 10.61± 0.56 %
μ = 424.4± 2.4
σ = 45.0± 2.1

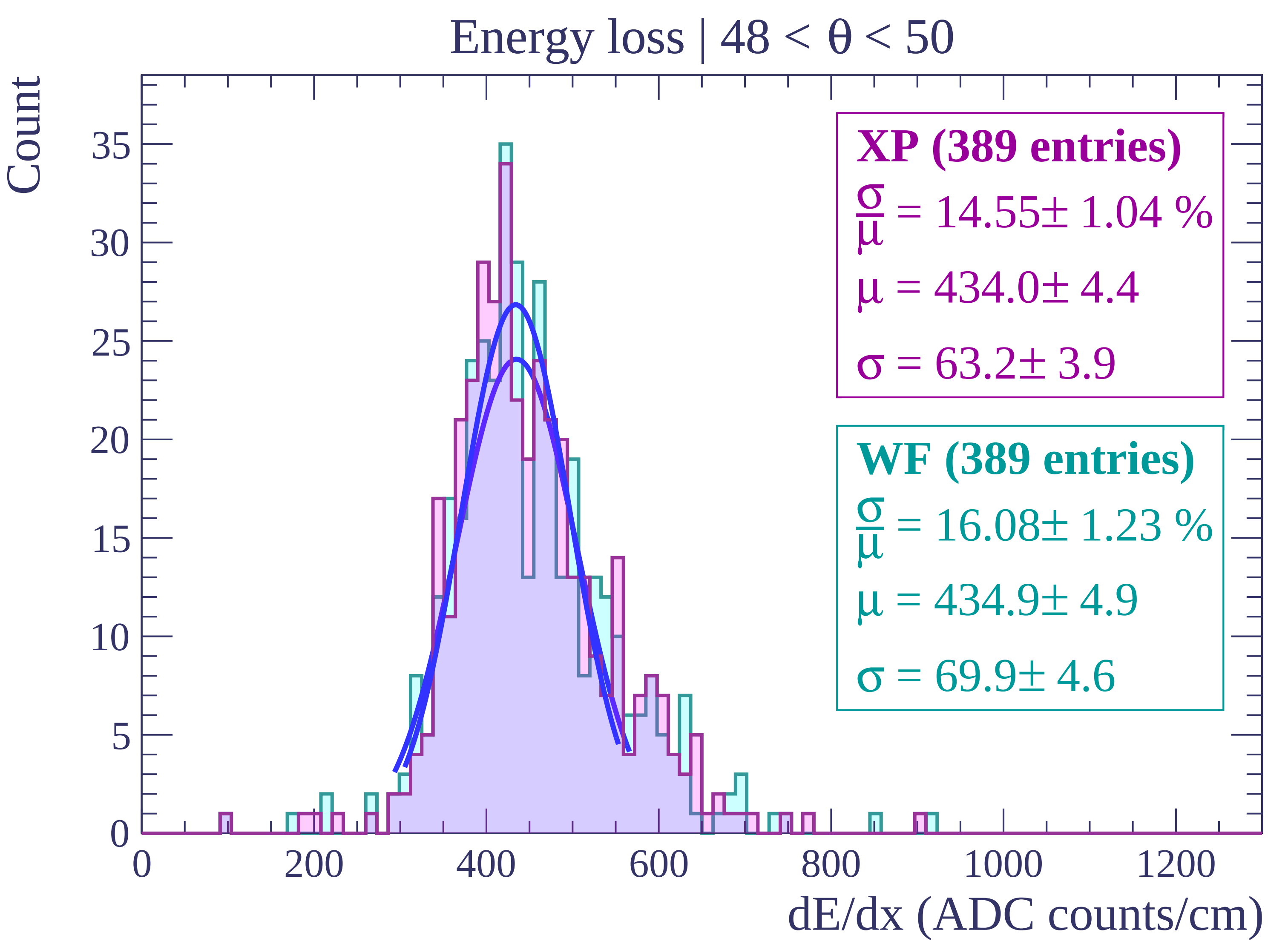

Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (611 entries)
σ/μ = 11.44± 0.57 %
μ = 427.8± 2.6
σ = 48.9± 2.1
WF (611 entries)
σ/μ = 13.11± 0.72 %
μ = 430.4± 2.9
σ = 56.4± 2.7



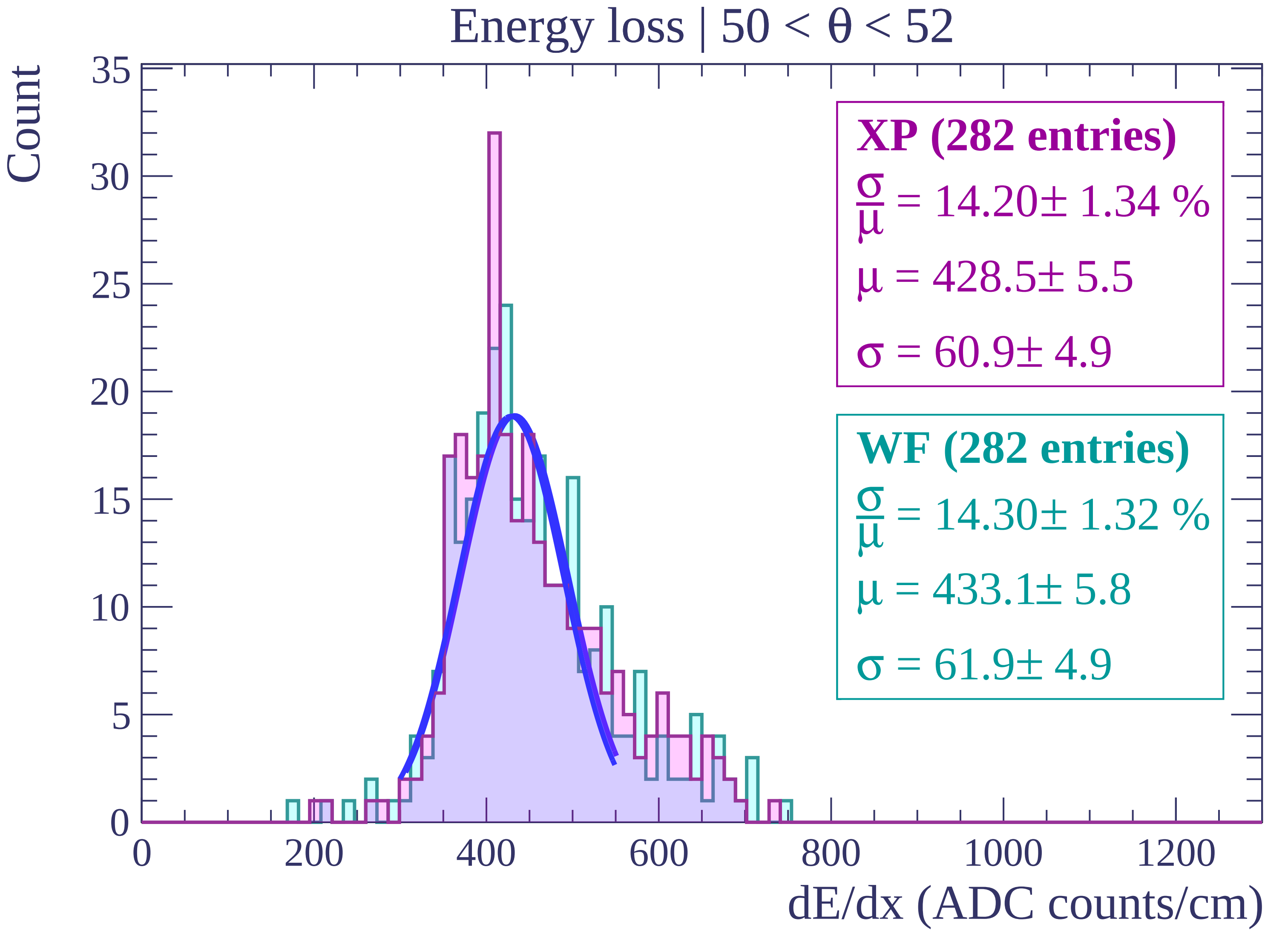
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (602 entries)
σ/μ = 9.63± 0.57 %
μ = 418.4± 2.3
σ = 40.3± 2.2
WF (602 entries)
σ/μ = 11.21± 0.65 %
μ = 423.7± 2.6
σ = 47.5± 2.5


Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP (503 entries)
σ/μ = 10.92± 0.74 %
μ = 416.0± 2.8
σ = 45.4± 2.8
WF (503 entries)
σ/μ = 12.31± 0.92 %
μ = 419.6± 3.2
σ = 51.7± 3.5

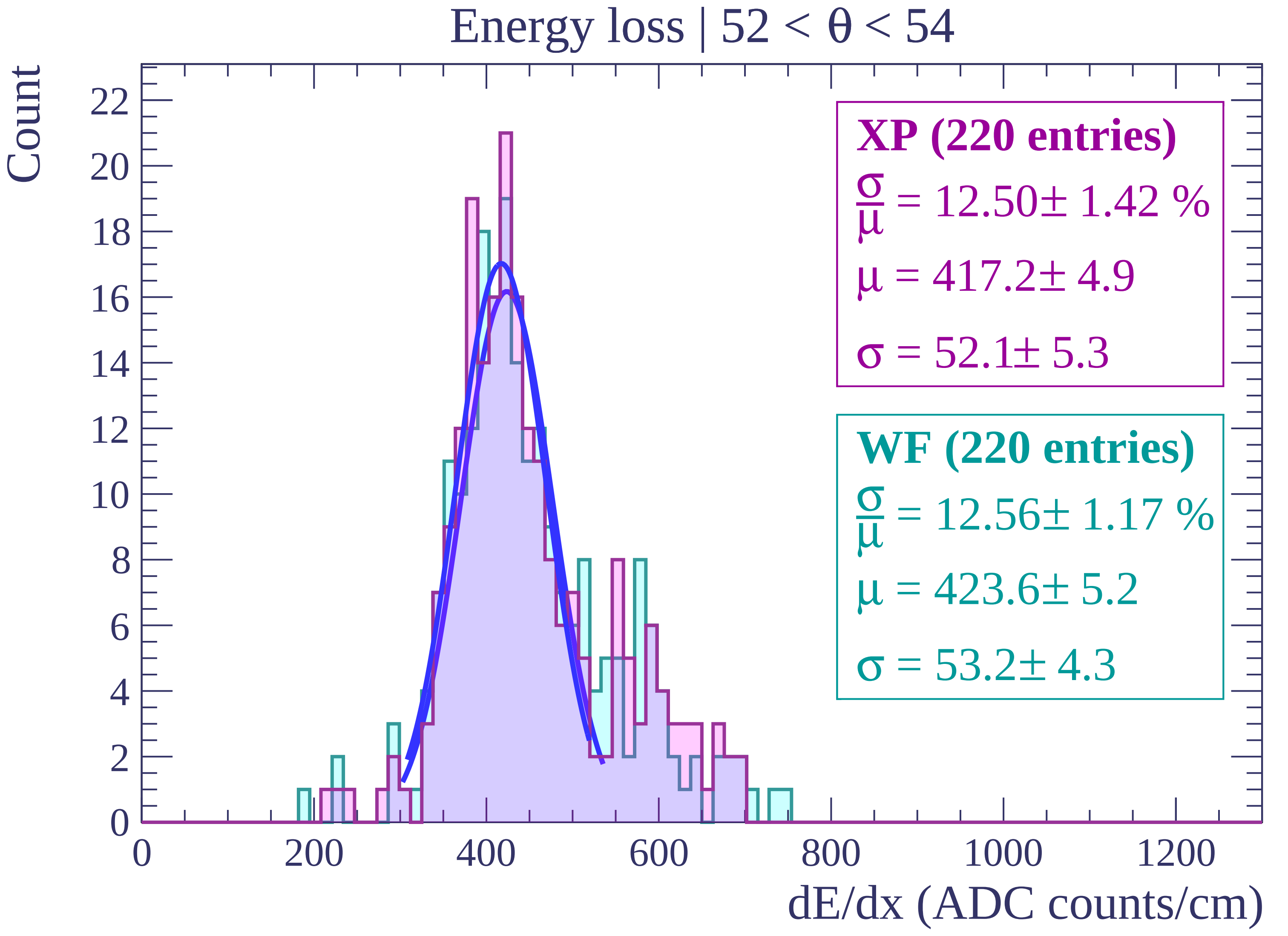

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (420 entries)
σ/μ = 11.58± 0.88 %
μ = 425.2± 3.8
σ = 49.2± 3.3
WF (420 entries)
σ/μ = 13.33± 0.99 %
μ = 433.1± 4.2
σ = 57.8± 3.7



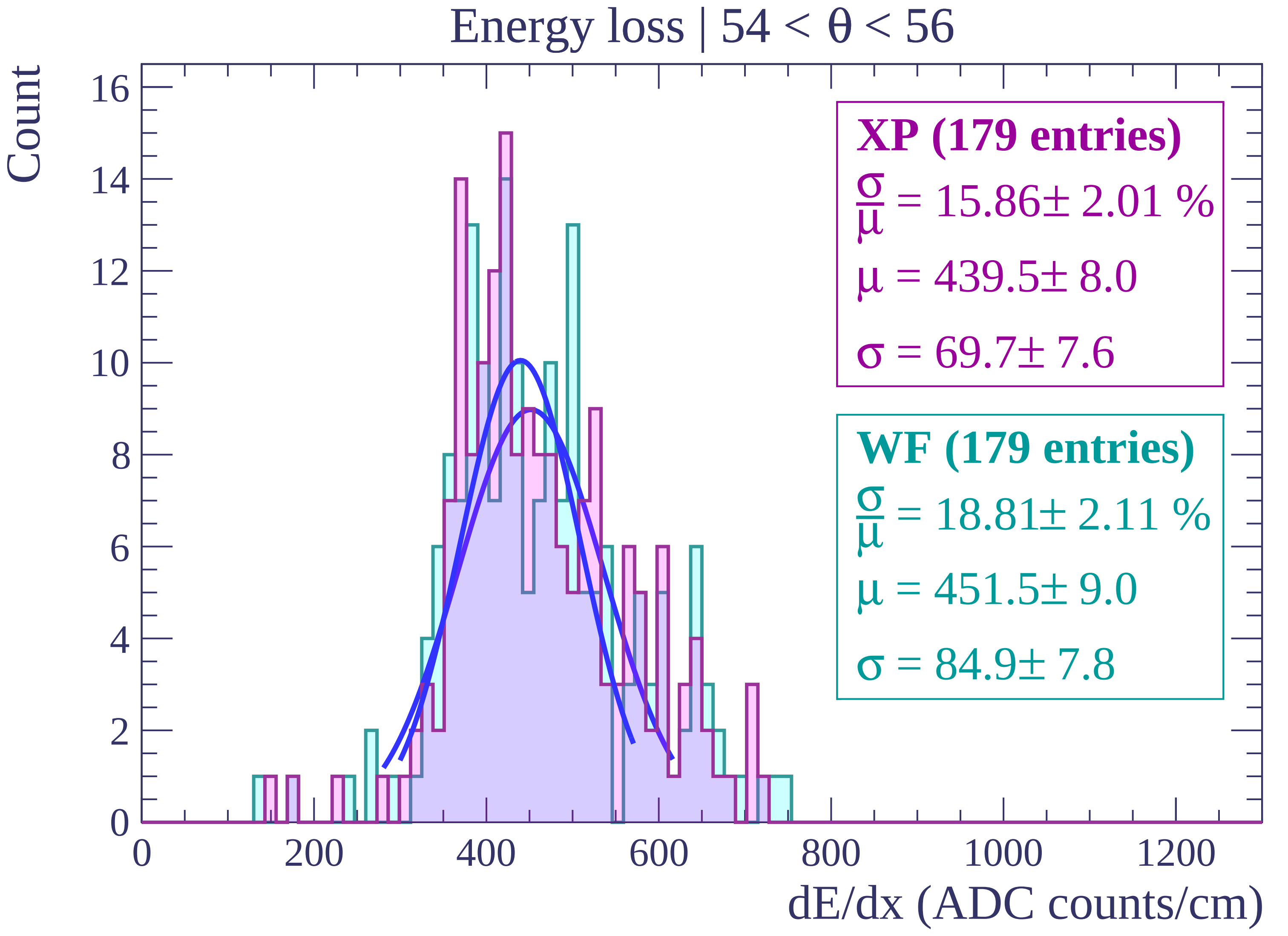
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP (389 entries)
$\frac{\sigma}{\mu}$ = 14.55± 1.04 %
μ = 434.0± 4.4
σ = 63.2± 3.9
WF (389 entries)
$\frac{\sigma}{\mu}$ = 16.08± 1.23 %
μ = 434.9± 4.9
σ = 69.9± 4.6



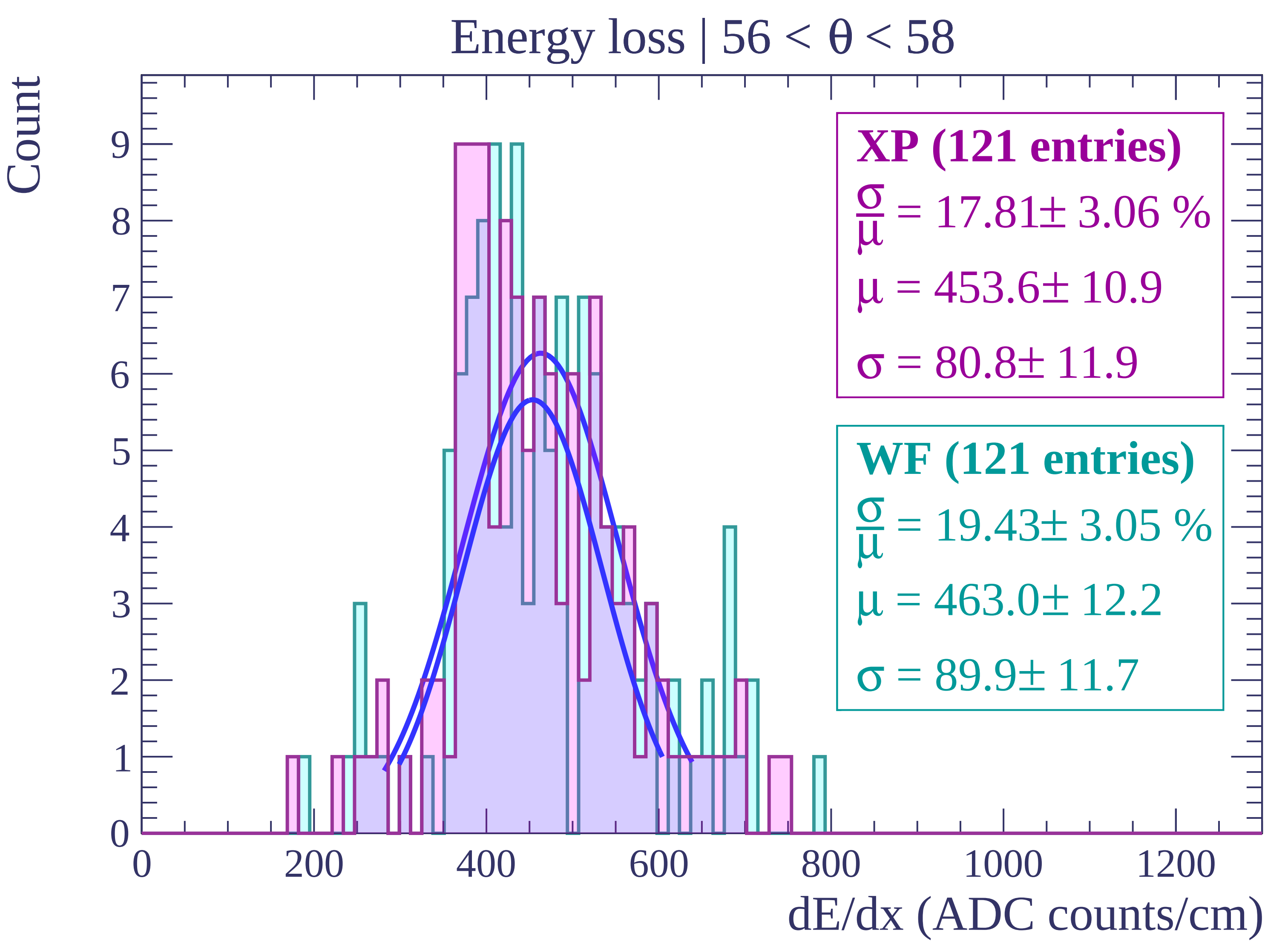
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (282 entries)
$\frac{\sigma}{\mu}$ = 14.20± 1.34 %
μ = 428.5± 5.5
σ = 60.9± 4.9
WF (282 entries)
$\frac{\sigma}{\mu}$ = 14.30± 1.32 %
μ = 433.1± 5.8
σ = 61.9± 4.9



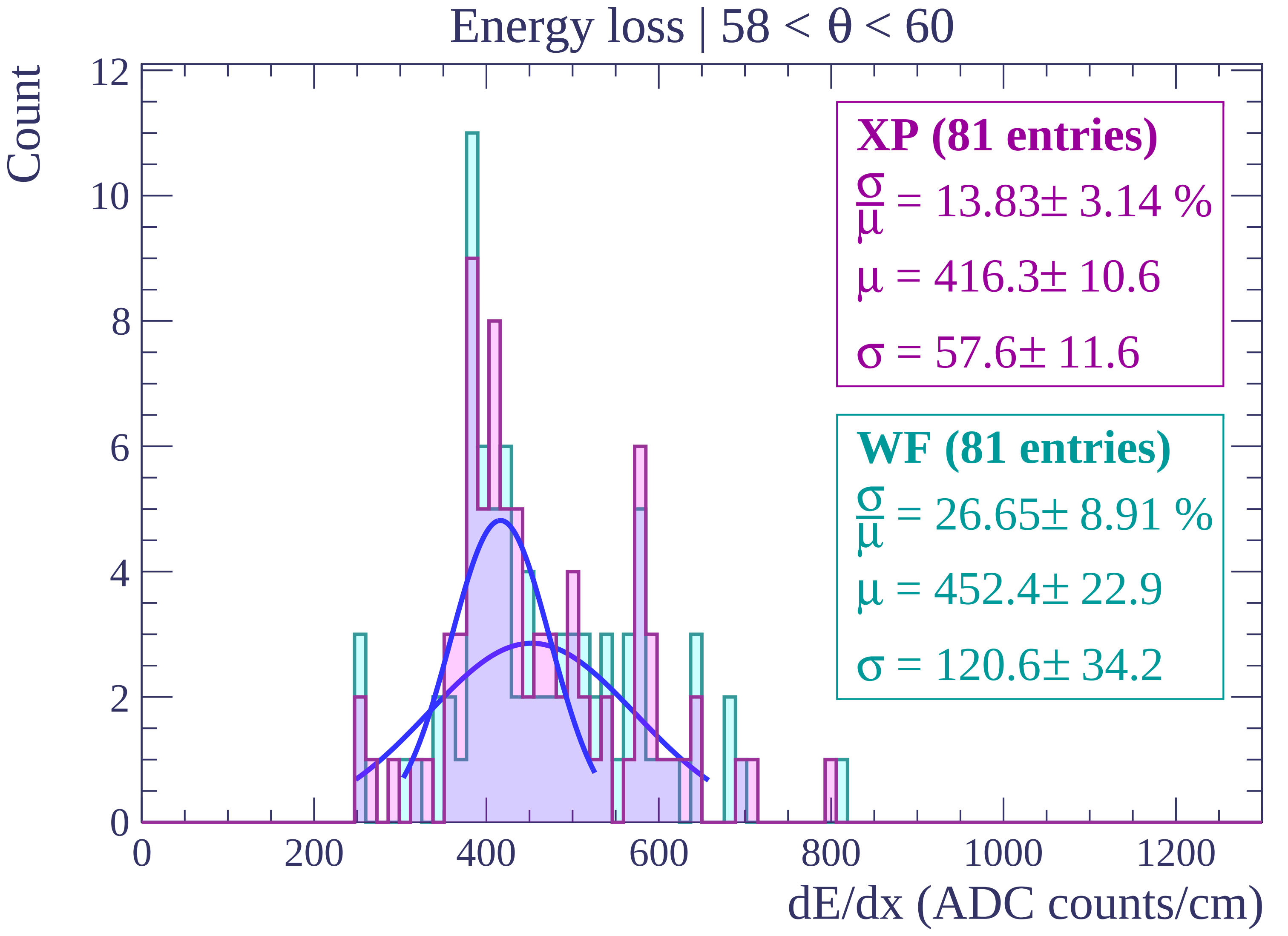
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP (220 entries)
σ/μ = 12.50± 1.42 %
μ = 417.2± 4.9
σ = 52.1± 5.3
WF (220 entries)
σ/μ = 12.56± 1.17 %
μ = 423.6± 5.2
σ = 53.2± 4.3
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22



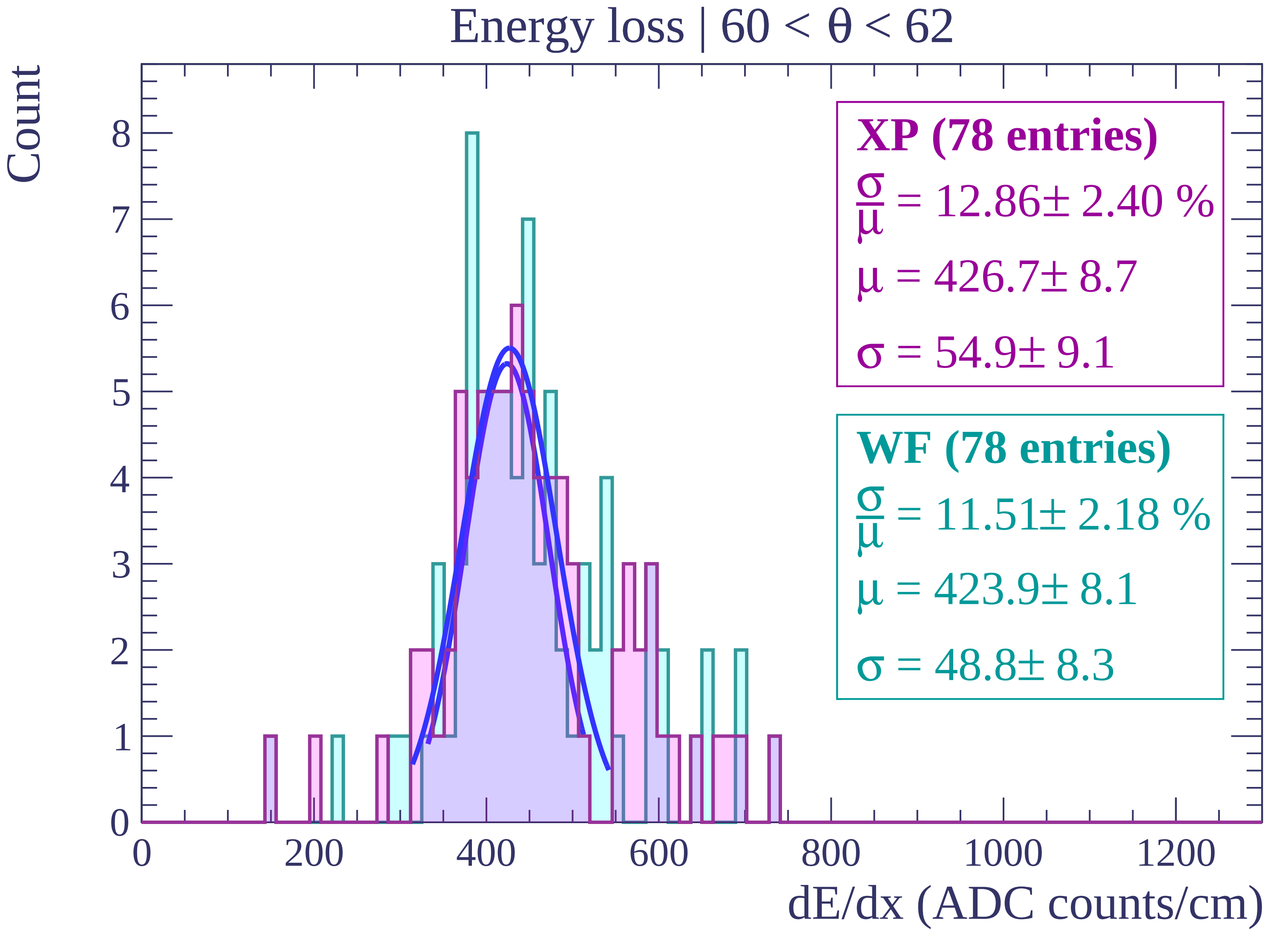
Energy loss | 54 < θ < 56
Count
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (179 entries)
σ/μ = 15.86± 2.01 %
μ = 439.5± 8.0
σ = 69.7± 7.6
WF (179 entries)
σ/μ = 18.81± 2.11 %
μ = 451.5± 9.0
σ = 84.9± 7.8



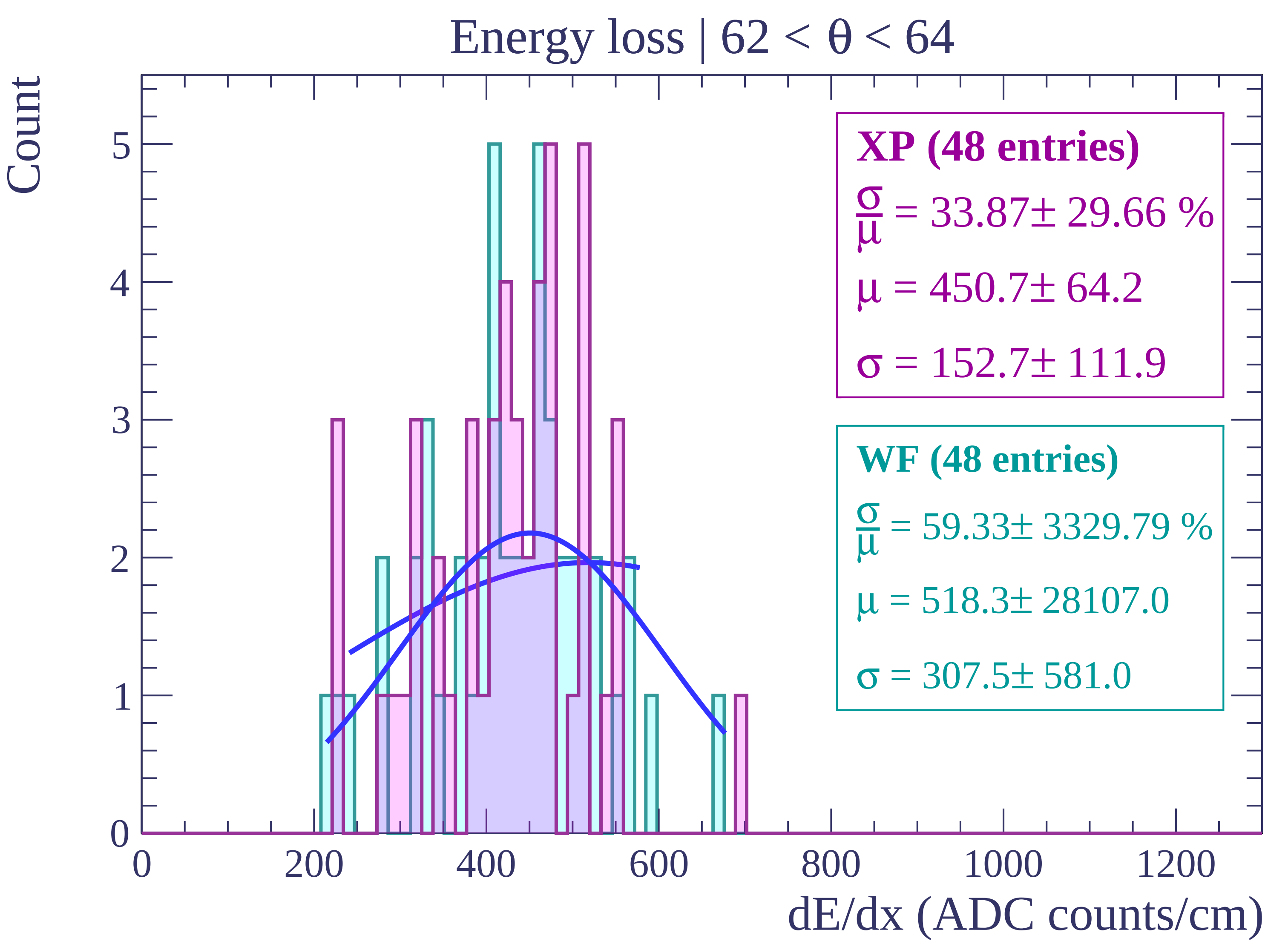
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP (121 entries)
$\frac{\sigma}{\mu}$ = 17.81± 3.06 %
μ = 453.6± 10.9
σ = 80.8± 11.9
WF (121 entries)
$\frac{\sigma}{\mu}$ = 19.43± 3.05 %
μ = 463.0± 12.2
σ = 89.9± 11.7



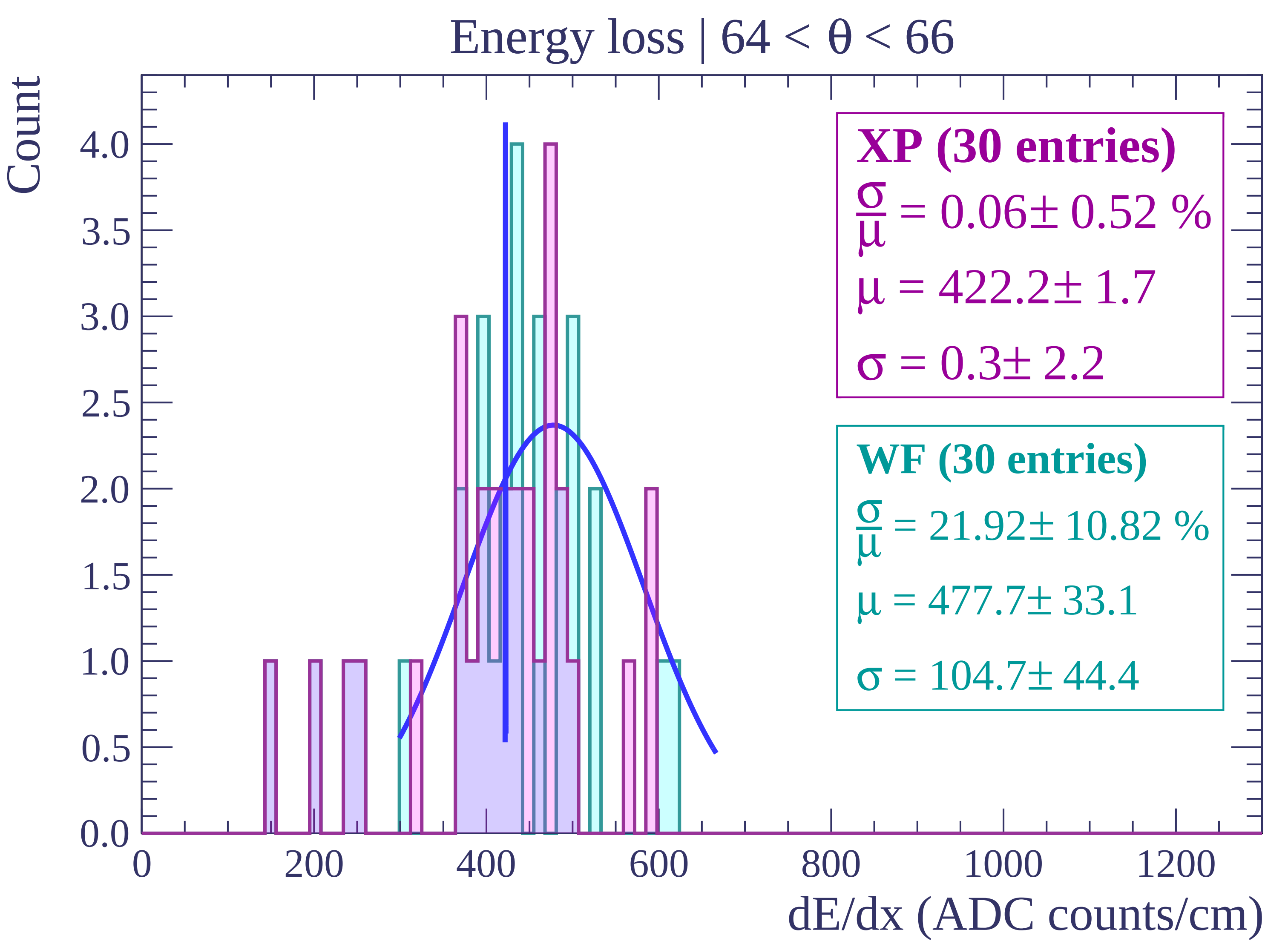
Energy loss | 58 < θ < 60
Count
XP (81 entries)
σ/μ = 13.83± 3.14 %
μ = 416.3± 10.6
σ = 57.6± 11.6
WF (81 entries)
σ/μ = 26.65± 8.91 %
μ = 452.4± 22.9
σ = 120.6± 34.2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



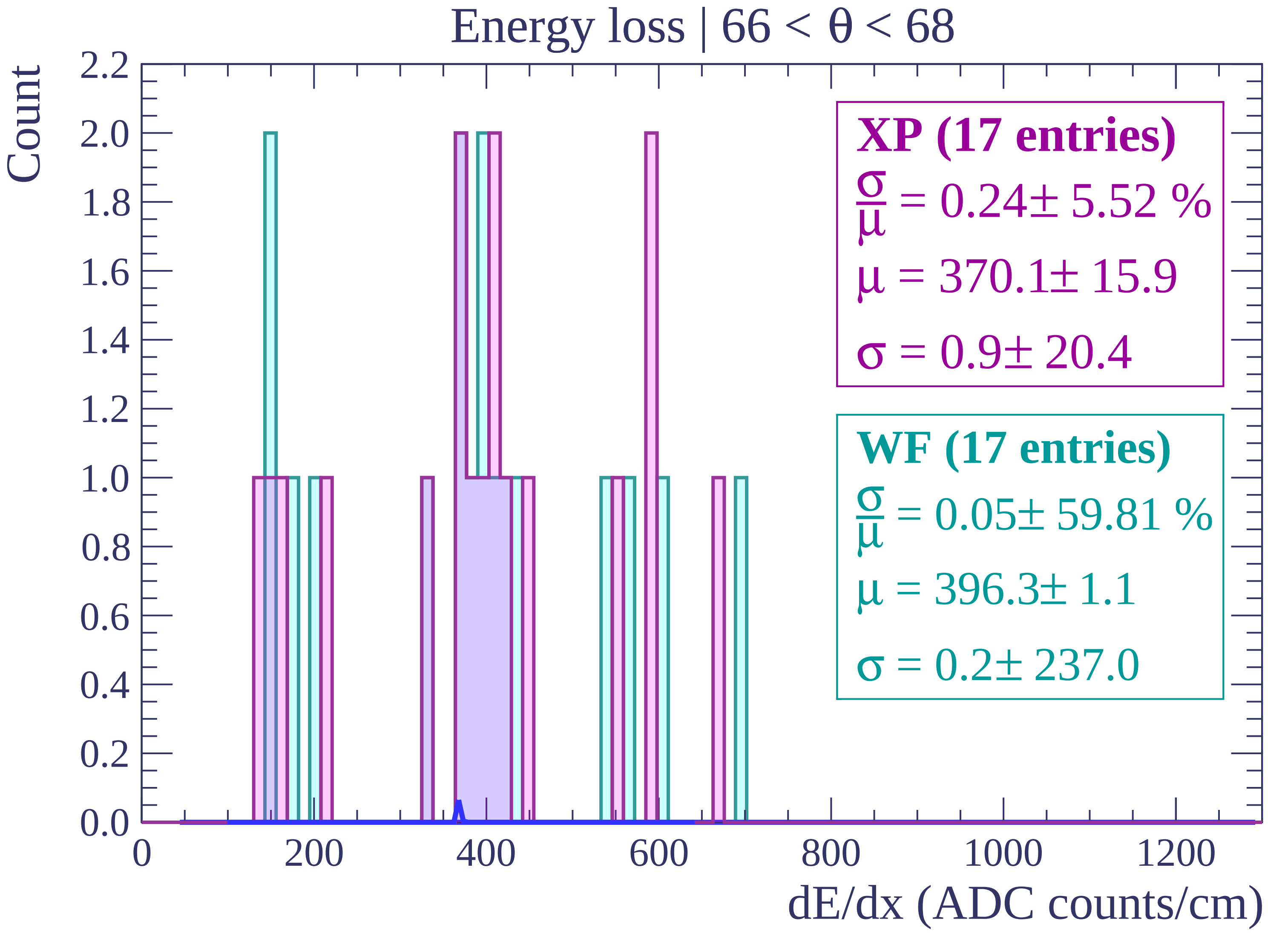
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP (78 entries)
σ/μ = 12.86± 2.40 %
μ = 426.7± 8.7
σ = 54.9± 9.1
WF (78 entries)
σ/μ = 11.51± 2.18 %
μ = 423.9± 8.1
σ = 48.8± 8.3



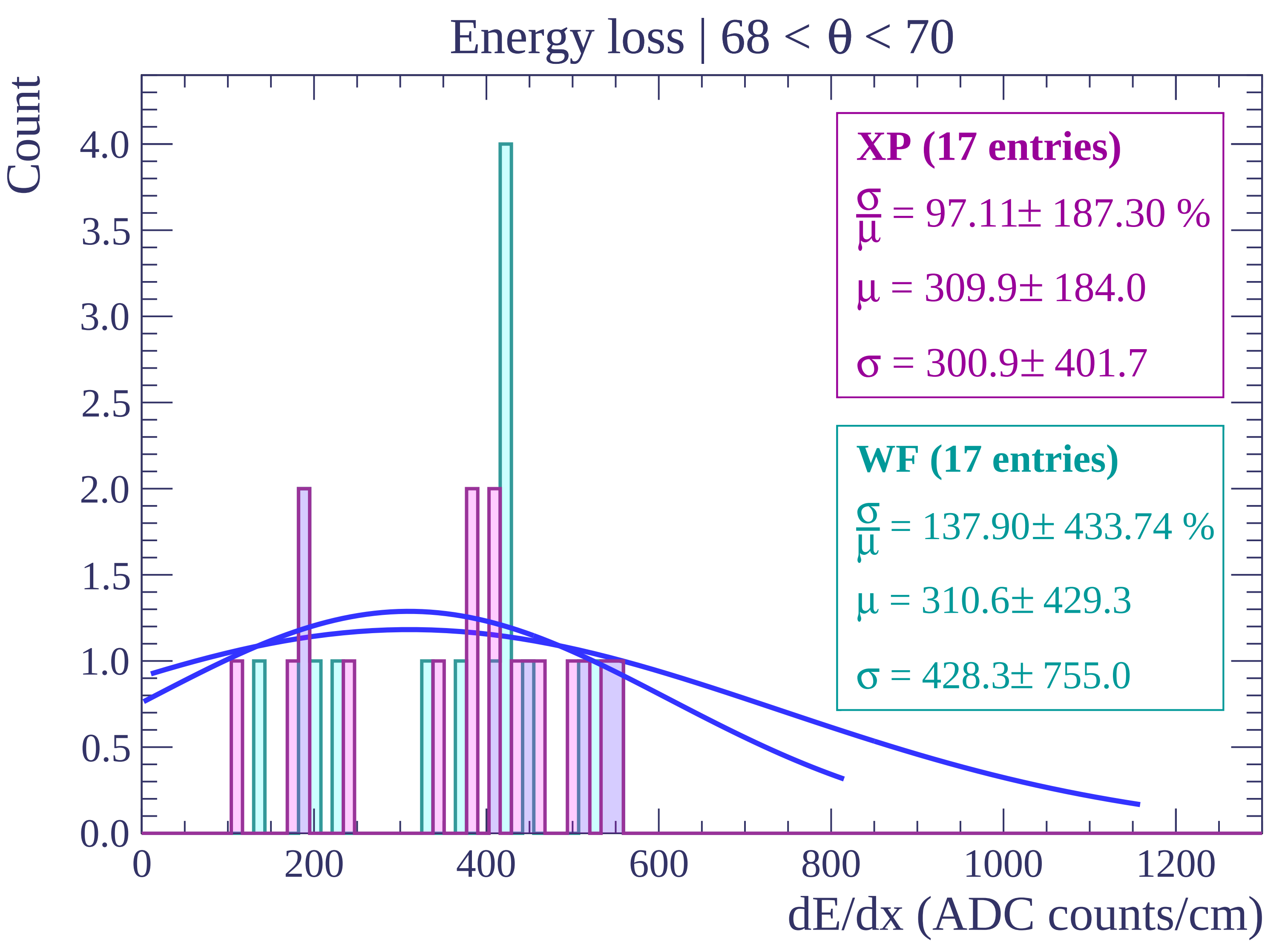
Energy loss | 62 < θ < 64
Count
XP (48 entries)
σ/μ = 33.87± 29.66 %
μ = 450.7± 64.2
σ = 152.7± 111.9
WF (48 entries)
σ/μ = 59.33± 3329.79 %
μ = 518.3± 28107.0
σ = 307.5± 581.0
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

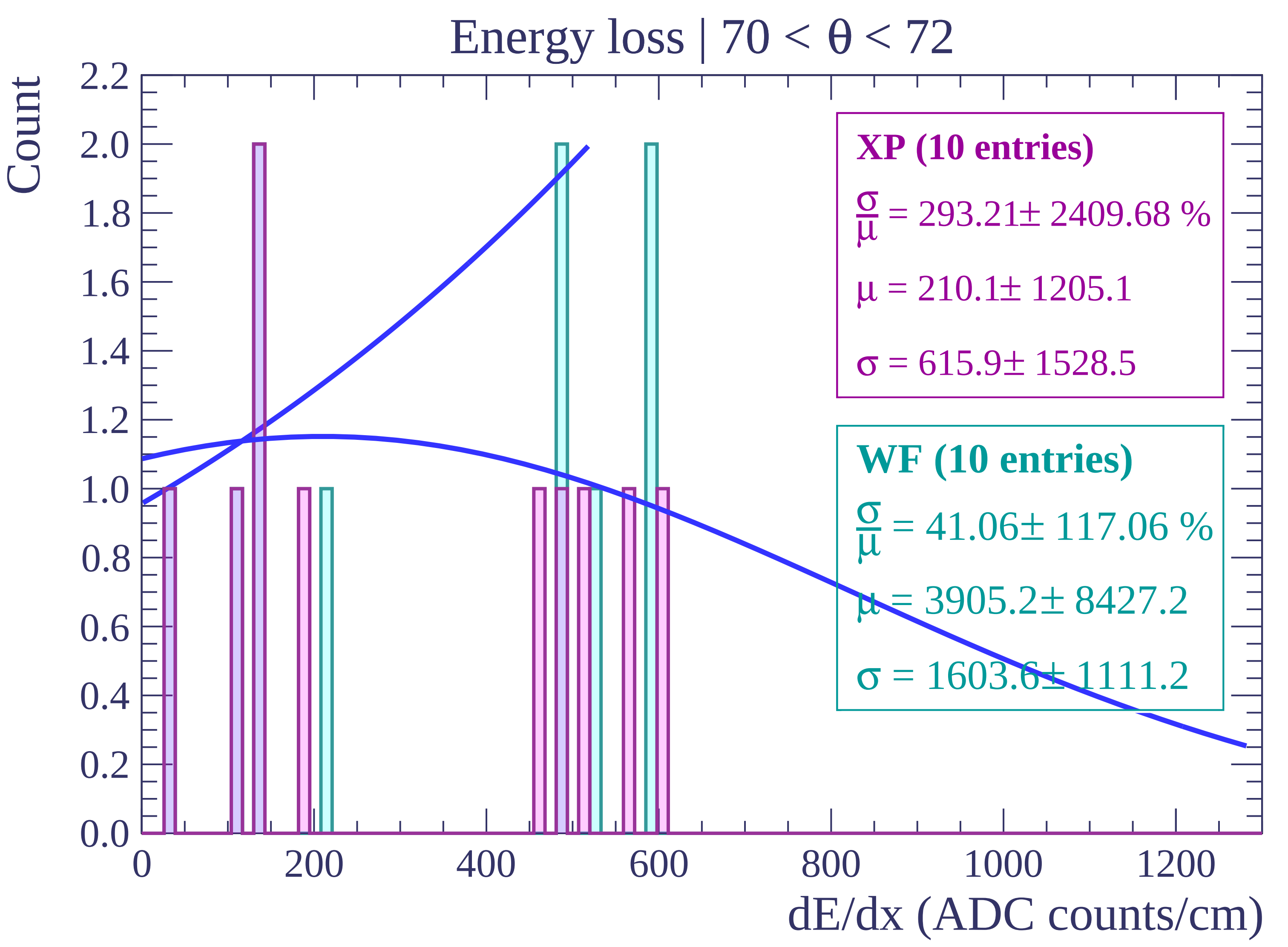


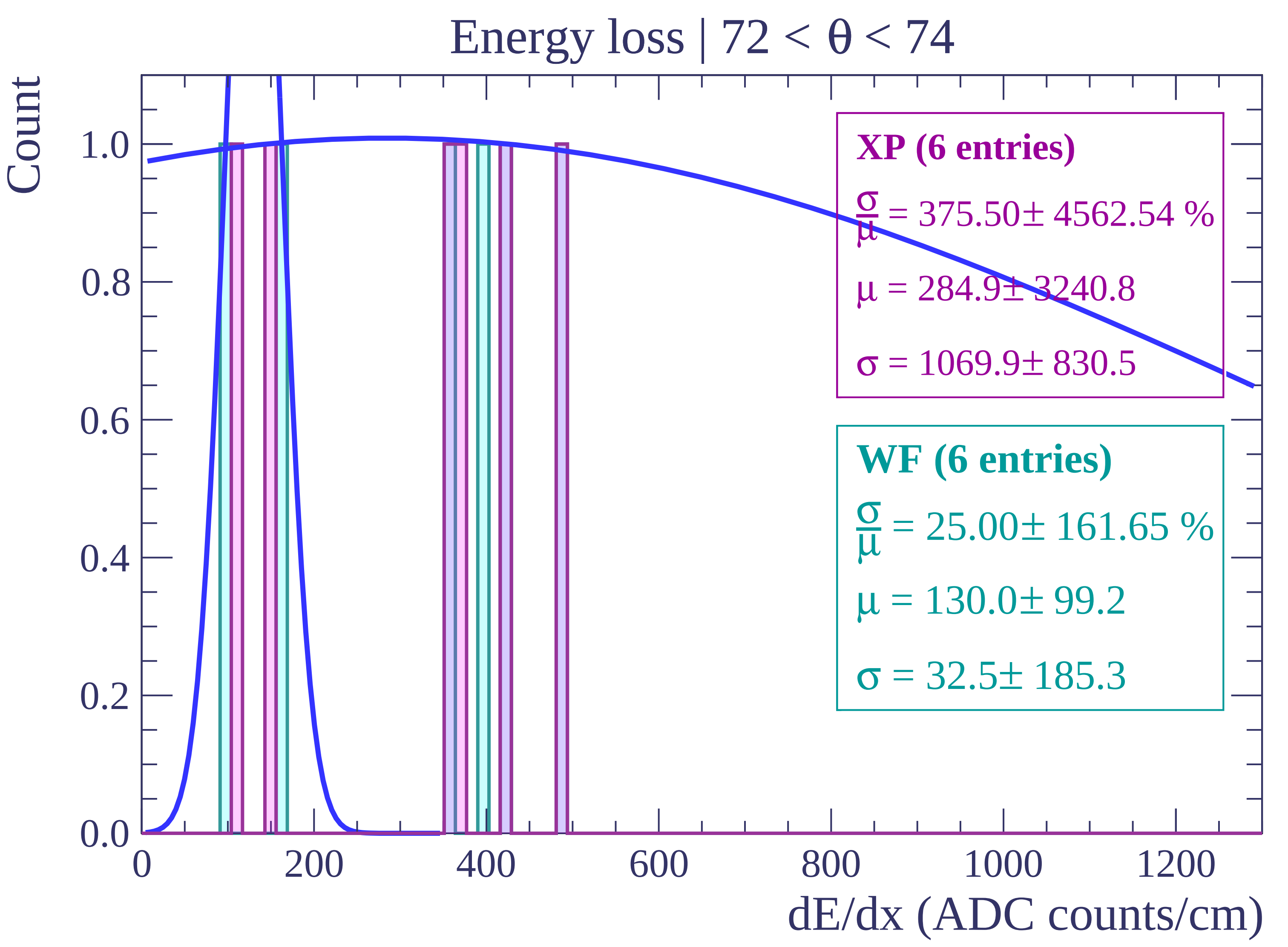
Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
XP (30 entries)
σ/μ = 0.06± 0.52 %
μ = 422.2± 1.7
σ = 0.3± 2.2
WF (30 entries)
σ/μ = 21.92± 10.82 %
μ = 477.7± 33.1
σ = 104.7± 44.4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200



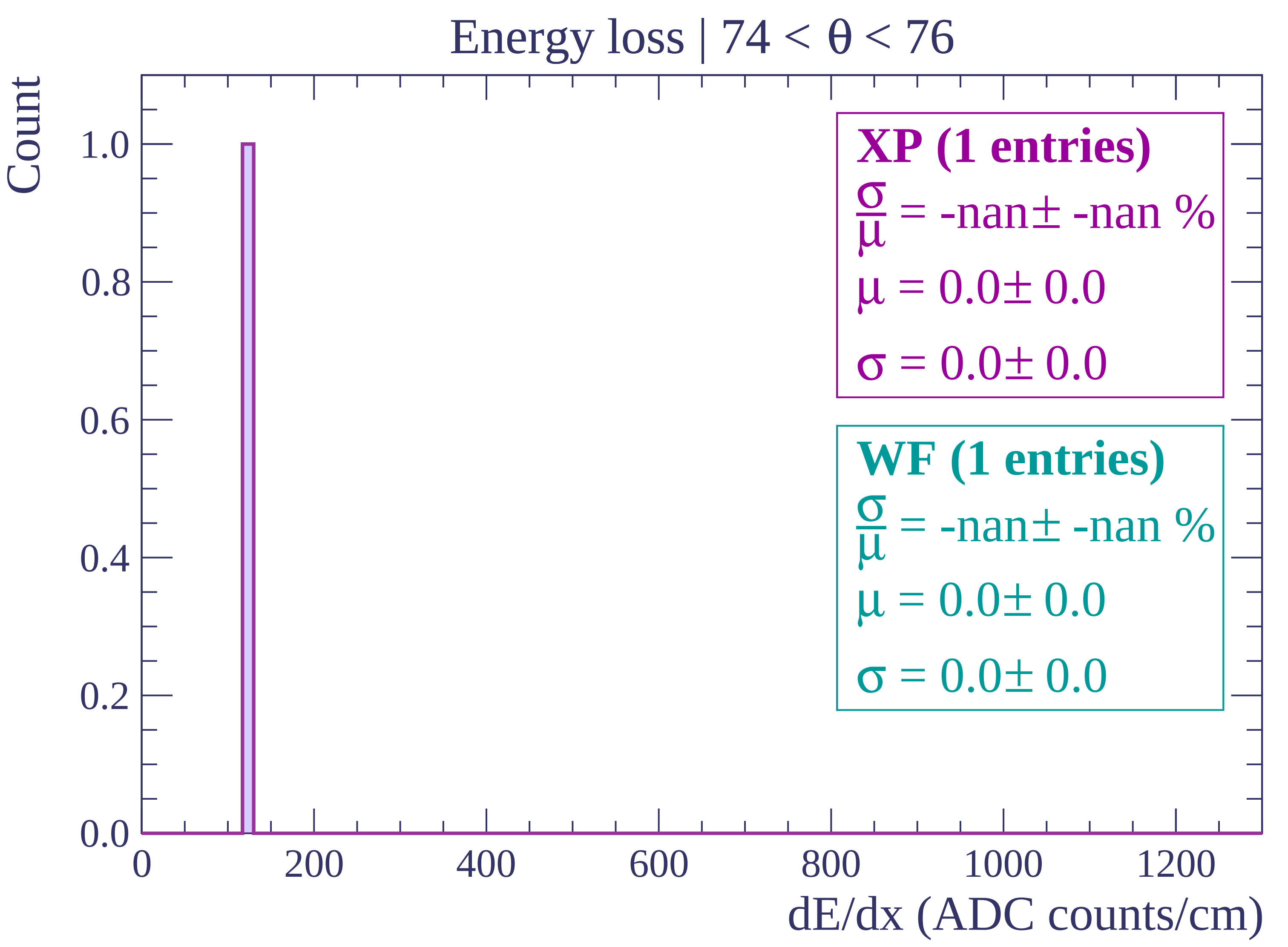
Energy loss | 66 < θ < 68
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (17 entries)
σ/μ = 0.24± 5.52 %
μ = 370.1± 15.9
σ = 0.9± 20.4
WF (17 entries)
σ/μ = 0.05± 59.81 %
μ = 396.3± 1.1
σ = 0.2± 237.0

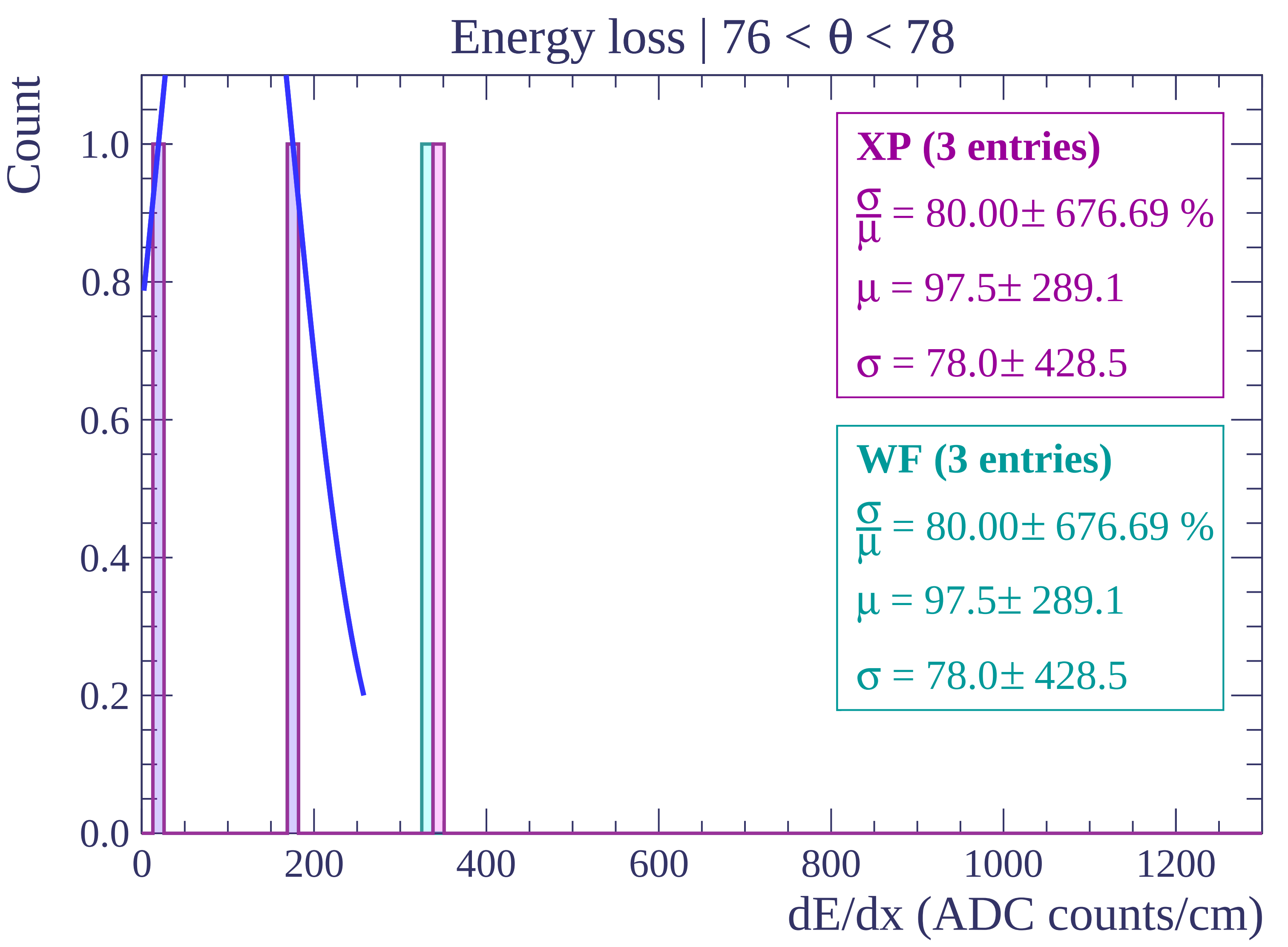

Energy loss | 68 < θ < 70
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (17 entries)
σ/μ = 97.11± 187.30 %
μ = 309.9± 184.0
σ = 300.9± 401.7
WF (17 entries)
σ/μ = 137.90± 433.74 %
μ = 310.6± 429.3
σ = 428.3± 755.0


Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
XP (10 entries)
σ/μ = 293.21± 2409.68 %
μ = 210.1± 1205.1
σ = 615.9± 1528.5
WF (10 entries)
σ/μ = 41.06± 117.06 %
μ = 3905.2± 8427.2
σ = 1603.6± 1111.2

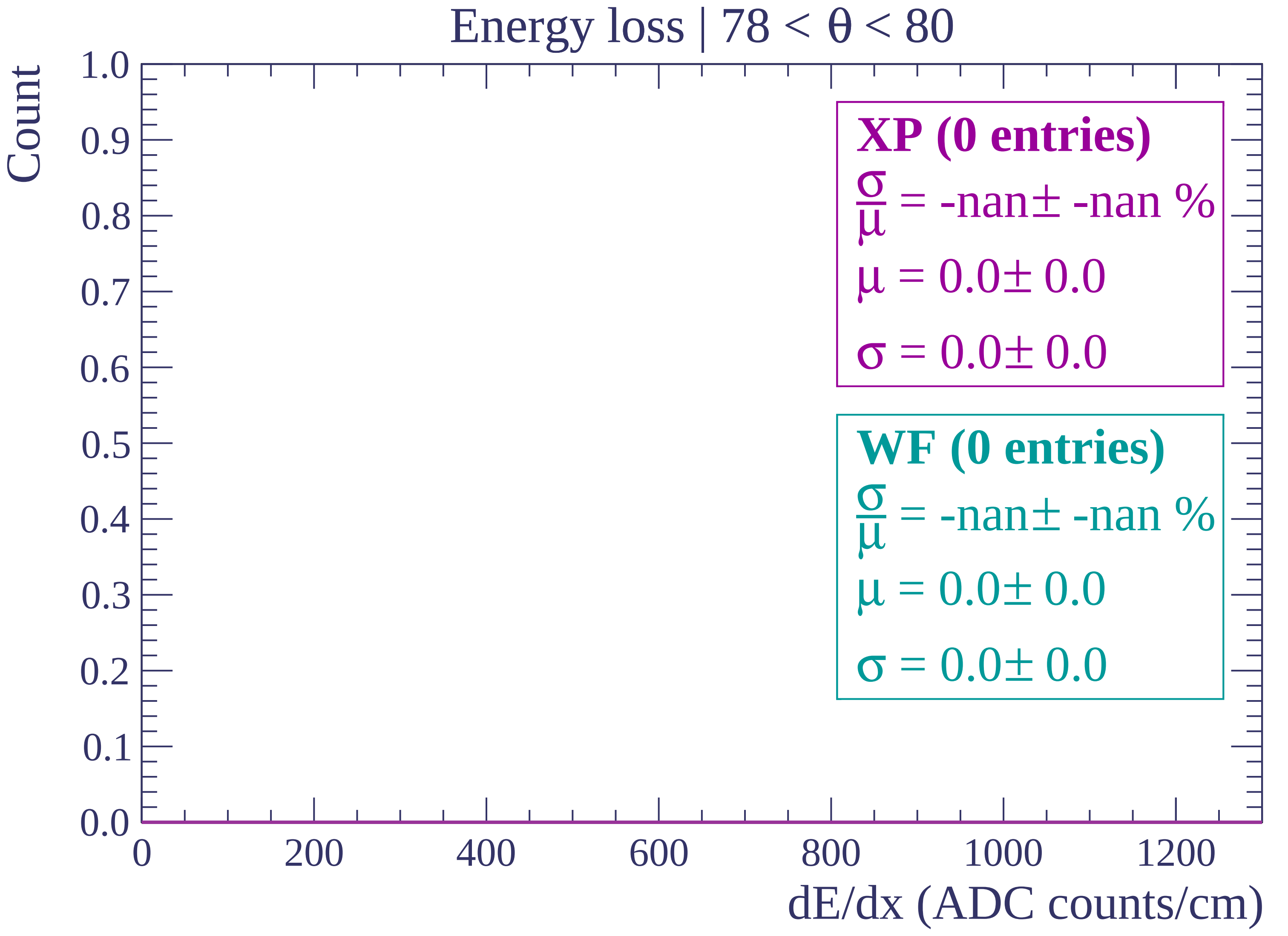


Energy loss | 72 < θ < 74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6 entries)
σ/μ = 375.50± 4562.54 %
μ = 284.9± 3240.8
σ = 1069.9± 830.5
WF (6 entries)
σ/μ = 25.00± 161.65 %
μ = 130.0± 99.2
σ = 32.5± 185.3

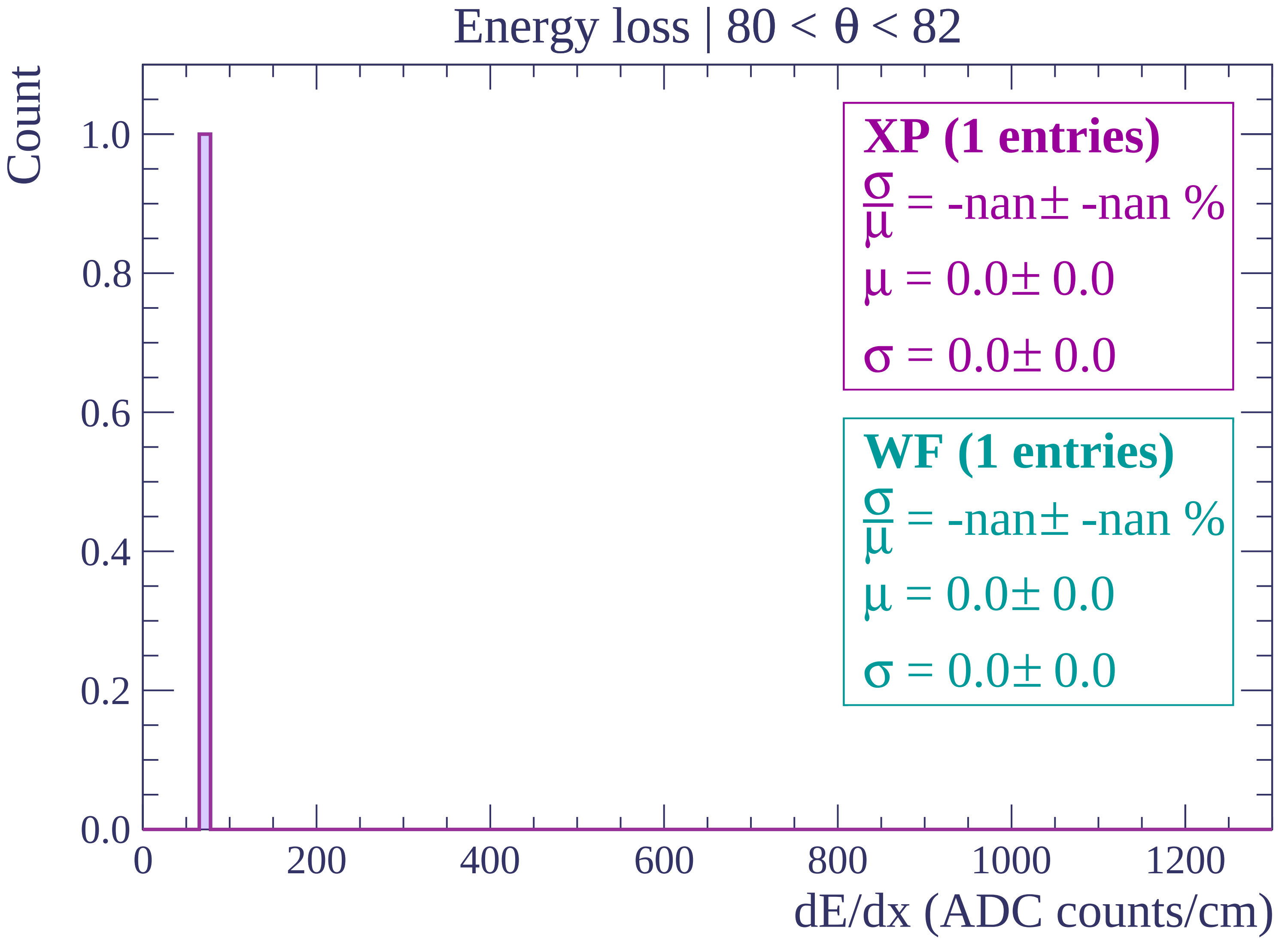

Energy loss | 74 < θ < 76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



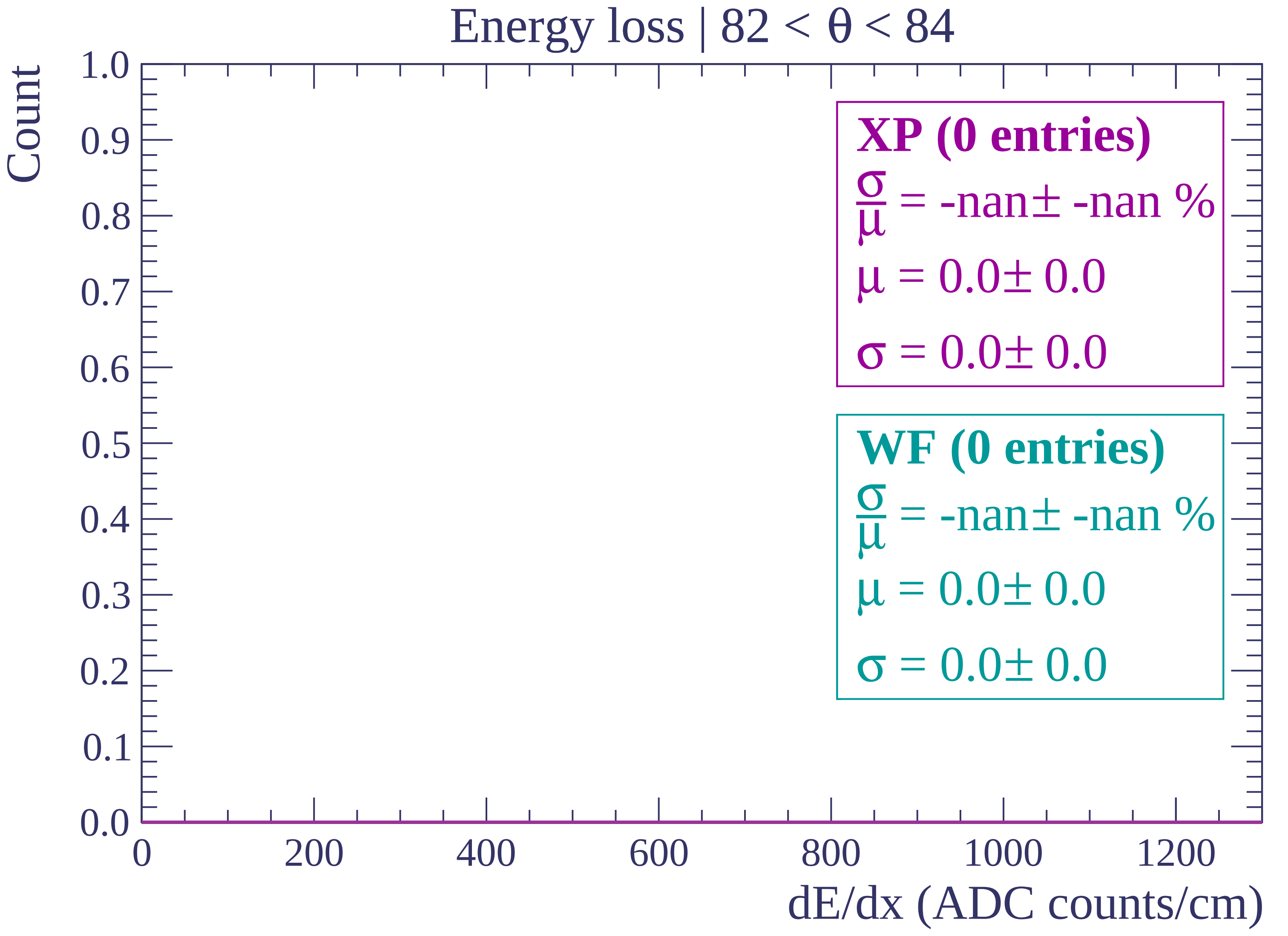
Energy loss | 76 < θ < 78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3 entries)
σ/μ = 80.00± 676.69 %
μ = 97.5± 289.1
σ = 78.0± 428.5
WF (3 entries)
σ/μ = 80.00± 676.69 %
μ = 97.5± 289.1
σ = 78.0± 428.5

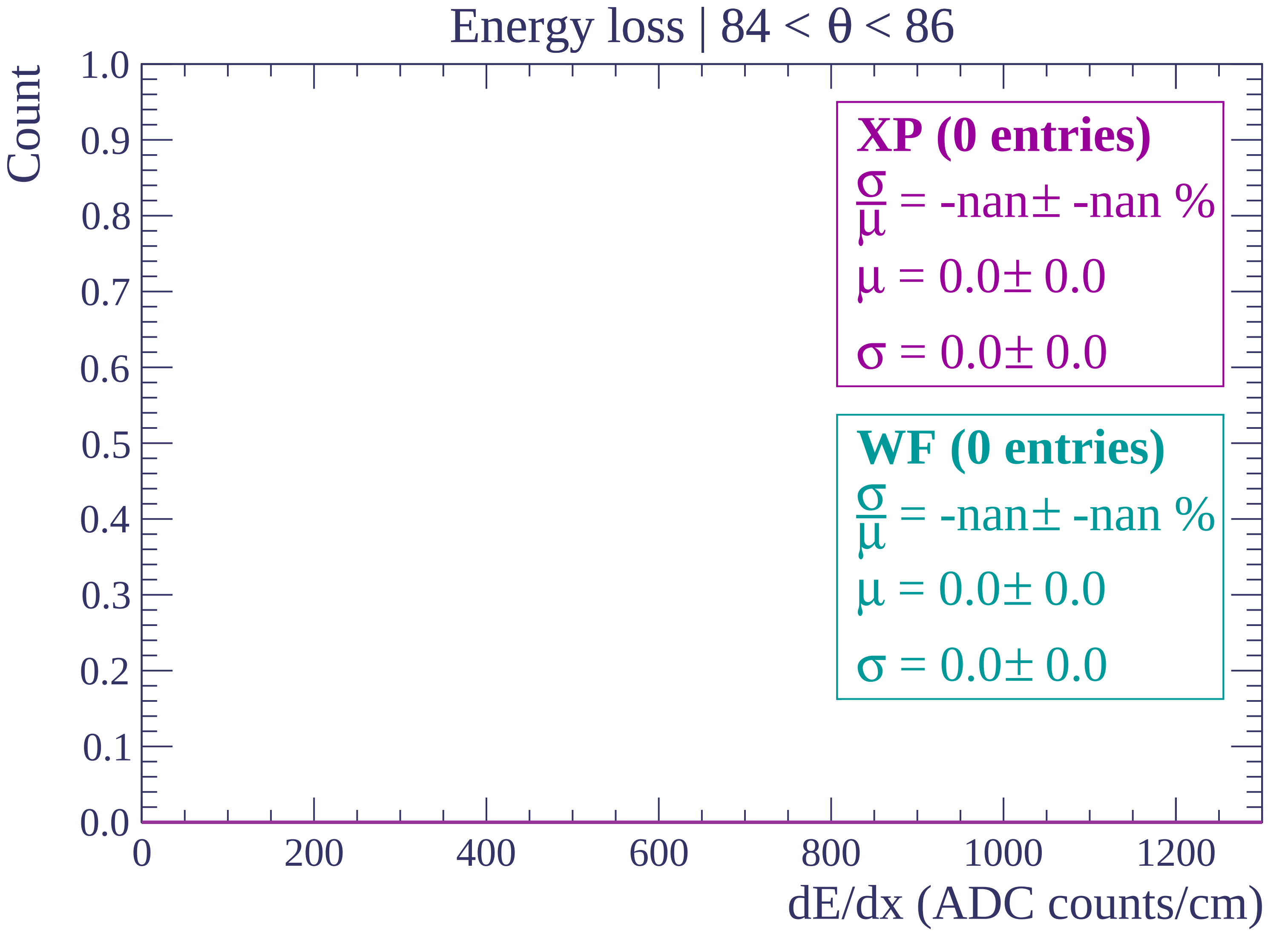


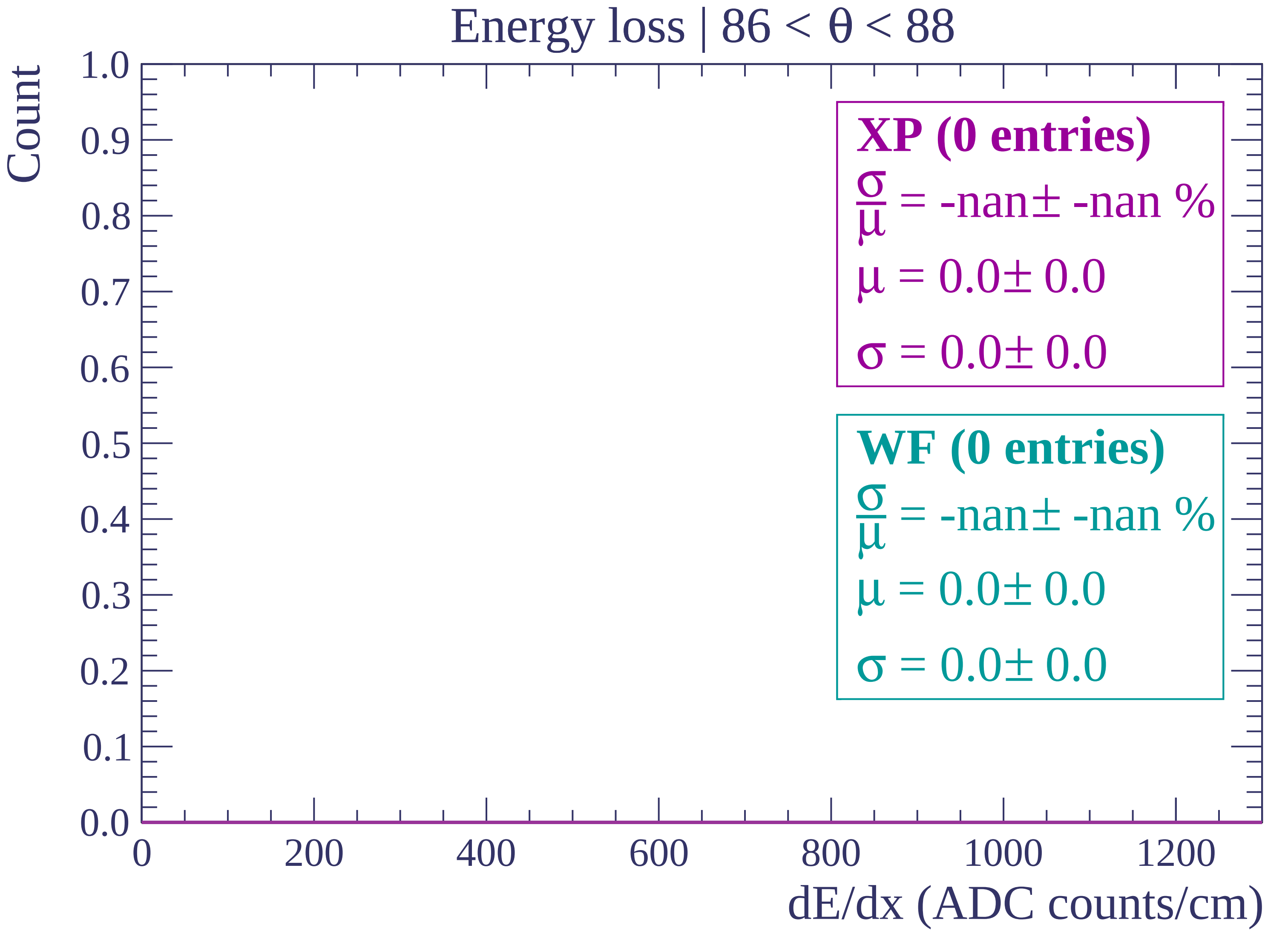
Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

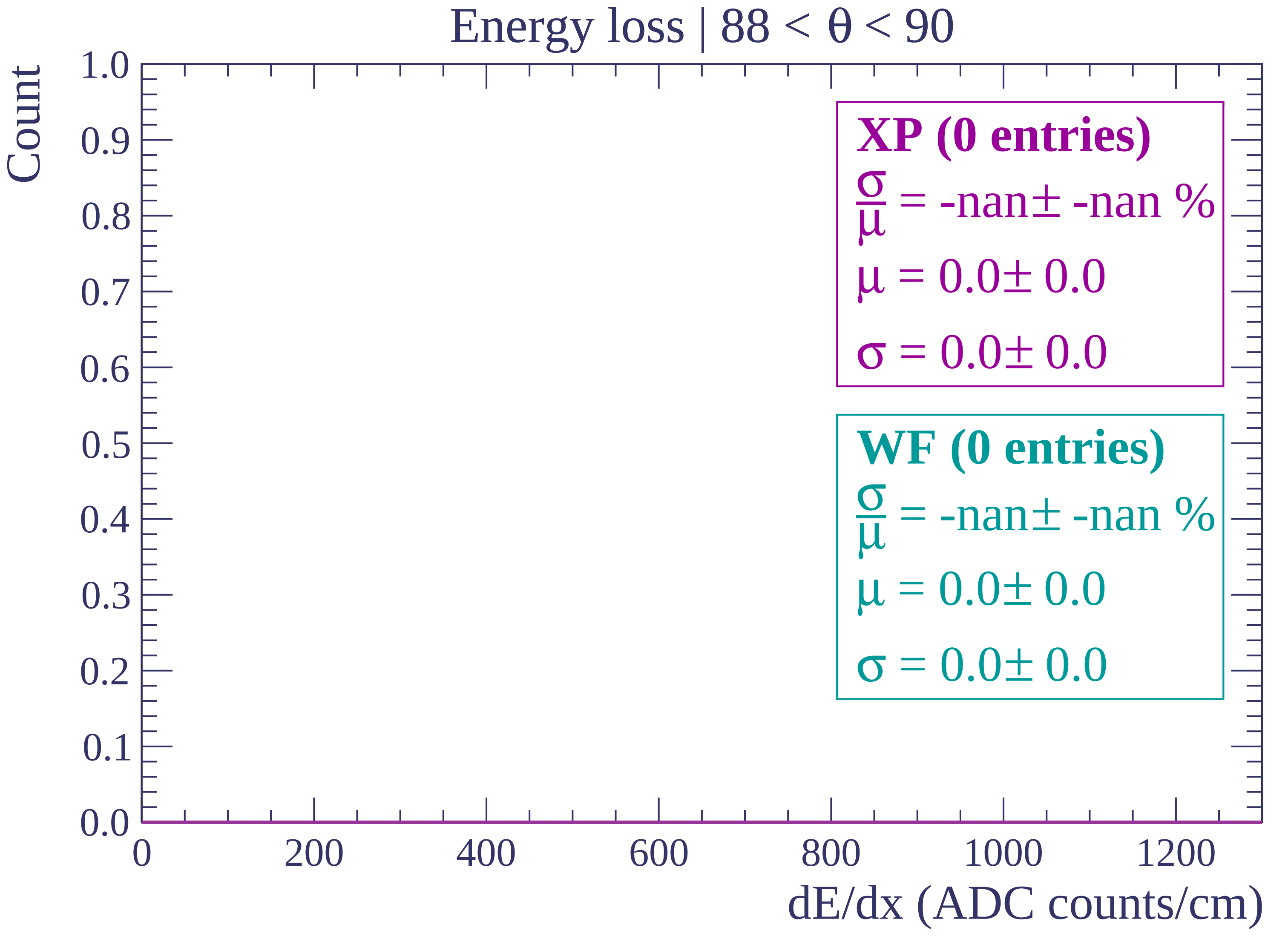


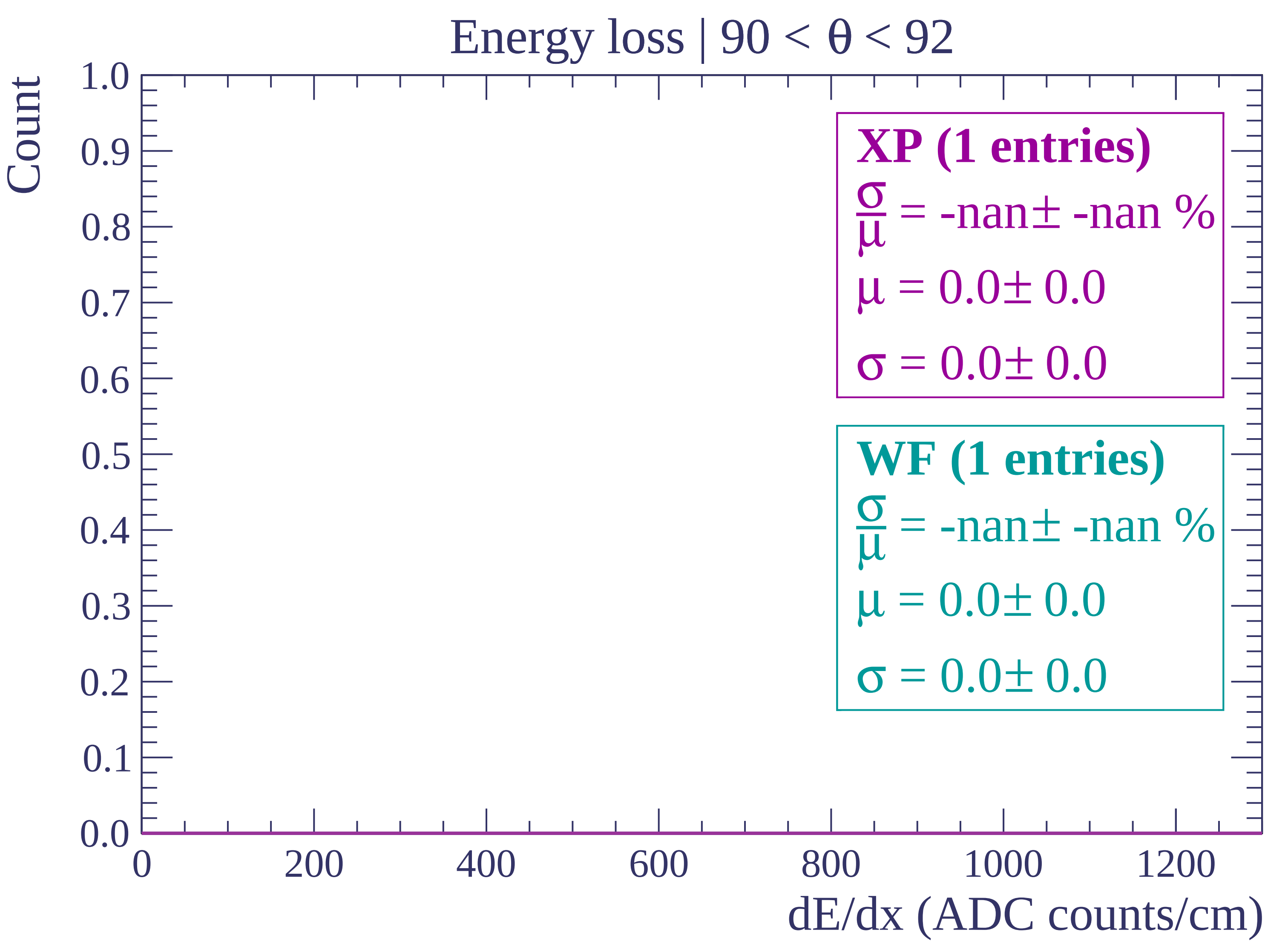
Energy loss | 80 < θ < 82
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

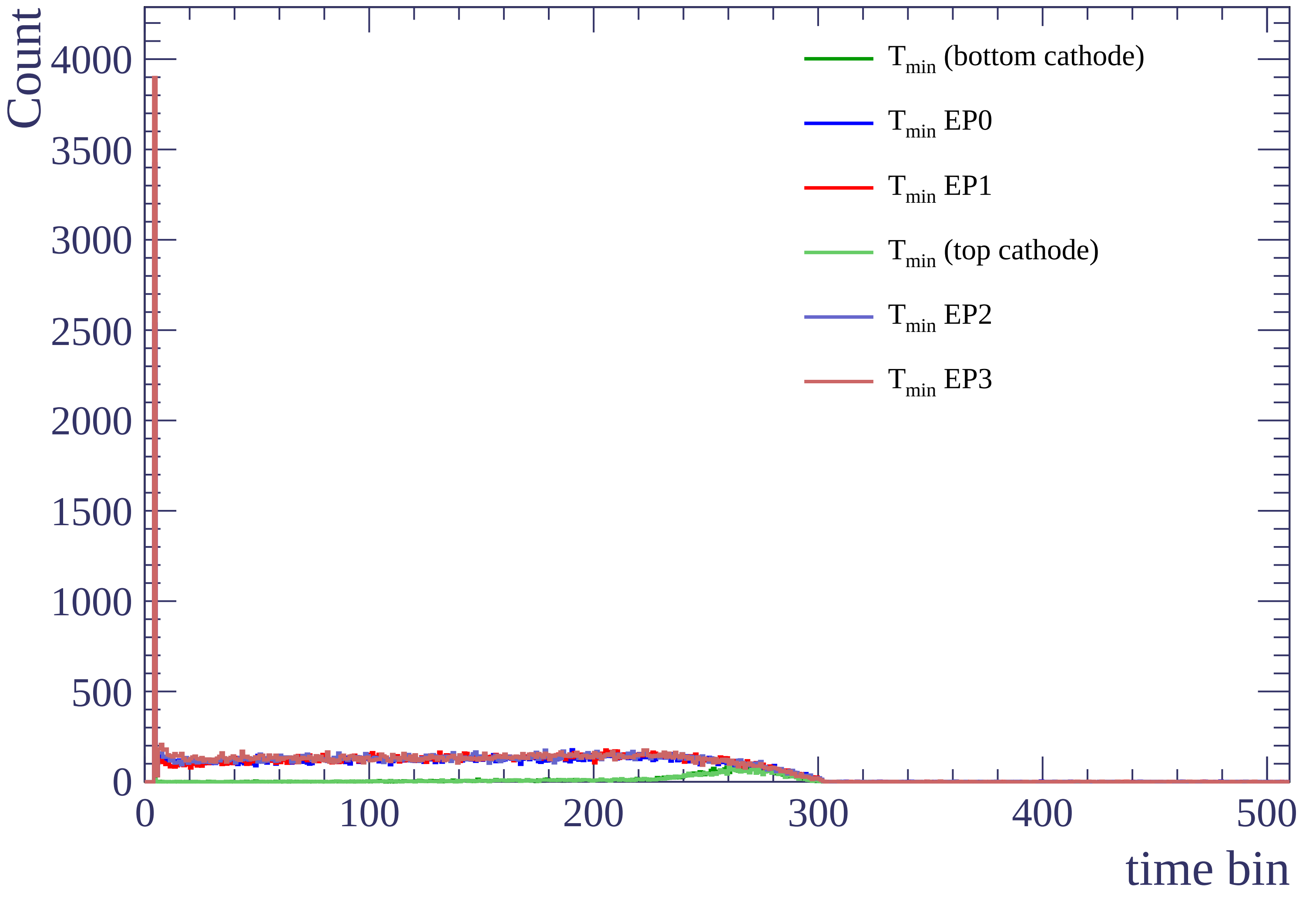

Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200


Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

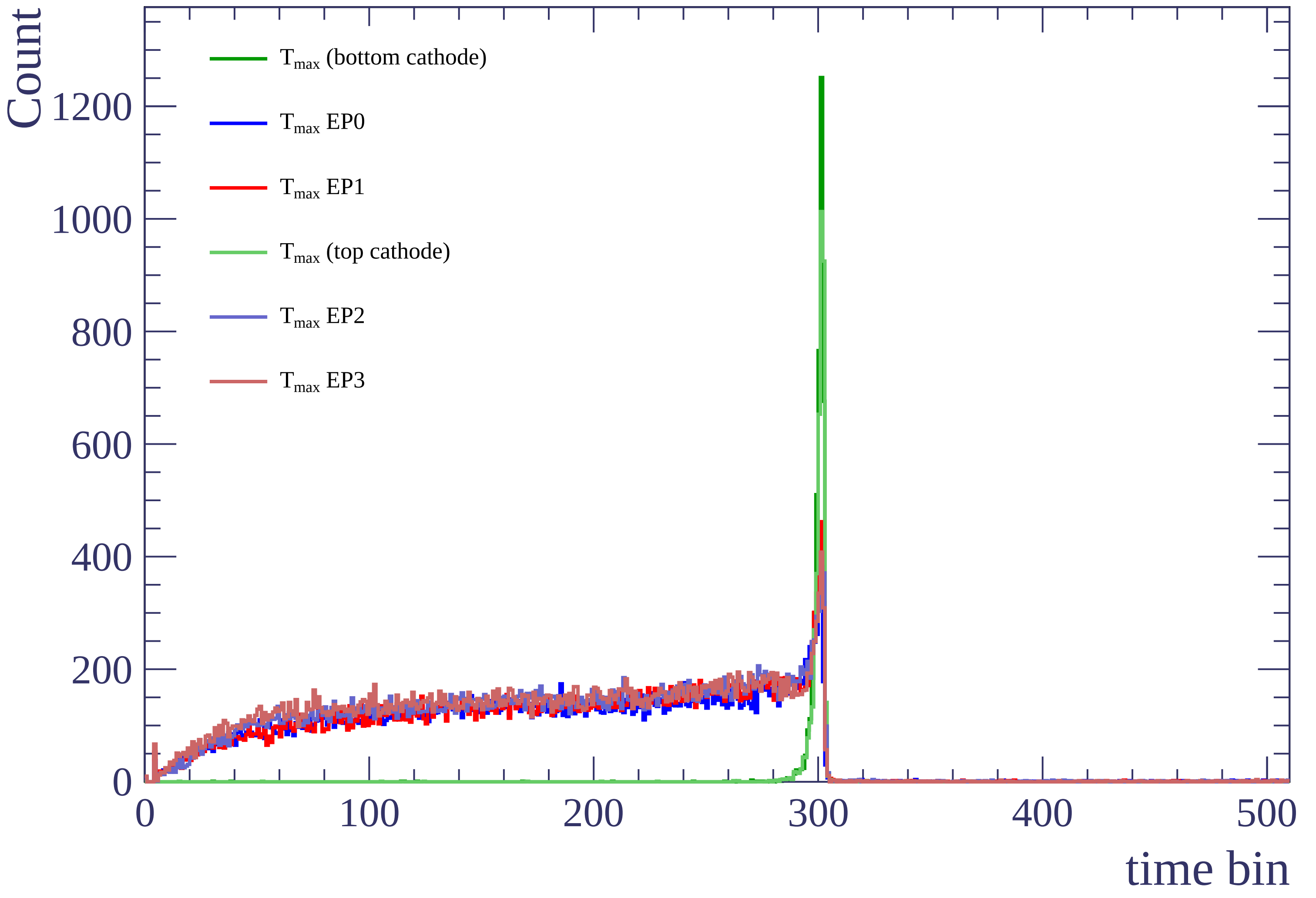

Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
500
1000
1500
2000
2500
3000
3500
4000
0
100
200
300
400
500



Count
1200
1000
800
600
400
200
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3